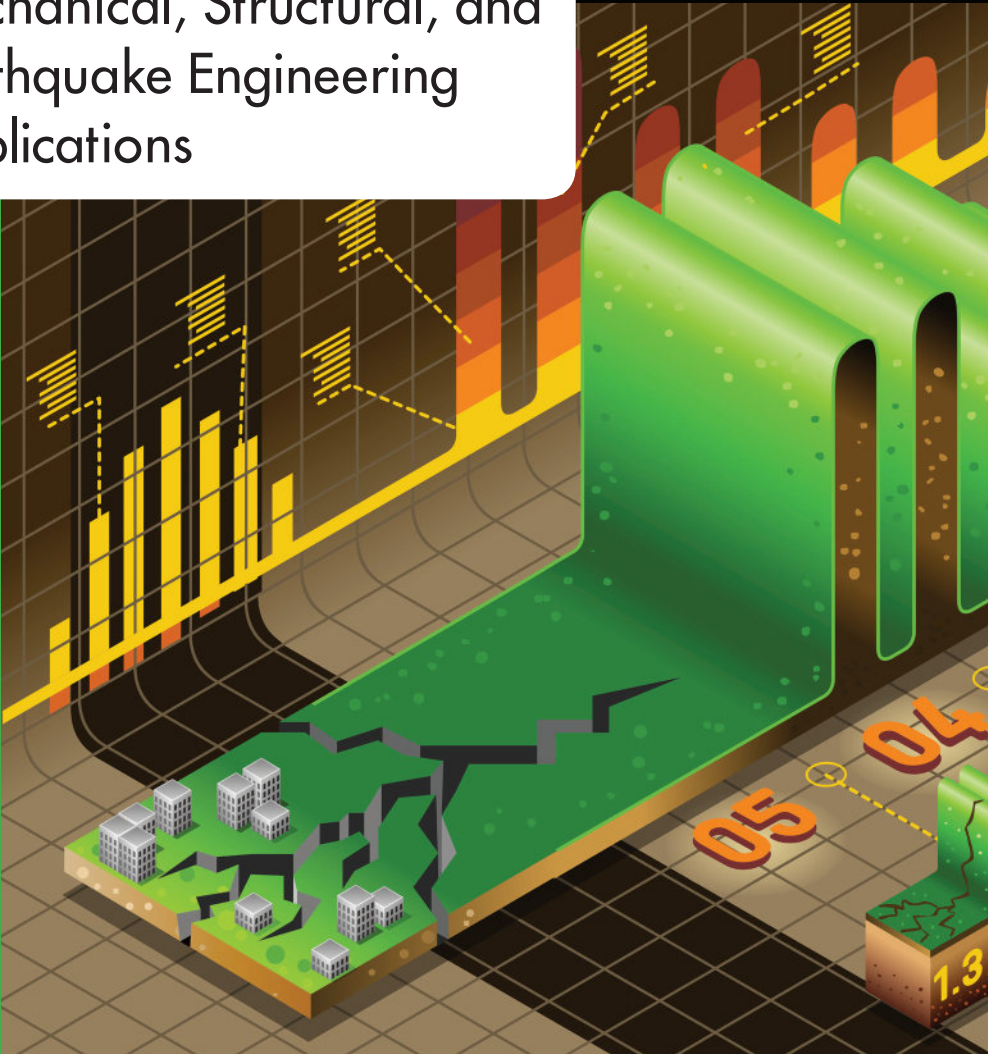


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Contents

Series Preface.....	xix
Preface.....	xxi
Acknowledgments.....	xxv
Series Editor.....	xxvii

SECTION I Basic Probability Theory

Chapter 1	Introduction	3
1.1	Background of Random Vibration	3
1.1.1	General Description	3
1.1.2	General Theory of Vibration.....	4
1.1.2.1	Concept of Vibration.....	4
1.1.3	Arrangement of Chapters	9
1.2	Fundamental Concept of Probability Theory.....	10
1.2.1	Set Theory	10
1.2.1.1	Basic Relationship (Operation)	11
1.2.2	Axioms of Probability	15
1.2.2.1	Random Tests and Classic Probability	15
1.2.2.2	Axiom of Probability	17
1.2.3	Conditional Probability and Independence.....	18
1.2.3.1	Conditional Probability.....	18
1.2.3.2	Multiplicative Rules.....	19
1.2.3.3	Independency	20
1.2.3.4	Total Probability and Bayes' Formula	21
1.2.3.5	Bayes' Formula	23
1.2.4	Engineering Examples	23
1.2.4.1	Additive Rules	23
1.2.4.2	Multiplication Rules.....	25
1.2.4.3	Independent Series.....	25
1.2.4.4	Return Period of Extreme Load.....	26
1.3	Random Variables	29
1.3.1	Discrete Random Variables and PMF.....	29
1.3.1.1	Single Random Variables	29
1.3.1.2	“Two-Dimensional” Approach	30
1.3.1.3	Probability Mass Function.....	30
1.3.1.4	Bernoulli Distribution (0–1 Distribution)....	30
1.3.1.5	Binomial Distribution	31
1.3.1.6	Poisson Distribution.....	32

1.3.1.7	Poisson Approximation.....	33
1.3.1.8	Summary of PMF $P_N(n)$	35
1.3.2	Continuous Random Variables and PDF.....	35
1.3.2.1	Continuous Random Variables	35
1.3.2.2	Probability Density Function.....	36
1.3.2.3	Uniform Distribution	38
1.3.2.4	Exponential Distribution.....	39
1.3.2.5	Rayleigh Distribution.....	42
1.3.3	Cumulative Distribution Functions	42
1.3.3.1	Probability of Cumulative Event	42
1.3.3.2	Cumulative Distribution Function (CDF)....	43
1.3.3.3	Certain Applications of PDF and CDF.....	44
1.3.4	Central Tendency and Dispersion	45
1.3.4.1	Statistical Expectations and Moments.....	45
1.3.4.2	Central Tendency, Mean Value.....	45
1.3.4.3	Variation, Variance, Standard Deviation, and Coefficient of Variation.....	46
1.3.4.4	Expected Values	47
1.3.4.5	Linearity of Expected Values	47
1.3.5	Normal Random Distributions.....	48
1.3.5.1	Standardized Variables Z	48
1.3.5.2	Gaussian (Normal) Random Variables.....	48
1.3.5.3	PDF of Normal Distribution.....	48
1.3.5.4	Cumulative Distribution Function of Normal Distribution.....	49
1.3.6	Engineering Applications.....	51
1.3.6.1	Probability-Based Design	51
1.3.6.2	Lognormal Distributions	53
1.3.6.3	Further Discussion of Probability- Based Design	55
Chapter 2	Functions of Random Variables	59
2.1	Systems and Functions	59
2.1.1	Dynamic Systems.....	59
2.1.1.1	Order of Systems	59
2.1.1.2	Simple Systems.....	60
2.1.2	Jointly Distributed Variables.....	61
2.1.2.1	Joint and Marginal Distributions of Discrete Variables.....	61
2.1.2.2	Joint and Marginal Distributions of Continuous Variables.....	63
2.1.3	Conditional Distribution and Independence.....	66
2.1.3.1	Discrete Variables.....	66
2.1.3.2	Continuous Variables.....	68
2.1.3.3	Variable Independence	68

2.1.4	Expected Value, Variance, Covariance, and Correlation.....	70
2.1.4.1	Expected Value of $g(X,Y)$	70
2.1.4.2	Conditional Expected Value	71
2.1.4.3	Variance.....	71
2.1.4.4	Covariance of X,Y	71
2.1.4.5	Correlation Coefficient	71
2.1.5	Linear Independence	72
2.1.5.1	Relationship between Random Variables X and Y	72
2.1.5.2	Expected Value of Sum of Random Variables X and Y	73
2.1.6	CDF and PDFs of Random Variables.....	73
2.1.6.1	Discrete Variables.....	74
2.1.6.2	Continuous Variables.....	76
2.2	Sums of Random Variables	80
2.2.1	Discrete Variables	80
2.2.2	Continuous Variables	81
2.2.2.1	Sums of Normally Distributed PDF	82
2.2.2.2	Sums of n Normally Distributed Variable ...	83
2.3	Other Functions of Random Variables	84
2.3.1	Distributions of Multiplication of X and Y	84
2.3.2	Distributions of Sample Variance, Chi-Square (χ^2) ...	85
2.3.2.1	Sample Variance	85
2.3.2.2	Chi-Square Distribution.....	86
2.3.2.3	CDF of Chi-Square, $n = 1$	86
2.3.2.4	PDF of Chi-Square, $n = 1$	86
2.3.2.5	Mean	86
2.3.2.6	Variance.....	86
2.3.2.7	PDF of Chi-Square, $n > 1$	86
2.3.2.8	Reproductive	87
2.3.2.9	Approximation	87
2.3.2.10	Mean of Y	87
2.3.2.11	Variance of Y	87
2.3.2.12	Square Root of Chi-Square (χ^2).....	88
2.3.2.13	Gamma Distribution and Chi-Square Distribution	88
2.3.2.14	Relation between Chi-Square χ_n^2 and Sample Variance S_X^2	89
2.3.3	Distributions of Ratios of Random Variables	90
2.3.3.1	Distribution of Variable Ratios	90
2.3.3.2	Student's Distribution	90
2.3.3.3	F Distribution.....	91
2.4	Design Considerations.....	92
2.4.1	Further Discussion of Probability-Based Design	92
2.4.2	Combination of Loads	95

2.5	Central Limit Theorems and Applications.....	97
2.5.1	Central Limit Theorems.....	98
2.5.1.1	Lyapunov Central Limit Theorem	98
2.5.1.2	Lindeberg–Levy Central Limit Theorem ...	99
2.5.1.3	De Moirve–Laplace Central Limit Theorem.....	100
2.5.2	Distribution of Product of Positive Random Variables.....	102
2.5.3	Distribution of Extreme Values.....	103
2.5.3.1	CDF and PDF of Distribution of Extreme Values.....	103
2.5.4	Special Distributions	104
2.5.4.1	CDF and PDF of Extreme Value of Rayleigh Distributions	104
2.5.4.2	Extreme Value Type I Distribution.....	104
2.5.4.3	Distribution of Minimum Values.....	107
2.5.4.4	Extreme Value Type II Distribution	108
2.5.4.5	Extreme Value Type III Distribution	109

SECTION II Random Process

Chapter 3	Random Processes in the Time Domain.....	115
3.1	Definitions and Basic Concepts.....	115
3.1.1	State Spaces and Index Sets	115
3.1.1.1	Definition of Random Process.....	115
3.1.1.2	Classification of Random Process	116
3.1.1.3	Distribution Function of Random Process...	117
3.1.1.4	Independent Random Process.....	121
3.1.2	Ensembles and Ensemble Averages.....	123
3.1.2.1	Concept of Ensembles.....	123
3.1.2.2	Statistical Expectations and Moments.....	124
3.1.3	Stationary Process and Ergodic Process	129
3.1.3.1	Stationary Process	129
3.1.3.2	Ergodic Process	133
3.1.4	Examples of Random Process.....	134
3.1.4.1	Gaussian Process	135
3.1.4.2	Poisson Process.....	136
3.1.4.3	Harmonic Process.....	142
3.2	Correlation Analysis.....	144
3.2.1	Cross-Correlation	144
3.2.1.1	Cross-Correlation Function	145
3.2.1.2	Cross-Covariance Function	146
3.2.2	Autocorrelation.....	147
3.2.2.1	Physical Meaning of Correlation.....	147

	3.2.2.2	Characteristics of Autocorrelation Function	148
	3.2.2.3	Examples of Autocorrelation Function	152
3.2.3		Derivatives of Stationary Process	154
	3.2.3.1	Stochastic Convergence	154
	3.2.3.2	Mean-Square Limit	155
	3.2.3.3	Mean-Square Continuity	157
	3.2.3.4	Mean-Square Derivatives of Random Process	159
	3.2.3.5	Derivatives of Autocorrelation Functions	159
	3.2.3.6	Derivatives of Stationary Process	161
	3.2.3.7	Derivatives of Gaussian Process	162
Chapter 4		Random Processes in the Frequency Domain	165
4.1		Spectral Density Function	165
4.1.1		Definitions of Spectral Density Functions	165
	4.1.1.1	Mean-Square Integrable of Random Process	165
	4.1.1.2	Stationary Process: A Review	169
	4.1.1.3	Autospectral Density Functions	170
	4.1.1.4	Spectral Distribution Function $\Psi(\omega)$	175
	4.1.1.5	Properties of Auto-PSD Functions	176
4.1.2		Relationship with Fourier Transform	179
	4.1.2.1	Fourier Transformation Random Process	179
	4.1.2.2	Energy Equation	179
	4.1.2.3	Power Density Functions	182
4.1.3		White Noise and Band-Pass Filtered Spectra	184
	4.1.3.1	White Noise	184
	4.1.3.2	Low-Pass Noise	186
	4.1.3.3	Band-Pass Noise	187
	4.1.3.4	Narrow-Band Noise	188
4.2		Spectral Analysis	188
4.2.1		Definition	188
	4.2.1.1	Cross-Power Spectral Density Function	188
	4.2.1.2	Estimation of Cross-PSD Function	190
4.2.2		Transfer Function	191
	4.2.2.1	Random Process through Linear Systems	191
	4.2.2.2	Estimation of Transfer Functions	197
	4.2.2.3	Stationary Input	199
4.2.3		Coherence Analysis	199
	4.2.3.1	Coherence Function	200
	4.2.3.2	Attenuation and Delay	201
	4.2.3.3	Sum of Two Random Processes	201
4.2.4		Derivatives of Stationary Process	202

4.3	Practical Issues of PSD Functions	203
4.3.1	One-Sided PSD.....	203
4.3.1.1	Angular Frequency versus Frequency	203
4.3.1.2	Two-Sided Spectrum versus Single-Sided Spectrum.....	204
4.3.1.3	Discrete Fourier Transform	204
4.3.2	Signal-to-Noise Ratios	205
4.3.2.1	Definition	205
4.3.2.2	Engineering Significances	207
4.4	Spectral Presentation of Random Process.....	208
4.4.1	General Random Process	208
4.4.2	Stationary Process.....	210
4.4.2.1	Dynamic Process in the Frequency Domain	210
4.4.2.2	Relationship between the Time and the Frequency Domains.....	210
4.4.2.3	Spectral Distribution and Representation....	213
4.4.2.4	Analogy of Spectral Distribution Function to CDF	216
4.4.2.5	Finite Temporal and Spectral Domains....	217
Chapter 5	Statistical Properties of Random Process	221
5.1	Level Crossings	222
5.1.1	Background	222
5.1.1.1	Number of Level Crossings	222
5.1.1.2	Correlations between Level Crossings.....	224
5.1.2	Derivation of Expected Rate	224
5.1.2.1	Stationary Crossing.....	224
5.1.2.2	Up-Crossing.....	225
5.1.2.3	Limiting Behavior.....	225
5.1.3	Specializations	227
5.1.3.1	Level Up-Crossing, Gaussian Process.....	227
5.1.3.2	Zero Up-Crossing	228
5.1.3.3	Peak Frequency.....	229
5.1.3.4	Bandwidth and Irregularity	230
5.1.4	Random Decrement Methods.....	232
5.1.4.1	Random Decrement (Level Up-Crossing)...	232
5.1.4.2	Lag Superposition (Zero Up-Crossing)	235
5.1.4.3	Lag Superposition (Peak Reaching)	237
5.1.5	Level Crossing in Clusters.....	238
5.1.5.1	Rice's Narrow-Band Envelopes	238
5.2	Extrema	243
5.2.1	Distribution of Peak Values.....	243
5.2.1.1	Simplified Approach.....	243
5.2.1.2	General Approach.....	245

5.2.2	Engineering Approximations	247
5.2.2.1	Background.....	247
5.2.2.2	Probability Distributions of Height, Peak, and Valley	249
5.3	Accumulative Damages.....	252
5.3.1	Linear Damage Rule: The Deterministic Approach	253
5.3.1.1	S–N Curves.....	253
5.3.1.2	Miner’s Rule	253
5.3.2	Markov Process	254
5.3.2.1	General Concept	255
5.3.2.2	Discrete Markov Chain.....	255
5.3.3	Fatigue.....	263
5.3.3.1	High-Cycle Fatigue.....	263
5.3.3.2	Low-Cycle Fatigue.....	266
5.3.4	Cascading Effect	270
5.3.4.1	General Background.....	270
5.3.4.2	Representation of Random Process	271
5.3.4.3	Occurrence Instance of Maximum Load	273

SECTION III *Vibrations*

Chapter 6	Single-Degree-of-Freedom Vibration Systems	279
6.1	Concept of Vibration	280
6.1.1	Basic Parameters	280
6.1.1.1	Undamped Vibration Systems	281
6.1.1.2	Damped SDOF System.....	291
6.1.2	Free Decay Response	297
6.1.2.1	Amplitude d and Phase ϕ	297
6.2	Periodically Forced Vibration	301
6.2.1	Harmonic Excitation	301
6.2.1.1	Equation of Motion.....	301
6.2.1.2	Harmonically Forced Response.....	302
6.2.1.3	Dynamic Magnification.....	307
6.2.1.4	Transient Response under Zero Initial Conditions	308
6.2.2	Base–Excitation and Force Transmissibility.....	313
6.2.2.1	Model of Base Excitation.....	313
6.2.2.2	Force Transmissibility	317
6.2.3	Periodic Excitations.....	319
6.2.3.1	General Response	319
6.2.3.2	The n th Steady-State Response	320
6.2.3.3	Transient Response	321

6.3	Response of SDOF System to Arbitrary Forces	321
6.3.1	Impulse Responses	321
6.3.1.1	Unit Impulse Response Function.....	322
6.3.2	Arbitrary Loading and Convolution.....	323
6.3.2.1	Convolution.....	323
6.3.2.2	Transient Response under Harmonic Excitation $f_0 \sin(\omega t)$	325
6.3.3	Impulse Response Function and Transfer Function...	327
6.3.4	Frequency Response and Transfer Functions	329
6.3.5	Borel's Theorem and Its Applications	330
6.3.5.1	Borel's Theorem.....	330
Chapter 7	Response of SDOF Linear Systems to Random Excitations.....	335
7.1	Stationary Excitations.....	335
7.1.1	Model of SDOF System	335
7.1.1.1	Equation of Motion.....	335
7.1.1.2	Zero Initial Conditions	335
7.1.1.3	Solution in Terms of Convolution	336
7.1.1.4	Nature of Forcing Function	336
7.1.1.5	Response	336
7.1.2	Mean of Response Process	336
7.1.3	Autocorrelation of Response Process.....	337
7.1.3.1	Autocorrelation	337
7.1.3.2	Mean Square	338
7.1.4	Spectral Density of Response Process	341
7.1.4.1	Auto-Power Spectral Density Function	341
7.1.4.2	Variance.....	342
7.1.5	Distributions of Response Process	342
7.2	White Noise Process.....	342
7.2.1	Definition.....	342
7.2.2	Response to White Noise	343
7.2.2.1	Auto-PSD Function.....	343
7.2.2.2	Variance.....	343
7.2.3	White Noise Approximation	345
7.3	Engineering Examples.....	346
7.3.1	Comparison of Excitations	346
7.3.1.1	Harmonic Excitation.....	346
7.3.1.2	Impulse Excitation	348
7.3.1.3	Random Excitation	351
7.3.1.4	Other Excitations	353
7.3.2	Response Spectra.....	355
7.3.2.1	Response Spectrum	355
7.3.2.2	Design Spectra	356
7.3.3	Criteria of Design Values	357
7.3.3.1	Pseudo Spectrum	357

	7.3.3.2	Correlation of Acceleration and Displacement	358
7.4		Coherence Analyses	359
	7.4.1	Estimation of Transfer Function.....	359
	7.4.2	Coherence Function.....	363
	7.4.3	Improvement of Coherence Functions	365
7.5		Time Series Analysis	365
	7.5.1	Time Series.....	366
	7.5.1.1	General Description.....	366
	7.5.1.2	Useful Models of Time Series	366
	7.5.2	Characters of ARMA Models	367
	7.5.2.1	Moving-Average Process $MA(q)$	367
	7.5.2.2	Autoregressive Process $AR(p)$	370
	7.5.2.3	$ARMA(p, q)$	372
	7.5.3	Analyses of Time Series in the Frequency Domain	376
	7.5.3.1	Z-Transform	376
	7.5.3.2	Sampling of Signals	377
	7.5.3.3	Transfer Function of Discrete Time System.....	378
	7.5.3.4	PSD Functions	379
	7.5.4	Time Series of SDOF Systems	380
	7.5.4.1	Difference Equations	380
	7.5.4.2	ARMA Models	382
	7.5.4.3	Transfer Functions	383
	7.5.4.4	Stability of Systems	385
Chapter 8		Random Vibration of MDOF Linear Systems	391
8.1		Modeling.....	391
	8.1.1	Background	391
	8.1.1.1	Basic Assumptions.....	391
	8.1.1.2	Fundamental Approaches	392
	8.1.2	Equation of Motion	392
	8.1.2.1	Physical Model.....	392
	8.1.2.2	Stiffness Matrix	395
	8.1.2.3	Mass and Damping Matrices	396
	8.1.3	Impulse Response and Transfer Functions.....	398
	8.1.3.1	Scalar Impulse Response Function and Transfer Function.....	398
	8.1.3.2	Impulse Response Matrix and Transfer Function Matrix	399
	8.1.3.3	Construction of Transfer Functions	399
	8.1.3.4	Principal Axes of Structures.....	400
8.2		Direct Model for Determining Responses.....	400
	8.2.1	Expression of Response.....	400
	8.2.2	Mean Values	401

	8.2.2.1	Single Coordinate	401
	8.2.2.2	Multiple Coordinates	402
8.2.3		Correlation Functions	404
8.2.4		Spectral Density Function of Response	405
	8.2.4.1	Fourier Transforms of $\mathbf{f}(t)$ and $\mathbf{x}(t)$	405
	8.2.4.2	Power Spectral Density Function	405
	8.2.4.3	Mean Square Response.....	407
	8.2.4.4	Variance	407
	8.2.4.5	Covariance	407
8.2.5		Single Response Variable: Spectral Cases	407
	8.2.5.1	Single Input.....	407
	8.2.5.2	Uncorrelated Input	408
8.3		Normal Mode Method	408
	8.3.1	Proportional Damping.....	408
		8.3.1.1 Essence of Caughey Criterion	408
		8.3.1.2 Monic System	409
	8.3.2	Eigen-Problems	410
		8.3.2.1 Undamped System	410
		8.3.2.2 Underdamped Systems	411
	8.3.3	Orthogonal Conditions	411
		8.3.3.1 Weighted Orthogonality	412
		8.3.3.2 Modal Analysis.....	413
	8.3.4	Modal Superposition	416
	8.3.5	Forced Response and Modal Truncation.....	418
		8.3.5.1 Forced Response	418
		8.3.5.2 Rayleigh Quotient	419
		8.3.5.3 Ground Excitation and Modal Participation Factor.....	420
		8.3.5.4 Modal Superposition, Forced Vibration ...	420
		8.3.5.5 Modal Truncation	422
	8.3.6	Response to Random Excitations	423
		8.3.6.1 Modal and Physical Response	423
		8.3.6.2 Mean	424
		8.3.6.3 Covariance	424
		8.3.6.4 Probability Density Function for $x_i(t)$	426
8.4		Nonproportionally Damped Systems, Complex Modes	428
	8.4.1	Nonproportional Damping	428
		8.4.1.1 Mathematical Background.....	428
		8.4.1.2 The Reality of Engineering	429
	8.4.2	State Variable and State Equation	429
	8.4.3	Eigen-Problem of Nonproportionally Damped System	430
		8.4.3.1 State Matrix and Eigen-Decomposition ...	430
		8.4.3.2 Eigenvectors and Mode Shapes	432
		8.4.3.3 Modal Energy Transfer Ratio	435

8.4.4	Response to Random Excitations	437
8.4.4.1	Modal and Physical Response	438
8.4.4.2	Mean	439
8.4.4.3	Covariance	440
8.4.4.4	Brief Summary	446
8.5	Modal Combination.....	446
8.5.1	Real Valued Mode Shape	446
8.5.1.1	Approximation of Real Valued Mode Shape.....	446
8.5.1.2	Linear Dependency and Representation...	447
8.5.2	Numerical Characteristics.....	449
8.5.2.1	Variance.....	449
8.5.2.2	Root Mean Square	449
8.5.3	Combined Quadratic Combination	451

SECTION IV Applications and Further Discussions

Chapter 9 Inverse Problems 459

9.1	Introduction to Inverse Problems	459
9.1.1	Concept of Inverse Engineering.....	459
9.1.1.1	Key Issues	459
9.1.1.2	Error.....	460
9.1.1.3	Applications	461
9.1.2	Issues of Inverse Problems	461
9.1.2.1	Modeling.....	461
9.1.2.2	Identification, Linear System.....	463
9.1.2.3	Identification, General System.....	463
9.1.2.4	Simulations	464
9.1.2.5	Practical Considerations	464
9.1.3	The First Inverse Problem of Dynamic Systems.....	465
9.1.3.1	General Description.....	465
9.1.3.2	Impulse Response	466
9.1.3.3	Sinusoidal Response	466
9.1.3.4	Random Response	466
9.1.3.5	Modal Model	468
9.1.4	The Second Inverse Problem of Dynamic Systems...	471
9.1.4.1	General Background.....	471
9.1.4.2	White Noise	472
9.1.4.3	Practical Issues	472
9.2	System Parameter Identification.....	472
9.2.1	Parameter Estimation, Random Set	473
9.2.1.1	Maximum Likelihood.....	473
9.2.1.2	Bias and Consistency	479

9.2.2	Confidence Intervals.....	481
9.2.2.1	Estimation and Sampling Distributions....	481
9.2.3	Parameter Estimation, Random Process	482
9.2.3.1	General Estimation	482
9.2.3.2	Stationary and Ergodic Process.....	487
9.2.3.3	Nonstationary Process	489
9.2.4	Least Squares Approximation and Curve Fitting	492
9.2.4.1	Concept of Least Squares	492
9.2.4.2	Curve Fitting.....	493
9.2.4.3	Realization of Least Squares Method.....	494
9.3	Vibration Testing	495
9.3.1	Test Setup	495
9.3.1.1	Mathematical Model.....	495
9.3.1.2	Numerical Model.....	496
9.3.1.3	Experimental Model	496
9.3.2	Equipment of Actuation and Measurement.....	497
9.3.2.1	Actuation.....	497
9.3.2.2	Measurement.....	501
9.3.3	Signal and Signal Processing	505
9.3.3.1	Data-Acquisition System	505
9.3.3.2	Single Processing and Window Functions...	507
9.3.4	Nyquist Circle.....	511
9.3.4.1	Circle and Nyquist Plot.....	511
9.3.4.2	Circle Fit	513
9.3.4.3	Natural Frequency and Damping Ratio....	515

Chapter 10 Failures of Systems 519

10.1	3σ Criterion	519
10.1.1	Basic Design Criteria	519
10.1.2	3σ Criterion	520
10.1.3	General Statement	520
10.1.3.1	General Relationship between S and R	520
10.1.3.2	System Failure, Further Discussion	522
10.2	First Passage Failure.....	522
10.2.1	Introduction	523
10.2.2	Basic Formulation	523
10.2.2.1	General Formulation.....	523
10.2.2.2	Special Cases	524
10.2.3	Largest among Independent Peaks.....	525
10.2.3.1	Exact Distribution	525
10.2.3.2	Extreme Value Distribution	527
10.2.3.3	Design Value Based on Return Period	529
10.3	Fatigue	529
10.3.1	Physical Process of Fatigue.....	529

10.3.2	Strength Models	530
10.3.2.1	High-Cycle Fatigue	530
10.3.2.2	Miner's Rule, More Detailed Discussion ...	531
10.3.3	Fatigue Damages	532
10.3.3.1	Narrowband Random Stress	532
10.3.3.2	Wideband Random Stress	538
10.3.4	Damages due to Type D Low Cycle	542
10.3.4.1	Fatigue Ductility Coefficient	543
10.3.4.2	Variation of Stiffness	543
10.4	Considerations on Reliability Design	545
10.4.1	Further Discussion of Probability-Based Design	545
10.4.1.1	Random Variable vs. Random Process	545
10.4.1.2	Necessity of Distinguishing Time-Invariant and Time-Variable Loads	546
10.4.1.3	Time-Variable Load at a Given Time Spot	547
10.4.1.4	Combination of Time-Variable Loads in a Given Time Period	547
10.4.1.5	Additivity of Distribution Functions	548
10.4.2	Failure Probability under MH Load	549
10.4.2.1	Failure Probability Computation	549
10.4.2.2	Time-Invariant and -Variable Loads	549
10.4.2.3	Principles of Determining Load and Load Combination	549
10.4.2.4	Total and Partial Failure Probabilities	550
10.4.2.5	Independent Events	550
10.4.2.6	Mutually Exclusive Failures, the Uniqueness Probabilities	552
10.4.3	General Formulations	556
10.4.3.1	Total Failure Probability	556
10.4.3.2	Occurrence of Loads in a Given Time Duration	556
10.4.3.3	Brief Summary	556
10.4.4	Probability of Conditions	557
10.4.4.1	Condition for Occurrence of Partial Failure Probabilities	558
10.4.4.2	Event of Single Type of Loads	558
10.4.5	Brief Summary	571
Chapter 11	Nonlinear Vibrations and Statistical Linearization	575
11.1	Nonlinear Systems	575
11.1.1	Examples of Nonlinear Systems	575
11.1.1.1	Nonlinear System	576
11.1.1.2	Memoryless Nonlinear System	578

11.1.2	General Nonlinear System, Volterra Model	579
11.1.3	Structure Nonlinearity	579
11.1.3.1	Deterministic Nonlinearity	579
11.1.3.2	Random Nonlinearity	585
11.2	Nonlinear Random Vibrations	594
11.2.1	General Concept of Nonlinear Vibration	594
11.2.1.1	The Phase Plane.....	595
11.2.1.2	Example of Nonlinear Vibration with Closed-Form Solution.....	596
11.2.1.3	System with Nonlinear Damping Only	601
11.2.1.4	System with Nonlinear Spring.....	601
11.2.2	Markov Vector.....	602
11.2.2.1	Itô Diffusion and Kolmogorov Equations ..	602
11.2.2.2	Solutions of FPK Equation	603
11.2.3	Alternative Approaches.....	605
11.2.3.1	Linearization.....	605
11.2.3.2	Perturbation	606
11.2.3.3	Special Nonlinearization	606
11.2.3.4	Statistical Averaging.....	606
11.2.3.5	Numerical Simulation.....	606
11.3	Monte Carlo Simulations.....	607
11.3.1	Basics of Monte Carlo Method.....	607
11.3.1.1	Applications	609
11.3.2	Monte Carlo and Random Numbers	613
11.3.2.1	Generation of Random Numbers.....	613
11.3.2.2	Transformation of Random Numbers	614
11.3.2.3	Random Process	617
11.3.3	Numerical Simulations.....	617
11.3.3.1	Basic Issues.....	617
11.3.3.2	Deterministic Systems with Random Inputs	619
11.3.3.3	Random Systems	619
	References	625

Series Preface

The editor takes pride in presenting another well-developed manuscript in the series. This excellent book is a result of the authors' more than two decades of extensive research and teaching on the subject of random vibrations related to earthquake structural response and multiple hazard mitigations of structural engineering. Since natural hazards rarely occur, the associated solutions must be based on probability criteria in the random process. Current random vibration books, however, are focused on conventional engineering problems, and consequently the authors coupled the traditional solution techniques with their research and teaching experiences to shape up this 11-chapter textbook. They intend to assist the reader in utilizing the traditional mathematical logic and then comprehensively and effectively solving more complex problems.

In earthquake engineering applications, the authors include some difficult and currently in vogue structural problems, such as load-and-resistance factor design under multiple hazard load effects, nonlinear vibration with random excitation, etc. These advanced topics are valuable and are believed to lead the researchers and practitioners to pursue further in the constructed facility community.

The book is aimed at one semester graduate class; in order for the reader to clearly grasp the concepts and mathematical formulation, the authors developed extensive homework problems for individual chapters accompanied by detailed solutions. The editor strongly suggests that the reader should patiently and gradually digest the materials in the book with the assistance of the solution manual. Note that a comprehensive solution manual is seldom available for other books of this nature that reflects the authors' admirable objective of preparing this manuscript, and the book is believed to be useful for years to come.

Preface

Understanding and modeling a vibration system and measuring and controlling its oscillation responses are important basic capacities for mechanical, structural, and earthquake engineers who deal with the dynamic responses of mechanical/structural systems. Generally speaking, this ability requires three components: the basic theories of vibrations, experimental observations, and measurement of dynamic systems and analyses of the time-varying responses.

Among these three efforts, the former two are comparatively easily learned by engineering students. However, the third component often requires a mathematical background of random processes, which is rather abstract for students to grasp. One course covering stochastic processes and random vibrations with engineering applications is already too much for students to absorb because it is mathematically intensive and requires students to follow an abstract thinking path through “pure” theories without practical examples. To carry out a real-world modeling and analysis of specific types of vibration systems while following through the abstract pure thinking path of mathematical logic would require an additional course; however, there is no room in curriculums for such a follow-up course. This has been the observation of the first author during many years of teaching random vibration. He frequently asked himself, How can one best teach the material of all three components in a one-semester course?

The authors, during the past 20 years, have engaged in an extensive research study to formulate bridge design limit states; first, for earthquake hazard and, subsequently, expanded to multiple extreme natural hazards for which the time-varying issue of rare-occurring extreme hazard events (earthquakes, flood, vehicular and vessel collisions, etc.) had to be properly addressed. This experience of formulating real-world failure probability-based engineering design criteria provided nice examples of using the important basic ideas and principles of random process (e.g., correlation analysis, the basic relationship of the Wiener–Khinchine formula to transfer functions, the generality of orthogonal functions and vibration modes, and the principles and approaches of dealing with engineering random process). We thus decided to emphasize the methodology of dealing with random vibration. In other words, we have concluded that it is possible to offer a meaningful course in random vibration to students of mechanical and structural engineering by changing the knowledge-based course approach into a methodology-based approach. The course will guide them in understanding the essence of vibration systems, the fundamental differences in analyzing the deterministic and dynamic responses, the way to handle random variables, and the way to account for random process. This is the basic approach that underlines the material developed in this book. By doing so, we give up coverage of the rigorous mathematical logic aspect and greatly reduce the portion of random process. Instead, many real-world examples and practical engineering issues are used immediately following the abstract concepts and theories. As a result, students might gain the basic methodology to handle the generality of engineering

projects and develop a certain capability to establish their own logic to systematically handle the issues facing the theory and application of random vibrations. After such a course, students are not expected to be proficient in stochastic process and to model a random process, but they will be able to design the necessary measurement and observation, to understand the basic steps and validate the accuracy of dynamic analyses, and to master and apply newly developed knowledge in random vibrations and corresponding system reliabilities.

With this approach, we believe it is possible to teach students the fundamental methodology accounting for random data and random process and apply them in engineering practice. This is done in this book by embedding engineering examples wherever appropriate to illustrate the importance and approach to deal with randomness. The materials are presented in four sections. The first is a discussion of the scope of random process, including engineering problems requiring the concept of probability to deal with. The second is the overview of random process, including the time domain approach to define time-varying randomness, the frequency domain approach for the spectral analysis, and the statistical approach to account for the process. The third section is dedicated specifically to random vibrations, a typical dynamic process with randomness in engineering practice. The fourth section is the application of the methodology. In recent years, we used typical examples of developing fatigue design limit states for mechanical components and reliability-based extreme event design limit states for bridge components in teaching this course. The nice performances and positive responses of the students have encouraged us to prepare this manuscript.

Section I consists of two chapters. Chapter 1 expresses the brief background and the objectives of this book, followed by a brief review of the theory of probability within the context at application to engineering. The attempt is to only introduce basic concepts and formulas to prepare for discussions of random process. The review of the theory of probability is continued in Chapter 2, with focus on treating random data measured as a function of certain basic random distributions for randomness in their actual applications. This will also help engineers to gain a deeper understanding of the randomness in sequences. In this section, the essence of probability as the chance of occurrence in sample space, the basic treatment to handle one-dimensional random variables by using two-dimensional deterministic probability distributions (PDF), and the tools to study the random variables of averaging (statistics) that changes quantities from random to deterministic are emphasized. Two important issues in engineering practice, the uncertainty of data and the probability of failure, are introduced.

Section II begins with Chapter 3, where the random (also called stochastic) process is introduced in the time domain. The nature of time-varying variables is first explained by joint PDF through the Kolmogorov extension. Because of the existence of the indices in both sample space and in the time domain, the averages should be well defined, in other words, the statistics must be used in rigorous conditions by identifying if the process is stationary as well as ergodic. Although the averaged results of mean and variance are often easily understandable, the essence of correlation analysis is explained through the concept of function/variable orthogonality. In Chapter 4, random process is further examined in the frequency domain. Based on

the Wiener–Khinchine relations, the spectral analyses on the frequency components of the deterministic power spectrum density function of random process are carried out. In these two chapters, several basic and useful models of random process are discussed. In Chapter 5, a new set of statistics for random processes that is different from averaging of the entire process is introduced, such as level crossing, peaks, and maxima. To further understand the important engineering problems of cumulated damage, the Markov process is introduced, which is a continuous approach of introducing random processes based on engineering motivations. Thus, due to the nature of random processes, which consists of a broad range of rather different types of mathematical models, to introduce each special process one by one is not an effective approach for students to learn. This book employs an approach to present important processes within the context of practical engineering problems, whereas the generality of dealing with randomness and the difference between random variables and processes are included. Necessary mathematical logic, such as limits, differentiation, and integration on random variables is only considered for the purposes of understanding the nature of randomness.

Section III of this book focuses on the topic of vibration problems. The basic concept is reviewed in Chapter 6, where the essence of vibration is emphasized based on energy exchange. The basic parameters of the linear single-degree-of-freedom (SDOF) system are discussed, followed by the key issues of dynamic magnification factors, convolutions, and transfer functions. The topic in Chapter 7 is on SDOF systems excited by random initial conditions and forcing functions. Together with the aforementioned correlation and spectral analyses, a new method of random process referred to as time series is also described. In Chapter 8, the discussion is extended to linear multi-degree-of-freedom (MDOF) systems. The statistical analyses of direct approach based on model decoupling of proportionally and non-proportionally damped systems are discussed, along with basic knowledge of eigenparameters, Rayleigh quotient, state variables, and equation and transfer function matrices. Engineering examples of how to deal with random excitations, such as earthquake response spectrum and various types of white noises are considered for students to further gain insight into random processes and particular random vibrations. Vibration is a special dynamic process and possesses time histories, whereas a random process is also a dynamic process. In the third section of this book, we not only present the generality of these dynamic processes but also treat the vibration response as the output of a second-order linear system due to the input of a random process.

Section IV and the last part of the book provides more materials on the applications of random process and vibration. Chapter 9 is especially dedicated to inverse problems, which are limited to system and excitation identifications. In engineering practice, inverse problems can be much more difficult to solve, both due to possible dimension reductions and noise contaminations. Measurement and testing, especially on vibration systems, should deal with uncertainties. Based on the methodology learned from previous chapters, statistical studies on random data and model identifications are discussed. In Chapter 10, the failure of systems is further discussed in a more systematic fashion, followed by the concept of reliability. For mechanical engineering applications, high cycle fatigue failure is further considered

as a continuation of the topic in Chapter 5. For structural engineering application, the example of load-and-resistance-factor-design under multiple hazard load effects are considered to explain how to deal with load combinations of several random processes, which is essentially different from the currently used bridge code based on bridge reliability. In Chapter 11, nonlinear vibration with random excitation is briefly considered, along with an introduction of linearization procedure. Again, the purpose of this chapter is not to systematically describe the system and response non-linearity. Rather, it is intended to explain the nonlinear phenomena and the general approach of linearization. In addition, a special method of Monte Carlo simulation is considered as a tool to study complex systems and their responses.

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Acknowledgments

In this book, certain materials are presented following the approach used in *Random Vibrations, Theory, and Practice* (Wirsching et al. 2006) and *Random Vibration of Mechanical and Structural Systems* (Soong and Grigoriu 1995). In preparing for this manuscript, the authors benefited greatly from discussions with their research collaborators (Drs. John Kulicki, Jerry Shen, Jianwei Song, and Chao Huang) on the development of bridge design guidelines for earthquake hazard effect and for multiple extreme hazard load effects for bridges. The research projects were funded by the Federal Highway Administration and the National Science Foundation. In particular, we express our appreciation to Dr. Phillip Wen-Huei Yen for his constant advice and support. We also thank Zhongwang Dou for his very helpful work in preparing the solution manual.

Series Editor



Dr. Franklin Cheng earned a BS (1960) at the National Cheng-Kung University, Taiwan, and an MS (1962) at the University of Illinois at Urbana-Champaign. He gained industrial experience with C. F. Murphy and Sargent & Lundy in Chicago, Illinois. Dr. Cheng then earned a PhD (1966) in civil engineering at the University of Wisconsin, Madison. Dr. Cheng joined the University of Missouri, Rolla (now named Missouri University of Science and Technology) as assistant professor in 1966 and then associate professor and professor in 1969 and 1974, respectively. In 1987, the board of curators of the univer-

sity appointed him curators' professor, the highest professorial position in the system comprising four campuses. He has been Curators' Professor Emeritus of Civil Engineering since 2000. In 2007, the American Society of Civil Engineers recognized Dr. Cheng's accomplishments by electing him to honorary membership, which is now renamed as distinguished membership. Honorary membership is the highest award the society may confer, second only to the title of ASCE president. Honorary members on this prestigious and highly selective list are those who have attained acknowledged eminence in a branch of engineering or its related arts and sciences. Until 2007, there have been 565 individuals who were elected to this distinguished grade of membership since 1853. For the year 2007, only 10 honorary members were selected from more than 14,000 members.

Dr. Cheng was honored for his significant contributions to earthquake structural engineering, optimization, nonlinear analysis, and smart structural control and for his distinguished leadership and service in the international engineering community, as well as for being a well-respected educator, consultant, author, editor, and member of numerous professional committees and delegations. His cutting-edge research helped recognize the vital importance of the possibilities of automatic computing in the future of civil engineering. He was one of the pioneers in allying computing expertise to the design of large and complex structures against dynamic loads. His research expanded over the years to include the important topics of structural optimization and design of smart structures. In fact, he is one of the foremost experts in the world on the application of structural dynamics and optimization to the design of structures. Due to the high caliber and breadth of his research expertise, Dr. Cheng has been regularly invited to serve on the review panels for the National Science Foundation (NSF), hence setting the direction of future structural research. In addition, he has been instrumental in helping the NSF develop collaborative research programs with Europe, China, Taiwan, Japan, and South Korea. Major industrial corporations and government agencies have sought Dr. Cheng's consultancy. He has consulted with Martin Marietta Energy Systems, Inc., Los Alamos National Laboratory, Kjaima Corporation, Martin & Huang International, Inc., and others.

Dr. Cheng received four honorary professorships from China and chaired 7 of his 24 NSF delegations to various countries for research cooperation. He is the author of more than 280 publications, including 5 textbooks: *Matrix Analysis of Structural Dynamics: Applications and Earthquake Engineering*, *Dynamic Structural Analysis*, *Smart Structures: Innovative Systems for Seismic Response Control*, *Structure Optimization—Dynamic and Seismic Applications*, and *Seismic Design Aids for Nonlinear Pushover Analysis of Reinforced Concrete and Steel Bridges*. Dr. Cheng has received numerous honors and awards, including Chi Epsilon, MSM-UMR Alumni Merit for Outstanding Accomplishments, Faculty Excellence Award, Halliburton Excellence Award, and recognitions in 21 biographical publications, such as *Who's Who in Engineering* and *Who's Who in the World*. He has twice received the ASCE State-of-the-Art Award, in 1998 and 2004.

Section I

Basic Probability Theory

1 Introduction

1.1 BACKGROUND OF RANDOM VIBRATION

1.1.1 GENERAL DESCRIPTION

This manuscript provides basic materials for an introductory course level on random vibrations, including a review of probability theory; concepts of summation, multiplication, and general functions of random variables; descriptions of random processes and their origins; responses of single and multiple degrees-of-freedom linear systems such as machines and structures due to transient and random excitations; and analyses in the time and frequency domains for reliability design, system identifications, vibration testing, and control in engineering applications.

The readers are assumed to be familiar with the concepts and theories of regular vibrations. Thus, these concepts and theories will not be systematically described. In a regular vibration system, although the signal is a function of time, that is, the vibration will vary from time to time, at a given time point, we can predict the exact value of the regular vibration.

However, fundamentally different from deterministic vibration, random vibration is a random process, which means, first of all, that the value of the vibration, no matter if it is denoted by displacement or other quantities, is unpredictable; and second, the vibration, for example, a displacement, will also be a function of time. It is also assumed that the readers are familiar with basic knowledge of random variables and the theory of probability. Therefore, a very brief review of certain portions of the entire theory of probability is given in this manuscript. That is, only foci on necessary ideas and models that closely relate to random process are given.

Assume that a set of random variables are of the same type, for example, a vibration displacement with identical units (e.g., inches). In other words, we can use a single dimension to place all these variables. The basic understanding of this set of random variables is to arrange them from the smallest to the largest, and at each given value of these variables, we try to find the occurrence probability. The corresponding idea is referred to as the probability density function (PDF), which is a deterministic function. Graphically speaking, a PDF is a two-dimensional plot to describe random distributions, that is, the abscissa is the value of the variables and the coordinate is the probability density. Thus, the basic treatment of random variables is to use a two-dimensional deterministic function to describe the one-dimensional random variable.

To deal with a random process, we can realize that it is more complex than random variables because at any interested moment, it should have not only an index of the value of the variable but also another index of time. In this sense, a collection of

random variables is a one-dimensional set, whereas a random process occupies two dimensions, and the occurrence is random.

This manuscript deals with random process, especially random vibrations, in which we need three-dimensional deterministic functions of random distribution. However, only determining a three-dimensional distribution is insufficient to realize how a system vibrates. We also need specific knowledge of the vibration. In this sense, the course of random vibration is a combination of the theory of vibration and the theory of random process.

1.1.2 GENERAL THEORY OF VIBRATION

In this chapter, we briefly describe the general idea of the theory of vibration. A detailed discussion of vibration will begin from Chapter 6.

1.1.2.1 Concept of Vibration

Vibration is a repetitive motion of objects to a stationary frame of reference or nominal position (usually equilibrium). It refers to mechanical oscillations about an equilibrium point. The oscillations may be periodic (such as the motion of a pendulum), transient (such as the impact response of vehicle collision), or random (such as the movement of a tire on a gravel road).

The common sense about vibration is that any object moves back and forth, such as a swing mass, a car driving on a bumpy road, an earthquake, a rocking boat, tree branches swaying in the wind, a man's heartbeat; vibration is everywhere.

Let us consider the generality of the above examples. What these examples have in common can be seen from the following vibrational responses:

First, all of them possess *mass*, *damping*, and *stiffness*. Second, potential energy and kinetic energy are exchanged, and third, a vibration system has frequency and damping ratio. In addition, vibration has a certain shape function. To see these general points, let us consider an example shown in Figure 1.1.

Vibration can also be seen as the responses of a system due to certain excitations. In Figure 1.1a, a flying airplane will be excited by flowing air and its wings will vibrate accordingly. Because of the uncertainty with how air flows, at a deterministic time point, it is difficult to predict the exact displacement of a specific location on the wing. Or we can say that the vibration of the wing is random.

In Figure 1.1, an airplane engine is also shown. Although the engine is working at a certain rotating speed, the corresponding vibration is mainly periodic, which is conceptually shown in Figure 1.1b. In this manuscript, we focus on random vibration responses.

In Figure 1.1a, the vibration of the airplane's wing is a function of a certain location in the air where different excitation input occurs. At virtually the same moment, the wing vibrates accordingly. The vibration will also be seen as a function of time, which is a more universal reference. Therefore, we use the time t , instead of other reference such as the position of the road to describe various vibrations. From either the location in the air or the moment in time, we can realize that the amplitude of vibration is not a constant but a variable. Generally speaking, it is a time or temporal variable. As a comparison, the vibration in the airplane engine shown in Figure 1.1b can be exactly seen as a deterministic function of time.

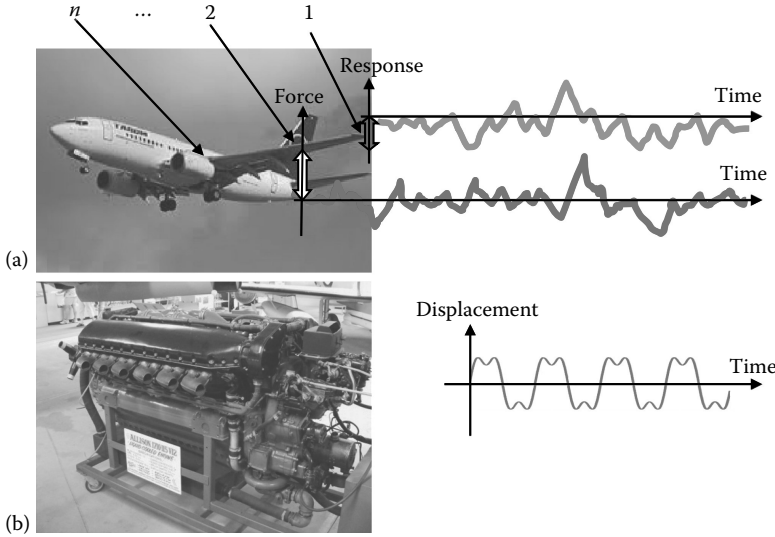


FIGURE 1.1 Different types of vibration. (a) Random vibration of airplane wings, (b) periodic vibration of airplane engine.

1.1.2.1.1 Deterministic Vibration

Let us now have a closer look at the temporal variable, which is conceptually shown in Figure 1.1 and referred to as vibration time histories. Specifically, the vibration time history of the engine vibration, described by displacement $x(t)$ can be represented by a Fourier series, that is,

$$x(t) = x_1 \cos(\omega_1 t + \theta_1) + x_2 \cos(\omega_2 t + \theta_2) \dots \quad (1.1)$$

where x_1 is the amplitude of the particular frequency component ω_1 , θ_1 is the phase shift of this component, and so on. The trigonometric function $\cos(\omega_1 t + \theta_1)$ indicates a back-and-forth movement, the vibration.

Once the excitation of a system is removed, the vibrational response will gradually decay because a realistic system will always dissipate the vibration energy. In this sense, we can add the energy-decaying term such as $e^{-\zeta_1 \omega_1 t}$ in the above equation so that the vibrational response is written as

$$x(t) = x_1 e^{-\zeta_1 \omega_1 t} \cos\left(\sqrt{1 - \zeta_1^2} \omega_1 t + \theta_1\right) + x_2 e^{-\zeta_2 \omega_2 t} \cos\left(\sqrt{1 - \zeta_2^2} \omega_2 t + \theta_2\right) \dots \quad (1.2)$$

From the above description, it can be seen that a temporal vibration function is described by three basic quantities or parameters: namely, the amplitude (x_i), the frequency (ω_i), and the phase (θ_i). If the energy dissipation is also considered, we should have another term ζ_i .

Now, let us compare the vibrations described in Equations 1.1 and 1.2, respectively. The essential difference between periodic vibration (Equation 1.1) and transient vibration (Equation 1.2) is twofold. First, periodic vibration (Equation 1.1) will,

theoretically, last “forever” whereas transient vibration (Equation 1.2) will sooner or later die out. Second, the vibration in Equation 1.1 will repeat itself periodically whereas the vibration in Equation 1.2 will not have repeatability.

Note that, if the ratio of frequency ω_1 and ω_2 is a rational number, the vibration (Equation 1.1) is periodic. If this is not satisfied, for example, $\omega_1/\omega_2 = \pi$, we will not have periodic vibration. Therefore, such a vibration is also referred to as transient vibration. In this sense, the duration of the vibration is not important. Thus, the major difference between Equations 1.1 and 1.2 is whether they can repeat themselves or not.

A closer look at both Equations 1.1 and 1.2 unveils their generality. Whenever we choose a given time, the value of $x(t)$ can be calculated if the frequencies ω_i and damping ratio ζ_i are known. Therefore, the regular vibration theory devotes most chapters to formulating methods of how to determine ω_i and damping ratio ζ_i , as well as how to find the response (Equations 1.1 or 1.2), which is the investigation of vibration systems.

1.1.2.1.2 Vibration Systems

In regular vibration, an object that vibrates is seen as a mass system, called a *vibration system*. In most cases, we assume the vibration system is linear. Any engineering object that is subjected to a certain load and has responses due to the load can be seen to have the relationship of “input-system-output.” When the input and output are functions of time, the system is dynamic. A vibration system is dynamic. On the other hand, if both the load and the response will not develop as time goes on, or if the development with time is sufficiently slow so that it can be treated as constant to time, the system is considered to be static.

The response of a dynamic system can be considered in two basic categories. The first is that the system response versus time can continue to grow until the entire system is broken, or the development of the response versus time will continue to decrease until the system response dies out. In the first case, the response starts at an origin and can continue to develop but will never come back to the origin. Such a dynamic system is not a vibration system.

The second type of dynamic system is that the response will sometimes grow but other times reduce when it reaches a certain peak value and then grow again, either along the same direction or the opposite direction. Furthermore, the growing response reaches the next peak value and starts to decrease. As mentioned previously, this repetitive motion is called vibration and thus the second type of dynamic system is the vibration system.

It is seen that the responses of a dynamic system will continue to vary, so that we need at least two quantities to describe the responses, namely, the amplitude of the responses and the time at which the amplitude reaches a certain level, such as the term x_i and t in the above equation, respectively.

The responses of vibrational dynamic system, however, need at least one additional quantity to express how fast the responses can go back and forth. This term is the frequency of vibration, such as the value of ω_i in Equation 1.1. From this discussion, we can realize that the response of a vibration system must go back and forth or it is not vibration. Therefore, the term that describes how fast the response changes values from growing to reducing is the fundamental quantity distinguishing if a system is

a vibration system. Thus, frequency is the most important parameter for a vibration system. In the viewpoint of vibration modal analysis for linear systems, frequency (or more precisely, natural frequency) is the most important modal parameter.

Also from the input-system-output viewpoint, we can understand that the reason the system will have a dynamic response is due to the input, or there must be an amount of energy input to the system. At the same time, a real-world system will dissipate energy, which can be understood through the second law of thermodynamics. Therefore, we need another term to describe the capability of a given system that dissipates energy. It can be seen that the larger the capacity of energy dissipation a system has, to have the same level of vibration, the more energy input is needed. A quantifiable term to describe the capacity of energy dissipation is called the damping ratio, which is the second most important modal parameter of a vibration system, such as the term ζ_i in the above equation.

When a mass system, such as a car, an airplane, a machine, or a structure vibrates, we often see that, at different locations of the system, the vibration level can be different. For example, the vibration at the driver's seat can be notably different from the rear passenger seat of a compact car. We thus need another parameter to express the vibration profile at different locations, such a vibration shape function is called the *mode shape*. Different from natural frequency and damping ratio, which are scalars, the mode shape function is a vector, which is the third most important parameter of modal analysis. This can be seen conceptually in Figure 1.1a. Suppose we can measure the vibration not only at location 1 but also at location 2 (through location n ; see Figure 1.1a), the vibration at these two different locations are likely not identical. In this case, let us assume that the vibration of the airplane wing is deterministic, which can be expressed by Equation 1.1. In this case of n vibration locations, the system free-decay responses in Equation 1.2 can be further written as

$$\begin{aligned} 1.x(t) &= \begin{bmatrix} x_{11} \\ x_{21} \\ x_{n1} \end{bmatrix} e^{-\zeta_1 \omega_1 t} \cos(\omega_{d1} t + \begin{bmatrix} \theta_{11} \\ \theta_{21} \\ \theta_{n1} \end{bmatrix}) + \begin{bmatrix} x_{12} \\ x_{22} \\ x_{n2} \end{bmatrix} e^{-\zeta_2 \omega_2 t} \cos(\omega_{d2} t + \begin{bmatrix} \theta_{12} \\ \theta_{22} \\ \theta_{n2} \end{bmatrix}) \dots\dots \\ 2.x(t) &= \begin{bmatrix} x_{11} \\ x_{21} \\ x_{n1} \end{bmatrix} e^{-\zeta_1 \omega_1 t} \cos(\omega_{d1} t + \begin{bmatrix} \theta_{11} \\ \theta_{21} \\ \theta_{n1} \end{bmatrix}) + \begin{bmatrix} x_{12} \\ x_{22} \\ x_{n2} \end{bmatrix} e^{-\zeta_2 \omega_2 t} \cos(\omega_{d2} t + \begin{bmatrix} \theta_{12} \\ \theta_{22} \\ \theta_{n2} \end{bmatrix}) \dots\dots \\ n.x(t) &= \begin{bmatrix} x_{11} \\ x_{21} \\ x_{n1} \end{bmatrix} e^{-\zeta_1 \omega_1 t} \cos(\omega_{d1} t + \begin{bmatrix} \theta_{11} \\ \theta_{21} \\ \theta_{n1} \end{bmatrix}) + \begin{bmatrix} x_{12} \\ x_{22} \\ x_{n2} \end{bmatrix} e^{-\zeta_2 \omega_2 t} \cos(\omega_{d2} t + \begin{bmatrix} \theta_{12} \\ \theta_{22} \\ \theta_{n2} \end{bmatrix}) \dots\dots \end{aligned} \quad (1.3)$$

where $\omega_{di} = \sqrt{1 - \zeta_i^2} \omega_i$ is the damped natural frequency. The amplitudes and phase angles at different locations, such as the ones specifically marked by using the solid and broken lines in the above equation, actually describe the vibration shapes, which are referred to as mode shapes, its j th component of the i th mode contains amplitude x_{ji} and phase θ_{ji} .

Again, suppose a system is linear. In this case, the three terms mentioned previously, natural frequency, damping ratio, and mode shape, are the most important parameters of a vibration system; the set of modal parameters. These parameters are, in most cases, deterministic values. In this manuscript, we assume that our vibration systems have deterministic modal parameters.

In a linear system, the ratio between the output and input measured at certain locations is a function of frequency (as well as damping ratio), which is referred to as

a *transfer function*, and is the most important parameter describing a linear vibration system. Therefore, the general relationship of input-system-output can be further written as *input-transfer function-output*.

1.1.2.1.3 Random Vibrations

Based on the theory of vibration discussed above, once the vibration system is known, with given forcing functions or initial conditions (or both), the responses of deterministic vibrations can be calculated as long as the forcing function and initial conditions are deterministic. Mathematically, the procedure is to find the solution of a governing equation of motion of a vibration system, namely, the parameters ω_i , ζ_i , x_{ij} , and θ_{ij} .

On the other hand, for random vibrations, we do not have these deterministic parameters ω_i , ζ_i , x_{ij} , and θ_{ij} in the closed forms of vibration responses described in Equations 1.1 or 1.2. The reason is, in general, we do not have deterministic time history of input or the deterministic initial conditions. In this case, even if the exact characteristics of the vibration system are known, we will not be able to predict the amplitude of vibration at a given time. Also, for a given value of vibration amplitude, we do not know when it will occur.

This does not mean that the vibration is totally uncertain. With the tools of basic random processes and statistics, we may be able to obtain the rate of occurrence of a particular value. We may predict the major vibration frequency of a linear system, with certain knowledge of the input statistics. We may understand the averaged value or root mean square value of the responses, and so on. In most engineering applications, these values can be sufficient to design or control the vibration systems, to predict the fatigue life of a machine or airplane, or to estimate the chance of the occurrence of some extreme values and the possibility of failures of certain systems. These are the major motivations for studying random vibration of a mass system.

A deterministic vibration is a dynamic process, and thus is a random vibration. Therefore, although the vibration responses cannot be written as Equations 1.1 through 1.3, what we do know is that the responses are a function of time. In other words, random vibration belongs to the category of random process.

Similar to deterministic vibration, in this circumstance, we still stay within the concept of a mass system. Additionally, in most cases, we have linear systems, and thus the basic concept or relationship of “input-transfer function-output” are continuously used. The only difference is, in the case of random vibrations, both inputs and outputs are random processes. Thus, instead of devoting more thought to transfer function as a deterministic vibration, we will focus more on the time histories of inputs and outputs. In addition, the main methodology to account for these time histories is through averaging.

Note that a thorough investigation of random processes can be mathematically intensive. For the purpose of understanding the engineering of random vibration and calculating commonly used values of random responses, the author minimizes the necessary knowledge of random process. Thus, this manuscript only emphasizes the introduction of stochastic dynamics. All the materials in this manuscript are managed to let mechanical and civil engineering students in their graduate levels to master the basics of random vibration in a one-semester course. Readers who are

interested in more advanced theory may further consult textbooks of random process, such as those by Soong and Grigoriu (1992).

1.1.3 ARRANGEMENT OF CHAPTERS

In Section 1.2, we study the fundamental concept of probability theory by briefly reviewing the necessary knowledge of random process and random vibrations. Thus, we only discuss the basics of set theory, axioms of probability, and conditional probability. In Section 1.3, we consider random variables. In particular, we emphasize the details of normal distribution. The focus is on single random variables, including continuous and discrete variables and the important probability density and mass functions.

In Chapter 2, we further discuss the functions of random variables, and the random distributions of input and output for vibration systems. In Chapter 3, the random processes in the time domain are introduced, including the basic definitions and classifications, the state spaces and index sets, the stationary process, and the conditions and calculations of ensemble and temporal averages. Correlation analysis is also discussed.

In Chapter 4, random processes in the frequency domain are discussed. The spectral density function and its relationship with correlation functions are the key issues. In addition, white noise and band-pass-filtered spectra are also discussed. In Chapter 5, the random process is further considered with certain statistical properties, such as the concepts of level crossings and distributions of extrema. By analyzing the level crossing, the readers can relate with the handling of time-varying randomness based on random processes. This chapter also provides knowledge bases to understand fatigue processes and engineering failure probabilities.

In Chapter 6, linear single degree of freedom (SDOF) vibration systems are considered with deterministic forcing functions and initial conditions, the emphasis is on the vibration system itself, including basic vibration models and basic vibration parameters. In addition, free vibration and simple forced vibration with harmonic excitation are also considered.

In Chapter 7, the response of linear SDOF systems to random forces will be discussed. Deterministic impulse responses are considered first followed by arbitrary loading convolutions. The relationship between impulse response function and the transfer function is also discussed as Borel's theorem is introduced. Here, the random environments are considered as excitations and are treated as random processes. The mean of the response process, the correlations and the spectral density functions of the response process, are also discussed.

In Chapter 8, we further extend the discussion to linear multi-degree of freedom (MDOF) systems. Proportional and nonproportional damping are discussed. The basic treatment of a MDOF system is to decouple it into SDOF systems. Because of different types of damping, the decoupling procedures are different. In Chapter 9, the concept of inverse problems is introduced, including the first and second inverse problems, to help engineers improve their estimations and to help them minimize measurement noises.

In Chapter 10, understanding the failures of signal components and total structures is introduced. The 3σ criterion, the first passage failure, and fatigue are discussed. In this chapter, the concept of failure is not only focused on materials but also on the

estimation of a system's reliability under multihazard environments. The probability of failure is the foundation of the design of limit state equations. Different from the theory of failure probability, which treats the loading and capacity of a system as random variables, they are viewed as random processes in this chapter.

In Chapter 11, we briefly introduce nonlinear vibration systems, which are also subjected to random excitations. Monte Carlo simulations are discussed to provide better response and parameter estimations under complex situations. In addition, typical nonlinear vibrations are specifically discussed for engineering practice. Finally, examples computing the nonlinear random responses are presented.

1.2 FUNDAMENTAL CONCEPT OF PROBABILITY THEORY

In this section, the basics of probability theory are reviewed for the purpose of summarizing the necessary knowledge base of random process.

1.2.1 SET THEORY

It is known that modern probability theory is based on the set theory. In the following, we briefly review basic concepts without further proofs. An *experiment* is a case that may lead to results referred to as outcomes. An *outcome* is the result of a single trial of an experiment, whereas an *event* is one or more outcomes of an experiment.

Set. A *set* is a collection of events. For example, the collection of all the vibration peak values can be a set, and all these values have the same units. The collection of the modal parameters, such as natural frequencies, damping ratios, and mode shapes of the first few modes of a vibration system can be another set. Here, these parameters can have different units. However, readers can still find what is in common with the second set.

Event. Now, consider the event in a set. If an event “a” occurs, it is denoted as ω_a ; note that if only “a” occurs, we have ω_a . When another event “b” occurs, we have ω_b .

Collect those ω_i 's, denoted by

$$A = \{\omega_a, \omega_b, \omega_c, \dots, \omega_n\} = \{\omega_i\}, i = 1, \dots, n \quad (1.4)$$

Equation 1.4 indicates

1. Set A is a collection of all the events ω_i 's.
2. If certain event ω_a in A exists (“a” occurs), then we will have A , or A occurs.
3. The language of set theory is based on a single fundamental relation, defined as a *membership*. ω_a is a member of set A , denoted by

$$\omega_a \in A \quad (1.5)$$

(ω_a is a member of A , ω_a is an element of A).

4. Relative to ω_a , A can also be seen as an event. To distinguish between ω_a and A , ω_a is called a *basic event*, A is called a *combined event* (collective event).

Space.

5. All the possible events consist of a space of basic events denoted by U . U is an event that must happen. It is also called a *universal set*, which contains all objects, including itself. Furthermore, in engineering, these events may also be called *samples*, and U is the space of all the samples. In the literature, the phrase “space” is often used for continuous variables only.
6. Impossible event Φ . The *empty set* is the set that has no elements (the empty set is uniquely determined by this property, as it is the only set that has no elements—this is a consequence of the understanding that sets are determined by their elements):

$$\Phi = \{\} \quad (1.6)$$

1.2.1.1 Basic Relationship (Operation)

To graphically illustrate the following relationships, a Venn diagram is a useful tool. In the following, the basic relationships between events are discussed with the expression of corresponding Venn diagrams (John Venn, 1834–1923; Figure 1.2, etc.).

1. *A union B (A or B, $A + B$)*

A union B is the collection of all events in both A and B , denoted by

$$A \cup B = A + B \quad (1.7)$$

See the gray area in Figure 1.2, Venn diagram of A union B .

Example 1.1

The union of $\{1, 2, 3\}$ and $\{2, 3, 5\}$ is the set $\{1, 2, 3, 5\}$.

2. *A intersection B (A and B)*

A intersection B is the portion shared by both A and B (Figure 1.3), denoted by

$$A \cap B = AB \quad (1.8)$$

See Figure 1.3 shaded area.

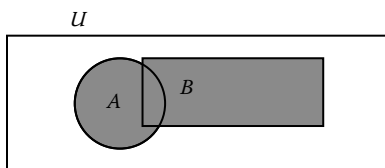


FIGURE 1.2 Venn diagram of A union B .

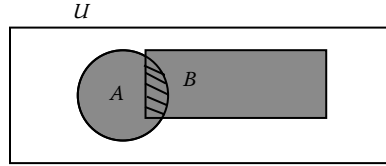


FIGURE 1.3 Venn diagram of A intersection B .

Example 1.2

The intersection of $\{1, 2, 3\}$ and $\{3, 4, 5\}$ is the set $\{3\}$.

The intersection of $\{1, 2, 3\}$ and $\{4, 5, 6\}$ is the set $\{\}$, or Φ .

(Note that, in the literature, $A \cap B$ is also denoted by " AB ")

3. A and B are mutually exclusive (A and B disjoint; Figure 1.4)

The case that A has nothing to do with B is denoted by

$$A \cap B = \Phi \quad (1.9)$$

4. A is contained inside set B , A is a subset of B

That A occurs must let B occur is denoted by

$$A \subset B \quad (1.10)$$

(A is included in B)

or

$$B \supset A \quad (1.11)$$

(B contains A , A is a subset of B ; Figure 1.5)

In this case, we have the following:

a. $A \subset A \quad (1.12)$

b. If $A \subset B, B \subset C$ then $A \subset C \quad (1.13)$

- c. A must be contained inside U , A is a subset of U

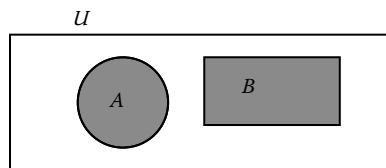


FIGURE 1.4 Venn diagram of A and B mutually exclusive.

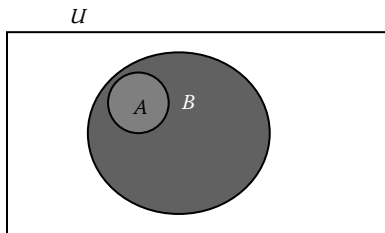


FIGURE 1.5 B contains A .

Example 1.3

$\{a,b\}$ is a subset of $\{a,b,c\}$, but $\{b,d\}$ is not.

$$d. A \supset \Phi \quad (1.14)$$

$$5. A = B \quad (1.15)$$

This is a special case, that set A and set B are identical, the condition is

$$\text{iff } A \subset B \text{ and } B \subset A$$

where the symbol “iff” means “if and only if.”

In this case, we have

$$a. A = A \quad (1.16a)$$

$$b. \text{ If } A = B \text{ and } B = C \text{ then } A = C \quad (1.16b)$$

$$c. \text{ If } A = B \text{ then } B = A \quad (1.16c)$$

6. *Cartesian product* of A and B , denoted by $A \times B$, is the set whose members are all possible *ordered pairs* (a,b) , where a is a member of A and b is a member of B .

A Cartesian product is a useful concept to denote a special relationship between A and B , which will be discussed in detail in Chapter 3.

7. A and B are mutually inverse, A and its complements

That “ A and B are mutually inverse” is denoted by

$$A \cap B = \Phi, \quad A + B = U \quad (1.17)$$

The complement of A is denoted by $\bar{A}$, which can be clearly seen in Figure 1.6. With the help of Equation 1.18, it is seen that $\bar{A}$ here is nothing but B described in Equation 1.17.

That is, the following about A and complement A are true

$$A + \bar{A} = U \quad (1.18a)$$

$$A \cap \bar{A} = \Phi \quad (1.18b)$$

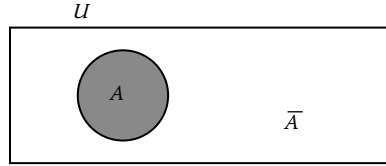


FIGURE 1.6 Venn diagram of A and its complements.

8. The event that A occurs and B does not occur
This case is denoted by

$$A - B \quad (1.19)$$

In Figure 1.7, the two dark gray areas are both $A - B$.
We can realize that

$$A - B = A - A \cap B = A \cap \bar{B} \quad (1.20)$$

9. Operation laws of events

The basic rule of event operation can be summarized as follows:
Commutativity:

$$A + B = B + A \quad (1.21a)$$

$$A \cap B = B \cap A \quad (1.21b)$$

Associativity:

$$(A + B) + C = A + (B + C) \quad (1.22a)$$

$$(A \cap B) \cap C = A \cap (B \cap C) \quad (1.22b)$$

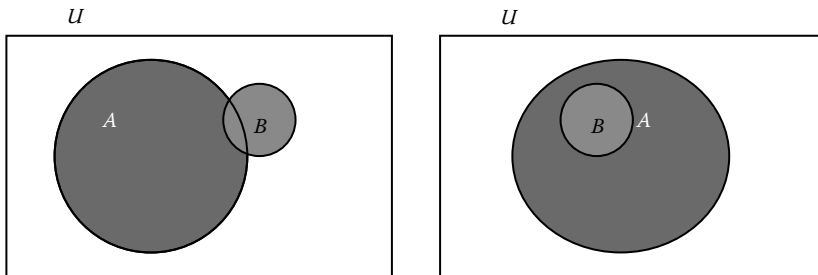


FIGURE 1.7 Venn diagram of $A - B$.

Distributive laws:

$$(A + B) \cap C = A \cap C + B \cap C \quad (1.23a)$$

$$A \cap B + C = (A + C) \cap (B + C) \quad (1.23b)$$

$$A + (B \cap C) = (A + B) \cap (A + C) \quad (1.23c)$$

$$A \cap (B + C) = (A \cap B) + (A \cap C) \quad (1.23d)$$

Duality (De Morgan's laws [Augustus De Morgan, 1806–1871]):

$$\overline{A_1 + A_2 + \dots + A_n} = \bar{A}_1 \bar{A}_2 \dots \bar{A}_n \quad (1.24a)$$

$$\overline{A_1 A_2 \dots A_n} = \bar{A}_1 + \bar{A}_2 + \dots + \bar{A}_n \quad (1.24b)$$

Example 1.4

The time history of a vibratory mass is taken at a time interval of 1 s.

Four samples are given in the following. Use X_i to denote that the i th sample of the vibration level is less than 1 cm. Let X_i represent the following cases:

1. None of the samples are greater than 1 cm
2. At least one of the samples is greater than 1 cm
3. Only one sample is greater than 1 cm
4. At least three samples are smaller than 1 cm

The answer to the above four cases are

1. $X_1 X_2 X_3 X_4$
2. $\overline{X_1 X_2 X_3 X_4}$
3. $\bar{X}_1 X_2 X_3 X_4 + X_1 \bar{X}_2 X_3 X_4 + X_1 X_2 \bar{X}_3 X_4 + X_1 X_2 X_3 \bar{X}_4$
4. $\bar{X}_1 \bar{X}_2 X_3 X_4 + X_1 \bar{X}_2 X_3 X_4 + X_1 X_2 \bar{X}_3 X_4 + X_1 X_2 X_3 \bar{X}_4 + X_1 X_2 X_3 X_4$

1.2.2 AXIOMS OF PROBABILITY

In most modern science, a theory is often described by several axioms and basic concepts, the probability can also be systematically established in a similar fashion. In the above, we actually introduced these basic concepts; in the following, let us consider the axioms.

1.2.2.1 Random Tests and Classic Probability

First of all, consider the classic expression of probability, starting with a random test.

1.2.2.1.1 Frequency $f_N(A)$

The frequency of occurrence of A can be viewed as that in N tests, A occurs n times, denoted by

$$f_N(A) = \frac{n}{N} \quad (1.25)$$

Equation 1.25 provides a very important starting point, which classically expresses the essence of probability. In other words, any case of probability can be seen as a ratio of n and N . If one can successfully and completely find n and N without any overlapping, then he or she locates the corresponding probability.

In Equation 1.26, N is referred to as the *sample space* and n is referred to as the total occurrence of the tested samples.

For the frequency of occurrence, we have the following basic relationships:

$$1. 0 \leq f_N(A) \leq 1 \quad (1.26)$$

$$2. f_N(U) = 1 \quad (1.27)$$

3. if $AB = \Phi$, then

$$f_N(A + B) = f_N(A) + f_N(B) \quad (1.28)$$

1.2.2.1.2 Probability

Now, with the help of the viewpoint of occurrence frequency, we have the classic definition of probability.

1.2.2.1.2.1 Classic Definition of Probability

The probability of A (occurrence of A) is

$$P(A) = \lim_{N \rightarrow N_U} \frac{n}{N} \quad (1.29)$$

Here,

N_U : in space U , the total possible number of tests

n : number of occurrence of A

1.2.2.1.2.2 Characteristics of Classic Probability

The classic random tests are characterized by

1. The total possible results are limited denoted by

$$\omega_1, \omega_2, \dots, \omega_N$$

2. The possibilities of occurrence of all the ω_i 's are equal, that is,

$$P(\omega_1) = P(\omega_2) = \dots = P(\omega_N) \quad (1.30)$$

1.2.2.2 Axiom of Probability

Having reviewed the classic thoughts of probability, let us introduce the axioms.

1.2.2.2.1 Axiomatic Theory of Probability

The mathematical approach of probability theory is based on the following three axioms

Axiom 1 (essence: nonnegative)

$$0 \leq P(A) \leq 1 \quad (1.31)$$

Axiom 2 (essence: normalization)

$$P(U) = 1 \quad (1.32)$$

Axiom 3 (essence: additivity)

A and B are mutually exclusive ($A \cap B = \Phi$), then

$$P(A \cup B) = P(A) + P(B) \quad (1.33)$$

In general

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \quad (1.34)$$

1.2.2.2.2 Properties of Axiomatic Probability

The expression above shows that three axioms can be quite clearly understood. We now consider the associated properties as follows:

$$1. P(\Phi) = 0 \quad (1.35)$$

2. If $A_1, A_2, \dots, A_m$ are mutually exclusive, then

$$P\left(\sum_{i=1}^m A_i\right) = \sum_{i=1}^m P(A_i) \quad (1.36)$$

3. For any event A

$$P(A) = 1 - P(\bar{A}) \quad (1.37)$$

4. For any two events A and B

$$P(A - B) = P(A) - P(A \cap B) \quad (1.38)$$

5. For any two events A and B , if

$$A \supset B$$

then

$$P(A - B) = P(A) - P(B) \quad (1.39)$$

1.2.3 CONDITIONAL PROBABILITY AND INDEPENDENCE

In the world of probability, any occurrence of an event requires certain conditions. In most cases, to find the probability of a certain occurrence in a complex situation is to break down the complicated cases into several “pure” conditions and find the corresponding probabilities of each one. In this subsection, let us consider the basic relationships of the conditional events.

1.2.3.1 Conditional Probability

A *conditional probability* is denoted as

$$P(A|B) = P(A \text{ occurs given that we know } B \text{ occurs}) \quad (1.40)$$

where the bar “|” stands for “given.”

The following is the basic formula for conditional probability:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad (1.41)$$

Equation 1.41 is a normalization to generate a new sample space, in which the probability of B is 1; namely, B always occurs (because B already occurs), that is, $P(B|B) = 1$.

Example 1.5

In 2010, we had twelve students from the Department of Mechanical and Aerospace Engineering (MAE), with one PhD candidate, and five students from the Department of Civil Engineering (CIE), with three PhD candidates in a cross-listed course, which is listed in Table 1.1.

1. Students that are masters known to be from MAE: $11 \rightarrow P(\text{master}|\text{MAE}) = 11/12$
2. Students that are from MAE: $12 \rightarrow P(\text{MAE}) = 12/17$
3. Students that are from MAE and are masters in our class: $11 \rightarrow P(\text{master} \cap \text{MAE}) = 11/17$

$$P(\text{master}|\text{MAE}) = \frac{11/17}{12/17} = 11/12$$

TABLE 1.1
Students in MAE536/CIE520

	PhD	Master	Subtotal
MAE	1	11	12
CIE	3	2	5
Subtotal	4	13	17

TABLE 1.2
MAE Students

	PhD	Master	Subtotal
MAE	1	11	12
CIE	3	2	5
Subtotal	4	13	17

TABLE 1.3
Master Students

	PhD	Master	Subtotal
MAE	1	11	12
CIE	3	2	5
Subtotal	4	13	17

4. “to be from MAE” → the space is shrunk from the “total classmates” to “MAE classmates”
The above can be expressed in Table 1.2 with bold numbers.

$$P(\text{MAE}|\text{master}) = \frac{11/17}{13/17} = 11/13$$

This can be expressed in Table 1.3 with bold numbers.

1.2.3.2 Multiplicative Rules

Based on the conditional probability, we have

$$P(A \cap B) = P(A) P(B|A) \tag{1.42a}$$

$$P(A \cap B) = P(B) P(A|B) \tag{1.42b}$$

Example 1.6

In the above example, assume $P(\text{MAE})$ is known, then

$$P(\text{MAE}) = 12/17;$$

and $P(\text{master}|\text{MAE})$ is also known, we have

$$P(\text{master}|\text{MAE}) = 11/12.$$

Therefore,

$$P(\text{MAE} \cap \text{master}) = P(\text{MAE}) P(\text{master}|\text{MAE}) = (12/17)(11/12) = 11/17$$

Note that

$$P(\text{MAE} \cap \text{master}) = 11/17 < P(\text{MAE}) = 12/17$$

and

$$P(\text{MAE} \cap \text{master}) = 11/17 < P(\text{master}|\text{MAE}) = 11/12.$$

That is, we can realize

$$P(A \cap B) \leq P(A) \quad (1.43a)$$

and we can prove that

$$P(A \cap B) \leq P(B) \quad (1.43b)$$

and

$$P(A \cap B) = P(A) \quad (1.44a)$$

if and only if

$$P(B) = 1 \quad (1.44b)$$

1.2.3.3 Independency

The following are useful concepts of variable independency:

1. Two events A and B , if

$$P(A \cap B) = P(A) P(B) \quad (1.45)$$

A and B are independent, which means the occurrence of A does not affect the occurrence of B .

In addition,

$$P(B|A) = P(B) \quad (1.46)$$

Proof:

From conditional probability

$$P(B|A) = P(A \cap B)/P(A) \quad (1.47)$$

Because A and B are independent, then Equation 1.45 holds. Substitution of Equation 1.45 into Equation 1.46 yields

$$P(B|A) = P(A) P(B)/P(A) = P(B)$$

2. If A and B are independent, then A and $\bar{B}$, $\bar{A}$ and $\bar{B}$, $\bar{A}$ and B are also independent

Example 1.7

If A and B are independent, then A and $\bar{B}$ are also independent.

From Equation 1.20, we have

$$A \cap \bar{B} = A - B$$

therefore

$$P(A \cap \bar{B}) = P(A - B)$$

From the 4th property of axiomatic probability shown in Equation 1.38

$$P(A - B) = P(A) - P(A \cap B)$$

From Equation 1.46, because A and B are independent, $P(A \cap B) = P(A)P(B)$, therefore

$$P(A \cap \bar{B}) = P(A - B) = P(A) - P(A)P(B) = P(A)(1 - P(B)) = P(A)P(\bar{B})$$

Thus, A and $\bar{B}$ are also independent.

1.2.3.4 Total Probability and Bayes' Formula

The concept of conditional probability can be further extended to the important concept of total probability and the corresponding Bayes' formula (Thomas Bayes, 1701–1761).

1.2.3.4.1 Total Probability

If $B_1, B_2, \dots, B_n$ are mutually exclusive, and $P(B_i) > 0$, in addition, A is included in one of the B_i 's, that is,

$$A \subset \sum_{i=1}^n B_i \quad (1.48)$$

then

$$P(A) = \sum_{i=1}^n P(B_i)P(A|B_i) \quad (1.49)$$

Proof:

Because $B_1, B_2, \dots, B_n$ are mutually exclusive, and so will be $A \cap B_1, A \cap B_2, \dots, A \cap B_n$.

From Equation 1.48

$$A = A \cap \sum_{i=1}^n B_i$$

Furthermore,

$$P(A) = P\left(A \cap \sum_{i=1}^n B_i\right) = P(A \cap B_1 + A \cap B_2 \dots + A \cap B_n) = \sum_{i=1}^n P(A \cap B_i)$$

From the multiplicative rules, $P(A \cap B) = P(B) P(A|B)$ (Equation 1.42b), the above equation

$$= \sum_{i=1}^n P(B_i) P(A|B_i)$$

Example 1.8

Consider an example of quality control

Quality inspectors a, b, and c check three different products in a factory, which are 20%, 35%, and 45% of the total production, respectively. Inspectors may make mistakes.

Suppose the probability of mistakes made by inspector a, b, and c are 5%, 7%, and 6%, respectively. Find the probability of a defective product from this factory.

Let

$B_a = \{\text{products passed by inspector a}\}$

$B_b = \{\text{products passed by inspector b}\}$

$B_c = \{\text{products passed by inspector c}\}$

$A = \{\text{a defective product taken from those that passed inspection}\}$

We see that

$$A = A \cap B_a + A \cap B_b + A \cap B_c$$

and

$$[A \cap B_a] \cap [A \cap B_b] = \Phi, [A \cap B_a] \cap [A \cap B_c] = \Phi, [A \cap B_b] \cap [A \cap B_c] = \Phi$$

Known

$$P(B_a) = 20\%$$

$$P(B_b) = 35\%$$

$$P(B_c) = 45\%$$

$$P(A|B_a) = 5\%$$

$$P(A|B_b) = 7\%$$

$$P(A|B_c) = 6\%$$

we can calculate

$$P(A) = 20\% (5\%) + 35\% (7\%) + 45\% (6\%) = 6.15\%$$

The essence of total probability is (i) to find $P(A)$, event A must join a group of mutually exclusive event B_i 's; (ii) identify this group of disjointed events $B_1, B_2, \dots, B_n$

$$B_i \cap B_j = \Phi \quad (1 \leq i < j \leq n) \quad (1.50)$$

1.2.3.5 Bayes' Formula (Thomas Bayes, 1701–1761)

If $B_1, B_2, \dots$ and B_n are mutually exclusive (see Equation 1.50) and $P(B_i) > 0$, in addition, A is included in one of the B_i 's, that is,

$$A \subset \sum_{i=1}^n B_i \quad (1.51)$$

then we have the Bayes' formula described in Equation 1.53.

$$P(B_i|A) = \frac{P(B_i)P(A|B_i)}{\sum_{i=1}^n P(B_i)P(A|B_i)} \quad (1.52)$$

Example 1.9

In the Example 1.8, one product is found to be defective, with the condition " A ," the question is, which inspector is more likely to be responsible? We have

$$P(B_a|A) = P(B_a) P(A|B_a) / [P(B_a) P(A|B_a) + P(B_b) P(A|B_b) + P(B_c) P(A|B_c)]$$

Therefore, we have

$$[P(B_a) P(A|B_a) + P(B_b) P(A|B_b) + P(B_c) P(A|B_c)] = P(A) = 6.15\%$$

$$P(B_a|A) = P(B_a) P(A|B_a) / [6.15\%] = 20\% (5\%) / [6.15\%] = 16.26\%$$

$$P(B_b|A) = P(B_b) P(A|B_b) / [6.15\%] = 35\% (7\%) / [6.15\%] = 39.84\%$$

$$P(B_c|A) = P(B_c) P(A|B_c) / [6.15\%] = 45\% (6\%) / [6.15\%] = 43.90\%$$

The essence of Bayes' theorem is

1. A has already occurred, to check the probability of B_i , that is, the probability of event " $B_i|A$ "
2. We also need to find the group of disjointed events $B_1, B_2, \dots, B_n$ (see Equation 1.50)

1.2.4 ENGINEERING EXAMPLES

Now, let us consider certain engineering examples of multiple occurrences.

1.2.4.1 Additive Rules

Consider two failure modes A and B that are treated as double extreme events to a single system, say a car or a bridge,

$$P(A \cup B) = P(A) + P(B),$$

if A and B are independent, find the failure probability of the system.

Practically, we have two or more possible cases: (1) failure modes in *parallel* and (2) failure modes in *series*, which can be seen in Figure 1.8a and b, respectively. Here, the terms $P_1, P_2, \dots$ are the probabilities of occurrences of event 1, 2, ...

In Figure 1.8a these events causing possible bridge failure occurs simultaneously. For example, the combined loads of “earthquake + vehicular collision,” etc. In Figure 1.8b, these events occur in consequence; say, first an earthquake occurs, and then a vehicular collision occurs.

Consider the case of combined loads on a bridge, which is listed as follows:

1. Single extreme event

First, the single case, which is omitted

2. Double extreme events

Second, we may have the following combinations:

Example of Collective Set: Combined Loads

Earthquake + vehicular collision

Earthquake + scour

Earthquake + debris flow/land slide

Earthquake + surge

Earthquake + fire

Earthquake + wind

Earthquake + vessel collision

Vehicular collision + scour

Vehicular collision + fire

Vehicular collision + wind

Scour + wind

Scour + surge

Scour + vessel collision

Scour + debris flow/land slide

Scour + fire

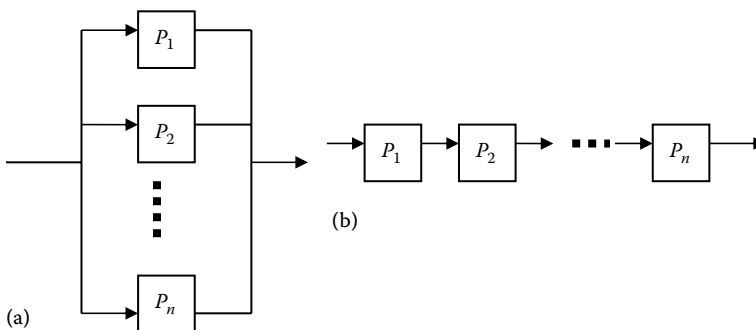


FIGURE 1.8 Relationship of (a) parallel and (b) series.

Debris flow/land slide + surge
 Debris flow/land slide + wind
 Debris flow/land slide + vessel collision

Surge + wind
 Surge + vessel collision

Fire + wind

3. Triple extreme events

Third, we may have triple events and so on, which are also omitted due to the limited space in the manuscript. In the above, each individual load will have its own occurrence probability. Now, the practical question is, what is the probability of the combined loads?

If we know the total sample space of all the combined loads, then based on the theory of total probability and Bayes' formula, we can determine for the combined probability.

1.2.4.2 Multiplication Rules

If an operation can be performed in p ways, and if for each of these ways, a second operation can be performed in q ways, then the two operations can be performed together in pq ways.

1.2.4.3 Independent Series

An independent series of tests satisfies

1. In each test, we have only two possible results, A and $\bar{A}$
2. The probability $P(A) = p$ and $P(\bar{A}) = 1 - p$ for each test are constant
3. All the results are mutually independent; these are also called *n-Bernoulli tests* (Bernoulli series, *n*-Binomial series)

In *n*-Bernoulli tests, the probability of A that occurs m times is expressed as

$$P_n(m) = C_n^m p^m (1 - p)^{n-m} \quad (1.53)$$

Here, C_n^m is the combination given by

$$C_n^m = \frac{n!}{m!(n-m)!} \quad (1.54)$$

Example 1.10

Suppose we have six bulbs. Two evens are shown in Figure 1.9 (six bulbs in series) and Figure 1.10 (three sets of two bulbs in series are in parallel), respectively. The probability of each bulb being broken is 0.2.

Question: What is the probability of total failure?

Let $\omega_i = \{\text{the } i\text{th bulb is broken}\}$, we have $P(A_i) = 20\%$

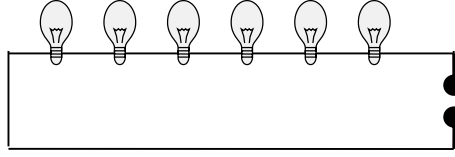


FIGURE 1.9 Six bulbs in series.

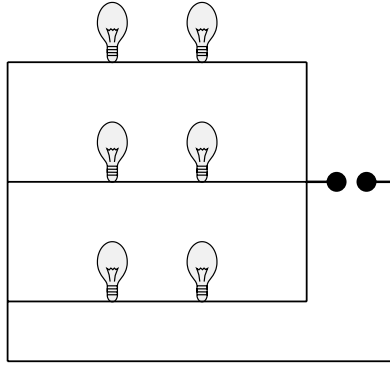


FIGURE 1.10 Three sets of two bulbs in series are in parallel.

Then,

$$\bar{\omega}_i = \{\text{the } i\text{th bulb works}\}, P(\bar{\omega}_i) = 1\% - 20\% = 80\%$$

$$1. \text{ Event } A = \omega_1 + \omega_2 + \omega_3 + \omega_4 + \omega_5 + \omega_6 = \overline{\bar{\omega}_1 \bar{\omega}_2 \bar{\omega}_3 \bar{\omega}_4 \bar{\omega}_5 \bar{\omega}_6}$$

$$P(A) = 1 - P(\bar{\omega}_1 \bar{\omega}_2 \bar{\omega}_3 \bar{\omega}_4 \bar{\omega}_5 \bar{\omega}_6) = 1 - \prod_{i=1}^6 P(\bar{\omega}_i) = 1 - 0.8^6 = 1 - 0.2621 = 0.7379$$

Thus, the failure probability is about 74%.

$$2. \text{ Event } A = (\omega_1 + \omega_2)(\omega_3 + \omega_4)(\omega_5 + \omega_6)$$

$$P(\omega_1 + \omega_2) = 1 - \overline{\bar{\omega}_1 \bar{\omega}_2} = 1 - 0.8^2 = 0.36 = P(\omega_3 + \omega_4) = P(\omega_5 + \omega_6)$$

$$P(A) = P(\omega_1 + \omega_2) P(\omega_3 + \omega_4) P(\omega_5 + \omega_6) = 0.36^3 = 0.0467$$

Compared with case I, the failure probability of about 5% is much smaller.

1.2.4.4 Return Period of Extreme Load

Now, consider the occurrence of extreme load, which is often denoted by the return period.

The basic description of return period, denoted by T_R , is

$$T_R = \frac{1}{p} \quad (1.55)$$

where p is the occurrence rate. For example, within a year, an earthquake with a return period of 2500 years has the probability

$$p = \frac{1}{T_R} = \frac{1}{2500} = 0.0004 \quad (1.56)$$

The probability of not seeing such an earthquake is

$$1 - p = 0.9996 \quad (1.57)$$

The probability of seeing the earthquake in n years, denoted by p_n , will be

$$p_n = 1 - (1 - p)^n \quad (1.58)$$

For example, the possibility in 100 years of seeing an earthquake with a return period of 2500 years is

$$P_{100} = 1 - \left(1 - \frac{1}{2500}\right)^{100} = 0.0392$$

The exposure or design service time is denoted by t_D . Therefore,

$$p_{t_D} = 1 - \left(1 - \frac{1}{T_R}\right)^{t_D} \quad (1.59)$$

When the return period becomes a large number, the following equation is used to approximate the probability of occurrence,

$$p_{t_D} = 1 - e^{-\frac{t_D}{T_R}} = 1 - e^{-p t_D} \quad (1.60)$$

Figure 1.11 shows this uniform distribution. The uniform probability distribution implies that we have no reason not to use such a distribution. That is, we have no reason in year i that the chance of seeing the event is greater than year j . In many cases,

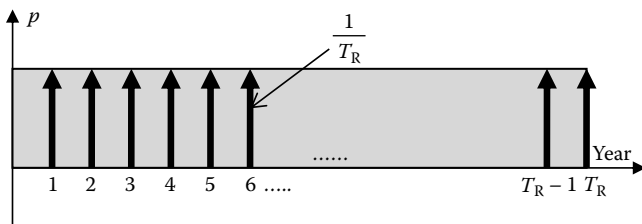


FIGURE 1.11 Uniform distribution of probability of occurrence in period T_R .

using uniform distribution reflects the fact that we have not yet clearly understood the nature of such events.

From Figure 1.11 and Equation 1.55, we can also see that

$$T_R p = 1 \quad (1.61)$$

However, from Equation 1.59, we can see that, when $t_D = T_R$,

$$p_n = 1 - \left(1 - \frac{1}{T_R}\right)^{T_R} < 1 \quad (1.62)$$

For example, when $t_D = T_R = 2500$,

$$p_n = 1 - \left(1 - \frac{1}{2500}\right)^{2500} = 0.6322 < 1 \quad (1.63)$$

1.2.4.4.1 Urn Problems (Ball-Picking Games)

The idea behind an urn test is that it is a method for “virtual experiments,” which assumes that certain balls with different colors represent different events. The total number of balls provides the sample space. By picking a specific type of ball, we can realize the corresponding probability. Using this virtual experiment is an important method to obtain a certain probability mass function (PMF).

Example 1.11

Suppose ten balls are in a bucket, two balls are black and eight are white (Figure 1.12).

1. For each test, pick one ball and return it to the bucket
2. Pick one ball and never return it
 - a. In the first test, the chance of getting a black ball is 2/10. In the second test, we also have a chance of 2/10.

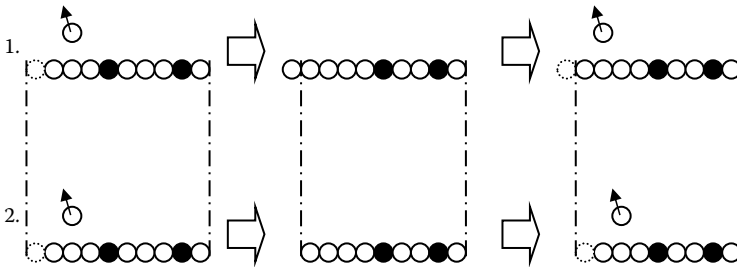


FIGURE 1.12 Ball picking.

- b. In the first test, the chance of getting a black ball is, again, $2/10$. In the second test, getting a black ball will depend on the first test. The probability of picking up a specially colored ball is then a variable.

Now, suppose the chance of occurrence of a hazard follows the possibility as described in the second ball game.

$$p_n = 1 - \left(1 - \frac{1}{T_R}\right) \left(1 - \frac{1}{T_R - 1}\right) \left(1 - \frac{1}{T_R - 2}\right) \cdots \left(1 - \frac{1}{T_R - n + 1}\right) \quad (1.64)$$

Equation 1.64 can be further written as

$$p_n = 1 - \left(\frac{T_R - 1}{T_R}\right) \left(\frac{T_R - 2}{T_R - 1}\right) \left(\frac{T_R - 3}{T_R - 2}\right) \cdots \left(\frac{T_R - n}{T_R - n + 1}\right) \quad (1.65)$$

Therefore, we have

$$p_n = \frac{n}{T_R} \quad (1.66)$$

1.3 RANDOM VARIABLES

In the above, the basic concepts of probability were reviewed. Now, let us consider random variables, based on the theory of probability.

1.3.1 DISCRETE RANDOM VARIABLES AND PMF

First of all, consider the single random variable, in the discrete cases. Then, we will discuss continuous variables.

1.3.1.1 Single Random Variables

That a variable x is random means that its value is uncertain or unpredictable.

Suppose we have m sets of random variables written in vector form as follows:

$$\begin{aligned} \mathbf{x}_1 &= [x_{11} x_{12} \cdots x_{1n}] \\ \mathbf{x}_2 &= [x_{21} x_{22} \cdots x_{2n}] \\ &\vdots \\ \mathbf{x}_m &= [x_{m1} x_{m2} \cdots x_{mn}] \end{aligned} \quad (1.67)$$

where, x_{ij} means the j th element of the i th set.

It is seen that it is possible to sort all the variable x_{ij} with a special arrangement. A one-dimensional axis can be used for this arrangement.

1.3.1.2 “Two-Dimensional” Approach

Suppose variable x_{ij} is associated with the probability of occurrence(s) written as

$$\begin{aligned} P(\mathbf{x}_1) &= [P(x_{11}), P(x_{12}) \dots P(x_{1n})] \\ &\dots \\ P(\mathbf{x}_m) &= [P(x_{m1}), P(x_{m2}) \dots P(x_{mn})] \end{aligned} \quad (1.68)$$

where $P(\cdot)$ stands for the probability of variable $(\cdot)$.

Now, in the above-mentioned axis, at each point x , we will have the probability $P(x)$, in this way, together with the above-mentioned one-dimensional approach for sorted values, we obtain the second dimension to represent the corresponding probabilities. Therefore, we now have a two-dimensional approach.

1.3.1.3 Probability Mass Function

Specifically, a variable set x_i is sorted in the following order

$$x_{i1} < x_{i2} < \dots < x_{in} \quad (1.69)$$

we can use integers to denote the position of x_{ij} , namely, j is an integer. For example,

$$\begin{array}{cccc} x_{i1}, & x_{i2}, & x_{i3} & \dots & x_{in} \\ 1 & 2 & 3 & \dots & n \end{array}$$

Here, the index j starts at 1, and so on.

Another example is to let j start from zero, namely,

$$\begin{array}{cccc} x_{i0}, & x_{i1}, & x_{i2} & \dots & x_{in} \\ 0 & 1 & 2 & \dots & n \end{array}$$

The corresponding distribution of $p(x_{ij})$'s is referred to as the probability mass function (PMF).

In the following, let us consider some typical PMFs. By comparing these closed-form PMFs with one's real-world problems, one can figure out one's specific PMF. To approximately formulate a specific PMF (and PDF in Section 1.3.2), one can use the technology of curve fitting.

1.3.1.4 Bernoulli Distribution (0–1 Distribution) (Jacob Bernoulli, 1655–1705)

The “true/false” or “yes/no” test of a special two-variable case yields a Bernoulli distribution. For example, if the occurrence of a binary, namely, 1 or 0 is random, we will have such a distribution.

Occurs/does not occur probability can be written as

$$P_K(k) = \begin{cases} p, & k = 1 \\ 1 - p, & k = 0 \\ 0, & \text{elsewhere} \end{cases} \quad (1.70)$$

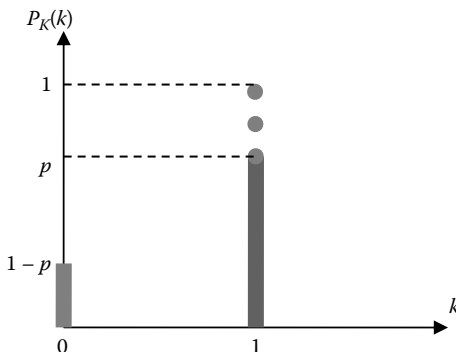


FIGURE 1.13 0-1 Distribution.

It is seen that the sum of these two cases is $p + (1 - p) + 0 = 1$.

Let us consider the sample space S versus the space U of all real numbers.

Suppose in Equation 1.70,

$$U = \text{real numbers (or nonnegative real)} \quad (1.71)$$

The sample space of a Bernoulli variable is

$$S = \{0, 1\} \quad (1.72)$$

We can also let $U = S = \{0, 1\}$, in this case, Equation 1.70 is replaced by

$$P_K(k) = \begin{cases} p, & k = 1 \\ 1 - p, & k = 0 \end{cases} \quad (1.73)$$

The distribution can be shown in Figure 1.13.

1.3.1.5 Binomial Distribution

In n tests, the probability of m event A with Bernoulli distribution ($P(A) = p$) is called the *Bernoulli test*, (see Section 1.2.4.3) which has sample space S given by

$$S = \{0, 1, 2, 3, 4, \dots, n\} \quad (1.74)$$

The corresponding PMF is also referred to as a *binomial distribution*, given by

$$P_n(m) = \begin{cases} C_n^m p^m (1 - p)^{n-m}, & m = 0, 1, 2, 3, \dots, n \\ 0, & \text{elsewhere} \end{cases} \quad (1.75)$$

Figure 1.14 shows the PMF with different values of p .

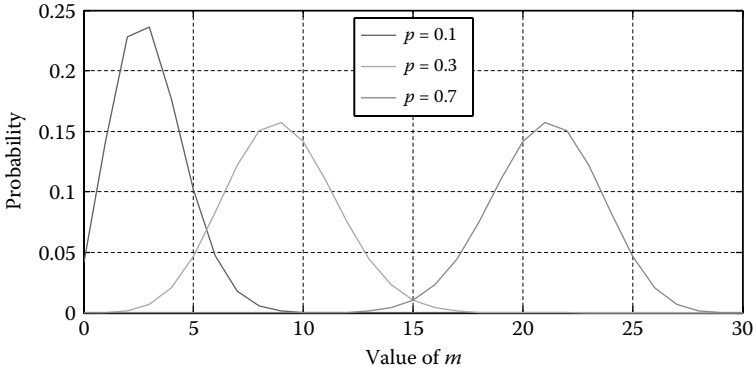


FIGURE 1.14 Binomial distributions.

Example 1.12: Mendel Model

A couple, a father and a mother, both have mixed genes. They have three children. Find the chance that one of them has the dominant gene.

Let the dominant gene be denoted by d and the recessive gene is denoted by r . We see that dd is the pure dominant, rr is the pure recessive, and rd is the mixed gene. Therefore,

$$dd = 1/4, rr = 1/4, rd = 1/2, \text{ the chance of } d: p = (1/4 + 1/2) = 3/4$$

Then, the chance can be calculated as

$$P_3(1) = C_3^1 p^1 (1-p)^{3-1} = 0.1406$$

1.3.1.6 Poisson Distribution (Simeon Denis Poisson, 1781–1840)

The sample space of a Poisson variable is a nonnegative integer, given by

$$S = \{0, 1, 2, 3, 4, \dots\} \quad (1.76)$$

The distribution is given by

$$P_{\Lambda}(k) = \begin{cases} \frac{\lambda^k e^{-\lambda}}{k!}, & k = 0, 1, 2, 3, \dots \\ 0, & \text{elsewhere} \end{cases} \quad (1.77)$$

Figure 1.15 plots the PMF with different values of parameter λ .

It is seen that

$$\sum_{k=0}^{\infty} P_{\Lambda}(k) = e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = e^{-\lambda} e^{\lambda} = 1 \quad (1.78)$$

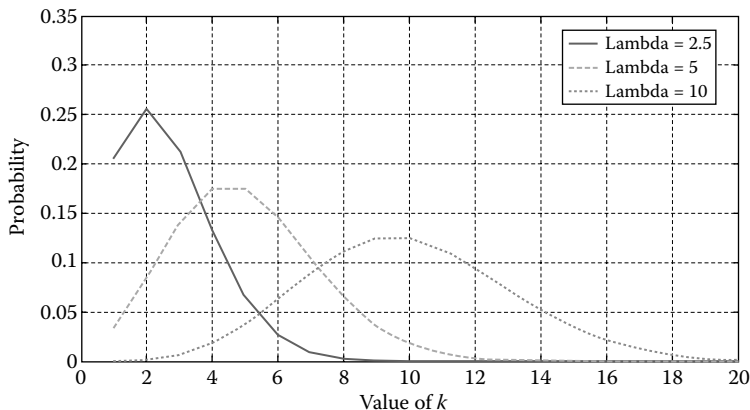


FIGURE 1.15 Poisson distributions.

Example 1.13

The number of times an oscillating mass travels across a certain level in a special time interval $(0, t)$ can be described as

$$P_N(n) = \begin{cases} \frac{(\lambda t)^n e^{-\lambda t}}{n!}, & n = 0, 1, 2, 3, \dots \\ 0, & \text{elsewhere} \end{cases} \quad (1.79)$$

Here the notation $(0, t)$, or more generally (a, b) , stands for an open interval excluding its two ends 0 and t or a and b ; whereas $[a, b]$ is a closed interval including a and b ; such as $[0, 1]$ in Equation 1.84a. These two notations will be used throughout this book.

1.3.1.7 Poisson Approximation

In a Bernoulli test, if n is sufficiently large and p is sufficiently small, we can use Poisson distribution to approximately calculate the corresponding probability, that is,

$$C_n^k p^k (1-p)^{n-k} \approx \frac{(\lambda)^k e^{-\lambda}}{k!} \quad (1.80)$$

where

$$\lambda = np \quad (1.81)$$

Example 1.14

In a workshop, there are 100 instruments working independently. The probability of failure is 0.01; and one technician can fix an instrument.

Question (1): How many technicians are needed to ensure that the probability of not fixing an instrument will be 0.005?

Question (2): If one technician handles 20 instruments, then find the probability of the instruments not being fixed.

Question (3): If three technicians are responsible for 80 instruments, then find the probability of an instrument not being fixed.

Answer:

1. Denote the number of broken instruments to be x , then $x \rightarrow$ Bernoulli distribution

$$P_N(x) = P_{100}(x) \text{ with probability } p = 0.01.$$

Now, assume y technicians are needed, that " $x > y$ " means $x - y$ instruments cannot be fixed. Thus, what we need is $P(x > y) \leq 0.005$.

$$\begin{aligned} P(x > y) &= \sum_{k=y+1}^{100} C_{100}^k p^k (1-p)^{100-k} = \sum_{k=y+1}^{100} C_{100}^k (0.01)^k (0.99)^{100-k} \\ &\approx \sum_{k=y+1}^{100} \frac{(\lambda)^k e^{-\lambda}}{k!} = \sum_{k=y+1}^{100} \frac{(1)^k e^{-1}}{k!} \end{aligned}$$

Solving

$$\sum_{k=y+1}^{100} \frac{(1)^k e^{-1}}{k!} \leq 0.005$$

$$y = 5$$

In fact, when $y = 5$, $P(x > y) = \sum_{k=6}^{100} C_{100}^k p^k (1-p)^{100-k} = 0.0034 < 0.005$, using

Poisson approximation

$$P(x > y) = \sum_{k=6}^{100} \frac{(1)^k e^{-1}}{k!} = 0.0037 < 0.005$$

2. In the group with 20 instruments, if only one is broken, and because a technician is available, it will be fixed. We need to consider the case of "more than 1." That is, the probability is $P(x \geq 2)$.

Using the Poisson distribution to approximate $P(0 \leq x \leq 1) = \sum_{k=0}^1 C_{20}^k p^k (1-p)^{20-k}$ and note that $\lambda = np = 20 (0.01) = 0.2$:

$$P(x \geq 2) = 1 - P(0 \leq x \leq 1) \approx 1 - \sum_{k=0}^1 \frac{(0.2)^k e^{-0.2}}{k!} = 0.0176$$

Note that, if the Bernoulli distribution is used,

$$P(x \geq 2) = 1 - P(0 \leq x \leq 1) = 1 - \sum_{k=0}^1 C_{20}^k p^k (1-p)^{20-k} = 0.0169$$

3. Consider $P(x \geq 4)$, here $\lambda = np = 80(0.01) = 0.8$. Using the Poisson distribution:

$$P(x \geq 4) = 1 - P(0 \leq x \leq 3) \approx 1 - \sum_{k=0}^3 \frac{(0.8)^k e^{-0.8}}{k!} = 0.0091 < 0.0176$$

Procedure for solving the problem:

- Denote x as the total number of broken instruments, denote y as the number of technicians $\rightarrow$ starting point
- Consider $P(x > y)$, etc. $\rightarrow$ describing the corresponding probability
- Find the formula of $P(x > y)$, etc.
- Compute $P(x > y)$, etc.

1.3.1.8 Summary of PMF $P_N(n)$

In Sections 1.3.1.4 through 1.3.1.7, we described several important PMF characteristics. Now, we consider their generality.

1.3.1.8.1 Essence of PMF

The reason we need to study the PMF can be realized by using the two-dimensional deterministic distribution to treat one-dimensional random variables. That is, we first sort the random variables, using certain integers to denote these “cases.” In so doing, we sort random variables into deterministic arrangement in x axis.

Then, we find the probability $p(x)$, using a fixed value to denote the probability at position j . In so doing, we turn random variables into deterministic values in the y axis. That is, the introduction of PMF allows us to use the two-dimensional deterministic relationship of variables and the corresponding probability to treat the original one-dimensional random variables. This is the essence of PMF.

1.3.1.8.2 Basic Property

The basic property of PMF is the basic property of generic probability, that is,

$$0 \leq P_N(n) \leq 1 \quad (1.82)$$

$$\sum P_N(n) = 1 \quad (1.83)$$

Therefore, any function that satisfies Equations 1.82 and 1.83 is a probability distribution function for discrete random variables.

1.3.2 CONTINUOUS RANDOM VARIABLES AND PDF

In Section 1.3.1, we discussed discrete random variables, which can be extended to continuous variables as follows.

1.3.2.1 Continuous Random Variables

1.3.2.1.1 Discrete versus Continuous

In many cases, variables are not discrete. For example, a given range can be any real number, instead of integers only.

The values of variables can be used to denote:

Discrete: 0, 1, 2, 3, 4 ... used to denote the j th cases

Continuous: x , which can be used to denote virtually any analog values

1.3.2.1.2 Sample Space

Compare the sample space. It is seen that

Discrete: Sample space $\neq$ space

Continuous: Sample space = space

Example 1.15

Consider the following continuous variables

$$\{\text{real number in the interval } [0 \ 1]\} \quad (1.84a)$$

$$\{\text{all nonnegative real numbers}\} \quad (1.84b)$$

$$\{\text{all real numbers}\} \quad (1.84c)$$

1.3.2.2 Probability Density Function

Similar to discrete variables, the random continuous variable can also have distributions. Mathematically, the expressions of discrete and continuous variables are slightly different.

1.3.2.2.1 Mass versus Density

Consider a 0 – 1 distribution as an example, at 0 (and/or 1), a value of p (or $1 - p$) is assigned.

For a continuous variable, which has an infinite number of those variables, the summation of these values will become infinity. Therefore, we use the concept of density. In physics, the integration of density becomes mass. It is seen that the integration of $f(x)$ along a line becomes area can be written as

$$A(a < X \leq b) = \int_a^b f(x) \, dx \quad (1.85)$$

1.3.2.2.2 Probability

The idea implied in Equation 1.85 can be used for the distribution function of continuous variables. Suppose the distribution can be expressed as a PDF $f_X(x)$ at location x . Consider the corresponding probability, which may also be an integral given by (see Figure 1.16)

$$P(a < X \leq b) = \int_a^b f_X(x) \, dx \quad (1.86)$$

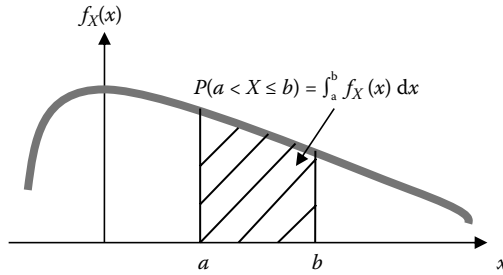


FIGURE 1.16 Probability density function.

1.3.2.2.3 Axiom Requirement

To realize that Equation 1.86 is indeed a probability, let us check the axiom requirement, that is,

1.
$$f_X(x) \geq 0 \quad (1.87)$$

and

2.
$$\int_{-\infty}^{\infty} f_X(x) dx = 1 \quad (1.88)$$

For the sample space in the interval $[a, b]$ (e.g., see Equation 1.84a), we alternatively have

$$\int_a^b f_X(x) dx = 1 \quad (1.89)$$

Because using the concept of density function can satisfy the basic axioms, we realize that the integral of the density function is indeed the probability. We thus called this the *probability density function* (PDF).

1.3.2.2.4 Probability Density versus Probability Mass

Now, let us compare the PDF and the PMF. In the continuous case, we have

$$P(x < X \leq x + dx) = \int_x^{x+dx} f_X(u) du = f_X(x) dx \quad (1.90a)$$

and in the discrete case, we have

$$P(X \approx x) = f_X(x) dx \quad (1.90b)$$

From now on, we will use both concepts of PMF and PDF as fundamental approaches to deal with random variables.

1.3.2.2.5 Property of PDF

Similar to PMF, PDF has basic properties.

First, it is impossible that $f_X < 0$. However, it is highly possible that $f_X > 1$. On the other hand, we do not have the chance that $f_X \rightarrow \infty$. Therefore, the basic property of PDF $f_X(x)$ can be expressed as

$$0 \leq f_X(x) < \infty \quad (1.91)$$

In the following, let us consider some important PDFs. Perhaps normal distribution is the most important one; however, it will be discussed in Section 1.3.5 separately.

1.3.2.3 Uniform Distribution

The uniform distribution has PDF given by

$$f_U(x) = \begin{cases} \frac{1}{b-a}, & a < x \leq b \\ 0, & \text{elsewhere} \end{cases} \quad (1.92)$$

For example, the possible phase angle of a sinusoidal signal is (see Figure 1.17)

$$f_\Theta(\theta) = \begin{cases} \frac{1}{2\pi}, & 0 < \theta \leq 2\pi \\ 0, & \text{elsewhere} \end{cases}$$

Example 1.16

Buses will arrive at a bus station at 8:00, 8:15, and 8:30. A passenger comes to the bus station in the time interval of 30 minutes from 8:00 to 8:30. Find that (1) he can take the bus within 5 minutes; (2) he must wait for more than 10 minutes.

1. Uniform distribution $p = \frac{1}{30}$

To wait for less than 5 minutes, he must be in between 8:10 and 8:15 or 8:25 and 8:30

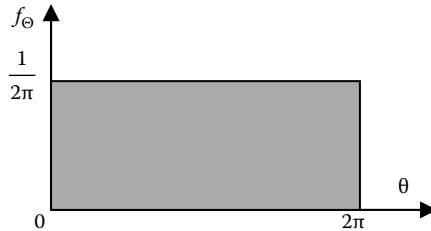


FIGURE 1.17 Uniform distribution.

$$P[(8:10 < x < 8:15) \cup (8:25 < x < 8:30)] = P(8:10 < x < 8:15) + P(8:25 < x < 8:30)$$

$$= \int_{10}^{15} \frac{1}{30} dx + \int_{25}^{30} \frac{1}{30} dx = 1/3$$

2. Only if he is in between 8:00 and 8:05 or 8:15 and 8:20, he needs to wait for more than 10 minutes

$$P[(8:00 < x < 8:05) \cup (8:15 < x < 8:20)] = P(8:00 < x < 8:05) + P(8:15 < x < 8:20)$$

$$= \int_0^5 \frac{1}{30} dx + \int_{15}^{20} \frac{1}{30} dx = 1/3$$

Again, the reason for employing the uniform distribution is as follows: we could not find a reason that the distribution is not uniform. For example, in between 8:00 and 8:30, we do not know the pattern or rule (if it does have a rule) when a bus can arrive at the station.

1.3.2.4 Exponential Distribution

The exponential distribution has PDF given by

$$f_{\lambda}(x) = \lambda e^{-\lambda x}, \quad x > 0 \quad (1.93)$$

λ is a possible constant.

Example 1.17

The total months of a car running without a mechanical problem is a discrete variable, but the total time becomes a continuous variable; with PDF (see Figure 1.18)

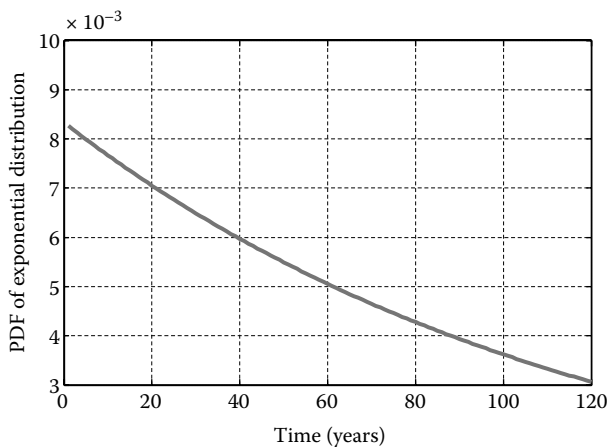


FIGURE 1.18 Exponential distribution.

$$f_{\Lambda}(x) = \lambda e^{-\frac{x}{120}}, \quad x > 0$$

Example 1.18

Consider the car described in Example 1.17.

Question (1): Before the occurrence of a problem, what is the probability that the car can run between 5 and 8 years?

Question (2): Find the probability that the car can only run within 120 months.

1. Because

$$\int_{-\infty}^{\infty} f_{\Lambda}(x) dx = 1 \rightarrow 1 = \int_0^{\infty} \lambda e^{-\frac{x}{120}} dx = 120 \lambda$$

We have

$$\lambda = \frac{1}{120}$$

Thus,

$$P_{\Lambda}(60 < x < 96) = \int_{60}^{96} \frac{1}{120} e^{-\frac{x}{120}} dx = -\left(e^{-\frac{96}{120}} - e^{-\frac{60}{120}}\right) = 0.1572$$

2. We also have

$$P_{\Lambda}(x < 120) = \int_0^{120} \frac{1}{120} e^{-\frac{x}{120}} dx = -\left(e^{-1} - e^{-\frac{0}{120}}\right) = 0.63$$

Compare with Poisson distribution for discrete variables (see Equation 1.77):

$$P_{\Lambda}(k) = \frac{\lambda^k}{k!} e^{-\lambda}, \quad k = 0, 1, 2, 3, \dots$$

where integer k denotes the k th event.

It is generally acknowledged that the number of telephone calls, the number of passengers in a bus station, and the number of airplanes landing in an airport during a period, with given length, can be modeled by discrete Poisson distributions.

Now, if we do not count a period of time duration $(0, t)$, but an event x occurs in a specific moment during $(0, t)$, then $x \sim P_{\Lambda}(\lambda t)$. This example shows the essential difference between discrete and continuous distributions. The reader may consider the question of what the difference is in terms of variables in this example.

Example 1.19: Waiting Period

Consider the example of a certain machine installed “waiting” to be used to deal with special jobs. In period $(0, t)$, the number x of the event “being used” yields discrete exponential distribution $f_{\Lambda}(\lambda t)$ (see Equation 1.73). Find the distribution function of the waiting time τ for the second job.

Assume the first job is at time $t = 0$.

Consider $P(\tau > t)$, first $\tau \geq 0$ and $t < 0$, $P(\tau \leq t) = 0$.

When $t > 0$, the event $\{\tau > t\} = \{\text{in duration } (0, t) \text{ no jobs}\} = \{x = 0\}$.

Thus, $P(\tau > t) = P(x = 0) = \frac{(\lambda t)^0}{0!} e^{-\lambda t} = e^{-\lambda t}$.

Furthermore,

$$P(\tau \leq t) = 1 - P(\tau > t) = 1 - e^{-\lambda t}$$

which is the total waiting time (see Figure 1.19).

The distribution function can be found as

$$f_{\Lambda}(t) = \lim_{\Delta t \rightarrow 0} \frac{\Delta P}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{P(\tau \leq t + \Delta t) - P(\tau \leq t)}{\Delta t} = \frac{d}{dt} P(\tau \leq t) = \lambda e^{-\lambda t}$$

That is, the waiting time yields exponential distribution.

Note that an important property of exponential distribution is being supposedly “memoryless.” That is, for any $\tau > 0$ and $t > 0$, we have

$$\begin{aligned} P[(\xi > \tau) \cup (t \mid \xi > \tau)] &= P[(\xi > \tau) + (t \mid \xi > \tau)] \\ &= \frac{P(\xi > \tau + t)}{P(\xi > \tau)} = \frac{e^{-\lambda(\tau+t)}}{e^{-\lambda\tau}} = e^{-\lambda t} = P(\xi > t) \end{aligned}$$

Let ξ denote the life span of a bridge (a car, a machine, and so on) if the bridge was built for τ years, that the probability distribution of lasting another t years has nothing to do with the first τ years. In other words, the first τ year is not memorized. Distribution of “always young.”

For any real-world buildings, bridges, machines, airplanes, and cars that are not “always young,” care must be taken to use exponential distributions.

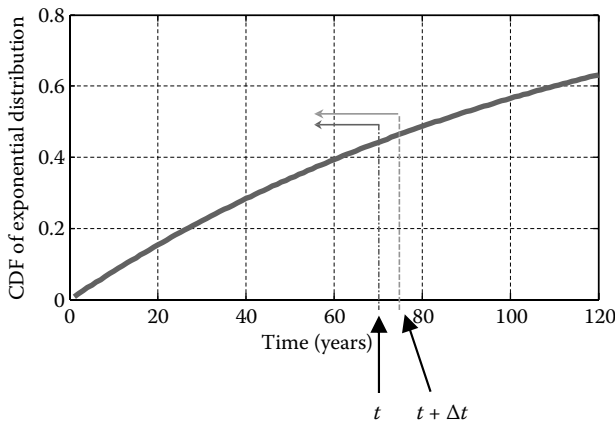


FIGURE 1.19 Waiting time.

Recall using Equation 1.60

$$p_{t_D} = 1 - e^{-\frac{t_D}{T_R}} = 1 - e^{-\rho t_D}$$

to approximate the probability of occurrence of an extreme event in year t_D , when the return period of such an event is T_R , see Equation 1.59.

The occurrence of such an event should be “memoryless.”

1.3.2.5 Rayleigh Distribution (Lord Rayleigh, 1842–1919)

The Rayleigh distribution has PDF given by (see Figure 1.20)

$$f_H(h) = \frac{h}{\sigma^2} e^{-\frac{1}{2}\left(\frac{h}{\sigma}\right)^2}, \quad h > 0 \quad (1.94)$$

Note that, symbol σ in Equation 1.94 is not the standard deviation of Rayleigh distribution.

1.3.3 CUMULATIVE DISTRIBUTION FUNCTIONS

Now, from the PDF, we can calculate the cumulative probabilities, which are discussed as follows.

1.3.3.1 Probability of Cumulative Event

In the above example of “waiting time,” namely,

$$P(\tau \leq t) = 1 - P(\tau > t) = 1 - e^{-\lambda t} \quad (1.95)$$

Consider the event $\tau \leq t$ (see Figure 1.21)

we have

$$\tau \leq t = (t_1 < t |_{\tau=t_1}) \cup (t_2 < t |_{\tau=t_2}) \cup (t_3 < t |_{\tau=t_3}) \dots \quad (1.96)$$

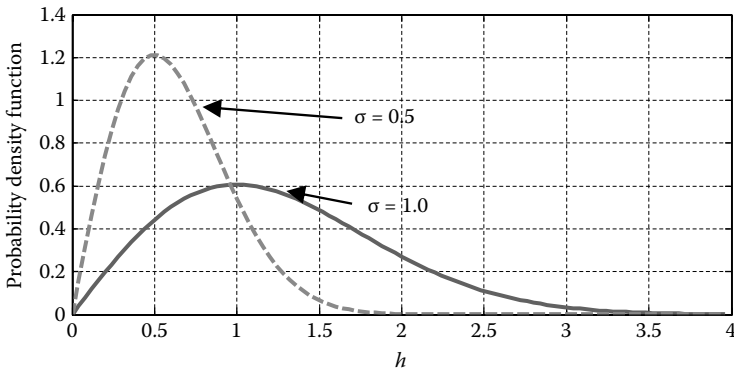


FIGURE 1.20 Rayleigh distribution.

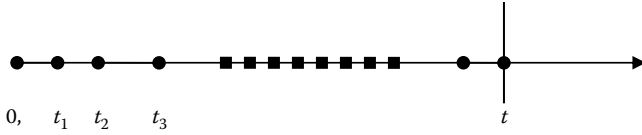


FIGURE 1.21 Event $\tau \leq t$.

$$P(\tau \leq t) = P[(t_1 < t |_{\tau=t_1}) \cup (t_2 < t |_{\tau=t_2}) \cup (t_3 < t |_{\tau=t_3}) \dots] \quad (1.97)$$

Note that event $(t_1 < t |_{\tau=t_1})$ and event $(t_2 < t |_{\tau=t_2})$ are not mutually exclusive so that using Equation 1.97 to calculate the probability $P(\tau \leq t)$ is extremely challenging. However, from another angle,

$$P(\tau \leq t) = \lim_{\Delta t \rightarrow 0} [f_{\Lambda}(\Delta t)\Delta t + f_{\Lambda}(2\Delta t)\Delta t + f_{\Lambda}(3\Delta t)\Delta t + \dots] \quad (1.98)$$

1.3.3.2 Cumulative Distribution Function (CDF)

From the previous discussion, we can further have the cumulative distribution function.

1.3.3.2.1 Definition of CDF

Equation 1.98 is the way to compute the probability of cumulative events, which is referred to as *cumulative distribution function* denoted by $F(t)$ and written as

$$F_{\Lambda}(t) = \int_0^t f_{\Lambda}(t) dt$$

In the above equation, the variable “ τ ” does not appear. Generally, we can write

$$F_X(x) = P(X \leq x) = \int_{-\infty}^x f_X(x) dx \quad (1.99)$$

1.3.3.2.2 Property of Cumulative Distribution Function

Now, consider the properties of CDF.

1. Range of CDF

$$0 \leq F_X(x) \leq 1 \quad (1.100)$$

2. Nondecreasing: if $x_2 \geq x_1$, then

$$F_X(x_2) \geq F_X(x_1) \quad (1.101)$$

1.3.3.2.3 Relation between PDF and Cumulative Distribution Function

$$f_x(x) = \frac{dF_x(x)}{dx}$$

(1.102)

1.3.3.3 Certain Applications of PDF and CDF

We now review the concepts of PDF and CDF and consider certain applications in the following.

1.3.3.3.1 Probability Computation

If the PDF is known, we can find the corresponding CDF through integration.

Example 1.20

Given earthquake records in the past 250 years, suppose we have the following data of PGA as

- Day d_1 Mon m_1 Yr y_1 , earthquake No. 1 with PGA_1
- Day d_2 Mon m_2 Yr y_2 , earthquake No. 2 with PGA_2
-
- Day d_n Mon m_n Yr y_n , earthquake No. n with PGA_n

Generate exceedance peak ground acceleration values (Table 1.4).
Using the data listed in Table 1.4, we can also obtain Figure 1.22.
Furthermore, by using Table 1.4, we can also generate Table 1.5 (the non-exceedance PGA)

TABLE 1.4

Exceedance Peak Ground Acceleration (U.S. Geological Survey)

	PGA > 0.01 g	PGA > 0.02 g	
Frequency (probability)	$p_{0.01}$	$p_{0.02}$	

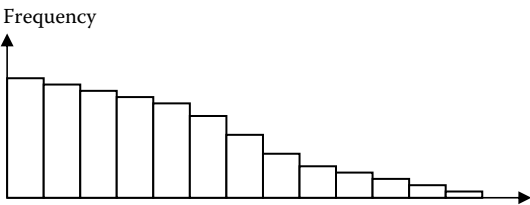


FIGURE 1.22 Frequency of PGA.

TABLE 1.5
Nonexceedance Peak Ground Acceleration

	PGA < 0.01 g	PGA < 0.02 g	
Frequency (probability)	$p_{0.01}$	$p_{0.01} + p_{0.02}$	

1.3.3.3.2 Find PDF from Cumulative Distribution Function

With the CDF known, we can also find the corresponding PDF through differentiation.

1.3.3.3.3 Curve Fit

To figure out a distribution through statistical data, we can use curve fit technology. Generally, using CDF can be easier than using PDF.

1.3.4 CENTRAL TENDENCY AND DISPERSION

1.3.4.1 Statistical Expectations and Moments

Central tendency is the observation of probability distributions when the collection of random variables follows a certain pattern. This kind of observation is important in understanding the nature of random process as well.

The following are statistical models of these observations.

1.3.4.2 Central Tendency, Mean Value

1.3.4.2.1 Discrete Variable Set X

Consider the mean value of a set of discrete variables, which can be written as

$$\mu_x = \sum_{\text{all } x_i} f_X(x_i)x_i \quad (1.103)$$

Recall a set

$$x = \{x_1, x_2, \dots, x_n\} \quad (1.104)$$

The mean value of x , denoted by $\bar{x}$ is given

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n} = \sum_{i=1}^n \frac{1}{n} x_i \quad (1.105)$$

In set X , each x_i has an equal chance to be considered; the weighting function for n is $1/n$; we thus have the probability of any value x_i showing up to be $1/n$, that is,

$$f_X(x) = \frac{1}{n} \quad (1.106)$$

1.3.4.2.2 Continuous Variable Set X

Similarly, consider the set of continuous variables. The mean value can be written as

$$\mu_x = \int_{-\infty}^{\infty} f_X(x)x dx \quad (1.107)$$

1.3.4.3 Variation, Variance, Standard Deviation, and Coefficient of Variation

Furthermore, consider the following measurement of dispersion.

1.3.4.3.1 Variance of Discrete Variable Set X, σ_X^2

The variance is defined as

$$\sigma_X^2 = \sum_{\text{all } x_i} f_X(x_i)(x_i - \mu_X)^2 \quad (1.108)$$

1.3.4.3.2 Variance of Continuous Variable Set X, σ_X^2

For continuous variables, we have

$$\sigma_X^2 = \int_{-\infty}^{\infty} f_X(x)(x - \mu_X)^2 dx \quad (1.109)$$

See Figure 1.23 for geometric description of mean and variance. Mean value is related to the first moment, centroid, or the center of mass. Variance is related to the second moment or moment of inertia.

1.3.4.3.3 Standard Deviation σ_X

The standard deviation is the square root of variance, that is,

$$\sigma_X = \sqrt{\sigma_X^2} \quad (1.110)$$

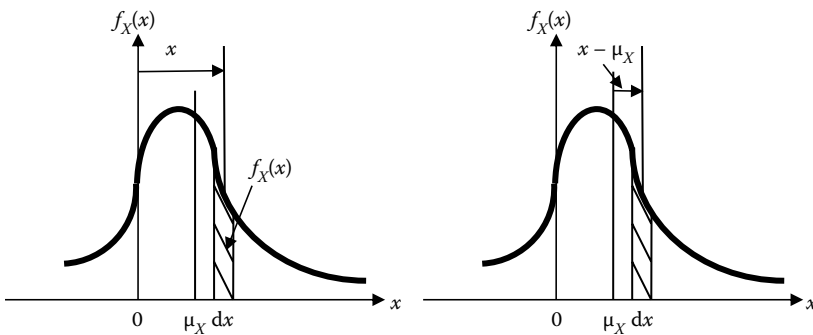


FIGURE 1.23 Moment and centroid, mean and variance.

Note that both the variance and the standard deviation can be used to denote the dispersion of a set of variables. However, the standard deviation has identical units as these variables.

1.3.4.3.4 Coefficient of Variation C_X

The coefficient of variation is the ratio of standard deviation and the mean value.

$$C_X = \frac{\sigma_X}{\mu_X} \quad (1.111)$$

1.3.4.4 Expected Values

Given a function of random variable $g(X)$, the expected value denoted by $E[g(X)]$ is written as

$$E[g(X)] = \int_{-\infty}^{\infty} f_X(x)g(x) dx \quad (1.112)$$

It is seen that the mean is the expected value of the set X .

$$\mu_X = E[X] \quad (1.113)$$

Thus, the expected value of a set of random variable $g(X)$, namely, $E[g(X)]$, can also be seen as a special operation of $g(X)$. It can be shown that

$$\sigma_X^2 = E[(X - \mu_X)^2] \quad (1.114)$$

Furthermore, we have

$$\sigma_X^2 = E[(X)^2] - \mu_X^2 \quad (1.115)$$

1.3.4.5 Linearity of Expected Values

One of the important properties of the operation of $E[(.)]$ is linearity, that is,

$$E[\alpha g(X) + \beta h(X)] = \alpha E[g(X)] + \beta E[h(X)] \quad (1.116)$$

where $g(X)$ and $h(X)$ are functions of X ; and α and β are deterministic scalars.

Example 1.21

Suppose Y is a linear variable of X , given by

$$Y = aX + b \quad (1.117)$$

where a and b are deterministic scalars; we have

$$\mu_Y = E[aX + b] = a\mu_X + b \quad (1.118)$$

1.3.5 NORMAL RANDOM DISTRIBUTIONS

In Section 1.3.4, we introduced several probability distributions. In the following, let us consider a particular distribution that plays an important role in understanding both random variables and random process.

1.3.5.1 Standardized Variables Z

First, consider a standardized variable defined as

$$Z = \frac{X - \mu_x}{\sigma_x} \quad (1.119)$$

the mean of Z is given by

$$\mu_Z = E\left[\frac{X - \mu_x}{\sigma_x}\right] = \frac{1}{\sigma_x} E[(X - \mu_x)] = \frac{1}{\sigma_x} \{E[X] - \mu_x\} = 0 \quad (1.120)$$

and the variance of Z can be expressed as

$$\sigma_Z^2 = E\left[\left(\frac{X - \mu_x}{\sigma_x}\right)^2\right] = \frac{1}{\sigma_x^2} E[(X - \mu_x)^2] = \frac{1}{\sigma_x^2} \sigma_x^2 = 1 \quad (1.121)$$

1.3.5.2 Gaussian (Normal) Random Variables (Carl F. Gauss, 1777–1855)

The normal random variables are denoted as

$$X \sim N(\mu_x, \sigma_x) \quad (1.122)$$

In the following, let us examine the distribution and additional important properties of normal random variables.

1.3.5.3 PDF of Normal Distribution

1.3.5.3.1 General Variable

The PDF of general normal random variables is given as

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma_x} e^{-\frac{(x-\mu_x)^2}{2\sigma_x^2}}, \quad -\infty < x < \infty \quad (1.123)$$

which is plotted in Figure 1.24.

1.3.5.3.2 Standardized Variable (See Equation 1.119)

For standardized variables, we have

$$f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}, \quad -\infty < z < \infty \quad (1.124)$$

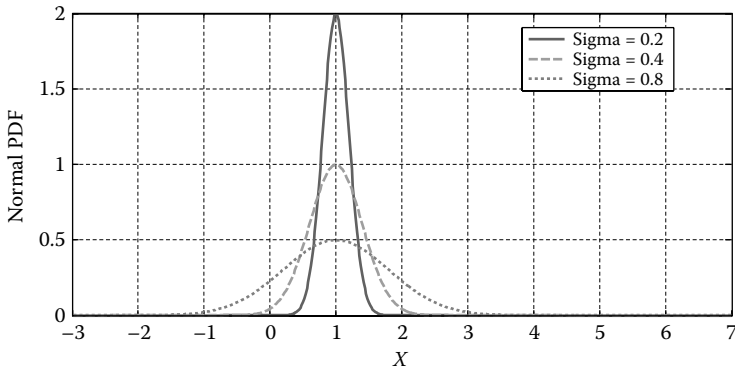


FIGURE 1.24 PDF of normal distribution with $\mu_X = 1$.

1.3.5.4 Cumulative Distribution Function of Normal Distribution

Now, consider the corresponding CDF.

1.3.5.4.1 General Variable

For general variables, we have

$$F_X(x) = \frac{1}{\sqrt{2\pi}\sigma_X} \int_{-\infty}^x e^{-\frac{(\xi-\mu_X)^2}{2\sigma_X^2}} d\xi \quad (1.125)$$

Figure 1.25 plots several CDF with different standard deviations.

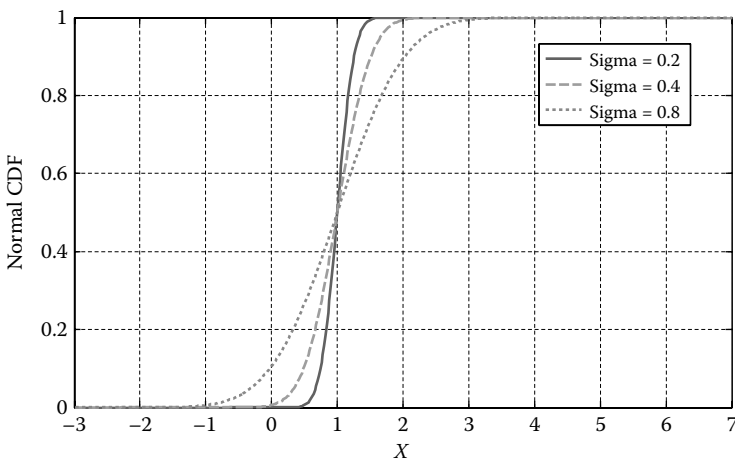


FIGURE 1.25 CDF of normal distribution with $\mu_X = 1$.

1.3.5.4.2 Standardized Variable

For standardized variables, we have

$$F_Z(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-\frac{\xi^2}{2}} d\xi \quad (1.126)$$

This CDF is very important, and is specially denoted as

$$\Phi(z) = F_Z(z) \quad (1.127)$$

Figure 1.26 plots both PDF and CDF for standardized normal variables.

Example 1.22

From a hotel to the Buffalo airport, one can take the local or expressway. Time needed through local road $t_L \sim N(26, 5)$. Time needed through expressway $t_L \sim N(31, 2)$.

Question (1): If one only has 30 minutes, then, which way is better? (Less chance to miss the flight)

Question (2): One has 35 minutes, which way is better?

1. Denote the time needed as t ,

$$P(t > 30) = 1 - P(t \leq 30) = 1 - \frac{1}{\sqrt{2\pi}(5)} \int_{-\infty}^{30} e^{-\frac{(\xi-26)^2}{2(5^2)}} d\xi = 1 - 0.7881 = 0.21$$

$$P(t > 30) = 1 - P(t \leq 30) = 1 - \frac{1}{\sqrt{2\pi}(2)} \int_{-\infty}^{30} e^{-\frac{(\xi-31)^2}{2(2^2)}} d\xi = 0.6915$$

Thus, using local road $\rightarrow$ less probability of “missing the flight.”

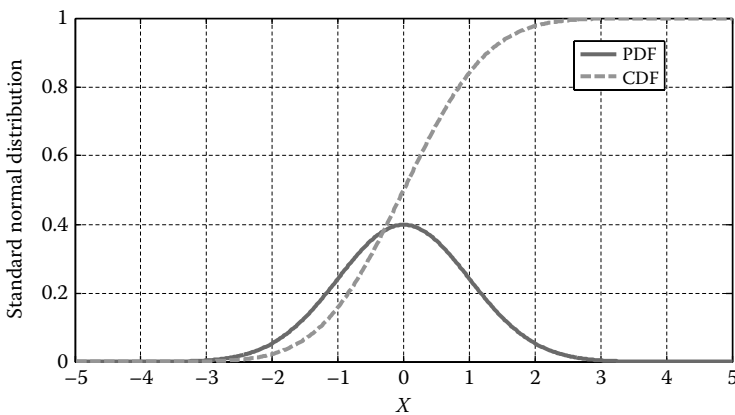


FIGURE 1.26 PDF and CDF for standardized normal variables.

Note that one can also use standard normal distribution. For example, for $t_L \sim N(26, 5)$.

Denote

$$T = \frac{t - 26}{5} \Big|_{t=30} = 0.8$$

$$P(t \leq 30) = \Phi(0.8) = 0.7881 \text{ and } 1 - \Phi(0.8) = 0.21$$

2. Local road

$$T = \frac{t - 26}{5} \Big|_{t=35} = 1.8$$

$$1 - \Phi(1.8) = 0.036$$

Expressway

$$T = \frac{t - 31}{2} \Big|_{t=35} = 2$$

$$1 - \Phi(2) = 0.023$$

Using the expressway lowers the chance of missing the flight.

1.3.6 ENGINEERING APPLICATIONS

1.3.6.1 Probability-Based Design

Consider the component of a system, which can be a structure, a machine, a vehicle, etc. Designing that component to resist a certain load, we basically have two different approaches, namely, the allowed stress design and the probability-based design.

1.3.6.1.1 Allowed Stress Design (Deterministic Design)

The allowed stress design is based on the following criterion:

$$R_N > Q_N \quad (1.128)$$

where

R_N and Q_N are, respectively, the values of nominal resistance and load.

Equation 1.128 can be realized by using a safety factor S , that is,

$$R_N = S Q_N \quad (1.129)$$

1.3.6.1.2 Load and Resistance Factor Design (Probability-Based Design)

Besides the allowed stress design, we also have the probability-based design, that is,

$$P(R_D \leq Q_D) < [p_r] \quad (1.130)$$

where R_D , Q_D values of nominal resistance and load. Equation 1.130 means that the probability of the event that the resistance is smaller/equal to the load, which is referred to as the failure probability p_f must be smaller than the allowed value $[p_r]$.

Or in general we can also write

$$p_f = 1 - p_r \quad (1.131)$$

where p_f and p_r are, respectively, the probability of failure and reliability.

In real-world design, we often use the nominal values. However, it is more direct to use the mean values in probability analysis, we thus have the following relationship:

$$Q_D = \gamma Q_N \quad (1.132)$$

and

$$R_D = \Phi R_N \quad (1.133)$$

Here, R_N and Q_N are, respectively, the mean values of resistance and load; the terms γ and Φ are, respectively, the load and resistance factors.

1.3.6.1.2.1 Load Factor γ

The load factor can be seen as follows

$$\gamma = \beta_Q(1 + \kappa_R C_Q) \quad (1.134)$$

where κ_R is the number of standard deviations; for example, $\kappa_R = 3$.

In Equation 1.134, the term β_Q is called the bias of load

$$\beta_Q = \frac{\mu_Q}{Q_N} \quad (1.135)$$

The term C_Q is the coefficient of variation, given by

$$C_Q = \frac{\sigma_Q}{\mu_Q} \quad (1.136)$$

1.3.6.1.2.2 Resistance Factor Φ

Now, consider Equation 1.133 and the resistance factor can be written as

$$\Phi = \beta_R(1 - \kappa_R C_R) \quad (1.137)$$

Here, the term β_R is the bias of resistance, that is,

$$\beta_R = \frac{\mu_R}{R_N} \quad (1.138a)$$

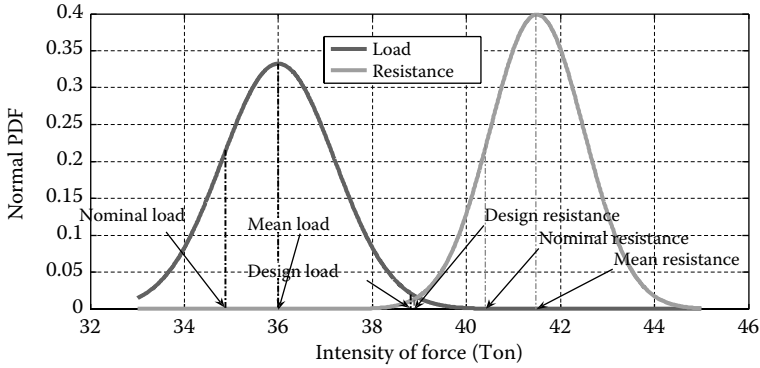


FIGURE 1.27 Load and resistance.

The term C_R is the coefficient of variation

$$C_R = \frac{\sigma_R}{\mu_R} \quad (1.138b)$$

The relationship between the PDF of the load and resistance relates to failure probability p_f can be shown in Figure 1.27, where the dark line is PDF of the demand f_Q and lighter line is the PDF of resistance f_R . Thus, the failure probability p_f can be written as

$$p_f = \int_{-\infty}^{\infty} f_Q(q) \int_{-\infty}^q f_R(r) dr dq \quad (1.139a)$$

If both the demand and the resistance are normal, then the random variable $R-Q$ is also normal; with the PDF denoted by $f_{R-Q}(z)$ the failure probability can be further written as

$$p_f = \int_{-\infty}^0 f_{R-Q}(z) dz \quad (1.139b)$$

1.3.6.2 Lognormal Distributions

In general, we do not have negative load or resistance; in these cases, we may consider lognormal distribution. A lognormal distribution of a random set has all of its variables whose logarithm is normally distributed. If x is a random variable with a normal distribution, then

$$y = e^x \quad (1.140)$$

is a lognormal distribution; likewise, y is lognormally distributed, then $x = \log(y)$ will be normally distributed.

1.3.6.2.1 Probability Density Function

The base of the logarithmic function does not matter for generic lognormal distribution. Using natural log:

$$f_X(y) = \frac{1}{\sqrt{2\pi} \sigma_X y} e^{-\frac{(\ln y - \mu_X)^2}{2\sigma_X^2}} \quad y > 0 \quad (1.141)$$

Here, μ_X and σ_X are the mean and standard deviation of the variable's natural logarithm, in which the variable's logarithm is normally distributed.

1.3.6.2.2 Cumulative Distribution Function

The CDF is given by

$$F_Y(y) = \Phi\left(\frac{\ln(y) - \mu_X}{\sigma_X}\right) \quad (1.142)$$

Here, function $\Phi(\cdot)$ is defined previously in Equation 1.127.

Denote M_Y as median value of Y (geometric mean value of Y)

$$P(Y \leq M_Y) = 0.5 \quad (1.143)$$

From Equation 1.142, we have

$$F_Y(M_Y) = \Phi\left(\frac{\ln(M_Y) - \mu_X}{\sigma_X}\right) = 0.5$$

Note that $\Phi^{-1}(0.5) = 0$, thus

$$\Phi^{-1}(0.5) = \left(\frac{\ln(M_Y) - \mu_X}{\sigma_X}\right) = 0 \quad (1.144)$$

Therefore, we have

$$\ln(M_Y) = \mu_X \quad (1.145)$$

Mean and standard deviation of a lognormal distribution as functions of the mean and standard deviation of the normally distributed variable set X are, respectively,

$$\mu_Y = e^{\mu_X + \frac{1}{2}\sigma_X^2} \quad (1.146a)$$

or

$$\mu_X = \frac{1}{2} \ln\left(\frac{\mu_Y^2}{1 + C_Y^2}\right) \quad (1.146b)$$

and

$$\sigma_Y = e^{\mu_X + \frac{1}{2}\sigma_X^2} \sqrt{e^{\sigma_X^2} - 1} \quad (1.147a)$$

or

$$\sigma_X^2 = \mu_Y^2 (e^{\mu_Y^2} - 1) \quad (1.147b)$$

Here, C_Y is the coefficient of variation of random variable Y .

1.3.6.3 Further Discussion of Probability-Based Design

It can be proven that the difference of $R_D - Q_D$, if both R_D and Q_D are normally distributed, is also normally distributed. That is, let

$$F = R_D - Q_D \quad (1.148)$$

We must have

$$F \sim N(\mu_F, \sigma_F) \quad (1.149)$$

The mean is

$$\mu_F = \mu_R - \mu_Q \quad (1.150)$$

where, μ_R and μ_Q are, respectively, the means of R_D and Q_D .

The standard deviation is

$$\sigma_F = (\sigma_R^2 + \sigma_Q^2)^{1/2} \quad (1.151)$$

where, σ_R and σ_Q are, respectively, the standard deviations of R_D and Q_D .

A critical state defined at

$$F = 0. \quad (1.152)$$

which is called the *limit state*.

Using standardized variables, we can write

$$\frac{F - \mu_F}{\sigma_F} = \frac{0 - \mu_F}{\sigma_F} = -\frac{\mu_R - \mu_Q}{\sqrt{\sigma_R^2 + \sigma_Q^2}} \equiv -\beta \quad (1.153)$$

Failure probability is then given by

$$p_f = P(F \leq 0) = P(R - Q \leq 0) = \Phi(-\beta) \quad (1.154)$$

Example 1.23

See Table 1.6

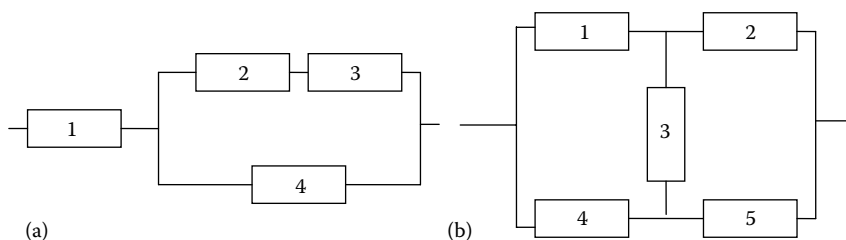
TABLE 1.6
Reliability Indices and Failure Probabilities

β	2	2.5	3	3.5
p_f	0.0228	0.0062	0.0013	0.0002

PROBLEMS

- Using a Venn diagram to show that $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ by the fact that the set $(A \cup B)$ can be seen as the union of disjointed sets $(A \cap \bar{B})$, $(\bar{A} \cap B)$, and $(A \cap B)$.
- Find the sample spaces of the following random tests:
 - The record of average score of midterm test of class CIE520/MAE536. (Hint, we have n people and the full score is 40).
 - To continuously have 10 products up to standard, the total number of checked products.
 - Inspecting products that are marked “C” if certified and marked “D” if defective; if two “D” are consequently checked, the inspection will be stopped. Or, if four products have been checked, the inspection will also be stopped; the records of the inspection.
 - The coordinates of point inside a circle.
- A and B denote two events, respectively.
 - If $\bar{A}B = A\bar{B}$ Prove $A = B$
 - Suppose either A or B occurs; find the corresponding probability
- For events A , B , and C , $P(A) = 1/2$, $P(B) = 1/3$, $P(C) = 1/5$, $P(A \cap B) = 1/10$, $P(A \cap C) = 1/15$, $P(B \cap C) = 1/20$, $P(A \cap B \cap C) = 1/30$, find
 - $P(A \cup B)$
 - $P(\bar{A} \cup \bar{B})$
 - $P(A \cup B \cup C)$
 - $P(\bar{A} \cap \bar{B} \cup C)$
 - $P(\bar{A} \cap \bar{B} \cap \bar{C})$
 - $P(\bar{A} \cap \bar{B} \cap C)$
- Calculate the probabilities of (a) $P(X < 3)$, (b) $P(X > 2)$, and (c) $P(2 < X < 3)$ with the following distributions
 - Poisson ($\lambda = 1$)
 - Uniform ($a = 1$, $b = 4$)
 - Rayleigh ($\sigma = 1$)
 - Normal ($\mu = 2$, $\sigma = 0.5$)
- Find the mean and standard deviation of the following distributions:
 - Bernoulli
 - Poisson
 - Rayleigh
 - Normal

7. Ten balls marked 1, 2, ..., 10, respectively, which are picked up by 10 people. Now, randomly choose three people, find the following probabilities:
 - a. The minimum number is 5.
 - b. The maximum number is 5.
8. The 11 letters of the word “probability” are written in 11 cards, respectively. Pick up seven cards. Find the probability of the seven cards being “ability.”
9. Box I contains 50 bearings, among which, 10 are grade A. Box II contains 30 bearings of the same type, among them, 18 are grade A. The rest of the bearings are grade B. If taking one of the boxes and picking up the bearings twice. In each case, only one bearing is picked and there are no returns. Find the probability that
 - a. The first bearing is grade A.
 - b. Suppose that the first one is grade A and that the second is also grade A.
10. Find the reliability of the following two systems:
 - a. Four independent elements, 1, 2, 3, and 4, with reliabilities at p_1 , p_2 , p_3 , and p_4 , respectively, are arranged as shown in Figure P1.1a.
 - b. Five independent elements, 1, 2, 3, 4, and 5, with identical reliabilities p are arranged as shown in Figure P1.1b.

**FIGURE P1.1**

2 Functions of Random Variables

In Chapter 1, the basic assumptions and theory of probability are briefly reviewed together with the concepts of random variables and their distribution. In this chapter, we will discuss several important functions of random variables, which are also random variables. Because the basic idea to treat random variables is to investigate their distributions, similarly, to study the functions of random variables, we also need to consider the corresponding probability density function (PDF) and cumulative distribution function (CDF).

2.1 SYSTEMS AND FUNCTIONS

In Chapter 1, it is shown that a vibrating object can be a dynamic system. In fact, in Chapter 6, we will further see that a single degree of freedom vibration system is a second-order dynamic system. In Chapter 9, we will see that an n -degree-of-freedom (n -DOF) vibration system will have $2n$ th order, which can be broken down into n second-order or $2n$ first-order systems. Figure 2.1a shows a typical block diagram of the relationship between input-system-output.

The function of random variables can be seen as the relationship between the original random variable and the response of a certain system. In these circumstances, the original variables are generally not functions of time (see Figure 2.1b). Furthermore, among random processes, we can have a similar relationship, except that the input and output become the functions of time.

In this manuscript, because of the limited space, we will not discuss the theory of the system in detail. Only the necessary concepts will be mentioned. For further knowledge of the system, especially dynamic systems, interested readers may consult corresponding textbooks, such as the works of Antsaklis and Michel (1997).

2.1.1 DYNAMIC SYSTEMS

2.1.1.1 Order of Systems

A dynamic system can be modeled as a differential equation. In this manuscript, the order of a system means the highest order of differentiation.

2.1.1.1.1 Zero Order System

If the relationship of input and output involves no differentiation or integration with respect to time, we have a zero order system.

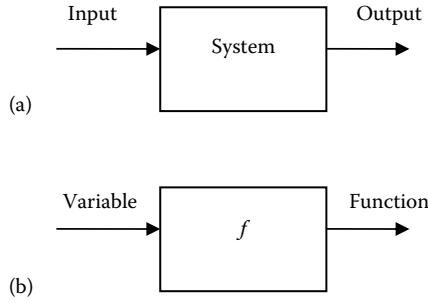


FIGURE 2.1 (a) System and (b) function.

2.1.1.1.2 First-Order System

If the output of a system is the first derivative of input with respect to time, we have a first-order system.

2.1.1.1.3 Second-Order System

If the output of a system is the second derivative of input with respect to time, we have a second-order system.

2.1.1.2 Simple Systems

2.1.1.2.1 Subsystems

A system of functions can be rather complex. However, in engineering applications, we can always break a complex system down into several subsystems. When a system is broken down, these subsystems can be in parallel or in series (see Figure 1.8a and b, respectively, where all the “ P_i ” symbols can be replaced by “subsystem i ”). In the following, let us consider basic subsystems.

2.1.1.2.2 Proportional System

A proportional system is zero ordered. Suppose X is a random set and scalar a is a proportional constant, we have

$$Y = aX \quad (2.1)$$

2.1.1.2.3 Differential System

Suppose X is a random set that is also a function of time and Y is the result of the first derivative of X with respect to time (in Chapter 5, we will discuss how to take a derivative of random variables), we have

$$Y = \frac{d}{dt} X \quad (2.2)$$

2.1.1.2.4 Integral System

Suppose X is a random set that is also a function of time and Y is the result of the integration of X with respect to time (in Chapter 5, we will discuss how to take an integration of random variables), we have

$$Y = \int X dt + C \quad (2.3)$$

Generally speaking, the integration constant C can also be a random variable.

2.1.1.2.5 Polynomial System

Suppose X is a random set and Y is the result of polynomials of X , we have

$$Y = a_0 + a_1X + a_2X^2 + \dots \quad (2.4)$$

where a_0 , a_1 , and a_2 are constants.

2.1.2 JOINTLY DISTRIBUTED VARIABLES

Besides the above-mentioned relationship between X and Y , in view of distributions, we have several special cases that are important for understanding the nature of random variables. Especially when considering the relationships between input and corresponding output variable, and between variables and the corresponding functions, the concept of jointly distributed variables plays an important role. They are then discussed as follows.

2.1.2.1 Joint and Marginal Distributions of Discrete Variables

Let us consider discrete variables first.

2.1.2.1.1 Joint Distributions

Suppose we have two sets of random variables, J and K , which are jointly considered. We can denote them as

$$(J, K) = \{j, k\}, \quad j, k = 1, 2, \dots \quad (2.5)$$

To describe the intersection of J and K , we can write

$$p_{JK}(j, k) = P[(J = j) \cap (K = k)] \quad (2.6)$$

where p_{JK} is the joint distribution, it satisfies:

$$(1) \quad p_{JK} \geq 0 \quad (2.7)$$

$$(2) \quad \sum_{\text{all } j} \sum_{\text{all } k} p_{JK} = 1 \quad (2.8)$$

See Table 2.1 for a list of joint distributions.

TABLE 2.1
Joint Distribution

<i>J</i>	<i>K</i>		
	<i>k</i> = 1	<i>k</i> = 2	
<i>i</i> = 1	p_{11}	p_{12}	
<i>i</i> = 2	p_{21}	p_{22}	
...			

Example 2.1

Among truck buyers, 50% purchase American pickups, 20% purchase Japanese, and 30% buy other country's pickups. Randomly find two customers, denote as *A* and *J* as the number of American and Japanese customers. Find the joint distributions.

The possible values of *A* and *J* are 0, 1, 2 (Table 2.2). When *A* = *a* (*a* people buy American); *J* = *j* (*j* people buy Japanese), 2 − *a* − *j* people buy others. Therefore,

$$P[(A = a) \cap (J = j)] = C_2^a C_{2-a}^j (0.5)^a (0.2)^j (0.3)^{2-a-j}$$

$$P[(A = 0) \cap (J = 0)] = C_2^0 C_2^0 (0.5)^0 (0.2)^0 (0.3)^2 = 0.09$$

$$P[(A = 0) \cap (J = 1)] = C_2^0 C_2^1 (0.5)^0 (0.2)^1 (0.3)^1 = 0.12$$

$$P[(A = 0) \cap (J = 2)] = C_2^0 C_2^2 (0.5)^0 (0.2)^2 (0.3)^0 = 0.04$$

$$P[(A = 1) \cap (J = 0)] = C_2^1 C_1^0 (0.5)^1 (0.2)^0 (0.3)^1 = 0.3$$

$$P[(A = 1) \cap (J = 1)] = C_2^1 C_1^1 (0.5)^1 (0.2)^1 (0.3)^0 = 0.2$$

$$P[(A = 1) \cap (J = 2)] = 0$$

$$P[(A = 2) \cap (J = 0)] = C_2^2 C_0^0 (0.5)^2 (0.2)^0 (0.3)^2 = 0.25$$

$$P[(A = 2) \cap (J = 1)] = 0$$

$$P[(A = 2) \cap (J = 2)] = 0$$

TABLE 2.2
Joint Distribution of *J* and *K*

<i>A</i>	<i>J</i>		
	<i>j</i> = 0	<i>j</i> = 1	<i>j</i> = 2
<i>a</i> = 0	0.09	0.12	0.04
<i>a</i> = 1	0.3	0.2	0
<i>a</i> = 2	0.25	0	0

TABLE 2.3
CDF of J and K

A	J			$p_A(a)$
	$j = 0$	$j = 1$	$j = 2$	
$a = 0$	0.09	0.12	0.04	0.25
$a = 1$	0.3	0.2	0	0.5
$a = 2$	0.25	0	0	0.25
$p_J(j)$	0.64	0.32	0.04	1

2.1.2.1.2 Marginal Distributions

From the above example, we have Table 2.3 to list the CDF of A and J . That is, between the two random variables, we only consider one of them, no matter what value the other variable is (which means that this variable will take any possible value)

$$\begin{aligned}
 p_J(j) &= P(J = j) = P_{JK}[(J = j) \cap (k \text{ takes all possible values})] \\
 &= \sum_{\text{all } k} P_{JK}[(J = j) \cap (K = k)] = \sum_{\text{all } k} p_{JK}, \quad j = 1, 2, 3, \dots
 \end{aligned} \quad (2.9)$$

$$\begin{aligned}
 p_K(k) &= P(K = k) = P_{JK}[(j \text{ takes all possible values}) \cap (K = k)] \\
 &= \sum_{\text{all } j} P_{JK}[(J = j) \cap (K = k)] = \sum_{\text{all } j} p_{JK}, \quad k = 1, 2, 3, \dots
 \end{aligned} \quad (2.10)$$

2.1.2.2 Joint and Marginal Distributions of Continuous Variables

Similar to the case of discrete variables, suppose we have two sets of continuous variables and consider the joint distributions. That is, we have

$$(X, Y) = \{x, y\} \quad (2.11)$$

Equation 2.11 can be graphically shown in Figure 2.2. Generally, the ranges of x and y are

$$-\infty < x < \infty, \quad -\infty < y < \infty \quad (2.12)$$

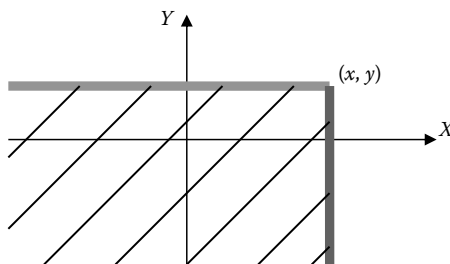


FIGURE 2.2 Ranges of variables.

2.1.2.2.1 Joint Distributions 1 (Bivariate PDF)

Now, the joint distribution of PDF can be written as

$$\begin{aligned} P[(X = x) \cap (Y = y)] &= P[(x < X \leq x + dx) \cap (y < Y \leq y + dy)] \\ &= f_{XY}(x, y) dx dy \end{aligned} \quad (2.13)$$

where $f_{XY}(x, y)$ is the bivariate density function, which satisfies

$$(1) f_{XY}(x, y) \geq 0 \quad (2.14)$$

$$(2) \iint f_{XY}(x, y) dx dy = 1 \quad (2.15)$$

2.1.2.2.2 Joint Distributions 2 (Bivariate CDF)

The joint distribution of CDF can be written as

$$F_{XY}(x, y) = P[(X \leq x) \cap (Y \leq y)] \quad (2.16)$$

which can be further written as

$$F_{XY}(x, y) = \int_{-\infty}^y \int_{-\infty}^x f_{XY}(u, v) du dv \quad (2.17)$$

The relationship of the PDF and CDF is

$$f_{XY}(x, y) = \frac{\partial^2}{\partial x \partial y} [F_{XY}(x, y)] \quad (2.18)$$

2.1.2.2.3 Marginal Distributions

Similar to case of discrete variables, marginal distributions can be obtained through joint distributions:

$$\begin{aligned} F_X(x) &= P(X \leq x) = P[(X \leq x) \cap (Y < \infty)] = F_{XY}(x, \infty) \\ &= \int_{-\infty}^x \int_{-\infty}^{\infty} f_{XY}(u, v) dv du \end{aligned} \quad (2.19)$$

and

$$\begin{aligned} F_Y(y) &= P(Y \leq y) = P(X < \infty \cap Y \leq y) = F_{XY}(\infty, y) \\ &= \int_{-\infty}^y \int_{-\infty}^{\infty} f_{XY}(u, v) du dv \end{aligned} \quad (2.20)$$

The PDF of X is

$$f_X(x) = \frac{dF_X(x)}{dx} = \int_{-\infty}^{\infty} f_{XY}(x, y) dy \quad (2.21)$$

The PDF of Y is

$$f_Y(y) = \frac{dF_Y(y)}{dy} = \int_{-\infty}^{\infty} f_{XY}(x, y) dx \quad (2.22)$$

Example 2.2

A two-dimensional random variable (X, Y) has the following PDF

$$f_{XY}(x, y) = ce^{-2(x+y)}, \quad 0 < x < \infty, 0 < y < \infty$$

Question (1): Find constant c .

Question (2): Find the CDF.

Question (3): Find the probability of area C shown in Figure 2.3.

$$\begin{aligned} (1) \quad \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{XY}(u, v) du dv &= \int_0^{\infty} \int_0^{\infty} ce^{-2(x+y)} dx dy \\ &= c \left[\int_0^{\infty} e^{-2(y)} dx \int_0^{\infty} e^{-2(y)} dy \right] = \frac{c}{4} \rightarrow c = 4 \end{aligned}$$

$$\begin{aligned} (2) \quad F_{XY}(x, y) &= \int_{-\infty}^y \int_{-\infty}^x f_{XY}(u, v) du dv = \int_{-\infty}^y \int_{-\infty}^x 4e^{-2(u+v)} du dv \\ &= (1 - e^{-2x})(1 - e^{-2y}), \quad 0 < x < \infty, 0 < y < \infty \end{aligned}$$

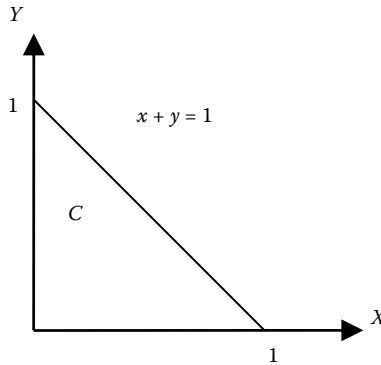


FIGURE 2.3 Probability of area C .

$$F_{XY}(x, y) = \begin{cases} (1 - e^{-2x})(1 - e^{-2y}), & 0 < x < \infty, 0 < y < \infty \\ 0, & \text{elsewhere} \end{cases}$$

$$(3) \quad P[(X, Y) \in C] = \int_0^1 \int_0^{1-x} f_{XY}(x, y) dx dy = 4 \left[\int_0^1 dx \int_0^{1-x} e^{-2(x+y)} dy \right] = 1 - 3e^{-2}$$

2.1.3 CONDITIONAL DISTRIBUTION AND INDEPENDENCE

2.1.3.1 Discrete Variables

From Section 2.1.2, it is seen that marginal distributions can be obtained through joint distributions. However, we cannot always determine joint distributions through marginal distributions. If we also know the probability of event $(X = x_i)$ under the condition $(Y = y_j)$, denoted by

$$P[(X = x_i)|(Y = y_j)] = p_{X|Y}(x_i|y_j) \quad (2.23)$$

then, from the conditional probability

$$P[(X = x_i) \cap (Y = y_j)] = P[(X = x_i)|(Y = y_j)] P(Y = y_j) \quad (2.24)$$

if

$$P(Y = y_j) \neq 0 \quad (2.25)$$

we have

$$p_{X|Y}(x_i|y_j) = P[(X = x_i)|(Y = y_j)] = \frac{P[(X = x_i) \cap (Y = y_j)]}{P(Y = y_j)} \quad (2.26)$$

Fixing j and letting i vary, we have a series $p_{X|Y}(x_i|y_j)$ ($i = 1, 2, 3, \dots$)

It is seen that

$$(1) \quad p_{X|Y}(x_i|y_j) \geq 0 \quad (i = 1, 2, 3, \dots) \quad (2.27)$$

and

$$(2) \quad \sum_{\text{all } i} p_{X|Y}(x_i|y_j) = 1 \quad (2.28)$$

That is, the series $p_{X|Y}(x_i|y_j)$ ($i = 1, 2, 3 \dots$) satisfies the two basic requirements of probability distributions. This implies that $p_{X|Y}(x_i|y_j)$ ($i = 1, 2, 3 \dots$) is a probability distribution, which describes the statistical property of random variable X , under the condition that $Y = y_j$.

Now, we can ask a question: is the probability distribution $p_{X|Y}(x_i|y_j)$ equal to $p_X(x_i)$? Generally speaking, it is not, because $p_{X|Y}(x_i|y_j)$ needs the condition

$$Y = y_j.$$

(2.29)

Here, $p_{X|Y}(x_i|y_j)$ is the *conditional distribution*. And we have the following

$$p_{Y|X}(y_j|x_i) = P[(Y = y_j)|(X = x_i)] = \frac{P[(X = x_i) \cap (Y = y_j)]}{P(X = x_i)}$$

(2.30)

Example 2.3

Recall the above-mentioned example shown in Table 2.4 again. Consider the conditional distribution under conditions $j = 0, j = 1$ and $j = 2$, we have Table 2.5.

TABLE 2.4
CDF of the Above-Mentioned Example

A	J			$p_A(a)$
	$j = 0$	$j = 1$	$j = 2$	
$a = 0$	0.09	0.12	0.04	0.25
$a = 1$	0.3	0.2	0	0.5
$a = 2$	0.25	0	0	0.25
$p_f(j)$	0.64	0.32	0.04	1

TABLE 2.5
Conditional Distributions

A	J		
	$P(A = a j = 0)$ $(p_j(0) = 0.64)$	$P(A = a j = 1)$ $(p_j(1) = 0.32)$	$P(A = a j = 2)$ $(p_j(2) = 0.04)$
$a = 0$	$0.09/0.64 = 0.14$	$0.12/0.32 = 0.375$	$0.04/0.04 = 1$
$a = 1$	$0.3/0.64 = 0.47$	$0.2/0.32 = 0.625$	$0/0.04 = 0$
$a = 2$	$0.25/0.64 = 0.39$	$0/0.32 = 0$	$0/0.04 = 0$
$\sum p_A(a)$	1	1	1

2.1.3.2 Continuous Variables

Similar to discrete variables, Equation 2.30 can be written as

$$P[(X = x)|(Y = y)] = f_{X|Y}(x|Y = y) dx = \frac{f_{XY}(x, y) dx dy}{f_Y(y) dy} \quad (2.31)$$

Thus, the conditional PDF of variable X is

$$f_{X|Y}(x|Y = y) = \frac{f_{XY}(x, y)}{f_Y(y)} \quad (2.32)$$

Similarly, we have

$$f_{Y|X}(y|X = x) = \frac{f_{XY}(x, y)}{f_X(x)} \quad (2.33)$$

2.1.3.3 Variable Independence

Variable independence is an important concept. Treating two independent sets can significantly reduce the efforts. On the other hand, if two sets of variables are not independent but we mistakenly treat them as independent, severe errors may be introduced.

2.1.3.3.1 Discrete Variables

First, consider discrete cases. Recall Equation 1.45: two events, A and B , are independent, if

$$P(A \cap B) = P(A)P(B) \quad (2.34)$$

furthermore, if

$$P(B|A) = P(B) \quad (2.35)$$

Now, consider variables. Similarly, random variable J and K are independent, if

$$p_{JK}(j, k) = p_J(j)p_K(k) \quad (2.36)$$

Because

$$p_{JK}(j, k) = P[(J = j) \cap (K = k)] \quad (2.37)$$

Because J and K are independent,

$$P[(J = j) \cap (K = k)] = P(J = j)P(K = k) = p_J(j)p_K(k) \quad (2.38)$$

Thus, we can also have

$$p_{X|Y}(x_i|y_j) = p_X(x_i) \quad (2.39)$$

and

$$p_{Y|X}(y_j|x_i) = p_Y(y_j) \quad (2.40)$$

2.1.3.3.2 Continuous Variables

Similarly, for the independence of continuous variables, we have

$$f_{XY}(x,y) = f_X(x) f_Y(y) \quad (2.41)$$

and

$$f_{X|Y}(x|Y=y) = f_X(x) \quad (2.42)$$

also

$$f_{Y|X}(y|X=x) = f_Y(y) \quad (2.43)$$

Example 2.4

Mr. A and Ms. B plan to meet in place C at 6:00 pm. The arriving times of each person are independent and are uniformly distributed between 6:00 pm and 7:00 pm. Find the probability that the first arrival must wait for more than 10 minutes.

Denote that Mr. A arrives at X minutes after 6:00 and Ms. B at Y minutes after 6:00. The probability is

$$P(X + 10 > Y) + P(Y + 10 > X)$$

Note that $P(X + 10 > Y) = P(Y + 10 > X)$

See Figure 2.4.

$$P(X+10>Y)=\iint_{X+10>Y} f_{XY}(x,y)dx dy=\int_{10}^{60} dy \int_0^{y-10} \left(\frac{1}{60}\right)^2 dx=\frac{25}{72}$$

So, $P = P(X + 10 > Y) + P(Y + 10 > X) = 0.69$.

Readers may consider the question what if the distribution of Mr. A is that shown in Figure 2.5.

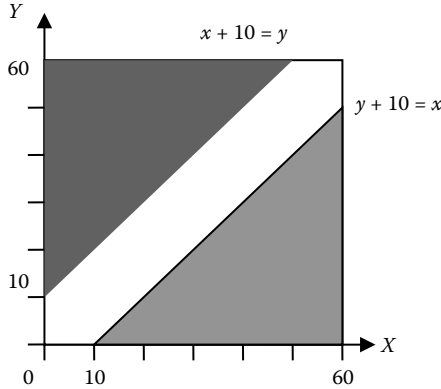


FIGURE 2.4 Regions of integrations.

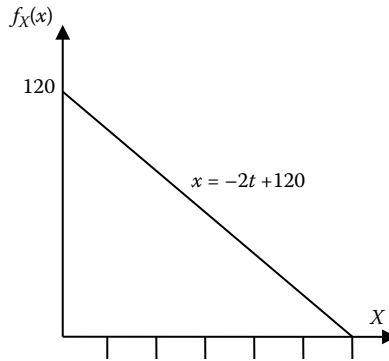


FIGURE 2.5 Distribution of Mr. A's probability.

2.1.4 EXPECTED VALUE, VARIANCE, COVARIANCE, AND CORRELATION

We now begin to consider a function of both random variables X and Y , denoted by $g(X,Y)$ and consider the central tendency, dispersion. We will also determine how these two sets of variables are related. Here, we only consider continuous variables. Readers can develop their own formulae for discrete variables. Or they can consult typical textbooks of probability, for example, see Saeed (2004).

2.1.4.1 Expected Value of $g(X,Y)$

The expected value can be written as

$$E[g(X,Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{XY}(x) g(x,y) dx dy \quad (2.44a)$$

Particularly, the expected value of variable X is

$$E[X] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{XY}(x) x dx dy \quad (2.44b)$$

2.1.4.2 Conditional Expected Value

The conditional expected value is given by

$$E[g(X,Y)|Y=y] = \int_{-\infty}^{\infty} f_{X|Y}(x|Y=y) g(x,y) dx \quad (2.45)$$

2.1.4.3 Variance

Because we have two sets of variables, we should have two variances to describe the dispersions, namely, for X , we have

$$D[X] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{XY}(x,y)(x-\mu_X)^2 dx dy \quad (2.46)$$

and for Y , we also have

$$D[Y] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{XY}(x,y)(y-\mu_Y)^2 dx dy \quad (2.47)$$

and

$$D[X] = E[X^2] - (E[X])^2 \quad (2.48)$$

In the following, we will continue to use $E[.]$ and $D[.]$ to denote the expected value and variance of the set $(.)$

2.1.4.4 Covariance of X,Y

The covariance of random variables X,Y is given as

$$\sigma_{XY} = E[(X-\mu_X)(Y-\mu_Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{XY}(x,y)(x-\mu_X)(y-\mu_Y) dx dy \quad (2.49)$$

The covariance can also be calculated as

$$\sigma_{XY} = E[(XY)] - \mu_X\mu_Y \quad (2.50)$$

2.1.4.5 Correlation Coefficient

Accordingly, the correlation coefficient is given by

$$\rho_{XY} = \frac{\sigma_{XY}}{\sigma_X\sigma_Y} \quad (2.51)$$

It is seen that the range of correlation coefficients is in between -1 and 1 , that is,

$$-1 \leq \rho_{XY} \leq 1 \quad (2.52)$$

Because the correlation coefficient can be written as

$$\rho_{XY} = E \left[\frac{(X - \mu_X)(Y - \mu_Y)}{\sigma_X \sigma_Y} \right] \quad (2.53)$$

Example 2.5

The joint distribution of $X \sim N(0, \sigma_X)$ and $Y \sim N(0, \sigma_Y)$ is given by

$$f_{XY}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho_{XY}^2}} \exp \left\{ -\frac{1}{2(1-\rho_{XY}^2)} \left[\frac{x^2}{\sigma_X^2} - 2\frac{\rho_{XY}}{\sigma_X\sigma_Y}xy + \frac{y^2}{\sigma_Y^2} \right] \right\}$$

where ρ_{XY} is a correlation coefficient.

2.1.5 LINEAR INDEPENDENCE

2.1.5.1 Relationship between Random Variables X and Y

For two sets of random variables, the amount of linear independence can be judged by the correlation coefficient as follows.

2.1.5.1.1 Linear Function

If we have a linear function given by

$$Y = aX + b \quad (2.54)$$

Then, the correlation coefficient is

$$\rho_{XY} = \pm 1 \quad (2.55)$$

Because

$$\begin{aligned} \rho_{XY} &= E \left[\frac{(X - \mu_X)(Y - \mu_Y)}{\sigma_X \sigma_Y} \right] = E \left[\frac{(X - \mu_X) \{ (aX + b) - (a\mu_X + b) \}}{\sigma_X (|a|\sigma_X)} \right] \\ &= \frac{a}{|a|} E \left[\frac{(X - \mu_X)^2}{(\sigma_X)^2} \right] = \frac{a}{|a|} \end{aligned} \quad (2.56)$$

That is,

$$\rho_{XY} = \begin{cases} 1, & a > 0 \\ -1, & a < 0 \end{cases} \quad (2.57)$$

2.1.5.1.2 Independence

Now, we further consider independence by examining the following cases.

1. If X and Y are independent, then X and Y are not correlated:

$$\rho_{XY} = 0 \quad (2.58)$$

Because

$$\rho_{XY} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}$$

and

$$\sigma_{XY} = E[XY] - \mu_X \mu_Y$$

Furthermore, the expected value of the production XY can be written as

$$\begin{aligned} E[XY] &= \\ \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{XY}(x, y)(x)(y) dx dy &= \int_{-\infty}^{\infty} f_X(x)x dx \int_{-\infty}^{\infty} f_Y(y)y dy = \mu_X \mu_Y \end{aligned} \quad (2.59)$$

Thus, $\sigma_{XY} = 0$ and Equation 2.58 holds

2. If X and Y are independent, then

$$E[XY] = E[X]E[Y] \quad (2.60)$$

3. If X and Y are independent, then

$$D[X + Y] = D[X] + D[Y] \quad (2.61)$$

2.1.5.2 Expected Value of Sum of Random Variables X and Y

$$E[X + Y] = E[X] + E[Y] \quad (2.62)$$

Note that Equation 2.62 holds for any pair of X and Y , no matter if it is independent or not.

2.1.6 CDF AND PDFs OF RANDOM VARIABLES

In many cases, we want to learn the random distribution of the output based on knowledge of the input. From Figure 2.1b, we may study the distributions of specific functions of random variables. For example, the random responses of a vibration

system can be seen as a function of the random input variables. Now, the CDF and PDF of functions Y of random variables X are discussed as follows. Here, $Y = f(X)$.

2.1.6.1 Discrete Variables

First, let us consider the following examples. Through these examples, we can realize how the probability mass functions (PMF) of discrete random variables are determined.

Example 2.6

The probability distribution of random variable X is shown in Table 2.6.
Let us find the distribution $Y = X^2$
Consider the possible values of Y :
Because $X = \{-2, -1, 0, 1, 2\}$ we have

$$Y = \{0, 1, 4\}$$

Therefore,

$$P(Y = 0) = P(X = 0) = 0.2 \quad (X = 0^{1/2})$$

$$P(Y = 1) = P(X = 1) + P(X = -1) = 0.3 \quad (X = \pm 1^{1/2} = \pm 1)$$

$$P(Y = 4) = P(X = 2) + P(X = -2) = 0.4 \quad (X = \pm 4^{1/2} = \pm 2)$$

The results are listed in Table 2.7.

Example 2.7

Let us further consider another example. The probability distribution of random variable X is shown in Table 2.8.

TABLE 2.6

Probability Distribution of Random Variable X

X	-2	-1	0	1	2
$p(X = x_i)$	0.3	0.2	0.2	0.1	0.2

TABLE 2.7

Probability Distribution of Random Variable X^2

Y	0	1	4
$p(Y = y_i)$	0.2	0.3	0.5

TABLE 2.8
Probability Distribution of Random Variable X

X	1	2	3	...	n
$p(X = x_i)$	$1/2$	$1/2^2$	$1/2^3$	...	$1/2^n$

Let us find the distribution of $Y = \sin(0.5\pi X)$.
Consider the possible values of Y :

$$\sin\left(\frac{\pi n}{2}\right) = \begin{cases} -1, & n = 4k - 1 \\ 1, & n = 4k - 3 \quad k = 1, 2, \dots \\ 0, & n = 2k \end{cases}$$

Thus, $\sin(0.5\pi X)$ can only have three values $(1, -1, 0)$

$$P(Y = -1) = \sum_{k=1}^{\infty} P[X = 4k - 1] = \frac{1}{2^3} + \frac{1}{2^7} + \frac{1}{2^{11}} + \dots = \frac{1}{8\left(1 - \frac{1}{16}\right)} = \frac{2}{15} \left(X = \frac{2}{\pi} \sin^{-1}(-1) \right)$$

$$P(Y = 0) = \sum_{k=1}^{\infty} P[X = 2k] = \frac{1}{2^2} + \frac{1}{2^4} + \frac{1}{2^6} + \dots = \frac{1}{4\left(1 - \frac{1}{4}\right)} = \frac{1}{3} \left(X = \frac{2}{\pi} \sin^{-1}(0) \right)$$

$$P(Y = 1) = \sum_{k=1}^{\infty} P[X = 4k - 3] = \frac{1}{2} + \frac{1}{2^5} + \frac{1}{2^9} + \dots = \frac{1}{2\left(1 - \frac{1}{16}\right)} = \frac{8}{15} \left(X = \frac{2}{\pi} \sin^{-1}(1) \right)$$

The results are listed in Table 2.9.
These two examples describe how to find the PMF of Y and the function of random variable X as follows:
First, list all the possible values of Y , arranging them from the smallest to the largest. Second, at each Y_i , find the sums of probabilities $P(Y_i) = \Sigma$ (all X_j corresponding to Y_i). Note that to systematically find those X_j , we have $X = f^{-1}(Y_i)$.

TABLE 2.9
Probability Distribution of Random Variable Y

Y	-1	0	1
$p(Y = y_i)$	$2/15$	$1/3$	$5/15$

2.1.6.2 Continuous Variables

For continuous variables and the corresponding functions, similar to the case of discrete variables, let us consider the following examples.

Example 2.8

The area of a circle has uniform distribution in $[a, b]$, the PDF is

$$f_X(x) = \begin{cases} \frac{1}{b-a}, & x \in [a, b] \\ 0, & \text{elsewhere} \end{cases}, \quad 0 < a < b$$

Find the PDF $f_Y(y) = d/dy [P(Y \leq y)]$ for circles with the radius $Y = \sqrt{\frac{X}{\pi}}$

1. Let us find the CDF $F_Y(y) = P(Y \leq y)$.

When $y < 0$, because

$$Y = \sqrt{\frac{X}{\pi}} \geq 0$$

We have

$$F_Y(y) = P(Y \leq y) = P(\Phi) = 0$$

and

$$f_Y(y) = d/dy [F_Y(y)] = 0$$

2. Considering $y \geq 0$, we have

$$\begin{aligned} F_Y(y) &= P(Y \leq y) = P\left[\sqrt{\frac{X}{\pi}} \leq y\right] = P[X \leq \pi y^2] = \int_{-\infty}^{\pi y^2} f_X(x) dx \\ &= \begin{cases} \int_{-\infty}^{\pi y^2} 0 dx = 0, & \pi y^2 < a \\ \int_{-\infty}^a 0 dx + \int_a^{\pi y^2} \frac{1}{b-a} dx = \frac{\pi y^2 - a}{b-a}, & a \leq \pi y^2 < b \\ \int_{-\infty}^a 0 dx + \int_a^b \frac{1}{b-a} dx + \int_b^{\pi y^2} 0 dx = 1, & \pi y^2 > b \end{cases} \end{aligned}$$

and

$$f_Y(y) = d/dy[F_Y(y)]$$

$$= \begin{cases} 0' = 0, & 0 \leq y < \sqrt{a/\pi} \\ \left(\frac{\pi y^2 - a}{b-a} \right)' = \frac{2\pi y}{b-a}, & \sqrt{a/\pi} \leq y \leq \sqrt{b/\pi} \\ 1' = 0, & y > \sqrt{b/\pi} \end{cases}$$

Thus,

$$f_Y(y) = \begin{cases} \frac{2\pi y}{b-a}, & \sqrt{a/\pi} \leq y \leq \sqrt{b/\pi} \\ 0, & \text{elsewhere} \end{cases}$$

2.1.6.2.1 General Approach

Having examined the above examples, more general cases can be considered.

Assume X are continuous random variables with PDF

$$f_X(x) = \begin{cases} f_X(x), & x \in (a, b) \\ 0, & \text{elsewhere} \end{cases} \quad (2.63)$$

and $g(x)$ is a differentiable monotonic function within range (a, b) , that is, when

$$x \in (a, b) \rightarrow g(x) \in (\alpha, \beta) \quad (2.64)$$

Denote the inverse function of $g(x)$ as $g^{-1}(y)$, so that $y \in (\alpha, \beta)$.

The PDF of $Y = g(X)$ can be written as

$$f_Y(y) = \begin{cases} f_X[g^{-1}(y)] \left| \frac{d[g^{-1}(y)]}{dy} \right|, & y \in (\alpha, \beta) \\ 0, & \text{elsewhere} \end{cases} \quad (2.65)$$

Proof:

To obtain the PDF $f_Y(y)$, find the corresponding CDF $F_Y(y)$.

$$(1) \quad \text{When } y \leq \alpha, F_Y(y) = P(Y \leq y) = P(\Phi) = 0, \rightarrow f_Y(y) = 0 \quad (2.66)$$

$$(2) \quad \text{When } y \geq \beta, F_Y(y) = P(Y \leq y) = P(U) = 1, \rightarrow f_Y(y) = 0 \quad (2.67)$$

$$(3) \quad \text{When } \alpha < y < \beta, F_Y(y) = P(Y \leq y) = P(g(X) \leq y) \quad (2.68)$$

Suppose $g(x)$ monotonically increasing,

$$F_Y(y) = P(Y \leq y) = P(g(X) \leq y) = P(X \leq g^{-1}(y)) = F_X[g^{-1}(y)] \quad (2.69)$$

Therefore,

$$f_Y(y) = d[F_Y(y)]/dy = d\{F_X[g^{-1}(y)]\}/dy = f_X[g^{-1}(y)] \left\{ \frac{d[g^{-1}(y)]}{dy} \right\} \quad (2.70)$$

Because $g(x)$ monotonically increases in range (a, b) , so does $g^{-1}(y)$ in range (α, β) . That is,

$$d[g^{-1}(y)]/dy \geq 0$$

So that

$$f_Y(y) = f_X[g^{-1}(y)] \left| \frac{d[g^{-1}(y)]}{dy} \right| \quad (2.71)$$

Suppose $g(x)$ is monotonically decreasing,

$$F_Y(y) = P(Y \leq y) = P(g(X) \leq y) = P(X \geq g^{-1}(Y)) = 1 - F_X[g^{-1}(y)] \quad (2.72)$$

Therefore,

$$f_Y(y) = d[F_Y(y)]/dy = d\{1 - F_X[g^{-1}(y)]\}/dy = -f_X[g^{-1}(y)] \left\{ \frac{d[g^{-1}(y)]}{dy} \right\} \quad (2.73)$$

Because $g(x)$ monotonically decreases in range (a, b) , so does $g^{-1}(y)$ in range (α, β) . That is,

$$d[g^{-1}(y)]/dy \leq 0$$

We also have

$$f_Y(y) = f_X[g^{-1}(y)] \left| \frac{d[g^{-1}(y)]}{dy} \right|$$

2.1.6.2.2 Linear Function of Normally Distributed Variables

Given random variable $X \sim N(\mu_X, \sigma_X)$, find the PDF of $Y = aX + b$
Let

$$g(X) = aX + b, \quad x \in (-\infty, \infty) \quad (2.74)$$

The inverse function

$$g^{-1}(y) = \frac{y-b}{a}, \quad \text{and} \quad y \in (-\infty, \infty)$$

$$\frac{d[g^{-1}(y)]}{dy} = \frac{d[(y-b)/a]}{dy} = \frac{1}{a}$$

Substitution of the above equation into Equation 2.65 yields

$$\begin{aligned} f_Y(y) &= f_X[g^{-1}(y)] \left| \frac{d[g^{-1}(y)]}{dy} \right| \\ &= \frac{1}{\sqrt{2\pi}\sigma_X} e^{-\frac{\left[\left(\frac{y-b}{a}\right) - \mu_X\right]^2}{2\sigma_X^2}} \left| \frac{1}{a} \right| = \frac{1}{\sqrt{2\pi}|a|\sigma_X} e^{-\frac{(y-a\mu_X-b)^2}{2a^2\sigma_X^2}} \quad y \in (-\infty, \infty) \end{aligned} \quad (2.75)$$

Thus, Y is also normally distributed and

$$Y \sim N(a\mu_X + b, |a|\sigma_X) \quad (2.76)$$

Example 2.9

$$Y = \sin(X)$$

Given X is uniformly distributed between $(0, 2\pi)$. Find the PDF of Y .

Note that $y = \sin(x)$ is not monotonic. However, from $-\pi/2$ to $\pi/2$, y is monotonically increasing, and from $\pi/2$ to $3\pi/2$, y is monotonically decreasing. We can have

$$F_Y(y) = \frac{1}{2} + \frac{\sin^{-1}(y)}{\pi}, \quad -1 \leq y \leq 1 \quad (2.77)$$

Furthermore, from Equation 2.77, $f_Y(y)$ can be determined.

2.2 SUMS OF RANDOM VARIABLES

Now, let us consider the sums of random variables, which play important roles in load combinations.

2.2.1 DISCRETE VARIABLES

First, consider discrete variables by studying some examples.

Example 2.10

X_1 and X_2 both have Bernoulli distributions with probability p and mutually independent variables (see Table 2.10). Find PDF of $Y = X_1 + X_2$.

The range of Y is $\{0, 1, 2\}$. We thus have

$$P(Y = 0) = P[(X_1 = 0) \cap (X_2 = 0)] = P(X_1 = 0) P(X_2 = 0) = (1 - p)^2$$

$$P(Y = 1) = P[(X_1 = 1) \cap (X_2 = 0)] + P[(X_1 = 0) \cap (X_2 = 1)] = 2(1 - p)p$$

$$P(Y = 2) = P[(X_1 = 1) \cap (X_2 = 1)] = P(X_1 = 1) P(X_2 = 1) = p^2$$

The results are listed in Table 2.11.
Generally, X and Y are independent random discrete variables with PDF as

$$p_X(x_i) = P(X = x_i) \tag{2.78}$$

and

$$p_Y(y_j) = P(Y = y_j) \tag{2.79}$$

TABLE 2.10
Probability Distribution of Random Variable X_i

X_i	0	1
$P(X_i = x_j)$	$1 - p$	p

TABLE 2.11
Probability Distribution of Sums

Y	0	1	2
$P(Y = y_k)$	$1 - p$	$2(1 - p)p$	p^2

The sum

$$Z = X + Y \quad (2.80)$$

has PDF as

$$p_Z(z_k) = \sum_{\text{all } i} P(X = x_i) P(Y = z_k - x_i), \quad k = 1, 2, \dots \quad (2.81)$$

To prove Equation 2.81, we see that $p_Z(z_k) = P(Z = z_k) = P(X + Y = z_k) = \sum_{\text{all } i} P(X = x_i, Y = z_k - x_i), k = 1, 2, \dots$

Because X and Y are independent, the above summation = $\sum_{\text{all } i} P(X = x_i) P(Y = z_k - x_i), k = 1, 2, \dots$

2.2.2 CONTINUOUS VARIABLES

Now, we extend the consideration to continuous variables. Similar to the case of discrete variables, when X and Y are independent random continuous variables with PDF $f_X(x)$ and $f_Y(y)$, the PDF of sum

$$Z = X + Y \quad (2.82)$$

has PDF as

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x) dx \quad (2.83a)$$

or

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(z - y) f_Y(y) dy \quad (2.83b)$$

Equations 2.83a and 2.83b are convolution integrals.

To prove Equation 2.83a,b, let us consider the CDF of the sum $Z, F_Z(z)$ as follows:

$$F_Z(z) = P(Z \leq z) = P(X + Y \leq z) = \iint_{x+y \leq z} f_{XY}(x, y) dx dy = \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f_{XY}(x, y) dy \right] dx$$

Denote $w = x + y$, then $y = w - x$; therefore,

$$F_Z(z) = \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f_{XY}(x, w-x) dw \right] dx = \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f_{XY}(x, w-x) dx \right] dw$$

Furthermore,

$$f_Z(z) = d/dz[F_Z(z)] = \int_{-\infty}^{\infty} f_{XY}(x, w-x) dx \quad (2.84)$$

This is the relationship for the sum of general variable X and Y . Note that when X and Y are independent, we have

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(w-x) dx$$

2.2.2.1 Sums of Normally Distributed PDF

2.2.2.1.1 Standardized Variables

First, consider the standardized variables, we have

$X \sim N(0,1)$ and $Y \sim N(0,1)$, their PDF are

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \quad (2.85)$$

and

$$f_Y(y) = \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} \quad (2.86)$$

The PDF of

$$Z = X + Y \quad (2.87)$$

is given by

$$f_Z(z) = \frac{1}{\sqrt{2\pi}\sqrt{2}} e^{-\frac{z^2}{2(\sqrt{2})^2}} \quad (2.88)$$

That is, we have

$$Z \sim N(0, \sqrt{2}) \quad (2.89)$$

2.2.2.1.2 General Variables

Now, consider the general case of $X \sim N(\mu_x, \sigma_x)$ and $Y \sim N(\mu_y, \sigma_y)$, their PDF are

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma_X} e^{-\frac{(x-\mu_X)^2}{2\sigma_X^2}} \quad (2.90)$$

$$f_Y(y) = \frac{1}{\sqrt{2\pi}\sigma_Y} e^{-\frac{(y-\mu_Y)^2}{2\sigma_Y^2}} \quad (2.91)$$

The PDF of

$$Z = X + Y \quad (2.92)$$

is

$$f_Z(z) = \frac{1}{\sqrt{2\pi}\sqrt{\sigma_X^2 + \sigma_Y^2}} e^{-\frac{(z - \mu_X - \mu_Y)^2}{2(\sigma_X^2 + \sigma_Y^2)}} \quad (2.93)$$

That is

$$Z \sim N(\mu_X + \mu_Y, \sqrt{\sigma_X^2 + \sigma_Y^2}) \quad (2.94)$$

2.2.2.2 Sums of n Normally Distributed Variable

The above can be expanded to more than two normally distributed variables:

2.2.2.2.1 Standard Variables

If n mutually independent standard variables X_i are normally distributed, that is,

$$X_i \sim N(0,1), \quad i = 1, 2, \dots, n \quad (2.95)$$

then, we have

$$\sum_{i=1}^n X_i \sim N(0, \sqrt{n}) \quad (2.96)$$

2.2.2.2.2 General Variables

If n mutually independent general variables X_i are normally distributed

$$X_i \sim N(\mu_{X_i}, \sigma_{X_i}), \quad i = 1, 2, \dots, n \quad (2.97)$$

then, we have

$$\sum_{i=1}^n X_i \sim N\left(\sum_{i=1}^n \mu_{X_i}, \sqrt{\sum_{i=1}^n \sigma_{X_i}^2}\right) \quad (2.98)$$

2.2.2.2.3 Linear Combination of Normally Distributed Variables

Suppose n mutually independent general variables X_i are normally distributed,

$$X_i \sim N(\mu_{X_i}, \sigma_{X_i}), \quad i = 1, 2, \dots, n \quad (2.99)$$

With a set of constant $c_1, c_2, \dots, c_n$, we have

$$\sum_{i=1}^n c_i X_i \sim N\left(\sum_{i=1}^n c_i \mu_{X_i}, \sqrt{\sum_{i=1}^n c_i^2 \sigma_{X_i}^2}\right) \quad (2.100)$$

2.3 OTHER FUNCTIONS OF RANDOM VARIABLES

We now consider other functions of random variables.

2.3.1 DISTRIBUTIONS OF MULTIPLICATION OF X AND Y

Generally, when X and Y are independent random continuous variables with PDF $f_X(x)$ and $f_Y(y)$, the PDF of the product

$$Z = XY \quad (2.101)$$

has PDF as

$$f_Z(z) = \int_{-\infty}^{\infty} \frac{f_X(x)f_Y\left(\frac{z}{x}\right)dx}{|x|} \quad (2.102)$$

or

$$F_Z(z) = \int_{-\infty}^z \int_{-\infty}^{\infty} \frac{f_X(x)f_Y\left(\frac{w}{x}\right)dx dw}{|x|} \quad (2.103)$$

When X and Y are not independent, they have joint PDF $f_{XY}(x,y)$,

$$f_Z(z) = \int_{-\infty}^{\infty} \frac{f_{XY}\left(x, \frac{z}{x}\right)dx}{|x|} \quad (2.104)$$

or

$$F_Z(z) = \int_{-\infty}^z \int_{-\infty}^{\infty} \frac{f_{XY}\left(x, \frac{w}{x}\right)dx dw}{|x|} \quad (2.105)$$

2.3.2 DISTRIBUTIONS OF SAMPLE VARIANCE, CHI-SQUARE (χ^2)

We now consider the distribution of sums of squares.

2.3.2.1 Sample Variance

Sample variance S_X^2 is used to estimate the variance σ_X^2 of random variable X

$$S_X^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{X})^2 \quad (2.106a)$$

It is also seen that

$$S_X^2 = \frac{1}{n-1} \left(\sum_{i=1}^n x_i^2 - n\bar{X}^2 \right) \quad (2.106b)$$

$\bar{X}$ is the mean value of X

X Normal $\rightarrow$ exact chi-square distribution

X not Normal $\rightarrow$ approximately OK

We can have the following:

Let Z be a standard normal random variable set

$$Z = \{z_1, z_2, z_3, \dots\} \quad (2.107)$$

2.3.2.2 Chi-Square Distribution

Chi-square distribution is defined as

$$\chi_n^2 = \sum_{i=1}^n z_i^2 \quad (2.108)$$

2.3.2.3 CDF of Chi-Square, $n = 1$

In the case of one degree of freedom, that is, $n = 1$, we have the CDF as

$$F_{\chi_1^2} = P(\chi_1^2 \leq u) = P(z_1^2 \leq u) = P(-\sqrt{u} < z_1 \leq \sqrt{u}) = \Phi(u) - \Phi(-u) \quad (2.109)$$

2.3.2.4 PDF of Chi-Square, $n = 1$

Furthermore, in the case of one degree of freedom, that is $n = 1$, we have the PDF as

$$f_{\chi_1^2} = \frac{1}{\sqrt{2\pi u}} e^{-\frac{u}{2}}, \quad u > 0 \quad (2.110)$$

2.3.2.5 Mean

The mean is given by

$$\mu_{\chi_1^2} = 1 \quad (2.111)$$

2.3.2.6 Variance

The variance is

$$\sigma_{\chi_1^2}^2 = 2 \quad (2.112)$$

2.3.2.7 PDF of Chi-Square, $n > 1$

Using the convolution integration, it can be proven that

$$f_{\chi_n^2}(u) = \frac{u^{n/2-1}}{2^{n/2} \Gamma(n/2)} e^{-\frac{u}{2}}, \quad u > 0 \quad (2.113)$$

where $\Gamma(n/2)$ is the gamma function of $n/2$. That is,

$$\Gamma(n/2) = \int_0^{\infty} t^{n/2-1} e^{-t} dt \quad (2.114)$$

2.3.2.8 Reproductive

The chi-square distribution is reproductive, which can be written as:

$$\chi_n^2 = \sum_{i=1}^n z_i^2 = \sum_{i=1}^k z_i^2 + \sum_{i=k+1}^n z_i^2 = \chi_k^2 + \chi_{n-k}^2 \quad (2.115)$$

2.3.2.9 Approximation

When n is sufficiently large, say

$$n > 25$$

The chi-square distribution approaches normal distribution

$$Y = \chi_n^2 \quad (2.116)$$

Then approximately

$$\frac{Y - n}{\sqrt{2n}} \sim N(0, 1) \quad (2.117)$$

2.3.2.10 Mean of Y

Consider the mean

$$\mu_{\chi_n^2} = n \quad (2.118)$$

2.3.2.11 Variance of Y

The variance is given by

$$\sigma_{\chi_n^2}^2 = 2n \quad (2.119)$$

2.3.2.12 Square Root of Chi-Square (χ^2)

Now, let us consider the square root of chi-square, denoted as χ

2.3.2.12.1 PDF of χ

First, the PDF is given as

$$f_{\chi_n} = \frac{1}{\Gamma(n/2)} v \left(\frac{v^2}{2} \right)^{n/2-1} e^{-\frac{v^2}{2}}, \quad v > 0 \quad (2.120)$$

Note that when $n = 2$, it reduces to a special Rayleigh distribution whose $s = 1$,

$$f_H(h) = \frac{h}{\sigma^2} e^{-\frac{1}{2}\left(\frac{h}{\sigma}\right)^2}, \quad h > 0 \quad (2.121)$$

2.3.2.12.2 Mean Value of χ

The mean of χ is

$$\mu_{\chi_n} = \sqrt{2} \frac{\Gamma\left(\frac{n+1}{2}\right)}{\Gamma\left(\frac{n}{2}\right)} \quad (2.122)$$

2.3.2.12.3 Variance of χ

The variance of χ is

$$\sigma_{\chi_n}^2 = n - \mu_{\chi_n} \quad (2.123)$$

2.3.2.13 Gamma Distribution and Chi-Square Distribution

Let us consider a new type of distribution called gamma distribution, which is closely related to the chi-square.

2.3.2.13.1 PDF

The PDF of gamma distribution is given by

$$f_X(x) = \frac{\lambda}{\Gamma(r)} (\lambda x)^{r-1} e^{-\lambda x}, \quad x > 0 \quad (2.124)$$

2.3.2.13.2 Mean

The mean of gamma distribution is

$$\mu_X = \frac{r}{\lambda} \quad (2.125)$$

2.3.2.13.3 Variance

The variance of gamma distribution is

$$\sigma_X^2 = \frac{r}{\lambda^2} \quad (2.126)$$

Here, λ and r are positive numbers.

2.3.2.14 Relation between Chi-Square χ_n^2 and Sample Variance S_X^2

In engineering statistics, we often calculate the mean values and variances of samples. Let us first consider the sample variance as follows.

Multiplying $\frac{n-1}{\sigma_X^2}$ on both sides of Equation 2.106a results in

$$\frac{n-1}{\sigma_X^2} S_X^2 = \frac{1}{\sigma_X^2} \sum_{i=1}^n (x_i - \bar{X})^2 \quad (2.127)$$

Insert $0 = -\mu_X + \mu_X$ in between x_i and $-\bar{X}$, we further have

$$\begin{aligned} \sum_{i=1}^n \left(\frac{x_i - \mu_X}{\sigma_X} - \frac{\bar{X} - \mu_X}{\sigma_X} \right)^2 &= \sum_{i=1}^n \left[\left(\frac{x_i - \mu_X}{\sigma_X} \right)^2 - 2 \left(\frac{x_i - \mu_X}{\sigma_X} \right) \left(\frac{\bar{X} - \mu_X}{\sigma_X} \right) + \left(\frac{\bar{X} - \mu_X}{\sigma_X} \right)^2 \right] \\ &= \sum_{i=1}^n \left(\frac{x_i - \mu_X}{\sigma_X} \right)^2 - n \left(\frac{\bar{X} - \mu_X}{\sigma_X} \right)^2 \end{aligned}$$

Thus,

$$\frac{n-1}{\sigma_X^2} S_X^2 = \sum_{i=1}^n \left(\frac{x_i - \mu_X}{\sigma_X} \right)^2 - n \left(\frac{\bar{X} - \mu_X}{\sigma_X} \right)^2 = \sum_{i=1}^n \left(\frac{x_i - \mu_X}{\sigma_X} \right)^2 - \sum_{i=1}^n \left(\frac{\bar{X} - \mu_X}{\sigma_X} \right)^2 \quad (2.128)$$

Note that

$$\sum_{i=1}^n \left(\frac{x_i - \mu_X}{\sigma_X} \right)^2 = \chi_n^2 \quad (2.129)$$

and it can be proven that (see Section 2.5.1.2, the Lindeberg–Levy theorem)

$$\sum_{i=1}^n \left(\frac{\bar{X} - \mu_X}{\sigma_X} \right)^2 = \chi_1^2 \quad (2.130)$$

and therefore

$$\frac{n-1}{\sigma_X^2} S_X^2 = \chi_n^2 - \chi_1^2 = \chi_{n-1}^2 \quad (2.131)$$

2.3.3 DISTRIBUTIONS OF RATIOS OF RANDOM VARIABLES

2.3.3.1 Distribution of Variable Ratios

Random variables X and Y have joint PDF $f_{XY}(x,y)$, the ratio

$$Z = \frac{X}{Y} \quad (2.132)$$

has the PDF

$$f_Z(z) = \int_{-\infty}^{\infty} |y| f_{XY}(yz, y) dy \quad (2.133)$$

If X and Y are independent, then

$$f_Z(z) = \int_{-\infty}^{\infty} |y| f_X(yz) f_Y(y) dy \quad (2.134)$$

where $f_X(x)$ and $f_Y(y)$ are, respectively, the PDF of variables X and Y .

2.3.3.2 Student's Distribution

2.3.3.2.1 Student's Random Variable

Random variable with Student's distribution, denoted by T_n is a ratio of a standard normal variable Z to the square root of a chi-square variable divided by its degree of freedom, that is,

$$T_n = \frac{Z}{\sqrt{\chi_n^2/n}} \quad (2.135)$$

2.3.3.2.2 PDF of Student's Distribution

We have the PDF of Student's distribution given by

$$f_T(t) = \frac{\Gamma\left(\frac{n+1}{2}\right)}{\sqrt{n\pi} \Gamma\left(\frac{n}{2}\right) \left(\frac{t^2}{n} + 1\right)^{(n+1)/2}} \quad (2.136)$$

2.3.3.2.3 Mean of Student's Distribution

The mean is

$$\mu_T = 0, \quad n > 1 \quad (2.137)$$

2.3.3.2.4 Variance of Student's Distribution

The variance is

$$\sigma_T^2 = \frac{n}{n-2}, \quad n > 2 \quad (2.138)$$

2.3.3.2.5 Relation with Standard Normal Distribution

$$\frac{\bar{X} - \mu_X}{\sigma_X / \sqrt{n}}$$

It is seen that the variable $Z = \frac{\bar{X} - \mu_X}{\sigma_X / \sqrt{n}}$, used to standardize the random variable X_i , is a standard normal distribution with variance σ_X^2/n . If the standard deviation is known, Z can be used to estimate how close the mean value can be. However, the standard deviation (or variance) is often unknown, and will be estimated by the sample variance S_X^2 . In this case, we have

$$\frac{\bar{X} - \mu_X}{S_X / \sqrt{n}} = \frac{(\bar{X} - \mu_X) / (\sigma_X / \sqrt{n})}{S_X / \sigma_X} = \frac{(\bar{X} - \mu_X) / (\sigma_X / \sqrt{n})}{\sqrt{\frac{(n-1)S_X^2}{(n-1)\sigma_X^2}}} = \frac{Z}{\sqrt{\frac{\chi_{n-1}^2}{(n-1)}}} = T_{n-1} \quad (2.139)$$

That is,

$$T_{n-1} = \frac{\bar{X} - \mu_X}{S_X / \sqrt{n}} \quad (2.140)$$

Here T_{n-1} is the student's t distributed with $n-1$ degrees of freedom.

2.3.3.3 F Distribution

We now consider another distribution of F random variables as follows

2.3.3.3.1 Definition of F Random Variable

First, consider the definition of F random variables given by

$$F(u, v) = \frac{\chi_u^2 / u}{\chi_v^2 / v} \quad (2.141)$$

2.3.3.3.2 PDF of F Random Variable

The PDF is

$$f_{F_{u,v}}(f) = \frac{\Gamma\left(\frac{u+v}{2}\right)}{\Gamma\left(\frac{u}{2}\right)\Gamma\left(\frac{v}{2}\right)} \frac{\left(\frac{u}{v}\right)^{u/2} f^{u/2-1}}{\left[\left(\frac{u}{v}\right)f + 1\right]^{(u+v)/2}}, \quad f > 0 \quad (2.142)$$

2.3.3.3.3 Mean of F Random Variables

The mean is

$$\mu_{F_{u,v}} = \frac{v}{v-2}, \quad v > 2 \quad (2.143)$$

2.3.3.3.4 Variance of F Random Variable

The variance is

$$\sigma_{F_{u,v}}^2 = \frac{2v^2(u+v-2)}{u(v-2)^2(v-4)}, \quad v > 4 \quad (2.144)$$

2.4 DESIGN CONSIDERATIONS

With the help of the above-mentioned random variables including definitions, means and variances, and PDF and CDF, let us consider the design under random loads. The focus is on the reliability or failure probability of these systems.

2.4.1 FURTHER DISCUSSION OF PROBABILITY-BASED DESIGN

From Equation 1.154, the failure probability of a system is given as

$$p_f = P(F \leq 0) = P(R - Q \leq 0) \quad (2.145)$$

By specifying the design value, we have

$$p_f = P(R_D - Q_D \leq 0) \quad (2.146)$$

By substitution of Equations 1.132 and 1.133 into Equation 2.146, we can have

$$p_f = P(\gamma Q_N - \phi R_N > 0) \quad (2.147)$$

Furthermore, if more than one load is applied, the limit state can be written as

$$F = -\left[\sum_i \gamma_i Q_i - \phi R_N\right] = 0 \quad (2.148)$$

Here, Q_i is the i th nominal load, and γ_i is the corresponding load factor. Assume all Q_i and R are normally distributed. That is,

$$Q_i \sim N(\mu_{Q_i}, \sigma_{Q_i}) \quad (2.149)$$

F will also be normally distributed.

$$F \sim N(\mu_F, \sigma_F) \quad (2.150)$$

It may be shown that

$$\mu_F = \sum_i \mu_{Q_i} \quad (2.151)$$

and

$$\sigma_F = \sqrt{\sum_i \sigma_{Q_i}^2} \quad (2.152)$$

Use the standard variable (see Equation 1.153)

$$\beta = \frac{\mu_F - \mu_R}{\sigma_F} = \frac{\mu_R - \sum_i \mu_{Q_i}}{\sqrt{\sigma_R^2 + \sum_i \sigma_{Q_i}^2}} \quad (2.153)$$

β is defined as the *reliability index*. Recall that Equation 1.154, the failure probability, can be computed as

$$p_f = \Phi(-\beta) \quad (2.154)$$

From Equation 2.154, if the allowed failure probability is given, then

$$[\beta] = -\Phi^{-1}([p_f]) \quad (2.155)$$

Here, $[.]$ stands for *allowed value* of $(.)$. From Equation 2.155, if only one load is considered and assuming the standard deviation σ_F can be determined, then

$$\mu_R - \mu_Q = [\beta]\sigma_F = \text{const.} \quad (2.156)$$

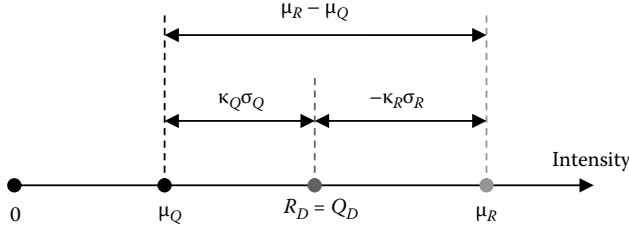


FIGURE 2.6 Relationship between mean values.

From Figure 2.6, when the distance between the two mean values μ_R and μ_Q is fixed, under the requirement of $[\beta]$, one can always find a middle point $R_D = Q_D$.

Recall Equations 1.132 through 1.139, we can write

$$Q_D = \mu_Q + \kappa_Q \sigma_Q = \beta_Q (1 + \kappa_R C_Q) Q_N = \gamma Q_N$$

and

$$R_D = \mu_R - \kappa_R \sigma_R = \beta_R (1 + \kappa_R C_R) R_N = \Phi R_N$$

From Equation 2.156, we further have

$$\mu_R - \kappa_R \sigma_R - (\mu_Q + \kappa_Q \sigma_Q) = [\beta] \sigma_F - \kappa_R \sigma_R - \kappa_Q \sigma_Q = \Phi R_N - \gamma Q_N \quad (2.157)$$

Let

$$[\beta] \sigma_F - \kappa_R \sigma_R - \kappa_Q \sigma_Q = [\beta] \sqrt{\sigma_R^2 + \sum_i \sigma_{Q_i}^2} - \kappa_R \sigma_R - \kappa_Q \sigma_Q = 0 \quad (2.158)$$

For a given allowed reliability index $[\beta]$, if the standard deviations σ_R and σ_Q are known, then by choosing the proper value of κ_R and κ_Q , the critical point is determined at $Q_D = R_D$ or

$$\gamma Q_N = \Phi R_N \quad (2.159)$$

That is, Equation 2.159 is determined by the required condition of allowed failure probability. This can be extended to multiple loads. Suppose that there exists n loads. The limit state is

$$\sum_{i=1}^n \gamma_i Q_i - \Phi R_N = 0 \quad (2.160)$$

must be determined by a preset allowed failure probability $[p_f]$. Here, γ_i and Q_i are, respectively, the i th load factor and the i th nominal load.

2.4.2 COMBINATION OF LOADS

With the help of Equation 2.160, if the allowed failure probability is given and suppose the resistance is also known, the summation $\sum_i \gamma_i Q_i$, namely, the load combination, can be determined.

Consider the inverse problem of given resistance and failure probability to determine the load factor γ_i s. In Equation 2.160, ΦR_N is given. Knowing all the nominal values of load Q_i s, we now have n unknowns (γ_i).

Consider the case of two loads.

$$\gamma_1 Q_1 + \gamma_2 Q_2 = \Phi R_N \quad (2.161)$$

When the value ΦR is specified, ΦR determines *equal- ΦR plan* or *ΦR -plan*, shown in Figure 2.7a as plan A-B-C-D, which is parallel to plan $O-Q_1-Q_2$.

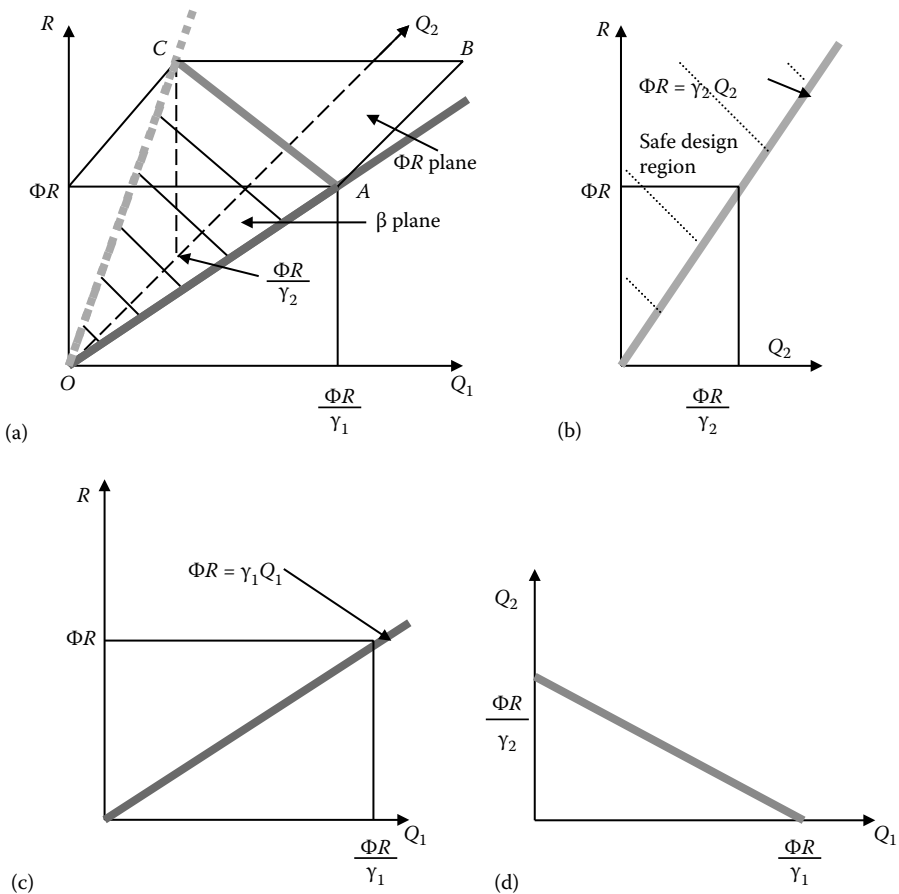


FIGURE 2.7 Loads and resistance case 1. (a) $R-Q_1-Q_2$ three dimensional plot, (b) $R-Q_2$ plan, (c) $R-Q_1$ plan, and (d) Q_1-Q_2 plan.

The design value of ΦR chosen above the β plane will yield a large value of β or a smaller value of failure probability. Thus, we can have a safe design region (see Figure 2.7b, for example). Now, let value ΦR be the design value ΦR_N for the designed resistance, and $\gamma_i Q_i$ be the i th design load.

Figure 2.7 shows a three-dimensional plot with two loads, Q_1 and Q_2 . In Figure 2.7a, the thick solid line in plan R - Q_1 , which is also shown in Figure 2.7c,

$$\Phi R = \gamma_1 Q_1 \quad (2.162)$$

Equation 2.162 is obtained under the condition $Q_2 = 0$, when the limit state $F = 0$ is reached. In Figure 2.7a, the thick break in plan R - Q_2 , which is also shown in Figure 2.7d, is

$$\Phi R = \gamma_2 Q_2 \quad (2.163)$$

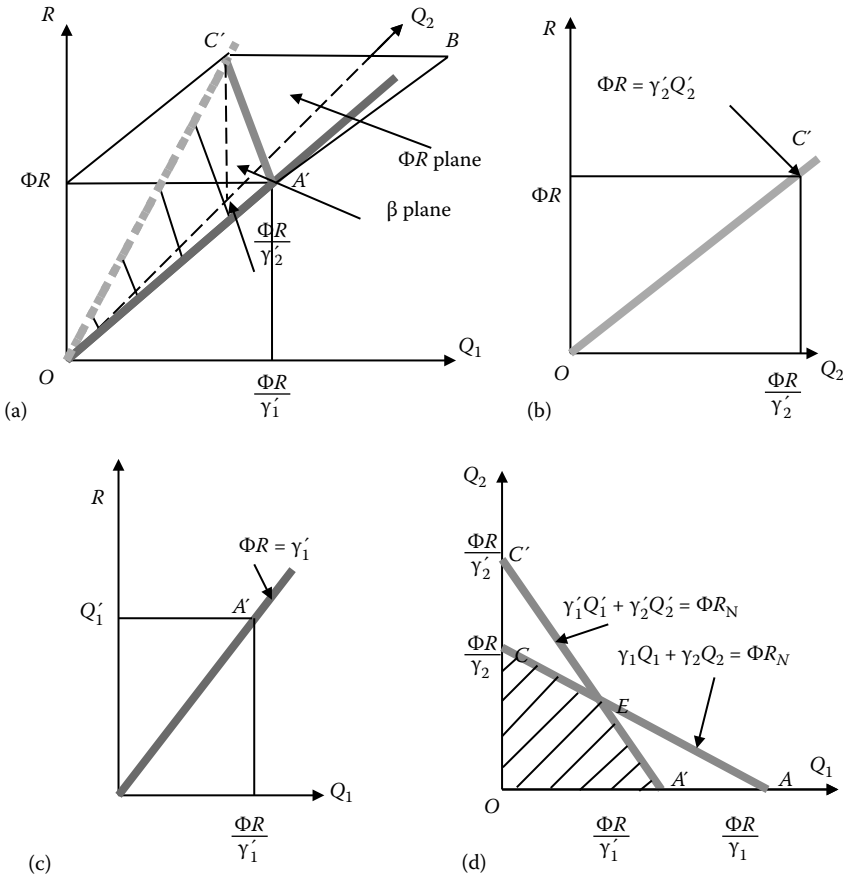


FIGURE 2.8 Loads and resistance case 2. (a) R - Q_1 - Q_2 three dimensional plot, (b) R - Q_2 plan, (c) R - Q_1 plan, and (d) Q_1 - Q_2 plan.

Equation 2.163 is obtained under the condition $Q_1 = 0$, when the limit state $F = 0$ is reached.

Because Equations 2.162 and 2.163 are determined based on a given allowed failure probability, namely, the given value of $[\beta]$, these two lines define a special plan called *equal β plan*, or simply *β plan*, shown in Figure 2.7a (plan O-A-C).

When we have different combinations of Q_1 and Q_2 , with given resistance ΦR , the load factors will be different. Let us consider two possible load combinations. The first case is denoted by Q_1 and Q_2 and the second is denoted by Q'_1 and Q'_2 . Correspondingly, we have γ_1 and γ_2 as well as γ'_1 and γ'_2 . The second case is shown in Figure 2.8.

The intersection of the β plan formed by Q_1 - Q_2 and the ΦR plan forms a straight line. Equation 2.161 is the corresponding equation (see Figures 2.7d and 2.8d). The intersection of an alternative β plan formed by Q'_1 - Q'_2 and the ΦR plan forms another straight line. Equation 2.164 is the corresponding equation (see Figure 2.8d).

$$\gamma'_1 Q'_1 + \gamma'_2 Q'_2 = \Phi R_N \quad (2.164)$$

Example 2.11

The Q_1 - Q_2 combination can be a large truck load combined with a small earthquake load, acting on a bridge. The Q'_1 - Q'_2 combination can be a small truck load combined with a large earthquake load, acting on the same bridge.

In Figure 2.8, only the area O-A'-E-C is the safe region.

2.5 CENTRAL LIMIT THEOREMS AND APPLICATIONS

For random events, although unpredictable specifically, an obvious pattern can be found through a large number of repeated observations. In this section, we study these patterns based on the theory of limits. There are two fundamental types of laws: the law of large numbers and the central limit theorems.

The law of large numbers unveils the stability of mean values of random events. In the literature, there are several theorems in this regard that describe the results of virtually the same experiments repeated a large number of times. In these situations, the average of the results tends to be close to the expected value, and will become closer as more experiments are carried out. The law of large numbers explains the stable long-term results for the averages of random events. In this section, however, we skip the corresponding formulae of these theorems. Interested readers may consult typical textbooks to study the details.

Practically speaking, quite often, when the pool of random variables becomes sufficiently large, many variables and/or their functions can be approximated by normal distributions. The corresponding theorems about these normal distributions are called central limit theorems.

2.5.1 CENTRAL LIMIT THEOREMS

Specifically, central limit theorems provide conditions under which the distribution of sums of a large number of independent random variables tends to become normal distributions. In this section, the focus is on the central limit theorem because they are more relevant to the main topics of random process.

2.5.1.1 Lyapunov Central Limit Theorem (Aleksandr Mikhailovich Lyapunov, 1857–1918)

$X_1, X_2, \dots, X_n$ are a sequence of independent random variables with mean and variance, $\mu_1, \mu_2, \dots, \mu_n$ and $\sigma_1^2, \sigma_2^2, \dots, \sigma_n^2$, respectively. S_n is the sums of X_i s:

$$S_n = \sum_{i=1}^n X_i \quad (2.165)$$

The mean and variance of S_n are given by

$$\mu_{S_n} = \sum_{i=1}^n \mu_i \quad (2.166)$$

and

$$\sigma_{S_n}^2 = \sum_{i=1}^n \sigma_i^2 \quad (2.167)$$

In the limit as n goes to infinity, the standardized variable of S_n , Z_n , has the standard normal distribution

$$Z_n = \frac{S_n - \mu_{S_n}}{\sigma_{S_n}} \quad (2.168)$$

$$\lim_{n \rightarrow \infty} f_{Z_n}(\xi) = \Phi(\xi) \quad (2.169)$$

That is,

$$\lim_{n \rightarrow \infty} P \left[\frac{\sum_{i=1}^n (X_i - \mu_i)}{\sqrt{\sum_{i=1}^n \sigma_i^2}} < \xi \right] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\xi} e^{-\frac{\zeta^2}{2}} d\zeta \quad (2.170)$$

$$E(S_n) = 0 \quad (2.171)$$

$$D(S_n) = 1 \quad (2.172)$$

The above statement has conditions, called Lyapunov conditions:

1. Individual terms in the sums make negligible contributions
2. It is very unlikely that any single term contributes disproportionately

2.5.1.2 Lindeberg–Levy Central Limit Theorem (Jarl W. Lindeberg, 1876–1932; Paul P. Lévy, 1886–1971)

$X_1, X_2, \dots, X_n$ are a sequence of independent random variables with identical mean and variance μ, σ^2 , S_n is the sum of X_i s:

$$S_n = \sum_{i=1}^n X_i \quad (2.173)$$

In the limit as n goes to infinity, the standardized variable of S_n , Z_n has the standard normal distribution

$$Z_n = \frac{S_n - n\mu}{\sqrt{n}\sigma} \quad (2.174)$$

That is,

$$\lim_{n \rightarrow \infty} F_{Z_n}(\xi) = \Phi(\xi) \quad (2.175)$$

or

$$\lim_{n \rightarrow \infty} P \left[\frac{\sum_{i=1}^n X_i - n\mu}{\sqrt{n}\sigma} < \xi \right] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\xi} e^{-\frac{\zeta^2}{2}} d\zeta \quad (2.176)$$

Again, we have

$$E(S_n) = 0 \quad (2.177)$$

and

$$D(S_n) = 1 \quad (2.178)$$

2.5.1.3 De Moivre–Laplace Central Limit Theorem (Abraham De Moivre, 1667–1754; Pierre-Simon Laplace, 1749–1827)

Let K be observation indicators of n -Bernoulli tests (also called n 0–1 tests, n binomial, etc., see Equation 1.53).

First, the random variable is

$$K = \{0, 1\} \quad (2.179)$$

The probability of $K = 0$ is

$$P(K = 0) = p \quad (2.180)$$

and the probability of $K = 1$ is

$$P(K = 1) = 1 - p \quad (2.181)$$

Let X be the sum of these observations

$$X = \sum_{i=1}^n K_i \quad (2.182)$$

As we see from Equation 2.182, X is the number of successes that is denoted by 1 in these n 0–1 tests. Therefore, X has the binomial distribution:

$$p_X(x) = C_n^x p^x (1-p)^{n-x}, \quad x = 0, 1, 2, \dots, n \quad (2.183)$$

Note that the mean and variance of n -Bernoulli distribution are

$$\mu_X = np \quad (2.184a)$$

and

$$\sigma_X^2 = np(1-p) \quad (2.184b)$$

When n becomes sufficiently large, the PMF of $P(X = x)$ can be written approximately as

$$p_X(x) = \frac{1}{\sqrt{2\pi\sigma_X}} e^{-\frac{(x-\mu_X)^2}{2\sigma_X^2}} \quad (2.185)$$

Although the proofs of Lyapunov and Lindeberg–Levy central limit theorems are skipped over, we show in the following how to prove Equation 2.185 as an example.

Denoting $x_k = \frac{x - np}{\sqrt{np(1-p)}}$ for convenience, which is a standardized variable, we can see that when $n \rightarrow \infty, x = np + \sqrt{np(1-p)}x_k \rightarrow \infty$ and $(n - x) \rightarrow \infty$. Furthermore, with Stirling's approximation

$$n! \approx \sqrt{2\pi n} n^{n+\frac{1}{2}} e^{-n}$$

we thus have

$$p_X(x) \approx \frac{1}{\sqrt{2\pi np(1-p)}} \left(\frac{np}{x}\right)^{x+\frac{1}{2}} \left[\frac{n(1-p)}{n-x}\right]^{n-x+\frac{1}{2}}$$

It can be proven that when $n \rightarrow \infty$,

$$\left(\frac{np}{x}\right)^{x+\frac{1}{2}} \approx e^{-x_k \sqrt{np(1-p)} - \frac{(1-p)}{2} x_k^2}$$

and

$$\left[\frac{n(1-p)}{n-x}\right]^{n-x+\frac{1}{2}} \approx e^{x_k \sqrt{np(1-p)} - \frac{p}{2} x_k^2}$$

Therefore,

$$\begin{aligned} p_X(x) &= C_n^x p^x (1-p)^{n-x} = \frac{1}{\sqrt{2\pi np(1-p)}} e^{-\frac{1}{2} x_k^2} = \frac{1}{\sqrt{2\pi np(1-p)}} e^{-\frac{(x-np)^2}{2np(1-p)}} \\ &= \frac{1}{\sqrt{2\pi \sigma_X}} e^{-\frac{(x-\mu_X)^2}{2\sigma_X^2}} \end{aligned}$$

This is the de Moivre–Laplace central limit theorem, which provides an approximation of the binomial PMF with a normal PDF when n becomes large.

Example 2.12

Suppose there are 170 identical and independent computers in a department and each of them has a 1% failure probability. Calculate the total failure probability of up to two computers.

1. Assume there are X malfunctioning computers. It is seen that X yields a typical binomial distribution.

$$P(X = x) = C_n^x p^x (1-p)^{n-x} = C_{170}^x 0.01^x 0.99^{170-x}$$

Therefore,

$$P(0 \leq X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2) = 1 \times 1 \times 0.1811 + 170 \times 0.0100 \times 0.1830 + 14,365 \times 0.0001 \times 0.1848 = 0.1811 + 0.3110 + 0.2655 = 0.7576$$

2. Use Poisson approximation (see Equation 1.80)

$$P(X = x) \approx C_n^k p^k (1-p)^{n-k} \approx \frac{(\lambda)^k e^{-\lambda}}{k!} \text{ with } \lambda = 170 \times 0.01 = 1.7$$

$$P(0 \leq X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2) = 1.7^0/0!e^{-1.7} + 1.7^1/1!e^{-1.7} + 1.7^2/2!e^{-1.7} = 0.1827 + 0.3106 + 0.2640 = 0.7573$$

3. Use the de Moivre–Laplace central limit theorem,

$$np = 1.7, \quad \sqrt{np(1-p)} = 1.2973$$

$$\begin{aligned} P(0 \leq X \leq 2) &\approx \Phi\left(\frac{2 - np}{\sqrt{np(1-p)}}\right) - \Phi\left(\frac{0 - np}{\sqrt{np(1-p)}}\right) = \Phi(0.2312) - \Phi(-1.3104) \\ &= 0.5914 - 0.0950 = 0.4964 \end{aligned}$$

Considering that the number of computers cannot be smaller than 0, we can also use $P(X \leq 2)$ to approximate the result, that is,

$$P(X \leq 2) \approx \Phi(0.2312) = 0.5914$$

Comparing the above approaches, we see that when we have a very small probability p (in this case $p = 0.01$), the Poisson approximation yields better results.

2.5.2 DISTRIBUTION OF PRODUCT OF POSITIVE RANDOM VARIABLES

We now consider the distribution of product and random variables, which are positive. That is,

$$Y = \prod_{i=1}^n X_i, \quad X_i > 0 \quad (2.186)$$

Taking the natural logarithm of Y , we have

$$Z = \ln(Y) = \sum_{i=1}^n \ln(X_i) \quad (2.187)$$

Thus,

$$Y = e^Z \quad (2.188)$$

As

$$n \rightarrow \infty \quad (2.189)$$

we finally have

$$\frac{Z - E(Z)}{\sqrt{D(Z)}} \sim N(0,1) \quad (2.190)$$

2.5.3 DISTRIBUTION OF EXTREME VALUES

2.5.3.1 CDF and PDF of Distribution of Extreme Values

Consider the distribution of the largest possible values of n independent samples taken from a random variable set.

Let X be the random set with a total of n variables. Y_n is the largest value in n independent samples.

The event $\{\max(X_1, X_2, \dots, X_n) \leq y\}$ is equivalent to $\{X_1 \leq y, X_2 \leq y, \dots, X_n \leq y\}$ or $\{\text{all } X_i \leq y\}$. Therefore, we can write

$$P(Y_n \leq y) = P(\text{all } X_i \leq y) = P[(X_1 \leq y) \cap (X_2 \leq y) \dots (X_n \leq y)] \quad (2.191)$$

Because X_i s are independent, with identical CDF $F_X(x)$

$$\begin{aligned} P(\text{all } X_i < y) &= P(X_1 \leq y)P(X_2 \leq y) \dots P(X_n \leq y) \\ &= F_{X_1}(y)F_{X_2}(y) \dots F_{X_n}(y) = [F_X(y)]^n \end{aligned} \quad (2.192)$$

Thus,

$$P(Y_n \leq y) = F_{Y_n}(y) = [F_X(y)]^n \quad (2.193)$$

$$f_{Y_n}(y) = \frac{d[F_{Y_n}(y)]}{dy} = n[F_X(y)]^{n-1} f_X(y) \quad (2.194)$$

2.5.4 SPECIAL DISTRIBUTIONS

2.5.4.1 CDF and PDF of Extreme Value of Rayleigh Distributions

2.5.4.1.1 CDF of Rayleigh Distribution

Recalling Equation 1.94 (Rayleigh distribution)

$$f_H(h) = \frac{h}{\sigma^2} e^{-\frac{1}{2}\left(\frac{h}{\sigma}\right)^2}, \quad h > 0$$

and with the help of this PDF, the CDF of Rayleigh distribution can be calculated as

$$F_H(h) = \int_0^h \frac{h}{\sigma^2} e^{-\frac{1}{2}\left(\frac{h}{\sigma}\right)^2} dh = \int_0^{\frac{h^2}{2\sigma^2}} e^{-u} du = 1 - e^{-\frac{y^2}{2\sigma^2}} \quad (2.195)$$

2.5.4.1.2 CDF

Furthermore, with the help of Equation 2.193, we can write the largest value in n independent samples of Rayleigh distribution as

$$F_{Y_n}(y) = \left[1 - e^{-\frac{y^2}{2\sigma^2}}\right]^n, \quad y \geq 0 \quad (2.196)$$

2.5.4.1.3 PDF

And the corresponding PDF is

$$f_{Y_n}(y) = n \left[1 - e^{-\frac{y^2}{2\sigma^2}}\right]^{n-1} \frac{y}{\sigma^2} e^{-\frac{y^2}{2\sigma^2}} \quad (2.197)$$

2.5.4.2 Extreme Value Type I Distribution

Many asymptotic distributions have the form of double exponential, known as the extreme value type I distribution (EVI, also known as Gumbel distribution) (Emil Julius Gumbel, 1891–1966),

2.5.4.2.1 CDF

The CDF of EVI is

$$F_{Y_n}(y) = e^{-e^{-\alpha(y-\beta)}} \quad (2.198)$$

2.5.4.2.2 PDF

The PDF of EVI is

$$f_{Y_n}(y) = \alpha e^{-\alpha(y-\beta)} - e^{-\alpha(y-\beta)} \quad (2.199)$$

Here, β is the characteristic largest value of Y_n and α is inversely related to the standard deviation of Y_n . β and α are related to the number of samples n and the deviation of X_i by

$$F_X(\beta) = 1 - \frac{1}{n} \quad (2.200)$$

or

$$\beta = F_X^{-1}\left(1 - \frac{1}{n}\right) \quad (2.201)$$

and

$$\alpha = n f_X(\beta) \quad (2.202)$$

2.5.4.2.3 Mean of the Asymptotic Distribution

With the help of these parameters α , β , and γ in EVI distributions, we can see that the mean of Y_n , the largest value in n independent samples, can be written as

$$\mu_{Y_n} = \beta + \frac{\gamma}{\alpha} \quad (2.203)$$

2.5.4.2.4 Variance of the Asymptotic Distribution

The variance of Y_n is

$$\sigma_{Y_n}^2 = \frac{\pi^2}{6\alpha^2} \approx \frac{1.645}{\alpha^2} \quad (2.204)$$

or the standard deviation is

$$\sigma_{Y_n} \approx \frac{1.282}{\alpha} \quad (2.205)$$

Here, γ is Euler's constant

$$\gamma = -\int_0^{\infty} e^{-x} \ln x dx = 0.5772157 \dots \approx 0.577 \quad (2.206)$$

2.5.4.2.5 Approximation of Distribution of Extreme Value of Rayleigh Distribution

We can also have the approximation of the distribution of the maximum value among n Rayleigh distributions.

2.5.4.2.5.1 Values of β and α First of all, the parameters β and α can be calculated as (see Gumbel 1958; Ang and Tang 1984)

$$\beta = \sigma \sqrt{2 \ln n} \quad (2.207)$$

and

$$\alpha = \frac{\sqrt{2 \ln n}}{\sigma} \quad (2.208)$$

respectively.

2.5.4.2.5.2 Mean The mean in this case is then calculated as

$$\mu_{Y_n} = \sigma \left(\sqrt{2 \ln n} + \frac{\gamma}{\sqrt{2 \ln n}} \right) \quad (2.209)$$

2.5.4.2.5.3 Variance It follows that the variance is given by

$$\sigma_{Y_n}^2 = \frac{\sigma^2 \pi^2}{12 \ln n} \quad (2.210)$$

Example 2.13

EVI distribution is often used to model peak annual flow of a river. It is measured that $\mu_Y = 2000 \text{ m}^3/\text{s}$ and $\sigma_Y = 1000 \text{ m}^3/\text{s}$.

Question (1): Find the CDF of EVI

Question (2): In a particular year, find the probability of peak flow exceeding $5000 \text{ m}^3/\text{s}$

1. We have

$$\alpha = \frac{1.282}{\sigma_Y} = 0.00128$$

$$\beta = \mu_Y - \frac{0.577}{\alpha} = 2000 - \frac{0.577}{0.00128} = 1549.8 \text{ (m}^3/\text{s)}$$

So that the CDF is

$$F_{Y_n}(y) = e^{-e^{-0.00128(y-1549.8)}}$$

2. We have

$$P(Y \geq 5000) = 1 - F_{Y_n}(5000) = 1 - e^{-e^{-0.00128(5000-1549.8)}} \approx 0.01$$

In other words, the corresponding return period T_R (see Equation 1.55)

$$T_R = 1/0.01 = 100$$

2.5.4.3 Distribution of Minimum Values

In Sections 2.5.3, 2.5.4.1, and 2.5.4.2, we discussed the distribution of the maximum values. Now, we consider the distribution of minimum values. That is, consider the distribution of the smallest possible values of n independent samples taken from a random variable set.

Let X be the random set with n total variables. Z_n is the smallest value in n independent samples.

The event $\{\min(X_1, X_2, \dots, X_n) \leq y\}$ has a complement event $\{\min(X_1, X_2, \dots, X_n) > y\}$, which is equivalent to $\{X_1 > y, X_2 > y, \dots, X_n > y\}$ or $\{\text{all } X_i > y\}$. Therefore, we can write

$$P(Z_n \leq y) = P(\text{all } X_i \leq y) = 1 - P(\text{all } X_i > y) = 1 - P[(X_1 > y) \cap (X_2 > y) \dots (X_n > y)] \quad (2.211)$$

Because X_i s are independent with identical CDF $F_X(x)$

$$\begin{aligned} P(\text{all } X_i \leq y) &= 1 - P(X_1 > y)P(X_2 > y) \dots P(X_n > y) \\ &= 1 - [1 - F_{X_1}(y)][1 - F_{X_2}(y)] \dots [1 - F_{X_n}(y)] = 1 - [1 - F_X(y)]^n \end{aligned} \quad (2.212)$$

Thus,

$$P(Z_n \leq y) = F_{Z_n}(y) = 1 - [1 - F_X(y)]^n \quad (2.213)$$

and

$$f_{Z_n}(y) = \frac{d[F_{Z_n}(y)]}{dy} = n[1 - F_X(y)]^{n-1} f_X(y) \quad (2.214)$$

Note that if X_i s are independent, each has its own CDF $F_{X_i}(x)$,

$$F_{Z_n}(y) = 1 - \prod_{i=1}^n [1 - F_{X_i}(y)] \quad (2.215)$$

2.5.4.4 Extreme Value Type II Distribution

In Section 2.5.4.2, we discussed EVI distribution. Now, we consider the extreme value type II distribution (EVII), also known as Fréchet distribution (Maurice Fréchet, 1878–1973), which is often used to model annual maximum wind velocity (see Gumbel 1958; Coles 2001).

2.5.4.4.1 CDF

First, consider the CDF. Suppose the variable X_i has CDF as

$$F_X(x) = 1 - \beta \left(\frac{1}{x} \right)^k, \quad x \geq 0 \quad (2.216)$$

Let X be the random set with n total variables. Y_n is the largest value in n independent samples, and $k < 0$ is the shape parameter. The asymptotic distribution of Y_n can be written as

$$F_{Y_n}(y) = e^{-\left(\frac{u}{y}\right)^k}, \quad y \geq 0 \quad (2.217)$$

2.5.4.4.2 PDF

The PDF of EVII is

$$f_{Y_n}(y) = \frac{k}{u} \left(\frac{u}{y} \right)^{k+1} e^{-\left(\frac{u}{y}\right)^k}, \quad y \geq 0 \quad (2.218)$$

2.5.4.4.3 Mean

The mean of EVII is

$$\mu_{Y_n} = u \Gamma \left(1 - \frac{1}{k} \right) \quad (2.219)$$

2.5.4.4.4 Variance

Finally, the variance of EVII is

$$\sigma_{Y_n}^2 = u^2 \left[\Gamma \left(1 - \frac{2}{k} \right) - \Gamma^2 \left(1 - \frac{1}{k} \right) \right] \quad (2.220)$$

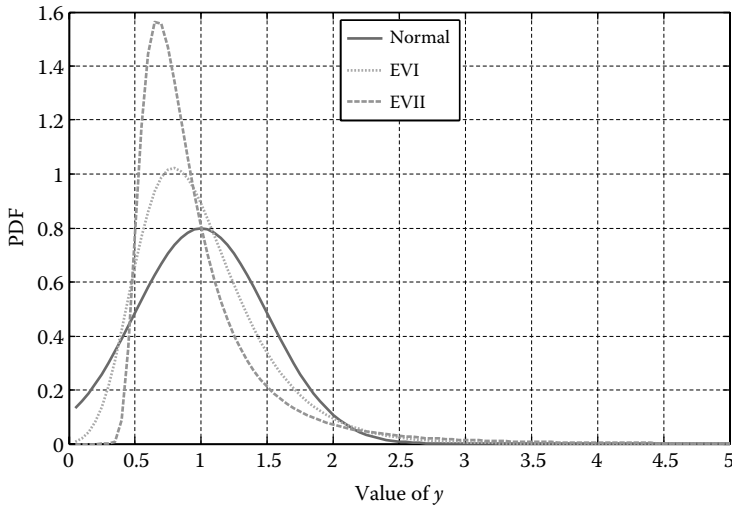


FIGURE 2.9 Several PDFs.

Figure 2.9 shows comparisons among normal, EVI, and EVII distributions. All these distributions have identical mean = 1 and standard deviation = 0.46. From Figure 2.9, we can realize that EVI and EVII have larger left tails, which means that the chance to have larger value of y , the extreme value, is comparatively greater.

2.5.4.5 Extreme Value Type III Distribution

Besides the extreme value type I and II distributions, we also use the third type of extreme value distribution, the extreme value type III distribution (EVIII, also known as Weibull distribution) (Ernst Hjalmar Waloddi Weibull, 1887–1979), which is also used to model annual maximum wind velocity (see Gumbel 1958; Coles 2001).

2.5.4.5.1 CDF

When the largest value of variables X_i falls off to a certain maximum value m , which has CDF as

$$F_X(x) = 1 - c(m - x)^k, \quad x \leq m, k > 0 \quad (2.221)$$

Let X be the random set with n total variables. Y_n is the largest value in n independent samples.

The distribution of Y_n is

$$F_{Y_n}(y) = e^{-\left(\frac{m-y}{m-u}\right)^k}, \quad y \leq m \quad (2.222)$$

2.5.4.5.2 PDF

The PDF of EVIII is

$$f_{Y_n}(y) = \frac{k}{m-u} \left(\frac{m-y}{m-u} \right)^{k-1} e^{-\left(\frac{m-y}{m-u} \right)^k}, \quad y \leq m \quad (2.223)$$

PROBLEMS

1. X and Y are continuous variables and $X > 0$, $Y > 0$. The joint PDF

$$f_{XY}(x, y) = \frac{y}{A} e^{-(x+y^2/2)}$$

- What is suitable value of parameter A ?
 - Determine the marginal PDFs
 - Find conditional PDF of X when $y = 1$
 - Find the covariance and correlation coefficient
2. Random variables X and Y are independent with exponential PDFs, respectively, given by

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

and

$$f_Y(y) = \begin{cases} \nu e^{-\nu y}, & y > 0 \\ 0, & y \leq 0 \end{cases}$$

Z is another random variable defined as

$$Z = \begin{cases} 1, & X \leq Y \\ 0, & X > Y \end{cases}$$

- Find the conditional PDF of $f_{X|Y}(x|y)$
 - Find the PDF (PMF) and CDF of Z
3. Suppose both $g(x)$ and $h(x)$ are PDFs. Showing that

$$f(x) = \alpha g(x) + (1 - \alpha)h(x), \quad 0 \leq \alpha \leq 1$$

is also a PDF.

- Derive the general normal PDF based on knowing the standard normal PDF
- The PMF of random variable X is shown in Table P2.1. Find the PMF of $Z = X^2$.

TABLE P2.1

X	-2	-1	0	1	5
p_k	1/5	1/6	1/5	1/15	11/30

6. Show that the sum of two correlated normal random variables is also normally distributed by using convolution [Hint: consider $a^2 \pm 2ab + b^2 = (a \pm b)^2$]
7. Suppose random variable X is uniformly distributed in $(0,1)$. Find the PDF of (a) $Y = e^X$ and (b) $Z = -2\ln(X)$

$$f(x) = \begin{cases} 1, & 0 < x < 1 \\ 0, & \text{elsewhere} \end{cases}$$

8. Random variable X has PDF given by

$$f_X(x) = \begin{cases} \frac{2x}{\pi^2}, & 0 < x < \pi \\ 0, & \text{elsewhere} \end{cases}$$

Find the PDF of $y = \sin(x)$.

9. An insurance company has 10,000 auto-insurances. The mean compensation is \$280 with a standard deviation of \$800. Calculate the probability of the compensation being greater than \$2,700,000.
10. Show that similar to the de Moivre–Laplace approximation, the normal distribution is the limiting case for the Poisson distribution as the expected value goes to infinity.

Section II

Random Process

3 Random Processes in the Time Domain

In Chapter 2, it was shown that for every number in random space, there is an occurrence probability creating a “two-dimensional” observation. In Chapter 3, an additional dimension is added, which is predominantly the dimension of time. The concept of expending the two-dimensional observation to one of three dimensions is based on the concept of random process.

3.1 DEFINITIONS AND BASIC CONCEPTS

To understand the main properties of the three-dimensional world, we must first have an in-depth understanding of random processes.

3.1.1 STATE SPACES AND INDEX SETS

3.1.1.1 Definition of Random Process

In Chapter 2, random events were expressed by one or more limited numbers of random variables. Although the idea of random series was mentioned, it was based on the assumption of independent variables. We now consider the case in which each variable will not only be associated with its probability density function (mass function) but also be indexed by an additional parameter.

First, we mathematically explain the essential difference between a two-dimensional random variable and a three-dimensional random process.

3.1.1.1.1 Random Variable

In a sample space of random test $\Omega = \{e\}$, where e denotes a sample point, if for any e there exists a unique real number $X(e)$, then $X(e)$ is referred to as the *random variable*.

Comparing the “random variable” $f(x)$ to the “function” $X(e)$, it is seen that

1. They are similar.
2. In function $f(x)$, x is an independent variable. In random variable $X(e)$, e is also an independent variable; however, it is the sample point.
3. The domain and range of $f(x)$ are of real axes, whereas the domain of a random variable is a sample space and the range is of a real axis.

3.1.1.1.2 Random Process

Now, let us consider the process of developments and variations of random variables, which can be infinite or uncountable.

In a sample space of random test $\Omega = \{e\}$, if at any moment $t \in T$, then there exists a random variable $X(t, e)$, referred to as the temporal set of random variables, where $\{X(t, e), t \in T, e \in \Omega\}$ is the *random process*. It is denoted by $\{X(t), t \in T\}$ or more simply $X(t)$.

The essence of a random process is that a random process must be of

1. A random sequence.
2. The sequence must be inside a set $\Omega = \{e\}$, sample space, which is seen as a state space, where e is an independent state used to denote an individual sample.
3. The order of sequence is trackable by index t , which is an element of the index set T .

By comparing a random process with a random variable, we find that

1. Both are random, thus the need to consider the distribution functions.
2. X is a time invariable, whereas $X(t)$ is a time variable, thus the distribution of X is fixed. In most cases, the first and second moments, mean, and variance, are sufficient to describe X . However, the distribution of $X(t)$ is a temporal function. In this case, moments alone are insufficient and a correlation analysis is needed.
3. The independent variable for random variable X is e , thus for the random process $X(t)$, we have (e, t) .

3.1.1.1.3 Basic Cases

There are four basic cases of $X(t)$:

1. Given e and t : $X(t)$ is deterministic.
2. Given t , e varying: $X(t)$ is a random variable.
3. Given e , t varying: $X(t)$ is a deterministic function of t (sample function).
4. Both e and t vary: $X(t)$ is a random process.

3.1.1.2 Classification of Random Process

The following are four basic classifications of a random process with examples:

1. Continuous state and continuous index: continuous random process
Example: The voltage difference of a resistor can have a fluctuation due to the random movement of free electrons; the voltage, denoted by $\{V(t, e)\}$, is called thermal noise.
2. Continuous state and discrete index: continuous random sequence
Example: Use Y_n to denote the n th observation of the maximum PGA of seismic ground motion in a certain location; $\{Y_n, n = 1, 2, 3, \dots\}$ is a continuous random sequence.
3. Discrete state and continuous index: discrete random process
Example: In time duration $[0, t]$, count the number of times that the peak acceleration of a car exceeds a preset level. For this example, this number will not be definite and will vary according to time t .

4. Discrete state and discrete index: discrete random sequence

Example: In an airport, count the number of passengers boarding and deboarding every 10 minutes, calculating the difference.

3.1.1.3 Distribution Function of Random Process

In Chapter 2, we established that the statistical property of random variables could be described by distribution functions. We now describe a random process.

In this case, we use a group of distributions with *limited dimensions* to describe the statistical pattern of random process, where limiting the dimension means the associated probability distribution is not infinite. Recall in Chapter 2 the case of a random variable in which the corresponding distributions were known, that the set of random variables would be understood fairly well. Now, for a sequence of random variables, the random process $X(t)$, one may feel that if the finite dimensional distributions at any t_i are all known, then the probabilities of $X(t_i)$ for any choice of e should be calculated. For engineering applications, this idea is approximately true.

3.1.1.3.1 One-Dimensional

It is important to note that the phrase “dimensional” differs from the one used at the beginning of Chapter 3. Previously, the phrase “dimensional” was used in a philosophical (or general) sense. Here, it is more mathematical (or specific). In other words, “one-dimensional” refers to only one group of variables $X_1(t) = X(t)$ being considered. Furthermore, “two-dimensional” will involve $X_1(t)$ and $X_2(t)$, where $X_1(t)$ and $X_2(t)$ are different groups of variables.

For each fixed time $t \in T$, the random process $X(t)$ becomes a random variable, its CDF

$$F_X(x; t) = P[X(t) \leq x] \quad (3.1)$$

is the one-dimensional distribution of random process $X(t)$.

If the partial derivative of $F_X(x; t)$, with respect to x , exists, then

$$f_X(x; t) = \frac{\partial F_X(x; t)}{\partial x} \quad (3.2)$$

is the one-dimensional density function of random process $X(t)$.

The one-dimensional distribution describes each individual time. It does not reflect the relationship among different time points. From Equations 3.1 and 3.2, we see that the basic approach in which to manage a random process is similar to that for random variables, considering the distributions. Doing so, “three dimensions” are reduced to “two dimensions.” Generally, the fundamental approach on random events is to reduce the number of “dimensions.”

3.1.1.3.2 Two Dimensional

We now consider “two dimensional” using two groups of variables: $X_1(t)$ and $X_2(t)$, taken from the same random process. Note that, in many cases, we will denote $X_1(t) = X(t_1)$ and $X_2(t) = X(t_2)$.

3.1.1.3.2.1 CDF For the relationship of a random process at arbitrary moment t_1 and t_2 , consider the distribution function of two-dimensional random variables $(X(t_1), X(t_2))$:

$$F_X(x_1, x_2; t_1, t_2) = P[(X(t_1) \leq x_1) \cap (X(t_2) \leq x_2)] \quad (3.3)$$

which is the two-dimensional distribution of random process $X(t)$.

3.1.1.3.2.2 PDF If a second-order partial derivative of $F_X(x; t)$ exists with respect to x_1 and x_2 , then

$$f_X(x_1, x_2; t_1, t_2) = \frac{\partial^2 F_X(x_1, x_2; t_1, t_2)}{\partial x_1 \partial x_2} \quad (3.4)$$

is the two-dimensional density function of random process $X(x; t)$.

3.1.1.3.2.3 Joint Distribution Generally, $X(t_1)$ and $X(t_2)$ are different sets of random variables. Thus, $X(t_1)$ and $X(t_2)$ will have a joint distribution:

$$f_{X(t_1)X(t_2)}(x_1, x_2) \quad (3.5)$$

3.1.1.3.3 N-Dimensional

Similar to joint distributions, we consider n -dimensional distributions.

3.1.1.3.3.1 CDF The N -dimensional distribution function of random process $X(t)$ is:

$$F_X(x_1, x_2, \dots, x_n; t_1, t_2, \dots, t_n) = P[(X(t_1) \leq x_1) \cap (X(t_2) \leq x_2) \dots \cap (X(t_n) \leq x_n)] \quad (3.6)$$

3.1.1.3.3.2 PDF The N -dimensional density function of random process $X(t)$ is:

$$f_X(x_1, x_2, \dots, x_n; t_1, t_2, \dots, t_n) = \frac{\partial^n F_X(x_1, x_2, \dots, x_n; t_1, t_2, \dots, t_n)}{\partial x_1 \partial x_2 \dots \partial x_n} \quad (3.7)$$

3.1.1.3.4 Property of N-Dimensional Distribution Functions (Finite-Dimensional)

The basic properties of finite-dimensional distribution functions of a random process are given in the following.

3.1.1.3.4.1 Symmetry From the arbitrary rearrangement of $(t_1, t_2, \dots, t_n)$, namely, $(t'_1, t'_2, \dots, t'_n)$

$$F_X(x_1, x_2, \dots, x_n; t_1, t_2, \dots, t_n) = F_X(x'_1, x'_2, \dots, x'_n; t'_1, t'_2, \dots, t'_n) \quad (3.8)$$

Let us use a simple, incomplete example to show this symmetry. Suppose, in two-dimensional cases, $x_1 = 1$, $x_2 = 2$, $t_1 = 0.1$, and $t_2 = 0.5$,

$$F_X(x_1, x_2; t_1, t_2) = F_X(1, 2, 0.1, 0.5) = P[(X(0.1) \leq 1) \cap (X(0.5) \leq 2)] = P[(X(0.5) \leq 2) \cap (X(0.1) \leq 1)] = F_X(2, 1, 0.5, 0.1) = F_X(x_2, x_1; t_2, t_1)$$

3.1.1.3.4.2 *Compatibility* When $m < n$,

$$F_X(x_1, x_2, \dots, x_m; t_1, t_2, \dots, t_m) = F_X(x_1, x_2, \dots, x_m, \infty|_{m+1}, \infty|_{m+2}, \dots, \infty|_n; t_1, t_2, \dots, t_m, t_{m+1}, \dots, t_n) \quad (3.9)$$

Proof:

$$F_X(x_1, x_2, \dots, x_m, \infty|_{m+1}, \infty|_{m+2}, \dots, \infty|_n; t_1, t_2, \dots, t_m, t_{m+1}, \dots, t_n) = P[(X(t_1) \leq x_1) \cap (X(t_2) \leq x_2) \dots \cap (X(t_m) \leq x_m) \cap (X(t_{m+1}) < \infty) \dots \cap (X(t_n) < \infty)] = P[(X(t_1) \leq x_1) \cap (X(t_2) \leq x_2) \dots \cap (X(t_m) \leq x_m)] = F_X(x_1, x_2, \dots, x_m; t_1, t_2, \dots, t_m)$$

Note that Equation 3.9 is also known as the consistency condition, which implies

$$F_X(x_1, x_2, \dots, x_{k-1}; t_1, t_2, \dots, t_{k-1}) = \lim_{x_k \rightarrow \infty} F_X(x_1, x_2, \dots, x_k; t_1, t_2, \dots, t_k)$$

3.1.1.3.5 *Kolmogorov Extension Theorem*

(Andrey N. Kolmogorov, 1903–1987)

If there exists a group of distribution functions F , which are symmetric and compatible, then there exists a random process whose n -dimensional distribution function is F .

It is seen that the two conditions (symmetric and compatible) required by this theorem are trivially satisfied by any random process, which means that the statement of the theorem is that no other conditions are required. Therefore, for any reasonable (i.e., consistent) group of finite-dimensional distributions, there must exist a random process with these distributions. The Kolmogorov extension theorem guarantees the existence of a random process with a given family of finite-dimensional probability distributions under the above-mentioned conditions.

This theorem is important: It states that we can construct a random process with given finite dimensional distributions. Mathematically speaking, a random process is denoted by $\{X(t, e), t \in T, e \in \Omega\}$. A reasonable question is, first, whether T is finite or infinite. Generally, we are dealing with an infinitely long time duration T , at least theoretically. That is, for finite duration T , the time development process is often referred to as a transient process, instead of a random process.

Having realized that $T \rightarrow \infty$, the next reasonable question is when the evaluation values, say, $x_{m+1}, \dots, x_n$, of a random process at certain time points, say, $t_{m+1}, \dots, t_n$, become infinite, how can we “handle” (or more precisely, “model”) this process? The Kolmogorov extension theorem answers this question.

Example 3.1

Suppose a random process $X(t, x)$ ($-\infty < t < \infty$) has only two sample functions

$$X(t, x_1) = 4 \text{ cost with } P(x_1) = 2/3$$

and

$$X(t, x_2) = -4 \text{ cost with } P(x_2) = 1/3$$

Let us find the following:

1. One-dimensional distribution function $F(0, x)$ and $F(\pi/3, x)$
2. Two-dimensional distribution function $F(0, \pi/3, x_1, x_2)$

1. $X(0)$ can be either -4 or 4 , thus we can have

$$P\{X(0) = -4\} = 1/3 \text{ and } P\{X(0) = 4\} = 2/3$$

The corresponding PMF is

$X(0)$	-4	4
P	$1/3$	$2/3$

and the CDF is

$$F(0, x) = \begin{cases} 0, & -\infty < x \leq -4 \\ 1/3, & -4 < x \leq 4 \\ 1, & x > 4 \end{cases}$$

Similarly, $X(\pi/3)$ has PMF as

$X(\pi/3)$	-2	2
P	$1/3$	$2/3$

and the CDF is

$$F(\pi/3, x) = \begin{cases} 0, & -\infty < x \leq -2 \\ 1/3, & -2 < x \leq 2 \\ 1, & x > 2 \end{cases}$$

2. Calculate the following probabilities first

$$P\{X(\pi/3) = -2 \mid X(0) = -4\} = 1$$

$$P\{X(\pi/3) = 2 \mid X(0) = -4\} = 0$$

$$P\{X(\pi/3) = -2 | X(0) = 4\} = 0$$

$$P\{X(\pi/3) = 2 | X(0) = 4\} = 1$$

Therefore, $P\{X(0) = -4, X(\pi/3) = -2\} = P\{X(0) = -4\} P\{X(\pi/3) = -2 | X(0) = -4\} = P\{X(0) = -4\} = 1/3$.

Similarly

$$P\{X(0) = 4, X(\pi/3) = 2\} = 2/3$$

$$P\{X(0) = -4, X(\pi/3) = 2\} = P\{X(0) = -4\} P\{X(\pi/3) = 2 | X(0) = -4\} = 0$$

and

$$P\{X(0) = 4, X(\pi/3) = -2\} = 0$$

The two-dimensional PMF of $\{X(0), X(\pi/3)\}$ is

	$X(\pi/3)$	-2	2
$X(0)$			
-4		1/3	0
4		0	2/3

and the two-dimensional CDF of $\{X(0), X(\pi/3)\}$ is

$$F(0, \pi/3, x_1, x_2) = \begin{cases} 0, & x_1 \leq -4, \text{ or } x_2 \leq -2 \\ 1/3, & x_1 > -4, -2 < x_2 \leq 2, \text{ or } x_2 > -2, -4 < x_1 \leq 4 \\ 1, & x_1 > 4, x_2 > 2 \end{cases}$$

3.1.1.4 Independent Random Process

The independence of a random process is an important concept. Although in the real world, two events can never be exactly independent; mathematically, we can assume such a case to describe two events as having a “tangled” relationship.

3.1.1.4.1 Independent Process

If

$$F_X(x_1, x_2, \dots, x_n; t_1, t_2, \dots, t_n) = F_X(x_1; t_1) F_X(x_2; t_2) \dots F_X(x_n; t_n) \quad (3.10)$$

or

$$f_X(x_1, x_2, \dots, x_n; t_1, t_2, \dots, t_n) = f_X(x_1; t_1) f_X(x_2; t_2) \dots f_X(x_n; t_n) \quad (3.11)$$

then the random process $X(t)$ is an independent process.

Equations 3.10 and 3.11 show that in an independent random process $X(t)$ at different moments: $t_1, t_2, \dots$, the distributions can be decoupled.

Note that decoupling is an important measure for dealing with complex events, functions, systems, and others.

3.1.1.4.2 Mutually Independent Processes

Two random processes, $X(t)$ and $Y(t)$, have a joint distribution function:

$$\begin{aligned} F_{XY}(x_1, \dots, x_m; y_1, \dots, y_n; t_1, \dots, t_m, t'_1, \dots, t'_n) \\ = P[(X(t_1) \leq x_1) \dots \cap (X(t_m) \leq x_m) \cap (Y(t'_1) \leq y_1) \dots \cap (Y(t'_n) \leq y_n)] \end{aligned} \quad (3.12)$$

If

$$\begin{aligned} F_{XY}(x_1, \dots, x_m; y_1, \dots, y_n; t_1, \dots, t_m, t'_1, \dots, t'_n) \\ = F_X(x_1, \dots, x_m; t_1, t_2, \dots, t_n) F_Y(y_1, \dots, y_n; t'_1, \dots, t'_n) \end{aligned} \quad (3.13)$$

then $X(t)$ and $Y(t)$ are mutually independent.

Example 3.2

For a random process $Z(t) = (X^2 + Y^2) t$, $t > 0$, where $X \sim N(0,1)$, $Y \sim N(0,1)$, and X, Y are independent, find the one-dimensional density function.

The joint PDF is

$$f_{XY}(x, y) = \frac{1}{2\pi} e^{-\frac{x^2+y^2}{2}}, \quad -\infty < x < \infty, \quad -\infty < y < \infty$$

when $z \geq 0$,

$$\begin{aligned} F_Z(z; t) &= P[Z(t) \leq z] = P[(x^2 + y^2) \leq z/t] \\ &= \iint_{x^2+y^2 \leq \frac{z}{t}} \frac{1}{2\pi} e^{-\frac{x^2+y^2}{2}} dx dy = \int_0^{2\pi} d\theta \int_0^{\sqrt{\frac{z}{t}}} \frac{r}{2\pi} e^{-\frac{r^2}{2}} dr = 1 - e^{-\frac{z}{2t}} \end{aligned}$$

Note that in the above derivation, the Cartesian coordinates are replaced by polar coordinates.

When

$$z < 0, \quad F_Z(z; t) = 0$$

Therefore,

$$F_Z(z; t) = \begin{cases} 1 - e^{-\frac{z}{2t}}, & z \geq 0 \\ 0, & z < 0 \end{cases}$$

and the PDF is

$$f_Z(z; t) = \frac{\partial f_Z(z; t)}{\partial z} = \begin{cases} \frac{1}{2t} e^{-\frac{z}{2t}}, & z \geq 0 \\ 0, & z < 0 \end{cases}$$

3.1.2 ENSEMBLES AND ENSEMBLE AVERAGES

It has been established that our basic measure to deal with random variables is by averaging. In Chapter 2, the average was carried over the range of X , in the instance of “two-dimensional” space. In Chapter 3, we are confronted with the problem that the average is performed over Ω or T .

From Figure 3.1, it can be seen that the average can be taken at any given $t = t_i$, based on the definition of CDF mentioned in Chapter 2.

3.1.2.1 Concept of Ensembles

To consider the above-mentioned average, we must introduce the concept of ensembles.

3.1.2.1.1 Definition

Ensembles: Set of all possible sample realizations of a random process. The essence of ensembles is that the process is viewed as a whole. The reason to employ ensembles in the study of a random process is to understand the joint probability distributions at all times.

Recall marginal distribution: the distribution of $X(t)$ at a given t , denoted by $f_{X(t)}(x)$ (see Figure 3.1).

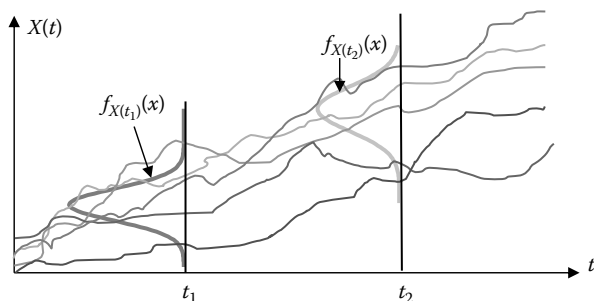


FIGURE 3.1 Statistics at different time points.

Generally, we have

$$f_{X(t_1)}(x) \neq f_{X(t_2)}(x), t_1 \neq t_2 \quad (3.14)$$

3.1.2.2 Statistical Expectations and Moments

Now, we consider the “should-be” average or the ensemble average as follows.

3.1.2.2.1 Mean

The mean value can be calculated as

$$\mu_X(t) = E[X(t)] = \int_{-\infty}^{\infty} x f_{X(t)} dx \quad (3.15)$$

3.1.2.2.2 Variance

The variance is

$$\sigma_X^2(t) = E[\{X(t) - \mu_X(t)\}^2] \equiv D[X(t)] = \int_{-\infty}^{\infty} [x - \mu_X(t)]^2 f_{X(t)} dx \quad (3.16)$$

Here, the symbol “ $\equiv$ ” stands for “is defined as.”

3.1.2.2.3 Autocorrelation Function

The autocorrelation function is given by

$$R_X(t_1, t_2) = E[X(t_1)X(t_2)] = \int_{-\infty}^{\infty} x_1 x_2 f_{X(t_1)X(t_2)}(x_1, x_2) dx \quad (3.17)$$

Equations 3.15 through 3.17 are averages over the ranges of X , namely, the whole space of Ω .

Example 3.3

$X(t) = A \cos 2\pi t$, $A > 0$, $t > 0$, in which A is a random variable with

$$\sigma_A^2 + \mu_A^2 = 1$$

Consider the joint PDFs of $X(t)$, at two pairs of time (τ_0, τ_1) and (τ_0, τ_2) .

For the PDF of A :

1. Find $f_A(a)$

Let us assume (note that this assumption is not necessary)

$$f_A(a) = ka, \quad 0 < a < x_0$$

then

$$\int_0^{x_0} (ka) da = k \frac{a^2}{2} \Big|_0^{x_0} = 1 \rightarrow k = \frac{2}{x_0^2}$$

and

$$f_A(a) = \frac{2}{x_0^2} a$$

2. Calculate μ_A and σ_A^2 and determine the value of x_0 .

The mean is given by

$$\mu_A = \int_0^{x_0} \frac{2a}{x_0^2} a da = \frac{2x_0}{3}$$

and the variance is given by

$$\sigma_A^2 = \int_0^{x_0} \frac{2a}{x_0^2} \left(a - \frac{2x_0}{3} \right)^2 da = \frac{x_0^2}{18}$$

because

$$\sigma_A^2 + \mu_A^2 = 1 = \left(\frac{2x_0}{3} \right)^2 + \frac{x_0^2}{18} = \frac{x_0^2}{2}$$

we have

$$x_0 = (2)^{1/2}$$

Thus, we can have

$$f_A(a) = \frac{2}{x_0^2} a = a$$

3. The autocorrelation function is

$$\begin{aligned} R_x(t_1, t_2) &= \int_{-\infty}^{\infty} x_1 x_2 f_{X(t_1)X(t_2)}(x_1, x_2) dx \\ &= \int_0^{\sqrt{2}} [a \cos(2\pi t_1)] [a \cos(2\pi t_2)] a da = \cos(2\pi t_1) \cos(2\pi t_2) \end{aligned}$$

3.1.2.2.4 Autocovariance Function

The autocovariance function is given by

$$\begin{aligned} \sigma_{xx}(t_1, t_2) &= E[\{X(t_1) - \mu_X(t_1)\} \{X(t_2) - \mu_X(t_2)\}] = E[X(t_1)X(t_2)] - \mu_X(t_1)\mu_X(t_2) \\ &= R_X(t_1, t_2) - \mu_X(t_1)\mu_X(t_2) \end{aligned} \quad (3.18)$$

Note that the second question in the above example can also be obtained by using Equation 3.18 as follows:

$$\begin{aligned}
 R_X(t_1, t_2) &= E[X(t_1)X(t_2)] = \sigma_{XX}(t_1, t_2) + \mu_X(t_1)\mu_X(t_2) \\
 &= \sigma_A^2 \cos(2\pi t_1) \cos(2\pi t_2) + \mu_A^2 \cos(2\pi t_1) \cos(2\pi t_2) \\
 &= [\sigma_A^2 + \mu_A^2] \cos(2\pi t_1) \cos(2\pi t_2) = \cos(2\pi t_1) \cos(2\pi t_2)
 \end{aligned}$$

3.1.2.2.5 Autocorrelation Coefficient Function

The autocorrelation coefficient function is given by

$$\rho_{XX}(t_1, t_2) = \frac{\sigma_{XX}(t_1, t_2)}{\sigma_X(t_1)\sigma_X(t_2)} \quad (3.19)$$

where

$$\sigma_X(.) = \sqrt{\sigma_X^2(.)}$$

3.1.2.2.6 Property of Autocorrelation Functions

3.1.2.2.6.1 Variance and Autocovariance

Let

$$t_1 = t_2 = t$$

we have

$$\sigma_{XX}(t, t) = E[\{X(t) - \mu_X(t)\}\{X(t) - \mu_X(t)\}] = \sigma_X^2(t) \quad (3.20)$$

That is, when

$$t = 0$$

the autocovariance is equal to the variance.

3.1.2.2.6.2 Autocorrelation Functions: Positive Semidefinite The autocorrelation function is positive semidefinite, which can be expressed as

$$\sum_{j=1}^n \sum_{k=1}^n \alpha_j \alpha_k R_X(t_j, t_k) \geq 0 \quad (3.21)$$

Example 3.4

1. Let $X(t) = C$, where C is a random variable with PDF: $f_C(c)$, $-\infty < c < \infty$, $t \in [0, T]$

In this case, $X(t)$ is not a function of time; therefore, the random process reduces to a random set.

That is, the mean is given by

$$\mu_X(t) = \mu_C = \text{const.}$$

The standard deviation is then

$$\sigma_X(t) = \sigma_C = \text{const.}$$

Furthermore, the autocorrelation is

$$R_X(t_1, t_2) = \mu_C^2 + \sigma_C^2 = \text{const.}$$

2. Let $X(t) = Bt$, where B is a random variable with PDF: $f_B(b)$, $-\infty < b < \infty$, $t \in [0, T]$;

The mean is

$$\mu_X(t) = \int_{-\infty}^{\infty} f_B(b)tb \, db = t \int_{-\infty}^{\infty} f_B(b)b \, db = \mu_B t$$

the variance is

$$\sigma_X^2(t) = \int_{-\infty}^{\infty} f_B(b)(bt - \mu_B t)^2 \, db = t^2 \int_{-\infty}^{\infty} f_B(b)(b - \mu_B)^2 \, db = \sigma_B^2 t^2$$

and the autocorrelation is

$$\begin{aligned} R_X(t_1, t_2) &= \int_{-\infty}^{\infty} x_1 x_2 f_{X(t_1)X(t_2)}(x_1, x_2) \, dx = \int_{-\infty}^{\infty} b_1 t_1 b_2 t_2 f_B(b) \, db \\ &= t_1 t_2 \int_{-\infty}^{\infty} b_1 b_2 f_B(b) \, db = t_1 t_2 (\mu_X^2 + \sigma_X^2) \end{aligned}$$

3. Let $X(t) = B + t$, where B is a random variable with PDF: $f_B(b)$, $-\infty < b < \infty$,

$$t \in [0, T]$$

The mean is

$$\mu_X(t) = \int_{-\infty}^{\infty} f_B(b)(t+b) \, db = \int_{-\infty}^{\infty} f_B(b)b \, db + t \int_{-\infty}^{\infty} f_B(b) \, db = \mu_B + t$$

the variance is

$$\sigma_X^2(t) = \int_{-\infty}^{\infty} f_B(b)(b+t - \mu_B - t)^2 \, db = \sigma_B^2$$

and the autocorrelation can be calculated as

$$\begin{aligned} R_X(t_1, t_2) &= \int_{-\infty}^{\infty} (b_1 + t_1)(b_2 + t_2)f_B(b)db = \int_{-\infty}^{\infty} b_1b_2f_B(b)db + \int_{-\infty}^{\infty} t_1b_2f_B(b)db \\ &\quad + \int_{-\infty}^{\infty} t_2b_1f_B(b)db + \int_{-\infty}^{\infty} t_1t_2f_B(b)db = \mu_B^2 + \sigma_B^2 + t_1\mu_B + t_2\mu_B + t_1t_2 \end{aligned}$$

4. Let $X(t) = \cos^2(\omega t + \Theta)$, where Θ is random variable $0 < \Theta \leq 2\pi$, with PDF $f_\Theta(\theta)$, $t \in [0, T]$

The mean is

$$\mu_\Theta(t) = \int_0^{2\pi} f_\Theta(\theta) \cos^2(\omega t + \theta) d\theta = \int_0^{2\pi} f_\Theta(\theta) \frac{1}{2} [1 + \cos(2\omega t + 2\theta)] d\theta = \frac{1}{2}$$

since

$$\int_0^{2\pi} [\cos(2\omega t + 2\theta)] d\theta \neq 1, f_\Theta(\theta) \neq \cos(2\omega t + 2\theta), \int_0^{2\pi} f_\Theta(\theta) [\cos(2\omega t + 2\theta)] d\theta = 0$$

The variance is

$$\begin{aligned} \sigma_\Theta^2(t) &= \int_0^{2\pi} f_\Theta(\theta) \left[\cos^2(\omega t + \theta) - \frac{1}{2} \right]^2 d\theta = \int_0^{2\pi} f_\Theta(\theta) \left[\frac{\cos(2\omega t + 2\theta)}{2} \right]^2 d\theta \\ &= \int_0^{2\pi} f_\Theta(\theta) \frac{1}{4} \left[\frac{1 + \cos(4\omega t + 4\theta)}{2} \right] d\theta = \frac{1}{8} \end{aligned}$$

The autocorrelation is

$$\begin{aligned} R_X(t_1, t_2) &= \int_0^{2\pi} \cos^2(\omega t_1 + \theta) \cos^2(\omega t_2 + \theta) f_\Theta(\theta) d\theta \\ &= \int_0^{2\pi} \left\{ \frac{1}{2} [1 + \cos(2\omega t_1 + 2\theta)] \right\} \left\{ \frac{1}{2} [1 + \cos(2\omega t_2 + 2\theta)] \right\} f_\Theta(\theta) d\theta = \frac{1}{8} \\ &\quad \left\{ \int_0^{2\pi} f_\Theta(\theta) d\theta + \int_0^{2\pi} \cos(2\omega t_1 + 2\theta) f_\Theta(\theta) d\theta + \int_0^{2\pi} \cos(2\omega t_2 + 2\theta) f_\Theta(\theta) d\theta \right. \\ &\quad \left. + \int_0^{2\pi} \cos(2\omega t_1 + 2\theta) \cos(2\omega t_2 + 2\theta) f_\Theta(\theta) d\theta \right\} = \frac{1}{8} \end{aligned}$$

It is important to note that in parts (1) and (4) of the above example, the means, variances, and autocorrelation functions are all constants, whereas in parts (2) and (3), the means, variances, and autocorrelation functions are time variables.

Example 3.5

For the random process given in the example of Section 3.1.1,

1. Find the function of mean $\mu_X(t)$
2. Find the autocorrelation function $R_X(t_1, t_2)$
3. Find the autocovariance function $\sigma_{XX}(t_1, t_2)$

a. The mean $\mu_X(t)$ is

$$\mu_X(t) = E[X(t)] = 2/3 \times 4 \cos(t) + 1/3 (-4 \cos t) = 4/3 \cos(t)$$

b. It is seen that the autocorrelation can be calculated as

$$\begin{aligned} R_X(t_1, t_2) &= E[X(t_1)X(t_2)] \\ &= \begin{cases} \frac{2}{3}(4)^2 \cos^2 t_1 + \frac{1}{3}(-4)^2 \cos^2 t_1, & t_1 = t_2 \\ \frac{2}{3}4 \cos t_1 4 \cos t_2 + \frac{1}{3}(-4 \cos t_1)(-4 \cos t_2), & t_1 \neq t_2 \end{cases} = \begin{cases} 16 \cos^2 t_1, & t_1 = t_2 \\ 16 \cos t_1 \cos t_2, & t_1 \neq t_2 \end{cases} \end{aligned}$$

$$\text{Thus, } R_X(t_1, t_2) = 16 \cos t_1 \cos t_2$$

c. The autocovariance function $\sigma_{XX}(t_1, t_2)$ is given by

$$\sigma_{XX}(t_1, t_2) = R_X(t_1, t_2) - \mu_X(t_1)\mu_X(t_2) = 16 \cos t_1 \cos t_2 - 16/9 \cos t_1 \cos t_2 = 128/9 \cos t_1 \cos t_2$$

3.1.3 STATIONARY PROCESS AND ERGODIC PROCESS

3.1.3.1 Stationary Process

Generally speaking, the mean and variance of a random process will depend on the index t , whereas the autocorrelation function will depend on t_1 and t_2 . However, there are instances in which for a random process, the average will not depend on time. In this case, the processes are said to be stationary. A stationary process is considerably simpler to deal with. In the following, we will consider various types of stationary processes.

3.1.3.1.1 Strictly Stationary Process

For any real number h , the n -dimensional distribution of random process $X(t)$ satisfies

$$F_X(x_1, x_2, \dots, x_n; t_1, t_2, \dots, t_n) = F_X(x_1, x_2, \dots, x_n; t_1 + h, t_2 + h; \dots, t_n + h) \quad (3.22)$$

$X(t)$ is a strictly stationary process.

If the PDF exists, then Condition 3.22 can be replaced by

$$f_X(x_1, x_2, \dots, x_n; t_1, t_2, \dots, t_n) = f_X(x_1, x_2, \dots, x_n; t_1 + h, t_2 + h; \dots, t_n + h) \quad (3.23)$$

Conditions 3.22 and 3.23 imply that the n -dimensional distribution does not evolve over the time intervals.

A strictly stationary process has the following properties:

1. If $X(t)$ is a strictly stationary process, then the joint PDF of $\{X(t_1), X(t_2), \dots, X(t_n)\}$ is identical to that of $\{X(t_1 + h), X(t_2 + h), \dots, X(t_n + h)\}$.
2. If $X(t)$ is a strictly stationary process, and if the expectation of $X(t)$ is

$$E[X(t)] < \infty \quad (3.24a)$$

and the variance of $X(t)$ is

$$D[X(t)] < \infty \quad (3.24b)$$

also the expectation of $X(t)^2$, the mean square value is

$$E[X(t)^2] < \infty \quad (3.24c)$$

Then,

$$E[X(t)] = \mu_X = \text{const.} \quad (3.25)$$

and

$$D[X(t)] = \sigma_X^2 = \text{const.} \quad (3.26)$$

Thus, the mean square is

$$E[X(t)^2] = \text{const.} \quad (3.27)$$

and the autocorrelation is

$$R_X(t + \tau, t) = R_X(\tau) \quad (3.28)$$

It is important to note that Condition 3.24 is not necessary for strictly stationary processes. Additionally, note that a strictly stationary process is defined by examining its distributions.

3.1.3.1.2 Weakly Stationary Process

Distribution functions of processes are sometimes difficult to obtain. One approach is to use the moments.

$X(t)$ is a weakly stationary process, if:

1. The mean square value of $X(t)$ is not infinite

$$E[X(t)^2] < \infty \quad (3.29)$$

2. The mean of $X(t)$ is constant

$$E[X(t)] = \mu_X = \text{const.} \quad (3.30)$$

3. The variance of $X(t)$ is constant

$$D[X(t)] = \sigma_X^2(t) = \sigma_X^2 = \text{const.} \quad (3.31)$$

4. The autocorrelation function of $X(t)$ depends only on the time difference τ

$$R_X(\tau) = R_X(t, t + \tau) = E[X(t)X(t + \tau)] \quad (3.32)$$

5. In this case, the autocovariance is equal to

$$\sigma_{XX}(\tau) = R_X(\tau) - \mu_X^2 \quad (3.33)$$

6. The autocorrelation coefficient function is

$$\rho_{XX}(\tau) = \frac{\sigma_{XX}(\tau)}{\sigma_X^2} \quad (3.34)$$

The main properties of real valued autocorrelation functions of weakly stationary processes will be listed in the next section. Here, we show the following points for general autocorrelation functions.

1. If the process is complex valued, then $R_X(\tau)$ is also complex valued and

$$R_X(-\tau) = R_X(\tau)^* \quad (3.35a)$$

Here, the symbol $(.)^*$ stands for taking the complex conjugate of $(.)$

2. The autocorrelation function is positive semi-definite, that is, for any complex number $\alpha_1, \alpha_2, \dots, \alpha_n$ and any real number $t_1, t_2, \dots, t_n$, we have

$$\sum_{j=1}^n \sum_{k=1}^n \alpha_j \alpha_k^* R_X(t_j - t_k) \geq 0 \quad (3.35b)$$

Example 3.6

A random sequence $X(t) = \sin(2\pi At)$ with $t = 1, 2, \dots$ and A is a uniformly distributed random variable in $[0, 1]$. Let us consider whether $X(t)$ is stationary.

The mean is given by

$$E[X(t)] = E[\sin(2\pi At)] = \int_0^1 \sin(2\pi At) da = \frac{-1}{2\pi t} \cos(2\pi At) \Big|_0^1 = 0$$

which satisfies the first condition described in Equation 3.30.

Second, check the autocorrelation function

$$R_X(t + \tau, t) = E\{\sin(2\pi A(t + \tau))\sin(2\pi A t)\} = E\{\sin(2\pi A t_1)\sin(2\pi A t_2)\}$$

$$= \int_0^1 \sin 2\pi A t_1 \sin 2\pi A t_2 da = \frac{1}{2} \int_0^1 [\cos 2\pi A(t_1 - t_2) - \cos 2\pi A(t_1 + t_2)] da = \begin{cases} 1/2, & t_1 = t_2 \\ 0, & t_1 \neq t_2 \end{cases}$$

Therefore, $X(t)$ is weakly stationary.

However, the PDF of $X(t)$ can be written as

$$f_X(x, t) = \begin{cases} \frac{1}{\pi t \sqrt{1 - x^2}}, & -1 < x < 1 \\ 0, & \text{elsewhere} \end{cases}$$

which is a function of time t ; therefore, $X(t)$ is not strictly stationary.

3.1.3.1.3 Strictly Process versus Weakly Process

1. A weakly process is when the first and second moments are specified. However, a weakly process is not necessarily a strictly process.
2. For a strictly process, the entire distribution must be stationary, but the first and/or second moments do not necessarily have to exist. Therefore, a strictly process is not necessarily a weakly process.
3. If the second moment of a strictly random process exists, then it is also a weakly process.
4. It is difficult to check the conditions of a strictly process. However, for a Gaussian process, its second moment completely defines the entire distribution. In this case, it is both a strictly and weakly process (the Gaussian process is further explained in the next subsection).
5. In the real world, purely stationary processes rarely exist. In engineering applications, it is acceptable within some limits to simplify a process to a weakly process (from now on, we will consider all stationary processes to be equal to weakly stationary processes in this book).

From the above-mentioned points, we can make further categorizations. For points (1) and (3) satisfied, the processes are stationary. Point (2) is not used often.

Example 3.7

A random process $Z(t) = X\cos(2\pi t) + Y\sin(2\pi t)$, where both X and Y are random variables and

$$EX = EY = 0$$

$$DX = DY = 1$$

As well as

$$EXY = 0$$

Prove $Z(t)$ is a stationary process.

First, the mean

$$E[Z(t)] = EX\cos(2\pi t) + EY\sin(2\pi t) = 0$$

Then, the autocorrelation function can be written as

$$\begin{aligned} R_X(t + \tau, t) &= E\{[X\cos(2\pi(t + \tau)) + Y\sin(2\pi(t + \tau))][X\cos(2\pi t) + Y\sin(2\pi t)]\} = \\ &EX^2\cos[2\pi(t + \tau)] \cos 2\pi t + EY^2\sin[2\pi(t + \tau)] \sin 2\pi t + EXY\cos[2\pi(t + \tau)] \sin 2\pi t + \\ &EXY\sin[2\pi(t + \tau)] \cos 2\pi t = \cos[2\pi(t + \tau)] \cos 2\pi t + \sin[2\pi(t + \tau)] \sin 2\pi t = \cos 2\pi\tau \end{aligned}$$

Therefore, the mean of $Z(t)$ is constant, and it can be readily shown that the variance of $Z(t)$ is also constant and the autocorrelation function of $Z(t)$ depends only on the time difference τ , which concludes that $Z(t)$ is a stationary process.

3.1.3.2 Ergodic Process

As mentioned previously, to find the moments, we take the ensemble average. In most instances, determining the ensemble average is difficult. On the contrary, the average over the time domain can be significantly simpler to compute.

In fact, in many engineering practices, the temporal average is used to calculate the mean and variance values. Before using the temporal average, there is a question that must first be asked. That is, under what conditions can the temporal average be used? Using the temporal average under the incorrect conditions may have severe computation errors. Mathematically, the correct condition is that the process must be ergodic. This issue, however, must be further discussed.

3.1.3.2.1 Ensemble Average and Temporal Average

First, let us consider the two types of averages.

All of the above noted averages are ensembles, denoted by $E[.]$

The temporal average is denoted by $\langle X(t, k) \rangle$. Here, $X(t, k)$ is the k th sample realization of random process $X(t)$.

$$\langle X(t, k) \rangle = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T X(t, k) dt \quad (3.36)$$

However, it is often more practical to use

$$\langle X(t, k) \rangle = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_0^{2T} X(t, k) dt \quad (3.37)$$

It is reasonable to deduce, $T \rightarrow \infty$ as taking a sufficiently long time. From this, we use Equation 3.37, because in the real world, there is no negative time.

3.1.3.2.2 Ergodicity

Ergodicity means the temporal average can be used to replace the ensemble average.

A process is ergodic in the mean, if

$$\langle X(t, k) \rangle = E[X(t)] = \mu_X \quad (3.38)$$

A process is ergodic in the variance, if

$$\langle \{X(t, k) - \mu_X\}^2 \rangle = D[X(t)] = E[\{X(t, k) - \mu_X\}^2] = \sigma_X^2 \quad (3.39)$$

Furthermore, a process is ergodic in the autocorrelation function if

$$\langle X(t, k) X(t + \tau, k) \rangle = E[X(t, k) X(t + \tau, k)] = R_X(\tau) \quad (3.40)$$

From Equations 3.38 through 3.40, it is established that an ergodic process must be stationary. However, it must be remembered that a stationary process is not necessarily an ergodic one.

A weakly ergodic process is one that satisfies these three conditions, a strongly ergodic process is one that satisfies all ensemble averages to be equal to temporal averages, whereas a nonergodic process is one that does not satisfy any of these three conditions.

3.1.3.2.2.1 Condition of Ergodicity A real stationary process is ergodic in mean, if and only if

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^{2T} \left(1 - \frac{\tau}{2T}\right) [R_X(\tau) - \mu_X^2] d\tau = 0 \quad (3.41)$$

A real stationary process is ergodic in autocorrelation function, if and only if

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^{2T} \left(1 - \frac{u}{2T}\right) \{E[X(t + \tau + u)X(t + \tau)X(t + u)X(t)] - R_X^2\} dt = 0 \quad (3.42)$$

In this example, $X(t) = C$ is stationary but nonergodic, and $X(t) = B + t$ is not stationary, and therefore is also nonergodic.

Ergodicity is important because we can use temporal averages to replace ensemble averages, which will be discussed in detail in the next section. Practically, however, we rarely have exact ergodic processes. Caution must be taken in using the temporal average. In many engineering applications, taking several temporal averages to see if the corresponding moments have converged to the correct values may be advantageous.

3.1.4 EXAMPLES OF RANDOM PROCESS

Now, let us consider several examples of random processes.

3.1.4.1 Gaussian Process (Carl F. Gauss, 1777–1855)

If the n -dimensional distribution function of a random process is normal, then $X(t)$ is a Gaussian process. Gaussian processes can be a good approximation for many processes. This in turn can greatly reduce the computational burden.

Example 3.8

Assuming both $X \sim N(0, 1)$ and $Y \sim N(0, 1)$ are mutually independent, let us consider

$$Z(t) = X + Yt$$

In the initial calculations, we have $E(X) = E(Y) = 0$; $D(X) = D(Y) = 1$; $E(XY) = 0$.

From further calculations, we receive the expected value of $Z(t)$ as

$$E[Z(t)] = E(X) + tE(Y) = 0$$

and the autocorrelation is

$$R(t_1, t_2) = E[(X + Yt_1)(X + Yt_2)] = E(X^2 + XYt_1 + XYt_2 + Y^2t_1t_2) = 1 + t_1t_2$$

Furthermore, the variance is

$$D[Z(t)] = 1 + t^2$$

and the CDF can be calculated as

$$F_Z(x; t) = \frac{1}{\sqrt{2\pi(1+t^2)}} \int_{-\infty}^x e^{-\frac{\xi^2}{2(1+t^2)}} d\xi$$

It is determined that $Z(t)$ is Gaussian but not stationary, and therefore nonergodic.

Note that, in Section 3.1.1.3, we use distribution functions to describe a random process. In this particular case, we have the Gaussian process.

Example 3.9

Given Gaussian processes $\{X(t) \mid -\infty < t < \infty\}$ and $\{Y(t) \mid -\infty < t < \infty\}$, which are independent, prove $Z(t) = X(t) + Y(t)$ $-\infty < t < \infty$ is also Gaussian.

Consider a nonzero vector $\mathbf{q} = [q_1, q_2, \dots, q_n]$ and the nonzero linear combination of $Z(t)$, that is,

$$\mathbf{q} \begin{Bmatrix} Z(t_1) \\ Z(t_2) \\ \dots \\ Z(t_n) \end{Bmatrix} = \mathbf{q} \begin{Bmatrix} X(t_1) \\ X(t_2) \\ \dots \\ X(t_n) \end{Bmatrix} + \mathbf{q} \begin{Bmatrix} Y(t_1) \\ Y(t_2) \\ \dots \\ Y(t_n) \end{Bmatrix}$$

Because $X(t)$ is a Gaussian process, then $[X(t_1), X(t_2), \dots, X(t_n)]$ is n -dimensional normal distribution; furthermore, $q_1X(t_1) + q_2X(t_2) + \dots + q_nX(t_n)$ is one-dimensional normal.

Similarly, $q_1 Y(t_1) + q_2 Y(t_2) + \dots + q_n Y(t_n)$ is also one-dimensional normal.

Now, because $X(t)$ and $Y(t)$ are independent, $q_1 X(t_1) + q_2 X(t_2) + \dots + q_n X(t_n)$ and $q_1 Y(t_1) + q_2 Y(t_2) + \dots + q_n Y(t_n)$ must also be independent. Due to the additivity of normal distribution, the following term

$$\mathbf{q} \begin{Bmatrix} Z(t_1) \\ Z(t_2) \\ \dots \\ Z(t_n) \end{Bmatrix}$$

must be normally distributed, so that $Z(t)$ is Gaussian. This example implies that the Gaussian process is addible. One of the nice properties of the Gaussian process is, if a process is Gaussian, then its derivatives and integrals are also Gaussian.

3.1.4.2 Poisson Process

Before specifically introducing the Poisson process, let us consider various general cases.

3.1.4.2.1 Independent Increments (Nonoverlapping)

The number of occurrences counted in disjointed intervals are independent from each other; therefore, $X(t)$ is a random process. For any $t_1 < t_2 \leq t_3 < t_4 \in T$, $X(t_2) - X(t_1)$ and $X(t_4) - X(t_3)$ are mutually independent, then $X(t)$ is an independent increment process.

3.1.4.2.2 Homogenous Process

The increment $X(t + \tau) - X(t)$ of the random process $X(t)$ in time interval $[t, t + \tau]$ is independent to t because it only depends on τ .

3.1.4.2.3 Poisson Process (Simeon-Denis Poisson, 1781–1840)

In the following, let us examine the Poisson process. Using the Poisson process as an example, we can realize how to model a random process in detail.

Definition

A Poisson process is a homogenous continuous process with independent increments. With zero initial condition $N(0) = 0$, the increment $N(t) - N(\tau)$ is a Poisson process distributed with parameter $\lambda(t - \tau)$.

This definition implies the following facts:

1. The number of arrivals in nonoverlapping intervals is an independent random process.
2. There exists a positive quantity $\lambda > 0$, such that in a short time interval Δt , the following are true:
 - a. The probability of exactly one arrival in Δt is proportional to the length of Δt , that is,

$$P[N(t + \Delta t) = n + 1 \mid N(t) = n] = \lambda \Delta t \quad (3.43)$$

Note that Equation 3.43 implies the meaning of factor λ , which is a proportional constant.

b. The probability of no arrivals in Δt is

$$P[N(t + \Delta t) = n \mid N(t) = n] = 1 - \lambda \Delta t \quad (3.44)$$

c. The probability of more than one arrival in Δt is negligible and vanishes as $\Delta t \rightarrow 0$

3. The zero initial condition is

$$N(0) = 0 \quad (3.45)$$

We now explain the nature of the Poisson process. Consider the following probability equation based on the above-mentioned condition,

$$p_N(n, t + \Delta t) = P[N(t + \Delta t) = n] = P\{[(N(t) = n) \cap (\text{no new arrival in } \Delta t)] \cup [(N(t) = n - 1) \cap (\text{one new arrival in } \Delta t)]\} = p_N(n, t) [1 - \lambda \Delta t] + p_N(n - 1, t) [\lambda \Delta t] \quad (3.46)$$

Rearranging Equation 3.46, we have

$$p_N(n, t + \Delta t) - p_N(n, t) = -p_N(n, t) [\lambda \Delta t] + p_N(n - 1, t) [\lambda \Delta t]$$

Furthermore,

$$\lim_{\Delta t \rightarrow 0} \frac{p_N(n, t + \Delta t) - p_N(n, t)}{\Delta t} = \frac{dp_N(n, t)}{dt} = -\lambda p_N(n, t) + \lambda p_N(n - 1, t) \quad (3.47)$$

Under the initial condition $n = 0$, $p_N(-1, t) = 0$, therefore,

$$\frac{dp_N(0, t)}{dt} = -\lambda p_N(0, t) \quad (3.48)$$

The solution for Equation 3.48 is given as follows:

$$p_N(0, t) = e^{-\lambda t}$$

Furthermore, we can have

$$p_N(n, t) = \frac{(\lambda t)^n}{n!} e^{-\lambda t}, \quad n \geq 0, t \geq 0 \quad (3.49)$$

This is the PMF of the Poisson process. Similar to the above-mentioned Gaussian process, we can use the distribution function to describe the Poisson process.

In addition, we consider the corresponding moments:

Mean

$$\mu_N(t) = \lambda t \quad (3.50)$$

Variance

$$\sigma_N^2(t) = \lambda t \quad (3.51)$$

Autocorrelation function

$$R_N(t_1, t_2) = \begin{cases} \lambda t_1 + \lambda^2 t_1 t_2, & 0 \leq t_1 \leq t_2 \\ \lambda t_2 + \lambda^2 t_1 t_2, & 0 \leq t_2 \leq t_1 \end{cases} \quad (3.52)$$

Furthermore, we can have

$$P\{[N(t) - N(\tau)] = k\} = \frac{[\lambda(t - \tau)]^k}{k!} e^{-\lambda(t - \tau)} \quad (3.53)$$

Example 3.10

The number of radiated particles during $[0, t]$ from a source is denoted by $N(t)$. $\{N(t), t \geq 0\}$ is a Poisson process with mean radiation rate λ . Assume each particle can be recorded with probability p , and the record of an individual particle is independent from other records, and also to the process $N(t)$. Denote the total number of particles during $[0, t]$ to be $M(t)$. Prove $\{M(t), t \geq 0\}$ is also a Poisson process with mean rate λp .

First, it is seen that $M(0) = 0$.

Second, let X_i be the record of the i th particle, the distribution is

$$X_i = \begin{cases} 1, & p \\ 0, & 1-p \end{cases}$$

These X_i s are mutually independent and with identical distributions. Consider a set of increments denoted as

$$M(t_2) - M(t_1), M(t_3) - M(t_2), \dots, M(t_n) - M(t_{n-1})$$

Because X_i s are mutually independent and have identical distributions, the increments of $M(t)$ must be independent from each other, thus, the process of $M(t)$ is an independent increment.

Now, consider $P[M(t_2) - M(t_1) = n]$, which can be written as

$$\begin{aligned} P[M(t_2) - M(t_1) = n] &= \left\{ \sum_{n=k}^{\infty} P[N(t_2) - N(t_1) = n] \right\} \left\{ P[M(t_2) - M(t_1) = k \mid N(t_2) - N(t_1) = n] \right\} \\ &= \sum_{n=k}^{\infty} \frac{[\lambda(t_2 - t_1)]^n}{n!} e^{-\lambda(t_2 - t_1)} C_n^k p^k (1-p)^{n-k} \\ &= e^{-\lambda p(t_2 - t_1)} \frac{[\lambda p(t_2 - t_1)]^k}{k!} \end{aligned}$$

Therefore, $M(t_2) - M(t_1)$ is a Poisson distribution with parameter $\lambda\rho(t_2 - t_1)$ and from the above statement, it is seen that $\{M(t), t \geq 0\}$ is a Poisson process with mean rate $\lambda\rho$.

Example 3.11

The process $\{N_1(t), t \geq 0\}$ is a Poisson with parameter λ_1 , whereas $\{N_2(t), t \geq 0\}$ is another Poisson process with parameter λ_2 and they are independent. Let

$$X(t) = N_1(t) + N_2(t)$$

and

$$Y(t) = N_1(t) - N_2(t)$$

Show that

1. $X(t)$ is Poisson with parameter $\lambda_1 + \lambda_2$
2. $Y(t)$ is not Poisson

1. First, $X(0) = N_1(0) + N_2(0) = 0$

Next, assume that $X(t_1), X(t_2), \dots, X(t_n)$ where $(t_1 < t_2 < \dots < t_n)$ is arbitrarily n random variables. We can see that the increments

$$X(t_2) - X(t_1), X(t_3) - X(t_2), \dots, X(t_n) - X(t_{n-1})$$

has joint distribution as

$$\begin{aligned} & P\{X(t_2) - X(t_1) = i_1, X(t_3) - X(t_2) = i_2, \dots, X(t_n) - X(t_{n-1}) = i_{n-1}\} \\ &= P\{N_1(t_2) - N_1(t_1) + N_2(t_2) - N_2(t_1) = i_1, N_1(t_3) - N_1(t_2) + N_2(t_3) - N_2(t_2) = i_2, \dots, \\ & \quad N_1(t_n) - N_1(t_{n-1}) + N_2(t_n) - N_2(t_{n-1}) = i_{n-1}\} \\ &= \sum_{j_1=0}^{i_1} P\{N_1(t_2) - N_1(t_1) = j_1, N_2(t_2) - N_1(t_1) = i_1 - j_1, N_1(t_3) - N_1(t_2) + N_2(t_3) - N_2(t_2) = i_2, \dots, \\ & \quad N_1(t_n) - N_1(t_{n-1}) + N_2(t_n) - N_2(t_{n-1}) = i_{n-1}\} \\ &= \sum_{j_1=0}^{i_1} \sum_{j_2=0}^{i_2} \dots \sum_{j_{n-1}=0}^{i_{n-1}} P\{N_1(t_2) - N_1(t_1) = j_1, N_2(t_2) - N_1(t_1) = i_1 - j_1, \dots, \\ & \quad N_1(t_n) - N_1(t_{n-1}) = j_{n-1}, N_2(t_n) - N_2(t_{n-1}) = i_{n-1} - j_{n-1}\} \\ &= \sum_{j_1=0}^{i_1} \sum_{j_2=0}^{i_2} \dots \sum_{j_{n-1}=0}^{i_{n-1}} P\{N_1(t_2) - N_1(t_1) = j_1\} P\{N_2(t_2) - N_1(t_1) = i_1 - j_1\}, \dots, \\ & \quad \dots P\{N_1(t_n) - N_1(t_{n-1}) = j_{n-1}\} P\{N_2(t_n) - N_2(t_{n-1}) = i_{n-1} - j_{n-1}\} \end{aligned}$$

On the other hand, we can see that

$$\begin{aligned}
 & P\{X(t_2) - X(t_1) = i_1\} \dots P\{X(t_n) - X(t_{n-1}) = i_{n-1}\} \\
 &= \sum_{j_1=0}^{i_1} P\{N_1(t_2) - N_1(t_1) = j_1\} P\{N_2(t_2) - N_1(t_1) = i_1 - j_1\} P\{X(t_3) - X(t_2) = i_2\} \\
 &\quad \dots P\{X(t_n) - X(t_{n-1}) = i_{n-1}\} \\
 &= \sum_{j_1=0}^{i_1} \sum_{j_2=0}^{i_2} \dots \sum_{j_{n-1}=0}^{i_{n-1}} P\{N_1(t_2) - N_1(t_1) = j_1\} P\{N_2(t_2) - N_1(t_1) = i_1 - j_1\}, \dots, \\
 &\quad \dots P\{N_1(t_n) - N_1(t_{n-1}) = j_{n-1}\} P\{N_2(t_n) - N_2(t_{n-1}) = i_{n-1} - j_{n-1}\}
 \end{aligned}$$

Therefore,

$$\begin{aligned}
 & P\{X(t_2) - X(t_1) = i_1, \dots, X(t_n) - X(t_{n-1}) = i_{n-1}\} \\
 &= P\{X(t_2) - X(t_1) = i_1\}, \dots, P\{X(t_n) - X(t_{n-1}) = i_{n-1}\}
 \end{aligned}$$

in which, $X(t_2) - X(t_1)$, $X(t_3) - X(t_2)$, ..., $X(t_n) - X(t_{n-1})$ are mutually independent, which means that $\{X(t), t \geq 0\}$ is an independent increment process.

Third, when $\tau < t$, we can have

$$\begin{aligned}
 & P\{X(t) - X(\tau) = k\} = P\{N_1(t) - N_1(\tau) + N_2(t) - N_2(\tau) = k\} \\
 &= \sum_{i=0}^k \frac{\lambda_1^i (t-\tau)^i}{i!} e^{-\lambda_1(t-\tau)} \frac{\lambda_2^{k-i} (t-\tau)^{k-i}}{(k-i)!} e^{-\lambda_2(t-\tau)} \\
 &= \sum_{i=0}^k \frac{\lambda_1^i \lambda_2^{k-i} (t-\tau)^k}{i! (k-i)!} e^{-(\lambda_1 + \lambda_2)(t-\tau)} \\
 &= \frac{(t-\tau)^k e^{-(\lambda_1 + \lambda_2)(t-\tau)}}{k!} \sum_{i=0}^k C_k^i \lambda_1^i \lambda_2^{k-i} \\
 &= \frac{[(\lambda_1 + \lambda_2)(t-\tau)]^k e^{-(\lambda_1 + \lambda_2)(t-\tau)}}{k!}
 \end{aligned}$$

Therefore, the process $X(t) - X(\tau)$ is Poisson with parameter $(\lambda_1 + \lambda_2)(t - \tau)$.

The above discussion illustrates that the Poisson processes are addible.

2. Now, consider $Y(t) = N_1(t) - N_2(t)$

$$\begin{aligned}
 & P\{Y(t) = -1\} = P\{N_1(t) - N_2(t) = -1\} = P\{N_2(t) - N_1(t) = 1\} \\
 &= \sum_{i=0}^{\infty} P\{N_1(t) = i\} P\{N_2(t) = i+1\} > P\{N_1(t) = 0\} P\{N_2(t) = 1\} > 0
 \end{aligned}$$

Therefore, $\{Y(t), t \geq 0\}$ is not Poisson, because if $Y(t)$ is Poisson, $P\{Y(t) = -1\} = 0$.

3.1.4.2.4 Application of Poisson Process

3.1.4.2.4.1 Arrival Time Time between arrivals in a Poisson process has exponential distribution, where Y is denoted as the random time of the first arrival.

$$P(Y > y) = P(N(y) = 0) = \frac{(\lambda y)^0}{0!} e^{-\lambda y} = e^{-\lambda y} \quad (3.54)$$

3.1.4.2.4.2 The CDF of Y

$$F_Y(y) = P(Y \leq y) = 1 - e^{-\lambda y} \quad (3.55)$$

Example 3.12

Suppose vehicles are passing a bridge with the rate of two per minute.

Question (1): In 5 minutes, what is the average number of vehicles?

Question (2): What is the variance in 5 minutes?

Question (3): What is the probability of at least one vehicle passing the bridge in that 5 minutes?

To determine the above, the Poisson process is assumed, where $V(t)$ is the number of vehicles in time interval $[0, t]$, with a rate of $\lambda = 2$.

$$P\{V(t) = n\} = \frac{(\lambda t)^n}{n!} e^{-\lambda t}, \quad n = 1, 2, 3, \dots$$

1. Substituting $t = 5$, we have

$$P\{V(t) = 5\} = \frac{(10)^k}{k!} e^{-10}$$

Therefore, the mean is

$$\mu_V(5) = 10$$

2. The variance is

$$\sigma_V^2(5) = 10$$

3. To calculate the probability, we can write

$$P[V(5) \geq 1] = 1 - P[V(5) = 0] = 1 - e^{-10} \approx 1.0$$

3.1.4.3 Harmonic Process

The concept of harmonic process is practically very useful in random variation.

3.1.4.3.1 Definition

$X(t)$ is a harmonic process given by

$$X(t) = A \cos \omega t + B \sin \omega t \quad (3.56)$$

where A and B are independent random variables with an identical PDF and with respect to the following conditions:

The mean is

$$\mu_A = \mu_B = 0 \quad (3.57)$$

The variance is

$$\sigma_A^2 = \sigma_B^2 = \sigma^2 \quad (3.58)$$

Furthermore, we have ω as a given frequency.

It is important to note that if A and B are normally distributed, $X(t)$ is a Gaussian process. Ordinarily, $X(t)$ is not a Gaussian process.

3.1.4.3.2 Mean

We see the mean is

$$\mu_X(t) = 0 \quad (3.59)$$

3.1.4.3.3 Autocorrelation Function

The autocorrelation function is calculated as

$$R_X(\tau) = E[X(t) X(t + \tau)] = E[\{A \cos \omega t + B \sin \omega t\} \{A \cos \omega(t + \tau) + B \sin \omega(t + \tau)\}] = E[A^2] \cos \omega t \cos \omega(t + \tau) + E[B^2] \sin \omega t \sin \omega(t + \tau)$$

Note that $E[AB] = 0$, therefore, we have $E[A^2] = E[B^2] = \sigma^2$.

Substituting for σ^2 into the above equation, we have

$$\sigma^2 \cos[\omega t - \omega(t + \tau)] = \sigma^2 \cos[-\omega \tau]$$

Consequently, resulting in

$$R_X(\tau) = \sigma^2 \cos \omega \tau \quad (3.60)$$

3.1.4.3.4 Sum of Harmonic Process

Now consider the sum of harmonic process. Let $A_1, A_2, \dots, A_m, B_1, B_2, \dots, B_m$ be independent random variables, and $\omega_1, \omega_2, \dots, \omega_m$ be m distinct frequencies. We then define a harmonic process as

$$X_k(t) = A_k \cos \omega_k t + B_k \sin \omega_k t \quad (3.61)$$

Furthermore, we define a new harmonic process as the sum of all the $X_k(t)$:

$$X(t) = \sum_{k=1}^m X_k(t) = \sum_{k=1}^m (A_k \cos \omega_k t + B_k \sin \omega_k t) \quad (3.62)$$

The variance is

$$\sigma^2 = \sum_{k=1}^m \sigma_k^2 \quad (3.63)$$

and the autocorrelation is

$$R_X(\tau) = \sum_{k=1}^m R_{X_k}(\tau) = \sum_{k=1}^m \sigma_k^2 \cos \omega_k \tau \quad (3.64)$$

Also, let $p(\omega_k)$ represent the portion of the total variance contributed by the process with frequency ω_k . In other words, let

$$p(\omega_k) = \frac{\sigma_k^2}{\sigma^2} \quad (3.65)$$

It is seen that

$$\sum_{k=1}^m p(\omega_k) = 1 \quad (3.66)$$

The autocorrelation function is then rewritten as

$$R_X(\tau) = \sigma^2 \sum_{k=1}^m p(\omega_k) \cos \omega_k \tau \quad (3.67)$$

Note that when the frequency interval between ω_{k+1} and ω_k for all k are equal, we can write

$$p(\omega_k) = \frac{1}{2\pi} g(\omega_k) \Delta\omega \quad (3.68)$$

In this case,

$$\Delta\omega = \omega_{k+1} - \omega_k, \quad k = 1, \dots, m-1 \quad (3.69)$$

and $g(\omega_k)$ is density function.

Under a certain condition, which will be explained in Chapter 4, the frequency interval can become infinitesimal, that is,

$$\omega_{k+1} - \omega_k \rightarrow d\omega \quad (3.70)$$

and the autocorrelation function becomes

$$R_X(\tau) = \sigma^2 \lim_{\Delta\omega \rightarrow 0} \left[\sum_{k=1}^m \frac{1}{2\pi} g(\omega_k) \Delta\omega \cos \omega_k \tau \right] = \frac{1}{2\pi} \int_0^\infty \sigma^2 g(\omega) \cos \omega \tau d\omega \quad (3.71)$$

Equation 3.71 is the Fourier cosine transform of function $\sigma^2 g(\omega)$. That is, $R_X(\tau)$ and $\sigma^2 g(\omega)$ is a Fourier pair, denoted by

$$R_X(\tau) \Leftrightarrow \sigma^2 g(\omega) \quad (3.72)$$

where the symbol “ $x(\tau) \Leftrightarrow g(\omega)$ ” denotes a Fourier pair of $x(\tau)$ and $g(\omega)$.

In Chapter 4, function $\sigma^2 g(\omega)$ is referred to as the spectral density function because it distributes the variance of $X(t)$ as a density across the spectrum in the frequency domain. Note that because the Fourier pair indicated in Equation 3.72 is unique, $g(\omega)$ contains precisely the same information as $R_X(\tau)$.

Comparing Equation 3.64 (in which a series consists of discrete harmonic terms $\cos \omega_k \tau$) and Equation 3.71 (in which an integral contains harmonic terms $\cos \omega \tau$), we see that both represent the autocorrelation functions. The autocorrelation described by Equation 3.71 has a continuous spectrum, with infinitesimal frequency resolution $d\omega$, which implies that at any frequency point ω , the resolution is identical. That is, the continuous spectrum has an infinite number of spectral lines. On the other hand, the autocorrelation described by Equation 3.64 has a discrete spectrum and the number of the spectral lines is m . However, at frequency ω_p and ω_q , the corresponding frequency intervals are not necessarily equal. That is, in general

$$\Delta\omega_p = \omega_{p+1} - \omega_p \neq \Delta\omega_q = \omega_{q+1} - \omega_q \quad (3.73)$$

The advantage of continuous spectrum and the disadvantage of equal frequency resolution will be further discussed in Chapter 4.

3.2 CORRELATION ANALYSIS

3.2.1 CROSS-CORRELATION

In Section 3.1, we introduced the concept of autocorrelation without discussing its physical meaning and engineering applications in detail. In this section, the concept of correlation of random processes in more specific ways is considered. The focus is given to stationary processes.

3.2.1.1 Cross-Correlation Function

3.2.1.1.1 Definition

Recalling Equation 3.17, the autocorrelation function is given by

$$R_X(t_1, t_2) = E[X(t_1)X(t_2)] = \int_{-\infty}^{\infty} x_1 x_2 f_{X(t_1)X(t_2)}(x_1, x_2) dx$$

Similar to the term $R_X(t_1, t_2)$, consider the case in which there exists a second process $Y(t_2)$. In this instance, we would have a *cross-correlation*, denoted by $R_{XY}(t_1, t_2)$, which is the measure of correlation between two random processes $X(t)$ and $Y(t)$.

$$R_{XY}(t_1, t_2) = E[X(t_1)Y(t_2)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_1 y_2 f_{XY}(x, y, t_1, t_2) dx dy \quad (3.74)$$

If both $X(t)$ and $Y(t)$ are stationary, the cross-correlation function depends only on the time lag τ , where

$$\tau = t_2 - t_1 \quad (3.75)$$

This is further illustrated in Figure 3.2.

Under these conditions, it is seen that

$$R_{XY}(\tau) = E[X(t) Y(t + \tau)] \quad (3.76)$$

and

$$R_{YX}(\tau) = E[Y(t) X(t + \tau)] \quad (3.77)$$

Observing that

$$R_{XY}(\tau) \neq R_{YX}(\tau) \quad (3.78)$$

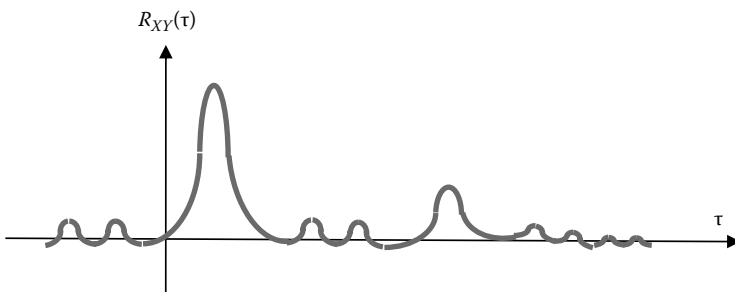


FIGURE 3.2 Conceptual cross-correlation function.

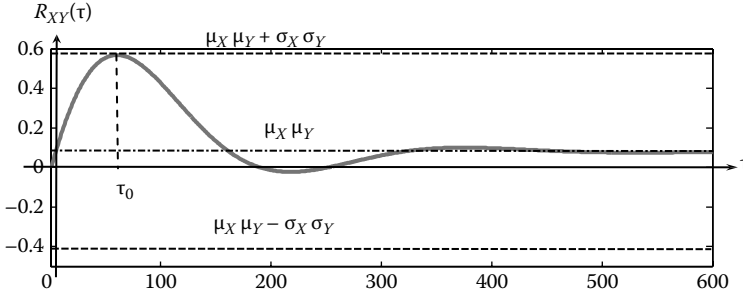


FIGURE 3.3 Bounds of cross-correlation function.

3.2.1.1.2 Skew Symmetry (Antisymmetric)

In the instance of skew symmetry, we have the following:

$$R_{XY}(\tau) = E[X(t) Y(t + \tau)] = E[X(s - \tau) Y(s)] = E[Y(s) X(s - \tau)] = R_{YX}(-\tau) \quad (3.79)$$

3.2.1.1.3 Bounds

The cross-correlation function has upper and lower bounds. Figure 3.3 shows conceptually the bounds of a cross-correlation function.

In Figure 3.3, $R_{XY}(\tau)$ is bounded by

$$-\sigma_X \sigma_Y + \mu_X \mu_Y \leq R_{XY}(\tau) \leq \sigma_X \sigma_Y + \mu_X \mu_Y \quad (3.80)$$

These bounds can be graphically shown in Figure 3.3.

3.2.1.2 Cross-Covariance Function

3.2.1.2.1 Definition

The cross-covariance function is given by

$$\sigma_{XY}(t_1, t_2) = E[\{X(t_1) - \mu_X(t_1)\} \{Y(t_2) - \mu_Y(t_2)\}] \quad (3.81)$$

It can be realized that

$$\sigma_{XY}(t_1, t_2) = E[X(t_1) Y(t_2)] - \mu_X(t_1) \mu_Y(t_2) \quad (3.82)$$

Thus, through substitution, we have

$$\sigma_{XY}(t_1, t_2) = R_{XY}(t_1, t_2) - \mu_X(t_1) \mu_Y(t_2) \quad (3.83)$$

3.2.1.2.2 Orthogonal Processes

If the cross-correlation function of two random processes $X(t)$ and $Y(t)$:

$$R_{XY}(t_1, t_2) = 0 \quad (3.84)$$

then $X(t)$ and $Y(t)$ are orthogonal.

3.2.1.2.3 Uncorrelated Processes

If the cross-covariance function of two random processes $X(t)$ and $Y(t)$:

$$\sigma_{XY}(t_1, t_2) = 0 \quad (3.85)$$

then $X(t)$ and $Y(t)$ are uncorrelated.

In the case of an uncorrelated process:

$$E[X(t_1) Y(t_2)] = E[X(t_1)] E[Y(t_2)] \quad (3.86)$$

1. If $X(t)$ and $Y(t)$ are mutually independent, then they are uncorrelated.
2. If $X(t)$ and $Y(t)$ are uncorrelated, they are not necessarily independent.
3. If $X(t)$ and $Y(t)$ are Gaussian processes, then the condition of being “mutually independent” is sufficient and a necessary condition of “uncorrelated.”

3.2.1.2.4 General Meaning of Correlation

In general, the value of a random process over duration T will vary randomly above and below the expected value. To count the number of instances the value steps above and/or below in a unit of time, the concept of frequency is used. In the case of the unit of time measured in seconds, the frequency would be measured in hertz.

If $X(t)$ and $Y(t)$ are correlated, then they share identical frequency components. If this holds untrue, then they are uncorrelated.

3.2.2 AUTOCORRELATION

By considering autocorrelation functions, the meaning of correlation will be further explored.

3.2.2.1 Physical Meaning of Correlation

3.2.2.1.1 Ensemble Average versus Temporal Average

Referring to Equation 3.74, let $Y(t)$ be $X(t)$ and remembering the cross-correlation function reduces to autocorrelation function (which was first introduced in Equation 3.17), Equations 3.17 and 3.74 are rewritten as follows:

$$R_X(t_1, t_2) = E[X(t_1)X(t_2)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_1 x_2 f_X(x, y, t_1, t_2) dx dy \quad (3.87)$$

Recalling Equation 3.40, we obtain

$$\langle X(t, k) X(t + \tau, k) \rangle = E[X(t, k) X(t + \tau, k)] = R_X(\tau) \quad (3.88)$$

From this equation, it is shown that if $X(t)$ is ergodic, it must also be stationary, then one can use temporal averages to replace the ensemble average as

$$R_X(t_1, t_2) = R_X(\tau) = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x(t)x(t + \tau) dt \quad (3.89)$$

In Equation 3.89

$$t_1 = t \quad (3.90a)$$

$$t_2 = t + \tau \quad (3.90b)$$

That is

$$\tau = t_2 - t_1 \quad (3.90c)$$

Generally, this is written as

$$R_X(\tau) = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x_k(t) x_k(t + \tau) dt = E[X_k(t) X_k(t + \tau)] \quad (3.91)$$

In this instance, the subscription k stands for the k th record. It can be shown that the notation “ k ” in Equation 3.91 is necessary.

3.2.2.1.2 Correlation Analysis

Suppose the time history $x_k(t)$ can be represented by Fourier series. Most engineering signals, if not all, can be written in such form. Then, we have

$$x_k(t) = a_0 + a_1 \cos(\omega_1 t) + b_1 \sin(\omega_1 t) + a_2 \cos(\omega_2 t) + b_2 \sin(\omega_2 t) + \dots \quad (3.92)$$

Readers may consider the condition that we can always have Equation 3.92.

Due to the orthogonality of the cosine and sines from Equation 3.91 in the integration of Equation 3.92, the result will cancel the “uncorrelated” frequency components. In this case, only the correlated terms will be left. This unveils the physical meaning of correlation analysis. From this point on, let us denote random processes as “signals.”

In Figure 3.4, several correlation functions of typical signals have been plotted. In the first case, a sinusoidal signal with an autocorrelation function that will never decay is shown. Note that a signal that does not contain a sine wave will always decay. Furthermore, the autocorrelation function of a signal that is closer to sinusoidal will have a slower decaying rate, or on the contrary, it will decay rather quickly. This is also shown in Figure 3.5.

3.2.2.2 Characteristics of Autocorrelation Function

In this section, we will further explore useful properties of autocorrelation functions of real valued processes.

3.2.2.2.1 Bounds

For the case: $X(t)$ is stationary.

$$R_X(\tau) = E[X(t)X(t + \tau)] = \sigma_{XX}(\tau) + \mu_X^2 = \rho_{XX}(\tau)\sigma_X^2 + \mu_X^2 \quad (3.93)$$

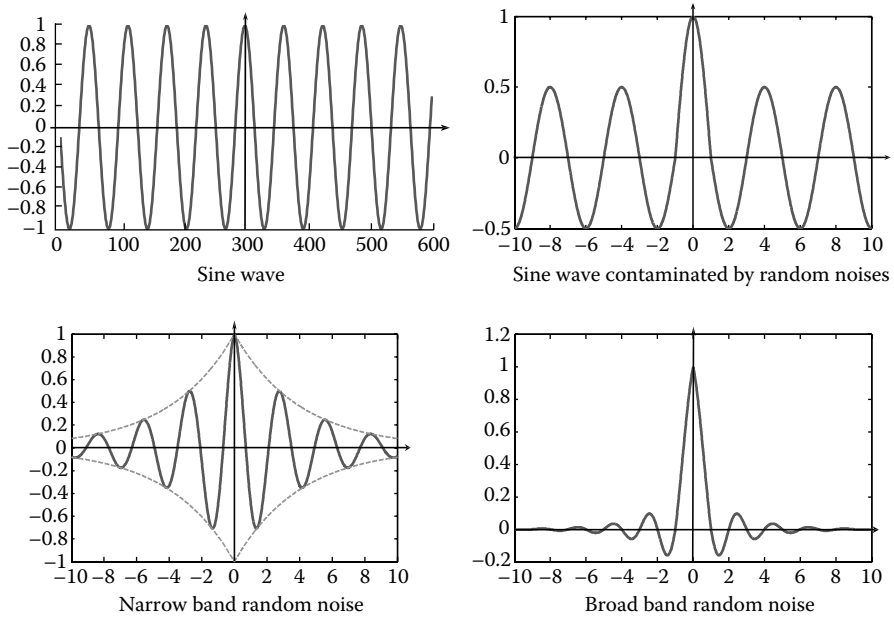


FIGURE 3.4 Examples of autocorrelation functions.

Note that

$$-1 \leq \rho_{XX}(\tau) \leq 1$$

Thus,

$$-\sigma_X \sigma_Y + \mu_X \mu_Y \leq R_{XY}(\tau) \leq \sigma_X \sigma_Y + \mu_X \mu_Y \quad (3.94)$$

Because

$$R_X(0) = E[X^2(t)] = \sigma_X^2 + \mu_X^2 \geq 0 \quad (3.95)$$

$R_X(0)$ is the maximum value and

$$|R_X(\tau)| \leq R_X(0) \quad (3.96)$$

3.2.2.2.2 Symmetry

If

$$R_X(\tau) = E[X(t) X(t + \tau)] = E[X(s - \tau) X(s)] = R_X(-\tau) \quad (3.97)$$

Then, it is symmetric at

$$\tau = 0 \quad (3.98)$$

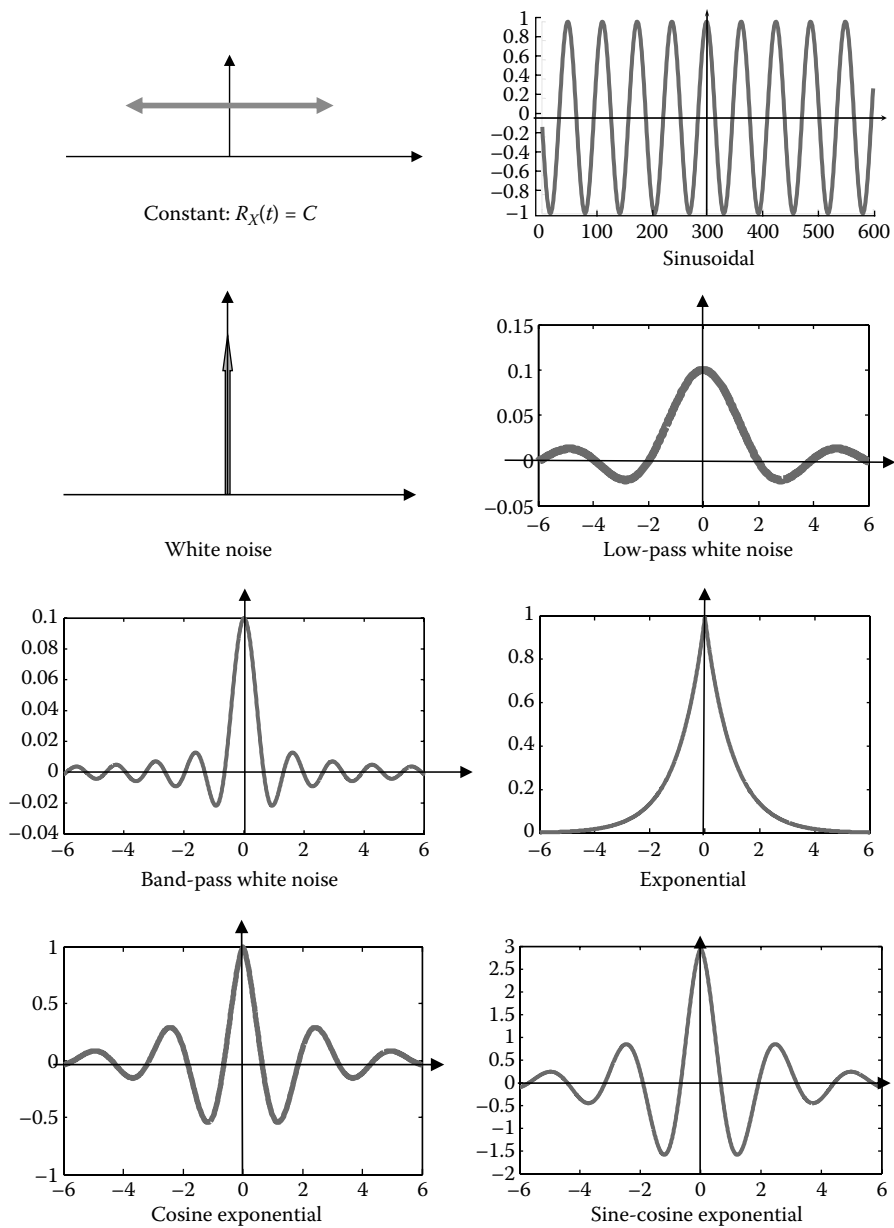


FIGURE 3.5 Typical autocorrelation functions.

3.2.2.2.3 Limiting Values

$$\lim_{\tau \rightarrow \infty} R_X(\tau) = \mu_X^2 \quad (3.99)$$

Equation 3.99 implies that when the time difference becomes sufficiently large, $X(t)$ and $X(t + \tau)$ becomes uncorrelated, and $\sigma_{XX}(\tau)$ vanishes.

Example 3.13

Given the autocorrelation of a stationary process $X(t)$ to be

$$R_X(\tau) = 36 + \frac{1}{1 + 36\tau^2}$$

find the mean and variance of $X(t)$.

1. According to Equation 3.99, it is seen that

$$R_X(\infty) = \mu_X^2 = 36$$

thus, the mean is

$$\mu_X = \pm 6$$

2. The variance can be written as

$$\sigma_X^2 = R_X(0) - \mu_X^2 = 37 - 36 = 1$$

3.2.2.2.4 Scales of Fluctuation

The autocorrelation function also measures the perseverance of the correlation. The faster the decay of $R_X(\tau)$, the less the sample realization of the process will remain correlated.

The scale of fluctuation θ is defined as

$$\theta = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \rho_{XX}(\tau) d\tau = \frac{1}{\sigma_X^2} \left[\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T R_X(\tau) d\tau \right] \quad (3.100)$$

Note that when θ is considerably longer than time lag τ , little correlation in the random process can be expected.

3.2.2.3 Examples of Autocorrelation Function

3.2.2.3.1 Low-Pass Random Process

In Figure 3.6, the plot of real and idealized low-pass filters is shown. A random process passing through a low-pass filter is referred to as a low-pass random process.

In this case, the corresponding autocorrelation function can be written as

$$R_X(\tau) = \sigma_X^2 \frac{\sin(\omega_C \tau)}{\omega_C \tau} \quad (3.101)$$

3.2.2.3.2 Delay and Attenuation

$X(t)$ is a random process. Let $Y(t)$ denote the delay of t by δ and the attenuation by factor α on $X(t)$, that is,

$$Y(t) = \alpha X(t - \delta) \quad (3.102)$$

In this example, consider the cross-correlation function of

$$R_{XY}(\tau) = E[X(t) Y(t + \tau)] = E[X(t) \{\alpha X(t - \delta + \tau)\}] = \alpha R_X(\tau - \delta) \quad (3.103)$$

Note that in this scenario

$$E[X(t) \{X(t - \delta)\}] = R_X(t - \delta)$$

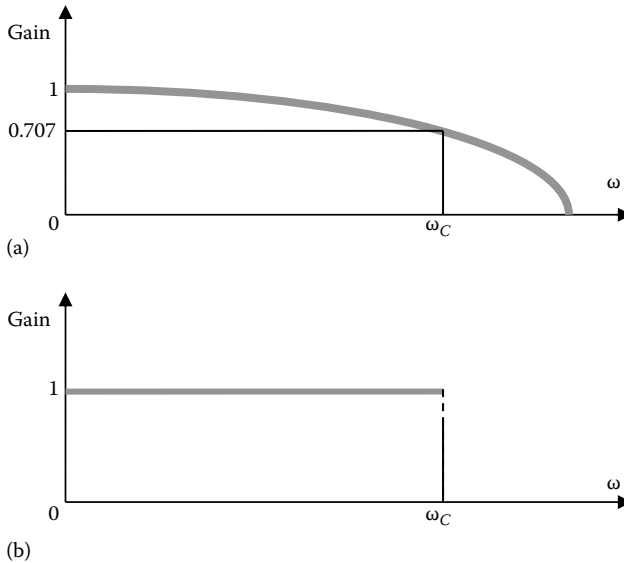


FIGURE 3.6 Low-pass filter. (a) Practical low-pass filter, (b) idealized low-pass filter.

3.2.2.3.3 Sum of Two Processes

If $X(t)$ and $Y(t)$ are stationary and the sum is given by

$$Z(t) = X(t) + Y(t) \quad (3.104)$$

then the cross-correlation $R_{ZX}(\tau)$ is

$$R_{ZX}(\tau) = E[Z(t) X(t + \tau)] = E[(X(t) + Y(t))\{X(t + \tau)\}] = R_X(\tau) + R_{XY}(\tau) \quad (3.105)$$

If $X(t)$ and $Y(t)$ are uncorrelated with zero mean, that is,

$$R_{XY}(\tau) = R_{YX}(\tau) = 0 \quad (3.106)$$

then

$$R_{ZX}(\tau) = R_X(\tau) \quad (3.107)$$

The autocorrelation function of $Z(t)$ is

$$\begin{aligned} R_Z(\tau) &= E[Z(t) Z(t + \tau)] = E[(X(t) + Y(t))\{X(t + \tau) + Y(t + \tau)\}] \\ &= R_X(\tau) + R_{XY}(\tau) + R_{YX}(\tau) + R_Y(\tau) \end{aligned} \quad (3.108)$$

Therefore, for the cases of $X(t)$ and $Y(t)$ are uncorrelated with zero mean

$$R_Z(\tau) = R_X(\tau) + R_Y(\tau) \quad (3.109)$$

3.2.2.3.4 Nondispersive Propagation

Suppose $X(t)$ is stationary and is transmitted as a signal, and the propagation is non-dispersive. The concept of dispersive and nondispersive is illustrated in Figure 3.7.

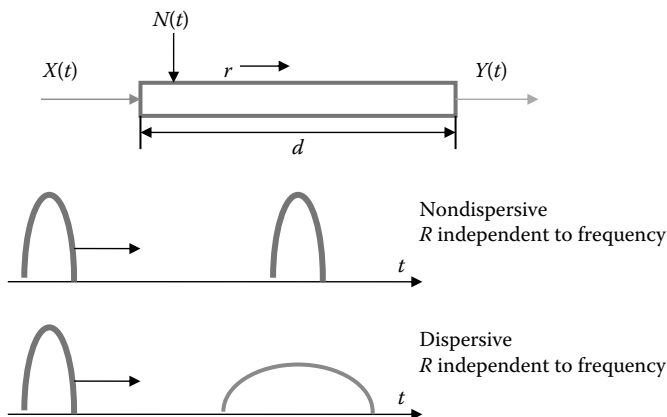


FIGURE 3.7 Dispersion.

Denoting d as the distance, r as the wave speed, and a as an attenuation factor, we obtain

$$Y(t) = aX\left(t - \frac{d}{r}\right) + N(t) \quad (3.110)$$

In Figure 3.7 and Equation 3.110, $N(t)$ is noise.
The autocorrelation function of $Y(t)$ is

$$R_Y(\tau) = a^2 R_X(\tau) + R_N(\tau) \quad (3.111)$$

The cross-correlation function of $X(t)$ and $Y(t)$ is

$$R_{XY}(\tau) = aR_X\left(\tau - \frac{d}{r}\right) \quad (3.112)$$

Example 3.14: Periodic Stationary Process

If a random process $X(t)$ satisfies

$$X(t) = X(t + T)$$

X is referred to as *periodic stationary*.

Show that, in the case of periodic stationary process, the autocorrelation function is also a periodic function, that is

$$R_X(T + \tau) = R_X(\tau)$$

It is seen that

$$R_X(T + \tau) = E[X(t) X(t + T + \tau)] = E[X(t) X(t + \tau)] = R_X(\tau)$$

3.2.3 DERIVATIVES OF STATIONARY PROCESS

We are familiar with derivatives of deterministic processes. Now, we will consider the case of stationary random process.

3.2.3.1 Stochastic Convergence

The first concept in the study of derivatives is convergence. Consider a sequence of random variables denoted by X_j , where $j = 0, 1, 2, \dots$

It is impossible to write

$$\lim_{j \rightarrow \infty} X_j = X_0$$

for X_j are random sets.

Using $f_{(X_j)}$ to denote the frequency, it is also problematic to have

$$\lim_{j \rightarrow \infty} f_{(X_j)} = p$$

This holds true because the above equation implies that there exists an $\varepsilon > 0$, for no matter how large a number $N > 0$, one can always find $n > N$, such that

$$|f_{(X_n)} - p| < \varepsilon$$

This is given that if we let $\varepsilon < p$, then

$$P\{f_{(X_n)} = 0\} = (1 - p)^n \neq 0$$

that is, the event $f_{(X_n)} = 0$ is possible. In other words, it is possible for $f_{(X_n)} \neq p$, which is against the observation based on the classical theory of probability. We thus consider the convergence from a different angle as follows.

3.2.3.1.1 Convergence with Probability 1

This angle is the consideration of chance, or probability, of convergence. First, understand the following:

$$\lim_{j \rightarrow \infty} P(X_j = X_0) = 1 \quad (3.113a)$$

3.2.3.1.2 Convergence in Probability

Because the above requirement is strong, we may in turn consider

$$\lim_{j \rightarrow \infty} P(|X_j - X_0| \geq \varepsilon) = 0 \quad (3.113b)$$

3.2.3.1.3 Convergence in Distribution

Another angle is the convergence in distribution function, namely,

$$\lim_{j \rightarrow \infty} F_{X_j}(x) = F_{X_0}(x) \quad (3.113c)$$

3.2.3.2 Mean-Square Limit

The second importance of derivatives in the temporal process is the limit. Similarly, because the process is random, we will need to consider some different approaches.

3.2.3.2.1 Definition

$\{X_n\}$ is a real series of random variables, where $N = 0, 1, 2, 3$. For $\{X_n\}$, its mean square values exist, given by

$$E[X_n^2] < \infty \quad (3.114)$$

If

$$\lim_{n \rightarrow \infty} \|X_n - X_0\| = 0 \quad (3.115)$$

or

$$\lim_{n \rightarrow \infty} E[(X_n - X_0)^2] = 0 \quad (3.116)$$

then, X_0 is the mean square limit of $\{X_n\}$, denoted by

$$\text{l.i.m}_{n \rightarrow \infty} X_n = X_0 \quad (3.117)$$

3.2.3.2.2 Property

$\{X_n\}$ and $\{Y_n\}$ are two real series of random variables, where both have a limited mean and variance:

$$\text{l.i.m}_{n \rightarrow \infty} X_n = X_0$$

and

$$\text{l.i.m}_{n \rightarrow \infty} Y_n = Y_0$$

Written with constants a and b , we have

$$1. \text{l.i.m}_{n \rightarrow \infty} (aX_n + bY_n) = aX_0 + bY_0 \quad (3.118)$$

$$2. E(X) = E\left[\text{l.i.m}_{n \rightarrow \infty} X_n\right] = \lim_{n \rightarrow \infty} E[X_n] \quad (3.119)$$

$$3. \lim_{\substack{n \rightarrow \infty \\ m \rightarrow \infty}} E[X_n Y_m] = E[X_0 Y_0] \quad (3.120)$$

For the case, $m = n$, we will have

$$\lim_{n \rightarrow \infty} E[X_n^2] = E[X_0^2] \quad (3.121)$$

Example 3.15

If, for random variables X and Y , we have $EX < \infty$ and $EY < \infty$, then the complex valued random variable $Z = X + jY$ has its mathematical expectation EZ given by

$$EZ = EX + jEY$$

We can further define the characteristic function of a real-valued random variable W , denoted by $\phi(t)$ and

$$\phi_W(t) \equiv E e^{itW} = E[\cos(tW)] + j E[\sin(tW)]$$

Note that $E[\cos(tW)] < \infty$ and $E[\sin(tW)] < \infty$, so that the characteristic function of the random variable W , $\phi_W(t)$, always exists.

It can be proven that the characteristic function and the PDF of a random variable are uniquely determined by each other. For example, for random variables whose distribution is Poisson, then

$$\phi_W(t) = e^{\lambda(e^{it} - 1)}$$

Now, let us consider showing the mean-square limit of a Poisson random sequence is Poisson random variable.

Let $\{X_n, n = 1, 2, \dots\}$ to denote the Poisson random sequence, and we have

$$\text{l.i.m}_{n \rightarrow \infty} X_n = X$$

In this case, it is seen that

$$\lim_{n \rightarrow \infty} E(X_n) = E(X)$$

which implies

$$\lim_{n \rightarrow \infty} \lambda_n = \lambda$$

Therefore, we further have

$$\phi_X(t) = \lim_{n \rightarrow \infty} \phi_{X_n}(t) = \lim_{n \rightarrow \infty} e^{\lambda_n(e^{it} - 1)} = e^{\lambda(e^{it} - 1)}$$

which implies that X is the random variable with Poisson distribution, for its characteristic function is Poisson.

3.2.3.3 Mean-Square Continuity

$X(t)$ is a real process, if for $t \in T$,

$$\text{l.i.m}_{h \rightarrow 0} X(t + h) = X(t) \quad (3.122a)$$

$X(t)$ is continuous in mean square at t .

If $X(t)$ is continuous in mean square at every point of $t \in T$, then $X(t)$ is continuous in mean square on T .

Stationary process $X(t)$ is continuous in mean square at t , if and only if $R_X(\tau)$ is continuous at $\tau = 0$.

Proof:

Because $X(t)$ is stationary, we have the following:

$$E[(X(t + \tau) - X(t))^2] = E[(X(t + \tau))^2] + E[(X(t))^2] - 2E[X(t + \tau)X(t)] = 2(R_X(0) - R_X(\tau))$$

$$\lim_{\tau \rightarrow 0} E[X(t + \tau) - X(t)] = \lim_{\tau \rightarrow 0} [2(R_X(0) - R_X(\tau))]$$

Example 3.16

A mutually independent sequence of random variables denoted as $\{X_n, n \geq 1\}$ has the following distribution

$$\begin{array}{ccc} X_n & 0 & n \\ P(X_n) & 1 - \frac{1}{n^2} & \frac{1}{n^2} \end{array}$$

Check whether X_n is a mean-square convergence.

Note that Equation 3.121 implies that

$$\lim_{h \rightarrow 0} E[X(t+h) - X(t)] = 0$$

Based on the above equation, let us check the expectation $E(|X_m - X_n|^2)$.

$$E(|X_m - X_n|^2) = E(X_m^2) + E(X_n^2) - 2E(X_n)E(X_m)$$

It is seen that

$$E(X_m) = m \frac{1}{m^2} = \frac{1}{m}, \quad E(X_m^2) = m^2 \frac{1}{m^2} = 1$$

and

$$E(X_n) = n \frac{1}{n^2} = \frac{1}{n}, \quad E(X_n^2) = n^2 \frac{1}{n^2} = 1$$

Therefore, when $m \neq n$

$$E(X_m X_n) = E(X_m)E(X_n) = \frac{1}{mn}$$

We now have

$$\lim_{\substack{m \rightarrow \infty \\ n \rightarrow \infty}} E(|X_m - X_n|^2) = \lim_{\substack{m \rightarrow \infty \\ n \rightarrow \infty}} \left(2 - 2 \frac{1}{mn} \right) = 2 \neq 0$$

Thus, if $\{X_n, n \geq 1\}$ is not continuous in mean square, it will not converge in a mean square.

3.2.3.4 Mean-Square Derivatives of Random Process

$X(t)$ is a random process. If $\dot{X}(t)$ exists such that, for $t \in T$,

$$\dot{X}(t) = \lim_{h \rightarrow 0} \frac{X(t+h) - X(t)}{h} \quad (3.122b)$$

then $X(t)$ is mean-square differentiable at t , and $\dot{X}(t)$ is the mean-square derivative of $X(t)$. If any $t \in T$, $X(t)$ is mean-square differentiable, then $X(t)$ is mean-square differentiable on T .

3.2.3.5 Derivatives of Autocorrelation Functions

In the following, suppose $X(t)$ is mean-square differentiable, we have

$$\frac{dR_X(\tau)}{d\tau} = \frac{dE[X(t)X(t+\tau)]}{d\tau} = E\left[X(t) \frac{dX(t+\tau)}{d\tau}\right] = E[X(t)\dot{X}(t+\tau) = R_{X\dot{X}}(\tau)] \quad (3.123)$$

Furthermore,

$$\frac{d^2 R_X(\tau)}{d\tau^2} = R_{\dot{X}\ddot{X}}(\tau) \quad (3.124)$$

It is equally true that

$$\begin{aligned} \frac{d}{d\tau} \left\{ \frac{dR_X(\tau)}{d\tau} \right\} &= \frac{d}{d\tau} \left\{ \frac{dE[X(t)X(t+\tau)]}{d\tau} \right\} = \frac{d}{d\tau} \left\{ \frac{dE[X(t-\tau)X(t)]}{d\tau} \right\} \\ &= E\left[-\frac{dX(t-\tau)}{d\tau} \frac{dX(t)}{d\tau} \right] = R_{\dot{X}}(-\tau) \end{aligned} \quad (3.125)$$

In addition, we have

$$\frac{d^3 R_X(\tau)}{d\tau^3} = -R_{\dot{X}\ddot{X}}(\tau) \quad (3.126)$$

and

$$\frac{d^4 R_X(\tau)}{d\tau^4} = R_{\ddot{X}\ddot{X}}(\tau) \quad (3.127)$$

$X(t)$ is mean-square differentiable on T , if and only if the following exists:

$$\frac{\partial^2 R_X(s, t)}{\partial s \partial t} = \lim_{\substack{h \rightarrow 0 \\ h' \rightarrow 0}} \frac{R_X(t+h, t+h') - R_X(t, t+h') - R_X(t+h, t) + R_X(t, t)}{hh'} \quad (3.128)$$

Example 3.17

The *ordinary random walk* (also called *binomial process*) is the simplest random process. Using Z_t to denote the increments from time $t - 1$ to time t , taking exclusively the values $+1$ or -1 , we have

$$Z_t = X_t - X_{t-1}$$

Assuming Z_t is independent from the initial value X_0 , we can write

$$X_t = X_0 + \sum_{k=1}^t Z_k, \quad t = 1, 2, \dots$$

Thus, $X_0, Z_1, Z_2, \dots$ are independent and for all k , we have

$$P(Z_k = 1) = p \quad \text{and} \quad P(Z_k = -1) = 1 - p$$

For a more general case, we can have the *binomial process*, if replacing 1 by f and replacing -1 by b (f stands for “walking” forward and b stands for backward), that is,

$$P(Z_k = f) = p \quad \text{and} \quad P(Z_k = b) = 1 - p \quad \text{for all } k$$

where f and b are constants.

Now consider the amount of charge in a given value and the location of cloud before a random time N is binomial whereas after time N it remains constant. Here N is Poisson random variable with parameter λ and is independent to the binomial process. Let X_n be the amount of charge at time n ; show that as $n \rightarrow \infty$, X_n is mean-square converged to X .

Letting $m \leq n$, we have

$$\begin{aligned} E(|X_n - X_m|^2) &= E[E\{(X_n - X_m)^2 | N\}] = P(N \leq m) [E\{(X_n - X_m)^2 | N \leq m\}] \\ &\quad + \sum_{k=m+1}^n P(N = k) E\{(X_n - X_m)^2 | N = k\} + \sum_{k=n+1}^{\infty} P(N = k) E\{(X_n - X_m)^2 | N = k\} \\ &= 0 + \sum_{k=m+1}^n \frac{\lambda^k}{k!} e^{-\lambda} (k - m) p [1 + (k - m - 1)p] + \sum_{k=n+1}^{\infty} \frac{\lambda^k}{k!} e^{-\lambda} (n - m) p [1 + (n - m - 1)p] \\ &\leq \sum_{k=m+1}^n \frac{\lambda^{k-2}}{(k-2)!} e^{-\lambda} \lambda^2 p + \sum_{k=n+1}^{\infty} \frac{\lambda^{k-2}}{(k-2)!} e^{-\lambda} \lambda^2 p \\ &= e^{-\lambda} \lambda^2 p \sum_{k=m+1}^{\infty} \frac{\lambda^{k-2}}{(k-2)!} = e^{-\lambda} \lambda^2 p \sum_{k=m-1}^{\infty} \frac{\lambda^k}{k!} \end{aligned}$$

Furthermore, considering a series S

$$S = \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = e^{\lambda}$$

and letting

$$S_n = \sum_{k=0}^n \frac{\lambda^k}{k!}$$

we can write

$$S - S_{m-2} = \sum_{k=m-1}^{\infty} \frac{\lambda^k}{k!} \rightarrow 0, \text{ as } m \rightarrow \infty$$

Therefore, when $m \leq n$, we have

$$\lim_{m \rightarrow \infty} e^{-\lambda} \lambda^2 p \sum_{k=m-1}^{\infty} \frac{\lambda^k}{k!} = 0$$

That is,

$$\lim_{\substack{m \rightarrow \infty \\ n \rightarrow \infty}} E(|X_n - X_m|^2) = 0$$

Therefore, $\{X_n, n \geq 1\}$ is mean-square converged to X .

3.2.3.6 Derivatives of Stationary Process

Because

$$R_X(0) = \max \quad (3.129)$$

we have the following:

$$\left. \frac{dR_X(\tau)}{d\tau} \right|_{\tau=0} = R_{\dot{X}\dot{X}}(0) = 0 \quad (3.130)$$

and

$$R_{\dot{X}\dot{X}}(\tau) \Big|_{\tau>0} > 0 \quad (3.131)$$

$$R_{\dot{X}\dot{X}}(\tau) \Big|_{\tau<0} < 0 \quad (3.132)$$

Combining Equations 3.130 through 3.132, we conclude that

$$R_{\dot{X}\dot{X}}(\tau) = \text{odd} \quad (3.133)$$

Therefore, $R_{\dot{X}\dot{X}}(\tau)$ is skew symmetric (antisymmetric).

3.2.3.7 Derivatives of Gaussian Process

Suppose $X(t)$ is Gaussian, then $X(t, k)$ denotes the k th sample realization

$$\dot{X}(t, k) = \lim_{h \rightarrow 0} \frac{X(t+h, k) - X(t, k)}{h} \quad (3.134)$$

Because $X(t, k)$ is Gaussian, we see that $[X(t+h, k) - X(t, k)]$ is also Gaussian and could therefore conclude that $\dot{X}(t, k)$ is Gaussian as well.

Example 3.18

A random sequence of Bernoulli distributed variables, which are mutually independent with identical PMF, is denoted by $\{Y(n), n \geq 1\}$. Let

$$X(t) = Y(i), \quad 2^{-i} < t < 2^{1-i}, \quad i = 1, 2, \dots$$

Consider if $\{X(t), 0 < t \leq 1\}$ is derivable.

First, when $2^{-i} < t < 2^{1-i}$, $i = 1, 2, \dots$, there must exist a small neighborhood of t , in which

$$X(t) = Y(i)$$

Therefore,

$$\dot{X}(t) = 0$$

Second, when $t = 1$, we can also realize that

$$\dot{X}_-(t) = 0$$

Third, when $t = 2^{-i}$ ($i = 1, 2, \dots$), we have $\dot{X}_-(t) = 0$.

However,

$$\dot{X}_+(t) = \lim_{h \rightarrow 0^+} \frac{X(t+h) - X(t)}{h} = \lim_{h \rightarrow 0^+} \frac{Y(i-1) - Y(i)}{h}$$

which means that the right mean-square derivative does not exist; therefore, at $t = 2^{-i}$, ($i = 1, 2, \dots$), $\{X(t), 0 < t \leq 1\}$ is not mean-square derivable.

PROBLEMS

1. Identify the state space and index set for the following random process
 - a. Temperature measured hourly at an airport. Continuous state and discrete index
 - b. Elevation of sea surface measured continuously at a wave staff. Continuous state and continuous index

- c. Daily closing price of a stock. Discrete state and discrete index
 - d. Number of significant defects in an optical fiber starting at one end, discrete state and continuous index
2. A random process $X(t)$ is given as

$$X(t) = \begin{cases} \cos \pi t & H \\ 2t, & T \end{cases} \quad -\infty < t < \infty \quad (\text{P3.1})$$

where H and T stands for the case of head and tail when a coin is tossed. Note that $P(H) = P(T) = 0.5$.

- a. Find the one-dimensional distribution $F(x; 0.5)$ and $F(x, 1)$ of the random process $X(t)$
 - b. Find the two-dimensional distribution $F(x_1, x_2, 0.5, 1)$ of $X(t)$
3. A deterministic square wave process $X_{\text{square}}(t)$ with period T . Find means, variations, and autocorrelations for the following random process variation of this function (Figure P3.1)

$$X_{\text{square}}(t) = \begin{cases} 1, & 0 < t < T/2 \\ -1, & T/2 < t < T \\ X_{\text{square}}(t+T), & t \leq 0 \\ X_{\text{square}}(t-T), & t \geq T \end{cases}$$

- a. Square-wave process with random amplitude
 - b. Square-wave process with random phase
(Hint: Denote the random variable set as Θ . Assume the range of Θ is from 0 to T , $f_{\Theta}(\theta) = 1/T$)
 - c. Square-wave process with random amplitude and phase
4. The mean and autocovariance of random process $X(t)$, $t \in T$ are denoted by $\mu_X(t)$ and $\sigma_{XX}(t_1, t_2)$. In addition, $g(t)$ is a deterministic function of t . Find the mean and covariance of random process $Y(t) = X(t) + g(t)$.
5. For the process $X(t) = a \cos(\omega t - \Theta)$, derive the stationary autocorrelation function from general autocorrelation function.

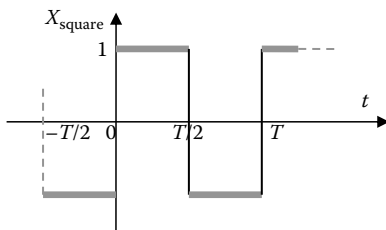


FIGURE P3.1

6. Find the scale of fluctuation of low-pass random process

$$R_X(\tau) = \sigma_X^2 \frac{\sin \omega_c \tau}{\omega_c \tau}$$

7. Find the autocorrelation and cross-correlation functions of $W(t)$ and $X(t)$
 $X(t)$ and $Y(t)$ are uncorrelated [Hint: $\sigma_{XY}(t) = 0$ and $W(t) = X(t) Y(t)$]
 8. Verify the mean, variance, and autocorrelation given for the Poisson process
 Poisson process $N(n, t)$: its density function is

$$p_N(n, t) = \frac{(\lambda t)^n}{n!} e^{-\lambda t}, \quad n \geq 0, \quad t \geq 0$$

9. Prove if $\lim_{n \rightarrow \infty} X_n = X$ then the characteristic function of X_n will converge to the characteristic function of X .
 10. Use MATLAB® to calculate the temporal average of the sample realization from the computer-generated process. Additionally, calculate the value of the sample autocorrelation function for $t_1 = 0$ and $t_2 = 1$.

4 Random Processes in the Frequency Domain

Chapter 3 introduced random processes, the variables of which are indexed by the temporal parameter t . A random process is defined as a certain kind of temporal function. For deterministic temporal functions, the corresponding spectra can be obtained by their Fourier transform, which provides a different angle to describe the natures of temporal functions.

In Chapter 4, it will be shown that Fourier transforms can also be used on many types of random processes. This allows random processes to be operated in the frequency domain, and also provides simplified formulas of the expected values. Mathematically, these processes need to be stationary. However, for engineering applications, this requirement may be relaxed to a certain degree, which will also be discussed hereafter.

4.1 SPECTRAL DENSITY FUNCTION

The spectral density function is a useful tool in the application of the Fourier transform and describes the frequency components of a stationary process.

4.1.1 DEFINITIONS OF SPECTRAL DENSITY FUNCTIONS

In this subsection, we will first discuss the necessary mathematical background useful to the main topic—the definitions of spectral density functions—which by themselves are powerful tools to study random processes in the frequency domain.

4.1.1.1 Mean-Square Integrable of Random Process

In Chapter 3, the properties of the derivatives of random processes were discussed. Here, the integral of random processes are now considered. This is necessary because Fourier transforms are based on integrations. However, similar to the difficulty of direct derivation of random process $X(t)$, special treatment of its integral is also necessary.

Consider $X(t)$ as a random process and $a, b \in T$. Divide the interval $[a, b]$ into an $n + 1$ moment such that

$$a = t_0 < t_1 < \dots < t_n = b \quad (4.1)$$

Denote

$$\Delta = \max_{0 \leq k \leq n-1} (t_{k+1} - t_k) \quad (4.2)$$

t'_k is a point inside $[t_k, t_{k+1}]$, that is,

$$t_k \leq t'_k \leq t_{k+1}, k = 0, 1, \dots, n-1 \quad (4.3)$$

If the term $\sum_{k=0}^{n-1} X(t'_k)(t_{k+1} - t_k)$ possesses a mean-square limit $\int_a^b X(t) dt$, namely,

$$\begin{aligned} & \text{l.i.m}_{n \rightarrow \infty} \left[\sum_{k=0}^{n-1} X(t'_k)(t_{k+1} - t_k) - \int_a^b X(t) dt \right] \\ &= \lim_{\Delta \rightarrow 0} E \left[\left| \sum_{k=0}^{n-1} X(t'_k)(t_{k+1} - t_k) - \int_a^b X(t) dt \right|^2 \right] = 0 \end{aligned} \quad (4.4)$$

then, $X(t)$ is a mean-square integrable in interval $[a, b]$ (Riemann integrable, Bernhard Riemann, 1826–1866).

Random process $X(t)$ is a mean-square integrable in interval $[a, b]$, if and only if $R_X(s, t)$ is Riemann integrable in space $[a, b] \times [a, b]$. Stationary process $X(t)$ is a mean-square integrable in interval $[a, b]$ if its autocorrelation function $R_X(\tau)$ is integrable, and

$$\begin{aligned} E \left[\int_a^b X(s) ds \int_a^b X(t) dt \right] &= \int_a^b \int_a^b E[X(s)X(t)] ds dt \\ &= \int_a^b \int_a^b R_X(s-t) ds dt = \int_a^b \int_a^b R_X(\tau) ds dt \end{aligned} \quad (4.5)$$

where

$$\tau = s - t \quad (4.6)$$

Example 4.1

Suppose $X(t)$ is a homogeneous continuous process with independent increments and with zero mean. If

$$|X(s) - X(t)| \sim N(0, \sigma^2 |s - t|) \quad (4.7)$$

where,

$$\sigma > 0 \quad (4.8)$$

then $X(t)$ is referred to as a *Wiener process* (also known as *Brownian motion process*; Norbert Wiener, 1894–1964; Robert Brown, 1773–1858). An instance of a Brownian motion process is illustrated in Figure 4.1.

For an independent increment random process $X(t)$, $t \geq 0$, if the probability distribution of its increment $\Delta X(t_1, t_2)$, $\Delta X(t_2, t_3)$, ..., $\Delta X(t_{n-1}, t_n)$ depends only on



FIGURE 4.1 Brownian motion.

the time difference $t_2 - t_1$, $t_3 - t_2$, $t_n - t_{n-1}$, then $X(t)$ is said to have a stationary independent increment (a Wiener process is a stationary independent increment). Consider now a case in which a Wiener process is a mean-square integrable.

By definition of the Wiener process,

$$E[X(t)] = 0 \quad (4.9)$$

Furthermore, the following is also true:

$$\sigma_X(s, t) = E[X(s)X(t)] = \begin{cases} \sigma^2 s, & s < t \\ \sigma^2 t, & t < s \end{cases}$$

or

$$\sigma_X(s, t) = E[X(s)X(t)] = \sigma^2 \min(s, t) = R_X(s, t) \quad (4.10)$$

Equations 4.9 and 4.10 can be realized as follows: when $t \geq \tau$,

$$E[X(s)X(t)] = E[X(s) \{X(t) - X(s) + X(s)\}] = E[X(s) \{X(t) - X(s)\}] + E[X(s)X(s)] = 0 + D[X(s)] = \sigma^2 s.$$

In interval $[0, u]$, it is seen that

$$\int_0^u \int_0^u E[X(s)X(t)] ds dt = \int_0^u \int_0^u R_X(s, t) dt ds = \int_0^u \int_0^u \sigma^2 \min(s, t) dt ds$$

For integral $\int_0^u \sigma^2 \min(s, t) dt$, there exists two possibilities: if $t < s$, then the limit of t is from 0 to s and we have $\int_0^s t dt$; if $t > s$, then the limit of t is from s to u , we have $\int_s^u s dt$. Therefore, $\int_0^u \int_0^u \sigma^2 \min(s, t) ds dt = \sigma^2 \int_0^u \left[\int_0^s t dt + \int_s^u s dt \right] ds = \frac{\sigma^2}{3} u^3$.

We can see that $\int_0^u X(t) dt$ exists.

Example 4.2

Using $Y(u)$ to denote the integral given by the above example, find its mean, auto-correlation function, and variance, where

$$Y(u) = \int_a^b X(t) dt = \int_0^u X(t) dt$$

For mean,

$$E[Y(u)] = \int_0^u E[X(t)] dt = 0$$

For autocorrelation function, assume the case in which a timepoint v exists such that

$$0 < v \leq u$$

This case is illustrated in Figure 4.2a, wherein the lower domain, denoted by D_l , the time point s can be either shorter or longer than t ; and in the upper domain, denoted by D_u , s is always smaller than t . Therefore, the correlation function can be written as

$$\begin{aligned} R_Y(u, v) &= \int_0^u \int_0^v E[X(s)X(t)] ds dt = \int_0^u \int_0^v \sigma^2 \min(s, t) ds dt \\ &= \iint_{D_l} \sigma^2 \min(s, t) ds dt + \iint_{D_u} \sigma^2 s ds dt = \underbrace{\int_0^v ds \int_0^s \sigma^2 t dt + \int_0^v dt \int_0^t \sigma^2 s ds}_{D_l} \\ &\quad + \underbrace{\int_0^v \sigma^2 s ds \int_v^u dt}_{D_u} \\ &= 2 \int_0^v \frac{\sigma^2 s^2}{2} ds + \int_0^v \sigma^2 (u - v) s ds = \frac{\sigma^2 s^2}{6} (3u - v) \end{aligned}$$

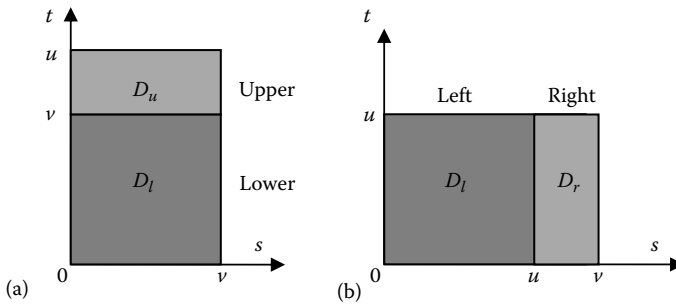


FIGURE 4.2 Integral domains. (a) Lower-upper. (b) Left-right.

Similarly, when, $0 < u \leq v$, see Figure 4.2b with domains D_l and D_r , we can write

$$R_Y(u, v) = \frac{\sigma^2 u^2}{6} (3v - u)$$

Additionally, it can be shown that

$$D[Y(u)] = R_Y(u, u) = \frac{\sigma^2 u^3}{3}$$

4.1.1.2 Stationary Process: A Review

In Chapter 3, we introduced a special group of random processes, the stationary process. Although it is not necessary for a random process to be stationary to analyze it in the frequency domain, stationary processes, especially stationary random vibrations, are more conveniently studied in the frequency domain than in the time domain.

In Chapter 3, we also roughly divided stationary processes into strictly stationary and weakly stationary processes. Recall that a strictly stationary process is defined by its finite dimensional distributions, whereas a weakly stationary process is characterized by its mean, variance, and autocorrelation functions, which are considered to be the first and second moments. In fact, we may have more detailed classifications by considering their n -order of moments. In Figure 4.3, a block diagram is used

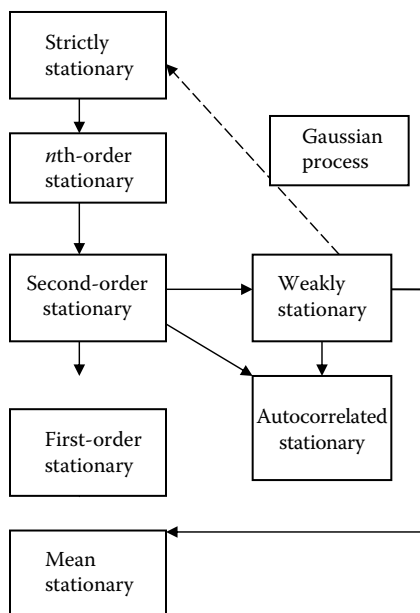


FIGURE 4.3 Stationary processes.

to show the relationships between strictly stationary processes and various weakly stationary processes.

It is noted that the stationary Gaussian process is both strictly and weakly stationary. In Figure 4.3, the second-order stationary process and specially the weakly stationary process are of great importance. Because in this situation, the autocorrelation functions are only a function of τ , the time difference $t_2 - t_1$. Namely, $R_X(t_1, t_2) = R_X(t_2 - t_1) = R_X(\tau)$. It will be shown that the Fourier transform of such correlation functions will have deterministic spectra, although the processes themselves are random.

Additionally, to operate random vibration, we need to consider both the input and output processes, namely, the excitation process $X(t)$ and the response process $Y(t)$. For a linear time-invariant system, if the excitation is stationary, then the response will also be stationary. In this circumstance, the cross-correlation functions will also be only functions of τ , the time lag, namely, $R_{XY}(t_1, t_2) = R_{XY}(t_2 - t_1) = R_{XY}(\tau)$. Therefore, we will see that the Fourier transform of the cross-correlation functions also have deterministic spectra.

Additionally, with the help of the Fourier transforms of these correlation functions, we can further obtain the transfer functions, which is one of the most fundamental concepts of vibrational systems. Practically speaking, the transfer functions obtained through the Fourier transforms of correlation functions will be notably more accurate than measurements through the direct definition of transfer functions.

4.1.1.3 Autospectral Density Functions

Suppose a stationary random process is mean-square integrable, then consider its spectral density functions, the function in the frequency domain. In the literature, this is more rigorously referred to as the power spectral density (PSD) function.

4.1.1.3.1 Further Discussion on Sum of Individual Harmonic Process

First, consider the sum of the harmonic process, the *Fourier series*, which will give us some insight into the above-mentioned Fourier transform (Jean B.J. Fourier, 1768–1830). Recall Equation 3.62:

$$X(t) = \sum_{k=1}^m X_k(t) = \sum_{k=1}^m (A_k \cos \omega_k t + B_k \sin \omega_k t) \quad (4.11)$$

Repeating Equation 3.64, we have

$$R_X(\tau) = \sum_{k=1}^m R_{X_k}(\tau) = \frac{1}{2\pi} \sum_{k=1}^m \sigma_k^2 \cos \omega_k \tau \quad (4.12)$$

When $m \rightarrow \infty$ (recall Equation 3.71),

$$R_X(\tau) = \sigma^2 \lim_{\Delta\omega \rightarrow 0} \left[\sum_{k=1}^m \frac{1}{2\pi} g(\omega_k) \Delta\omega \cos \omega_k \tau \right] = \frac{\sigma^2}{2\pi} \int_0^\infty g(\omega) \cos \omega \tau d\omega \quad (4.13)$$

That is, $R_X(\tau)$ and $\sigma^2 g(\omega)$ are a Fourier pair, denoted by (recall Equation 3.72)

$$R_X(\tau) \Leftrightarrow \sigma^2 g(\omega) \quad (4.14)$$

The relationship between $R_X(\tau)$ and $\sigma^2 g(\omega)$ can be extended to general cases. This is one of the fundamental approaches in dealing with random processes.

Example 4.3

Given the following density function $\sigma^2 g(\omega)$ taken from a stationary process

$$1. \quad \sigma^2 g(\omega) = \sum_{p=1}^n \frac{a_p}{\omega^2 + b_p^2}, \quad b_p > 0, \quad p = 0, 1, 2, \dots, n$$

and

$$2. \quad \sigma^2 g(\omega) = \begin{cases} a^2, & \omega_1 \leq |\omega| \leq 2\omega_1 \\ 0, & \text{elsewhere} \end{cases}$$

find the corresponding autocorrelation functions $R_X(\tau)$.

1. It is known that the Fourier pair exists

$$\frac{a_p}{\omega^2 + b_p^2} \Leftrightarrow \frac{a_p}{2b_p} e^{-b_p|\tau|}$$

Therefore,

$$R_X(\tau) = \sum_{p=1}^n \frac{a_p}{2b_p} e^{-b_p|\tau|}$$

2. Let

$$\sigma^2 g_1(\omega) = \begin{cases} a^2, & |\omega| \leq 2\omega_1 \\ 0, & \text{elsewhere} \end{cases}$$

and

$$\sigma^2 g_2(\omega) = \begin{cases} a^2, & |\omega| < \omega_1 \\ 0, & \text{elsewhere} \end{cases}$$

we have

$$\sigma^2 g(\omega) = \sigma^2 g_1(\omega) - \sigma^2 g_2(\omega)$$

Furthermore,

$$\begin{aligned} R_1(\tau) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} a^2 e^{j\omega\tau} d\omega = \frac{a^2}{2\pi} \int_{-2\omega_1}^{2\omega_1} (\cos\omega\tau + j\sin\omega\tau) d\omega \\ &= \frac{2\omega_1 a^2}{\pi} \left(\frac{\sin 2\omega_1 \tau}{2\omega_1 \tau} \right) = \frac{a^2 \sin 2\omega_1 \tau}{\pi \tau} \end{aligned}$$

and

$$\begin{aligned} R_2(\tau) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} a^2 e^{j\omega\tau} d\omega = \frac{a^2}{2\pi} \int_{-\omega_1}^{\omega_1} (\cos\omega\tau + j\sin\omega\tau) d\omega \\ &= \frac{\omega_1 a^2}{\pi} \left(\frac{\sin \omega_1 \tau}{\omega_1 \tau} \right) = \frac{a^2 \sin \omega_1 \tau}{\pi \tau} \end{aligned}$$

Therefore, we can obtain

$$\sigma^2 g(\omega) = \frac{a^2 (\sin 2\omega_1 \tau - \sin \omega_1 \tau)}{\pi \tau}$$

4.1.1.3.2 Wiener–Khinchine Relations (Norbert Wiener, 1894–1964; Aleksandr Y. Khinchine, 1894–1959)

We now extend the relationship between the autocorrelation function and the density function into a broader view. In fact, the Fourier pair of the autocorrelation function is called the auto-PSD function.

Formally, the auto-PSD function was defined as

$$S_X(\omega) = \int_{-\infty}^{\infty} R_X(\tau) e^{-j\omega\tau} d\tau \quad (4.15)$$

with the inverse transform as

$$R_X(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_X(\omega) e^{j\omega\tau} d\omega \quad (4.16)$$

The Wiener–Khinchine relation is now written as

$$R_X(\tau) \Leftrightarrow S_X(\omega) \quad (4.17)$$

Note that in Equation 4.15, the auto-PSD function is introduced as a “definition.” In Section 4.4.2, we will compare Equation 4.14 [$R_X(\tau) \Leftrightarrow \sigma^2 g(\omega)$] with Equation 4.17 [$R_X(\tau) \Leftrightarrow S_X(\omega)$]. As described in Chapter 3, the autocorrelation function $R_X(\tau)$ of a stationary random process becomes a deterministic function of time lag τ only. From Equation 4.17, we can further realize that the auto-PSD function is also deterministic.

Example 4.4

A stationary process $\{X(t), -\infty < t < \infty\}$ is zero-mean. It has the PSD function as

$$S_X(\omega) = \frac{6\omega^2}{\omega^4 + 5\omega^2 + 4}$$

Find the correlation function $R_X(\tau)$ and variance $D(t)$

To obtain the correlation function, let

$$S_X(\omega) = \frac{6\omega^2}{\omega^4 + 5\omega^2 + 4} = \frac{A}{\omega^2 + 4} + \frac{B}{\omega^2 + 1}$$

We can obtain

$$6\omega^2 = A\omega^2 + A + B\omega^2 + 4B$$

By comparing the coefficients on both sides of the above equation,

$$A + B = 6$$

and

$$A + 4B = 0$$

Therefore, we have $A = 8$ and $B = -2$, and then

$$S_X(\omega) = \frac{8}{\omega^2 + 4} + \frac{-2}{\omega^2 + 1}$$

Furthermore,

$$\begin{aligned} R_X(\tau) &= \mathcal{F}^{-1}[S_X(\omega)] = \mathcal{F}^{-1}\left[\frac{8}{\omega^2 + 4}\right] + \mathcal{F}^{-1}\left[\frac{-2}{\omega^2 + 1}\right] \\ &= 2e^{-2|\tau|} - e^{-|\tau|} \end{aligned}$$

The variance can be further calculated as

$$D[X(t)] = R(0) - \mu_X^2(t) = 1$$

4.1.1.3.3 Existence of Fourier Transform

To have the Wiener–Khinchine relationship, as stated in Equation 4.17, the Fourier transform must exist. Therefore, the corresponding condition should be considered:

$R_X(\tau)$ must be absolutely integrable, that is,

$$\int_{-\infty}^{\infty} |R_X(\tau)| d\tau < \infty \quad (4.18)$$

For a non-zero mean process, $X'(t)$, when $\tau \rightarrow \infty$, Equation 3.99 will be rewritten with the help of the mean-square limit:

$$\lim_{\tau \rightarrow \infty} R_{X'}(\tau) = \mu_{X'}^2 \quad (4.19)$$

This makes $R_{X'}(\tau)$ not absolutely integrable. Now let

$$X(t) = X'(t) - \mu_{X'} \quad (4.20)$$

so that

$$E(X(t)) = 0 \quad (4.21)$$

In this operation, the Fourier transform of $X(t)$ will not be divergent.

Example 4.5

$\{W(t), t \geq 0\}$ is a Wiener process with parameter σ^2 . Let

$$X(t) = W(t + s) - W(t), s > 0$$

Show that the increment $\{X(t), t \geq 0\}$ is a stationary process and find the autocorrelation function as well as auto-PSD function.

First,

$$\mu_X(t) = E[X(t)] = E[W(t + s) - W(t)] = 0$$

Second,

$$\begin{aligned} R_X(t, t + \tau) &= E[\{W(t + s) - W(t)\} \{W(t + s + \tau) - W(t + \tau)\}] \\ &= R_W(t + s, t + s + \tau) - R_W(t + s, t + \tau) - R_W(t, t + s + \tau) + R_W(t, t + \tau) \\ &= \sigma^2 \min\{t + s, t + s + \tau\} - \sigma^2 \min\{t + s, t + \tau\} - \sigma^2 \min\{t, t + s + \tau\} + \sigma^2 \min\{t, t + \tau\} \\ &= \sigma^2 \min\{s, s + \tau\} - \sigma^2 \min\{s, \tau\} - \sigma^2 \min\{0, s + \tau\} + \sigma^2 \min\{0, \tau\} = R_X(\tau) \end{aligned}$$

Third,

$$E[X^2(t)] = R(t, t) = s\sigma^2 < \infty$$

Therefore, the increment $\{X(t), t \geq 0\}$ of the Wiener process is stationary. It is noted that a Wiener process itself is not stationary. To find the auto-PSD function, consider

$$\begin{aligned} R_X(\tau) &= \sigma^2 \min\{s, s + \tau\} - \sigma^2 \min\{s, \tau\} - \sigma^2 \min\{0, s + \tau\} + \sigma^2 \min\{0, \tau\} \\ &= \begin{cases} 0, & \tau < -s \\ \sigma^2(\tau + s), & -s \leq \tau < 0 \\ \sigma^2(s - \tau), & 0 \leq \tau \leq s \\ 0, & \tau > s \end{cases} = \begin{cases} \sigma^2(s - |\tau|), & |\tau| \leq s \\ 0, & \text{elsewhere} \end{cases} \end{aligned}$$

Then

$$S_X(\omega) = \mathcal{F}[R_X(\tau)] = \frac{1}{2\pi} \int_{|\tau| \leq s} \sigma^2(s - |\tau|) e^{-j\omega\tau} d\tau = \frac{4\sigma^2 \sin^2 \frac{s\omega}{2}}{\omega^2}$$

4.1.1.4 Spectral Distribution Function $\Psi(\omega)$

We now define the direct integral of the auto-PSD function $S_X(\omega)$ as a *spectral distribution function*, to be used in a later section.

$$\Psi(\omega) = \int_{-\infty}^{\omega} S_X(\varpi) d\varpi = \int_{-\infty}^{\omega} R_X(\tau) \frac{e^{-j\omega\tau} - 1}{j\tau} d\tau \quad (4.22)$$

It can be proven that any stationary random process can be regarded as the superposition of mutually uncorrelated harmonic oscillations of various frequencies and with random phases and amplitudes. In this regard, the spectral distribution function $\Psi(\omega)$ unveils a different insight into the random time series because $\Psi(\omega)$ is also deterministic.

Note that in Equation 4.22, the integral limit starts from $-\infty$, which may limit certain existence of $\Psi(\omega)$. However, in engineering applications, the lowest frequency of a PSD function starts from zero. Namely, for engineering applications, $\Psi(\omega)$ always exist.

Example 4.6

A stationary process has auto-PSD function given by the following equation. Find the corresponding spectral distribution function, $\Psi(\omega)$.

$$S_X(\omega) = \begin{cases} S_0 & |\omega_C| < \omega \\ S_0/2 & |\omega_C| = \omega \\ 0 & |\omega_C| > \omega \end{cases}$$

It may be shown that

$$\Psi(\omega) = \int_{-\infty}^{\omega} S_X(\varpi) d\varpi = \int_{-\omega_C}^{\omega} S_0 d\varpi = S_0(\omega + \omega_C)$$

4.1.1.5 Properties of Auto-PSD Functions

The important properties of the auto-PSD function are as follows:

4.1.1.5.1 Symmetry

$$S_X(-\omega) = S_X(\omega) \quad (4.23)$$

Proof:

Because both $R_X(\tau)$ and $\cos(\omega\tau)$ are even functions, in this case, we have

$$S_X(\omega) = 2 \int_0^{\infty} R_X(\tau) \cos(\omega\tau) d\tau \quad (4.24)$$

From Equation 4.24, it is easy to see that Equation 4.23 holds; furthermore,

$$R_X(\tau) = \frac{1}{\pi} \int_0^{\infty} S_X(\omega) \cos(\omega\tau) d\omega \quad (4.25)$$

4.1.1.5.2 Real and Positive Values

Generally speaking, a Fourier transform of a function is complex valued. However, the autopower spectrum function is real, and in addition, positive.

Proof:

Recall Equation 3.35b

$$\sum_{j=1}^n \sum_{k=1}^n \alpha_j \alpha_k R_X(t_j - t_k) \geq 0 \quad (4.26)$$

We thus have

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(s) g(t) R_X(t - s) ds dt \geq 0 \quad (4.27)$$

By denoting

$$q(u) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(s)g(u+t)R_X(u+t-s)ds dt \geq 0 \quad (4.28)$$

we can see that

$$\begin{aligned} Q(\omega) &= \int_{-\infty}^{\infty} q(u)e^{-j\omega u} du \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(s)g(u+t)R_X(u+t-s) ds dt e^{-j\omega u} du \\ &= 2\pi |G(\omega)|^2 S_X(\omega) \geq 0 \end{aligned} \quad (4.29)$$

4.1.1.5.3 Mean-Square Value

A useful expression of mean-square value in terms of integration of the auto-PSD function can be seen as follows:

$$R_X(0) = E[X^2(t)] = \sigma_X^2 = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_X(\omega)e^0 d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_X(\omega) d\omega \quad (4.30a)$$

This can be further simplified as

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} S_X(\omega) d\omega = \sigma_X^2 \quad (4.30b)$$

Evaluating this with Equation 4.22, we have

$$\sigma_X^2 = \frac{1}{2\pi} \Psi(\infty) \quad (4.31)$$

Example 4.7

Check if the following functions are auto-PSD functions. If the function is an auto-PSD function, find the corresponding autocorrelation function and mean-square value.

$$1. \quad S_1(\omega) = \frac{\omega^2 + 9}{(\omega^2 + 4)(\omega + 1)^2}$$

An auto-PSD function must be an even function.

$$S_1(-\omega) = \frac{\omega^2 + 9}{(\omega^2 + 4)(-\omega + 1)^2} \neq S_1(\omega)$$

$S_1(\omega)$ is not an auto-PSD function.

$$2. \quad S_2(\omega) = \frac{\omega^2 + 4}{\omega^4 - 10\omega^2 + 3}$$

An auto-PSD function must be greater than zero.

$$S_2(1) = -0.8333$$

Therefore, $S_2(\omega)$ is not an auto-PSD function.

$$3. \quad S_3(\omega) = \frac{e^{-j\omega^2}}{\omega^2 + 6}$$

An auto-PSD function must be real valued. S_3 is complex valued, therefore, $S_3(\omega)$ is not an auto-PSD function.

$$4. \quad S_4(\omega) = \frac{\omega^2 + 1}{\omega^4 + 5\omega^2 + 8}$$

It is seen that $S_4(\omega)$ is an auto-PSD function.

$$\begin{aligned} S_4(\omega) \equiv S_X(\omega) &= \frac{\omega^2 + 1}{\omega^4 + 5\omega^2 + 8} = \frac{-1}{\omega^2 + 2} + \frac{2}{\omega^2 + 3} \\ &= -\frac{1}{2\sqrt{2}} \frac{2\sqrt{2}}{\omega^2 + (\sqrt{2})^2} + \frac{1}{\sqrt{3}} \frac{2\sqrt{3}}{\omega^2 + (\sqrt{3})^2} \end{aligned}$$

The autocorrelation function is

$$\begin{aligned} R_X(\tau) &= \mathcal{F}^{-1}[S_X(\omega)] = \mathcal{F}^{-1} \left[-\frac{1}{2\sqrt{2}} \frac{2\sqrt{2}}{\omega^2 + (\sqrt{2})^2} \right] + \mathcal{F}^{-1} \left[\frac{1}{\sqrt{3}} \frac{2\sqrt{3}}{\omega^2 + (\sqrt{3})^2} \right] \\ &= -\frac{1}{2\sqrt{2}} e^{-\sqrt{2}|\tau|} + \frac{1}{\sqrt{3}} e^{-\sqrt{3}|\tau|} \end{aligned}$$

The mean square is given by

$$E\left[|X(t)|^2\right] = -\frac{1}{2\sqrt{2}} + \frac{1}{\sqrt{3}}$$

4.1.2 RELATIONSHIP WITH FOURIER TRANSFORM

4.1.2.1 Fourier Transformation Random Process

Fourier transform is an important tool in engineering applications for analyzing time-varying processes in the frequency domain. One reasonable question is, can a random process have Fourier transform as given by the following equations?

$$X(\omega) = \int_{-\infty}^{\infty} X(t)e^{-j\omega t} dt \text{ and } X(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega)e^{j\omega t} d\omega$$

Generally speaking, the answer is negative. The primary reason is that the absolute integrable requirement for Fourier transform to exist, namely,

$$\int_{-\infty}^{\infty} |X(t)| dt < \infty \quad (4.32)$$

often does not satisfy random processes. To deal with this problem, we introduced the power spectrum instead. In the following, let us discuss this issue in detail.

4.1.2.2 Energy Equation

First, consider the amount of energy contained in a dynamic process $X(t)$.

4.1.2.2.1 Parseval Equation

The following Parseval equation implies energy conservation:

$$\int_{-\infty}^{\infty} X^2(t) dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega \quad (4.33)$$

The left-hand side is the total energy in $(-\infty, \infty)$, which is the time domain. Remember that $X(t)$ is a random process in the time domain.

The Parseval equation is important in signal analyses. Nevertheless, for random processes, there may be two problems. The primary issue is that in the domain $(-\infty, \infty)$, the energy can become infinite so that the energy integration does not exist.

4.1.2.2.2 Average Power

Because Equation 4.32 does not satisfy many engineering temporal functions by having an infinite amount of energy in $(0, T)$ when $T \rightarrow \infty$, instead of energy, the concept of average power will be used:

$$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T X(t)^2 dt$$

This is why the PSD function is used (this approach is described in the following section). The power spectrum is used because when “defining” the Fourier transformation of a random process $X(t)$ as

$$X(\omega) = \int_{-\infty}^{\infty} X(t)e^{-j\omega t} dt \quad (4.34)$$

$X(t)$ continues forever, and Equation 4.18 will not be satisfied, thus the spectrum $X(\omega)$, as described in Equation 4.34, does not exist. As a result, we need an alternative approach to consider the average power spectrum.

4.1.2.2.3 Finite Fourier Transform and Corresponding Complex Valued Process

We try to solve the above-mentioned problem as follows:

First, define a special function $X_T(t)$ as

$$X_T(t) = \begin{cases} X(t), & |t| \leq T \\ 0, & |t| > T \end{cases} \quad (4.35)$$

As seen above, the Fourier transform exists as

$$X(\omega, T) = \int_{-\infty}^{\infty} X_T(t) e^{-j\omega t} dt = \int_0^T X(t) e^{-j\omega t} dt \quad (4.36)$$

Let us denote a function $Y(\omega)$ to represent a case with a limited time duration T ,

$$Y(\omega) = X(\omega, T) \quad (4.37)$$

By focusing on Y , we can calculate the mean, variance as follows

4.1.2.2.3.1 Mean By visual inspection,

$$\mu_Y(\omega) = 0 \quad (4.38)$$

4.1.2.2.3.2 Variance Determining the variance, we have

$$\begin{aligned} \sigma_Y^2(\omega) &= E[Y^2(\omega)] = E[X^2(\omega, T)] = E[X(\omega, T)X^*(\omega, T)] \\ &= E \left[\int_0^T X(t) e^{-j\omega t} dt \int_0^T X(s) e^{j\omega s} ds \right] = E \left[\int_0^T \int_0^T X(t) X(s) e^{-j\omega(t-s)} ds dt \right] \end{aligned} \quad (4.39)$$

To further evaluate $\sigma_Y^2(\omega)$, change the variables in Equation 4.39 by letting (see Figure 4.4)

$$\tau = t - s \quad (4.40)$$

Note that the autocorrelation function $R_X(\tau)$ is even. Exchanging the order of mathematical expectation and integration, we have

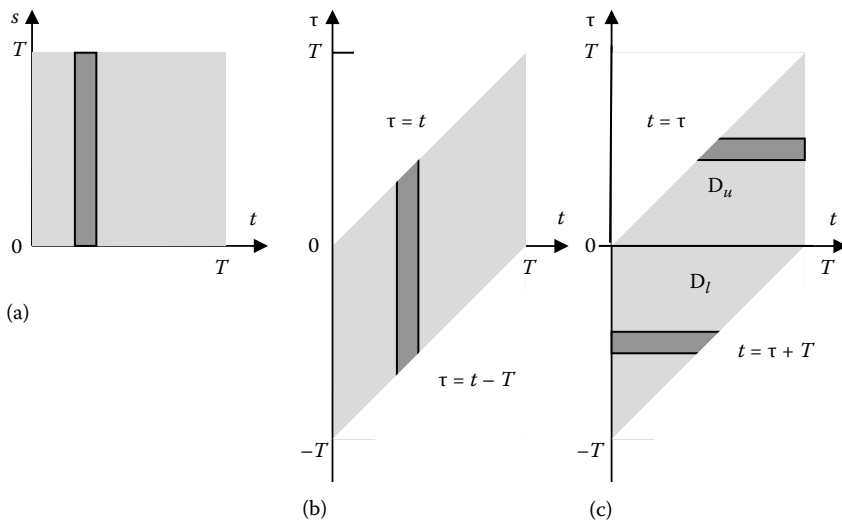


FIGURE 4.4 Integration domains of variance function. (a) Original coordinates (t, s) . (b) Transformed coordinates (t, τ) . (c) In (t, τ) integrating first on t .

$$\begin{aligned}
 \sigma_Y^2(\omega) &= E \left[\int_0^T \int_{t-T}^t X(t)X(t-\tau)e^{-j\omega\tau} d\tau dt \right] = \int_0^T \int_{t-T}^t E[X(t)X(t-\tau)]e^{-j\omega\tau} d\tau dt \\
 &= \int_0^T \int_{t-T}^t R_X(\tau)e^{-j\omega\tau} d\tau dt \quad (4.41)
 \end{aligned}$$

In Equation 4.41, it is seen that when $s = 0$, $\tau = t$, and when $s = T$, $\tau = t - T$. Additionally, because $ds = -d\tau$, the limits of the section integral should be exchanged, namely, from $t - T$ to t (see Figure 4.4b).

To specifically evaluate integral on τ first in Equation 4.41 is difficult because $R_X(\tau)$ is a generic term. However, in Equation 4.41, the integrand is independent from time t , yet the integration limits of τ depend on t . Thus, we change the order of the double integration, switching first on t and then on τ . In Figure 4.4c, for the integral domain D_l , $\tau \in (-T, 0)$, the interval of integration of t is 0 to $\tau + T$; and for domain D_u , $\tau \in (0, T)$, the interval is τ to T ; explicitly, we can write

$$\begin{aligned}
 \sigma_Y^2(\omega) &= \underbrace{\int_{-T}^0 R_X(\tau)e^{-j\omega\tau} \left[\int_0^{\tau+T} dt \right] d\tau}_{D_l} + \underbrace{\int_0^T R_X(\tau)e^{-j\omega\tau} \left[\int_{\tau}^T dt \right] d\tau}_{D_u} \\
 &= \int_{-T}^0 R_X(\tau)e^{-j\omega\tau}(\tau+T)d\tau + \int_0^T R_X(\tau)e^{-j\omega\tau}(T-\tau)d\tau = \int_{-T}^T R_X(\tau)e^{-j\omega\tau}(T-|\tau|)d\tau \quad (4.42)
 \end{aligned}$$

In the above expression of the Wiener–Khinchine relations, the autopower density function is defined. Also, it suggests that the variance of function $Y(\omega)$ does relate to the autocorrelation function of $X(t)$. In the following, a more detailed fusion process is examined.

4.1.2.3 Power Density Functions

To mathematically have Fourier transform, because the integration interval is $(-\infty, \infty)$, we need to further let $T \rightarrow \infty$ in Equation 4.42. This attempt, however, is not always doable because in Equation 4.37, beyond time T , the process $\{X(t), t > T\}$ is not defined. Because $X(t)$ is random, when $t > T$, we may have unexpected circumstances. In the following, however, we will limit the topic for those processes with doable $T \rightarrow \infty$, and in Section 4.4, we will provide a more rigorous expression of the required conditions. Now, in Equation 4.42, when $T \rightarrow \infty$,

$$\sigma_Y^2(\omega) \Big|_{T \rightarrow \infty} \rightarrow \infty$$

To force the integrand to be finite, we must divide by T on both sides of Equation 4.42, resulting in

$$\frac{1}{T} \sigma_Y^2(\omega) = \int_{-T}^T R_X(\tau) e^{-j\omega\tau} \left(1 - \frac{|\tau|}{T}\right) d\tau \quad (4.43)$$

The operation of dividing by T indicates that the resulting term is power, instead of energy. That is why, in the literature, this function is often called the “power” spectral density function.

Note that τ is the time difference between s and t , and both are elements inside $(0, T)$. Thus, when $T \rightarrow \infty$, it results in

$$\left(1 - \frac{|\tau|}{T}\right) \Big|_{T \rightarrow \infty} = 1 \quad (4.44)$$

Therefore,

$$\begin{aligned} \lim_{T \rightarrow \infty} \left[\frac{1}{T} \sigma_Y^2(\omega) \right] &= \lim_{T \rightarrow \infty} \left[\int_{-T}^T R_X(\tau) e^{-j\omega\tau} d\tau \right] \\ &= \int_{-\infty}^{\infty} R_X(\tau) e^{-j\omega\tau} d\tau = S_X(\omega) \end{aligned} \quad (4.45)$$

Note that Equation 4.45 is derived from the equation $\sigma_Y^2(\omega) = E[X(\omega, T)X^*(\omega, T)]$. Thus, we can have a very useful formula to obtain the auto-PSD:

$$S_X(\omega) = \lim_{T \rightarrow \infty} \frac{1}{T} E \left[|X(\omega, T)|^2 \right] \quad (4.46)$$

Practically, this can be written as though the following average for each Fourier transform, say, the k th, denoted by $X_k(\omega, T)$ was taken from a sample realization of $X_T(t)$:

$$\hat{S}_X(\omega, T, n) = \frac{1}{T} \frac{1}{n} \sum_{k=1}^n \left[|X_k(\omega, T)|^2 \right] \quad (4.47)$$

4.1.2.3.1 Parseval Equation, Further Discussion (Marc-Antoine Parseval, 1755–1836)

With the gain of the specially defined function $X_T(t)$ in Equation 4.35, the Parseval equation can now be written as

$$\int_{-\infty}^{\infty} X_T^2(t) dt = \int_{-T}^T X^2(t) dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |Y(\omega)|^2 d\omega \quad (4.48)$$

4.1.2.3.1.1 Average Power Spectrum To have the power function, we must first divide by $2T$ on both sides of Equation 4.49, then take the limit as

$$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T X^2(t) dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} \lim_{T \rightarrow \infty} \frac{1}{2T} |Y(\omega)|^2 d\omega \quad (4.49)$$

Additionally, by taking the expected value, it results in

$$\lim_{T \rightarrow \infty} E \left[\frac{1}{2T} \int_{-T}^T X^2(t) dt \right] = \frac{1}{2\pi} \lim_{T \rightarrow \infty} E \left[\int_{-\infty}^{\infty} \frac{1}{2T} |Y(\omega)|^2 d\omega \right] \quad (4.50)$$

Because the mean of the stationary process is zero, we are able to write

$$\lim_{T \rightarrow \infty} E \left[\frac{1}{2T} \int_{-T}^T X^2(t) dt \right] = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-\infty}^{\infty} EX^2(t) dt = R_X(0) = EX^2(t) \quad (4.51)$$

Thus, the average power is the value of autocorrelation function at $\tau = 0$. Additionally, from Equation 4.50, the following can be concluded.

4.1.2.3.1.2 Mean Square (Further Discussion)

$$R_X(0) = \lim_{T \rightarrow \infty} \int_{-\infty}^{\infty} \frac{1}{4\pi T} E \left[|X(\omega, T)|^2 \right] d\omega \quad (4.52)$$

From Equation 4.50, the integrand on the right-hand side of Equation 4.51 is $S_X(\omega)$; therefore,

$$R_X(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_X(\omega) d\omega \quad (4.53)$$

More detailed issues on practical applications will be discussed in Chapter 9.

4.1.3 WHITE NOISE AND BAND-PASS FILTERED SPECTRA

Having introduced the general definition of $S_X(\omega)$, in this section, additional specific PSD functions will be introduced.

4.1.3.1 White Noise

First, consider the white noise process as shown in Figure 4.5.

The auto-PSD function is

$$S_X(\omega) = S_0 \quad (4.54)$$

Thus, the autocorrelation function is given by

$$R_X(\tau) = S_0 \delta(\tau) \quad (4.55)$$

Example 4.8

A white noise sequence $\{X_n, n = 0, \pm 1, \pm 2, \dots\}$ has autocorrelation function given by

$$R_X(n) = \begin{cases} \sigma^2, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

Find the autopower spectral power density function

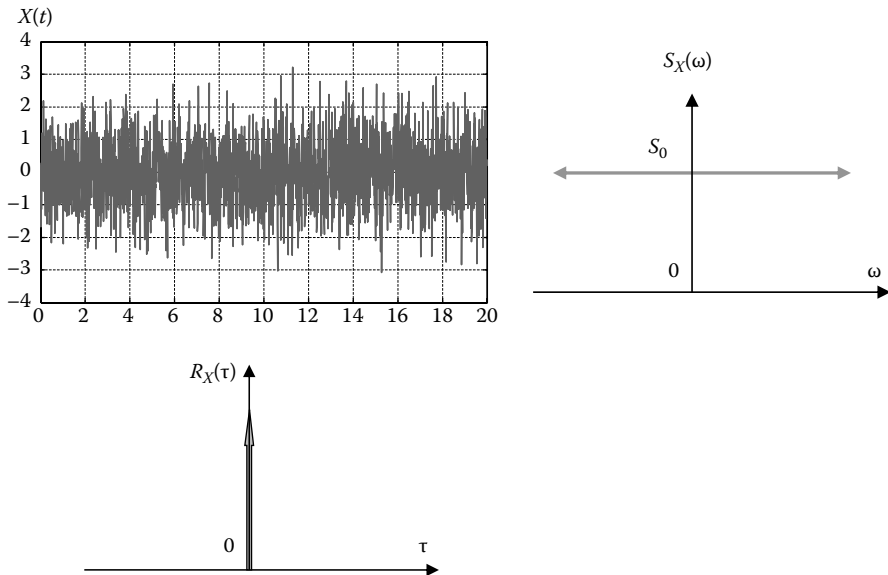


FIGURE 4.5 White noise.

Because X_n is a white noise sequence, we have

$$E(X_n) = 0$$

$$D(X_n) = \sigma^2 < \infty$$

Therefore,

$$S_X(\omega) = \sum_{n=-\infty}^{\infty} R_X(n) e^{-jn\omega} = R_X(0)e^0 = \sigma^2$$

In the above example, the autocorrelation function $R_X(n)$ of discrete time sequence $X(n)$ can be realized with the help of Equation 4.55. It is seen that only if $\tau = 0$, which means that $n = 0$, $\delta(\tau) \neq 0$ so that the variance is not zero. Generally speaking, Equation 4.55 implies that a white noise $X(t)$, also referred to as a *white process*, possesses the following properties:

1. Zero mean

$$E[X(t)] = 0 \quad (4.56a)$$

2. Orthogonality of $X(t_1)$ and $X(t_2)$ at different time points, that is,

$$E[X(t_1) X(t_2)] = 0, \quad t_1 \neq t_2 \quad (4.56b)$$

Because zero mean, $X(t_1)$ and $X(t_2)$ must be uncorrelated.

Note that white noise is not necessarily Gaussian, as long as it has the auto-PSD as shown in Equation 4.54. However, if it is Gaussian, then $X(t_1)$ and $X(t_2)$ must also be independent.

As seen from Equation 4.55, the average power of white noise is infinity; therefore, it cannot be realized in realistic practical observations. However, in engineering measurements, we can observe the output of measuring equipment with a white noise input because the output is a filtered (results of convolution) signal. In the following, we will discuss mathematical models of these filtered signals.

Example 4.9

Consider the spectral distribution function $\Psi(\omega)$ of the white noise process.

Recall Equation 4.22. We have

$$\Psi(\omega) = \int_{-\infty}^{\omega} S_X(\varpi) d\varpi = \int_{-\infty}^{\omega} S_0 d\varpi = S_0 \varpi \Big|_{-\infty}^{\omega} = S_0(\omega + \infty) = \infty$$

It is shown that, for white noise, the spectral distribution function does not exist.

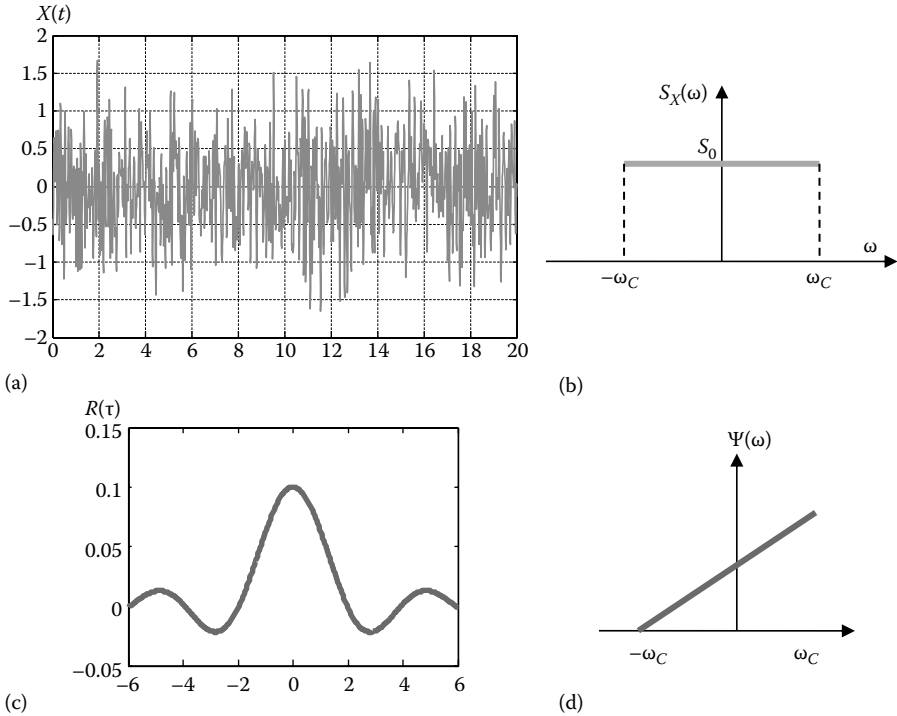


FIGURE 4.6 Low pass noise. (a) Random process, abscissa : time. (b) Auto-PSD, abscissa : frequency. (c) Autocorrelation abscissa : time lag τ . (d) Spectral distribution function abscissa : frequency.

4.1.3.2 Low-Pass Noise

When the white noise is low-pass filtered (as seen in Figure 4.6), we have the auto-PSD function given by

$$S_X(\omega) = \begin{cases} S_0 & |\omega| < \omega_C \\ S_0/2 & |\omega| = \omega_C \\ 0 & |\omega| > \omega_C \end{cases} \quad (4.57)$$

Additionally, the autocorrelation function is

$$R_X(\tau) = \sigma_X^2 \frac{\sin(\omega_C \tau)}{\omega_C \tau} \quad (4.58)$$

Specifically, the variance is

$$\sigma_X^2 = 2\omega_C S_0 \quad (4.59)$$

Recall Equation 4.22, and from the following example, we can see that the spectral distribution function is

$$\Psi(\omega) = S_0(\omega + \omega_c)$$

4.1.3.3 Band-Pass Noise

When the white noise is band-pass filtered (as seen in Figure 4.7), we have the auto-PSD function given by

$$S_X(\omega) = \begin{cases} S_0 & \omega_L < |\omega| < \omega_U \\ S_0/2 & |\omega| = \omega_L, \omega_U \\ 0 & \text{elsewhere} \end{cases} \quad (4.60)$$

The autocorrelation function is

$$R_X(\tau) = \sigma_X^2 \frac{\sin(\Delta\omega\tau/2)}{\Delta\omega\tau/2} \cos \omega_0\tau \quad (4.61)$$

In this case:

$$\Delta\omega = \omega_U - \omega_L \quad (4.62)$$

$$\sigma_X^2 = 2 \Delta\omega S_0 \quad (4.63)$$

$$\omega_0 = (\omega_U + \omega_L)/2 \quad (4.64)$$

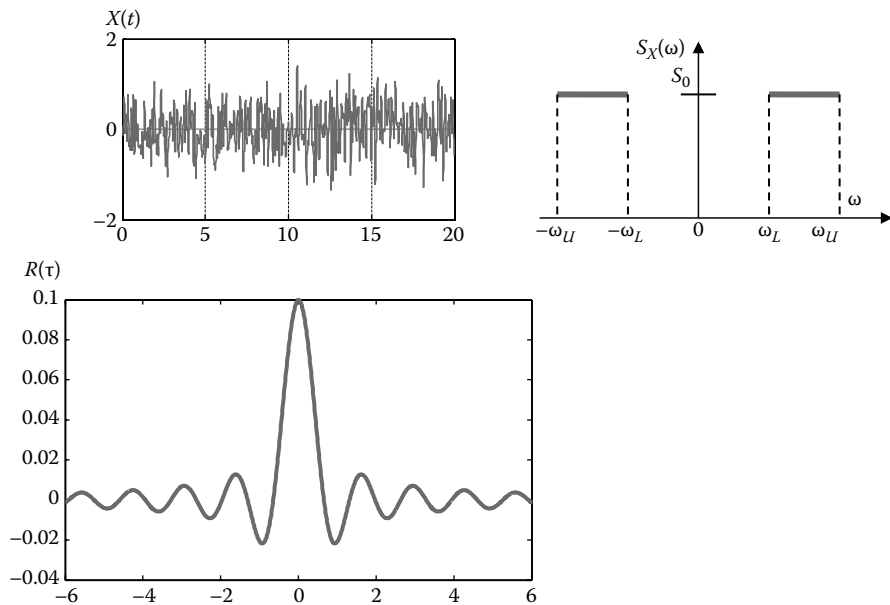


FIGURE 4.7 Band-pass noise.

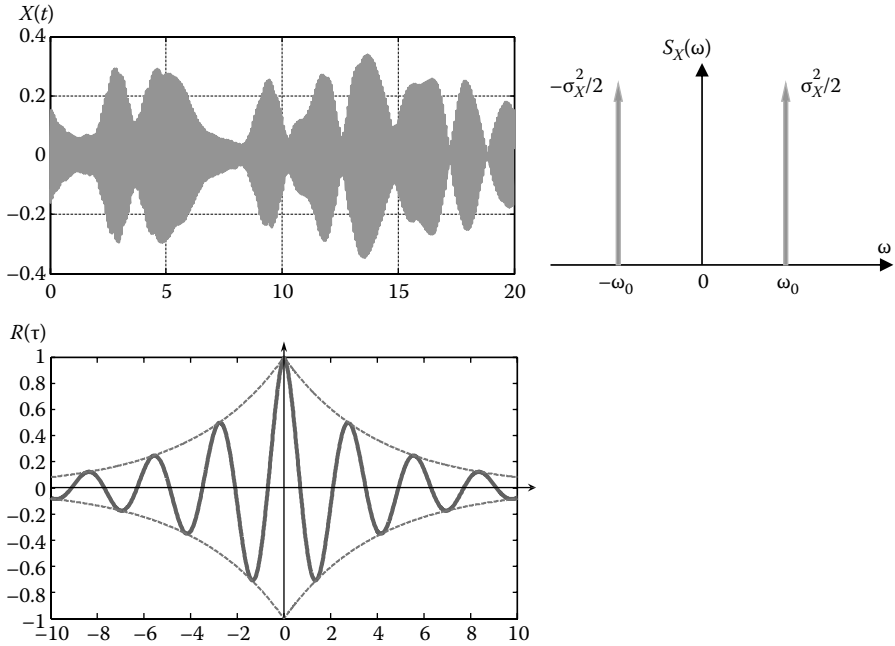


FIGURE 4.8 Narrow band noise.

4.1.3.4 Narrow-Band Noise

When the white noise is narrow-band filtered (as shown in Figure 4.8), we can obtain the following auto-PSD function as

$$S_X(\omega) = \pi \sigma_X^2 [\delta(\omega + \omega_0) + \delta(\omega - \omega_0)] \quad (4.65)$$

Additionally, the autocorrelation function is

$$R_X(\tau) = \sigma_X^2 \cos(\omega_0 \tau) \quad (4.66)$$

4.2 SPECTRAL ANALYSIS

In the second section of this chapter, the auto-PSD function and cross-PSD function, which are also related to the Wiener–Khinchine formula, are discussed. The focus is on the spectral analysis of vibration systems.

4.2.1 DEFINITION

4.2.1.1 Cross-Power Spectral Density Function

4.2.1.1.1 Wiener–Khinchine Relation

Defining the cross-power spectral density function $S_{XY}(\omega)$ through the Wiener–Khinchine relations:

$$S_{XY}(\omega) = \int_{-\infty}^{\infty} R_{XY}(\tau) e^{-j\omega\tau} d\tau \quad (4.67)$$

Similarly,

$$S_{YX}(\omega) = \int_{-\infty}^{\infty} R_{YX}(\tau) e^{-j\omega\tau} d\tau \quad (4.68)$$

Additionally,

$$R_{XY}(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XY}(\omega) e^{j\omega\tau} d\omega \quad (4.69)$$

$$R_{YX}(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{YX}(\omega) e^{j\omega\tau} d\omega \quad (4.70)$$

4.2.1.1.2 Properties of Cross-Power Spectral Density Function

Having defined the cross-PSD functions, select properties can now be examined.

4.2.1.1.3 Symmetry

4.2.1.1.3.1 Skew Symmetry Unlike the auto-PSD functions, the cross-PSD functions are not symmetric. However, cross spectral density functions do have the following relationship referred to as skew symmetry:

$$S_{XY}(\omega) = S_{YX}(-\omega) \quad (4.71)$$

and

$$S_{YX}(\omega) = S_{XY}(-\omega) \quad (4.72)$$

4.2.1.1.3.2 Hermitian Symmetry Cross-PSD functions exhibit Hermitian symmetry, which is given by

$$S_{YX}(-\omega) = S_{YX}^*(\omega) \quad (4.73)$$

In this instance, $S_{YX}^*(\omega)$ is the complex conjugate of $S_{YX}(\omega)$, and so on.

$$S_{YX}(-\omega) = S_{YX}^*(\omega) \quad (4.74)$$

4.2.1.1.3.3 Real and Imaginary Portions Furthermore, we have

$$\text{Re}[S_{XY}(\omega)] = \text{Re}[S_{YX}(\omega)] \quad (4.75)$$

and

$$\text{Re}[S_{YX}(\omega)] = \text{Re}[S_{YX}(\omega)] \quad (4.76)$$

also

$$\text{Im}[S_{XY}(\omega)] = -\text{Im}[S_{XY}(-\omega)] \quad (4.77)$$

Furthermore,

$$\text{Im}[S_{YX}(\omega)] = -\text{Im}[S_{YX}(-\omega)] \quad (4.78)$$

4.2.1.1.3.4 Bounds The cross-PSD function has bounds described by

$$[S_{XY}(\omega)]^2 \leq S_X(\omega)S_Y(\omega) \quad (4.79)$$

4.2.1.2 Estimation of Cross-PSD Function

Similar to Equation 4.47, a practical approach on cross-PSD functions is needed; it can be proven by

$$S_{XY}(\omega) = \lim_{T \rightarrow \infty} \frac{1}{T} E[X^*(\omega, T)Y(\omega, T)] \quad (4.80)$$

and

$$S_{YX}(\omega) = \lim_{T \rightarrow \infty} \frac{1}{T} E[X(\omega, T)Y^*(\omega, T)] \quad (4.81)$$

Similar to Equation 4.48, when the Fourier transforms of $X(t)$ and $Y(t)$ are obtained through the practicality of the k th measurement, namely, $X_k(\omega, T)$, $Y_k(\omega, T)$, the estimated cross-correlation function can be written as

$$\hat{S}_{XY}(\omega, T, n) = \frac{1}{T} \frac{1}{n} \sum_{k=1}^n [X_k^*(\omega, T)Y_k(\omega, T)] \quad (4.82)$$

and

$$\hat{S}_{YX}(\omega, T, n) = \frac{1}{T} \frac{1}{n} \sum_{k=1}^n [Y_k^*(\omega, T)X_k(\omega, T)] \quad (4.83)$$

Equations 4.47, 4.82, and 4.83 enable us to obtain the cross-PSD functions practically, which will be discussed further in Chapters 7 and 9.

4.2.2 TRANSFER FUNCTION

The transfer function is an important concept in linear systems, given that it completely describes the dynamic behavior of the system. In this section, two basic issues are considered. The first is, given the transfer function and input random excitations, to find the statistical properties of the random output. The second is, by measuring both the random input and output, to find the transfer function of a linear system.

4.2.2.1 Random Process through Linear Systems

Let us consider the first question, the nature of the random output. Generally, Figure 4.9 shows the relationship of the input-system-output. From Figure 4.9, the output can be seen as the result of the input being transferred through mapping $T[\cdot]$, that is,

$$Y(t) = T[X(t)] \quad (4.84)$$

The mapping of $T[\cdot]$ can be analyzed in both the time and the frequency domain. Let us consider the operation in the time domain first.

4.2.2.1.1 Linearity (Linear Superposition)

The transfer function exists only if the system is linear. Linearity can be explained as follows:

Suppose there exists a random process $X(t)$, $Y(t)$, and $Z(t)$, with constants a and b . In the event that the system is linear, then the following properties must also hold true:

1. Foldable

$$X(t) + Y(t) + Z(t) = [X(t) + Y(t)] + Z(t) = X(t) + [Y(t) + Z(t)] \quad (4.85)$$

2. Interchangeable

$$X(t) + Y(t) = Y(t) + X(t) \quad (4.86)$$

3. Superposition

$$T[aX_1(t) + bX_2(t)] = a T[X_1(t)] + b T[X_2(t)] \quad (4.87)$$

4.2.2.1.2 Input–Output Relationship in the Time Domain

First, consider the input–output relationship in the time domain. In this instance, the output is seen as the result of convolution between the impulse response function

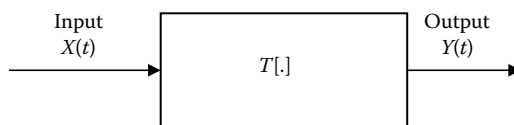


FIGURE 4.9 System and input–output.

and the input forcing function. Thus, based on the convolution integral, the statistical properties (mean values, etc.) of the output as well as correlation between input and output can also be considered. In Section 4.3, the properties describing the frequency domain will be further explored.

4.2.2.1.2.1 Linear Filtering and Convolution When the unit impulse response function, $h(t)$, is known, the random process $X(t)$ being mean square integrable through the corresponding linear time-invariant system can be described by convolution:

$$Y(t) = \int_{-\infty}^{\infty} X(\tau)h(t-\tau) d\tau = X(t) * h(t) \quad (4.88)$$

or

$$Y(t) = \int_{-\infty}^{\infty} X(t-\tau)h(\tau) d\tau = h(t) * X(t) \quad (4.89)$$

Here the symbol $*$ denotes convolution.

The system denoted by $h(t)$ can be seen as a linear filter. Filtering is one of the central concepts that will be discussed further in Chapter 6. Along with filtering convolution, the impulse response function will also be discussed in more detail.

4.2.2.1.2.2 Mean Consider the mean value of the output

$$\mu_Y(t) = E[Y(t)] = E\left[\int_{-\infty}^{\infty} X(t-\tau)h(\tau) d\tau\right] \quad (4.90)$$

If $E[Y(t)]$ is integrable, then,

$$\mu_Y(t) = \int_{-\infty}^{\infty} E[X(t-\tau)h(\tau)] d\tau = \int_{-\infty}^{\infty} h(\tau)E[X(t-\tau)] d\tau \quad (4.91)$$

Because

$$E[X(t-\tau)] = \mu_X(t-\tau) \quad (4.92)$$

we have

$$\mu_Y(t) = \int_{-\infty}^{\infty} h(\tau) \mu_X(t-\tau) d\tau = \mu_X(t) * h(t) \quad (4.93)$$

4.2.2.1.2.3 Autocorrelation Consider the autocorrelation functions $R_X(t, u)$ and $R_Y(t, u)$.

$$R_Y(t, u) = E[Y(t)Y(u)] = E\left[\int_{-\infty}^{\infty} X(t-\zeta)h(\zeta) d\zeta \int_{-\infty}^{\infty} X(u-\xi)h(\xi) d\xi\right] \quad (4.94)$$

For a deterministic linear system, we are able to rewrite Equation 4.94 as

$$\begin{aligned} R_Y(t, u) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} E[X(t - \zeta)X(u - \xi)] h(\zeta) h(\xi) d\zeta d\xi \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} R_X(t - \zeta, u - \xi) h(\zeta) h(\xi) d\zeta d\xi \end{aligned} \quad (4.95)$$

For the stationary system,

$$R_X(t - \zeta, u - \xi) = R_X(\tau + \xi - \zeta) \quad (4.96)$$

Furthermore, substitution of Equation 4.96 into Equation 4.95 yields

$$R_Y(\tau) = h(\tau)^* R_X(\tau)^* h(-\tau) = h(\tau)^* h(-\tau)^* R_X(\tau) \quad (4.97)$$

Explicitly, the autocorrelation function of the output process $Y(t)$ can be written as the convolution of the three terms $h(\tau)$, $h(-\tau)$, and $R_X(\tau)$.

4.2.2.1.2.4 Cross-Correlation Consider the cross-correlation of the input and output processes $X(t)$ and $Y(t)$

$$\begin{aligned} R_{XY}(t, u) &= E[X(t)Y(u)] \\ &= E[X(t)]E\left[\int_{-\infty}^{\infty} h(\xi)X(u - \xi) d\xi\right] = \int_{-\infty}^{\infty} h(\xi)E[X(t)X(u - \xi)] d\xi \\ &= \int_{-\infty}^{\infty} h(\xi)R_X(t, u - \xi) d\xi \end{aligned} \quad (4.98)$$

For the stationary process

$$R_X(t, u - \xi) = R_X(\tau - \xi) \quad (4.99)$$

Then

$$R_{XY}(\tau) = R_X(\tau)^* h(\tau) \quad (4.100)$$

Because

$$R_{XY}(\tau) = R_{YX}(-\tau) \quad (4.101)$$

Using $-\tau$ to replace τ , Equation 4.100 can be written as

$$R_{YX}(\tau) = R_X(\tau)^* h(-\tau) \quad (4.102)$$

Here, $R_X(-\tau)$ is even.

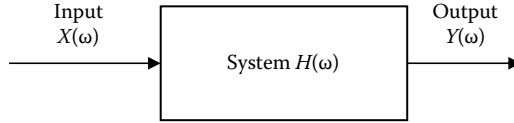


FIGURE 4.10 System in the frequency domain.

4.2.2.1.3 Input–Output Relationship in the Frequency Domain (Linear Time Invariable)

Switching attention from the time domain to the frequency domain, the input–output relationship can be re-examined. This is illustrated in Figure 4.10. When mathematically recalling Equation 4.32, it is known that the following does not exist for random input.

$$X(\omega) = \int_{-\infty}^{\infty} X(t)e^{-j\omega t} dt$$

With the aid of Equation 4.35, this can be redefined as

$$X(\omega) \equiv X(\omega, T) = \int_0^T X(t)e^{-j\omega t} dt \quad (4.103)$$

In this instance, the symbol “ $\equiv$ ” stands for “define.”

Similarly, for output, we have

$$Y(\omega) \equiv Y(\omega, T) = \int_0^T Y(t)e^{-j\omega t} dt \quad (4.104)$$

By taking the Fourier transforms on both sides of Equation 4.88, it results in Borel’s theorem

$$Y(\omega) = H(\omega)X(\omega) \quad (4.105)$$

where $H(\omega)$ is the Fourier transform of the unit impulse response function $h(t)$ (in Chapter 6, we will discuss $h(t)$ in more detail):

$$H(\omega) = \int_0^T h(t)e^{-j\omega t} dt \quad (4.106)$$

It will be shown later in this section that $H(\omega)$ is nothing more than the transfer function.

4.2.2.1.3.1 Laplace and Fourier Transforms of Mean Assuming zero initial conditions, take the Laplace transform on both sides of Equation 4.83. This yields

$$\mathcal{L}[\mu_Y(t)] = H(s)\mathcal{L}[\mu_X(t)] \quad (4.107)$$

Here $\mathcal{L}[(.)]$ denotes the Laplace transform of $(.)$ otherwise written as

$$\mu_Y(t) = \mathcal{L}^{-1}\{H(s)\mathcal{L}[\mu_X(t)]\} \quad (4.108)$$

For a stationary process, the mean is constant

$$\mu_X(t) = \mu_X \quad (4.109)$$

Shown from Equation 4.108.

$$\mu_Y(t) = \mathcal{L}^{-1}\{H(s)\}\mu_X = \mu_Y = \text{constant} \quad (4.110)$$

Through the Fourier transform, this can also be written as

$$\mu_Y = H(j0) \mu_X \quad (4.111)$$

Here, $H(j0)$ is the gain of the filter at $\omega = 0$, namely, the DC gain.

4.2.2.1.3.2 Auto-Power Spectral Density Function Taking the Fourier transform on both sides of Equation 4.97,

$$\mathcal{F}[R_Y(\tau)] = S_Y(\omega) = \mathcal{F}[(h(\tau) * h(-\tau))]\mathcal{F}[R_X(\tau)] = H(\omega)H^*(\omega)S_X(\omega) \quad (4.112a)$$

where $\mathcal{F}[(.)]$ denotes the Fourier transform of $(.)$.

Or, we can use the Laplace transform on both sides of Equation 4.97,

$$\mathcal{L}[R_Y(\tau)] = S_Y(s) = \mathcal{L}[(h(\tau) * h(-\tau))]\mathcal{L}[R_X(\tau)] = H(s)H^*(s)S_X(s) \quad (4.112b)$$

Thus, the auto-PSD function of $Y(t)$ can be obtained as

$$S_Y(\omega) = |H(\omega)|^2 S_X(\omega) \quad (4.113a)$$

By using Laplace transform, we have

$$S_Y(s) = |H(s)|^2 S_X(s) \quad (4.113b)$$

Example 4.10

Suppose a linear system has an impulse response of

$$h(t) = 2 e^{-t}, \quad t > 0$$

A random process $X(t)$ is applied from $t = -\infty$, which has an autocorrelation of

$$R_X(\tau) = e^{-2|\tau|}$$

Find the autocorrelation function $R_Y(\tau)$.

The autospectral density function of $X(t)$ is (refer to Equation 4.15, the Wiener–Khinchine theorem)

$$S_X(\omega) = \int_{-\infty}^{\infty} e^{-2|\tau|} e^{-j\omega\tau} d\tau = \frac{4}{(\omega^2 + 4)}$$

Furthermore,

$$H(\omega) = \int_{-\infty}^{\infty} e^{-\tau} e^{-j\omega\tau} d\tau = \frac{1}{(j\omega + 1)}$$

Thus,

$$S_Y(\omega) = \left| \frac{1}{(j\omega + 1)} \right|^2 \frac{4}{(\omega^2 + 4)} = \frac{1/\pi}{(\omega^2 + 1)(\omega^2 + 4)}$$

4.2.2.1.3.3 Cross-Power Spectral Density Function Taking the Fourier transform on both sides of Equation 4.100

$$\mathcal{F}[R_{XY}(\tau)] = \mathcal{F}[R_X(\tau)]\mathcal{F}[h(\tau)]$$

So that

$$S_{XY}(\omega) = H(\omega)S_X(\omega) \quad (4.114a)$$

By using Laplace transform, we have

$$S_{XY}(s) = H(s) S_X(s) \quad (4.114b)$$

Similarly, we apply Laplace transform on both sides of Equation 4.100, that is,

$$\mathcal{L}[R_{XY}(\tau)] = \mathcal{L}[R_X(\tau)]\mathcal{L}[h(\tau)]$$

The above equation is obtained with known PSD function and transfer function; we thus used inverse Fourier and/or inverse Laplace transform to calculate the cross-correlation function.

Similarly, by again taking the Fourier transform on both sides of Equation 4.102, we have

$$S_{YX}(\omega) = H^*(\omega) S_X(\omega) \quad (4.115a)$$

By using Laplace transform, we have

$$S_{YX}(s) = H^*(s) S_X(s) \quad (4.115b)$$

Example 4.11

A linear system has unit impulse response function $h(t) = e^{-t}$, $t > 0$. Its input is a random process and has autocorrelation function given by $R_X(\tau) = 1/2 e^{-2|\tau|}$ applied from $t = -\infty$. In the following, we describe the process of determining the cross-correlation function $R_{YX}(\tau)$.

Upon checking the autocorrelation function, which is only a function of time lag τ , the input is a stationary process. Using inverse Laplace transform, we have

$$\begin{aligned}
 R_{YX}(\tau) &= \mathcal{L}^{-1}[H^*(s)S_X(s)] \stackrel{\tau > 0}{=} \mathcal{L}^{-1}[\mathcal{L}\{e^{-t}\}\mathcal{L}(1/2e^{-2t})] \\
 &= \mathcal{L}^{-1}\left[\frac{1}{s+1} \cdot \frac{1}{2}\left(\frac{1}{s+2} + \frac{1}{-s+2}\right)\right] = \mathcal{L}^{-1}\left[\frac{2}{(s+1)(s+2)(-s+2)}\right] = \mathcal{L}^{-1}\left[\frac{-\frac{2}{3}}{s+1} + \frac{-\frac{1}{2}}{s+2} + \frac{\frac{1}{6}}{-s+2}\right] \\
 &= \left(-\frac{2}{3}e^{-\tau} - \frac{1}{2}e^{-2\tau}\right)u(\tau) + \frac{1}{6}e^{2\tau}u(-\tau)
 \end{aligned}$$

where $u(\tau)$ is a Heaviside step function and

$$u(\tau) = \begin{cases} 1, & \tau \geq 0 \\ 0, & \tau < 0 \end{cases}$$

4.2.2.2 Estimation of Transfer Functions

Now, consider the second issue, the estimation of the transfer function by measuring the input and output.

4.2.2.2.1 Definition

Dividing by $X(\omega)$ on both sides of Equation 4.105 yields

$$H(\omega) = \frac{Y(\omega)}{X(\omega)} \quad (4.116)$$

$H(\omega)$ is referred to as the *transfer function*.

4.2.2.2.2 Estimation

Stemming from Equation 4.114a,

$$H(\omega) = \frac{S_{XY}(\omega)}{S_X(\omega)} \quad (4.117)$$

On the right-hand side of Equation 4.114, both the numerator and the denominator can be multiplied by $X^*(\omega)$, such that

$$H(\omega) = \frac{Y(\omega)X^*(\omega)}{X(\omega)X^*(\omega)} \quad (4.118)$$

Practically speaking, it is beneficial to let $X_k(\omega, T)$ and $Y_k(\omega, T)$ denote the k th sample pair. By taking the average of n total sample realizations, Equation 4.118 can be rewritten as

$$\hat{H}(\omega) = \frac{\frac{1}{2nT} \sum_{k=1}^n Y_k(\omega, T) X_k^*(\omega, T)}{\frac{1}{2nT} \sum_{k=1}^n X_k(\omega, T) X_k^*(\omega, T)} = \frac{\hat{S}_{XY}(\omega)}{\hat{S}_X(\omega)} \quad (4.119)$$

In addition, on the right-hand side of Equation 4.114, both the numerator and the denominator can be multiplied by $Y^*(\omega)$, such that

$$H(\omega) = \frac{Y(\omega)Y^*(\omega)}{X(\omega)Y^*(\omega)} \quad (4.120)$$

Once more, let $X_k(\omega, T)$ and $Y_k(\omega, T)$ denote the k th sample pair, and take the average of n total sample realizations. This allows Equation 4.118 to be rewritten as

$$\hat{H}(\omega) = \frac{\frac{1}{2nT} \sum_{k=1}^n Y_k(\omega, T) Y_k^*(\omega, T)}{\frac{1}{2nT} \sum_{k=1}^n X_k(\omega, T) Y_k^*(\omega, T)} = \frac{\hat{S}_Y(\omega)}{\hat{S}_{YX}(\omega)} \quad (4.121)$$

Both Equations 4.121 and 4.119 can be used to measure the transfer function in practical measurements. To distinguish these two cases, an estimation of the transfer function through Equation 4.121 will be denoted as $H_1(\omega)$ and that through Equation 4.119 as $H_2(\omega)$.

Formally, the functions $H_1(\omega)$ and $H_2(\omega)$ are defined as follows:

$$H_1(\omega) = \frac{S_{XY}(\omega)}{S_X(\omega)} \quad (4.122)$$

$$H_2(\omega) = \frac{S_Y(\omega)}{S_{YX}(\omega)} \quad (4.123)$$

4.2.2.2.3 Amplitude and Phase of Transfer Functions

In general, a transfer function is a complex value, which can be expressed as follows

$$H(\omega) = \text{Re}[H(\omega)] + j \text{Im}[H(\omega)] \quad (4.124)$$

A transfer function can also be stated by its amplitude and phase angle as

$$H(\omega) = |H(\omega)|e^{j\theta} \quad (4.125)$$

Here, the amplitude is

$$|H(\omega)| = \{\text{Re}^2[H(\omega)] + \text{Im}^2[H(\omega)]\}^{1/2} \quad (4.126)$$

and the phase angle is

$$\theta = \angle H(\omega) = \tan^{-1} \frac{\text{Im}[H(\omega)]}{\text{Re}[H(\omega)]} \quad (4.127)$$

The phase angle, θ , is the phase difference between the input $X(t)$ and the output $Y(t)$, which can be directly obtained through Equation 4.114. Referring to Equation 4.122, the expression of $H_2(\omega)$, $S_Y(\omega)$ is recognized as being real valued, which does not contain information on the phase. Thus, information on the phase is contained in the cross-PSD function $S_{YX}(\omega)$.

4.2.2.3 Stationary Input

Based on the above discussion, we now consider whether an output is stationary when the input is stationary. In most cases of engineering applications, we do not distinguish whether the input is strictly stationary or weakly stationary. However, in this situation, they will be different.

4.2.2.3.1 Strictly Stationary Input

When a strictly stationary random process is input into a linear time-invariant system, we have shown that the mean of output is a constant. Namely, the mean is time-invariant. In addition, the autocorrelation function is only a function of time difference $\tau = t_2 - t_1$.

These two facts imply that when a strictly stationary random process is input into a linear time-invariant system, the output is also a strictly stationary random process.

4.2.2.3.2 Weakly Stationary Input

When a weakly stationary random process is input into a linear time-invariant system, however, it is not guaranteed to have time-invariant mean. Additionally, it is not guaranteed to have an autocorrelation function to be the function of time difference only. Therefore, the output is not necessarily stationary.

4.2.3 COHERENCE ANALYSIS

In this section, the coherence function will first be defined. It will then be explained how the coherence function is related to the aforementioned transfer functions and, finally, the application of coherence analysis. The practical measurement of coherence functions and applications will be discussed further in Chapter 7.

4.2.3.1 Coherence Function

4.2.3.1.1 Definition

The coherence function of two random processes is defined as

$$\gamma_{XY}^2(\omega) = \frac{|S_{XY}(\omega)|^2}{S_X(\omega)S_Y(\omega)} = \frac{S_{XY}(\omega)S_{XY}^*(\omega)}{S_X(\omega)S_Y(\omega)} \quad (4.128)$$

4.2.3.1.2 Property of Coherence Function

$$-1 \leq \gamma_{XY}^2(\omega) \leq 1 \quad (4.129)$$

On the right-hand side of Equation 4.128, it is seen that

$$\frac{S_{XY}(\omega)}{S_X(\omega)} = H_1(\omega) \quad (4.130)$$

and

$$\frac{S_{XY}^*(\omega)}{S_Y(\omega)} = \frac{S_{YX}(\omega)}{S_Y(\omega)} = \frac{1}{H_2(\omega)} \quad (4.131)$$

Substitution of Equations 4.130 and 4.131 into Equation 4.128,

$$\gamma_{XY}^2(\omega) = \frac{H_1(\omega)}{H_2(\omega)} \quad (4.132)$$

Furthermore, from Equation 4.129,

$$\left| \frac{H_1(\omega)}{H_2(\omega)} \right| \leq 1 \quad (4.133)$$

Thus,

$$|H_1(\omega)| \leq |H_2(\omega)| \quad (4.134)$$

Because $\gamma_{XY}^2(\omega)$ is real valued

$$\angle H_1(\omega) = \angle H_2(\omega) \quad (4.135)$$

4.2.3.1.3 Application of Coherence Function

When a system is applied to a forcing function, regardless of being deterministic or random, the input will always contain a certain amount of noise, which is random. In other words, in the real world, there will always be random input. Consequently, the output will also always be random.

From the previous discussion on measurement of transfer functions, H_1 and H_2 estimations were defined. This implies that the measured transfer function will always be altered due to undisputable types of noises. In addition, in the attempt to measure transfer function, it must always be assumed that the system is perfectly linear, which is not always true.

The presentation of noises and nonlinearity will irrefutably introduce errors. Thus, there is a need to judge if the error in measurement is practically significant. The coherence function, as a ratio of the H_2 and H_1 functions, can be used in this purpose. The higher the value of the coherence function, the better the measurement is. Generally speaking, the value of coherence function should be greater than 60% to be considered a dependable measurement. In certain applications, the requirement will be to have the value greater than 70% or possibly 80%.

4.2.3.2 Attenuation and Delay

The following examples are used to show how the statistical properties in the frequency domain on certain random processes can be analyzed. The first case considered is that of attenuation/delay. The instance of the sum of two random processes is then discussed.

Recall the attenuation and delay, $X(t)$ to be a random process and

$$Y(t) = aX(t - \delta) \quad (4.136)$$

The auto-PSD function is

$$S_Y(\omega) = a^2 S_X(\omega) \quad (4.137)$$

The cross-PSD function between input and output is

$$S_{XY}(\omega) = a S_X(\omega) e^{j\omega\delta} \quad (4.138)$$

The transfer functions are

$$H_2(\omega) = S_{XY}(\omega)/S_X(\omega) = a e^{j\omega\delta} = H_1(\omega) \quad (4.139)$$

Finally, the coherence function is

$$\gamma_{XY}^2(\omega) = 1 \quad (4.140)$$

Note that, in this case, we do not have any noises.

4.2.3.3 Sum of Two Random Processes

Now, consider two random processes, not necessarily independent, denoted by $X(t)$ and $Y(t)$. Their sum is denoted by $Z(t)$, thus,

$$Z(t) = X(t) + Y(t) \quad (4.141)$$

The cross-PSD function between $Z(t)$ and $X(t)$ is

$$S_{ZX}(\omega) = S_X(\omega) + S_{YX}(\omega) \quad (4.142)$$

If $X(t)$ and $Y(t)$ is not correlated, the auto-PSD of $Z(t)$ is

$$S_Z(\omega) = S_X(\omega) + S_Y(\omega) \quad (4.143)$$

In this case, finally, the coherence function is

$$\gamma_{ZX}^2(\omega) = \frac{S_X(\omega)}{S_X(\omega) + S_Y(\omega)} \quad (4.144)$$

4.2.4 DERIVATIVES OF STATIONARY PROCESS

In Chapter 3, the derivative of the autocorrelation function of a random process $X(t)$ resulted in cross-correlation function of $X(t)$ and its derivative $\dot{X}(t)$, given by

$$\frac{d}{d\tau} R_X(\tau) = R_{X\dot{X}}(\tau) \quad (4.145)$$

Here, if the random process $X(t)$ is seen as displacement, then respectively its derivatives $\dot{X}$ and $\ddot{X}$ can be seen as velocity and acceleration.

The Fourier transform of the derivative of a temporal function $g(t)$ is

$$\mathcal{F}\left\{\frac{d}{d\tau} g(t)\right\} = j\omega G(\omega) \quad (4.146)$$

Thus,

$$S_{X\dot{X}}(\omega) = \mathcal{F}\{R_{X\dot{X}}(\tau)\} = \mathcal{F}\left\{\frac{d}{d\tau} R_X(\tau)\right\} = j\omega S_X(\omega) \quad (4.147)$$

It is known that

$$S_{\dot{X}}(\omega) = \omega^2 S_X(\omega) \quad (4.148)$$

We thus have

$$S_{X\ddot{X}}(\omega) = -\omega^2 S_X(\omega) \quad (4.149)$$

and

$$S_{\ddot{X}}(\omega) = \omega^4 S_X(\omega) \quad (4.150)$$

Now, the variance functions of the velocity and the acceleration can be written as

$$\sigma_{\dot{X}}^2 = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{\dot{X}}(\omega) d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} \omega^2 S_X(\omega) d\omega \quad (4.151)$$

and

$$\sigma_{\ddot{X}}^2 = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{\ddot{X}}(\omega) d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} \omega^4 S_X(\omega) d\omega \quad (4.152)$$

4.3 PRACTICAL ISSUES OF PSD FUNCTIONS

In this section, several practical issues regarding spectral PSD functions are discussed.

4.3.1 ONE-SIDED PSD

Initially, in real-world measurements, it is difficult to have negative frequency, in particular, as the lower integral limit approaches $-\infty$. For this reason, the one-sided Fourier transform is used.

4.3.1.1 Angular Frequency versus Frequency

Frequency is more commonly expressed in hertz than in radians per second. As a result, we have

$$\omega = 2\pi f \quad (4.153)$$

$$\omega \sim (\text{rad/s}); \quad f \sim (\text{Hz})$$

The period is

$$T = 1/f \quad (4.154)$$

and

$$T \sim (\text{s})$$

The relationship between the period and the angular frequency is

$$T = 2\pi/\omega \quad (4.155)$$

By drawing on the above notations, the one-sided auto-PSD function can be written as

$$G_X(f) = 2\pi S_X(\omega) \quad (4.156)$$

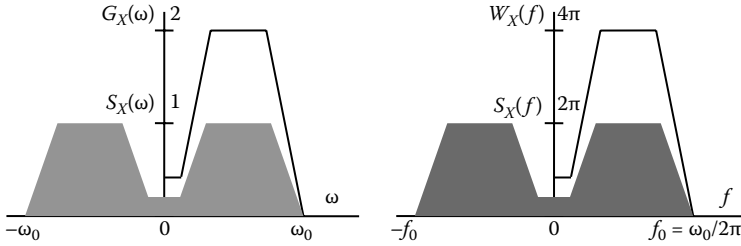


FIGURE 4.11 Double-sided and single-sided spectra.

4.3.1.2 Two-Sided Spectrum versus Single-Sided Spectrum

Let us now compare the two-sided and the one-sided functions. If angular frequency ω is used, then

$$G_X(\omega) = \begin{cases} 2S_X(\omega), & \omega \geq 0 \\ 0, & \omega < 0 \end{cases} \quad (4.157)$$

If frequency f is used, then alternatively, we have the one-sided auto-PSD $W_X(f)$ written as

$$W_X(f) = \begin{cases} 4\pi S_X(\omega), & \omega, f \geq 0 \\ 0, & \omega, f < 0 \end{cases} \quad (4.158)$$

Figure 4.11 graphically describes the relationships shown in Equations 4.157 and 4.158.

4.3.1.3 Discrete Fourier Transform

In practice, the signal picked up through a data acquisition system is digitized with a given sampling rate, Δt , and a limited duration, T . The corresponding Fourier transform will thus become discrete. In this subsection, the issues related to the discrete type of operations will be discussed.

4.3.1.3.1 Scale of Frequency

Let us first consider the scale of the frequency domain. In this case, a computer program is used to conduct the fast Fourier transform (FFT), which, in other terms, is an efficient algorithm to compute the discrete Fourier transform (DFT), the total scale of the frequency domain is universally asked about.

As seen in Figure 4.12,

$$T = n \Delta t \quad (4.159)$$

where n is the total number of samples.

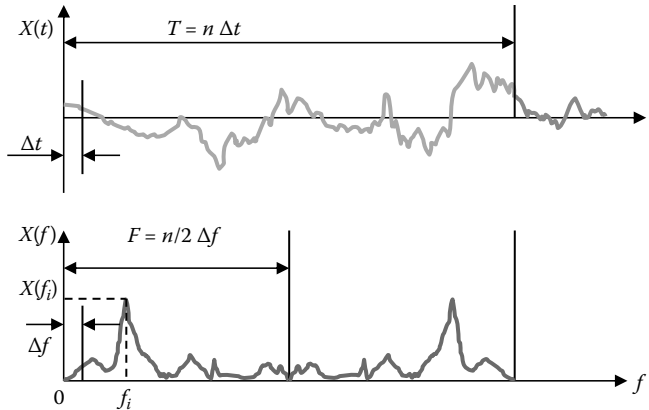


FIGURE 4.12 Relationship between T and F .

The frequency resolution can then be written as

$$\Delta f = 1/T = \frac{1}{n \Delta t} \quad (4.160)$$

Thus, the frequency limit F is

$$F = n/2 \Delta f \quad (4.161)$$

4.3.1.3.2 Scale of Amplitude

The second question of computational FFT is the scale of the amplitude. By denoting $X(f)$ as the resulting DFT of time history $x(t)$, we have

$$X(f) = \frac{2}{n} [\text{FFT}(x(t))] \quad (4.162)$$

In this case, $\text{FFT}(\cdot)$ means the fast Fourier transform of $x(t)$.

4.3.2 SIGNAL-TO-NOISE RATIOS

Signal-to-noise ratio (S/N) is one of the fundamental concepts in dealing with signals. Nevertheless, it is difficult to measure the exact value of a S/N because it is almost impossible to measure a random process of noises.

4.3.2.1 Definition

Let us consider the definition of S/N as follows.

4.3.2.1.1 Deterministic

In deterministic cases, the signal-to-noise ratio, r , is equal to

$$r = \frac{p_s}{p_n} \quad (4.163)$$

where p_s and p_n are the powers of signal and noise, respectively.

4.3.2.1.2 Random Process

In random cases, the ratio will depend on the time delay τ , namely,

$$r(\tau) = \frac{R_S(\tau)}{R_N(\tau)} \quad (4.164)$$

where, R_S and R_N are autocorrelation functions of the signal and noise, respectively.

4.3.2.1.3 Random Process in the Frequency Domain

The S/N can also be described in the frequency domain. That is,

$$r(\omega) = \frac{S_X(\omega)}{S_N(\omega)} \quad (4.165)$$

If the noise is close to white noise, then

$$S_N(\omega) \approx S_0 = \text{const.} \quad (4.166)$$

In this case, the signal-to-noise ratio is proportional to $S_X(\omega)$

$$r(\omega) \propto S_X(\omega) \quad (4.167)$$

In most instances, the region is very close to the region between the so-called “half-power points” (Figure 4.13), which will be discussed in Chapter 6.

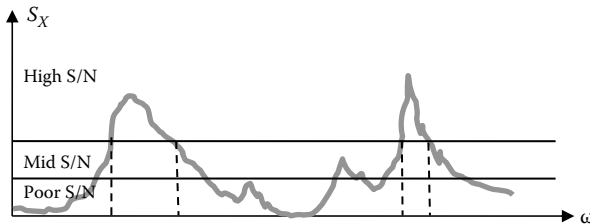


FIGURE 4.13 S/N ratios and PSD.

4.3.2.2 Engineering Significances

Practically speaking, there exists other issues that must be taken notice of as follows.

4.3.2.2.1 Confidence of Measurement

When signals are picked up, we always need to know how reliable a specific measurement is. In general, the confidence factor can be defined as

$$C = r\sqrt{n} \quad (4.168)$$

where n is the size of measurement and r is the S/N ratio.

An excepted judgment is

$$C \geq C_{\text{preset}} \quad (4.169)$$

where C_{preset} is a preset value of confidence.

If m parameters are estimated, the overall confidence factor can be written as

$$C_m = r\sqrt{\frac{n}{m}} \quad (4.170)$$

Similarly, there is also the following judgment

$$C_m \geq C_{m,\text{preset}} \quad (4.171)$$

Here, the second subscript preset indicates a preset value.

4.3.2.2.2 Sensitivity Analysis in Dealing with Normal Uncertainty

An additional uncertainty relating to measurement is the stability or repeatability of the corresponding signal processing. Often, the method of sensitivity testing is used in determining this unknown.

4.3.2.2.2.1 Perturbation To explain the essence of sensitivity analysis, the perturbation phenomenon must first be considered. A deterministic process $A(t)$, due to noise $N(t)$, is commonly measured to be random.

$$X(t) = A(t) + N(t) \quad (4.172)$$

When

$$\|A(t)\| \gg \|N(t)\| \quad (4.173)$$

$N(t)$ can be seen as a small term “added” to the main process. Suppose $N(t)$ can be represented by the Taylor series such that

$$X(t) = A(t) + \varepsilon_0 N_0(t) + \varepsilon_1 N_1(t) + \dots \quad (4.174)$$

where ε_0 and ε_1 , and so on, are small numbers and

$$\varepsilon_0 > \varepsilon_1 > \dots \quad (4.175)$$

4.3.2.2.2 Stability of Realization Assume $N_0(T)$ is Gaussian so that it is “known.” In this case, $\varepsilon_1 N_1(t)$ is considered to be a perturbation (first order perturbation). By adding small amounts of perturbation in the region of $\varepsilon = (1-5)\%$ of white noise, whose level is equal to the peak level of $X(t)$, the mean and standard deviation of $X(t)$ will vary. If the variation is within $(1-5)\%$, the measurement or realization is considered to be “stable.”

Practically, this can be expressed. First, let the initial condition

$$X_0 = \|A(t)\| \quad (4.176)$$

Then, generate an artificial Gaussian process $\varepsilon(t)$ and let

$$\|\varepsilon(t)\| \leq (1-5)\% \quad (4.177)$$

Subsequently, by adding noise to the process $X(t)$, it results in

$$Y(t) = X(t) + X_0 \varepsilon(t) \quad (4.178)$$

By checking the mean

$$\frac{\|\mu_Y(t) - \mu_X(t)\|}{\|\mu_X(t)\|} \leq (1-5)\% \quad (4.179)$$

and the variance

$$\frac{\|\sigma_Y^2(t) - \sigma_X^2(t)\|}{\|\sigma_X^2(t)\|} \leq (1-5)\% \quad (4.180)$$

or the standard deviation

$$\frac{\|\sigma_Y(t) - \sigma_X(t)\|}{\|\sigma_X(t)\|} \leq (1-5)\% \quad (4.181)$$

The above concept may be summarized as though the amount of noise is often unknown; artificial “noises” can be added, which are known. Thus, if a small amount of “random” noise, namely, a perturbation, is added to the total measured data and it is determined that the outcome of the data analysis was not varied in a statistically significant manner, then the total system of the signal pickup and processing is considered to be stable.

4.4 SPECTRAL PRESENTATION OF RANDOM PROCESS

4.4.1 GENERAL RANDOM PROCESS

To understand an object, one of the most common methods is to mathematically model the target, which is also referred to as a mathematical construction. Recall the case of random variables. The modeling is done by the distribution functions,

denoted by $F_X(x)$, which is essentially a calculation of probabilities. That is, the probability of all the chances that X is smaller than x through averaging.

To deal with a set of random variables, the averaging is simple because the computation is done among the variables themselves, namely, in the sample space. In the case of a random process $X(t,e)$, however, the average can be far more complex because we will have not only the sample space $\Omega = \{e\}$ but also another index t , “time.” As a result, the distribution is defined in an n -dimensional way, denoted by $F_X(x_1, x_2, \dots, x_n; t_1, t_2, \dots, t_n)$. Therefore, only if the entire n -dimensional distribution is evaluated would we understand the global properties of $X(t,e)$. It is understandable that this task can be very difficult.

On the other hand, in many cases of engineering applications, we may only need two or even one dimensional distribution, in which the corresponding averages cannot provide global information. Instead, these averages provide local parameters, such as autocorrelation and cross-correlation functions, variance, and mean.

In Figure 4.14, the global and local properties of random processes are illustrated by a conceptual block diagram. In showing the relationships between these properties, we also realize the major topics of random process, discussed mainly in Chapter 3. In Figure 4.14, inside the frame with broken lines is a special relationship between correlation functions and PSD functions, which is the main topic in this chapter and is shown in detail in Figure 4.15.

In Figure 4.15, through Fourier transform (practically, also including Laplace transform), which is a powerful mathematical tool, the correlation functions in the time domain are transferred into the frequency domain. Analyzing vibration signals, which is one of the major topics in this manuscript, can be carried out in the frequency domain. Such analysis is a necessary and powerful tool and can provide insight into vibration systems, which cannot be obtained through the time domain only. In Section 4.4.2, we discuss the frequency distributions and spectral presentations of random process in a more rigorous fashion.

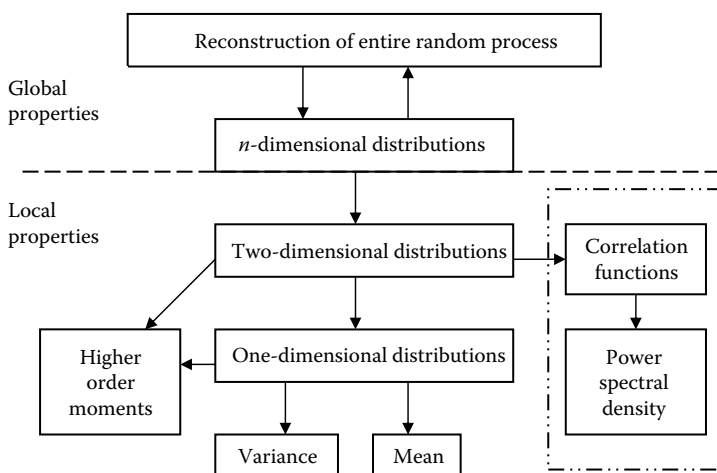


FIGURE 4.14 Properties of random process.

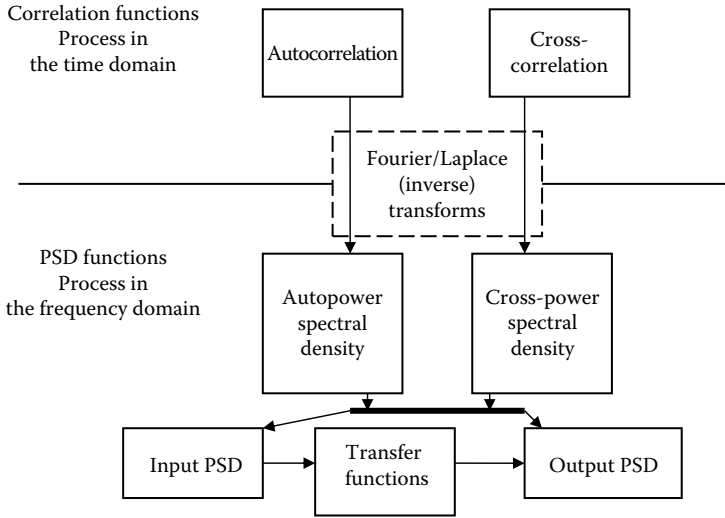


FIGURE 4.15 Relationship between analyses of the time and the frequency domain.

4.4.2 STATIONARY PROCESS

4.4.2.1 Dynamic Process in the Frequency Domain

By definition, in general, a stationary process will not grow to infinity nor will it die out. As a dynamic process, one can realize that the instance value of the process will fluctuate, namely, it will be up at a certain time point and down at other points. Such an up-and-down process in the time domain can be represented by various sinusoidal terms, $\sin(\omega_i t)$'s and $\cos(\omega_i t)$'s.

Such an up-and-down process will contain a certain group of frequency components. To view such processes in the frequency domain is in fact to list these frequency components as a spectrum, which unveils important information on the profile of frequencies.

For a nonstationary process, the Fourier spectrum often does not exist. Therefore, we cannot have the spectrum that a stationary process does. However, for nonstationary processes, one can also perform frequency analysis by introducing the finite Fourier transform (recall Equation 4.35). In this circumstance, when the value of T is not sufficiently long, the corresponding spectra are not deterministic.

4.4.2.2 Relationship between the Time and the Frequency Domains

The above viewpoint that a stationary process can be represented by harmonic oscillations is the key to further exploring random vibrations, the main topic of this manuscript. To realize this point clearly, let us reconsider the autocorrelation $R_x(\tau)$ as the result of inverse Fourier transform of autopower spectrum density function (recall Equation 4.17), that is,

$$R_x(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S(\omega) e^{j\omega\tau} d\omega \quad (4.182)$$

Furthermore, based on the formula of spectral distribution function $\Psi(\omega)$, it is seen that

$$S_X(\omega) = \frac{d\Psi(\omega)}{d\omega} \quad (4.183)$$

Thus,

$$d\Psi(\omega) = S_X(\omega)d\omega \quad (4.184)$$

Substitution of Equation 4.184 into Equation 4.182 results in

$$R_X(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{j\omega\tau} d\Psi(\omega) \quad (4.185)$$

Equation 4.185 is called a *Fourier–Stieltjes integral* or *Fourier–Stieltjes transform* (Thomas J. Stieltjes, 1856–1894).

In Equation 4.185, functions $R_X(\tau)$, $S_X(\omega)$, and $\Psi_X(\omega)$ are all deterministic; therefore, as long as the auto-PSD function is an absolute integrable, namely,

$$\int_{-\infty}^{\infty} |S(\omega)| d\omega < \infty$$

then Equations 4.182 through 4.185 exist.

Now, if Equation 4.34 exists, that is, if

$$X(\omega) = \int_{-\infty}^{\infty} X(t)e^{-j\omega t} dt$$

then the random process $X(t)$ can be seen as an inverse Fourier transform of function $X(\omega)$, that is,

$$X(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega)e^{j\omega t} d\omega$$

Similar to the above-mentioned derivation of Equation 4.185, the Fourier transform of $X(t)$, namely, $X(\omega)$, can be seen as a derivative of function $Z(\omega)$ (see Equation 4.183), that is,

$$X(\omega) = \frac{dZ(\omega)}{d\omega}$$

Additionally, the function $Z(\omega)$ is the integration of $X(\omega)$,

$$Z(\omega) = \int_{-\infty}^{\omega} X(\varpi) d\varpi$$

Similar to Equation 4.185, the random process $X(t)$ can be written as a Fourier–Stieltjes integral:

$$X(t) = \int_{-\infty}^{\infty} e^{j\omega t} dZ(\omega) \quad (4.186)$$

If Equation 4.186 does exist, then it implies that a random process can be replaced by its spectral representation. In fact, $Z(\omega)$ does exist under certain conditions, and it is defined as the *spectral representation function* of random process. Now, let us consider the required conditions. In Chapter 3, we noted that to use Fourier transform to replace the Fourier series of a random process is not always doable because it requires conditions. In Sections 4.1 and 4.2, we further point out that to let $T \rightarrow \infty$ also needs conditions. These two issues are closely related.

To see this point, recall the sum of harmonic process (Equation 3.62) with zero mean, which can be written in an alternative form given by

$$X(t) = \sum_{k=-m/2}^{m/2} X_k(t) = \sum_{k=-m/2}^{m/2} C_k e^{j\omega_k t} \quad (4.187)$$

where C_k are complex-valued uncorrelated random variables with zero mean, and have variance σ^2 . Because $X(t)$ must be real-valued, we have symmetry for the pair of frequencies ω_k and $-\omega_k$, $C_{-k} = C_k^*$.

With help from the notation in Equation 4.187, consider the autocorrelation function

$$\begin{aligned} R_X(\tau) &= E[X(t) X(t + \tau)] = E[X(t)^* X(t + \tau)] \\ &= E \left\{ \left[\sum_{p=-m/2}^{m/2} C_p e^{j\omega_p t} \right]^* \left[\sum_{q=-m/2}^{m/2} C_q e^{j\omega_q (t+\tau)} \right] \right\} = E \left\{ \left[\sum_{p=1}^m C_p^* e^{-j\omega_p t} \right] \left[\sum_{q=1}^m C_q e^{j\omega_q (t+\tau)} \right] \right\} \\ &= \sum_{p=-m/2}^{m/2} \sum_{q=-m/2}^{m/2} E[C_p^* C_q] e^{j[-\omega_p t + \omega_q (t+\tau)]} \end{aligned} \quad (4.188)$$

Note that X is zero-mean, and $E[C_p C_q] = 0$ for $p \neq q$. Also, based on Euler equation $e^{j\theta} = \cos\theta + j \sin\theta$, the above equation can be rewritten as

$$R_X(\tau) = \text{Re} \left\{ \sum_{k=-m/2}^{m/2} E[C_k^2] e^{j\omega_k \tau} \right\} = \text{Re} \left\{ \sum_{k=1}^m E[C_k^2] e^{j\omega_k \tau} \right\} \quad (4.189)$$

where the symbol $\text{Re}(\cdot)$ means taking the real portion of function $(\cdot)$ only.

When the frequency resolution becomes infinitely small, namely, in between ω_k and ω_{k+1} , we insert an infinitely large number of frequency components. The series in Equation 4.187 will therefore be replaced by an integral

$$X(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{j\omega\tau} Z(d\omega) \quad (4.190)$$

where

$$Z(\varpi) = \sum_{\substack{p=-m/2 \\ m \rightarrow \infty}}^{\varpi} C_p \quad (4.191)$$

and Equation 4.189 can be replaced by

$$R_X(\tau) = \text{Re} \left\{ \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{j\omega\tau} \Upsilon(d\omega) \right\} = \frac{1}{2\pi} \int_{-\infty}^{\infty} \cos \omega\tau \Upsilon(d\omega) \quad (4.192a)$$

Here Υ is a special function, which will be expressed in detail in Section 4.4.2.3.

In more general cases, a random process $X(t)$ can be complex valued, so that the corresponding autocorrelation function is complex in general. In such a situation, Equation 4.192a can be rewritten as

$$R_X(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{j\omega\tau} \Upsilon(d\omega) \quad (4.192b)$$

4.4.2.3 Spectral Distribution and Representation

In Figure 4.16a and b, we first show the operation to use finer frequency resolutions, in which the magnitude M_p is used to represent either the amplitude of function $\Upsilon(\varpi)$ or the function $Z(\varpi)$. Note that M_p is the area at frequency ω_p . Additionally, rectangular-shaped M_p can be seen as a product of the height $\frac{\sigma^2}{2\pi} g_p$ and the frequency interval $\Delta\omega$. That is,

$$M_p = \frac{\sigma^2}{2\pi} g_p \Delta\omega \quad (4.193)$$

Thus, in Equation 4.193 using M_p to represent the magnitude of function $\Upsilon(\varpi)$ results in

$$\begin{aligned} \Upsilon(\varpi) &= E[C_p^2] = \text{l.i.m}_{n \rightarrow \infty} \frac{1}{n} \sum_{p=-m/2}^{\varpi} C_p^2 \\ &= \lim_{m \rightarrow \infty} M_p = \frac{\sigma^2}{2\pi} \lim_{\substack{m \rightarrow \infty \\ \Delta\omega \rightarrow 0}} g_p \Delta\omega \end{aligned} \quad (4.194)$$

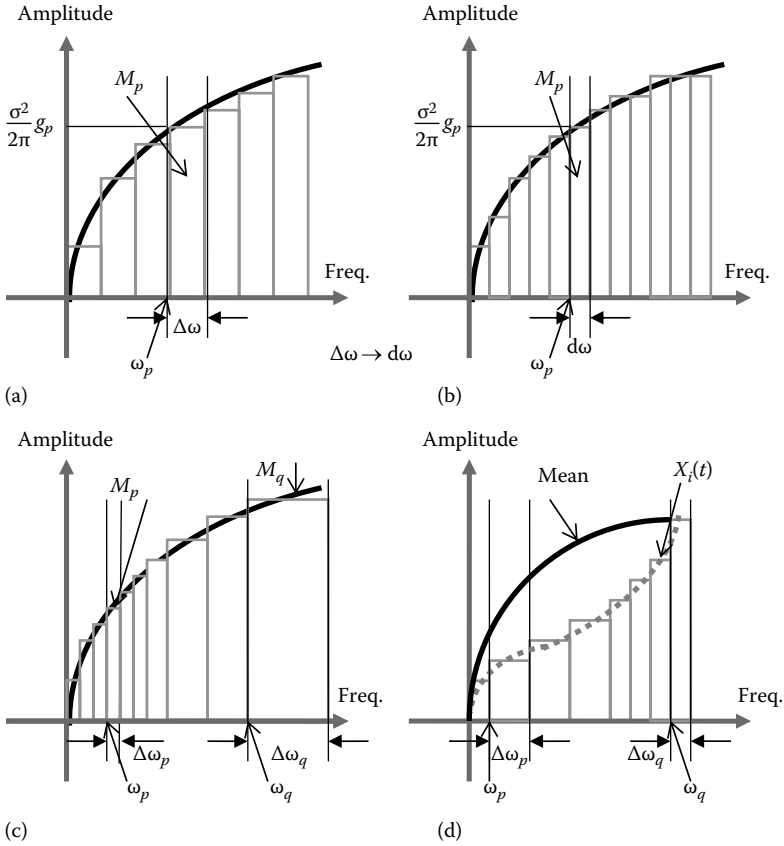


FIGURE 4.16 Mass and density functions with various frequency intervals. (a) Equal frequency interval $\Delta\omega$. (b) Equal frequency interval $d\omega$. (c) Unequal frequency interval $\Delta\omega$. (d) Unequal frequency interval $d\omega$.

To conceptually show the operation described in Equation 4.194, in Figure 4.16a, we have a less spectral line whereas in Figure 4.16b, there are considerably more spectral lines. Although in b, the spectrum is still discrete, its frequency interval is marked as $d\omega$ for comparison, one can realize that when the resolution $\Delta\omega$ becomes finer and finer, until it reaches an infinitesimally small value $d\omega$, a continuous spectrum will be obtained. Note that in the process, the frequency interval approach becomes finer and finer, with all of them having equal length.

We see that M_p is the “mass” function and height $\frac{\sigma^2}{2\pi} g_p$ is the density function. For the specific case of function $Y(\omega)$, we use M_p to represent the accumulative average $E[C_p^2]$ and we can compare $Y(\omega)$ to CDF, the accumulative probabilities in a later discussion. Because M_p is the magnitude and g_p is the “height” shown in Figure 4.16, when $\Delta\omega \rightarrow d\omega$, $g(\omega)$ is used to replace g_p and $g(\omega)$ becomes a density function. Here, the term $\frac{1}{2\pi}$ is used only for mathematical convenience.

Note that when we replace the harmonic random series by integrations (Equation 4.11), it is not necessary to have equally spaced frequency intervals. That is, recall Equation 3.73, that is, $\Delta\omega_p = \omega_{p+1} - \omega_p \neq \Delta\omega_q = \omega_{q+1} - \omega_q$. The equal and unequal frequency interval can also be conceptually shown in Figure 4.16.

In Figure 4.16a and b, we show equal frequency intervals whereas in Figure 4.16c and d, the frequency interval is unequal. From Figure 4.16c, it is seen that to have unequal frequency interval $\Delta\omega_p$ and $\Delta\omega_q$ at point ω_p and ω_q , respectively, does have advantages because at point ω_p the curve has a steeper slope and at point ω_q the curve is flatter.

Mathematically, when equal frequency intervals are used, it implies that the original function, say, $R_X(\tau)$ and/or $X(t)$, is periodic with period T , namely,

$$\frac{2\pi}{T} = \Delta\omega = \omega_T \quad (4.195)$$

In this case, the function, say, $R_X(\tau)$, can be represented by Fourier series with harmonic terms $\cos n\omega_T\tau$, $\sin n\omega_T\tau$ and/or $e^{jn\omega_T\tau}$, and others, and ω_T is the fundamental frequency. However, we may also use nonperiodic cases, in this situation, we use the relationship (Equation 4.192)

$$R_X(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{j\omega\tau} \Upsilon(d\omega)$$

instead of using Fourier transform

$$R_X(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{j\omega\tau} S_X(\omega) d\omega$$

The following is a more general description. A Fourier series, the discrete form, only works for periodic functions. A numerical realization of Fourier transform in discrete format, with limited recorded length, will inherit certain drawbacks of the Fourier series. In addition, when we use $Y(\omega) = X(\omega, T)$ in Equation 4.37, letting $T \rightarrow \infty$ is not always legitimate. To be mathematically consistent, the above-mentioned Wiener-Khinchine equation defines auto-PSD function $S_X(\omega)$, instead of introducing the concept of PSD first and then proving that it is the Fourier transform of $R_X(\tau)$.

Compared with the definition of the formula for spectral distribution function in Equation 4.22, namely, $\Psi(\omega) = \int_{-\infty}^{\omega} S_X(\varpi) d\varpi$, we can realize that if periodic functions do exist, $\Upsilon(\varpi)$ is nothing but the spectral distribution function Ψ , that is,

$$\Upsilon(\varpi) \equiv \Psi(\varpi) \quad (4.196)$$

and

$$d[\Upsilon(\varpi)] = d[\Psi(\varpi)] = S_X(\varpi)d\varpi \quad (4.197)$$

Equation 4.197 indicates that the autocorrelation function $R_X(\tau)$ and function $\frac{d[\Psi(\omega)]}{d\omega}$ are Fourier pairs, that is,

$$R(\tau) \Leftrightarrow \frac{d[\Psi(\omega)]}{d\omega} = S_X(\omega) \quad (4.198)$$

Furthermore, compare Equation 4.192a with Equation 4.13, for continuous frequency domain,

$$\Upsilon(d\omega) = \Psi(d\omega) = \frac{\sigma^2}{2\pi} g(\omega)d\omega \quad (4.199)$$

Therefore, in this case

$$\frac{\sigma^2}{2\pi} g(\omega) = S_X(\omega) \quad (4.200)$$

The quantity C_p in Equation 4.188 is the magnitude of the frequency component at frequency ω_p , so that it is the magnitude of the spectral line of the specific point of $X(t)$'s spectrum. Therefore, the summation function $Z(\varpi)$ is an accumulated spectral value up to frequency ϖ . In the literature, $Z(\varpi)$ is referred to as the *spectral representation function*.

Although in Figure 4.16, M_p can represent both the magnitude of $Z(\omega)$ for Equation 4.191, or that of $\Upsilon(\omega)$ for Equation 4.194, the nature of $Z(\omega)$ in Equation 4.191 and the nature of $\Upsilon(\omega)$ in Equation 4.194 are completely different. The former is a random value where the latter is deterministic. As shown in Figure 4.16d, the solid curve stands for a determine function, say, the mean of the random process $X(t)$, whereas the dotted line represents a random function, say, $X(t)$. For a random process, its spectral amplitude, if it exists in this case, is C_p , which can be random. Therefore, using unequal frequency $\Delta\omega_p$ and $\Delta\omega_q$ at point ω_p and ω_q , respectively, will show another advantage accounting for general random processes.

It can be proven that the variance of $Z(\omega)$ is

$$D[Z(\omega)] = \Upsilon(\omega) \quad (4.201)$$

The covariance of $Z(\omega_\alpha)$ and $Z(\omega_\beta)$ is

$$\sigma_{Z_\alpha Z_\beta} = \Upsilon(\omega_\alpha \cap \omega_\beta) \quad (4.202)$$

and

$$Z(-\omega) = Z(\omega)^* \quad (4.203)$$

4.4.2.4 Analogy of Spectral Distribution Function to CDF

The spectral distribution $\Upsilon(\omega)$ is essentially a rescaled CDF. First, it is easy to see that

$$\Upsilon(0) \equiv \Psi(0) = 0 \quad (4.204)$$

Second, $\Upsilon(\omega)$ does not integrate to unity, but integrates to the 2π times the variance of the process because σ_X^2 , which can be seen from Equation 4.31. We thus have, however,

$$\Upsilon(\infty) = \Psi(\infty) = 2\pi\sigma_X^2 = 2\pi R_X(0) \quad (4.205)$$

Both the spectral distribution and the CDF of a random variable are right-continuous, nondecreasing bounded functions with countable jumps (in the case of mixtures of discrete and continuous random variables).

4.4.2.5 Finite Temporal and Spectral Domains

Up to now, infinite time t including time lag τ is used most of the time. That is, the domain of time is $(-\infty, \infty)$. Similarly, in the frequency domain, the domain of frequency is also $(-\infty, \infty)$. Although we realized that letting $T \rightarrow \infty$ was not always legitimate and PSD was introduced through direct definition to avoid a mathematical dilemma, insights about this problem have not yet been discussed. The only improvement is to use the frequency distribution and representation functions to deal with nonperiodic processes. Although the case of infinite domains is beyond the scope of this manuscript, in the following, let us consider these functions and the corresponding Fourier–Stieltjes integral in the case of finite domains of time and frequency. Because in practical applications, when the time becomes sufficiently large, the temporal functions often tend to be sufficiently small (see Figure 4.12 for examples).

Consider that in Equation 4.33, the Parseval energy equation, the domains of the integrands $X^2(t)$ and $|X(\omega)|^2$ become finite, that is, $(-T, T)$, $T < \infty$ and $(-\omega_0, \omega_0)$, $\omega_0 < \infty$. We have

$$\int_{-T}^T X^2(t) \, d\tau \quad \text{and/or} \quad \frac{1}{2\pi} \int_{-\omega_0}^{\omega_0} |X(\omega)|^2 \, d\omega \quad (4.206)$$

In the following, we discuss the finite temporal and frequency domains of random and transient processes, which are sometimes treated as measured signals.

4.4.2.5.1 Finite Time Duration

Generally speaking, with limited duration $(0, T)$, a dynamic process $X(t)$ is better classified as a transient instead of a random process. However, the amplitude of a transient process at different time points t_i and t_j , $0 \leq t_i < t_j \leq T$, can be random variables. For example, an earthquake ground motion is often treated as a random signal.

In practical computations, the Fourier integral of the limited time domain is treated as a period signal with period T , that is, the transient signal will be repeated continuously with multiple duration $[(n-1)T, nT]$, $n = 1, 2, \dots$. Very often, a random process is sampled with limited time duration (see Figure 4.12). In this case, we have forced finite time duration. Such a sampling may introduce errors referred to as power leakage, which will be discussed in Chapter 9.

4.4.2.5.2 Finite Frequency Domain

In engineering applications, we always have dynamic processes with finite frequency domains. That is, the highest frequency is limited. This is partly because when a signal is picked up, due to the limit of frequency response of instrumentation, frequencies higher than a certain level cannot be measured. Another reason is to avoid signal aliasing, lower pass filters are used so that frequencies higher than the cut-off threshold will be removed.

Additionally, to measure a random signal, the total number of samples and the duration T are also limited. Suppose in a measurement, n samples are taken, then, with the help of Equations 4.37 and 4.39, the upper limit of the frequency domain or the maximum frequency $f_{\max}$ can be determined by

$$f_{\max} = \frac{n}{2T} \quad (4.207)$$

PROBLEMS

1. A random process $\{X(t), -\infty < t < \infty\}$ is given by $X(t) = At^2 + Bt + C$ with A , B , and C to be independent random variables and $A \sim N(0,1)$, $B \sim N(0,1)$, and $C \sim N(0,1)$. Find if $X(t)$ is mean-square continuous, mean-square differentiable, or mean-square integrable.
2. Derive autocorrelation functions for the following processes:
 - a. White noise

$$S_X(\omega) = S_0$$

- b. Low pass

$$S_X(\omega) = \begin{cases} S_0 & |\omega_C| < \omega \\ S_0/2 & |\omega_C| = \omega \\ 0 & |\omega_C| > \omega \end{cases}$$

- c. Band pass

$$S_X(\omega) = \begin{cases} S_0 & \omega_L < |\omega| < \omega_U \\ S_0/2 & |\omega| = \omega_L, \omega_U \\ 0 & \text{elsewhere} \end{cases}$$

- d. Narrow band

$$S_X(\omega) = \sigma_X^2 [\delta(\omega + \omega_0) + \delta(\omega - \omega_0)]/2$$

3. Derive PSD functions

a. $R_X(\tau) = e^{-\alpha|\tau|}$

b. $R_X(\tau) = e^{-\alpha|\tau|} \cos(\omega_0 \tau)$

4. A low-pass random process $X(t)$ has a cut-off frequency ω_c (or $2\pi f_c$). It is proposed to estimate the PSD function of this process using Equation 4.58 and sample records of length $T = 10/f_c$. Is T long enough? What if T is 10 times longer? How long would you make it and why? Hint: Consider the rate at which the following ratio approaches unity.

$$\frac{\int_{-T}^T R_X(\tau) e^{-j\omega\tau} \left(1 - \frac{|\tau|}{T}\right) d\tau}{\int_{-\infty}^{\infty} R_X(\tau) e^{-j\omega\tau} d\tau} = 1$$

5. Let $W(t) = X(t)Y(t)$, X and Y uncorrelated random process. Find the PSD function of $W(t)$ and the cross-PSD function and coherence of $W(t)$ and $X(t)$.
6. Consider a random binary function with random phasing Θ , which is uniformly distributed between 0 and T , shown in Figure P4.1
- Model this random process $Y(t)$
 - Find the autocorrelation and auto-PSD
 - Is the process stationary? Ergodic? Hint: It still depends on whether or not t_1 and t_2 are in the same time interval, but this now depends on Θ
7. A local average process $Y_T(t)$ is defined by

$$Y_T(t) = \frac{1}{T} \int_{t-T/2}^{t+T/2} X(u) du$$

$X(t)$ is a random process. Show that the PSD function is given by

$$S_{Y_T}(\omega) = S_X(\omega) \left| \frac{\sin(\omega T/2)}{\omega T/2} \right|^2$$

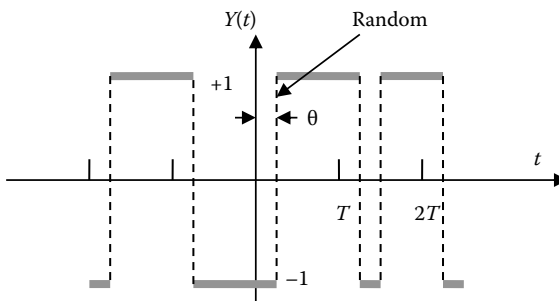


FIGURE P4.1 Random phasing Θ .

8. A stationary process $\{X(t), -\infty < t < \infty\}$ has PSD given by

$$S_X(\omega) = \frac{\omega^2 + 33}{\omega^4 + 10\omega^2 + 9}$$

Find its autocorrelation function and variance.

9. $X(t)$ and $Y(t)$ are non-zero-mean and non-cross-correlated stationary random processes

$$Z(t) = X(t) + Y(t)$$

- a. Is $Z(t)$ stationary?
 - b. Find the cross-PSD $S_{ZY}(\omega)$ and $S_{XZ}(\omega)$.
10. Θ is a uniformly distributed random variable in a period of frequency ω_T . The parameters a_i , b_i , and ω are all constant. The summation

$$\sum_{i=1}^{\infty} (a_i^2 + b_i^2) < \infty$$

A random process is given as

$$X(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \{a_n \cos[n\omega_T(t + \Theta)] + b_n \sin[n\omega_T(t + \Theta)]\}$$

Find the auto-PSD of $X(t)$.

5 Statistical Properties of Random Process

Up to now, we have reviewed the basic knowledge on the theory of probability, which is the fundamental tool for the study of random processes. In addition, we have learned the approach of using two-dimensional (2D) deterministic probability distribution functions to handle one-dimensional (1D) random events. We also introduced random processes in Chapter 3, the time domain approach, where the nature of such a dynamic process was unveiled. From the viewpoint of probability distributions, a random process can have many, if not infinite, pieces of random distributions, instead of a single one as a set of random variables does. Although it is not necessary, in most cases, we will use time as new indices to describe and handle these multiple distribution functions and refer to the methodology as a three-dimensional (3D) approach. Furthermore, in Chapter 4, the functions in the time domain, either the time-varying process itself, which is random, or the correlation functions, which become deterministic through the operation of mathematical expectation, were transferred into the frequency domain. Although it is still a “3D” approach, the corresponding Fourier transforms provided spectral analysis. The latter is a powerful tool to describe the frequency components of a random process.

Different from typical textbooks on random process, which deals with important models of the processes in detail through rigorous mathematical derivations, the previous chapters only provide materials on general descriptions and explanations. Only a few specific random processes are discussed in a more systematic fashion such as how to master the nature of dynamic process, how to model a process with both state and time indices, how to use a 3D approach to understand and calculate the mean, variance, and correlation functions, and how to use the concept of random process to understand our random world. To achieve the above objectives, the previous chapters attempt to outline an overall picture instead of dealing with particular features of individual process. It is also noted that certain useful mathematic tools, such as detailed Fourier and Laplace transforms, how to determine integral limits for sample space, and others, which can occupy many sections in typical textbooks, are also not described.

To compensate for these drawbacks, before systematically presenting the main topic of this manuscript (random vibrations), three unique issues of random processes, level crossing, peak values, and fatigue will further be considered. Here, especially with fatigue, the focus is given on when to account for the occurrence process, instead of how to evaluate and what the results of failures caused by the random process will be (this will be discussed in detail in Chapter 10). To study the time-varying developments, certain important random processes, such as Rayleigh process and Markov chains, will be studied as tools not only to handle these engineering

problems but also to describe the methodology of how to understand specific random process in detail. However, the emphasis is still given to practical application instead of mathematical rigorousness. Due to its randomness, statistical surveys are used to carry out the analyses. In so doing, the “3D” processes are reduced into “2D” variables and furthermore into “1D” parameters, in general.

5.1 LEVEL CROSSINGS

To analyze a random time history, a specific preset level is first established. The objective of this is then to examine the probability of when the value of the process is greater than the preset level. Note that the objective is now reduced to a 1D problem. This approach may be referred to as a special parameterization (see Rice 1944, 1945 and Wirsching et al. 2006).

5.1.1 BACKGROUND

5.1.1.1 Number of Level Crossings

For a time history $x(t)$, in an arbitrary time interval $(t, t + \Delta T)$, with an arbitrary level

$$x = a \quad (5.1)$$

the total number for which $x(t) > a$, namely, the level being crossed, can be expressed as

$$n_a = N_a(t, t + \Delta t) \quad (5.2)$$

where n_a is a random variable. For convenience, we do not use capital letters to denote the random variable to avoid the risk of confusing Figures 5.1 and 5.2.

From Figure 5.1, a and b are points where the curves cross the zero line, whereas c and d are points of line a crossing. In addition, f is the peak value. Generally speaking, $N_a(0, t)$ is a nonstationary random process beginning with

$$N_a(0, 0)|_{t=0} = 0 \quad (5.3)$$

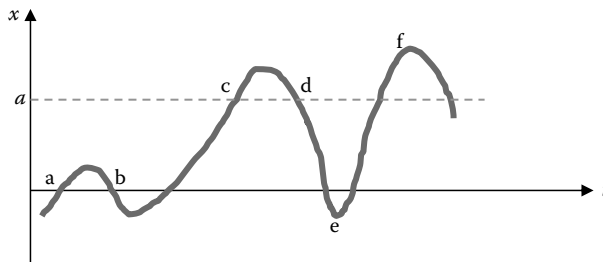


FIGURE 5.1 Level a and crossing points c and d .

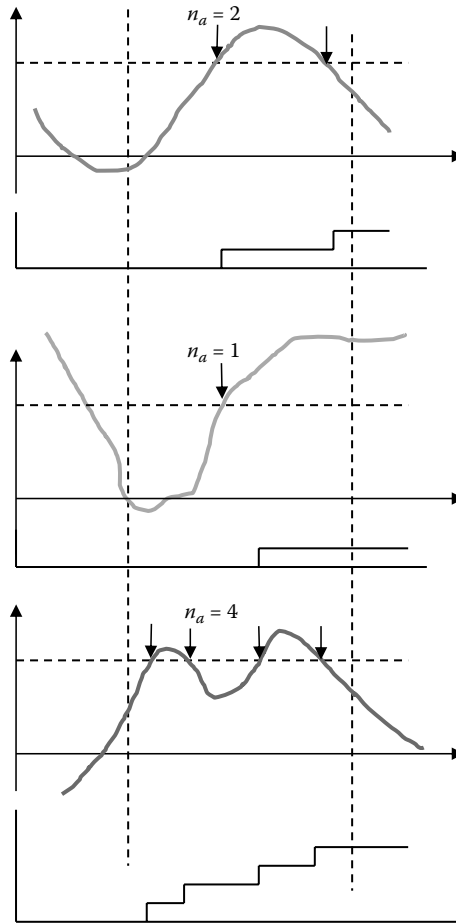


FIGURE 5.2 Number crossing.

If n_a , which is considered to be the number of “arrivals,” is independent and the arrival rate λ is constant, then

$$\lambda = \text{const.} \quad (5.4)$$

It follows that $N_a(t)$ is a Poisson process.

The waiting time until the first arrival, denoted by Y , is a random variable, exponentially distributed. The mean can then be written as

$$\mu_Y = 1/\lambda \quad (5.5)$$

Furthermore, the mean of the random process $N_a(0, t)$ is

$$\mu_{N_a}(t) = E[N_a(t)] = \lambda t \quad (5.6)$$

and the variance is

$$\sigma_{N_a}^2(t) = \lambda t \quad (5.7)$$

5.1.1.2 Correlations between Level Crossings

Now, let us consider the correlations between the level crossings as follows.

5.1.1.2.1 Crossing Pair

It is frequently seen that an up-crossing is followed by a down-crossing, as shown in Figure 5.2.

5.1.1.2.2 Cluster Crossing

As shown in Figure 5.3, an additional case called cluster crossing may also occur. In this instance, the time history is a narrow-band noise. Unlike the crossing pair, in this case, many pairs may follow the initial pair.

5.1.2 DERIVATION OF EXPECTED RATE

First, let us consider the case of crossing pairs.

5.1.2.1 Stationary Crossing

If the probability distributions of random processes $N_a(t_1, t_2)$ and $N_a(t_1 + s, t_2 + s)$ are identical, the random process is said to have stationary increments. It is seen that a Poisson process has stationary increments.

Let $X(t)$ be a zero mean stationary random process in which, for simplicity, $X(t)$ is interpreted as displacement. Therefore, $\dot{X}(t)$ and $\ddot{X}(t)$ are considered to be the velocity and the acceleration, respectively. For these conditions, the expected value is

$$E[N_a(t, t + \Delta t)] = E[N_a(\Delta t)] \quad (5.8)$$

In addition,

$$E[N_a(\Delta t)] = v_a \Delta t \quad (5.9)$$

where v_a is the expected rate of level crossing per unit time. The unit of v_a is hertz (Hz).

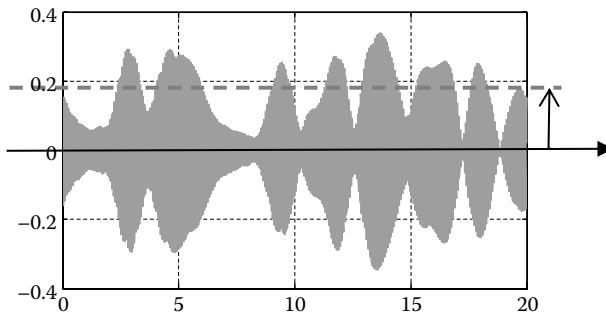


FIGURE 5.3 Cluster crossing.

The concept of crossing rate is helpful. Because $X(t)$ is a random process, we cannot predict where or when a crossing occurs. What can be done is, statistically, the calculation of how often crossing happens. Intuitively, it should be related to the oscillating frequency of $X(t)$ and the level of a .

5.1.2.2 Up-Crossing

Considering the case of up-crossing only, the rate is

$$v_{a^+} = \frac{1}{2} v_a \quad (5.10)$$

The next objective is to find the rate v_{a^+} .

5.1.2.3 Limiting Behavior

When

$$\Delta t \rightarrow 0$$

there will be either zero or one up-crossing in the interval Δt .

Consider event A denoted by

$$A = \{x = a \text{ is crossed with positive slope in } dt\} \quad (5.11)$$

To have the subsequent probability:

$$P[N_{a^+}(dt)] = \begin{cases} P(A), & N_{a^+}(dt) = 1 \\ 1 - P(A), & N_{a^+}(dt) = 0 \\ 0, & \text{elsewhere} \end{cases} \quad (5.12)$$

Then,

$$E[N_{a^+}(dt)] = 1 \times P(A) + 0 \times [1 - P(A)] = P(A) \quad (5.13)$$

From Equation 5.9, where $\Delta t \rightarrow dt$, the following can be written:

$$E[N_{a^+}(dt)] = v_{a^+} dt$$

such that

$$v_{a^+} dt = P(A) \quad (5.14)$$

To further calculate v_{a+} , consider, to have an up-crossing of $x = a$ from t to $t + \Delta T$, the conditions must be

$$1. X(t) < a \quad (5.15)$$

$$2. \dot{X}(t) > 0 \quad (5.16)$$

$$3. X(t + \Delta t) > a \quad (5.17)$$

From Figure 5.4, the resultant relationship is illustrated:

$$X(t) + \dot{X}(t)dt > a \quad (5.18a)$$

or

$$X(t) > a - \dot{X}(t)dt \quad (5.18b)$$

The event in which the above three conditions are met has the single probability $P(A)$:

$$P(A) = P\left[(a - \dot{X}(t)dt < X(t) < a) \cap (\dot{X}(t) > 0)\right] \quad (5.19)$$

Figure 5.5 shows the integral domain with the $x, \dot{x}$ coordinates. The probability can then be written as

$$P(A) = \int_{\dot{x}}^{\infty} \int_{a-vdt}^a f_{X\dot{X}}(u, v) du dv \quad (5.20)$$

When $dt \rightarrow 0$, the starting point of x must be very close to line a , that is, $u \rightarrow a$, in this case, $f_{X\dot{X}}(u, v) \rightarrow f_{X\dot{X}}(a, v)$ and

$$\int_{a-vdt}^a f_{X\dot{X}}(a, v) du = f_{X\dot{X}}(a, v) \int_{a-vdt}^a du = f_{X\dot{X}}(a, v)[a - (a - vdt)] = f_{X\dot{X}}(a, v)vdt$$

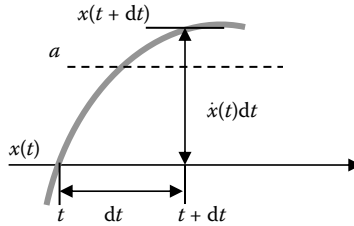


FIGURE 5.4 Up-crossing.

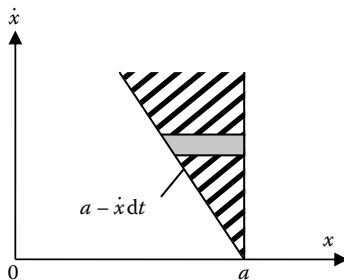


FIGURE 5.5 Integral domain.

Thus, the probability is

$$P(A) = \int_0^{\infty} f_{X\dot{X}}(a, v) (|v| dt) dv \quad (5.21)$$

The absolute value of v indicates that the slope of $X(t)$ must be positive. Thus, from Equation 5.15, the closed form formula of the rate v_{a^+} is:

$$v_{a^+} = \frac{P(A)}{dt} = \int_0^{\infty} v f_{X\dot{X}}(a, v) dv \quad (5.22)$$

5.1.3 SPECIALIZATIONS

To find the rate v_{a^+} in Equation 5.22, the joint density function $f_{X\dot{X}}(a, v)$ must be known. The latter is generally unknown, unless $X(t)$ is Gaussian. In the event that $X(t)$ is Gaussian, then $\dot{X}(t)$ will also be Gaussian. Practically speaking, the assumption of Gaussian process is reasonably valid.

5.1.3.1 Level Up-Crossing, Gaussian Process

Suppose $X(t)$ is Gaussian. It can be proven that if $X(t)$ and $\dot{X}(t)$ are uncorrelated, then to calculate the joint distribution, they can be treated as independent processes. Then, the rate can be written as

$$v_{a^+} = \int_0^{\infty} v f_{X\dot{X}}(a, v) dv = \int_0^{\infty} v f_X(a) f_{\dot{X}}(v) dv = f_X(a) \int_0^{\infty} v f_{\dot{X}}(v) dv \quad (5.23)$$

Here, the variable v represents the “velocity” $\dot{X}(t)$, which is zero-mean, that is,

$$\int_0^{\infty} v f_{\dot{X}}(v) dv = \frac{\sigma_{\dot{X}}}{\sqrt{2\pi}}$$

and based on Equation 5.23, we have

$$v_{a^+} = \frac{1}{2\pi} \frac{\sigma_{\dot{X}}}{\sigma_X} e^{-\frac{1}{2} \frac{a^2}{\sigma_X^2}} \quad (5.24)$$

Note that when the crossing threshold “ a ” increases, the level up-crossing rate decreases.

For a given σ_X , whereas the RMS velocity $\sigma_{\dot{X}}$ increases, the crossing rate also increases.

5.1.3.2 Zero Up-Crossing

When the level of interest is zero, up-crossing will likewise be zero

$$a = 0 \quad (5.25)$$

Substitution of Equation 5.25 into Equation 5.24 results in

$$v_{0^+} = \frac{1}{2\pi} \frac{\sigma_{\dot{X}}}{\sigma_X} \quad (5.26)$$

Note that v_{a^+} is the rate of level up-crossing, namely, in a unit time, the number of crossings from which the angular frequency of crossing is determined to be

$$\omega_{0^+} = 2\pi v_{0^+} = \frac{\sigma_{\dot{X}}}{\sigma_X} \quad (5.27)$$

In Equations 5.26 and 5.27, the terms of standard deviations can be expressed as follows:

$$\sigma_X^2 = \int_{-\infty}^{\infty} S_X(\omega) d\omega = \int_0^{\infty} W_X(f) df \quad (5.28)$$

and

$$\sigma_{\dot{X}}^2 = \int_{-\infty}^{\infty} \omega^2 S_X(\omega) d\omega = 4\pi^2 \int_0^{\infty} f^2 W_X(f) df \quad (5.29)$$

Substitution of the above formulas into Equations 5.26 and 5.27, respectively, results in

$$v_{a^+} = \sqrt{\frac{\int_0^{\infty} f^2 W_X(f) df}{\int_0^{\infty} W_X(f) df}} \quad (5.30)$$

and

$$\omega_{0^+} = \sqrt{\frac{\int_{-\infty}^{\infty} \omega^2 S_X(\omega) d\omega}{\int_{-\infty}^{\infty} S_X(\omega) d\omega}} \quad (5.31)$$

5.1.3.3 Peak Frequency

5.1.3.3.1 General Process

The same level crossing analysis can also be applied on the velocity, $\dot{X}(t)$. A zero down-crossing of $\dot{X}(t)$ results in a change of velocity from positive to negative at the peak of $X(t)$, so the velocity is zero. Comparing this case with the “displacement crossing,” we replace a by v and let $v = 0$ for the velocity and replace v by ϖ for the “acceleration” in Equation 5.22. In this case, the “acceleration” is negative, so that the peak frequency v_p can be written as

$$v_p = \int_{-\infty}^0 -\varpi f_{\dot{X}\ddot{X}}(0, \varpi) d\varpi \quad (5.32)$$

5.1.3.3.2 Gaussian Process

A special case to have the term $f_{\dot{X}\ddot{X}}(0, \varpi)$ is when the process is Gaussian. In case the process is Gaussian, similar to the approach for the development of the formula for v_{a^+} ,

$$v_p = \frac{1}{2\pi} \frac{\sigma_{\ddot{X}}}{\sigma_{\dot{X}}} \quad (5.33)$$

and

$$\omega_p = 2\pi v_p = \frac{\sigma_{\ddot{X}}}{\sigma_{\dot{X}}} \quad (5.34)$$

Additionally, through substitution

$$v_p = \sqrt{\frac{\int_0^{\infty} f^4 W_X(f) df}{\int_0^{\infty} f^2 W_X(f) df}} \quad (5.35)$$

$$\omega_p = \sqrt{\frac{\int_{-\infty}^{\infty} \omega^4 S_X(\omega) d\omega}{\int_{-\infty}^{\infty} \omega^2 S_X(\omega) d\omega}} \quad (5.36)$$

5.1.3.4 Bandwidth and Irregularity

Now, let us consider the following special cases of narrow-band and non-narrow-band processes.

5.1.3.4.1 Narrow-Band Gaussian

First, suppose the random process $X(t)$ is a narrow-band Gaussian process (refer to Equation 4.65). For this case, the frequency of zero up-crossing as well as peak frequency will be examined.

For the narrow-band Gaussian process, the auto-spectral density function can be written as

$$S_X(\omega) = \sigma_X^2 [\delta(\omega + \omega_m) + \delta(\omega - \omega_m)]/2 \quad (5.37)$$

where ω_m is the midband (normal) frequency of the process. For further clarification, see description of ω_0 in Figure 4.8.

Substitution of Equation 5.37 into Equation 5.31 yields

$$\omega_{0^+} = \sqrt{\frac{\int_{-\infty}^{\infty} \omega^2 [\delta(\omega + \omega_m) + \delta(\omega - \omega_m)]/2 d\omega}{\int_{-\infty}^{\infty} [\delta(\omega + \omega_m) + \delta(\omega - \omega_m)]/2 d\omega}} = \sqrt{\frac{\omega_m^2}{1}}$$

Thus,

$$\omega_{0^+} = \omega_m \quad (5.38)$$

Similarly, it is also true that

$$\omega_p = \omega_m \quad (5.39)$$

In conclusion, for narrow-band processes, the zero up-crossing and the peak frequency are identical and equal to the midband frequency.

5.1.3.4.2 Non-Narrow-Band

If the frequency band is not narrow, then the logical subsequent question is “how wide” it can be. To measure the width of the band-pass filtering, an irregular factor is introduced.

5.1.3.4.2.1 Irregularity Factor The measure of the bandwidth can be described by an irregularity factor α , defined as the ratio of zero-up-crossing and the peak frequency (Ortiz 1985), that is,

$$\alpha = \frac{\nu_{0^+}}{\nu_p} = \frac{\omega_{0^+}}{\omega_p} \quad (5.40)$$

By definition, this can be further written as

$$\alpha = \frac{E[N_{0^+}(\Delta t)]}{E[N_p(\Delta t)]} \quad (5.41)$$

It is understandable that the irregular factor has the following range

$$0 < \alpha < 1 \quad (5.42)$$

When $\alpha = 1$, there will be one peak for every zero up-crossing (see Figure 5.6a). This implies that the random process only contains a single frequency, which is the case with the narrow band. Otherwise, if the process contains higher frequency, whose mean special values are often considerably smaller than that of the lowest frequency (called fundamental frequency), we will have $\nu_p > \nu_{0^+}$, so that $0 < \alpha < 1$.

Specifically, when $\alpha \rightarrow 0$, there will be an infinite number of peaks for every zero up-crossing–high-frequency dithering. This is illustrated in Figure 5.6b.

5.1.3.4.2.2 Special Width Parameter ε With the help of the irregular factor, the special width parameter is further defined as

$$\varepsilon = \sqrt{1 - \alpha^2} \quad (5.43)$$

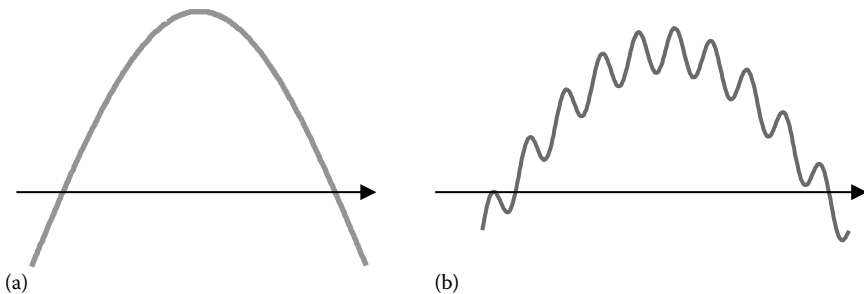


FIGURE 5.6 Number of peaks. (a) Single peak. (b) Multiple peaks.

5.1.3.4.3 Gaussian

If the process is Gaussian, the expression of the irregular factor can be simplified as follows.

Substitutions of Equations 5.26 and 5.33 into Equation 5.40 yields

$$\alpha = \frac{\sigma_{\ddot{X}}^2}{\sigma_X \sigma_{\ddot{X}}} \quad (5.44)$$

Given that the process is Gaussian, then

$$E[X(t)\ddot{X}(t)] = \sigma_X \sigma_{\ddot{X}} \quad (5.45)$$

Thus,

$$E[X(t)\ddot{X}(t)] = -\sigma_X^2 \quad (5.46)$$

and

$$\alpha = \frac{-\sigma_{X\ddot{X}}}{\sigma_X \sigma_{\ddot{X}}} = -\rho_{X\ddot{X}} \quad (5.47)$$

Observe that the irregularity factor, α , is equal to the minus correlation coefficient between the displacement and the acceleration.

5.1.4 RANDOM DECREMENT METHODS

As an application of level up and zero up-crossing, consider a special measurement that can generate free-decay time-history through a random process. To identify vibration parameters such as natural frequencies, damping ratios, and mode shapes through random vibration responses, we basically have two approaches. The first approach is to use correlation analysis, which is discussed in Chapter 3. The second approach is to use free-decay vibrations. However, the random responses are usually not transient signals, that is, they will not decay as time becomes longer. The random decrement method (Cole 1968) is a way to obtain free-decay time history through random signals.

5.1.4.1 Random Decrement (Level Up-Crossing)

Suppose a random process is the response of a linear time-invariable system due to unknown excitations. To analyze the random process, in this section, the method of random decrement is discussed as follows. First we suppose that a zero-mean random process $Y(t)$ is stationary. This can be seen as a response time history, which is conceptually shown in Figure 5.7. To obtain a free-decay time history, in Figure 5.7, line a is drawn. Here, $a = 0.5$.

To use the random decrement method, select the measurement duration T . For example, in Figure 5.7, $T = 3.15$ seconds. The first initial up-crossing point is denoted by t_1 . This can be seen in Figure 5.7, where $t_1 = 0.1$ seconds. A time history of $Y(t)$, $t_1 \leq t \leq T$ is denoted by $Y_1(t)$.

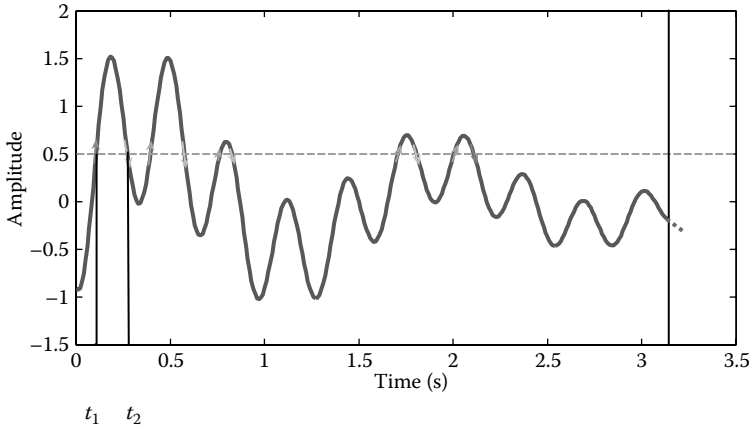


FIGURE 5.7 Random decrement method.

Furthermore, the second crossing point, first down-crossing, for example, is denoted by t_2 ; the third crossing point, second up-crossing, is denoted by t_3 .

The time history taken from $Y(t_2)$, $t_2 \leq t \leq T + (t_2 - t_1)$, is denoted by $Y_2(t)$. Note that $Y_2(t)$ and $Y_1(t)$ have the same measurement length, $T - t_1$, and so on. The time history taken from $Y(t_3)$, $t_3 \leq t \leq T + (t_3 - t_1)$, is denoted by $Y_3(t)$, and so on. Suppose the case in which there are a total of n time histories is taken.

A new time history $Z(t)$ is generated as the sum of all the $Y_i(t)$, that is,

$$Z(t) = \frac{1}{n} \sum_{i=1}^n Y_i(t) \quad (5.48)$$

$Z(t)$ will be a free decay time history, with the initial condition

$$\dot{y}(0) = 0 \quad (5.49)$$

and

$$y(0) = a \quad (5.50)$$

$Y_i(t)$ can be seen as a response of the system due to three kinds of excitations:

1. A random input $F_i(t)$ with zero mean, where the corresponding portion of $Y_i(t)$ is denoted as $Y_{F_i}(t)$.
2. An initial velocity v_i , which is equal to the slope of $Y_i(t)$ at t_i , where the corresponding portion of $Y_i(t)$, is denoted as $Y_{v_i}(t)$.
3. An initial displacement a , where the corresponding portion of $Y_i(t)$ is denoted as $Y_{D_i}(t)$.

Explicitly, this can be written as

$$Y_i(t) = Y_{F_i}(t) + Y_{v_i}(t) + Y_{D_i}(t) \quad (5.51)$$

Substitution of Equation 5.51 into Equation 5.48 yields

$$Z(t) = \frac{1}{n} \sum_{i=1}^n Y_{F_i}(t) + \frac{1}{n} \sum_{i=1}^n Y_{v_i}(t) + \frac{1}{n} \sum_{i=1}^n Y_{D_i}(t) \quad (5.52)$$

Initially, let us consider the first term: the system's impulse response function is a linear time invariant for a stationary process, which is denoted by $h(t)$. This can be written as

$$\sum_{i=1}^n Y_{F_i}(t) = \sum_{i=1}^n h(t) * F_i(t) = h(t) * \left[\sum_{i=1}^n F_i(t) \right] = h(t) * \{0\} = \{0\} \quad (5.53a)$$

Here, $\{0\}$ is null with the dimension of $T - t_1$.

Next, consider the second term: from Figure 5.6, it is seen that the following is approximately true

$$v_1 \approx -v_2$$

and it is understandable that

$$v_i \approx -v_{i+1} \quad (5.53b)$$

Consequently, the responses due to the initial velocity v_i and v_{i+1} will cancel each other. Therefore, this will be reduced to the following:

$$\sum_{i=1}^n Y_{v_i}(t) = \{0\} \quad (5.53c)$$

Lastly, consider the response due to the initial displacement a . With the same initial displacement a , the same response should result, which is a free decay time history under the excitation of the step function

$$u(t) = a \quad (5.54)$$

As a result, the following is achieved

$$Z(t) = \frac{1}{n} \sum_{i=1}^n Y_{D_i}(t) \quad (5.55)$$

where n is the number of fraction pieces of selected time histories.

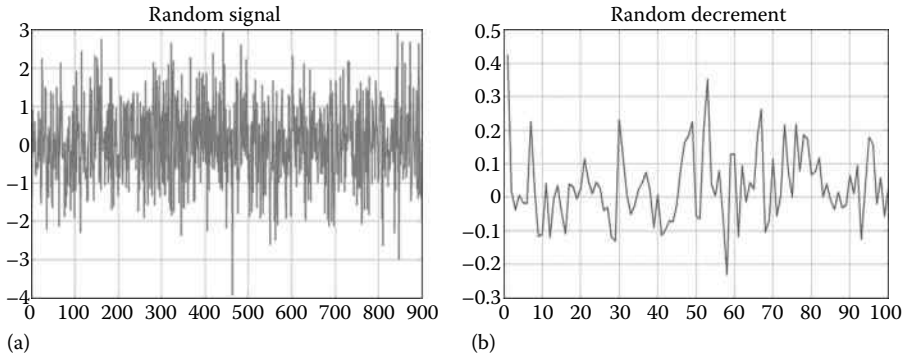


FIGURE 5.8 Example of random decrement. (a) Random process. (b) Free-decay process.

Example 5.1

In this example, we show the original random signal and the free-decay time history of using the random decrement method. Figure 5.8a plots the original signal and Figure 5.8b is the re-generated time history.

5.1.4.2 Lag Superposition (Zero Up-Crossing)

The above-mentioned random decrement method provides a free decay time history by manipulating the random process, which is crucial for many applications of vibration analysis. However, it requires a significantly large magnitude of measurements. The resulting free decay time history can be too short to obtain good results. The following method of lag superposition can significantly reduce the requirement of measurements, while being able to yield virtually the same accuracy.

With the same random time history $Y(t)$, first denote t_i , the i th time point, where $Y(t_i) > 0$ with $Y(t_{i-1}) < 0$. That is, the zero crossing happens in between t_{i-1} and t_i .

Next, take $Y_i(t)$, for the range $t_i \leq t \leq T + t_i$.

Last, let

$$Z_+(t) = \frac{1}{m_+} \sum_{i=1}^{m_+} Y_i(t) \quad (5.56a)$$

where the subscript $+$ stands for zero up-crossing and m_+ is the number of fraction pieces of selected time histories.

Slightly different from the case of random decrement based on level up-crossing, $Z(t)$ can be seen to have two kinds of excitations. The first decrement is due to the random force and the summation that cancels the response. The second decrement is due to the initial velocities at time t_i , because $Y_i(t)$ is taken from the points in time for which $Y(t_i) > 0$.

Directly from the above discussion, the case in which $Y_j(t)$, $t_j \leq t \leq T + t_j$ is taken from $Y(t_{j-1}) > 0$ and $Y(t_j) < 0$ can also be considered. By changing the sign and placing the time history inside the sum, the result is

$$Z_{-}(t) = \frac{1}{m_{-}} \sum_{j=1}^m Y_j(t) \quad (5.56b)$$

where the subscript $-$ stands for zero down-crossing and m_{-} is the number of fraction pieces of selected time histories.

Based on the rate of zero up-crossing given by Equation 5.26 and the rate of level up-crossing given by Equation 5.24, we can calculate the ratio of $v_{0^{+}}/v_{a^{+}}$

$$v_{0^{+}}/v_{a^{+}} = \left[\frac{1}{2\pi} \frac{\sigma_{\dot{x}}}{\sigma_x} \right] \bigg/ \left[\frac{1}{2\pi} \frac{\sigma_{\dot{x}}}{\sigma_x} e^{-\frac{1}{2} \frac{a^2}{\sigma_x^2}} \right] = e^{\frac{1}{2} \frac{a^2}{\sigma_x^2}} \quad (5.57)$$

since $\frac{1}{2} \frac{a^2}{\sigma_x^2} > 0$

$$v_{0^{+}} > v_{a^{+}} \quad (5.58)$$

This ratio is always greater than 1, and it can be seen that such a ratio can be quite large.

Example 5.2

In this example, we have shown the original random signal and the free-decay time history of using the leg superposition method. Figure 5.9a plots the original signal and Figure 5.9b is the re-generated time history.

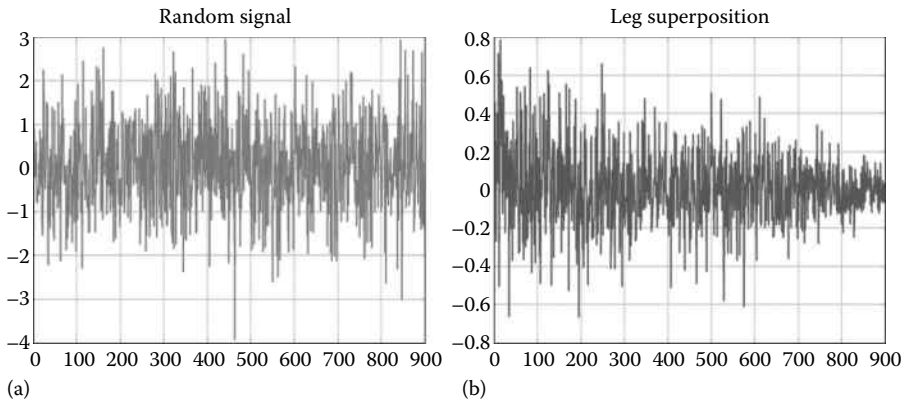


FIGURE 5.9 Example of lag superposition. (a) Random process. (b) Free-decay process.

5.1.4.3 Lag Superposition (Peak Reaching)

The above-mentioned lag superposition can have a considerably large amount of averaging based on the same piece of random time history $Y(t)$, compared with the random decrement method because the crossing rate v_{0+} is generally much higher than the crossing rate v_{a+} . In addition, because we can use both Equations 5.56a and 5.56b, the total number of selected time histories is doubled.

We can also have an additional choice by picking up the pieces of time histories from each peak of the process $Y(t)$. Namely, with the same random time history $Y(t)$, first denote t_i , the i th time point, where $Y(t_{i-1}) < Y(t_i)$ and $Y(t_i) > Y(t_{i+1})$. Namely, the peak is reached between t_{i-1} and t_{i+1} .

Next, take $Y_i(t)$, for the range $t_i \leq t \leq T + t_i$.

Last, let

$$Z_+(t) = \frac{1}{m_+} \sum_{i=1}^{m_+} Y_i(t) \quad (5.59a)$$

where the subscript + stands for the positive value of the peak and m_+ is the number of fraction pieces of selected time histories.

Similarly, we can also pick up the pieces of time histories from each valley of the process $Y(t)$. Namely, with the same random time history $Y(t)$, first denote t_i , the i th time point, where $Y(t_{i-1}) > Y(t_i)$ and $Y(t_{i+1}) > Y(t_i)$. Namely, the valley is reached between t_{i-1} and t_{i+1} .

Next, take $Y_i(t)$, for the range $t_i \leq t \leq T + t_i$.

Last, let

$$Z_-(t) = \frac{1}{m_-} \sum_{i=1}^{m_-} Y_i(t) \quad (5.59b)$$

where the subscript – stands for the negative value of the valley and m_- is the number of fraction pieces of selected time histories.

Similar to the case of random decrement based on level up-crossing, $Z(t)$ also has three kinds of excitations. The first one is due to the random force and the summation that cancels the response. The second is due to the initial velocities. This portion of $Z(t)$ due to the initial velocities will also, respectively, cancel each other. The third decrement is the summation of the responses due to multiple levels of step functions; each is excited by a specific level of the initial displacement at the peaks.

Now, let us consider the numbers of useful pieces of selected time histories based on peak reaching and zero-crossing. It is seen that this ratio is the reciprocal of the irregularity factor α , that is,

$$\frac{v_p}{v_{0+}} = \frac{1}{\alpha} \quad (5.60)$$

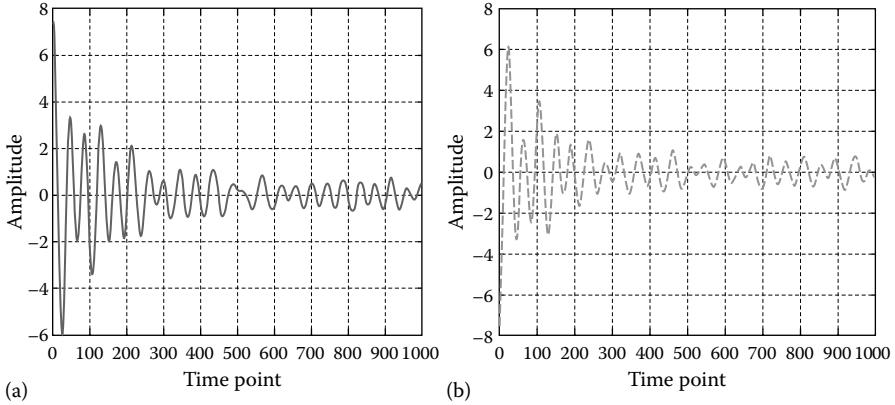


FIGURE 5.10 Free-decay time histories generated from lag superposition (peak reaching). (a) Peak reaching. (b) Valley reaching.

Because in most $\alpha < 1$, this ratio is usually greater than 1. Additionally, it is seen that such a ratio can be rather large.

$$v_p > v_{0+} \quad (5.61)$$

Example 5.3

The following example shows a free-decay time history generated through lag supposition methods based on peak/valley reaching methods. The results are plotted in Figure 5.10a and b.

5.1.5 LEVEL CROSSING IN CLUSTERS

In Sections 5.1.1 through 5.1.4, the main focus was on the crossing pair. Now, cluster crossing will be considered.

5.1.5.1 Rice's Narrow-Band Envelopes (Stephen O. Rice, 1907–1986)

Before discussing cluster crossing, a useful approach to deal with narrow-band processing is described. It is used in comparing the aforementioned to “crossing pairs” because we likely will have two arrivals per cycle in a cluster. Subsequently, there can be many arrivals. However, between clusters, there will be zero arrivals. This results in waiting times not being exponentially distributed (see Figure 5.11).

For the purpose of engineering design, often the first time that $X(t)$ crosses the level $x = a$ is of interest. From this, the random process of the “envelope” will be defined and the up-cross rate of the envelope will be determined.

For the case, $X(t)$ is stationary narrow-band Gaussian. Suppose,

$$X(t) = R(t) \cos[\omega_0 t + \Theta(t)] \quad (5.62)$$

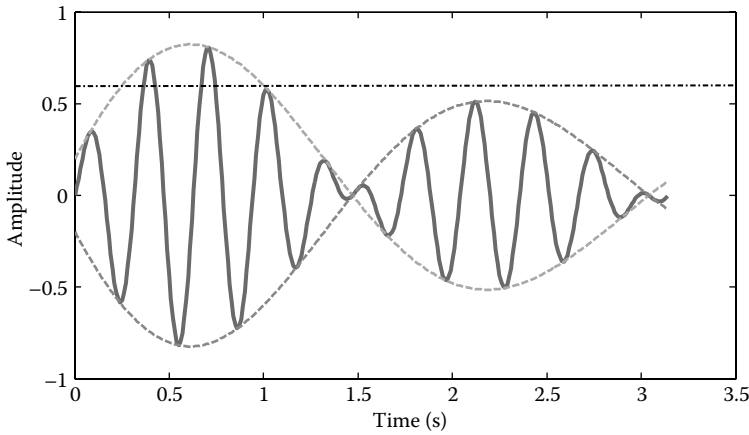


FIGURE 5.11 Example of Rice's narrow-band envelope.

Here, ω_0 is the center frequency of the narrow band, referring back to Equation 4.66 and the frequency ω_0 ; the random process $R(t)$ is the envelope, and

$$R(t) > 0 \quad (5.63)$$

The phase angle $\Theta(t)$ is an additional random process, where

$$0 < \Theta(t) \leq 2\pi \quad (5.64)$$

In this case, it is assumed that the variation of both $R(t)$ and $\Theta(t)$ are slower in comparison with $X(t)$.

5.1.5.1.1 Joint Density Function

To use the level-up-crossing formula in Equation 5.23, the joint density function of $R(t)$ and $\dot{R}(t)$ are needed, and they will be discussed in the following.

5.1.5.1.1.1 Trigonometry Equation First, $X(t)$ will be rewritten as

$$X(t) = C(t) \cos(\omega_m t) + S(t) \sin(\omega_m t) \quad (5.65)$$

$C(t)$ and $S(t)$ are independent, identical Gaussian, with a zero mean and variance of σ_X^2 .

$$C(t) = R(t) \cos[\Theta(t)] \quad (5.66)$$

$$S(t) = R(t) \sin[\Theta(t)] \quad (5.67)$$

From Equations 5.66 and 5.67, $C(t)$ and $S(t)$ are determined to be zero-mean.

The derivatives of $C(t)$ and $S(t)$ are

$$\dot{C}(t) = \dot{R}(t) \cos[\Theta(t)] + R(t) \sin[\Theta(t)] \dot{\Theta}(t) \quad (5.68)$$

and

$$\dot{S}(t) = \dot{R}(t) \sin[\Theta(t)] + R(t) \cos[\Theta(t)] \dot{\Theta}(t) \quad (5.69)$$

From Equations 5.68 and 5.69, the derivatives of $C(t)$ and $S(t)$ are also determined to be zero-mean.

5.1.5.1.1.2 Joint PDF of $C(t)$, $S(t)$ and $\dot{C}(t)$, $\dot{S}(t)$ If the one-sided spectral density function, $W_X(f)$, is symmetric about the midband frequency ω_m , then $C(t)$, $S(t)$ and $\dot{C}(t)$ and $\dot{S}(t)$ are all independent. Suppose $\dot{C}(t)$ and $\dot{S}(t)$ have a variance σ_R^2 , which will be determined later, then the joint density function of $C(t)$, $S(t)$, $\dot{C}(t)$ and $\dot{S}(t)$ is given by

$$f_{C\dot{C}S\dot{S}}(c, s, \dot{c}, \dot{s}) \frac{1}{4\pi^2 \sigma_X^2 \sigma_R^2} = \exp \left[-\frac{1}{2} \left(\frac{c^2 + s^2}{\sigma_X^2} + \frac{\dot{c}^2 + \dot{s}^2}{\sigma_R^2} \right) \right] \quad (5.70)$$

where

$$c, s > 0 \quad (5.71)$$

and

$$-\infty < \dot{c}, \dot{s} < \infty \quad (5.72)$$

5.1.5.1.1.3 Joint PDF of $R(t)$, $\Theta(t)$, $\dot{R}(t)$, and $\dot{\Theta}(t)$ By variable transformation, the joint probability density function of $R(t)$, $\Theta(t)$, $\dot{R}(t)$, and $\dot{\Theta}(t)$ is

$$f_{R\Theta\dot{R}\dot{\Theta}}(r, \theta, \dot{r}, \dot{\theta}) = \frac{r^2}{4\pi^2 \sigma_X^2 \sigma_R^2} \exp \left[-\frac{1}{2} \left(\frac{r^2}{\sigma_X^2} + \frac{\dot{r}^2 + r^2 \dot{\theta}^2}{\sigma_R^2} \right) \right] \quad (5.73)$$

where $r > 0$,

$$0 < \theta \leq 2\pi \quad (5.74)$$

and

$$-\infty < \dot{r}, \dot{\theta} < \infty \quad (5.75)$$

If $R(t)$ has a Rayleigh distribution and $\Theta(t)$ has a uniform distribution, then the joint PDF of $R(t)$ and $\dot{R}(t)$ is

$$f_{R\dot{R}}(r, \dot{r}) = \frac{r}{\sqrt{2\pi}\sigma_X^2\sigma_R} \exp\left[-\frac{1}{2}\left(\frac{r^2}{\sigma_X^2} + \frac{\dot{r}^2}{\sigma_R^2}\right)\right], \quad r > 0, \quad -\infty < \dot{r} < \infty \quad (5.76)$$

The joint density function is the product of the two marginal PDF, given by

$$f_R(r) = \frac{r}{\sigma_X^2} \exp\left[-\frac{1}{2}\left(\frac{r^2}{\sigma_X^2}\right)\right], \quad r > 0 \quad (5.77)$$

and

$$f_{\dot{R}}(\dot{r}) = \frac{1}{\sqrt{2\pi}\sigma_R} \exp\left[-\frac{1}{2}\left(\frac{\dot{r}^2}{\sigma_R^2}\right)\right], \quad -\infty < \dot{r} < \infty \quad (5.78)$$

As a result, $R(t)$ and $\dot{R}(t)$ are independent.

5.1.5.1.1.4 Determination of Variance σ_R^2 As previously noted, the variance σ_R^2 should be determined.

The derivative of $X(t)$ is

$$\dot{X}(t) = \dot{C}(t)\cos(\omega_m t) - C(t)\omega_m \sin(\omega_m t) - \dot{S}(t)\sin(\omega_m t) - S(t)\omega_m \cos(\omega_m t) \quad (5.79)$$

Equation 5.79 shows a linear combination of four independent processes. The variance of $\dot{X}(t)$ is determined by

$$\sigma_{\dot{X}}^2 = \sigma_R^2 + \omega_m^2 \sigma_X^2 \quad (5.80)$$

The above equation can be rewritten as

$$\sigma_R^2 = \sigma_{\dot{X}}^2 - \omega_m^2 \sigma_X^2 \quad (5.81)$$

By further analysis and substitution of Equation 5.27, this becomes

$$\sigma_{\dot{X}}^2 = \sigma_X^2 \omega_{0^+}^2 \quad (5.82)$$

where

$$\sigma_R^2 = \sigma_X^2 \omega_{0^+}^2 - \omega_m^2 \sigma_X^2 = \sigma_X^2 (\omega_{0^+}^2 - \omega_m^2) \quad (5.83)$$

Substitutions of Equations 5.28 and 5.29 into Equation 5.80 results in

$$\sigma_R^2 = \int_{-\infty}^{\infty} (\omega^2 - \omega_m^2) S_X(\omega) d\omega \quad (5.84)$$

Equation 5.84 indicates that the term σ_R^2 can be seen as the moment of the special density function $S_X(\omega)$ taken about ω_m , the midband frequency of the process.

When $X(t)$ approaches its narrow-band limit, σ_R^2 tends to zero; this indicates that no variation exists. Distinctively, each envelope of the narrow-band process does not vary, thus becoming deterministic. Likewise, $X(t)$ reduces to a pure deterministic sine wave.

5.1.5.1.2 Expected Rate of Level Up-Crossing

Finally, the rate of level up-crossing for the envelope can be stated as follows. Substitution of the joint PDF of $R(t)$ and $\dot{R}(t)$ into the formula of level-up-crossing, noted by Equation 5.23, results in

$$\begin{aligned} v_{R=a^+} &= \int_{-\infty}^{\infty} \dot{r} f_{R\dot{R}}(a, \dot{r}) d\dot{r} \\ &= \int_{-\infty}^{\infty} \frac{r\dot{r}}{\sqrt{2\pi\sigma_X^2\sigma_R}} \exp\left[-\frac{1}{2}\left(\frac{r^2}{\sigma_X^2} + \frac{\dot{r}^2}{\sigma_R^2}\right)\right] d\dot{r} \end{aligned} \quad (5.85)$$

Consequently,

$$v_{R=a^+} = \frac{a\sigma_R}{\sqrt{2\pi\sigma_X^2}} \exp\left[-\frac{1}{2}\left(\frac{a^2}{\sigma_X^2}\right)\right] \quad (5.86)$$

To eliminate σ_1 , Equation 5.83 is substituted into Equation 5.86, producing an equation of rate

$$v_{R=a^+} = \frac{a}{\sqrt{2\pi\sigma_X}} \sqrt{\omega_{0^+}^2 - \omega_m^2} \exp\left[-\frac{1}{2}\left(\frac{a^2}{\sigma_X^2}\right)\right] \quad (5.87)$$

5.1.5.1.3 Average Clump Size

In using Equation 5.87, the expected rates of envelope crossing of $R(t)$ and crossing of the original Gaussian process $X(t)$ can be compared, through the use of their ratio, which is referred to as the *average clump size* (R.H. Lyon, 1961) given by

$$\langle cs(a) \rangle = \frac{v_{X=a^+}}{v_{R=a^+}} \quad (5.88)$$

Substitution of the two values of crossing rate expressed in Equations 5.23 and 5.86 into Equation 5.87 yields

$$\langle cs(a) \rangle = \frac{\sigma_x}{\sqrt{2\pi}a} \frac{\omega_{0+}}{\sqrt{\omega_{0+}^2 - \omega_m^2}} = \frac{\sigma_x}{a\sqrt{2\pi}\sqrt{1 - \omega_m^2/\omega_{0+}^2}} \quad (5.89)$$

The average clump size can be used to estimate the waiting time of envelope crossing. It is seen that

$$\begin{aligned} E[\text{time of first crossing of } x = a \text{ by } R(t)] \\ = \langle cs(a) \rangle E[\text{time of first crossing of } x = a \text{ by } X(t)] \end{aligned} \quad (5.90)$$

Equation 5.90 shows that if the size is large, there will be a significant waiting time.

5.2 EXTREMA

In the second section of this chapter, the extreme values of certain random processes are considered. By collecting the extreme values, which are no longer processes but variables, and determining the corresponding distributions, the 3D problems are reduced to 2D distributions. Namely, the targets are the probability density functions for specific cases.

5.2.1 DISTRIBUTION OF PEAK VALUES

5.2.1.1 Simplified Approach

5.2.1.1.1 General Process

To study the phenomena of peaks, consider the case of double line crossing (see Powell 1958 and Wirsching et al. 2006).

Seen in Figure 5.12, in crossing the first level

$$x = z \quad (5.91)$$

is denoted by event A and will have a probability of $P(A)$, where

$$v_{z+} dt = P(A) \quad (5.92)$$

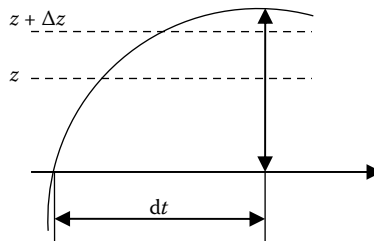


FIGURE 5.12 Double line crossing.

In crossing the second level

$$x = z + \Delta z \quad (5.93)$$

$$E[\text{rate of peaking in } (z, z + \Delta z)] = v_{z^+} - v_{z+\Delta z^+} \quad (5.94)$$

$$\lim_{\Delta z \rightarrow 0} (v_{z^+} - v_{z+\Delta z^+}) = -\frac{d}{dz} v_{z^+} dz \quad (5.95)$$

$$E[\text{rate of peaking in } (z, z + \Delta z)] = E[\text{total rate of peaking}] P[\text{peak in } (z, z + \Delta z)] \quad (5.96)$$

$$\lim_{\Delta z \rightarrow 0} P[\text{peak in } (z, z + \Delta z)] = f_Z(z) dz \quad (5.97)$$

Combining Equations 5.95 and 5.97, results in

$$f_Z(z) dz = \frac{-\frac{d}{dz} v_{z^+} dz}{v_p}$$

Thus, simplifying the above equation produces

$$f_Z(z) = \frac{-\frac{d}{dz} v_{z^+}}{v_p} \quad (5.98)$$

5.2.1.1.2 Narrow-Band Process

5.2.1.1.2.1 General Narrow-Band Process If it is a narrow-band process, the peaking frequency can be replaced by the zero-up-crossing frequency:

$$f_Z(z) = \frac{-\frac{d}{dz} v_{z^+}}{v_{0^+}} \quad (5.99)$$

5.2.1.1.2.2 Gaussian Narrow-Band Process, PDF of Peaks If $X(t)$ is narrow-band Gaussian, then replace a with the variable z in Equation 5.24, and take the derivative with respect to z . Furthermore, substituting Equation 5.126 into the results, we can have the PDF of the peaks written as

$$f_Z(z) = \frac{z}{\sigma_X^2} \exp \left[-\frac{1}{2} \left(\frac{z^2}{\sigma_X^2} \right) \right], \quad z > 0 \quad (5.100)$$

Note that Equation 5.100 is a Rayleigh distribution.

5.2.1.1.2.3 *Gaussian Narrow-Band Process, PDF of Height of the Rise* The height of rise

$$H = 2Z \quad (5.101)$$

can be calculated, for zero-mean narrow-band process, by replacing z with h in Equation 5.100, thus the PDF of the height is given by

$$f_H(h) = \frac{h}{4\sigma_x^2} \exp \left[-\frac{1}{2} \left(\frac{h^2}{4\sigma_x^2} \right) \right], \quad h > 0 \quad (5.102)$$

5.2.1.2 General Approach

5.2.1.2.1 Conditional Probability

Recall the conditional probability:

$$P(B|C) = \frac{P(B \cap C)}{P(C)} \quad (5.103)$$

$$P(\text{peak} = Z | X(t) \text{ is a peak}) = \frac{P[(\text{peak} = Z) \cap (X(t) \text{ is a peak})]}{P(X(t) \text{ is a peak})} \quad (5.104)$$

5.2.1.2.2 Events in Terms of $X(t)$, $\dot{X}(t)$, and $\ddot{X}(t)$

Conditions for a peak of any magnitude are the same as for a zero down-crossing of $\dot{X}(t)$, specifically, at the start of this interval:

$$1. \dot{X}(t) > 0 \quad (5.105)$$

$$2. \ddot{X}(t) < 0 \quad (5.106)$$

3. Correspondingly at the end of the interval,

$$\dot{X}(t + dt) < 0 \quad (5.107)$$

In terms of $\ddot{X}(t)$, the following is true

$$\dot{X}(t) + \ddot{X}(t)dt < 0 \quad (5.108)$$

or

$$\dot{X}(t) < 0 - \ddot{X}(t)dt \quad (5.109)$$

Let C denote the event of having a peak of any magnitude, then $P(C)$ can be written as the combination of the above statements. Specifically, this can be written as

$$P(C) = P\left\{[0 < \dot{X}(t) < 0 - \ddot{X}(t)dt] \cap [\ddot{X}(t) < 0]\right\} \quad (5.110)$$

Let B denote the additional constrain that $X(t) = Z$, that is,

$$z < X(t) \leq z + dz \quad (5.111)$$

Recognizing B to be a subset of C , the joint probability is

$$P(B \cap C) = P\left\{[z < X(t) \leq z + dz] \cap [0 < \dot{X}(t) < 0 - \ddot{X}(t)dt] \cap [\ddot{X}(t) < 0]\right\} \quad (5.112)$$

5.2.1.2.3 General Application

Denote the joint PDF of $X(t)$, $\dot{X}(t)$, and $\ddot{X}(t)$ to be $f_{X\dot{X}\ddot{X}}(u, v, w)$.

$$\begin{aligned} P(\text{peak} = Z \mid X(t) \text{ is a peak}) &= f_Z(z)dz = \frac{\int_{-\infty}^0 \int_0^{0-wdt} \int_z^{z+dz} f_{X\dot{X}\ddot{X}}(u, v, w) du dv dw}{\int_{-\infty}^0 \int_0^{0-wdt} \int_{-\infty}^{\infty} f_{X\dot{X}\ddot{X}}(u, v, w) du dv dw} \\ &= \frac{\int_{-\infty}^0 dz(-w dt) f_{X\dot{X}\ddot{X}}(z, 0, w) dw}{\int_{-\infty}^0 (-w dt) f_{\dot{X}\ddot{X}}(0, w) dw} \\ &= \frac{\int_{-\infty}^0 (-w) f_{X\dot{X}\ddot{X}}(z, 0, w) dw}{\int_{-\infty}^0 (-w) f_{\dot{X}\ddot{X}}(0, w) dw} dz \end{aligned} \quad (5.113)$$

Because the resulting denominator is the peaking frequency v_p , dz can be divided on both sides, resulting in

$$f_Z(z) = \frac{\int_{-\infty}^0 -w f_{X\dot{X}\ddot{X}}(z, 0, w) dw}{v_p} \quad (5.114)$$

Equation 5.114 is a workable solution for zero-mean process with arbitrary bandwidth and distribution.

5.2.1.2.4 Gaussian

The joint PDF of $f_{\ddot{X}\ddot{X}}(u, v, w)$ is, in general, quite complex. However, in the event the displacement $X(t)$ is Gaussian, then the velocity and the acceleration will also be Gaussian. Additionally, suppose each of the three processes are independent, then the joint PDF of $f_{\ddot{X}\ddot{X}}(u, v, w)$ can be simplified to $f_{\ddot{X}\ddot{X}}(u, v, w) = f_{\ddot{X}}(u) f_{\ddot{X}}(v) f_{\ddot{X}}(w)$. Substitution of the PDF into Equation 5.114 yields

$$f_Z(z) = (1 - \alpha^2) \frac{1}{\sqrt{2\pi(1 - \alpha^2)\sigma_X}} \exp\left[-\frac{1}{2} \frac{z^2}{(1 - \alpha^2)\sigma_X^2}\right] + \alpha \Phi\left(\frac{\alpha z}{(1 - \alpha^2)\sigma_X}\right) \frac{z}{\sigma_X^2} \exp\left[-\frac{1}{2} \frac{z^2}{\sigma_X^2}\right], \quad -\infty < z < \infty \quad (5.115)$$

Here, α is the irregularity factor and $\Phi(\cdot)$ is the cumulative of the standard normal distribution.

5.2.2 ENGINEERING APPROXIMATIONS

5.2.2.1 Background

In the above discussion, it was assumed that the PDF was known. Realistically, this assumption is seldom true. When the PDF is not known, the approximate distributions are then used (see Rice [1964], Yang [1974], Krenk [1978], and Tayfun [1981]). Random rises and falls are useful in studying fatigue problems, but the corresponding exact PDF in most cases is not known. Fortunately, the average rise can be easily obtained. In this subsection, the issue of rise and fall will be examined. For engineering applications, a certain percentage error may be tolerated to find workable solutions. In the following, comparative loss assumptions are made.

5.2.2.1.1 Basic Concept

Rise H_i is the difference between $X(t)$, denoted by $X(t_i)$ at the valley and $X(t)$, denoted by $X(t'_i)$ at the next peak. The value of h_i can be expressed as (see Figure 5.13)

$$h_i = \max X(t'_i) - \min X(t_i) \quad (5.116)$$

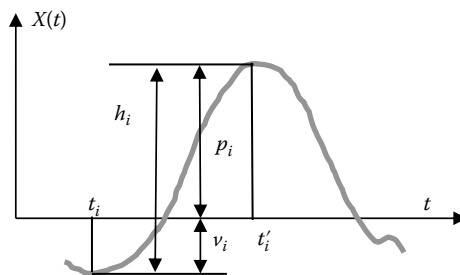


FIGURE 5.13 Peaks and valleys.

In this case, t_i is used to denote the event when $X(t)$ reaches a valley. To denote the height H is once more used.

$$H = \{h_i\} \quad (5.117)$$

Here, H is a random set.

For stationary process, the PDF of the rise and fall is a symmetric function. Typically, the peak distribution of a Gaussian process is known, but the PDF of the rise is unknown. To determine the time of the next peak is a first-passenger time problem, given

$$\dot{X}(0) = 0 \quad (5.118)$$

To find the next zero crossing is more difficult.

5.2.2.1.2 Average Rise

In the duration of Δt , the average traveling distance of $|X(t)|$ is equal to $E[|\dot{X}(t)|]\Delta t$. This distance is also equal to the average height $2\mu_H$ times the number of ups and down, denoted by $\nu_p\Delta t$. This can be expressed as follows:

$$\mu_H = \frac{E[|\dot{X}(t)|]}{2\nu_p} \quad (5.119)$$

It can be proven that if the process is Gaussian, then the average height is

$$\mu_H = \alpha\sqrt{2\pi}\sigma_X \quad (5.120)$$

5.2.2.1.3 Shape Function

Now, consider the trajectory between a valley and the subsequent peak, referred to as the shape or trajectory function. With a simple shape function, the analysis becomes easier.

5.2.2.1.3.1 Displacement Trajectory The trajectory function is assumed to be sinusoidal, which is one of the simplest functions but can be a good approximation of the trajectory path, especially for narrow-band processes, that is,

$$\Psi(t) = -H/2 \cos(\omega_p t) \quad (5.121)$$

where ω_p is the frequency of the assumed sinusoidal fluctuation of $X(t)$. Note that in this case,

$$H > 0$$

5.2.2.1.3.2 *Velocity* To solve for the velocity, the derivative of Equation 5.121 is taken.

$$\dot{\Psi}(t) = \omega_p (H/2) \sin(\omega_p t) \quad (5.122)$$

At the valley, the conditional distribution of $X(t)$ is $f_z(-z)$, and

$$\dot{X}(t_i) = 0 \quad (5.123)$$

5.2.2.1.3.3 *Acceleration* In addition, the acceleration can be calculated as

$$\ddot{\Psi}(t) = \omega_p^2 (H/2) \cos(\omega_p t) = A \cos(\omega_p t) \quad (5.124)$$

Here,

$$A = \frac{\omega_p^2 H}{2} \quad (5.125)$$

where A is a random variable representing the amplitude of the acceleration.

5.2.2.2 Probability Distributions of Height, Peak, and Valley

Thus far, the distribution of the height, peak, and valley have been considered, based on the above-mentioned simplification of shape functions.

5.2.2.2.1 PDF of $\dot{X}(t_i)$

5.2.2.2.1.1 *PDF of A , General Process* The distribution of $\ddot{X}(t_i)$, the acceleration at the valley, can be found by using the same approach as used in Section 5.1. For this approach, first define the conditions for $X(t_i)$, then calculate the PDF for $\ddot{X}(t_i)$.

It can be proven that

$$f_A(a)da = \frac{\int_a^{a+da} \int_{0-w}^0 \int_{-\infty}^{\infty} f_{\ddot{X}\ddot{X}\ddot{X}}(u, v, w) du dv dw}{\int_0^{\infty} \int_{0-w}^0 \int_{-\infty}^{\infty} f_{\ddot{X}\ddot{X}\ddot{X}}(u, v, w) du dv dw}, \quad a > 0 \quad (5.126)$$

Thus, in simplifying

$$f_A(a) = \frac{a f_{\ddot{X}\ddot{X}}(0, a)}{v_p}, \quad a > 0 \quad (5.127)$$

The PDF of A for a Gaussian process is

$$f_A(a) = \frac{a}{\sigma_{\ddot{x}}^2} e^{-\frac{a^2}{2\sigma_{\ddot{x}}^2}}, \quad a > 0 \quad (5.128)$$

Note that Equation 5.128 is a Rayleigh distribution with $\sigma_{\ddot{x}}$ as the parameter σ . For further explanation, refer to Equation 1.94. This results in

$$\sigma_{\ddot{x}} = \sigma \quad (5.129)$$

From Equation 5.125,

$$H = \frac{2A}{\omega_p^2} \quad (5.130)$$

Thus,

$$f_H(h) = \frac{h}{\theta_H^2} e^{-\frac{h^2}{2\theta_H^2}}, \quad h > 0 \quad (5.131)$$

where,

$$\theta_H = \frac{2\sigma_{\ddot{x}}}{\omega_p^2} \quad (5.132)$$

The above parameter can also be written as

$$\theta_H = 2\sigma_{\dot{x}}a \quad (5.133)$$

Furthermore, $f_H(h)$ can be written as

$$f_H(h) = \frac{h}{(2\sigma_{\dot{x}}a)^2} e^{-\frac{h^2}{2(2\sigma_{\dot{x}}a)^2}}, \quad h > 0 \quad (5.134)$$

For Equation 5.134, when

$$a = 1 \quad (5.135)$$

which is the case of a narrow-band process, refer to Equation 5.102.

5.2.2.2.2 Joint Distribution of Height, Peak, and Valley

5.2.2.2.2.1 *Peak and Valley* As seen in Figure 5.14, the random variable valley V and peak P can be denoted, explicitly

$$V = \{v_i\} \quad (5.136)$$

and

$$P = \{p_i\} \quad (5.137)$$

The midpoint between adjoining valleys and peaks is denoted by

$$M = 1/2(V + P) = \{1/2(v_i + p_i)\} = \{m_i\} \quad (5.138)$$

For convenience, the subscript i will be omitted in the following equations: consider the example in which the joint distribution of the height, peak, and valley can be used to count the fatigue cycles that rise above a floor level set by a crack opening stress (Perng 1989).

5.2.2.2.2.2 Joint PDF of H and V

$$\begin{aligned} f_{HV}(h, v) &= \frac{1}{\sqrt{2\pi(1-\alpha^2)\sigma_x}} \exp\left[-\frac{1}{2} \frac{(v+h/2)^2}{(1-\alpha^2)\sigma_x^2}\right] \frac{h}{4\alpha^2\sigma_x^2} \exp\left[-\frac{1}{2} \frac{h^2}{4\alpha^2\sigma_x^2}\right] = \\ &= \frac{1}{\sqrt{1-\alpha^2}\sigma_x} \Phi\left(\frac{v+h/2}{\sqrt{1-\alpha^2}\sigma_x}\right) \frac{h}{4\alpha^2\sigma_x^2} \exp\left[-\frac{1}{2} \frac{h^2}{4\alpha^2\sigma_x^2}\right], \quad 0 < h < \infty, -\infty < v < \infty \end{aligned} \quad (5.139)$$

5.2.2.2.2.3 Joint PDF of H and P

$$\begin{aligned} f_{HP}(h, p) &= \frac{1}{\sqrt{2\pi(1-\alpha^2)\sigma_x}} \exp\left[-\frac{1}{2} \frac{(p-h/2)^2}{(1-\alpha^2)\sigma_x^2}\right] \frac{h}{4\alpha^2\sigma_x^2} \exp\left[-\frac{1}{2} \frac{h^2}{4\alpha^2\sigma_x^2}\right] = \\ &= \frac{1}{\sqrt{1-\alpha^2}\sigma_x} \Phi\left(\frac{p-h/2}{\sqrt{1-\alpha^2}\sigma_x}\right) \frac{h}{4\alpha^2\sigma_x^2} \exp\left[-\frac{1}{2} \frac{h^2}{4\alpha^2\sigma_x^2}\right], \quad 0 < h < \infty, -\infty < p < \infty \end{aligned} \quad (5.140)$$

5.2.2.2.2.4 Joint PDF of H and M

$$\begin{aligned} f_{HM}(h, m) &= \frac{1}{\sqrt{2\pi(1-\alpha^2)\sigma_x}} \exp\left[-\frac{1}{2} \frac{m^2}{(1-\alpha^2)\sigma_x^2}\right] \frac{h}{4\alpha^2\sigma_x^2} \exp\left[-\frac{1}{2} \frac{h^2}{4\alpha^2\sigma_x^2}\right] = \\ &= \frac{1}{\sqrt{1-\alpha^2}\sigma_x} \Phi\left(\frac{m}{\sqrt{1-\alpha^2}\sigma_x}\right) \frac{h}{4\alpha^2\sigma_x^2} \exp\left[-\frac{1}{2} \frac{h^2}{4\alpha^2\sigma_x^2}\right], \quad 0 < h < \infty, -\infty < m < \infty \end{aligned} \quad (5.141)$$

Notice that the term

$$\frac{1}{\sqrt{1-\alpha^2}\sigma_x}\Phi\left(\frac{m}{\sqrt{1-\alpha^2}\sigma_x}\right) \quad (5.142)$$

does not contains variable h .

Furthermore, the term

$$\frac{h}{4\alpha^2\sigma_x^2}\exp\left[-\frac{1}{2}\frac{h^2}{4\alpha^2\sigma_x^2}\right] \quad (5.143)$$

does not contains variable m .

Hence, the H and M are independent.

5.2.2.2.2.5 Joint PDF of P and V

$$\begin{aligned} f_{pV}(p, v) &= \frac{1}{\sqrt{2\pi(1-\alpha^2)}\sigma_x} \exp\left[-\frac{(p+v)^2}{8(1-\alpha^2)\sigma_x^2}\right] \frac{p-v}{4\alpha^2\sigma_x^2} \exp\left[-\frac{(p-v)^2}{8\alpha^2\sigma_x^2}\right] = \\ &= \frac{1}{\sqrt{2\pi(1-\alpha^2)}\sigma_x} \Phi\left(\frac{p+v}{2\sqrt{(1-\alpha^2)}\sigma_x}\right) \frac{p-v}{4\alpha^2\sigma_x^2} \exp\left[-\frac{(p-v)^2}{8\alpha^2\sigma_x^2}\right], \quad -\infty < v < p < \infty \end{aligned} \quad (5.144)$$

In the equation above, each joint PDF is the product of a Gaussian term and a Rayleigh term.

5.3 ACCUMULATIVE DAMAGES

In Sections 5.1 and 5.2, we examined certain important statistical properties rather than typical mean, variance, and correlation functions only. These studies imply that, for certain types of random processes, we can have issues beyond typical statistics. It is known that, for a general process, the previous conclusion may not be sufficiently accurate; they may not even be workable. Therefore, specific processes such as the stationary Gaussian process are assumed. In the following, we will use an engineering problem as an example to show that if certain statistical conclusions are needed, we need to select a proper model of random process. In this case, the Markov process will be used.

Accumulative damages are often seen in engineering structures with repeated loading, such as vibration displacements or unbalanced forces, which are closely related to the phenomena of level crossing. Material fatigue is a typical example of damage accumulation. As mentioned at the beginning of this chapter, the focus here is given to the time-varying developments of the accumulative damage, instead of the resulting damage itself. In Chapter 10, such damages will

be further analyzed to study the nature of accumulative failures. To discuss the time-varying process, two major approaches are considered: deterministic models and random process.

The theory based on deterministic approach assumes that a stress cycle with an alternating stress above the endurance limit inflicts a measurable permanent damage. It also assumes that the total damage caused by a number of stress cycles is equal to the summation of damages caused by the individual stress cycles.

Although the deterministic approach is simple and widely used with fairly accepted accuracy, it cannot assess uncertainty in fatigue life and dependency between the current and future damages (the cascading effect). In this section, we will first introduce the Markov process, a useful model of accumulative damage. Then, the problem of fatigue based on the approach of random process will be discussed. Finally, the concept of cascading damage will be briefly mentioned.

In the literature, there are many excellent works published. Among them, Dowling (1993) summarized basic approaches of fatigues. Collins (1981) comprehensively reviewed the models used to estimate fatigue life.

5.3.1 LINEAR DAMAGE RULE: THE DETERMINISTIC APPROACH

Let us first review the widely accepted damage rule based on a deterministic approach for the purpose of comparison.

5.3.1.1 S–N Curves

When a component of a machine or a structure is subjected to high-cycle loading, although the load level is smaller than its yielding threshold, after certain cycles, it may fail to take additional loads. The number of cycles is referred to as fatigue lifetime. Such fatigue is called high-cycle fatigue, or simply fatigue.

Generally speaking, S – N curves are used in a high-cycle fatigue study. An S – N curve for a material defines alternating stress values versus the number of duty cycles required to cause failure at a given stress ratio. A typical S – N curve is shown in the figure. The y axis represents the alternating stress (S) and the x axis represents the number of cycles (N). An S – N curve is based on a stress ratio or mean stress. One can define multiple S – N curves with different stress ratios for a material. The software uses linear interpolation to extract data when you define multiple S – N curves for a material.

S – N curves are based on mean fatigue life or a given probability of failure. Generating an S – N curve for a material requires many tests to statistically vary the alternating stress, mean stress (or stress ratio), and count the number of duty cycles.

5.3.1.2 Miner's Rule

In 1945, M.A. Miner popularized a rule that had first been proposed by A. Palmgren in 1924, which is variously called *Miner's rule* or the *Palmgren–Miner linear damage hypothesis*. Consider the S – N curve shown in Figure 5.14. Suppose that it takes N_1 duty cycles at an alternating stress S_1 to cause fatigue failure; then the theory

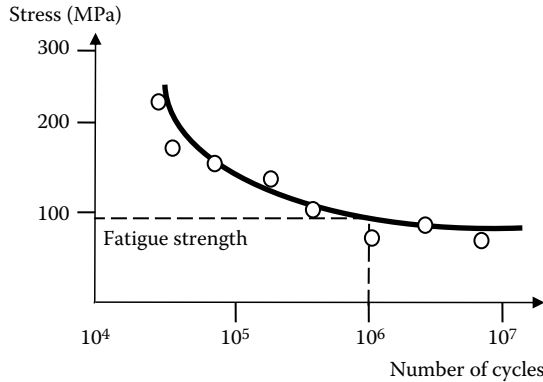


FIGURE 5.14 Conceptual S – N curve.

states that each cycle causes a damage factor D_1 that consumes $1/N_1$ of the life of the structure.

Moreover, if a structure is subjected to N_1 duty cycles at S_1 alternating stress and N_2 duty cycles at S_2 alternating stress, then the total damage factor D is calculated as

$$D = (n_1/N_1 + n_2/N_2) \quad (5.145)$$

where N_1 is the number of cycles required to cause failure under S_1 , and N_2 is the number of cycles required to cause failure under S_2 .

The damage factor D , also called usage factor, represents the ratio of the consumed life of the structure. A damage factor of 0.35 means that 35% of the structure's life is consumed. Failure due to fatigue occurs when the damage factor reaches 1.0.

The linear damage rule does not consider the effects of load sequence. In other words, it predicts that the damage caused by a stress cycle is independent of where it occurs in the load history. It also assumes that the rate of damage accumulation is independent of the stress level. Observed behavior indicates that cracks initiate in a few cycles at high stress amplitudes, whereas almost all the life is spent on initiating the cracks at low stress amplitudes.

The linear damage rule is used in its simple form when you specify that fatigue events do not interact with each other in the properties of the study. When you set the interaction between events to random, the program uses the ASME code to evaluate the damage by combining event peaks.

5.3.2 MARKOV PROCESS

To better understand when the accumulative damage occurs as a random process, let us describe a useful model, the Markov process (Andrey A. Markov, 1856–1922).

A random process whose present state is dependent on its past history, the Markov process is one of the most important processes that plays an important role in engineering applications. The aforementioned Poisson and Wiener processes (Brownian motion) are all Markovian. Many physical phenomena, including

communication networks, signal processes, transportation arrangements, structural failures by multiple natural hazards, and others, can be approximated by Markov processes.

In this subsection, we introduce and discuss the basic concepts and properties of Markov processes, mainly focusing on the discrete Markov chain, which can be mathematically intensive. However, to study the main objective of this section, accumulative damage, readers can skip these mathematical descriptions and directly consider the resulting conclusions in Sections 5.3.3 and 5.3.4.

5.3.2.1 General Concept

5.3.2.1.1 Definition

A random process $X(t)$, $0 \leq t \leq T$ is said to be a *Markov process*, if for every n and for $t_1 < t_2 < \dots < t_n \leq T$, we can have the distribution given by

$$F(x_n, t_n | x_{n-1}, \dots, x_1; t_{n-1}, \dots, t_1) = F(x_n, t_n | x_{n-1}, t_{n-1}) \quad (5.146)$$

If the process is continuous, then we can have PDF given by

$$f(x_n, t_n | x_{n-1}, \dots, x_1; t_{n-1}, \dots, t_1) = f(x_n, t_n | x_{n-1}, t_{n-1}) \quad (5.147)$$

The above equations imply that a Markov process represents a set of trajectories whose conditional probability distribution at a selected instance, given all past observations, only depends on the most recent ones. For example, the fatigue damage at a given time point t_2 depends only on the state of time t_1 ; anything before t_1 , however, has no influence on the damage level at t_2 .

Equation 5.146 is equivalent to the following conditional probability

$$\begin{aligned} P\{X(t_n) < x_n, | X(t_1) < x_1, X(t_2) < x_2, \dots, X(t_{n-1}) < x_{n-1}\} \\ = P\{X(t_n) < x_n, | X(t_{n-1}) < x_{n-1}\} \end{aligned} \quad (5.148)$$

5.3.2.2 Discrete Markov Chain

5.3.2.2.1 Definition

If the process is a discrete random sequence (discrete state, discrete time), it is referred to as a *discrete Markov chain*. If the process has continuous time but the state is discrete, it is called a *continuous Markov chain*.

Example 5.4

Suppose $\{X(n), n = 1, 2, \dots\}$ is a mutually independent random sequence, and

$$Y(t_q) = \sum_{k=1}^q X_k, \quad t_1 < t_2 < \dots$$

Show that $Y(t)$ is a Markov process.

To prove that $Y(t)$ is Markovian, write the sequence in a recursive form given by

$$Y(t_q) = Y(t_{q-1}) + X_q, \quad q = 2, 3, \dots$$

Because X_q are independent variables, the properties of $Y(t)$ at t_q are a function of those at t_{q-1} only. Therefore, $Y(t)$ is Markovian. This example implies that a mutually independent random process is Markovian. Furthermore, an independent increment random process is also a Markov process. Denote $X(t)$, $t > 0$ with $P\{X(0) = 0\} = 1$. At any $t_q \geq 0$, we have

$$X(t_q) = X(t_q) - X(0) = \sum_{k=1}^q [X(t_k) - X(t_{k-1})] = \sum_{k=1}^q \Delta X(t_{k-1}, t_k), \quad t_0 = 0$$

Note that $\Delta X(t_{k-1}, t_k)$, $k = 1, \dots, j$ are independent random variables. Therefore, the above equation shows that their sum, the independent increment process, is Markovian. The Poisson process is an independent increment, so that it is a Markov process.

5.3.2.2.2 Transition Probability

Suppose $\{X(n), n = 0, 1, 2, \dots\}$ is a discrete Markov chain. The following probability

$$p_{ij}(n, k) = P\{X(n+k) = j \mid X(n) = i\}, \quad n \geq 0, k \geq 1 \quad (5.149)$$

is called the k steps (k th power) transition probability of $\{X(n), n = 0, 1, 2, \dots\}$ at instant n .

Furthermore, the k steps transition probability matrix of $\{X(n), n = 0, 1, 2, \dots\}$ at instant n is defined as

$$\mathbf{P}(n, k) = [p_{ij}(n, k)]_{i, j \in \Omega} \quad (5.150)$$

where

$$\Omega = \{x: X(t) = x, 0 < t < T\} \quad (5.151)$$

is called the state space of the Markov process.

Particularly, when $k = 1$, at time instant n , the one-step transition probability and transition probability matrix are denoted by $p_{ij}(n)$ and $\mathbf{P}(n)$.

Example 5.5

$\{X(n), n = 1, 2, \dots\}$ is an independent random sequence of positive integers, whose distribution is given by

$X(k)$	1	2	3	...	k
PMF	p_{k1}	p_{k2}	p_{k3}	...	p_{kk}

Denote

$$Y(n) = \sum_{k=1}^n X(k), n = 1, 2, \dots$$

Show that $\{Y(n), n = 1, 2, \dots\}$ is a Markov chain and find its transition probability matrix.

From the equation that defines sequence $Y(n)$, it is seen that the increment $[Y(n) - Y(n - 1)]$, that is, $X(n)$, and increment $[Y(m) - Y(m - 1)]$, that is, $X(m)$, are independent because $X(n)$ and $X(m)$ are independent. Therefore, $\{Y(n), n = 1, 2, \dots\}$ is an independent increment process, and thus it is a Markov chain.

Furthermore, the entry of the corresponding transition probability matrix can be written as

$$\begin{aligned} P_{ij}(m, k) &= P\{Y(m+k) = j \mid Y(m) = i\} = \\ &= \frac{P\{Y(m) = i, Y(m+k) = j\}}{P\{Y(m) = i\}} = \frac{P\{Y(m) = i\}P\{Y(m+k) - Y(m) = j - i\}}{P\{Y(m) = i\}} \\ &= P\left\{\sum_{r=m+1}^{m+k} X(r) = j - i\right\} = \sum_{i_1 + i_2 + \dots + i_k = j - i} P_{m+1, i_1} P_{m+2, i_2} \dots P_{m+k, i_k} \end{aligned}$$

where

$$m \leq i \leq \frac{m(m+1)}{2}, m+k \leq j \leq \frac{(m+k)(m+k+1)}{2}, j - i \geq k$$

5.3.2.2.3 Probability Distribution

The initial distribution of a discrete Markov chain $\{X(n), n = 0, 1, 2, \dots\}$ is denoted by $\tilde{\mathbf{P}}_0$, given by

$$\tilde{\mathbf{P}}_0 = \{p_i = P[X(0) = i], i \in \Omega\} \quad (5.152)$$

In addition, absolute distribution of a discrete Markov chain $\{X(n), n = 0, 1, 2, \dots\}$ is denoted by $\tilde{\mathbf{P}}_n$, given by

$$\tilde{\mathbf{P}}_n = \{p_j = P[X(n) = j], j \in \Omega\} \quad (5.153)$$

5.3.2.2.4 Homogeneity

If the one-step transition probability $p_{ij}(n)$ is not related to the initial time n , denoted by $p(n)$, for a discrete Markov chain $\{X(n), n = 0, 1, 2, \dots\}$, then such a discrete Markov chain is homogeneous, whose k steps transition the probability matrix denoted as $p_{ij}(k)$ and $\mathbf{P}(k)$, respectively; and the corresponding one-step transition probability matrix is denoted by $\mathbf{P}$.

Example 5.6

$\{X(n), n = 1, 2, \dots\}$ is a random sequence with independent and identical distributions, and

$$Y(n) = \sum_{k=1}^n X(k)$$

Show that the following $\{Y(n), n = 1, 2, \dots\}$ are homogeneous Markov chains.

1. $\{X(n), n = 1, 2, \dots\}$ is a Bernoulli random sequence with $P\{X(n) = 0\} = q$, $P\{X(n) = 1\} = p$, $0 < q < 1$, $p + q = 1$ and $n = 1, 2, \dots$ and
2. $X(n) \sim N(\mu, \sigma^2)$, $n = 1, 2, \dots$

First, for question (1), consider any instant $0 < m_1 < m_2 < \dots < m_n$, and $0 \leq i_1 \leq i_2 \leq \dots \leq i_n$ that satisfies for any $1 \leq k \leq n-1$, we have $i_k \leq i_{k+1} \leq i_k + m_{k+1} - m_k$, then

$$\begin{aligned}
 P\{Y(m_n) = i_n | Y(m_1) = i_1, \dots, Y(m_{n-1}) = i_{n-1}\} &= P\left\{ \sum_{k=1}^{m_n} X(k) = i_n \left| \sum_{k=1}^{m_1} X(k) = i_1, \dots, \sum_{k=1}^{m_{n-1}} X(k) = i_{n-1} \right. \right\} \\
 &= P\left\{ \sum_{k=m_{n-1}+1}^{m_n} X(k) = i_n - i_{n-1} \left| \sum_{k=1}^{m_1} X(k) = i_1, \sum_{k=m_1+1}^{m_2} X(k) = i_2 - i_1, \dots, \sum_{k=m_{n-2}+1}^{m_{n-1}} X(k) = i_{n-1} - i_{n-2} \right. \right\} \\
 &= P\left\{ \sum_{k=m_{n-1}+1}^{m_n} X(k) = i_n - i_{n-1} \right\} P\{Y(m_n) = i_n | Y(m_{n-1}) = i_{n-1}\} \\
 &= P\left\{ \sum_{k=1}^{m_n} X(k) = i_n \left| \sum_{k=1}^{m_{n-1}} X(k) = i_{n-1} \right. \right\} \\
 &= P\left\{ \sum_{k=m_{n-1}+1}^{m_n} X(k) = i_n - i_{n-1} \left| \sum_{k=1}^{m_{n-1}} X(k) = i_{n-1} \right. \right\} \\
 &= P\left\{ \sum_{k=m_{n-1}+1}^{m_n} X(k) = i_n - i_{n-1} \right\}
 \end{aligned}$$

The above equation implies that

$$\begin{aligned}
 &P\{Y(m_n) = i_n | Y(m_1) = i_1, \dots, Y(m_{n-1}) = i_{n-1}\} \\
 &= P\{Y(m_n) = i_n | Y(m_{n-1}) = i_{n-1}\}
 \end{aligned}$$

Therefore, $\{Y(n), n = 1, 2, \dots\}$ is a Markov chain, with transition probability being

$$\begin{aligned} p_{ij}(n, k) &= P\{Y(n+K) = j \mid Y(n) = i\} = P\{Y(n+K) - Y(n) = j - i \mid Y(n) = i\} \\ &= P\{Y(k) = j - i\} = P\left\{\sum_{m=1}^k X(m) = j - i\right\} \end{aligned}$$

It is seen that this probability is not related to n ; therefore, $\{Y(n), n = 1, 2, \dots\}$ is a homogeneous Markov chain.

Second, for question (2), consider any instant $0 < m_1 < m_2 < \dots < m_n$, and $0 \leq i_1 \leq i_2 \leq \dots \leq i_n \in \Omega$ that

$$\begin{aligned} P\{Y(m_n) = i_n \mid Y(m_1) = i_1, \dots, Y(m_{n-1}) = i_{n-1}\} &= P\left\{\sum_{k=1}^{m_n} X(k) < i_n \mid \sum_{k=1}^{m_1} X(k) = i_1, \dots, \sum_{k=1}^{m_{n-1}} X(k) = i_{n-1}\right\} \\ &= P\left\{\sum_{k=m_{n-1}+1}^{m_n} X(k) < i_n - i_{n-1} \mid \sum_{k=1}^{m_1} X(k) = i_1, \sum_{k=m_1+1}^{m_2} X(k) = i_2 - i_1, \dots, \sum_{k=m_{n-2}+1}^{m_{n-1}} X(k) = i_{n-1} - i_{n-2}\right\} \\ &= \int_0^{i_n - i_{n-1}} f \sum_{k=m_{n-1}+1}^{m_n} X(k) \left| \sum_{k=1}^{m_1} X(k), \sum_{k=m_{n-2}+1}^{m_{n-1}} X(k) \right. (x_n \mid i_1, \dots, i_{n-1} - i_{n-2}) dx_n \\ &= \int_0^{i_n - i_{n-1}} \frac{f(i_1, \dots, i_{n-1} - i_{n-2}, x_n)}{f(i_1, \dots, i_{n-1} - i_{n-2})} dx_n \\ &= \int_0^{i_n - i_{n-1}} f(x_n) dx_n \end{aligned}$$

And furthermore, we have

$$\begin{aligned} P\{Y(m_n) < i_n \mid Y(m_{n-1}) = i_{n-1}\} &= P\left\{\sum_{k=1}^{m_n} X(k) < i_n \mid \sum_{k=1}^{m_{n-1}} X(k) = i_{n-1}\right\} \\ &= P\left\{\sum_{k=m_{n-1}+1}^{m_n} X(k) < i_n - i_{n-1} \mid \sum_{k=1}^{m_{n-1}} X(k) = i_{n-1}\right\} \\ &= \int_0^{i_n - i_{n-1}} f \sum_{k=m_{n-1}+1}^{m_n} X(k) \left| \sum_{k=1}^{m_{n-1}} X(k) \right. (x_n \mid i_{n-1}) dx_n \\ &= \int_0^{i_n - i_{n-1}} \frac{f(i_{n-1}, x_n)}{f(i_{n-1})} dx_n \\ &= \int_0^{i_n - i_{n-1}} f(x_n) dx_n \end{aligned}$$

Therefore, $\{Y(n), n = 1, 2, \dots\}$ is a Markov chain. In addition, we have

$$\begin{aligned}
 & P\left\{Y(n+m) < i \mid Y(n) = i\right\} P\left\{\sum_{k=n+1}^{m+n} X(k) < i \mid \sum_{k=1}^n X(k) = j\right\} \\
 &= P\left\{\sum_{k=n+1}^{m+n} X(k) < i - j \mid \sum_{k=1}^n X(k) = j\right\} \\
 &= \int_0^{i_n - i_{n-1}} \frac{\sum_{k=n+1}^{m+n} X(k) \mid \sum_{k=1}^n X(k) = (X_{n+m}, j)}{f_n \sum_{k=1}^n X(k)} dx_{n+m} \\
 &= \int_0^{i-j} \frac{f_{n+m} \sum_{k=n+1}^{m+n} X(k)}{f_n \sum_{k=1}^n X(k)} dx_{n+m} = \int_0^{i-j} \frac{1}{2\pi\sigma\sqrt{m}} e^{-\frac{x^2}{2m\sigma^2}} dx \\
 &= \Phi\left(\frac{i-j}{\sigma\sqrt{m}}\right)
 \end{aligned}$$

It is thus seen that this probability does not relate to the starting time point n ; therefore, $\{Y(n), n = 1, 2, \dots\}$ is a homogeneous Markov process.

5.3.2.2.5 Ergodicity

For a homogeneous discrete Markov chain $\{X(n), n = 0, 1, 2, \dots\}$, if for any state $i, j \in \Omega$, there exists a limit independent to i , such that

$$\lim_{n \rightarrow \infty} p_{ij}(n) = \pi_j > 0, i, j \in \Omega \quad (5.154)$$

then, such a Markov chain is ergodic.

It is seen that a discrete Markov chain can be defined by its state probabilities and transition probabilities. The probability distribution of the initial state is described by n dimensional vector

$$\mathbf{p} = [\pi_1, \pi_2, \dots, \pi_j, \dots], \quad j \in \Omega \quad (5.155a)$$

For convenience, Equation 5.155a can also be denoted as

$$\mathbf{p} = \{\pi_j, j \in \Omega\} \quad (5.155b)$$

Note that based on the nature of probability

$$\sum_{j \in \Omega} \pi_j = 1 \quad (5.156)$$

if the vector $\mathbf{p}$ denotes a probability distribution, it must satisfy Equation 5.156.

5.3.2.2.6 Stationary Distribution of a Homogeneous Discrete Markov Chain

A homogeneous discrete Markov chain, $\{X(n), n = 0, 1, 2, \dots\}$, is stationary, provided that there exists $\{v_j, j \in \Omega\}$ such that

$$1. v_j \geq 0 \quad (5.157)$$

$$2. \sum_{j \in \Omega} v_j = 1 \quad (5.158)$$

$$3. v_j = \sum_{i \in \Omega} v_i p_{ij} \quad (5.159)$$

In this case, the vector

$$\mathbf{v} = \{v_j, j \in \Omega\} \quad (5.160)$$

is the stationary distribution of this Markov chain, and we can write

$$\mathbf{vP} = \mathbf{v} \quad (5.161)$$

5.3.2.2.7 Main Properties of a Homogeneous Discrete Markov Chain

1. C–K Equation

For the Markov chain, the Chapman–Kolmogorov (C–K) equation is an identity relating the joint probability distributions of different sets of coordinates on a random process (Sydney Chapman, 1888–1970), given by

$$p_{ij}(k+q) = \sum_{r \in \Omega} p_{ir}(k) p_{rj}(q) \quad (5.162)$$

$$2. \mathbf{P}(n) = \mathbf{P}^n \quad (5.163)$$

3. The absolute distribution is determined by the initial distribution and transition probability, and the following equation is satisfied

$$p_j(n) = \sum_{i \in \Omega} p_i p_{ij}(n) \quad (5.164)$$

Equation 5.164 can be expressed in matrix form, that is,

$$\tilde{\mathbf{P}}_n = \tilde{\mathbf{P}}_0 \mathbf{P}^n \quad (5.165)$$

4. The limited dimensional distributions can be determined by the initial distribution and transition probability, and the following equation is satisfied

$$P\{X(n_1) = i_1, X(n_2) = i_2, \dots, X(n_k) = i_k\} \\ = \sum_{i \in \Omega} p_i p_{i i_1}(n_1) p_{i_1 i_2}(n_2 - n_1) \dots p_{i_{k-1} i_k}(n_k - n_{k-1}) \quad (5.166)$$

5. Suppose the state space $\Omega = \{1, 2, \dots, s\}$ of a homogeneous discrete Markov chain is a limited set. If there exists an integer $n_0 > 0$, such that for any $i, j \in \Omega$, we have

$$p_{ij}(n_0) > 0 \quad (5.167)$$

then this Markov chain is ergodic, and its limited distribution $p_j (j \in \Omega)$ is a unique solution of the equation

$$\pi_j = \sum_{i=1}^s \pi_i p_{ij}, \quad j = 1, 2, \dots, s \quad (5.168)$$

under the conditions

$$\pi_i > 0, \quad j = 1, 2, \dots, s \quad (5.169)$$

and

$$\sum_{i=1}^s \pi_i = 1 \quad (5.170)$$

6. The limit distribution of an ergodic homogeneous discrete Markov chain is stationary
 7. Denote the stationary limit distribution of an ergodic homogeneous discrete Markov chain to be $\mathbf{v} = \{v_j, j \in \Omega\}$. To any integer n , we have

$$\mathbf{v} = \mathbf{vP}^n \quad (5.171)$$

Example 5.7

$\Omega = \{1, 2, 3\}$ is the state space of homogeneous Markov chain $\{X(n), n = 1, 2, \dots\}$, which has the following transition probability matrix

$$\mathbf{P} = \begin{bmatrix} 1/2 & 1/3 & 1/6 \\ 1/3 & 1/3 & 1/3 \\ 1/3 & 1/2 & 1/6 \end{bmatrix}$$

and with the initial distribution being

$$\begin{array}{ccc} X(0) & 1 & 2 & 3 \\ p & 2/5 & 2/5 & 1/5 \end{array}$$

1. Calculate the second transition probability matrix.
2. Find the probability distribution of $X(2)$.
3. Find the stationary distribution.

$$1. \mathbf{P}^2 = \begin{bmatrix} 5/12 & 13/36 & 2/9 \\ 7/18 & 7/18 & 2/9 \\ 7/18 & 13/36 & 1/4 \end{bmatrix}$$

2. $\tilde{\mathbf{P}}_2 = \tilde{\mathbf{P}}_0 \tilde{\mathbf{P}}^2 = (2/5, 67/180, 41/180)$
3. Because

$$\mathbf{v} = \mathbf{vP}$$

and

$$\sum v_i = 1$$

The stationary distribution is

$$\mathbf{v} = (2/5, 13/35, 8/35)$$

5.3.3 FATIGUE

5.3.3.1 High-Cycle Fatigue

With the help of the above-mentioned discrete Markov chain, let us consider the process of accumulative damage, failure due to a material's high-cycle fatigue (see Soong and Grigoriu, 1992).

Suppose that, during a fatigue test, a sufficient number of specimens are used under cyclic loading. Assume that the damage probability can be described by Equation 5.155 but, in this case, rewritten as

$$\mathbf{p}_0 = [\pi_1, \pi_2, \dots, \pi_j, \dots, \pi_{f-1}, 0]_{1 \times f} \quad (5.172)$$

where the term π_j is the probability that the test specimens are in the damage state j at time zero and the probability at the failure state at the initial time is assumed to be zero. It is seen that these probabilities satisfy Equation 5.156, with the dimension $1 \times f$, where f states a final failure and the states can be denoted by

$$\Omega = \{1, 2, \dots, f\} \quad (5.173)$$

Having denoted the failure probabilities in state j at time zero, let us further denote time x to be

$$\mathbf{p}_x = [\pi_x(1), \pi_x(2), \dots, \pi_x(j), \dots, \pi_x(f)], \quad x = 0, 1, 2, \dots \quad (5.174)$$

It is seen that

$$\sum_{j=1}^f \pi_x(j) = 1 \quad (5.175)$$

Based on Equation 5.165, we have

$$\mathbf{p}_x = \mathbf{p}_0 \mathbf{P}^x \quad (5.176)$$

with $\mathbf{P}$ being the first transition probability matrix. Now, assume that the damage state does not change (no increase) by more than one unit during the duty cycle, the transition probability matrix $\mathbf{P}$ can be written as

$$\mathbf{P} = \begin{bmatrix} \pi_1 & 1-\pi_1 & 0 & 0 & \dots & 0 & 0 \\ 0 & \pi_2 & 1-\pi_2 & 0 & \dots & 0 & 0 \\ 0 & 0 & \pi_3 & 1-\pi_3 & \dots & 0 & 0 \\ & & & \dots & & & \\ 0 & 0 & 0 & 0 & \dots & \pi_{f-1} & 1-\pi_{f-1} \\ 0 & 0 & 0 & 0 & \dots & 0 & 0 \end{bmatrix} \quad (5.177)$$

It is noted that this case is a stationary Markov chain because the transition probability matrix $\mathbf{P}$ is time-invariant and also note that any π_j in $\mathbf{P}$ is smaller than 1 but greater than 0.

Furthermore, the cumulative distribution function of time T_f to failure state f is $\pi_x(f)$, that is,

$$F_{T_f}(x) = P\{T_f \leq x\} = \pi_x(f) \quad (5.178)$$

When the time becomes sufficiently long, namely, let $x \rightarrow \infty$, the failure probability approaches unity, that is,

$$\lim_{x \rightarrow \infty} \pi_x(f) = 1 \quad (5.179)$$

Let us denote the reliability function as

$$R_{T_f}(x) = 1 - F_{T_f}(x) \quad (5.180)$$

The mean and standard deviation of the fatigue problem can be calculated as

$$\mu_{T_f} = E[T_f] = \sum_{x=0}^{\infty} R_{T_f}(x) \quad (5.181)$$

and

$$\sigma_{T_f} = \left\{ E[T_f^2] - \mu_{T_f}^2 \right\}^{1/2} = \left\{ \sum_{x=0}^{\infty} [xR_{T_f}(x) + \mu_{T_f}] - \mu_{T_f}^2 \right\}^{1/2} \quad (5.182)$$

We can also directly calculate the mean and standard deviation in terms of the quantity f and the probabilities π_q and $1 - \pi_q$. Suppose all new specimens have a probability of 1 at the initial state 1, that is,

$$\pi_1 = 1, \pi_2 = 0, \pi_3 = 0, \dots \quad (5.183)$$

The fatigue failure time $T_{f|1}$ can then be written as

$$T_{f|1} = T_1 + T_2 + \dots T_q + \dots T_{f-1} \quad (5.184)$$

where T_q stands for the time within the duty cycle in state q . It can be proven that all T_q are mutually independent with the following distribution

$$P\{T_q = x\} = (1 - \pi_q) p_q^{x-1}, q = 1, 2, \dots, f-1, x = 1, 2, \dots \quad (5.185)$$

The mean and standard deviation of the fatigue problem can be calculated based on the distribution described in Equation 5.185. It can be proven that

$$\mu_{T_{f|1}} = E[T_{f|1}] = f - 1 + \sum_{i=1}^{f-1} r_i \quad (5.186)$$

and

$$\sigma_{T_{f|1}} = \left\{ E[T_{f|1}^2] - \mu_{T_{f|1}}^2 \right\}^{1/2} = \left\{ \sum_{i=1}^{f-1} r_i (1 + r_i) \right\}^{1/2} \quad (5.187)$$

where the ratio is defined as

$$r_i = \frac{\pi_i}{1 - \pi_i} \quad (5.188)$$

When the ratio becomes constant, that is,

$$r_i = r$$

we will have the simplest case as follows:

$$\mu_{T_{fl}} = (f-1)(1+r) \quad (5.189)$$

and

$$\sigma_{T_{fl}} = \sqrt{(f-1)r(1+r)} \quad (5.190)$$

5.3.3.2 Low-Cycle Fatigue

When the stress is sufficiently high, plastic deformation will occur. Accounting for the loading in terms of stress is less useful and the strain in the material can be a simpler and more accurate description. In this case, we witness the *low-cycle fatigue*. One of the widely accepted theories is the *Coffin–Manson relation* (see Sornette et al. 1992) given by

$$\Delta \epsilon_p = f(\epsilon'_f, N, c) \quad (5.191)$$

In Equation 5.191, $\Delta \epsilon_p$ is the range of plastic strain, which is a function of ϵ'_f , an empirical constant; N , the number of half-reversals to failure (N cycles); and c , an empirical constant known as the *fatigue ductility exponent*. In Chapter 10, we will further explore the essence of Equation 5.191. In the following paragraphs, let us qualitatively yet briefly introduce low-cycle fatigue.

5.3.3.2.1 Cyclic Test

Unlike high-cycle fatigue, when the load applied on a component or a test specimen is greater than its yielding point, nonlinear displacement will occur. To study the material behaviors, a forced cyclic test is often performed. In this case, one type of material will retain its stiffness for several cycles until reaching the final stage of broken failure (see Figure 5.15a), where $B(t)$ stands for the broken point. Steel is a typical material of this kind. Another type of material will reduce its stiffness continuously until the stiffness is below a certain level at which the total failure is defined (see Figure 5.15b), where a preset level marked by the dot-dashed line stands for the failure level. When this level is reached at the n th cycle shown in Figure 5.15b, the corresponding amount of force is denoted by $B(n)$. Reinforced concrete is typically seen with the overloading cycles. In both cases, the number of cycles when failure occurs is considerably smaller than the above-mentioned high-cycle fatigue.

Note that during a low-cycle fatigue test, if the stiffness is reduced, the amount of loading will often be reduced as well; otherwise, the corresponding displacement can be too large to realize with common test machines. Therefore, instead of applying

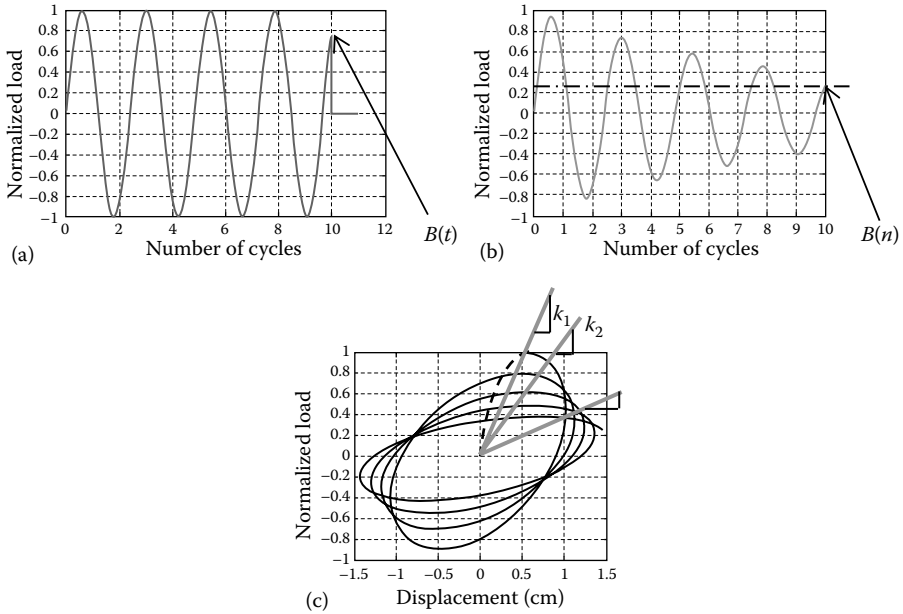


FIGURE 5.15 Low-cycle fatigue. (a) Failure history without significant change in stiffness of materials. (b) Failure history with decrease of stiffness of materials. (c) Stiffness variation.

equal amplitudes of force, equal displacement is used, which is referred to as the test with the *displacement control*. On the other hand, during a cyclic test, if the level of force is controlled, it is called the *force control*. There is another type of low-cycle fatigue test, which is *uncontrolled*. Tests with ground excitations on a vibration system whose stiffness is contributed by the test specimen can be carried out to study uncontrolled low-cycle fatigue. Figure 5.15c shows conceptually an uncontrolled cyclic test, from which we can see that the secant stiffness, which is the ratio of the peak force and corresponding displacement, marked as k_1 , k_2 , and so on, is continuously reduced.

5.3.3.2.2 Remaining Stiffness

From Figure 5.15a, we can see that the first type of low-cycle fatigue occurs without decaying stiffness, because when using displacement control, the force applied on a test specimen will remain virtually constant in several cycles until sudden failure occurs. We may classify the failure of this type of material under overloading conditions with constant stiffness as “type C” low-cycle fatigue.

On the other hand, as seen in Figure 5.15b, the overload failure of materials with decaying stiffness can be called “type D” low-cycle fatigue. To model these two types of low-cycle fatigue, we will have rather different random processes.

Consider type D low-cycle fatigue first. We will see that, under controlled cyclic tests, the remaining stiffness at the q th cycle may be caused by the accumulated deformation in the previous cycles. The corresponding forces under displacement control at cycle q measured from different specimens are likely different, that is, random. On the other hand, under force control, the displacement at cycle q is also

random. Thus, both the displacement and the force can be seen as random processes. In the following paragraphs, we can see that the sums of these random quantities at cycle q can be approximated as Markov processes.

Experimental studies shows that the force or the displacement of type C material before failure will remain constant. Therefore, the corresponding test process cannot be characterized as a random process. However, the amount of force at the broken point is rather random. In addition, the specific broken cycle is also random. That is, Figure 5.15a conceptually shows the point to be 10 cycles, which is only realized in the test. In reality, it can happen at 9, 11, or other cycles. In addition, because the level of force is random, the exact time point of the sudden failure is also random. In the following examples, we can see that using $B(t)$ to mark the failure force $B(t)$ is not Markovian.

5.3.3.2.3 Type C Low-Cycle Fatigue

Suppose the sudden failure of type C low-cycle fatigue occurs at time point t_f and the amount of force is B , and $B(t)$ is used to denote the process. Also, suppose the failure happens at cycle q and the cyclic test uses sinusoidal displacement control and the period of the loading process is T . The reason the above-mentioned process $B(t, f)$ is not Markovian is that its previous force measured at $t-T$, $t-2T$, and so on, are all constant. Also, the peak value of the force before the failure point is also constant. Therefore, we cannot say that $B(t)$ depends on the nearest recent cycle. In fact, the failure should depend on all the previous overloading process. In this case, the corresponding materials can memorize the previous process.

Liu et al. (2005) reported that, under their test conditions, the fatigue point only depends on the total number of previous cycles and the amount of peak forces. That is, the failure point has little relation with how long the test force is kept at a loading level. The material memory has little relation with the temporal loading path, which is conceptually shown in Figure 5.16a, where the solid lines are sinusoidal loading paths and the break lines are triangular loading paths. Under Liu's test conditions, as long as the peak value of the load is controlled to be constant, different load paths result in little change. However, the level of peak load plays much more significant roles.

More importantly, when using different overloading levels $F_1, F_2, \dots$ at cycle 1, 2, ..., the fatigue failure $B(t)$ can be rather different between the increasing load test ($F_1 < F_2 < \dots$) and the decreasing load test ($F_1 > F_2 > \dots$). Figure 5.16b conceptually shows increasing and decreasing loading processes. More detailed observations unveil the reason for such a phenomena through metallographic examinations. It is seen that when the specimen is overloaded to yield inelastic deformation, the local metallographic microstructure will be altered. Therefore, although no obvious cracks were found on the specimen, the specimen memorizes the metallographic defection and the larger the inelastic deformation is, the heavier the metallographic defection will be. Where a large deformation has previously occurred, the microstructure becomes susceptible to damage when subjected to further metallographic defections. On the other hand, when the overloading is increased cycle by cycle, the previous smaller defections will result in fewer alterations of the metallographic microstructure.

Because the loading path has few effects on fatigue life, one can use the above-mentioned method with ground excitations for uncontrolled low-cycle fatigue tests, which is conceptually shown in Figure 5.16c, where the upper and lower broken lines

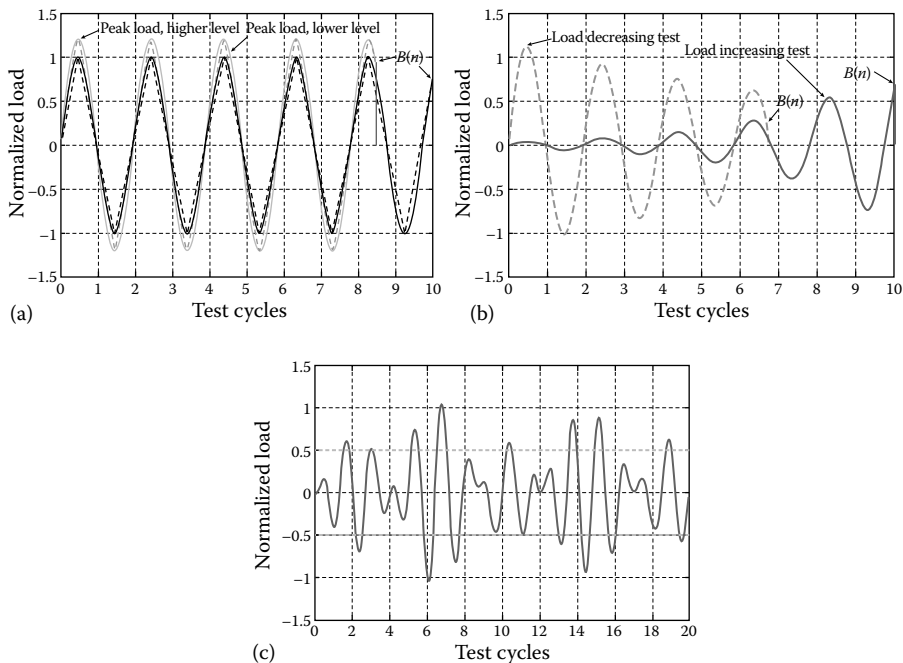


FIGURE 5.16 Different loadings for type C tests. (a) Constant load amplitude. (b) Monotonic increasing/decreasing load amplitude. (c) Random loading amplitude.

specify the yielding load. Comparing Figure 5.16c with Figure 5.1, we can realize that the study on type C low-cycle fatigue can be carried out based on the aforementioned conclusions to the engineering problem of level-crossing.

5.3.3.2.4 Type D Low-Cycle Fatigue

Now, consider the second kind of fatigue, the type D low-cycle test. One useful model to indicate the decrease of stiffness is by relating stiffness to the accumulative inelastic displacement $N(n)$.

That is (see Chapter 10, Section 10.3.4)

$$k = f(N(n)) \quad (5.192)$$

Assume that in the n th cycle, the total displacement is $D(n)$, which can be $N(n)$ longer than elastic displacement $L(n)$. Note that the inelastic distance is treated as a random process because of the possibility that when the material is damaged, the allowed linear displacement may vary from cycle to cycle. That is,

$$N(n) = D(n) - L(n) \quad (5.193)$$

That is, $D(n)$ is the distance of inelastic displacement in cycle n . The accumulated total displacement, up to cycle n , is denoted by $Z(n)$. To simplify the process, let the

allowed linear displacement be constant, that is, $L(n) = L$, and assume that, in each cycle, the forced displacement is greater than L . In this case, we can write

$$Z(n) = \sum_{q=1}^n N(q) = \sum_{q=1}^n D(q) - nL \quad (5.194)$$

It is seen that $Z(n)$ is a continuous state Markov chain. To simplify the analysis, however, let us consider another Markov chain, $Y(n)$, that has only a deterministic difference with $Z(n)$ by nL , such that,

$$Y(n) = Z(n) + nL = \sum_{q=1}^n D(q) \quad (5.195)$$

Assume the displacement $D(q)$ has a normal distribution with zero mean, that is,

$$D(n) \sim N(0, \sigma) \quad (5.196)$$

From the above-mentioned example on sums of random sequence with a normal distribution, it is seen that the probability distribution of $Y(n)$ is normally given by

$$f_{Y(n)}(n, d) = \frac{1}{2\pi\sigma\sqrt{n}} e^{-\frac{d^2}{2n\sigma^2}} \quad (5.197)$$

5.3.4 CASCADING EFFECT

5.3.4.1 General Background

When a system or a structure subjects multiple loads applied in sequence, either of the same kind or with different types of forces, the previous ones may cause certain damage and the consequent ones will make further damage that can be far more severe than a single load. This consequent loading and damaging is referred to as a cascading effect.

Among load-resilient designs, the cascading effect is one of the most unclear issues. This is because the normal training for engineers is to design systems to be successful, instead of being failures. However, many real-world experiences have witnessed severe failure under cascading effects. Examples can be found such as mountain sliding after strong earthquakes, bridge scour failure after heavy floods, structural failure due to overload after fatigue, and so on.

Because both the magnitudes and the acting time are random, the cascading effect can be treated as a random process. Although fewer systematic researches on such effects have been carried out, in this manuscript, we discuss possible approaches.

Again, the discussion of such a topic serves only to encourage readers to develop a knowledge and methodology to understand the nature and essence of random processes. It will also serve the purpose of opening the window to engineers who have been trained in the deterministic world.

5.3.4.2 Representation of Random Process

The cascading effect consists of two components, the magnitude and the occurrence time, which are also two of the basic elements of a random process $X(t, e)$, $e \in \Omega$, and $0 < t < T$. Here, Ω is the total state space and T is the total service time of the system. It is often helpful to separate the variables in the state space and in the time domain.

Generally speaking, any record of a random process can be seen as a deterministic temporal function, which can be represented by summations of orthogonal functions with corresponding parameters. The aforementioned Fourier series is one of the popular examples using orthogonal sinusoidal temporal functions, namely, $\sin(n\omega_T t)$ and $\cos(n\omega_T t)$. In this regard, the random process itself can also be represented by certain sets of orthogonal functions under conditions.

Specifically, under the mean square convergence conditions, a random process $X(t)$ can be represented in $a \leq t \leq b$, by the following series:

$$X(t) = \sum_{i=0}^{\infty} A_i \phi_i(t), \quad t \in [a, b] \quad (5.198)$$

such that

$$\lim_{n \rightarrow \infty} \left[\sum_{i=0}^n A_i \phi_i(t) - X(t) \right] = \lim_{n \rightarrow \infty} \left\{ E \left[\sum_{i=0}^n A_i \phi_i(t) - X(t) \right]^2 \right\} = 0 \quad (5.199)$$

In Equations 5.198 and 5.199, $\{A_i\}$ a set of random variables and $\{\phi_i(t)\}$ is a set of deterministic temporal functions, which is called a base or coordinate functions.

The essence of Equation 5.198 is the *methodology of variable separation*, because in each individual product $A_i \phi_i(t)$, the random variable A_i in state space and the temporal variable $\phi_i(t)$ in the time domain are separated. This is similar to the method of variable separation to solve partial differential equations, where the spatial and temporal variables are first separated to form a set of ordinary equations.

A common representation that satisfies Equation 5.199 takes the following form

$$X(t) = \mu(t) + \sum_{i=1}^{\infty} A_i \phi_i(t), \quad t \in [a, b] \quad (5.200)$$

where

$$\mu(t) = E[X(t)] \quad (5.201)$$

and for all i, j the set $\{A_i\}$ satisfies

$$1. E[A_i] = 0 \quad (5.202)$$

$$2. E[A_i A_j] = \sigma_i^2 \delta_{ij} \quad (5.203)$$

where

$$\sigma_i^2 = E[A_i^2] \quad (5.204a)$$

and δ_{ij} is a Koronic delta function such that

$$\delta_{ij} = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases} \quad (5.204b)$$

The complete set of temporal functions $\{\phi_i(t)\}$ should satisfy the following:

1. The covariance of $X(t)$ can be represented by

$$\sigma_{XX}(t_1, t_2) = \sum_{i=1}^{\infty} \sigma_i^2 \phi_i(t_1) \phi_i(t_2), \quad t_1, t_2 \in [a, b] \quad (5.205)$$

$$2. \int_a^b |\phi_i(t_1)|^2 dt < \infty \quad (5.206)$$

and

$$3. \int_a^b \phi_i(t_1) \phi_i(t_2) dt = \delta_{ij} \quad (5.207)$$

If $X(t)$ is taken to be a measured time history of the random process (see Chapter 9, Inverse Problems), then the coefficient A_i is no longer random and can then be calculated by

Note that the coefficient A_i can then be calculated by

$$A_i = \int_a^b [X(t) - \mu(t)] \phi_i(t) dt \quad (5.208)$$

Practically speaking, only the first n terms in Equation 5.200 are used to approximate the random process $X(t)$, that is

$$\hat{X}(t) = \mu(t) + \sum_{i=1}^n A_i \phi_i(t) \approx X(t) \quad (5.209)$$

In this case, $\hat{X}(t)$ is the approximated process and

$$\mu_{\hat{X}}(t) = E[\hat{X}(t)] \quad (5.210)$$

$$\sigma_{\hat{X}}^2(t) = \sum_{i=1}^n \sigma_i^2 \phi_i^2(t) \quad (5.211)$$

and

$$R_{\hat{X}}(t_1, t_2) = \sum_{i=1}^n \sigma_i^2 \phi_i(t_1) \phi_i(t_2), \quad t_1, t_2 \in [a, b] \quad (5.212)$$

To represent or reconstruct a random process, the sample range $[a, b]$ must be realized. Suppose those orthogonal function $\phi_i(t)$ are known, the following integral can be used as a trial-and-error approach to specify $[a, b]$

$$\int_a^b R_{\hat{X}}(t_1, t_2) \phi_i(t_2) dt_2 = \lambda_i \phi_i(t_1) \quad t_1, t_2 \in [a, b] \quad (5.213)$$

where λ_i is called the corresponding eigenvalue and

$$\lambda_i \propto E[A_i^2] = \sigma_i^2 \quad (5.214)$$

From Equation 5.214, the calculated parameter λ_i is a constant if the range $[a, b]$ is chosen correctly. Furthermore, if λ_i has a drastic variation over a period of time and/or after the targeted system undergoes significant loading, then the system may have a cascading damage.

5.3.4.3 Occurrence Instance of Maximum Load

The peak value of loading magnitude must be calculated not only when considering the cascading effect but also for most system designs that resist external loads. In the cases of cascading loads as well as low-cycle fatigue, both are treated as random processes; *when* the maximum load occurs, it is an important issue. For example, in case of random loading in a system with type C low-cycle fatigue, the moment of earlier or later maximum load will cause rather different results. In this subsection, let us consider this issue through a simplified Markovian model.

Suppose a system is subjected to random loads from time to time. The amplitude of these loads possesses state space denoted by $\{1, 2, \dots, a\}$ with uniform distributions. Here, using uniform distribution simplifies the analysis. Generally speaking, lognormal distributions can be more realistic. If one can measure these loads in every unit time, then the maximum values of these loads can be recorded as a random sequence $\{X(n), n \geq 1\}$.

We shall see that $\{X(n), n \geq 1\}$ is Markovian. We will analyze its one-step transition probability matrix and find the average time that the maximum value m is recorded. Denote the k th record of the amplitude to be Y_k ($k = 1, 2, \dots$). It is seen that $Y_1, Y_2, \dots, Y_n, \dots$ are mutually independent and have identical distributions. This is because $X(k)$ is the maximum value recorded in the first k measurement, that is, $X(k) = \max_{1 \leq i \leq k} \{Y_i\}$ and $Y_n(n > k)$ and $X(k) = \max_{1 \leq i \leq k} \{Y_i\}$ are mutually independent.

Now, consider any instant $0 < m_1 < m_2 < \dots < m_n < m_{n+1}$, and $i_1, i_2, \dots, i_{n+1} \in \Omega$. When

$$P\{[X(m_1) = i_1] \cap [X(m_2) = i_2] \cap \dots \cap [X(m_n) = i_n]\} > 0 \quad (5.215)$$

we have

$$\begin{aligned}
 & P\{X(m_{n+1})=i_{n+1} | [X(m_1)=i_1] \cap [X(m_2)=i_2] \cap \dots \cap [X(m_n)=i_n]\} \\
 &= P\{\max[X(m_n), Y_{m_n+1}, Y_{m_n+2}, \dots, Y_{m_{n+1}}] = i_{n+1} | [X(m_1)=i_1] \cap [X(m_2)=i_2] \cap \dots \cap [X(m_n)=i_n]\} \\
 &= P\{\max[i_n, Y_{m_n+1}, Y_{m_n+2}, \dots, Y_{m_{n+1}}] = i_{n+1} | [X(m_1)=i_1] \cap [X(m_2)=i_2] \cap \dots \cap [X(m_n)=i_n]\} \\
 &= \begin{cases} 0, & i_{n+1} < i_n \\ P\{\max[Y_{m_n+1}, Y_{m_n+2}, \dots, Y_{m_{n+1}}] < i_{n+1}\}, & i_{n+1} = i_n \\ P\{\max[Y_{m_n+1}, Y_{m_n+2}, \dots, Y_{m_{n+1}}] = i_{n+1}\}, & i_{n+1} > i_n \end{cases}
 \end{aligned} \tag{5.216}$$

Furthermore, we also have

$$\begin{aligned}
 & P\{X(m_{n+1})=i_{n+1} | X(m_n)=i_n\} \\
 &= P\{\max[X(m_n), Y_{m_n+1}, Y_{m_n+2}, \dots, Y_{m_{n+1}}] = i_{n+1} | X(m_n)=i_n\} \\
 &= P\{\max[i_n, Y_{m_n+1}, Y_{m_n+2}, \dots, Y_{m_{n+1}}] = i_{n+1} | X(m_n)=i_n\} \\
 &= \begin{cases} 0, & i_{n+1} < i_n \\ P\{\max[Y_{m_n+1}, Y_{m_n+2}, \dots, Y_{m_{n+1}}] < i_{n+1}\}, & i_{n+1} = i_n \\ P\{\max[Y_{m_n+1}, Y_{m_n+2}, \dots, Y_{m_{n+1}}] = i_{n+1}\}, & i_{n+1} > i_n \end{cases}
 \end{aligned} \tag{5.217}$$

Therefore, we can write

$$\begin{aligned}
 & P\{X(m_{n+1})=i_{n+1} | [X(m_1)=i_1] \cap [X(m_2)=i_2] \cap \dots \cap [X(m_n)=i_n]\} \\
 &= P\{X(m_{n+1})=i_{n+1} | X(m_n)=i_n\}
 \end{aligned} \tag{5.218}$$

which indicates that $\{X(n), n \geq 1\}$ is Markovian. The one-step transition probability is

$$\begin{aligned}
 & P\{X(n+1)=j | X(n)=i\} \\
 &= P\{\max[X(n), Y_{n+1}] = j | X(n)=i\} = P\{\max[i, Y_{n+1}] = j | X(n)=i\} \\
 &= \begin{cases} 0, & j < i \\ P\{Y_{n+1} < j\}, & j = i \\ P\{Y_{n+1} = j\}, & j > i \end{cases} = \begin{cases} 0, & j < i \\ \frac{i}{a}, & j = i \\ \frac{1}{a}, & j > i \end{cases}
 \end{aligned} \tag{5.219}$$

It is seen that the transition probability is not related to instant n ; therefore, $\{X(n), n \geq 1\}$ is a homogeneous Markov chain.

In addition, because

$$p_{ij} = \begin{cases} 0, & \text{elsewhere} \\ \frac{i}{a}, & j = i \\ \frac{1}{a}, & a \geq j > i \end{cases} \quad (5.220)$$

the one-step transition probability matrix is

$$\mathbf{P} = \begin{bmatrix} 1/a & 1/a & 1/a & \dots & 1/a \\ 0 & 2/a & 1/a & \dots & 1/a \\ 0 & 0 & 3/a & \dots & 1/a \\ & & \dots & & \\ 0 & 0 & 0 & \dots & 1 \end{bmatrix}$$

Now, to consider the averaged time, denote the first time to record the maximum value a to be T_a . We see that

$$\begin{aligned} P\{T_a = k\} &= \sum_{i=1}^{a-1} P\{[X(1) = i] \cap [T_a = k]\} \\ &= \sum_{i=1}^{a-1} P[X(1) = i] P[T_a = k | X(1) = i] = \sum_{i=1}^{a-1} \frac{1}{a} \left(\frac{a-1}{a} \right)^{k-2} \frac{1}{a} = \frac{(a-1)^{k-1}}{a^k} \end{aligned} \quad (5.221)$$

In this case, the averaged time, denoted by T_E , is

$$E(T_E) = \sum_{k=1}^{\infty} k P[T_a = k] = \sum_{k=1}^{\infty} k \frac{(a-1)^{k-1}}{a^k} = \frac{1}{a} \sum_{k=1}^{\infty} k \left(1 - \frac{1}{a} \right)^{k-1} = a \quad (5.222)$$

Thus, the larger the maximum value is, the longer the average record time will be.

PROBLEMS

1. Show that the peak frequency of a random process can be given by the following equation and find the formula for the Gaussian specialization

$$\omega_p = 2\pi\nu_p = \sqrt{\frac{\int_{-\infty}^{\infty} \omega^4 S_X(\omega) d\omega}{\int_{-\infty}^{\infty} \omega^2 S_X(\omega) d\omega}}$$

2. Show that the velocity of Gaussian process $X(t)$ at a zero up-crossing has a Rayleigh distribution
3. The RMS velocity and displacement of a narrow-band Gaussian vibration with zero mean are, respectively, 2.0 m/s and 0.05 m. Calculate
 - a. the rate of level up-crossing with the level to be 0.03 m
 - b. zero up-crossing rate

Here, $a = 0.03$; $\sigma_{\dot{X}} = 2.0$ and $\sigma_X = 0.05$

4. A narrow-band Gaussian process with zero mean has RMS displacement of 3.5 cm. Calculate and plot the distribution density function of
 - a. amplitude for this process and
 - b. height for this process
5. The joint PDF of a narrow-band process $X(t)$ and its first derivative is given by a joint Laplace distribution (Lin 1976)

$$f_{X\dot{X}}(x, \dot{x}) = \frac{1}{4ab} e^{-\left(\frac{|x|}{a} + \frac{|\dot{x}|}{b}\right)}, \quad -\infty < x < \infty, -\infty < \dot{x} < \infty$$

where $a > 0$, $b > 0$

Show that the PDF for the peak magnitude can be approximated by

$$f_Z(z) = \frac{1}{a} e^{-\frac{z}{a}}, \quad z \geq 0$$

6. Suppose $X(t)$ is a narrow-band process. What is the value of the peak with a 1% probability of being exceeded?
7. Show that the probability of exceedance $P(Z > z_0)$ of Rice's distribution of peaks is approximately a times the probability found from the Rayleigh distribution
8. $\{X(n), n = 0, 1, 2, \dots\}$ is a Markov chain. Show that the inverse sequence of $X(n)$ is also a Markov chain, that is,

$$P\{X(1) = x_1 | X(2) = x_2, X(3) = x_3, \dots, X(n) = x_n\} = P\{X(1) = x_1 | X(2) = x_2\}$$

Section III

Vibrations

6 Single-Degree-of-Freedom Vibration Systems

In previous chapters, we showed that random process may not always necessarily be a time history operation. Time histories occurring physically in the real world have specific reasons and they are limited by certain conditions. In previous chapters, we mainly focused on time-varying development, instead of a clear understanding of why time history exists. At most, we studied several important conditions of those time histories. Yet, these studies were limited to how a process behaves, such as if it is stationary, what the corresponding statistical parameters are, as well as what frequency spectra they have listed. Generally speaking, the previous chapters were limited to the mathematical models of random processes, instead of what the causes of these models were.

Many time-varying processes, or time histories, are purely artificial. For example, one can use computational software to generate a random signal. These purely artificial “stochastic” processes, although they seem very random, are actually controlled by man-made signal generators and therefore we should have prior knowledge of how they behave.

On the other hand, most real-world temporal developments are not purely or directly man-made, and thus are not that easily controllable; some of them cannot be easily measured. Accounting for all of these time-varying processes would be a huge task—most of them are far beyond the scope of this manuscript. Here, we focus only on a special type of process, the vibration signals, and mainly on mechanical vibrations.

To classify linear vibration signals by their degree of certainty, there are typically three essentially different types. The first is periodic vibration, such as harmonic steady-state vibration, which contains a limited number of frequency components and periodically repeats identical amplitudes. The second type is transient vibration, such as free decay vibration, which is caused only by initial conditions. Although we could also have a combination of harmonic steady-state vibration and free decay vibration, the resulting signal is often not treated as the third type because we study the first two types of vibration separately and simply add them together. Moreover, both of them are deterministic. The third type is random vibration. Based on the knowledge gained in the previous chapters, the nature of random signals is that, at any future moment, their value is uncertain. Therefore, for the first two types of vibration, with the given initial conditions and known input, we can predict their future, including amplitudes, frequencies, and phases. However, we cannot predict

the response of random vibrations. For a simple vibrational system, even if we know the bounds of a random input, the output bound is unpredictable.

Therefore, to handle random signals, we need a basic tool, that is, averaging. However, to account for random vibrations, we can do something more than the statistical measure. Namely, we need to study the nature of a vibration system itself. Here, the main philosophy to support our action is that any real-world vibration signal must be a result of a certain convolution. That is, vibration is a response caused by the combination of external excitation and the vibration system. Without any external excitations, of course, there would be no responses. However, without vibration systems, the presence of an excitation only will not cause any response either. We thus need to study the nature of vibration systems to further understand how their responses behave.

In this chapter, the basics of a single-degree-of-freedom (SDOF) vibration system that is linear and time-invariant will be described. The periodic and transient responses of the SDOF system under harmonic and impulse excitations will also be examined, respectively. For a more detailed description of vibrations, readers may consult Meirovitch (1986), Weaver et al. (1990), Chopra (2001), and Inman (2008), as well as Liang et al. (2012).

6.1 CONCEPT OF VIBRATION

The dynamic behavior of a SDOF system can be characterized by key parameters through free vibration analysis, such as the natural frequency and the damping ratio.

6.1.1 BASIC PARAMETERS

Generally speaking, the background knowledge needed for the study of SDOF vibration systems can be found in a standard vibration textbook. Examples of vibration textbooks that provide an ample understanding include Inman's *Engineering Vibration* and Chopra's *Dynamics of Structures*. Background knowledge that should be gained in reading one of these texts include: what vibration is, what the essence of vibration is versus another form of motion, and why vibration should be studied.

Vibration is a unique type of motion of an object, which is repetitive and relative to its nominal position. This back-and-forth motion can be rather complex; however, the motion can often be decoupled into harmonic components. A single harmonic motion is the simplest motion of vibration given by

$$x(t) = d\sin(\omega t) \quad (6.1a)$$

In this case, d is the amplitude, ω is the frequency, and t is the time. Thus, $x(t)$ is a deterministic time history. Typically, Equation 6.1a is used to describe the vibration displacement.

The velocity, with the amplitude $v = d\omega$, is the derivative of the displacement, then given by

$$\dot{x}(t) = d\omega \cos(\omega t) = v \cos(\omega t) \quad (6.1b)$$

Subsequently, the acceleration, with the amplitude $a = -d\omega^2$, is the derivative of the velocity, given by

$$\ddot{x}(t) = -d\omega^2 \sin(\omega t) = a \sin(\omega t) \quad (6.1c)$$

6.1.1.1 Undamped Vibration Systems

The previous examples can be seen as the responses of a SDOF system. In the following, the system will be examined using physical models.

6.1.1.1.1 Equation of Motion

Figure 6.1 shows an undamped SDOF system consisting of a mass m and a spring k .

Consider the motion in the x direction only (the vertical gravity force mg and supporting force n are then ignored). Thus, there are two forces in balance, the inertial force $f_m = m\ddot{x}$ and the spring force $f_k = kx$. This can be expressed as

$$m\ddot{x} + kx = 0 \quad (6.2)$$

6.1.1.1.2 Natural Frequency

Defined as the *angular natural frequency*, this is also often simply referred to as the *natural frequency*; its unity is in radians per second.

$$\omega_n = \sqrt{\frac{k}{m}} \quad (6.3)$$

6.1.1.1.3 Monic Equation

By dividing m from both sides of Equation 6.2, with the help of the notation in Equation 6.3, results in the monic equation.

$$\ddot{x} + \omega_n^2 x = 0 \quad (6.4)$$

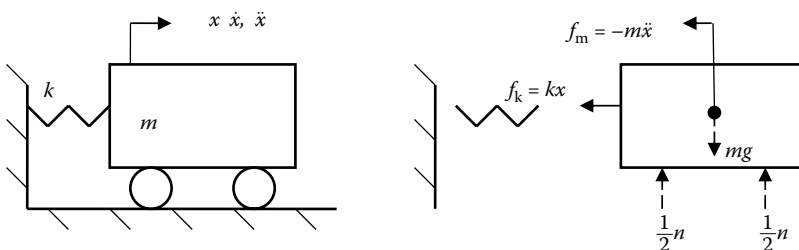


FIGURE 6.1 SDOF system.

Example 6.1

An undamped SDOF system has a mass of 10 kg and a stiffness of 10,000 N/m, find the angular natural frequency.

$$\omega_n = \sqrt{\frac{1000}{10}} = 10 \text{ (rad/s)}$$

Note that the natural frequency of cycles per second, denoted by f_n can be written as

$$f_n = \frac{\omega_n}{2\pi} = 1.5915 \text{ (Hz)}$$

6.1.1.1.4 Solutions

To solve the above equation, the so-called semidefinite method is used. In this approach, it is first assumed that

$$x(t) = d_c \cos \omega_n t + d_s \sin \omega_n t \quad (6.5)$$

Then, Equation 6.5 is substituted into Equation 6.4. If the proper parameters d_c and d_s can be determined, the parameters can be either infinite nor zero, then the assumption is valid and Equation 6.5 is one of the possible solutions.

Noticeably, with initial conditions x_0 and $\dot{x}_0$, the parameters are

$$d_c = x_0 \quad (6.6)$$

and

$$d_s = \frac{\dot{x}_0}{\omega_n} \quad (6.7)$$

Accordingly, $x(t) = d_c \cos \omega_n t + d_s \sin \omega_n t$ is indeed a solution, which implies that the vibration displacement of the mass is harmonic, with frequency ω_n and certain amplitudes. Here, it is also realized why ω_n is called the natural frequency.

Example 6.2

Suppose a monic undamped vibration system with nature frequency $\omega_n = 10$ has an initial displacement $x(0) = 1$, and initial velocity $\dot{x}(0) = -2$, calculate the response.

From Equation 6.15, it is seen that

$$x(0) = d_c \cos \omega_n(0) + d_s \sin \omega_n(0)$$

therefore

$$d_c = 1$$

Furthermore, taking the derivative on both sides of Equation 6.15 with respect to time t and then letting $t = 0$, we have

$$\dot{x}(0) = d_c \omega_n \sin \omega_n 0 + d_s \omega_n \cos \omega_n 0 = d_s \omega_n$$

Therefore,

$$d_s = -2/10 = -0.2$$

6.1.1.1.5 Essence of Vibration

To answer the important question: “Why is there vibration?,” consider the following discussion.

6.1.1.1.5.1 Conservation of Energy First, the energy terms in the SDOF system are defined.

Potential energy

$$U(t) = \frac{1}{2} kx(t)^2 \quad (6.8)$$

Kinetic energy

$$T(t) = \frac{1}{2} m\dot{x}(t)^2 \quad (6.9)$$

Total energy conservation

$$\frac{d}{dt} [T(t) + U(t)] = 0 \quad (6.10)$$

6.1.1.1.5.2 Energy Exchange During the vibration, an exchange of potential and kinetic energy occurs. Furthermore,

$$T_{\max} = U_{\max} \quad (6.11)$$

Explicitly, this can be expressed as

$$\frac{1}{2} kx_{\max}^2 = \frac{1}{2} m\dot{x}_{\max}^2 \quad (6.12)$$

Notice that Equation 6.2 can also be obtained through Equation 6.10. From Equation 6.12, it is determined that

$$\dot{x}_{\max} = \omega_n x_{\max} \quad (6.13)$$

Substitution of Equation 6.13 into Equation 6.12 results in

$$\frac{1}{2} k x_{\max}^2 = \frac{1}{2} m \omega_n^2 x_{\max}^2 \quad (6.14)$$

From Equation 6.14, Equation 6.3 can be calculated. This procedure implies that the ratio of k and m is seen as a measurement of normalized vibration energy.

Example 6.3

Based on Equation 6.14, we can find the natural frequency of a complex system, which consists of more than one mass (moment of inertia) but is described by a single variable x (or rotation angle θ). Using such an energy method can simplify the procedure for natural frequency analysis.

As shown in Figure 6.2, a system has three gears and a rack that is connected to the ground through a stiffness k . In this system, the pitch radii of gear 1 to gear 2 are, respectively, R_1 and R_2 . The pitch radii of gear 2 to gear 3 (also of gear 3 to the rack) are, respectively, r_2 and r_3 . To simplify the problem, let the teeth and shafts of gears as well as the rack have infinitely strong stiffness and the system is frictionless.

To use the energy method, $T_{\max} = U_{\max}$, we need to find both the potential and the kinetic energies. The kinetic energies T_{gear} are the functions of the moment of inertia, J_i , the gear ratio, r_i , and the displacement x . That is $T_i = T_i(J_i, r_i, x)$. The kinetic energy T_{rack} is the function of the mass of the rack as well as the displacement x . Here, x is the only parameter to denote the motion; therefore, the system is of SDOF.

The potential energy is given by

$$U = 1/2 k x^2$$

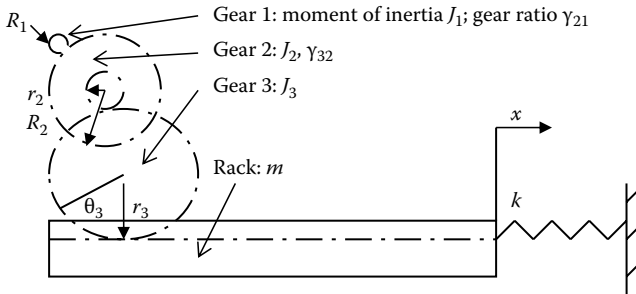


FIGURE 6.2 A complex system.

The total kinetic energy T are contributed by these three gears, denoted by T_{gear1} , T_{gear2} , and T_{gear3} , respectively, and by the rack, denoted by T_{rack} . That is,

$$T = T_{\text{gear1}} + T_{\text{gear2}} + T_{\text{gear3}} + T_{\text{rack}}$$

Denote θ_i as the rotation angle of gear i . Then, the relationship between the translational displacement x and the rotational angle is $x = r_3\theta_3$ so that for translational velocity $\dot{x}$ and the rotational angle velocity $\dot{\theta}_3$, we have $\dot{x} = r_3\dot{\theta}_3$. Therefore,

$$\theta_3 = x/r_3$$

$$\dot{\theta}_3 = \dot{x}/r_3$$

Gear 3 to gear 2 has gear ratio γ_{32} given by

$$\gamma_{32} = r_3/r_2$$

So that the rotation angle of gear 2 is

$$\theta_2 = \gamma_{32}\theta_3$$

Furthermore,

$$\theta_2 = (r_3/r_2)x/r_3 = x/r_2$$

and

$$\dot{\theta}_2 = \dot{x}/r_2$$

Similarly, the gear 2 to gear 1 ratio is

$$\gamma_{21} = R_2/R_1$$

So that the rotation angle of gear 1 is

$$\theta_1 = \gamma_{21}\theta_2$$

and

$$\theta_1 = (R_2/R_1)x/r_2$$

or

$$\theta_1 = x(R_2/r_2R_1)$$

Therefore,

$$\dot{\theta}_1 = \frac{R_2\dot{x}}{r_2R_1}$$

With the above defined notations, kinetic energy can be summarized as

$$T = 1/2 m \dot{x}^2 + 1/2 (J_3 \dot{\theta}_3^2 + J_2 \dot{\theta}_2^2 + J_1 \dot{\theta}_1^2) = 1/2 \dot{x}^2 \left(m + \frac{J_1}{r_3^2} + \frac{J_2}{r_2^2} + \frac{J_3 R_2^2}{R_1^2 r_2^2} \right)$$

Note that T is represented by variable x only. Furthermore, the maximum velocity is

$$\dot{x}_{\max} = \omega_n x_{\max}$$

Then, based on Equation 6.14, we have

$$1/2 \dot{x}_{\max}^2 \left(m + \frac{J_1}{r_3^2} + \frac{J_2}{r_2^2} + \frac{J_3 R_2^2}{R_1^2 r_2^2} \right) = 1/2 k x_{\max}^2$$

The natural frequency can be written as

$$\omega_n = \sqrt{\frac{k}{m + \frac{J_1}{r_3^2} + \frac{J_2}{r_2^2} + \frac{J_3 R_2^2}{R_1^2 r_2^2}}}$$

Alternatively, we can also use Equation 6.10. We have

$$\frac{d}{dt} \left[1/2 \dot{x}^2 \left(m + \frac{J_1}{r_3^2} + \frac{J_2}{r_2^2} + \frac{J_3 R_2^2}{R_1^2 r_2^2} \right) + 1/2 k x^2 \right] = 0$$

Thus,

$$\ddot{x} \left(m + \frac{J_1}{r_3^2} + \frac{J_2}{r_2^2} + \frac{J_3 R_2^2}{R_1^2 r_2^2} \right) + k \dot{x} = 0$$

So that

$$\ddot{x} \left(m + \frac{J_1}{r_3^2} + \frac{J_2}{r_2^2} + \frac{J_3 R_2^2}{R_1^2 r_2^2} \right) + k x = 0$$

The differential equation is then given by

$$\ddot{x} \left(m + \frac{1}{r_3^2} + \frac{1}{r_3^2} + \frac{R_2^2}{R_1^2 r_2^2} \right) + k x = 0$$

Furthermore, the natural frequency is

$$\begin{aligned}\omega_n &= \sqrt{\frac{\text{stiffness}}{\text{mass}}} = (\text{coefficient of acc./coefficient of disp.})^{1/2} \\ &= \sqrt{\frac{k}{m + \frac{J_1}{r_3^2} + \frac{J_2}{r_2^2} + \frac{J_3 R_2^2}{R_1^2 r_2^2}}}\end{aligned}$$

If there are rotational stiffnesses associated with each gear, k_1 , k_2 , k_3 , then, due to the rotational deformation of each gears' shaft, there will be more potential energies:

$$U_{\text{gear1}} = 1/2 k_1 \theta_1^2 = 1/2 k_1 \frac{R_2^2}{R_1^2 r_3^2}$$

$$U_{\text{gear2}} = 1/2 k_2 \theta_2^2 = 1/2 k_2 \frac{1}{r_2^2}$$

$$U_{\text{gear3}} = 1/2 k_3 \theta_3^2 = 1/2 k_3 \frac{1}{r_1^2}$$

Thus, the total potential energy is given by

$$U = 1/2 x^2 \left(k + k_3 \frac{1}{r_1^2} + k_2 \frac{1}{r_2^2} + k_1 \frac{R_2^2}{R_1^2 r_3^2} \right)$$

The natural frequency can finally be calculated as

$$\omega_n = \sqrt{\frac{k + \frac{k_1}{r_3^2} + \frac{k_2}{r_2^2} + \frac{k_3 R_2^2}{R_1^2 r_2^2}}{m + \frac{J_1}{r_3^2} + \frac{J_2}{r_2^2} + \frac{J_3 R_2^2}{R_1^2 r_2^2}}}$$

6.1.1.1.5.3 Force and Momentum Equation 6.3 can be further obtained using additional approaches. For example, consider the momentum q , where

$$q(t) = m\dot{x}(t) \quad (6.15)$$

Figure 6.3 conceptually shows the relationships among the forces and momentum mentioned above, where f_X , f_M , and q_0 represent maximum restoring, inertia, and momentum, respectively.

Here, Equation 6.19 implies that

$$\text{Natural frequency} = \frac{\text{maximum restoring force}}{\text{maximum momentum}} \quad (6.20)$$

6.1.1.1.5.4 Rate of Energy Exchange Additionally, consider the rate of energy exchange and, for convenience, denote

$$x(t) = de^{\lambda t} \quad (6.21)$$

Furthermore, let $\lambda = \pm j\omega_n$, this will yield

$$x(t) = de^{\pm j\omega_n t} \quad (6.22)$$

Note that

$$\dot{x}(t) = \lambda x(t) \quad (6.23)$$

with given

$$\begin{aligned} T(t) &= \frac{1}{2} m \dot{x}(t)^2 \\ \frac{dT(t)}{dt} &= \frac{2}{2} m \dot{x}(t) \frac{d}{dt} \dot{x}(t) = m \dot{x}(t) \frac{d}{dt} [\lambda D e^{\lambda t}] = m \dot{x}(t) \lambda \frac{d}{dt} [x(t)] \\ &= 2\lambda \frac{1}{2} m \dot{x}(t)^2 = 2\lambda T(t) \end{aligned} \quad (6.24)$$

Results in the following:

$$\frac{dT(t)}{dT(t)} \frac{dT(t)}{dt} = \lambda \quad (6.25)$$

Furthermore, it is seen that

$$\omega_n = |\lambda| = \frac{1}{2} \left| \frac{dT(t)}{dT(t)} \frac{dT(t)}{dt} \right| \quad (6.26)$$

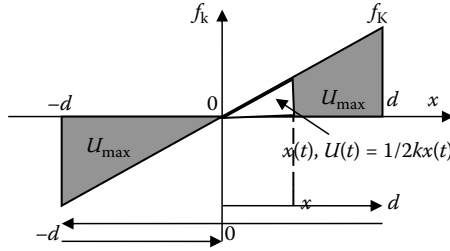


FIGURE 6.4 Potential energy in one vibration cycle.

and

$$\omega_n = |\lambda| = \frac{1}{2} \left| \frac{\frac{dU(t)}{dt}}{U(t)} \right| \quad (6.27)$$

Equations 6.26 and 6.27 indicate that the angular natural frequency is a unique ratio of energy exchange. It is characterized by the absolute value of one-half of the rate of energy exchanged over the kinetic (potential) energy. Furthermore, the higher the rate is, the larger the value of natural frequency will be. Readers may consider why it is one-half or twice the kinetic (potential) energy needed.

Figure 6.4 conceptually shows the potential energy in one vibration cycle.

6.1.1.1.6 Natural Frequency: A Brief Review

Natural frequency is one of the most important concepts in vibration, given that it unveils the essence of repetitive motion both qualitatively and quantitatively. There exist several angles in which to examine the natural frequency ω_n .

$$\omega_n = \sqrt{\frac{k}{m}} \quad (6.28)$$

$$\omega_n = |\lambda| \quad (6.29)$$

$$\omega_n = \frac{v}{d} = \frac{a}{v} = \sqrt{\frac{a}{d}} \quad (6.30)$$

$$\omega_n = \frac{f_K}{q_0} \quad (6.31)$$

$$\omega_n = \frac{1}{2} \left| \frac{\frac{dT(t)}{dt}}{T(t)} \right| = \frac{1}{2} \left| \frac{\frac{dU(t)}{dt}}{U(t)} \right| \quad (6.32)$$

Readers may consider if all of the above approaches always apply. Here, we just emphasize that, as seen in Equation 6.28, if either $m = 0$ or $k = 0$, then the natural frequency will not exist. In this case, a stable system given by Equation 6.12 is linear, SDOF, and undamped. This will not always be true, when $k > 0$. From Equation 6.8, however, if we have negative stiffness, then the potential energy $U(t) \propto k$ will become negative, which means that a certain source will continuously input energy to the system and the response will continuously be increasing, which makes a system unstable. Therefore, taking the absolute value of the ratio does not mean that k can be smaller than zero. Furthermore, if $c \neq 0$, we will have a damped system. For the existence of a stably damped vibration system, not only are the conditions $m > 0$, $k > 0$ needed but also the condition regarding c is needed, which will be discussed as follows.

6.1.1.2 Damped SDOF System

From Equation 6.5, the response of an undamped system will vibrate indefinitely. In the real world, energy dissipation will always exist, causing the free vibration to eventually die out. This energy dissipation mechanism is referred to as damping.

6.1.1.2.1 Viscous Damping

The viscous damping force is

$$f_c = c\dot{x} \quad (6.33)$$

where c is the proportional coefficient, defined as the *damping coefficient*. The parameter c is always greater than or equal to zero: semipositive or nonnegative. In Figure 6.5, a damper c is added to the SDOF system and the resulting balance of force is

$$\sum_x f_{(x)} = 0 \rightarrow f_m + f_c + f_k = 0 \quad (6.34)$$

As a result, the equation of motion becomes

$$m\ddot{x} + c\dot{x} + kx = 0 \quad (6.35)$$

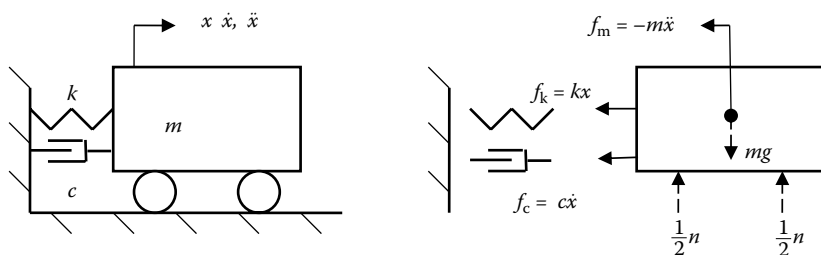


FIGURE 6.5 SDOF damped system.

6.1.1.2.2 Semidefinite Method

Similarly, to solve Equation 6.35, let

$$x = de^{\lambda t} \quad (6.36)$$

$$\dot{x} = d\lambda e^{\lambda t} \quad (6.37)$$

$$\ddot{x} = d\lambda^2 e^{\lambda t} \quad (6.38)$$

Substitution of Equations 6.36 through 6.38 into Equation 6.35 yields

$$md\lambda^2 e^{\lambda t} + cd\lambda e^{\lambda t} + kde^{\lambda t} = 0$$

Given that $m \neq 0$, $d \neq 0$, and $e^{\lambda t} \neq 0$ will result in the characteristic equation.

6.1.1.2.3 Characteristic Equation

The characteristic equation of a SDOF system is defined by

$$m\lambda^2 + c\lambda + k = 0 \quad (6.39)$$

To find the solution of Equation 6.39, the following is used:

$$\lambda_{1,2} = \frac{-c \pm \sqrt{c^2 - 4mk}}{2m} \quad (6.40)$$

6.1.1.2.4 Damping Ratio

To analyze Equation 6.40, the concept of *critical damping ratio* must be introduced. By dividing m from both sides, Equation 6.41 yields

$$\lambda^2 + \frac{c}{m}\lambda + \frac{k}{m} = 0 \quad (6.41)$$

Given that

$$\frac{k}{m} = \omega_n^2$$

and let

$$\frac{c}{m} = 2\zeta\omega_n \quad (6.42)$$

in which both m and c are positive, or in the case of c , semipositive; ω_n should also be positive, such that ζ is greater than or equal to zero. The *critical damping ratio*, ζ , or simply referred to as the *damping ratio*, is a semipositive number.

$$\zeta = \frac{c}{2m\sqrt{\frac{k}{m}}} = \frac{c}{2\sqrt{mk}} \quad (6.43)$$

In Equation 6.40, if $c^2 = 4mk$ or $c = 2\sqrt{mk}$, then

$$\sqrt{c^2 - 4mk} = 0 \quad (6.44)$$

In addition, the two roots are equal:

$$\lambda_1 = \lambda_2 = -\frac{c}{2m} = -\frac{2\sqrt{mk}}{2m} = -\sqrt{\frac{k}{m}} = -\omega_n \quad (6.45)$$

Thus,

$$c_c = 2\sqrt{mk} \quad (6.46)$$

Twice of the geometric average of mk is referred to as the *critical damping coefficient*.

In the case of

$$\zeta = \frac{c}{c_c} \quad (6.47)$$

Then, we can write

$$x(t) = de^{\lambda_{(1)}t} = de^{-\frac{c}{2m}t} = de^{-\omega_n t} \quad (6.48)$$

In this instance, Equation 6.48 no longer describes a motion of vibration, such that $c = c_c = 2\sqrt{mk}$ is a critical point. A system is a vibratory system only if

$$c < 2\sqrt{mk} \quad (6.49)$$

For this reason, ζ is called the critical damping ratio.

Condition 6.49 can be rewritten as

$$\zeta < 1 \quad (6.50a)$$

In this case, we have an underdamped system.

Whereas when

$$\zeta = 1 \quad (6.50b)$$

In this case, we have a critically damped system

And when

$$\zeta > 1 \quad (6.50c)$$

In this case, we have an overdamped system.

However, if

$$c < 0 \quad (6.51)$$

we have negative damping, which means a certain energy is continuously input to the system so that the response will continuously increase and the system becomes unstable. This phenomenon is similar to negative stiffness. The difference is that with $k < 0$, the input energy is proportional to displacement whereas with $c < 0$, the input is proportional to the velocity.

From Equation 6.43, we see that the damping ratio is proportional to the damping coefficient, so that if Equation 6.52 holds, then for the unstable condition, we can also write

$$\zeta < 0 \quad (6.52)$$

Example 6.4

A car has mass 2000 kg and total stiffness of the suspension system is 2840 kN/m. The design damping ratio is 0.12; find the total damping coefficient of its suspension system. Suppose five people weighing 5 kN are sitting in this car. Calculate the resulted damping ratio ($g = 9.8 \text{ m/s}^2$).

Based on Equation 6.43, the damping coefficient c can be calculated as

$$c = 2\zeta\sqrt{mk} = 18.1 \text{ kN/m-s}$$

With additional mass $\Delta m = 5000/9.8 = 510.2 \text{ kg}$, the new damping ratio is

$$\zeta_{\text{new}} = \frac{c}{2\sqrt{(m + \Delta m)k}} = 0.11$$

It is seen that when the mass is increased by about 1/4, the reduction of the damping ratio is only about 10%.

6.1.1.2.5 Eigenvalue λ

Rewrite Equation 6.40 to

$$\begin{aligned}\lambda_{1,2} &= \frac{-c \pm \sqrt{c^2 - 4mk}}{2m} = -\frac{c}{2m} \pm \sqrt{(-1) \left\{ \frac{4mk}{4m^2} - \left(\frac{c}{2m} \right)^2 \right\}} \\ &= -\frac{c}{2m} \pm j \sqrt{\left\{ \frac{k}{m} - \left(\frac{c}{2m} \right)^2 \right\}}\end{aligned}\quad (6.53)$$

With the substitution of ζ , this results in

$$\frac{c}{2m} = \frac{2\zeta\sqrt{mk}}{2m} = \zeta\sqrt{\frac{k}{m}} = \zeta\omega_n \quad (6.54)$$

Thus, the eigenvalue λ can be expressed as

$$\lambda_{1,2} = -\zeta\omega_n \pm j\sqrt{1 - \zeta^2}\omega_n \quad (6.55)$$

Note that

$$\lambda_2 = \lambda_1^* \quad (6.56)$$

where $(\cdot)^*$ denotes complex conjugate of $(\cdot)$. Figure 6.6 illustrates the eigenvalues.

From Figure 6.6, it is shown that

$$\lambda_1\lambda_2 = \lambda_1\lambda_1^* = \frac{k}{m} = \omega_n^2 \quad (6.57)$$

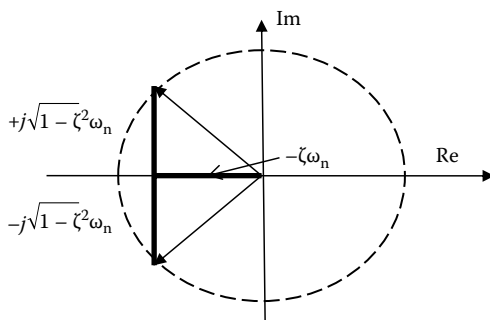


FIGURE 6.6 Eigenvalues of an underdamped system.

and

$$\lambda_1 + \lambda_2 = -\frac{c}{m} = -2\zeta\omega_n \quad (6.58)$$

6.1.1.2.6 Damped Natural Frequency

Equation 6.55 can be further simplified to

$$\lambda_{1,2} = -\zeta\omega_n \pm j\omega_d$$

where

$$\omega_d = \omega_n \sqrt{1 - \zeta^2} \quad (6.59)$$

Here, ω_d is called the damped natural frequency.

Because $0 \leq \zeta < 1$,

$$\omega_d = \omega_n \sqrt{1 - \zeta^2} \leq \omega_n \quad (6.60)$$

Example 6.5

A system has an eigenvalue of $\lambda = -3.0000 + 9.5394j$; find the corresponding undamped, damped natural frequencies, and damping ratio.

Damped natural frequency ω_d

$$\omega_d = \text{Im}(\lambda) = 9.5394 \text{ (rad/s)}$$

Undamped natural frequency ω_n

$$\omega_n = |\lambda| = [\text{Re}(\lambda)^2 + \text{Im}(\lambda)^2]^{1/2} = 10.0 \text{ (rad/s)}$$

Damping ratio ζ

$$\zeta = -\text{Re}(\lambda)/\omega_n = 0.3$$

6.1.1.2.7 Energy Dissipation

In Figure 6.7a, the area of the damping force–displacement loop, ΔE , is the energy dissipated during a cycle. If the amplitude of the steady-state displacement remains constant, then there must be an energy input equal to ΔE . How to input this amount of energy will be discussed in Section 6.2.

In this case, the energy input is equal to zero, so that with the capacity of energy dissipation, the amplitude of the response will continuously decay. This is shown in Figure 6.7b. Note that for the decaying response, the curve is clockwise.

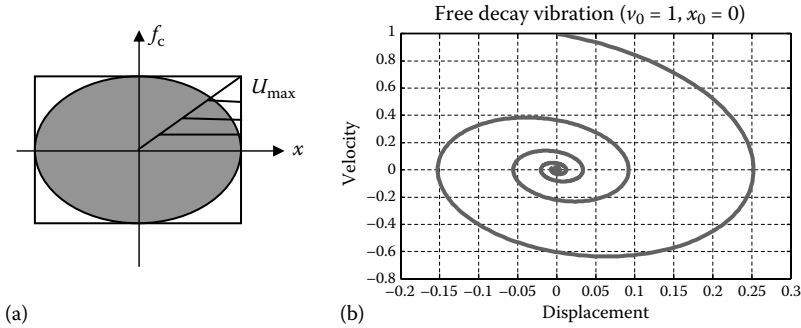


FIGURE 6.7 Energy dissipations: (a) steady-state response; (b) free decay response.

6.1.1.2.8 Essence of Symbol j

First, compare Equation 6.36 to the following:

$$x(t) = de^{\pm j\omega_n t}$$

Given this, Equation 6.36 can be rewritten as

$$x(t) = de^{-\zeta\omega_n t} e^{\pm j\sqrt{1-\zeta^2}\omega_n t} = de^{-\zeta\omega_n t} e^{\pm j\omega_d t} \quad (6.61)$$

Both equations that describe vibrations share a similar term, $e^{\pm j\omega_n t}$ or $e^{\pm j\omega_d t}$; therefore, the “ j ” term must be related to dynamic oscillations. In fact, this term implies energy exchanges between potential and kinetic energies. If this term is eliminated, then there will be no energy exchange and no vibration.

6.1.2 FREE DECAY RESPONSE

Now consider the vibration due to initial condition only, the free decay response, in detail. We can have an alternative form of the solution of Equation 6.35, which can be written as

$$x(t) = de^{-\zeta\omega_n t} \sin(\omega_d t + \phi) \quad (6.62)$$

Readers may consider how this form compares to that of Equation 6.5.

6.1.2.1 Amplitude d and Phase ϕ

Similar to undamped systems, with initial conditions

$$x(0) = x_0 \quad \text{and} \quad \dot{x}(0) = \dot{x}_0$$

the amplitude and phase can be calculated as follows:

When $t = 0$,

$$x(0) = de^{-\zeta\omega_n 0} \sin(\omega_d 0 + \phi) = d \sin(\phi) = x_0 \quad (6.63)$$

Rewriting the above equation results in

$$d = \frac{x_0}{\sin \phi} \quad (6.64)$$

Taking the derivative of Equation 6.63,

$$\dot{x}(t) = \frac{d}{dt} x(t) = -\zeta\omega_n de^{-\zeta\omega_n t} \sin(\omega_d t + \phi) + \omega_d de^{-\zeta\omega_n t} \cos(\omega_d t + \phi) \quad (6.65)$$

Substituting the initial conditions in

$$\begin{aligned} \dot{x}(0) &= -\zeta\omega_n de^{-\zeta\omega_n 0} \sin(\omega_d 0 + \phi) + \omega_d de^{-\zeta\omega_n 0} \cos(\omega_d 0 + \phi) \\ &= d(-\zeta\omega_n \sin \phi + \omega_d \cos \phi) \end{aligned} \quad (6.66)$$

Furthermore, substitution of Equation 6.64 into Equation 6.66 yields

$$\dot{x}(0) = \frac{x_0}{\sin \phi} (-\zeta\omega_n \sin \phi + \omega_d \cos \phi) = x_0 \left(-\zeta\omega_n + \omega_d \frac{\cos \phi}{\sin \phi} \right) = \dot{x}_0 \quad (6.67)$$

Consequently,

$$\frac{\cos \phi}{\sin \phi} = \cot \phi = \frac{\dot{x}_0 + x_0(\zeta\omega_n)}{x_0(\omega_d)} \quad (6.68)$$

and

$$\phi = \tan^{-1} \left(\frac{x_0 \omega_d}{\dot{x}_0 + x_0 \zeta \omega_n} \right) \quad (6.69)$$

Refer to Figure 6.8 for reference.

Further calculation yields

$$\sin \phi = \frac{x_0 \omega_d}{\sqrt{(\dot{x}_0 + x_0 \zeta \omega_n)^2 + (x_0 \omega_d)^2}} \quad (6.70)$$

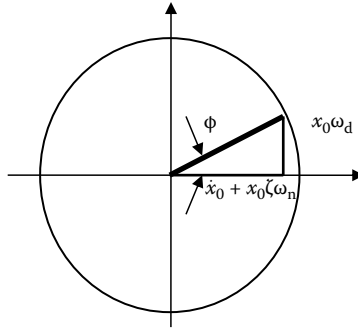


FIGURE 6.8 Phase angle.

Thus, resulting in

$$d = \frac{\sqrt{(\dot{x}_0 + x_0 \zeta \omega_n)^2 + (x_0 \omega_d)^2}}{\omega_d} \quad (6.71)$$

and

$$\phi = \tan^{-1} \left(\frac{\omega_d x_0}{\dot{x}_0 + \zeta \omega_n x_0} \right) + h_\phi \pi \quad (6.72)$$

The reason we have a term $h_\phi \pi$ in Equation 6.72 is to account for the cases of $\dot{x}_0 + x_0 \zeta \omega_n = 0$ as well as $\dot{x}_0 + x_0 \zeta \omega_n < 0$. The period of the tangent function is π ; therefore the arctangent function has multiple values. The period of the sine and cosine functions is 2π . Consequently, the Heaviside function, h_ϕ , cannot be chosen arbitrarily. Based on the fact that most computational programs, such as MATLAB®, calculate the arctangent by limiting the values from $-\pi/2$ to $+\pi/2$, h_ϕ is defined as

$$h_\phi = \begin{cases} 0, & v_0 + \zeta \omega_n x_0 > 0 \\ 1, & v_0 + \zeta \omega_n x_0 < 0 \end{cases} \quad (6.73)$$

As shown in Figure 6.8, there can be four instances of the phase angle ϕ . This is a result of the possible combinations of $\omega_d x_0$ and $v_0 + \zeta \omega_n x_0$, which have either a positive or negative value. Regardless of the values of $\omega_d x_0$, it is shown from Figure 6.9 and Equation 6.69 that the sign of $v_0 + \zeta \omega_n x_0$ determines the value of h_ϕ .

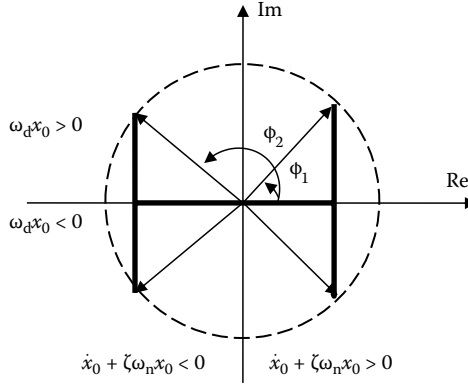


FIGURE 6.9 Determination of phase angle ϕ .

Example 6.6

A linear system with mass = 100 kg, stiffness = 1000 kN/m, and damping ratio = 0.5 is excited by initial condition $x_0 = 0.01$ m and $v_0 = -2$ m/s; calculate and plot the free-decay displacement.

The undamped and damped natural frequencies are given by

$$\omega_n = (k/m)^{1/2} = 100 \text{ (rad/s)}, \quad \text{and}$$

$$\omega_d = (1 - \zeta^2)^{1/2} \omega_n = 86.6 \text{ (rad/s)}$$

$$= 100 \text{ (rad/s)}$$

The amplitude is

$$d = \frac{\sqrt{(\dot{x}_0 + x_0 \zeta \omega_n)^2 + (x_0 \omega_d)^2}}{\omega_d} = 0.02$$

Because $\dot{x}(0) + \zeta \omega_n x(0) = -1.5 < 0$.

The phase angle is

$$\phi = \tan^{-1} \left(\frac{\omega_d x_0}{\dot{x}_0 + \zeta \omega_n x_0} \right) + \pi = 2.62$$

Therefore, the response is

$$x(t) = d e^{-\zeta \omega_n t} \sin(\omega_d t + \phi) = 0.02 e^{-50 t} \sin(86.6 t + 2.62)$$

The time history is plotted in Figure 6.10.

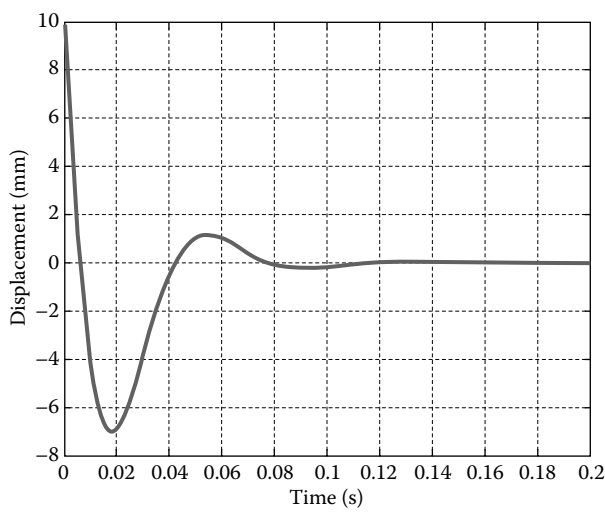


FIGURE 6.10 Free decay vibration.

6.2 PERIODICALLY FORCED VIBRATION

This section describes periodically forced vibration. A periodic forcing function can be represented by a Fourier series, which consists of a group of harmonic functions. We first consider the harmonic excitation followed by using linear combinations to obtain the responses of periodic forced vibrations.

6.2.1 HARMONIC EXCITATION

6.2.1.1 Equation of Motion

From the graphic description of damped SDOF systems shown in Figure 6.11, the following equation of motion is obtained with initial conditions $\dot{x}(0)$ and $x(0)$.

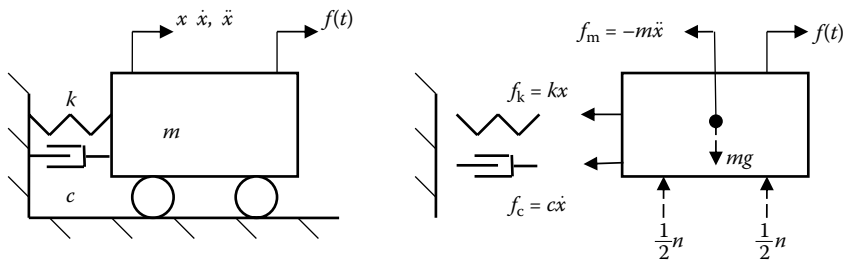


FIGURE 6.11 Damped SDOF system with excitation force.

$$\begin{cases} m\ddot{x} + c\dot{x} + kx = f(t) \\ \dot{x}(0) = v_0 \\ x(0) = x_0 \end{cases} \quad (6.74)$$

Equation 6.74 is a complete form to illustrate a SDOF vibration, often referred to as the *m-c-k* equation.

6.2.1.2 Harmonically Forced Response

To examine the solution of the *m-c-k* equation, we first consider the instance of a sinusoidal excitation, which is referred to as the harmonically forced response.

6.2.1.2.1 Forcing Function

The forcing function for harmonic excitation is sinusoidal, specifically

$$f(t) = f_0 \sin(\omega t) \quad (6.75a)$$

or

$$f(t) = f_0 \cos(\omega t) \quad (6.75b)$$

In the above, f_0 is the deterministic amplitude of force and ω is a driving frequency. Thus, this results in

$$m\ddot{x} + c\dot{x} + kx = f(t) = f_0 \sin(\omega t) \quad (6.76)$$

6.2.1.2.2 Solution, Forced Response

The general solution of Equation 6.76 contains two portions:

$$x(t) = x_h(t) + x_p(t) \quad (6.77)$$

in which $x_h(t)$ is the response due to the initial displacement and velocity and $x_p(t)$ is the particular solution due to the force excitation.

The particular solution can be expressed as

$$x_p(t) = x_{pt}(t) + x_{ps}(t) \quad (6.78)$$

where $x_{pt}(t)$ is the transient response due to the force $f(t)$ and $x_{ps}(t)$ is the steady-state solution. The total transient response, denoted by $x_t(t)$, is

$$x_t(t) = x_h(t) + x_{pt}(t) \quad (6.79)$$

Now, we first consider the steady-state response.

The condition of steady state signifies the work done by the external force during a cycle, ΔW , is equal to the energy dissipated by the vibration system, ΔE , that is,

$$\Delta W = \Delta E \quad (6.80)$$

and

$$x_{ps}(t) = x_{p0} \sin(\omega t + \phi) \quad (6.81)$$

In this case, the amplitude is (refer to Figure 6.7a)

$$x_{p0} = \text{const}$$

6.2.1.2.3 Complex Response Method

There are several methods to solve x_{p0} . In the complex response method, it is assumed that

$$f(t) = f_0(\cos(\omega t) + j\sin(\omega t)) = f_0 e^{j\omega t} \quad (6.82)$$

Equation 6.82 linearly combines the two cases, expressed by Equations 6.75a and 6.75b, by using complex functions. This case does not exist in the real world. Because the real and the imaginary domain are orthogonal, the response due to the real and the imaginary portions of the excitation will also be orthogonal. Suppose the response can be written as

$$x_{ps}(t) = x_{p0} e^{j(\omega t + \phi)} \quad (6.83)$$

The response due to the real force, $f_0 \cos(\omega t)$, denoted by $x_{ps}^{(R)}(t)$, is

$$x_{ps}^{(R)}(t) = \text{Re}[x_{p0} e^{j(\omega t + \phi)}] = x_{p0} \cos(\omega t + \phi) \quad (6.84a)$$

and the response due to the imaginary force, $f_0 \sin(\omega t)$, denoted by $x_{ps}^{(I)}(t)$, is

$$x_{ps}^{(I)}(t) = \text{Im}[x_{p0} e^{j(\omega t + \phi)}] = x_{p0} \sin(\omega t + \phi) \quad (6.84b)$$

In addition, Equation 6.83 can be written as

$$x_{ps}(t) = x_{p0} e^{j\phi} e^{j\omega t} = \tilde{x} e^{j\omega t} \quad (6.85)$$

where $\tilde{x}$ is a complex valued amplitude and

$$\tilde{x} = x_{p0} e^{j\phi} \quad (6.86)$$

Taking the first and the second order derivatives of Equation 6.85 with respect to t , yields

$$\dot{x}_{ps}(t) = j\omega \tilde{x} e^{j\omega t} \quad (6.87a)$$

and

$$\ddot{x}_{\text{ps}}(t) = -\omega^2 \tilde{x} e^{j\omega t} \quad (6.87b)$$

Substitution of Equations 6.82, 6.85, 6.86, 6.87a, and 6.87b into Equation 6.74 results in

$$-\omega^2 m \tilde{x} e^{j\omega t} + j\omega c \tilde{x} e^{j\omega t} + k \tilde{x} e^{j\omega t} = f_0 e^{j\omega t} \quad (6.88)$$

This gives a solution of

$$\tilde{x} = \frac{f_0}{(-\omega^2 m + j\omega c + k)} \quad (6.89)$$

or

$$\begin{aligned} \tilde{x} &= \frac{f_0 \div k}{(-\omega^2 m + j\omega c + k) \div k} = \frac{f_0}{k} \frac{1}{(-\omega^2/\omega_n^2 + j2\zeta\omega/\omega_n + 1)} \\ &= \frac{f_0}{k} \frac{1}{(-r^2 + j2\zeta r + 1)} \end{aligned} \quad (6.90)$$

where r is the frequency ratio

$$r = \frac{\omega}{\omega_n} \quad (6.91)$$

Equation 6.89 can also be expressed as

$$\tilde{x} = \frac{f_0}{k} \frac{1}{\sqrt{(1-r^2)^2 + (2\zeta r)^2}} e^{-j \tan^{-1} \frac{2\zeta r}{1-r^2}} \quad (6.92)$$

The absolute value of the complex valued amplitude is the amplitude of the steady-state response $x_{\text{ps}}(t)$, that is,

$$x_{\text{p0}} = |\tilde{x}| = \left| \frac{f_0}{k} \frac{1}{\sqrt{(1-r^2)^2 + (2\zeta r)^2}} e^{-j \tan^{-1} \frac{2\zeta r}{1-r^2}} \right| = \frac{f_0}{k} \frac{1}{\sqrt{(1-r^2)^2 + (2\zeta r)^2}} \quad (6.93)$$

The phase angle of the complex valued amplitude is the phase angle of the steady-state response $x_{ps}(t)$, that is,

$$\phi = \angle(\tilde{x}) = \angle\left(e^{-j \tan^{-1} \frac{2\zeta r}{1-r^2}}\right) = -\tan^{-1} \frac{2\zeta r}{1-r^2} \quad (6.94)$$

Because the tangent function is periodic with a period π , the inverse of the tangent function is multiplied in value. A more precise description of the phase angle is given by

$$\phi = \angle(\tilde{x}) = \angle\left(e^{-j \tan^{-1} \frac{2\zeta r}{1-r^2}}\right) = -\tan^{-1} \frac{2\zeta r}{1-r^2} + h_\phi \pi \quad (6.95)$$

where h_ϕ is the Heaviside step function given by

$$h_\phi = \begin{cases} 0, & \omega < \omega_n \\ 1, & \omega > \omega_n \end{cases} \quad (6.96)$$

Example 6.7

An m - c - k system with mass = 10 kg, $c = 15$ N/m^{-s}, and $k = 2000$ N/m is excited by a harmonic force $f_1(t) = 100 \sin(4t)$ under zero condition. Calculate and plot the response of displacement. If the excitation changes to $f_2(t) = 100 \sin(14t)$, how does the response change accordingly?

First, the natural frequency and damping ratio are, respectively, calculated to be 14.14 rad/s and 0.05.

For $f_1(t)$ and $f_2(t)$, the frequency ratios are, respectively, $r_1 = \omega_1/\omega_n = 0.282$, and $r_2 = \omega_2/\omega_n = 0.990$.

Let us now consider the steady-state solution $x_{ps}(t)$. Its amplitude, x_{p0} , can be calculated by taking the absolute value of $\tilde{x}$. That is,

$$x_{p0} = |\tilde{x}| = \left| \frac{f_0}{k} \frac{1}{\sqrt{(1-r^2)^2 + (2\zeta r)^2}} e^{j \tan^{-1} \frac{2\zeta r}{1-r^2}} \right| = \frac{f_0}{k} \frac{1}{\sqrt{(1-r^2)^2 + (2\zeta r)^2}}$$

The amplitude x_{p0} due to $f_1(t)$ is $100/2000/0.9205 = 0.05$

The amplitude x_{p0} due to $f_2(t)$ is $100/2000/0.9205 = 0.468$

The phase angle of the steady-state solution $x_{ps}(t)$ can be calculated by using the angle of $\tilde{x}$, that is,

$$\phi = \angle(\tilde{x}) = \angle\left(e^{-j \tan^{-1} \frac{2\zeta r}{1-r^2}}\right) = -\tan^{-1} \frac{2\zeta r}{1-r^2} + 0\pi$$

The phase ϕ due to $f_1(t)$ is $-\tan^{-1}(0.03/0.92) = -0.0326$

The phase ϕ due to $f_2(t)$ is $-\tan^{-1}(0.105/0.02) = -1.383$

Therefore, based on Equation 6.81, for the steady-state response due to $f_1(t)$, we have

$$x_{p1}(t) = x_{p0}\sin(\omega t + \phi) = 0.05\sin(4t - 0.033)$$

and for the steady-state response due to $f_2(t)$, we have

$$x_{p2}(t) = 0.468\sin(14t - 1.383)$$

The results are plotted in Figure 6.12, where the dotted line is $x_{p1}(t)$ and the solid line is $x_{p2}(t)$.

From Figure 6.12, it is seen that with driving frequency = 14 rad/s, comparatively much larger amplitude of the response is shown; this phenomenon is resonance, which will be further discussed in detail in Section 6.2.1.3.1. Also from Figure 6.12, we see the amplitude of the responses jump to their peak values in the first quarter cycle. At least for the excitation $f_2(t)$ and corresponding resonance, in our experience, this direct jump is not typical. Because a resonance is a cumulative effect, namely, to reach the peak value in the steady state, the amplitude is gradually increased with certain duration, and there must be a term in the total solution to describe the transient phenomenon. This is why we have to consider Equation 6.78 with the transient term $p_{pt}(t)$. In the following, using the concept of dynamic magnification factors and the semidefinite method, we can determine the transient response. In addition, we can also use convolution of the input force and unite impulse response functions to derive the transient response.

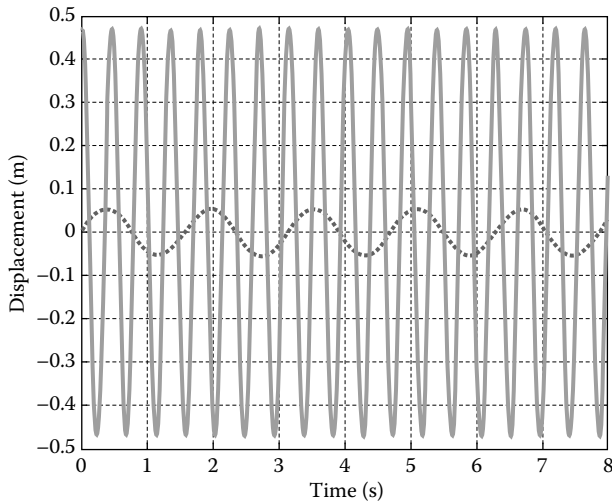


FIGURE 6.12 Responses under $f_1(t)$ and $f_2(t)$.

6.2.1.3 Dynamic Magnification

Equation 6.90 implies that the amplitude of $\tilde{x}$ is a function of r and ζ . This function unveils an important phenomenon: the amplitude of the vibration response can be magnified or reduced, dependent upon the frequency range and the damping capacity.

6.2.1.3.1 Dynamic Magnification Factor

In comparing Equation 6.90 with Equation 6.86, the absolute value of $\tilde{x}$ can alternatively be written as

$$x_{p0} = \frac{f_0}{k} \frac{1}{\sqrt{(1-r^2)^2 + (2\zeta r)^2}} = \frac{f_0}{k} \beta_D \quad (6.97)$$

In the above, the term β_D is referred to as the *dynamic magnification factor* because the amplitude of the response is magnified from the static response: $\frac{f_0}{k}$. This is shown in Figure 6.13a and the dynamic magnification factor can be written as

$$\beta_D = \frac{x_{p0}k}{f_0} = \frac{1}{\sqrt{(1-r^2)^2 + (2\zeta r)^2}} \quad (6.98)$$

In the term β_D , the subscript D stands for displacement.

In Figure 6.13a, it is seen that when the ratio r of the driving frequency and the natural frequency is approaching unity, the value of the dynamic magnification factor β_D will be comparatively much larger than that of the rest of the frequency regions, which is referred to as *resonance*, especially when the damping ratio is small. In the previous example, we have seen this phenomenon in the time domain. Now, the plot in Figure 6.13a shows the variation of the dynamic magnification factor β_D in the frequency domain. At the resonant point, β_D reaches its peak value. The frequency

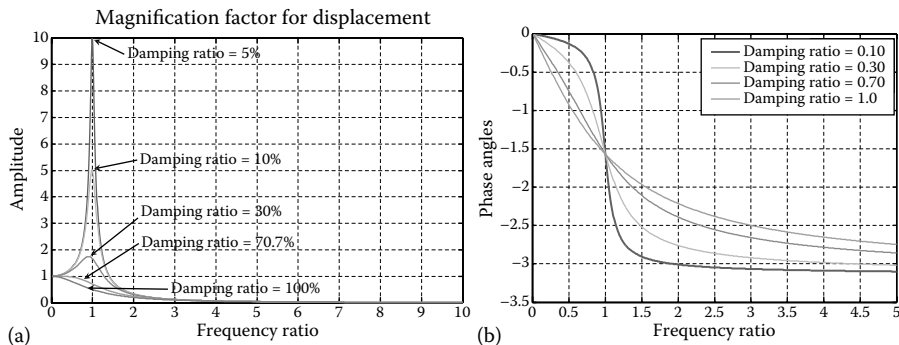


FIGURE 6.13 Plot of dynamic magnification factors: (a) amplitudes; (b) phases.

band where the value of β_D is greater than 70.7% of the peak value is defined as the *resonance region*.

It is seen that when the frequency ratio is much smaller than unity, namely, when the driving frequency is comparatively much smaller than the natural frequency, the value of β_D is close to unity and its value will gradually increase when the ratio r becomes larger.

When the frequency ratio is larger than unity, the value of β_D will become smaller. When the frequency ratio is much larger than unity, namely, when the driving frequency is comparatively much larger than the natural frequency, the value of β_D is approaching zero.

The phase angle between the forcing function and the response is

$$\phi = \tan^{-1} \frac{2\zeta r}{1 - r^2} + h_\phi \pi \quad (6.99)$$

In this instance, the Heaviside function is given in Equation 6.96. From Figure 6.13b, we can see that when the frequency ratio varies, so will the phase angle. As the frequency ratio becomes larger, the phase angle decreases from zero toward $-\pi$ (or -180°). At exactly the resonant point, the phase angle becomes $-\pi/2$.

6.2.1.4 Transient Response under Zero Initial Conditions

With zero initial conditions, we have particular solutions. The steady-state response is only a portion of the particular solution. To obtain the transient response, it is not necessary to use the above-mentioned dynamic magnification factor. However, with the help of such a concept, we can have a deeper insight into how the steady state is reached, especially into how resonance is accumulated.

Now, let us consider the transient part of the particular solution $x_{pt}(t)$ based on Equations 6.76 and 6.78, namely,

$$x_p(t) = x_{pt}(t) + x_{ps}(t)$$

From the above discussion, it is known that

$$x_{ps}(t) = f_0/k\beta_D \sin(\omega t + \phi)$$

We also know that a transient response is a free-decay vibration, which may take the following form,

$$x_{pt}(t) = ae^{-\zeta\omega_n t} \sin(\omega_d t + \theta_t) \quad (6.100)$$

where both the amplitude a and the phase θ_t are to be determined.

Now, the total particular solution of displacement may be written as

$$x_p(t) = ae^{-\zeta\omega_n t} \sin(\omega_d t + \theta_t) + f_0/k\beta_D \sin(\omega t + \phi) \quad (6.101a)$$

We use the semidefinite method to see if it is a true response. To do so, we first determine parameter a and θ_t , then check if Equation 6.101a can satisfy the m - c - k equation with a sinusoidal excitation $f_0 \sin(\omega t)$ with zero-initial conditions.

Take the derivative on both sides of Equation 6.101a with respect to time t , the possible velocity can be written as

$$\dot{x}_p(t) = ae^{-\zeta\omega_n t} \left[-\zeta\omega_n \sin(\omega_d t + \theta_t) + \omega_d \cos(\omega_d t + \theta_t) \right] + f_0/k\beta_D \omega \cos(\omega t + \phi) \quad (6.101b)$$

With zero initial conditions, we thus have the following two equations:

$$x_p(0) = 0 = a \sin(\theta_t) + f_0/k\beta_D \sin(\phi) \quad (6.102a)$$

and

$$\dot{x}_p(0) = 0 = a \left[-\zeta\omega_n \sin(\theta_t) + \omega_d \cos(\theta_t) \right] + f_0/k\beta_D \omega \cos(\phi) \quad (6.102b)$$

From Equation 6.102a, we can write

$$a = -f_0/k\beta_D \sin(\phi)/\sin(\theta_t) \quad (6.103)$$

Substituting Equation 6.103 into Equation 6.102b results in

$$\frac{f_0/k\beta_D \sin(\phi)}{\sin(\theta_t)} [-\zeta\omega_n \sin(\theta_t) + \omega_d \cos(\theta_t)] + f_0/k\beta_D \omega \cos(\phi) = 0$$

Therefore, we have

$$-\zeta\omega_n \sin(\phi) + \sqrt{1-\zeta^2} \omega_n \cot(\theta_t) + \omega \cos(\phi) = 0 \quad (6.104)$$

Because

$$\tan(\phi) = \frac{2\zeta r}{1-r^2}$$

We have

$$\sin(\phi) = \frac{2\zeta r}{\sqrt{(1-r^2)^2 + (2\zeta r)^2}} = \beta_D (2\zeta r) \quad (6.105a)$$

as well as

$$\cos(\phi) = \beta_D(1 - r^2) \quad (6.105b)$$

Substitution of Equations 6.105a and 6.105b into Equation 6.104 yields

$$\tan(\theta_t) = \frac{2\zeta\sqrt{1-\zeta^2}}{2\zeta^2 + r^2 - 1} \quad (6.106a)$$

Furthermore,

$$\sin(\theta_t) = \frac{2\zeta\sqrt{1-\zeta^2}}{\sqrt{(2\zeta r)^2 + (r^2 - 1)^2}} = \beta_D \left(2\zeta\sqrt{1-\zeta^2} \right) \quad (6.106b)$$

Comparing Equations 6.106b with 6.106a, we can write

$$\theta_t = (-1)^{h_{\theta_t}} \sin^{-1} \left\{ \beta_D \left(2\zeta\sqrt{1-\zeta^2} \right) \right\} + h_{\theta_t} \pi \quad (6.107a)$$

where the Heaviside step function h_{θ_t} is given by

$$h_{\theta_t} = \begin{cases} 0, & 2\zeta^2 + r^2 - 1 > 0 \\ 1, & 2\zeta^2 + r^2 - 1 < 0 \end{cases} \quad (6.107b)$$

Substitution of Equation 6.106a into Equation 6.104 results in

$$a = \frac{f_0/k\beta_D^2(2\zeta r)}{\beta_D \left(2\zeta\sqrt{1-\zeta^2} \right)} = \frac{f_0}{k} \frac{r\beta_D}{\sqrt{1-\zeta^2}} \quad (6.108)$$

Finally, to complete the semidefinite approach, we can substitute $x_p(t)$, $\dot{x}_p(t)$, and $\ddot{x}_p(t)$ into Equation 6.76 to see if it is balanced. With a positive result, it can be shown that Equation 6.101 is indeed a particular solution and thus Equation 6.100 is the transient part of the particular responses. In Figure 6.14a, the normalized amplitudes of the transient solution with zero initial conditions (namely, $f_0/k = 1$) versus the frequency ratio are plotted. It is seen that when $r = 1$, the amplitude reaches the peak value. Similar to the amplitude of the steady-state responses, when the damping ratio becomes smaller, the larger peak value can be seen. In Figure 6.14b, the phase angles versus the frequency ratio are also plotted. It is seen that with different damping ratios, the curves of the phase angle can be rather different.

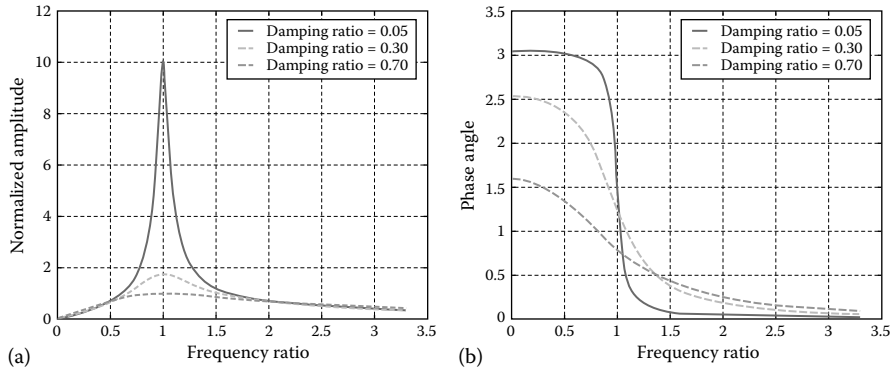


FIGURE 6.14 Amplitudes and phase angles of $x_p(t)$: (a) normalized amplitudes; (b) phases.

It is noted that unlike the free-decay vibration caused purely by initial velocity or initial displacement (or both). The response therefore exists without other conditions; the transient part of a particular solution cannot exist without the presentation of a steady-state solution.

6.2.1.4.1 Transfer Function Method

To study the steady-state response, we can also use the transfer function method, which was described in Chapter 4 and will be further discussed as follows.

6.2.1.4.1.1 Transfer Function When all the initial conditions are zero, the following Laplace transforms (Pierre-Simon Laplace, 1749–1827) exist:

$$\mathcal{L}[x(t)] = X(s) \quad (6.109)$$

and

$$\mathcal{L}[\dot{x}(t)] = sX(s) \quad (6.110)$$

Furthermore,

$$\mathcal{L}[\ddot{x}(t)] = s^2 X(s) \quad (6.111)$$

As well as

$$\mathcal{L}[f(t)] = F(s) \quad (6.112)$$

In general, s is the Laplace variable, where

$$s = \sigma + j\omega \quad (6.113)$$

Taking the Laplace transform on both sides of Equation 6.76 yields

$$\mathcal{L}[m\ddot{x}(t) + c\dot{x}(t) + kx(t)] = \mathcal{L}[f(t)] \quad (6.114)$$

and as a result:

$$ms^2X(s) + csX(s) + kX(s) = F(s) \quad (6.115)$$

or

$$[ms^2 + cs + k]X(s) = F(s) \quad (6.116)$$

The *transfer function* for the SDOF system can be defined as

$$H(s) = \frac{X(s)}{F(s)} = \frac{1}{ms^2 + cs + k} \quad (6.117)$$

6.2.1.4.1.2 Frequency Response Function In the case of steady-state response $\sigma = 0$ in Equation 6.113, that is,

$$s = j\omega$$

the frequency response function is

$$H(j\omega) = \frac{X(\omega)}{F(\omega)} = \frac{1}{m(j\omega)^2 + c(j\omega) + k} = \frac{1}{-m\omega^2 + j\omega c + k} \quad (6.118)$$

$$H(j\omega) = \frac{1}{k} \frac{1}{-\frac{m}{k}(\omega)^2 + \frac{c}{k}(j\omega) + 1} = \frac{1}{k} \frac{1}{1 - r^2 + 2j\zeta r} = \frac{1}{k} \beta_D e^{j\phi} \quad (6.119)$$

This results in

$$H(j\omega) = \left(\frac{1}{k} \beta_D \right) e^{j\phi} \quad (6.120)$$

$$|H(j\omega)| = \frac{1}{k} \beta_D \quad (6.121)$$

and

$$\angle H(j\omega) = \phi \quad (6.122)$$

The transfer function and the frequency response function are important concepts in describing the dynamic behavior of a linear vibration system.

6.2.2 BASE-EXCITATION AND FORCE TRANSMISSIBILITY

6.2.2.1 Model of Base Excitation

We now consider a special case of base excitation modeled by

$$m\ddot{x}_A + c\dot{x} + kx = 0 \quad (6.123)$$

where x_A is the absolute displacement. Let x_g be the base displacement, and x is the displacement of the mass relative to the base. This is given by

$$x_A = x_g + x \quad (6.124)$$

$$\dot{x}_A = \dot{x}_g + \dot{x} \quad (6.125)$$

$$\ddot{x}_A = \ddot{x}_g + \ddot{x} \quad (6.126)$$

Consequently, the case of base excitation is written as

$$m\ddot{x} + c\dot{x} + kx = -m\ddot{x}_g \quad (6.127)$$

In comparing the above equation with Equation 6.76, the term $-m\ddot{x}_g$ can be seen as a forcing function noted by

$$f(t) = -m\ddot{x}_g \quad (6.128)$$

Taking the Laplace transform on both sides of Equation 6.127 with zero initial conditions results in

$$[ms^2 + cs + k]X(s) = -ms^2X_g(s) \quad (6.129)$$

The transfer function between the ground displacement and the relative displacement of the base-isolator is

$$H_{Dr}(s) = \frac{X(s)}{X_g(s)} = \frac{-s^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (6.130a)$$

Here, the first subscript D stands for displacement, whereas the second stands for relative. In the instance of a frequency response function, it is given by

$$H_{Dr}(\omega) = \frac{X(\omega)}{X_g(\omega)} = \frac{(\omega^2) \div \omega_n^2}{(-\omega^2 + 2j\zeta\omega\omega_n + \omega_n^2) \div \omega_n^2} = \frac{r^2}{1 - r^2 + 2jr} \quad (6.130b)$$

Note the transfer function between the ground acceleration and the relative acceleration of the base-isolator, given by Equation 6.131, is identical to Equation 6.130b.

$$H_{Ar}(j\omega) = \frac{\ddot{X}(\omega)}{\ddot{X}_g(\omega)} = \frac{\omega^2 X(\omega)}{\omega^2 X_g(\omega)} = \frac{r^2}{1 - r^2 + 2jr} \quad (6.131)$$

In Equation 6.131, the first subscript A stands for acceleration.

In terms of dynamic magnification factor and phase angle,

$$H_{Ar}(j\omega) = \beta_{Ar} e^{j\Phi} \quad (6.132)$$

where $\|(\cdot)\|$ denotes the magnitude of function $(\cdot)$ and

$$\beta_{Ar} = \frac{\|\ddot{X}\|}{\|\ddot{X}_g\|} = \left| \frac{r^2}{1 - r^2 + 2jr} \right| = \frac{r^2}{\sqrt{(1 - r^2)^2 + (2\zeta r)^2}} \quad (6.133)$$

In the above, β_{Ar} is the dynamic magnification factor for the relative acceleration $\ddot{x}(t)$ excited by harmonic base acceleration $\ddot{x}_g(t)$. Its value is exactly the dynamic magnification factor for the relative displacement $x(t)$ excited by harmonic base displacement $y(t)$, denoted by β_{Dr} . Namely,

$$\beta_{Dr} = \beta_{Ar} \quad (6.134)$$

Additionally,

$$\Phi = \tan^{-1} \left(\frac{2\zeta r}{1 - r^2} \right) + h_\phi \pi \quad (6.135)$$

where h_ϕ is the Heaviside step function given by Equation 6.76. The transfer function between the ground acceleration and the absolute acceleration of the base-isolator is

$$H_{Aa}(s) = \frac{\ddot{X}(s) + \ddot{X}_g(s)}{\ddot{X}_g(s)} = \frac{r^2}{1 - r^2 + 2jr} + 1 = \frac{1 + 2jr}{1 - r^2 + 2jr} \quad (6.136)$$

Here, the second subscript a stands for absolute. In terms of the dynamic magnification factor and phase angle, as well as for the steady-state solution, let $s = j\omega$.

$$H_{Aa}(j\omega) = \beta_{Aa} e^{j\phi} \quad (6.137)$$

where

$$\beta_{Aa} = \frac{\|\ddot{x}_A\|}{\|\ddot{x}_g\|} = \left| \frac{1 + 2j\zeta r}{1 - r^2 + 2j\zeta r} \right| = \sqrt{\frac{1 + (2\zeta r)^2}{(1 - r^2)^2 + (2\zeta r)^2}} \quad (6.138)$$

In Equation 6.138, $\frac{\|\ddot{x}\|}{\|\ddot{x}_g\|}$ is the ratio of amplitudes of the absolute and ground acceleration and

$$\Phi = \tan^{-1} \left[\frac{2\zeta r^3}{1 - (1 - 4\zeta^2)r^2} \right] + h_\Phi \pi \quad (6.139)$$

where h_Φ is the Heaviside step function given by

$$h_\Phi = \begin{cases} 0, & 1 - (1 - 4\zeta^2)r^2 > 0 \\ 1, & 1 - (1 - 4\zeta^2)r^2 < 0 \end{cases} \quad (6.140)$$

Therefore, suppose the normalized ground excitation is

$$\ddot{x}_g(t) = \sin(\omega t) \quad (6.141)$$

The steady-state absolute acceleration can then be written as

$$\ddot{x}_A(t) = \beta_{Aa} \sin(\omega t - \Phi) \quad (6.142)$$

The dynamic magnification factor and phase angle versus frequency ratio are plotted in Figure 6.15a and b, respectively. Compared with the dynamic magnification factors, which can be seen as normalized amplitude of forced responses shown in Figure 6.13a, we can observe that they both have a similar trend of ups and downs. However, for the case of based isolation, when the amplitude is reduced from the resonance value toward zero, no matter what the damping ratio is, the curves will reach unity at exactly $r = 1.4142$.

Note that in Equation 6.81, the phase angle for the sine function is $\sin(\omega t + \phi)$. However, for the cases of base excitation, the phase angle for the sine function is $\sin(\omega t - \Phi)$, namely, with a minus sign.

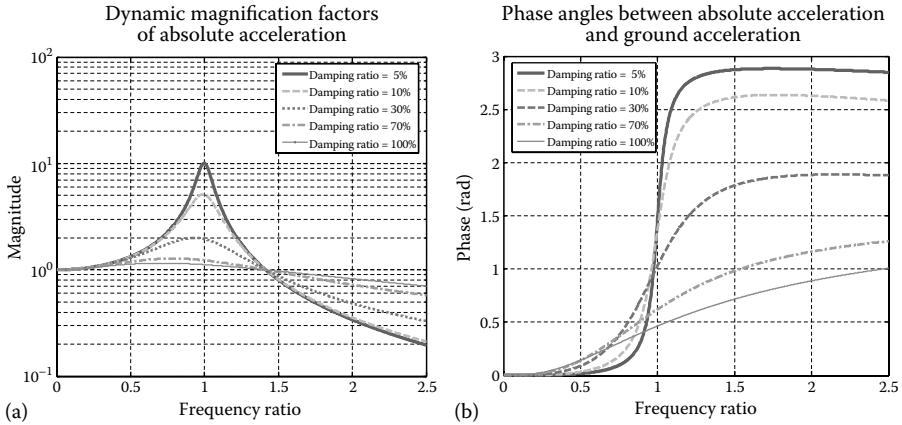


FIGURE 6.15 Magnitude (a) and phase (b) of absolute acceleration due to ground excitation.

Example 6.8

A machine with mass = 1500 kg is supported by a stiffness $k = 1480.450$ kN/m with a damping ratio of 0.01 and is excited by ground acceleration with a driving frequency of 7 Hz and an amplitude of 1.5 g (1 g = 9.8 m/s²). If the absolute acceleration needs to be reduced to less than 1.2 g and the relative displacement must be limited to less than 14.0 mm, design the required stiffness and damping ratio.

The ground displacement is $y = 1.5 * 9.8 / (5 * 2\pi)^2 = 14.9$ mm.

The natural frequency is $(k/m)^{1/2} / 2\pi = 5.0$ (Hz) so that the frequency ratio is $r = 7/5 = 1.40$. Then, the dynamic magnification factors for the absolute acceleration and the relative displacement are, respectively, given by

$$\beta_{Aa} = \frac{1 + (2\zeta r)^2}{\sqrt{(1 - r^2)^2 + (2\zeta r)^2}} = 1.0875$$

and

$$\beta_{Dr} = \beta_{Ar} = \frac{r^2}{\sqrt{(1 - r^2)^2 + (2\zeta r)^2}} = 2.048$$

Thus, the absolute acceleration is 1.5 g * 1.0875 = 1.631 g and the relative displacement is 14.9 mm * 2.048 = 30.51 mm, neither of which satisfies the requirements.

It is seen that the displacement is more than 20 mm, and the acceleration is more than 1.2 g. To reduce the acceleration, we need the dynamic magnification factor to be

$$\beta_{Aa} \leq 1.2 / 1.5 \leq 0.8$$

If we keep the same damping ratio, the new frequency ratio can be calculated as

$$0.8^2 \leq \frac{1 + (2\zeta r)^2}{(1 - r^2)^2 + (2\zeta r)^2}$$

so that we have $r > 1.45$. Let us choose $r = 1.46$.

However, in this case, β_{Dr} is calculated to be 1.9 so that the displacement is $14.9 \times 1.9 \text{ mm} > 14.0 \text{ mm}$. Next, we choose the damping ratio to be 0.68. In this case, β_{Dr} is calculated to be 0.93 so that the displacement is $13.9 < 14 \text{ mm}$. The final design is natural frequency $5/1.46 = 3.42 \text{ (Hz)}$, and the damping ratio is 0.68.

6.2.2.2 Force Transmissibility

Suppose a forcing function $f_0 \sin(\omega t)$ is applied on an m - c - k structure; find the amplitude of the steady-state force transferred from the structure to the ground. In Figure 6.11, the dynamic force transferred to the ground, denoted by $f_g(t)$, is the sum of damping and the stiffness forces, namely,

$$f_g(t) = c\dot{x} + kx \quad (6.143)$$

From the above discussion, it is seen that the steady-state displacement is given by

$$x(t) = f_0/k\beta_D \sin(\omega t + \phi)$$

and the velocity is given by

$$\dot{x}(t) = f_0/k\beta_D \omega \cos(\omega t + \phi)$$

Therefore,

$$f_g(t) = f_0/k\beta_D [k \sin(\omega t + \phi) + c\omega \cos(\omega t + \phi)] \quad (6.144)$$

Let the amplitude be denoted by f_G , which can be written as

$$\begin{aligned} f_G &= |f_g(t)| = f_0/k\beta_D [k^2 + (c\omega)^2]^{1/2} \\ &= f_0/k \left[\frac{k^2 + c^2\omega^2}{(1 - r^2)^2 + (2\zeta r)^2} \right]^{1/2} = f_0 \sqrt{\frac{1 + (2\zeta r)^2}{(1 - r^2)^2 + (2\zeta r)^2}} \end{aligned} \quad (6.145)$$

Denoting the dynamic magnification factor for the force transmissibility as β_T , the amplitude of ground force can be written

$$f_G = \beta_T f_{-0} \quad (6.146)$$

where

$$\beta_T = \sqrt{\frac{1 + (2\zeta r)^2}{(1 - r^2)^2 + (2\zeta r)^2}} \quad (6.147)$$

and thus

$$\beta_T = \beta_{Aa} \quad (6.148)$$

The term is often simply referred to as the *force transmissibility*.

To find the phase between the forcing function and the steady-state ground force $f_g(t)$, let us again apply the complex method by letting the forcing function be $f(t) = f_0 e^{j\omega t}$. In this case, the ground force can be written as

$$f_g(t) = k\tilde{x}e^{j\omega t} + jc\omega\tilde{x}e^{j\omega t} = (k + jc\omega)\tilde{x}e^{j\omega t} \quad (6.149)$$

where the amplitude of the complex value displacement $\tilde{x}$ is given by Equation 6.92. As a result, the steady-state ground force can be further written as

$$f_g(t) = \frac{f_0}{k} \frac{(k + jc\omega)}{\sqrt{(1 - r^2)^2 + (2\zeta r)^2}} e^{-j \tan^{-1} \frac{2\zeta r}{1 - r^2}} e^{j\omega t} \quad (6.150)$$

From the absolute value of the ground force described in Equation 6.150, we can realize that dynamic magnification of the force transmissibility is indeed the one given in Equation 6.147. Furthermore, the phase difference can be written as

$$\begin{aligned} \Phi &= \angle \left[\frac{f_0}{k} \frac{(k + jc\omega)}{\sqrt{(1 - r^2)^2 + (2\zeta r)^2}} e^{j \tan^{-1} \frac{2\zeta r}{1 - r^2}} \right] \\ &= \angle \left[(1 + j2\zeta r)\beta_D \left\{ \cos \left(\tan^{-1} \frac{2\zeta r}{1 - r^2} \right) - j\beta_D \sin \left(\tan^{-1} \frac{2\zeta r}{1 - r^2} \right) \right\} \right] \\ &= \angle \left[(1 + j2\zeta r)\{\beta_D(1 - r^2) + j\beta_D(2\zeta r)\} \right] \\ &= \angle \left[(1 - r^2) + (2\zeta r)^2 + j\{(2\zeta r)(1 - r^2) - (1 - r^2)\} \right] \\ &= \tan^{-1} \frac{2\zeta r^3}{1 - r^2 + (2\zeta r)^2} + h_\Phi \pi \end{aligned} \quad (6.151)$$

where the Heaviside function h_Φ is given by

$$h_\Phi = \begin{cases} 0, & 1 - r^2 + (2\zeta r)^2 > 0 \\ 1, & 1 - r^2 + (2\zeta r)^2 < 0 \end{cases} \quad (6.152)$$

The ground force can then be written as

$$f_g(t) = f_0 \beta_T \sin(\omega t + \Phi) \quad (6.153)$$

Comparing Equations 6.151 and 6.139, we realize that the phase difference of the absolute acceleration excited by the ground and the phase difference of the ground force due to force applied on a system are identical, similar to the corresponding magnifications factors.

6.2.3 PERIODIC EXCITATIONS

6.2.3.1 General Response

Consider now a linear m - c - k system excited by a periodic forcing function $f(t)$ of period T , with initial conditions x_0 and v_0 . That is,

$$\begin{cases} m\ddot{x} + c\dot{x} + kx = f(t) \\ x(0) = x_0 \\ \dot{x}(0) = v_0 \end{cases} \quad (6.154)$$

Because the excitation is periodic, a Fourier series with a basic frequency of

$$\omega_T = 2\pi/T \quad (6.155)$$

is used to represent the forcing functions. Suppose $f(t)$ can be represented by the Fourier series

$$f(t) = \frac{f_{A0}}{2} + \sum_{n=1}^N [f_{An} \cos(n\omega_T t) + f_{Bn} \sin(n\omega_T t)] \quad (6.156)$$

where f_{A0} , f_{An} , and f_{Bn} are Fourier coefficients. Because the system is linear, the responses are first considered individually due to the forcing function

$$f_A = \frac{f_{A0}}{2} \quad (6.157)$$

with the initial conditions, denoted by $x_0(t)$ and the steady-state responses due to the forcing functions

$$f_{an}(t) = f_{An} \cos(n\omega_T t) \quad (6.158)$$

$$f_{bn}(t) = f_{Bn} \sin(n\omega_T t) \quad (6.159)$$

denoted by $x_{an}(t)$ and $x_{bn}(t)$.

The total response can then be seen as the summation of $x_0(t)$ and all the $x_{an}(t)$ and $x_{bn}(t)$. Thus,

$$x(t) = x_0(t) + \sum_{n=1}^N [x_{an}(t) + x_{bn}(t)] = x_0(t) + \sum_{n=1}^N [x_n(t)] \quad (6.160)$$

where

$$x_n(t) = x_{an}(t) + x_{bn}(t) \quad (6.161)$$

6.2.3.2 The n th Steady-State Response

The steady-state response corresponding to the n th excitation component described by Equation 6.161 can also be written as

$$x_n(t) = \frac{f_N}{k} \beta_n \sin(n\omega_T t + \phi_n) \quad (6.162)$$

Here,

$$f_N = \sqrt{f_{An}^2 + f_{Bn}^2} \quad (6.163)$$

is the amplitude of the n th forcing function and the dynamic magnification β_n as well as phase angle ϕ_n are

$$\beta_n = \frac{1}{\sqrt{\left(1 - \frac{n^2 \omega_T^2}{\omega_n^2}\right)^2 + \left(2\zeta \frac{n\omega_T}{\omega_n}\right)^2}} = \frac{1}{\sqrt{(1 - n^2 r^2)^2 + (2\zeta nr)^2}} \quad (6.164)$$

$$\phi_n = \tan^{-1} \left[\frac{2\zeta nr}{n^2 r^2 - 1} \right] + \tan^{-1} \left(\frac{f_{An}}{f_{Bn}} \right) + (h_\phi + h_\Phi) \pi \quad (6.165)$$

where

$$h_{\phi} = \begin{cases} 0 & n^2 r^2 > 1 \\ -1 & n^2 r^2 \leq 1 \end{cases} \quad (6.166)$$

and

$$h_{\Phi} = \begin{cases} 0 & f_{Bn} \geq 0 \\ 1 & f_{Bn} < 0 \end{cases} \quad (6.167)$$

In Equations 6.164 through 6.166, r is the frequency ratio and

$$r = \frac{\omega_T}{\omega_n} \quad (6.168)$$

6.2.3.3 Transient Response

Assume that the transient response due to the initial conditions and the force constant f_{A0} is

$$x_0(t) = e^{-\zeta\omega_n t} [A \cos(\omega_d t) + B \sin(\omega_d t)] + \frac{f_{A0}}{k} \quad (6.169)$$

The first term in Equation 6.169 is mainly generated by the initial conditions and the second term is a particular solution due to the step input described in Equation 6.157.

It can be proven that the coefficients A and B are

$$A = x_0 - \frac{f_{A0}}{k} - \sum_{n=1}^N \frac{f_N}{k} \beta_n \sin \phi_n \quad (6.170)$$

$$B = \frac{1}{\omega_d} \left(v_0 + A \zeta \omega_n - \sum_{n=1}^N \frac{f_N}{k} \beta_n n \omega_T \cos \phi_n \right) \quad (6.171)$$

6.3 RESPONSE OF SDOF SYSTEM TO ARBITRARY FORCES

In this section, arbitrary excitations will be considered.

6.3.1 IMPULSE RESPONSES

In the case when a very large force is applied to a SDOF system for a very short duration, the excitation can be seen as an impulse process. This simple excitation can be

modeled by a delta function multiplied by the amplitude of the impulse, which is the foundation of the study of arbitrary excitations.

6.3.1.1 Unit Impulse Response Function

With zero initial conditions and impulse excitations, the equation of motion can be given as

$$\begin{cases} m\ddot{x} + c\dot{x} + kx = f_0\delta(t) \\ x(0) = \dot{x}(0) = 0 \end{cases} \quad (6.172)$$

Here, f_0 is the amplitude of an impulse. In the case $f_0 = 1$, the response can be calculated as follows.

First, when the initial displacement is zero, the mass is considered to be at rest just shortly prior to the application of the impulse f_0 . At the moment when f_0 is applied, the momentum of the system gains mv_0 . That is,

$$f_0 = f(t)\Delta t = mv_0 - 0$$

Thus,

$$v_0 = \frac{f(t)\Delta t}{m} = \frac{f_0}{m} \quad (6.173)$$

Then, the effect of an impulse applied to the SDOF m - c - k system is identical to the case of a free vibration with zero initial displacement and initial velocity equal to that described in Equation 6.62, with $\dot{x}(0) = v_0 = f_0/m$ and $x(0) = 0$. Furthermore, with unit impulse $f_0 = 1$ from Equation 6.71, the amplitude d is

$$d = \frac{1}{m\omega_d}$$

and from Equation 6.172, the phase angle is

$$\phi = 0$$

Therefore, when $I_0 = 1$, the response due to unit impulse is

$$x(t) = \frac{1}{m\omega_d} \left[e^{-\zeta\omega_n t} \sin(\omega_d t) \right] \quad (6.174)$$

This expression is quite important. A special notation is used to represent this unit impulse response, denoted by $h(t)$. Thus, let

$$h(t) = x(t) \Big|_{\substack{\text{unit impulse} \\ \text{zero initial condition}}} = \frac{1}{m\omega_d} \left[e^{-\zeta\omega_n t} \sin(\omega_d t) \right] \quad (6.175)$$

where the quantity $h(t)$ is known as the *unit impulse response function*.

Substitution of Equation 6.175 into Equation 6.172 yields

$$m\ddot{h} + c\dot{h} + kh = \delta(t) \quad (6.176)$$

In this case, $h(t)$ is called unit impulse response function. Generally, when $f_0 \neq 1$, this results in

$$x(t) = f_0 h(t) \quad (6.177)$$

6.3.2 ARBITRARY LOADING AND CONVOLUTION

6.3.2.1 Convolution

It is known that the response of an SDOF system under an arbitrary excitation $f(t)$ can be expressed as the following convolution:

$$x(t) = \int_0^t f(\tau) h(t - \tau) d\tau \quad (6.178)$$

Figure 6.16 graphically shows the essence of Equation 6.178. With the help of Figure 6.16a, let us consider a special instance $t = t_i$. The amplitude of the force $f(t_i)$ can be seen as a result of the sampling effect by the delta function $\delta(t - t_i)$ (see Figure 6.16b). That is, a response will occur starting from time $t - t_i$ as well as afterward (Figure 6.16c). It can be regarded as a unit impulse response times the amplitude of the instantaneous force. That is, the response is $[f(t_i) \Delta t][h(t - t_i)]$. However, before the instance t_i , we already had many other impulse responses. Each response can be seen as an impulse that occurs at $t_{i-1}, t_{i-2}, \dots, 0$. Thus, the total response can be regarded as a sum of all of these impulse responses. Note that the consideration of the response is not just ending at t_i . It may last until time t (Figure 6.16d). In this instance, we just show an additional response at t_{i+1} . Then, we will have the summation of those impulse responses, which are functions of t_i , and t_i starting at 0 and ending at t . When letting $\Delta t \rightarrow dt$, the summation becomes an integral and we thus have Equation 6.178.

Additionally, Equation 6.178 can be rewritten as

$$x(t) = \int_{-\infty}^t f(\tau) h(t - \tau) d\tau = \int_{-\infty}^{\infty} f(\tau) h(t - \tau) d\tau \quad (6.179)$$

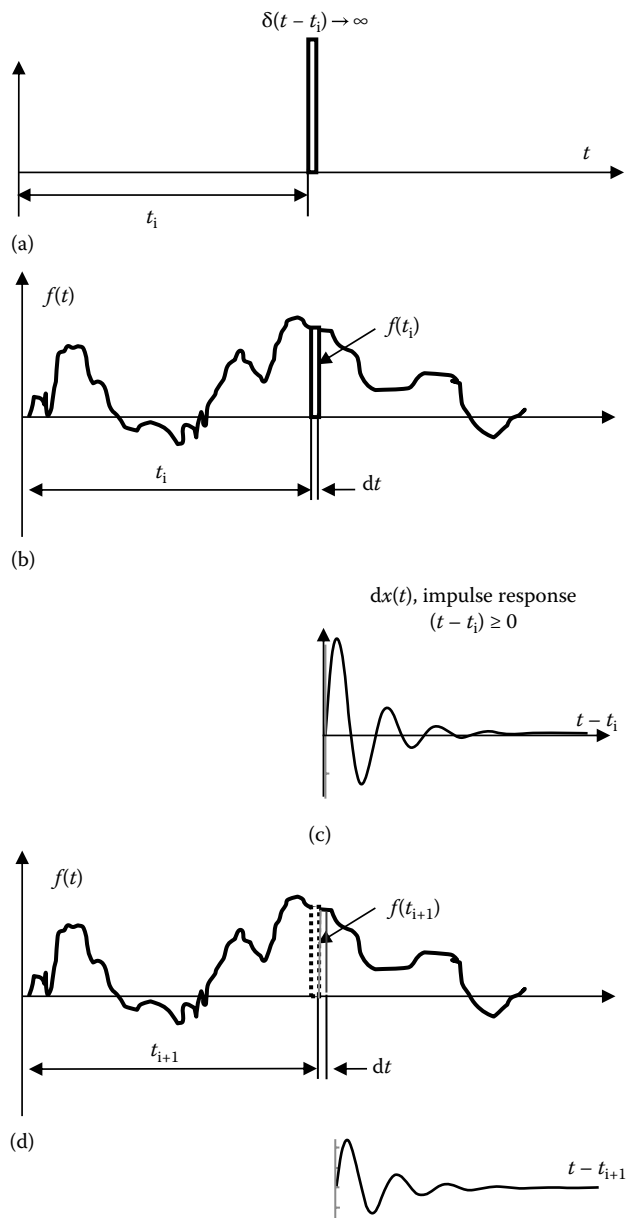


FIGURE 6.16 Convolution integral. (a) impulse; (b) forcing function; (c) impulse response; (d) additional impulse and response.

and

$$x(t) = \int_{-\infty}^{\infty} f(t - \tau)h(\tau) d\tau \quad (6.180)$$

Combining Equations 6.38 and 6.39, it can be simply denoted by

$$x(t) = f(t) * h(t) = h(t) * f(t) \quad (6.181)$$

6.3.2.2 Transient Response under Harmonic Excitation $f_0 \sin(\omega t)$

With the help of Equation 6.180, we consider the transient response under a unit harmonic excitation $f(t) = \sin \omega t$, namely, the term $x_{pt}(t)$ in Equation 6.78, under zero initial conditions.

Substituting the forcing function $\sin \omega(t - \tau)$ into Equation 6.180 and with the help of the unit impulse response function of SDOF vibration systems (see Equation 6.176), we have

$$x(t) = \frac{e^{-\zeta\omega_n t}}{m\omega_d} \int_{-\infty}^{\infty} e^{-\zeta\omega_n \tau} \sin[\omega_d(t - \tau)] \sin\omega\tau d\tau \quad (6.182)$$

Note that the response $x(t)$ is due to the forcing function only, that is, it is $x_p(t)$. Furthermore, we can write

$$x_p(t) = \frac{e^{-\zeta\omega_n t}}{m\omega_d} \int_0^t e^{-\zeta\omega_n \tau} \sin[\omega_d(t - \tau)] \sin\omega\tau d\tau \quad (6.183)$$

Evaluating Equation 6.183, we can have the solution (see Equation 6.78) repeated as follows:

$$x_p(t) = x_{pt}(t) + x_{ps}(t) \quad (6.184)$$

whereas $x_{ps}(t)$ is the steady-state response for the particular solution $x_p(t)$ described above, the transient response, $x_{pt}(t)$ can be calculated as

$$x_{pt}(t) = \frac{r\beta_D}{k\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_d t + \theta_t) \quad (6.185)$$

where the phase angle θ_t is given by

$$\theta_t = \tan^{-1} \frac{2\zeta\sqrt{1-\zeta^2}}{2\zeta^2 + r^2 - 1} + h_\theta\pi \quad (6.186a)$$

where h_θ is the Heaviside step function given by

$$h_\theta = \begin{cases} 0, & 2\zeta^2 + r^2 - 1 > 0 \\ 1, & 2\zeta^2 + r^2 - 1 < 0 \end{cases} \quad (6.186b)$$

Note that in Equation 6.185, generally speaking, $x_{pt}(t) \neq 0$. This implies that even under zero initial conditions and under the zero initial force (because $\sin\omega 0 = 0$), we still have the nonzero transient response. Equation 6.185 is exactly the same with the formula obtained through Equations 6.108 and 6.186a, which is equivalent to Equation 6.107a. However, based on the method of convolution, we can obtain a complete solution. Through a semidefinite method, we do have a solution, but we cannot prove that it is the only solution.

Example 6.9

Reconsider the above-mentioned example of the m - c - k system with mass = 10 kg, $c = 15 \text{ N/m}^s$ and $k = 2000 \text{ N/m}$ excited by a harmonic force $f_1(t) = 100\sin(4t)$ under zero condition $x_0 = 2 \text{ m}$ and $v_0 = 1 \text{ m/s}$, calculate and plot the response of displacement. If the excitation changes to $f_2(t) = 100\sin(14t)$, how does the response change accordingly?

The natural frequency and damping ratio are, respectively, calculated to be 14.14 and 0.05. We then can calculate the following parameters, where the subscripts 1 and 2 stand for the cases of $f_1(t)$ and $f_2(t)$, respectively

$$r_1 = 0.2828, \quad r_2 = 0.9899$$

$$\beta_{D1} = 1.0864, \quad \beta_{D2} = 9.3556$$

and

$$\theta_{t1} = 3.0263, \quad \theta_{t2} = 1.7057$$

With the above parameters, the transient solutions for the particular responses can be computed. Furthermore, in the previous example, the steady-state responses were calculated (Figure 6.12). Therefore, the total particular responses $x_p(t) = x_{pt}(t) + x_{ps}(t)$ can also be calculated.

The results are plotted in Figure 6.17a with driving frequency = 4 rad/s, and in Figure 6.17b with driving frequency 14, where the solid lines are the transient responses and the broken lines are the total particular responses. Compared with Figure 6.12, the steady-state responses, it is seen that with the transient portions, the total response can be rather different.

Furthermore, consider the initial conditions $v_0 = 1 \text{ m/s}$ and $x_0 = 2 \text{ m}$, we can calculate the homogeneous solutions $x_h(t)$ based on Equation 6.62, which is not affected by the forcing function for $f_1(0) = f_2(0) = 0$. Including the homogeneous solution, the total responses are plotted in Figure 6.18a with driving frequency = 4 rad/s and Figure 6.18b with driving frequency 14 rad/s.

Compare the total responses with the particular solutions shown in Figure 6.17 and with the steady-state ones shown in Figure 6.12, we can see the differences

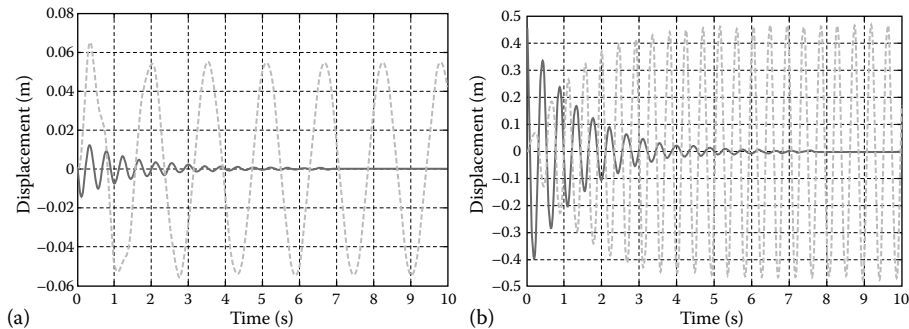


FIGURE 6.17 Particular responses: (a) driving frequency = 4 rad/s; (b) driving frequency = 14 rad/s.

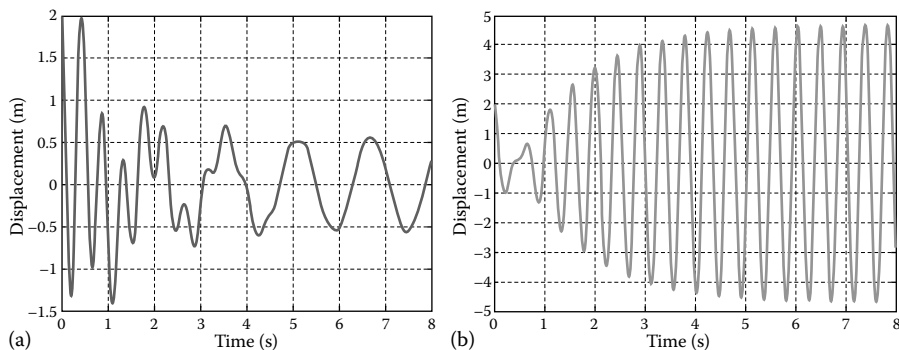


FIGURE 6.18 Total forced responses: (a) $f_1(t) = 100\sin(4t)$; (b) $f_2(t) = 100\sin(14t)$.

once more. In the course of deterministic vibration, often the transient responses, caused by both the initial conditions and the driving force, are ignored. This is because when the time is sufficiently long, the transient responses will die out. Only the steady-state response remains. However, in the case of random excitation, the transient portion of the responses should be carefully examined because with a random excitation, the transient portion will not die out.

6.3.3 IMPULSE RESPONSE FUNCTION AND TRANSFER FUNCTION

Recalling Equations 6.116 and 6.117, and additionally, Equation 6.129, for base isolation,

$$[ms^2 + cs + k]X(s) = F(s) \quad (6.187)$$

From Equation 6.187, the transfer function is given by

$$H(s) = \frac{X(s)}{F(s)} = \frac{1}{ms^2 + cs + k} \quad (6.188)$$

Furthermore, it is seen that

$$X(s) = H(s)F(s) \quad (6.189)$$

In the case of unit impulse response,

$$[ms^2 + cs + k]\mathcal{L}[h(t)] = \mathcal{L}[\delta(t)] = 1 \quad (6.190)$$

Thus,

$$\mathcal{L}[h(t)] = H(s)(1) = H(s) \quad (6.191)$$

Specifically, it can be stated that the unit impulse response function and the transfer function is a Laplace pair.

$$h(t) \Leftrightarrow H(s) \quad (6.192)$$

Generally speaking, for a harmonic excitation, when the response reaches steady state, let $s = j\omega$, the unit impulse response function and the transfer function also become a Fourier pair:

$$h(t) \Leftrightarrow H(j\omega) \quad (6.193a)$$

For convenience, Equation 6.193a is often rewritten as

$$h(t) \Leftrightarrow H(\omega) \quad (6.193b)$$

Here, the unit impulse response function is given by

$$h(t) = \frac{e^{-\zeta\omega t}}{m\omega_d} \sin(\omega_d t) \quad (6.194)$$

Now, consider the case of harmonic excitation. Let

$$f(t) = e^{j\omega t} \quad (6.195)$$

Then,

$$x(t) = f(t) * h(t) = \int_{-\infty}^{\infty} h(\tau)f(t-\tau) d\tau = \int_{-\infty}^{\infty} h(\tau)e^{j\omega(t-\tau)} d\tau = e^{j\omega t} \int_{-\infty}^{\infty} h(\tau)e^{-j\omega\tau} d\tau \quad (6.196)$$

Because

$$\int_{-\infty}^{\infty} h(\tau)e^{-j\omega\tau} d\tau = \mathcal{F}[h(t)] = H(\omega) \quad (6.197)$$

As a result, we obtain a useful relationship representing the harmonic response.

$$x(t) = H(\omega)e^{j\omega t} \quad (6.198)$$

For a general response, take the Fourier transform on both sides of Equation 6.181,

$$\mathcal{F}[x(t)] = \mathcal{F}[f(t) * h(t)] = \mathcal{F}[f(t)]\mathcal{F}[h(t)] \quad (6.199)$$

$$X(\omega) = F(\omega)H(\omega) = H(\omega)F(\omega) \quad (6.200)$$

Furthermore, taking the inverse Fourier transform on both sides of Equation 6.200 results in

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega)H(\omega)e^{j\omega t} d\omega \quad (6.201)$$

6.3.4 FREQUENCY RESPONSE AND TRANSFER FUNCTIONS

We now compare Equations 6.200 and 6.198. Equation 6.198 is obtained through harmonic excitation. Because

$$\dot{x}(t) = j\omega x(t), \quad \ddot{x}(t) = -\omega^2 x(t)$$

and accordingly,

$$(-m\omega^2 + j\omega c + k)x(t) = e^{j\omega t}$$

and

$$x(t) = \frac{1}{-m\omega^2 + j\omega c + k} e^{j\omega t} = H(\omega)e^{j\omega t}$$

Therefore,

$$H(\omega) = x(t)e^{-j\omega t} = \left. \frac{x(t)}{f(t)} \right|_{\text{condition: } f(t)=e^{j\omega t}} \quad (6.202)$$

Equation 6.202 is obtained through arbitrary excitation, in which

$$H(\omega) = \frac{X(\omega)}{F(\omega)} \Big|_{\text{condition} = \text{steady state response}} \quad (6.203)$$

In the literature, the transfer function obtained through the steady-state response of harmonic excitation is specifically defined as the frequency response function. Mathematically, obtaining the frequency response function is equivalent to replacing s with $j\omega$ in the generic form of the transfer function.

6.3.5 BOREL'S THEOREM AND ITS APPLICATIONS

6.3.5.1 Borel's Theorem

In rewriting the general form as described in Equation 6.200, produces Borel's theorem, which is the convolution theorem described in the Laplace domain:

$$X(s) = H(s)F(s) \quad (6.204)$$

In this instance, the response $X(s)$ can be seen as the input $F(s)$ being transferred through $H(s)$. Because Borel's theorem does not specify the input force, $F(s)$ can be any forcing function for which a Fourier transform exists.

6.3.5.1.1 Forward Problem

Borel's theorem can be used to find the solution of a SDOF system. In the event, $H(s)$ and $F(s)$ are known. First, $X(s)$ must be found, then through the inverse Laplace transform, $x(t)$ can be found.

$$X(t) = \mathcal{L}^{-1}[X(s)] = \mathcal{L}^{-1}[H(s)F(s)] \quad (6.205)$$

6.3.5.1.2 First Inverse Problem

An additional application is the case when $X(s)$ and $F(s)$ are known. They can be used to find the transfer function, referred to as the *first inverse problem*, using the equation

$$H(s) = \frac{X(s)}{F(s)} \quad (6.206)$$

Furthermore, for the transfer function to find the physical parameters m , c , k :

$$H(s) = H(m, c, k) = \frac{1}{ms^2 + cs + k} \quad (6.207)$$

6.3.5.1.3 Second Inverse Problem

The second inverse problem is referred to when $X(s)$ and $H(s)$ are known. Here, to find the forcing function:

$$F(s) = \frac{X(s)}{H(s)} \quad (6.208)$$

The inverse problem will be discussed in further detail in Chapter 9.

PROBLEMS

1. A system is excited by an initial displacement $x_0 = 5$ cm only, shown in Figure P6.1.

Find (a) natural period, (b) natural frequency in Hertz (f_n), and natural frequency in radians per second (rad/s), (c) mass in kilograms, (d) damping ratio (ζ), (e) damping coefficient c , (f) stiffness k ($g = 10$ m/s²).

2. An SDOF system is shown in Figure P6.1. Suppose the base has a peak ground acceleration $\ddot{x}_g = 0.6$ g with 2.0 Hz driving frequency; and the natural frequency is 2.5 Hz, damping ratio is 0.064.

Find (a) the dynamic magnification factor for the absolute acceleration in the base excitation problem, (b) the amplitude of absolute acceleration,

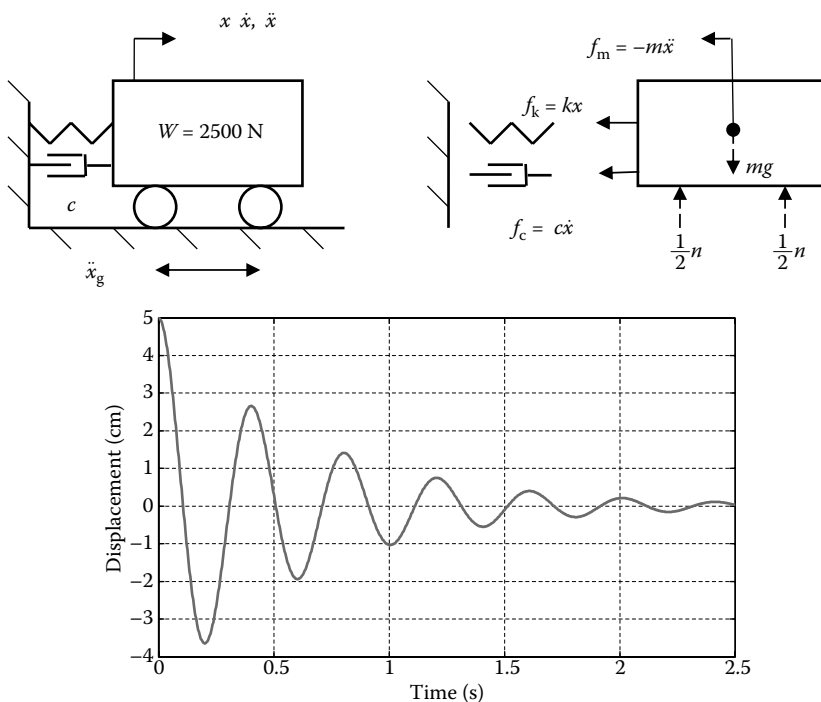


FIGURE P6.1

- (c) the dynamic magnification factor for relative displacement, (d) the amplitude of relative displacement, and (e) compute the amplitude of the ground force $f_c + f_k$.
3. A structure is shown in Figure P6.2 with mass = 12 kg the structure itself is weightless
 - a. Determine the damping ratio and stiffness
 - b. The system is excited by vertical base motion and amplitude = A ; the driving frequency = 6 Hz. The absolute displacement is measured to be 6.1 mm. Find the value of A .
 4. An SDOF structure has weight = 6000 lb, damping ratio = 0.08, and stiffness is k with natural period = 1.15 seconds. The structure has a ground excitation with amplitude = 0.25 g and driving period is 1.80 seconds. Find the amplitude of the relative displacement.

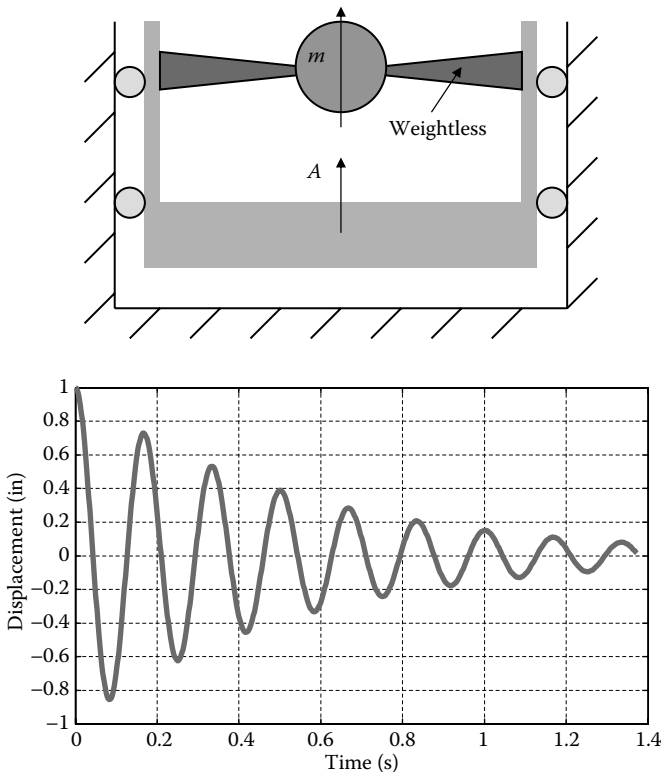


FIGURE P6.2

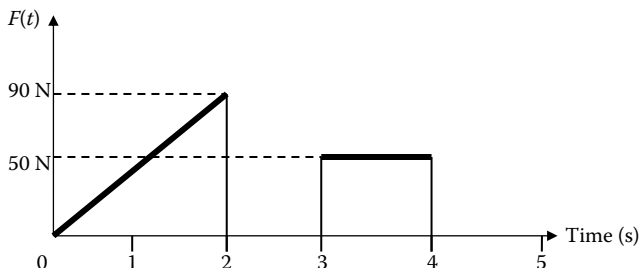


FIGURE P6.3

5. An SDOF system with mass = 1.5 kg; $c = 8$ N/m-s, and $k = 120$ N/m, is excited by a forcing function shown in Figure P6.3. The initial conditions are $v_0 = -1.5$ m/s and $x_0 = 0.05$ m. Calculate the displacement.
6. If the relationship between the log decrement δ and the damping ratio ζ is approximated to be $\delta = 2\pi\zeta$, what values of ζ could you calculate if the allowable error is 12%.
7. If the amplitude of response of a system under bounded input grows into infinity, the system is unstable. Consider an inverted pendulum as shown in Figure P6.4, where $k_1 = 0.7 k$, both installed at exactly the middle point of the pendulum. Find the value of k such that the system becomes unstable. Assume that the damper c is installed on the pendulum parallel to the two springs. How does this affect the stability properties of the pendulum?
8. In Problem 7, $m = 12$ kg, $\ell = 2.1$ m, $k = 110$ kN/m, $c = 1.1$ kN/m-s, the bar is massless. Suppose the initial angle θ is 2.0° , calculate the response.
9. Consider the system in Figure P6.5; write the equation of motion, and calculate the response assuming that the system is initially at rest for the slope angle = 30° , $k = 1200$ N/m, $c = 95$ N/m-s, and $m = 42$ kg. The amplitude of the vertical force $f(t)$ is 2.1 N with driving frequency = 1.1 Hz.

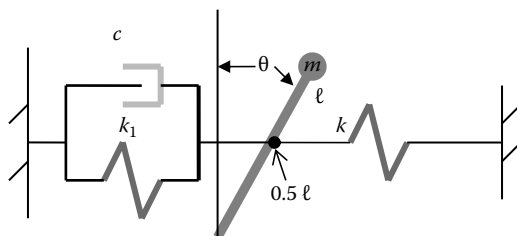


FIGURE P6.4

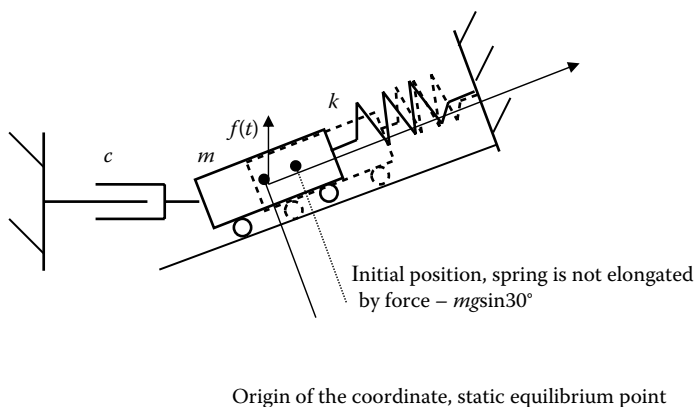


FIGURE P6.5

10. A mechanism is modeled as shown in Figure P6.6, with $k = 3400 \text{ N/m}$, $c = 80 \text{ kg/s}$, $m = 45 \text{ kg}$, and the ground has a motion along the 45° line with displacement $x_g(t) = 0.06 \cos(\pi t)$; compute the steady-state vertical responses of both relative displacement and absolute acceleration, assuming the system starts from a horizontal position. Assume the rotation angle is small.

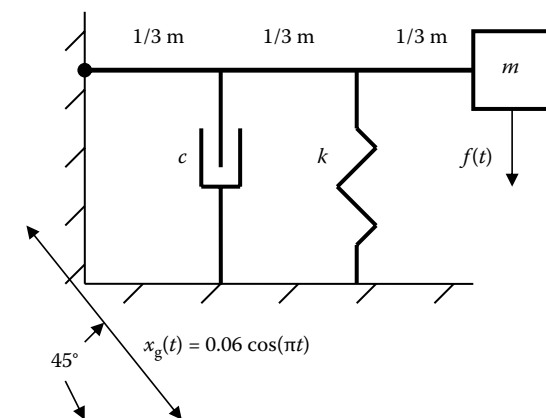


FIGURE P6.6

7 Response of SDOF Linear Systems to Random Excitations

In Chapter 6, the linear single-degree-of-freedom (SDOF) system is discussed in terms of deterministic forcing functions. In this chapter, random excitations will be considered. That is, the vibration response is a random process. Analyses in both the time and frequency domain discussed in Chapters 3 and 4 will be applied here. In addition, a special random process of the time series will also be described for response modeling.

In this chapter and in Chapter 8, which is concerned with multi-degree-of-freedom (MDOF) vibrations, we focus on linear systems. In Chapter 11, the general concept of nonlinear vibration and selected nonlinear systems will be introduced. General references can be found in Schueller and Shinnozuka (1987), Clough and Penzien (1993), Wirsching et al. (2006), Chopra (2003), and Liang et al. (2012).

7.1 STATIONARY EXCITATIONS

The simplest cases of random excitations occur when a forcing process is stationary, for which a weak stationary process is specifically used.

7.1.1 MODEL OF SDOF SYSTEM

7.1.1.1 Equation of Motion

First, we recall the equation of motion

$$\left\{ \begin{array}{l} m\ddot{x} + c\dot{x} + kx = f(t) \\ \dot{x}(0) = v_0 \\ x(0) = x_0 \end{array} \right. \quad (7.1)$$

7.1.1.2 Zero Initial Conditions

In most practical cases, zero initial conditions are assumed.

$$x(0) = 0 \quad (7.2)$$

$$\dot{x}(0) = 0 \quad (7.3)$$

Furthermore, the forced transient response will also be assumed to be zero.

7.1.1.3 Solution in Terms of Convolution

Here, the basic approach remains the method of convolution.

Recalling the convolution equation (Equation 6.180):

$$X(t) = \int_{-\infty}^{\infty} f(t - \tau)h(\tau) d\tau \quad (7.4)$$

In this chapter, unless specially announced, $f(t)$ is the realization of a random process.

7.1.1.4 Nature of Forcing Function

7.1.1.4.1 Initial Force

The initial force, $f(t)$, is initially zero, that is,

$$f(t) = 0, \quad t \leq 0 \quad (7.5)$$

7.1.1.4.2 Stationary Random Process

Furthermore, $f(t)$ is a stationary random process. Thus,

$$\mu_F(t) = \mu_F = \text{const.} \quad (7.6)$$

and

$$R_F(t, s) = R_F(\tau) \quad (7.7)$$

where

$$\tau = t - s \quad (7.8)$$

7.1.1.5 Response

Given $f(t)$, $x(t)$ is also a realization of the stationary random process. Equation 7.4 can be seen as a case of random process through an SDOF linear system.

7.1.2 MEAN OF RESPONSE PROCESS

Similar to a general random process as discussed in Chapter 3, for a random response $x(t)$, the numerical characteristics of the response will first be examined. Initially, the mean of the response in general is considered. Note that we now use lowercase letters to denote random process in the time domain for convenience.

$$\begin{aligned} \mu_X(t) &= E[x(t)] = E\left[\int_{-\infty}^{\infty} f(t - \tau)h(\tau) d\tau\right] \\ &= \int_{-\infty}^{\infty} E[f(t - \tau)]h(\tau) d\tau = \mu_F \int_{-\infty}^{\infty} h(\tau) d\tau \end{aligned} \quad (7.9)$$

Recalling Equation 6.193:

$$H(\omega) = \int_{-\infty}^{\infty} h(\tau) e^{-j\omega\tau} d\tau \quad (7.10)$$

Because,

$$\lim_{\omega \rightarrow 0} (e^{-j\omega\tau}) = 1 \quad (7.11a)$$

we have

$$\int_{-\infty}^{\infty} h(\tau) d\tau = H(0) \quad (7.11b)$$

and therefore

$$\mu_X(t) = \mu_F H(0) = \text{const.} \quad (7.12)$$

Thus,

$$H(\omega) \Big|_{\omega \rightarrow 0} = \frac{1}{-m\omega^2 + jc\omega + k} \Big|_{\omega \rightarrow 0} = \frac{1}{k} \quad (7.13)$$

Finally, the mean of the response can be written as

$$\mu_X = \frac{\mu_F}{k} \quad (7.14)$$

7.1.3 AUTOCORRELATION OF RESPONSE PROCESS

Assuming a mean-square convergence, consider the autocorrelation function of the response.

7.1.3.1 Autocorrelation

The autocorrelation function is given as

$$\begin{aligned} R_X(t, s) &= E[X(t)X(s)] \\ &= E \left[\int_{-\infty}^t f(t-u)h(u) du \int_{-\infty}^s f(s-v)h(v) dv \right] \\ &= E \left[\int_{-\infty}^t du \int_{-\infty}^s f(t-u) f(s-v) h(u)h(v) dv \right] \\ &= \int_{-\infty}^t du \int_{-\infty}^s R_F(t-u, s-v) h(u)h(v) dv, \quad t, s \geq 0 \end{aligned} \quad (7.15)$$

7.1.3.2 Mean Square

In this section, we consider mean square values in general cases and in stationary processes.

7.1.3.2.1 General Case

In the general case, by using autocorrelation, the mean square value of the response in general can be further written as

$$E[X^2(t)] = R_X(t, t) = \int_{-\infty}^t du \int_{-\infty}^t R_F(t-u, t-v)h(u)h(v)dv, \quad t \geq 0 \quad (7.16)$$

7.1.3.2.2 Stationary Process

Now, consider the special case where the excitation is stationary, that is

$$R_F(t, s) = R_F(s - t) \quad (7.17a)$$

For this event, because

$$R_X(t, s) = \int_{-\infty}^t du \int_{-\infty}^s R_F(s-t-(u-v))h(u)h(v)dv, \quad t, s \geq 0 \quad (7.17b)$$

the variance, which is also the mean square value for a zero-mean process, is

$$\sigma_X^2(t) = R_X(t, t) = \int_{-\infty}^t du \int_{-\infty}^t R_F(u-v)h(u)h(v)dv, \quad t \geq 0 \quad (7.18)$$

It is observed that

$$\lim_{t \rightarrow \infty} \frac{\sigma_X^2(t)}{W_0/4kc} = 1 \quad (7.19)$$

when

$$t, s \rightarrow \infty \quad (7.20)$$

By denoting

$$\tau = s - t \quad (7.21)$$

then

$$R_X(\tau) = \int_{-\infty}^{\infty} du \int_{-\infty}^{\infty} R_F(\tau-(u-v))h(u)h(v)dv \quad (7.22)$$

Equation 7.20 implies the practical consideration of a stationary process when the time t and s are sufficiently long.

Furthermore, the mean square value in this case is

$$E[X(t)^2] = R_X(0) = \int_{-\infty}^{\infty} du \int_{-\infty}^{\infty} R_F(u-v)h(t)h(v)dv \quad (7.23)$$

Example 7.1

In the previous discussion, zero initial conditions are assumed. In this example, we assume *random initial conditions*. In Chapter 6, we knew that without a forcing function, we would have free-decay vibrations. The response is given by Equation 6.62. That is,

$$x(t) = de^{-\zeta\omega_n t}[\sin(\omega_d t + \phi)]$$

which can be further written as

$$x(t) = e^{-\zeta\omega_n t}[d_1 \cos(\omega_d t) + d_2 \sin(\omega_d t)]$$

With the given initial conditions (see Equation 7.1), we can solve the coefficient d_1 and d_2 as

$$d_1 = x_0$$

and

$$d_2 = \frac{v_0 + \zeta\omega_n x_0}{\omega_d}$$

Therefore, we have

$$x(t) = e^{-\zeta\omega_n t} \left[x_0 \cos(\omega_d t) + \frac{v_0 + \zeta\omega_n x_0}{\omega_d} \sin(\omega_d t) \right]$$

and furthermore

$$x(t) = e^{-\zeta\omega_n t} \left[x_0 \left\{ \cos(\omega_d t) + \frac{\zeta}{\sqrt{1-\zeta^2}} \sin(\omega_d t) \right\} + \frac{v_0}{\omega_d} \sin(\omega_d t) \right]$$

Suppose the initial conditions X_0 and V_0 are random variables. We can examine the statistical properties of the free-decay response. For convenience, in the following examples, we use lowercase letters to denote random variables.

$$\mu_X(t) = E[X(t)] = e^{-\zeta\omega_n t} \left[E(X_0) \left\{ \cos(\omega_d t) + \frac{\zeta}{\sqrt{1-\zeta^2}} \sin(\omega_d t) \right\} + \frac{E(V_0)}{\omega_d} \sin(\omega_d t) \right]$$

Denote $E(X_0) = \mu_{X_0}$ and $E(V_0) = \mu_{V_0}$, we write

$$\mu_X(t) = \mu_{X_0} e^{-\zeta\omega_n t} \left[\cos(\omega_d t) + \frac{\zeta}{\sqrt{1-\zeta^2}} \sin(\omega_d t) \right] + \mu_{V_0} \left[\frac{e^{-\zeta\omega_n t}}{\omega_d} \sin(\omega_d t) \right]$$

$$\begin{aligned} R_X(t_1, t_2) &= E[X(t_1)X(t_2)] \\ &= e^{-\zeta\omega_n(t_1+t_2)} \left[E(X_0^2) \left\{ \cos(\omega_d t_1) + \frac{\zeta}{\sqrt{1-\zeta^2}} \sin(\omega_d t_1) \right\} \left\{ \cos(\omega_d t_2) + \frac{\zeta}{\sqrt{1-\zeta^2}} \sin(\omega_d t_2) \right\} \right] \\ &\quad + \frac{E(X_0 V_0)}{\omega_d} e^{-\zeta\omega_n(t_1+t_2)} \left[\left\{ \cos(\omega_d t_1) + \frac{\zeta}{\sqrt{1-\zeta^2}} \sin(\omega_d t_1) \right\} \sin(\omega_d t_2) \right. \\ &\quad \left. + \left\{ \cos(\omega_d t_2) + \frac{\zeta}{\sqrt{1-\zeta^2}} \sin(\omega_d t_2) \right\} \sin(\omega_d t_1) \right] \\ &\quad + \frac{E(V_0^2)}{\omega_d^2} e^{-\zeta\omega_n(t_1+t_2)} \sin(\omega_d t_1) \sin(\omega_d t_2) \end{aligned}$$

Denote $E(X_0^2) = \sigma_{X_0}^2$ and $E(V_0^2) = \sigma_{V_0}^2$, and $E(X_0 V_0) = \sigma_{XV_0}$, we write

$$\begin{aligned} R_X(t_1, t_2) &= E[X(t_1)X(t_2)] \\ &= e^{-\zeta\omega_n(t_1+t_2)} \left[\sigma_{X_0}^2 \left\{ \cos(\omega_d t_1) + \frac{\zeta}{\sqrt{1-\zeta^2}} \sin(\omega_d t_1) \right\} \left\{ \cos(\omega_d t_2) + \frac{\zeta}{\sqrt{1-\zeta^2}} \sin(\omega_d t_2) \right\} \right] \\ &\quad + \frac{\sigma_{XV_0}}{\omega_d} e^{-\zeta\omega_n(t_1+t_2)} \left[\left\{ \cos(\omega_d t_1) + \frac{\zeta}{\sqrt{1-\zeta^2}} \sin(\omega_d t_1) \right\} \sin(\omega_d t_2) \right. \\ &\quad \left. + \left\{ \cos(\omega_d t_2) + \frac{\zeta}{\sqrt{1-\zeta^2}} \sin(\omega_d t_2) \right\} \sin(\omega_d t_1) \right] \\ &\quad + \frac{\sigma_{V_0}^2}{\omega_d^2} e^{-\zeta\omega_n(t_1+t_2)} \sin(\omega_d t_1) \sin(\omega_d t_2) \end{aligned}$$

Variance

$$\begin{aligned}\sigma_X^2(t) &= \sigma_{X_0}^2 e^{-2\zeta\omega_n t} \left\{ \cos(\omega_d t) + \frac{\zeta}{\sqrt{1-\zeta^2}} \sin(\omega_d t) \right\}^2 \\ &\quad + \frac{2\sigma_{XV_0}}{\omega_d} e^{-2\zeta\omega_n t} \left[\left\{ \cos(\omega_d t) + \frac{\zeta}{\sqrt{1-\zeta^2}} \sin(\omega_d t) \right\} \sin(\omega_d t) \right] \\ &\quad + \frac{\sigma_{V_0}^2}{\omega_d^2} e^{-2\zeta\omega_n t} \sin^2(\omega_d t)\end{aligned}$$

7.1.4 SPECTRAL DENSITY OF RESPONSE PROCESS

Thus far, numerical characteristics in the time domain have been considered. Now, the properties of the SDOF response in the frequency domain will be taken into consideration.

7.1.4.1 Auto-Power Spectral Density Function

The auto-power spectral density (PSD) function of the response is seen as

$$\begin{aligned}S_X(\omega) &= \int_{-\infty}^{\infty} R_X(t) e^{-j\omega\tau} d\tau = \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} du \int_{-\infty}^{\infty} R_F(\tau+u-v) h(u) h(v) dv \right] e^{-j\omega\tau} d\tau \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(u) h(v) \left[\int_{-\infty}^{\infty} R_F(\tau+u-v) e^{-j\omega\tau} d\tau \right] du dv\end{aligned}\quad (7.24)$$

where the autocorrelation function $R_X(\tau)$ is represented by using Equation 7.22. Now, let

$$\theta = \tau + u - v \quad (7.25)$$

Then, we have

$$\tau = \theta - u + v \quad (7.26)$$

From this, it can be seen that

$$\begin{aligned}S_X(\omega) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(u) h(v) \left[\int_{-\infty}^{\infty} R_F(\theta) e^{-j\omega\theta} d\theta \right] e^{j\omega u} e^{-j\omega v} du dv \\ &= \left[\int_{-\infty}^{\infty} h(u) e^{j\omega u} du \right] \left[\int_{-\infty}^{\infty} h(v) e^{-j\omega v} dv \right] \left[\int_{-\infty}^{\infty} R_F(\theta) e^{-j\omega\theta} d\theta \right] \\ &= H(-\omega) H(\omega) S_F(\omega)\end{aligned}\quad (7.27)$$

Therefore,

$$S_X(\omega) = |H(\omega)|^2 S_F(\omega) \quad (7.28)$$

Note that Equation 4.113a from Chapter 4 provided the same information as given in Equation 7.28.

Furthermore, the frequency expressed in units of hertz, yields

$$W_X(f) = |H(f)|^2 W_F(f) \quad (7.29)$$

7.1.4.2 Variance

From Equation 7.28, the variance can be written as

$$\sigma_X^2 = \int_{-\infty}^{\infty} |H(\omega)|^2 S_F(\omega) d\omega \quad (7.30)$$

Additionally, the variance, in terms of hertz for the frequency, becomes

$$\sigma_X^2 = \int_{-\infty}^{\infty} |H(f)|^2 W_F(f) df \quad (7.31)$$

7.1.5 DISTRIBUTIONS OF RESPONSE PROCESS

To consider the distributions of the responses, the process must first be checked if it is stationary and ergodic. For an ergodic excitation, when the duration of excitation process is sufficiently long, the response is also ergodic. In this case, the temporal average can be used to estimate its distribution.

For a nonergodic response, if $F(t)$ is Gaussian, then the response of the linear SDOF system $X(t)$ is also Gaussian. Contrary to this, if $F(t)$ is non-Gaussian, then it is likely that $X(t)$ is unknown. In studying $X(t)$, convolution can be used.

$$X(t) = \int_{-\infty}^t F(t - \tau) h(\tau) d\tau \quad (7.32)$$

In comparing Equation 7.32 with Equation 7.4, both $F(t)$ and $X(t)$ are found to be generic random processes. From Equation 7.32, it is understood that if $F(t)$ is Gaussian, then $X(t)$ will also be Gaussian.

7.2 WHITE NOISE PROCESS

7.2.1 DEFINITION

A white noise random process is defined by its auto-PSD function given by Equation 4.54 and repeated as follows

$$S_F(\omega) = S_0, \quad -\infty < \omega < \infty \quad (7.33)$$

The autocorrelation function is given by Equation 4.55, repeated as follows

$$R_F(\tau) = \mathcal{F}^{-1}[S_F(\omega)] = S_0 \delta(\tau) \quad (7.34)$$

7.2.2 RESPONSE TO WHITE NOISE

7.2.2.1 Auto-PSD Function

By applying Equation 7.28, the auto-PSD function $S_X(\omega)$ can be written as

$$S_X(\omega) = |H(\omega)|^2 S_F(\omega) = \left| \frac{1}{-m\omega^2 + jc\omega + k} \right|^2 S_0 = \frac{S_0}{(k - m\omega^2)^2 + (c\omega)^2} \quad (7.35)$$

Recalling the equations:

$$\frac{1}{\omega_n^2} = \frac{m}{k} \quad (7.36)$$

$$\frac{c}{k} = \frac{2\zeta\omega_n m}{k} = \frac{2\zeta}{\omega_n} \quad (7.37)$$

and

$$\frac{\omega}{\omega_n} = r \quad (7.38)$$

With these notations, Equation 7.35 can be expressed as

$$S_X(\omega) = \frac{S_0/k^2}{(1 - r^2)^2 + (2\zeta r)^2} \quad (7.39)$$

7.2.2.2 Variance

In the following, we describe the variance.

7.2.2.2.1 General Description

First, recall the equation:

$$\sigma_X^2 = \int_{-\infty}^{\infty} S_X(\omega) d\omega \quad (7.40)$$

Substitution of Equation 7.35 into Equation 7.40 yields

$$\sigma_X^2 = \int_{-\infty}^{\infty} \left| \frac{1}{-m\omega^2 + jc\omega + k} \right|^2 S_0 d\omega = S_0 \int_{-\infty}^{\infty} \left| \frac{1}{-m\omega^2 + jc\omega + k} \right|^2 d\omega \quad (7.41)$$

7.2.2.2.2 Alternative Form of Transfer Function

On the right-hand side of Equation 7.41, it is essential to evaluate the integral.

First, denote

$$I = \int_{-\infty}^{\infty} \left| \frac{1}{-m\omega^2 + jc\omega + k} \right|^2 d\omega = \int_{-\infty}^{\infty} |H(\omega)|^2 d\omega \quad (7.42)$$

for Equation 7.41. Then, consider an alternative form of transfer function as

$$H^{(n)} = \frac{B_0 + (j\omega)B_1 + (j\omega)^2 B_2 + \dots + (j\omega)^{n-1} B_{n-1}}{A_0 + (j\omega)A_1 + (j\omega)^2 A_2 + \dots + (j\omega)^n A_n} \quad (7.43)$$

Note that the number n is determined by the nature of the system. For example, for a SDOF vibration system, $n = 2$.

Thus, the integral can be expressed as

$$I^{(n)} = \int_{-\infty}^{\infty} |H^{(n)}(\omega)|^2 d\omega \quad (7.44)$$

It can be calculated:

For $n = 1$, we have

$$I^{(1)} = \pi \frac{B_0^2}{A_0 A_1} \quad (7.45)$$

For $n = 2$, we further have

$$I^{(2)} = \pi \frac{A_0 B_1^2 + A_2 B_0^2}{A_0 A_1 A_3} \quad (7.46)$$

The coefficients $A_{(i)}$ and $B_{(i)}$, are computed using the equation:

$$H^{(2)} = \frac{B_0 + (j\omega)B_1}{A_0 + (j\omega)A_1 + (j\omega)^2 A_2} = \frac{1}{k + j\omega c - m\omega^2} \quad (7.47)$$

The results obtained are

$$B_0 = 1, B_1 = 0, A_0 = k, A_1 = c, \text{ and } A_2 = m \quad (7.48)$$

Substitution of Equation 7.48 into Equation 7.46 yields

$$I^{(2)} = \frac{\pi}{kc} \quad (7.49)$$

such that

$$\sigma_X^2 = \frac{\pi S_0}{kc} \quad (7.50)$$

In terms of the engineering spectral density function W_0 ,

$$\sigma_X^2 = \frac{W_0}{4kc} = \frac{0.785 f_n W_0}{k^2 \zeta} \quad (7.51)$$

For the case $n = 3$, we let

$$I_3 = \pi \frac{A_0 A_3 (2B_0 B_2 - B_1^2) - A_0 A_1 B_2^2 - A_2 A_3 B_0^2}{A_0 A_3 (A_0 A_3 - A_1 A_2)} \quad (7.52)$$

Finally, for the case $n = 4$, we can have

$$I_4 = \frac{A_0 B_3^2 (A_0 A_3 - A_1 A_2) + A_0 A_1 A_4 (2B_1 B_3 - B_2^2) - A_0 A_3 A_4 (B_1^2 - 2B_0 B_2) + A_4 B_0^2 (A_1 A_4 - A_2 A_3)}{A_0 A_4 (A_0 A_3^2 + A_1^2 A_4 - A_1 A_2 A_3)} \pi \quad (7.53)$$

7.2.3 WHITE NOISE APPROXIMATION

Now, we consider approximations of variance. In general,

$$S_F(\omega) \neq \text{const.} \quad (7.54)$$

However, in the resonance region, the region in between the half-power points,

$$S_F(\omega) \approx \text{const.} \quad \omega_1 < \omega < \omega_2 \quad (7.55)$$

Then, the exact variance given in Equation 7.30 can be approximated by that in Equation 7.51, where the constant W_0 is used to estimate the exact variance, that is,

$$\sigma_X^2 \Big|_{\text{(exact)}} \approx \sigma_X^2 \Big|_{\text{(approx)}} = \frac{0.785 f_n W_0}{k^2 \zeta} \quad (7.56)$$

In Figure 7.1, the broad line is one of the realizations of the response and, in the resonance region, the variance is close to constant. Notice the frequency band between the half-power points as illustrated

$$\Delta F = (f_2 - f_1) = 2\zeta f_n \quad (7.57)$$

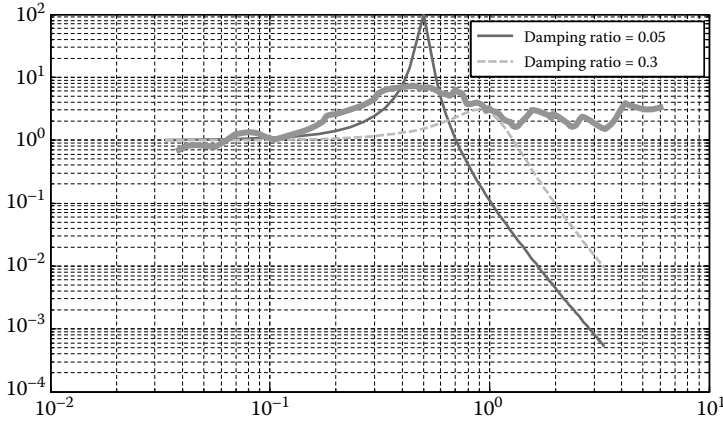


FIGURE 7.1 Approximation of variance.

The requirements of the approximation are obtained as

1. $W_F(f)$ is relatively smooth in the resonance region
2. f_n is comparatively low
3. ζ is small

7.3 ENGINEERING EXAMPLES

In this section, specific practical applications of random responses are discussed. As a comparison, typical deterministic excitations will also be discussed.

7.3.1 COMPARISON OF EXCITATIONS

First, we review the commonly used excitations.

7.3.1.1 Harmonic Excitation

As mentioned in Chapter 6, the harmonic forcing function can be used to measure the transfer function of a linear system.

7.3.1.1.1 Sine Dwell

When the driving frequency of sinusoidal excitation is fixed, the forcing function is

$$f(t) = f_0 e^{i\omega t} \quad (7.58)$$

Note that in the real world, only either $f(t) = f_0 \cos(\omega t)$ or $f(t) = f_0 \sin(\omega t)$ will exist. The latter can be written as $j f(t) = j f_0 \sin(\omega t)$. Thus, Equation 7.58 is a combination of these two cases, which is used solely for mathematical convenience. Under the forcing function described by Equation 7.58, the response is given by

$$x(t) = H(\omega) f_0 e^{i\omega t} \quad (7.59)$$

7.3.1.1.2 Sine Sweep

To use Equation 7.59 to measure the transfer function of a SDOF system, for all the interested frequencies, for example, n pieces of frequencies, n forcing functions are needed. A commonly used method is sine sweep. Namely, the driving frequency can be arranged from the smallest to the highest, from the highest to the smallest or both.

7.3.1.1.2.1 Forcing Function Consider the case of a forcing function of sine sweep.

$$f(t) = f_0 \sum_{n=0}^{\infty} \delta(t - nT) \sin(n\omega_0 t) \quad (7.60)$$

Using Figure 7.2 for $f_0 \sum_{n=-\infty}^{\infty} \delta(t - nT)$, the Fourier transform is shown to be

$$\mathcal{F} \left[f_0 \sum_{n=-\infty}^{\infty} \delta(t - nT) \right] = f_0 \omega_0 \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_0) \quad (7.61)$$

7.3.1.1.2.2 Waiting Time Let the waiting time be expressed by the variable, T . In terms of the number of cycles, T can be written as

$$T = k \frac{2\pi}{\omega} \quad (7.62)$$

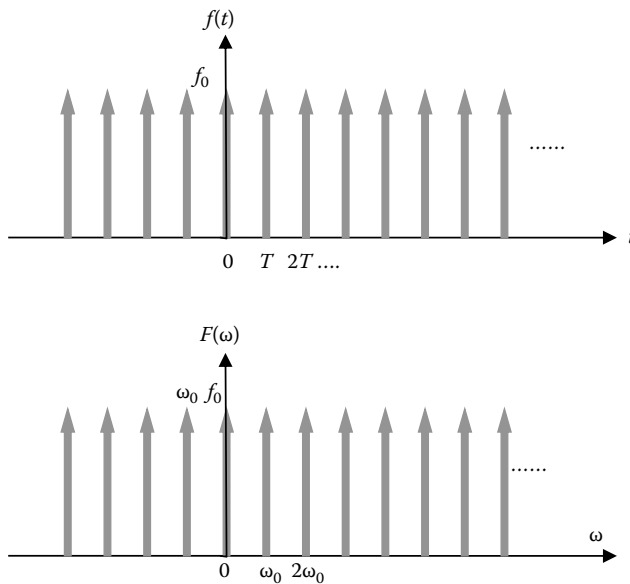


FIGURE 7.2 Impulse series.

Suppose the waiting time for a 10% decay of transit response is to be calculated.
Then

$$e^{-\zeta\omega T} = 10\%$$

and

$$k = \frac{\ln(0.1)}{2\pi\zeta} \quad (7.63)$$

Practically speaking, because the response due to initial conditions can be considerably smaller than that caused by the force, k , the number of cycles can be smaller.

7.3.1.1.2.3 Number of Cycles At each driving frequency, p cycles are needed. The integer p is determined by the criterion of the response as it reaches a steady state. Readers may consider the question of how to judge the steady states.

7.3.1.1.2.4 Auto-PSD The auto-PSD function is

$$S_F(\omega) = S_0 = f_0^2 \quad (7.64)$$

7.3.1.1.2.5 Transfer Function The transfer function is given by

$$H(j\omega) = \frac{f_0}{k} \frac{1}{1 - \left(\frac{\omega}{\omega_n}\right)^2 + 2j\zeta\left(\frac{\omega}{\omega_n}\right)} = \frac{1}{\sqrt{\left[1 - \left(\frac{\omega}{\omega_n}\right)^2\right]^2 + \left[2\zeta\left(\frac{\omega}{\omega_n}\right)\right]^2}} \exp \left[j \tan^{-1} \frac{2\zeta\left(\frac{\omega}{\omega_n}\right)}{1 - \left(\frac{\omega}{\omega_n}\right)^2} \right] \quad (7.65)$$

7.3.1.2 Impulse Excitation

Another commonly used method is impulse excitation.

7.3.1.2.1 Impact Force

First, the impulse or the impact force must be considered.

7.3.1.2.1.1 Ideal Case As mentioned in Chapter 6, for an ideal case

$$f(t) = f_0\delta(t) \quad (7.66)$$

Given that the unit impulse response is

$$h(t) = 1/m\omega_d \left[e^{-\zeta\omega_n t} \sin(\omega_d t) \right], \quad \text{for } t > 0 \quad (7.67)$$

the response is repeated as

$$x(t) = f_0 h(t) \quad (7.68)$$

Here, the auto-PSD function is

$$S_F(\omega) = S_0 = f_0^2 \quad (7.69)$$

Mathematically, the transfer function is equal to the Fourier transform of the unit impulse response function.

$$H(\omega) = \mathcal{F}[h(t)] \quad (7.70)$$

7.3.1.2.1.2 Real Case In the real world, because the duration of the impact force cannot be infinitely short, the impact time history is approximated to be close to a half sine wave. This is shown in Figure 7.3. Specifically written as

$$f(t) = \begin{cases} f_0 \sin\left(\frac{\pi t}{T}\right), & 0 < t < T \\ 0, & \text{elsewhere} \end{cases} \quad (7.71)$$

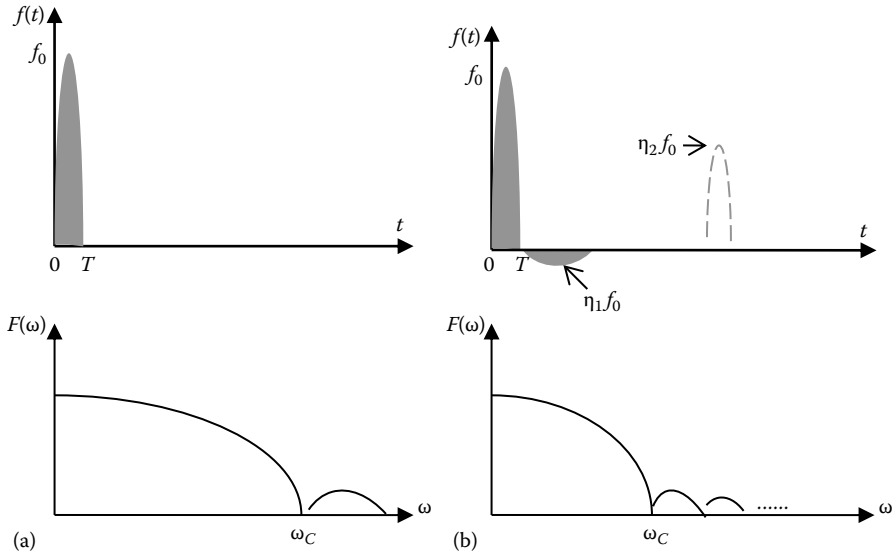


FIGURE 7.3 Concept of realistic impulse.

The Fourier transform is

$$F(\omega) = e^{j\frac{\omega T}{2}} \frac{2T}{\pi} \left(\frac{\cos \frac{\omega T}{2}}{1 - \frac{\omega^2 T^2}{\pi^2}} \right) \quad (7.72)$$

Here, referring to Figure 7.4 for frequency ω_C ,

$$\omega_C = \frac{\pi}{T} \quad (7.73)$$

A practical issue (referring to the case given in Figure 7.4b) is that the history of impact force can differ from a half sine wave. For example, the history may have the supposed “double hit,” resulting in

$$\omega_C < \frac{\pi}{T} \quad (7.74)$$

Note that the auto-PSD function in this case is written as

$$S_F(\omega) = F(\omega)F(\omega)^* \quad (7.75)$$

7.3.1.2.2 Step Force

The step function is another frequently used excitation. This function can be realized through quick-cable-releasing, weight-dropping, and so on.

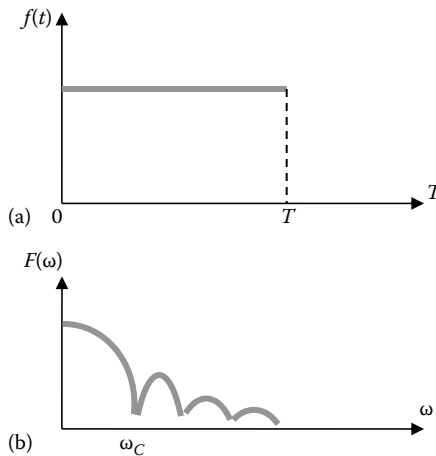


FIGURE 7.4 Concept of realistic step function.

7.3.1.2.2.1 Ideal Case First, consider the idealized case, where the forcing function is modeled as a Heaviside step function:

$$f(t) = f_0 u(t) \quad (7.76)$$

The Fourier transform is

$$F(\omega) = \mathcal{F}[f(t)] = f_0 \left[\pi \delta(\omega) + \frac{1}{j\omega} \right] \quad (7.77)$$

7.3.1.2.2.2 Real Case In the real world, the excitation as described by Equation 7.76 can be simulated with reasonable accuracy. Equation 7.76 is commonly rewritten as

$$f(t) = \begin{cases} f_0, & 0 < t < T \\ 0, & \text{elsewhere} \end{cases} \quad (7.78)$$

The corresponding Fourier transform is

$$F(\omega) = \mathcal{F}[f(t)] = f_0 \left[T \frac{\sin\left(\frac{\omega T}{2}\right)}{\frac{\omega T}{2}} e^{j\frac{\omega T}{2}} \right] \quad (7.79)$$

Similar to impact force, the frequency ω_c exhibits one of two conditions:

$$\omega_c = \frac{\pi}{T} \quad (7.80)$$

and

$$\omega_c < \frac{\pi}{T} \quad (7.81)$$

It is important to note that the time duration T is considerably longer than impact force; therefore, the corresponding value of ω_c is much smaller. Again, the auto-power spectral function is

$$S_F(\omega) = F(\omega)F(\omega)^* \quad (7.82)$$

7.3.1.3 Random Excitation

As a comparison, consider a random forcing function, which is in general among these commonly used excitations.

7.3.1.3.1 White Noise (Ideal Case)

First, look at the case of white noise excitation $f(t)$. The auto-PSD function is found to be

$$S_F(\omega) = S_0 \quad (7.83)$$

Readers may consider how the measurement of $|H(\omega)|^2$ can be carried out practically. From Equation 7.35,

$$S_X(\omega) = |H(\omega)|^2 S_0 \quad (7.84)$$

Rearranging the terms in the equation yields

$$\frac{S_X(\omega)}{S_0} = |H(\omega)|^2 = \frac{1}{(k - m\omega^2)^2 + (c\omega)^2} \quad (7.85)$$

Readers may also consider how to estimate S_0 practically.

In conducting a vibration test, the auto-spectral density function S_0 will be known for both a deterministic and white noise random process. Measuring the auto-PSD, $S_X(\omega)$, at $\omega = 0$, will result in

$$\frac{S_X(0)}{S_0} = \frac{1}{k^2} \quad (7.86)$$

Equation 7.86 is used in vibration testing to determine unknown stiffness.

In tangible applications, the value of S_0 is often unknown. Consider the example in which a bridge is excited by ambient vibrations. If a sufficiently long measurement is taken, then the auto-spectral density function can be treated as a constant in an interested frequency range. Additionally, if the static stiffness k can be obtained through finite element analysis and a measurement of the auto-spectral density function of the response is obtained, S_0 can be estimated as

$$S_0 = k^2 S_X(0) \quad (7.87)$$

Readers may consider the question: How would the measurement of a transfer function be accomplished?

Similarly, the transfer function can be measured through a vibration test. Mathematically speaking,

$$H(\omega) = \frac{X(\omega)}{F(\omega)} \quad (7.88)$$

As mentioned in Chapter 4, Equations 4.122 and 4.123, the transfer function can be approximated by

$$H_1(\omega) = \frac{S_{FX}(\omega)}{S_F(\omega)} = \frac{S_{FX}(\omega)}{S_0} \quad (7.89)$$

or

$$H_2(\omega) = \frac{S_X(\omega)}{S_{XF}(\omega)} \quad (7.90)$$

7.3.1.3.2 Band-Pass White Noise (Real Case)

Generally, excitations as broadband white noise do not occur. Alternatively, the natural excitation is closer to band-pass white noise, where the excitation and the frequency range is limited from ω_L to ω_U .

7.3.1.3.2.1 Band Width Recall from Chapter 4, the band width is repeated

$$\Delta\omega = \omega_U - \omega_L \quad (7.91)$$

7.3.1.3.2.2 Variance The variance is repeated as

$$\sigma_F^2 = 2\Delta\omega S_0 \quad (7.92)$$

7.3.1.3.2.3 Autocorrelation and Auto-PSD The corresponding autocorrelation function is repeated as

$$R_F(\tau) = \sigma_F^2 \frac{\sin(\Delta\omega\tau/2)}{\Delta\omega\tau/2} \cos\omega_0\tau \quad (7.93)$$

The auto-spectral density function is repeated as

$$S_F(\omega) = \begin{cases} S_0 & \omega_L < |\omega| < \omega_U \\ S_0/2 & |\omega| = \omega_L, \omega_U \\ 0 & \text{elsewhere} \end{cases} \quad (7.94)$$

Readers may consider the question: In the case of random excitations, why is $R_F(\tau)$ and $S_F(\omega)$ used in the place of $F(\omega)$?

Because the excitation is limited to a limited frequency band, outside the boundaries of ω_L and ω_U , there will be no excitation energy. In other words, the estimated transfer function is also band-limited.

7.3.1.4 Other Excitations

Lastly, we consider additional excitations.

7.3.1.4.1 Fast Sine Sweep (Chirp)

For a fast sine sweep excitation, linear and exponential chirps are commonly used.

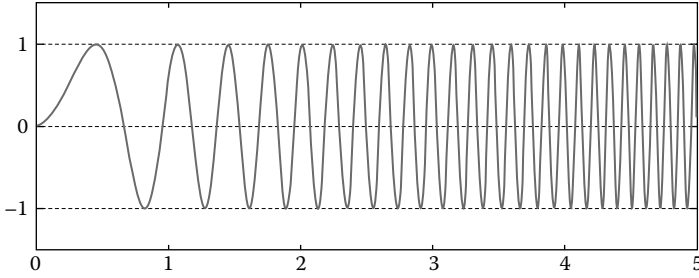


FIGURE 7.5 Linear chirp.

7.3.1.4.1.1 Linear Chirp As shown in Figure 7.5, we formulated the linear chirp as

$$f(t) = f_0 \sin[(\omega_0 + k\pi)t] \quad (7.95)$$

7.3.1.4.1.2 Exponential Chirp As shown in Figure 7.6, we formulated the exponential chirp as

$$f(t) = f_0 \sin \left[\omega_0 \frac{k^t - 1}{\ln(k)} t \right] \quad (7.96)$$

7.3.1.4.2 Pink Noise etc.

In addition to white noise, colored noise can exist. This can be seen in Figure 7.6. *Pink noise* is defined as

$$S_F(\omega) = \frac{S_0}{\omega} \quad (7.97)$$

Generally speaking, the PSD for a color noise can also be written as

$$S_F(\omega) = \frac{S_0}{\omega^\alpha} \quad (7.98)$$

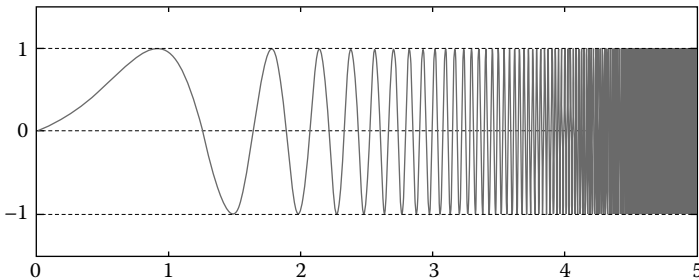


FIGURE 7.6 Exponential chirp.

By varying the parameter α in Equation 7.98, the color of the noise is varied. For white noise, $\alpha = 0$; for pink noise, $\alpha = 1$; and for red (brown) noise, $\alpha = 2$. An alternative way to write the PSD for a special color noise is

$$S_F(\omega) = \frac{2\eta S_0}{\omega^2} + \eta^2 \quad (7.99)$$

and the autocorrelation function is

$$R_F(\tau) = S_0 e^{-\eta|\tau|} \quad (7.100)$$

The corresponding random process $f(t)$ can be seen as a transient solution of the following first-order differential equation

$$\frac{d}{dt} f(t) = \eta f(t) + \sqrt{2\eta} w(t) \quad (7.101)$$

where $w(t)$ is a white noise process with auto-PSD S_0 . That is, the color noise can be seen as the response of a first-order system excitation by a white noise. The solution can be represented by a convolution given by

$$f(t) = \int_0^\infty e^{-\eta\tau} \sqrt{2\eta} w(t - \tau) d\tau \quad (7.102)$$

7.3.1.4.3 Ambient Excitation $F(t)$

The case of ambient excitation is a common occurrence initiated by instances such as a bumpy road, car-riding, ground tremor and bridge response, and others. In ambient excitation,

1. The forcing function $f(t)$ is often a random process
2. The time history of the forcing function is difficult to measure
3. Measurements of the responses, however, are comparatively much easier
4. Correlation analyses in both the time and the frequency domain are the primary form of measures
5. However, random decrement analysis is also used
6. In ambient excitation, a narrow band response for a lightly damped SDOF system will often behave as a narrow-band filter

7.3.2 RESPONSE SPECTRA

7.3.2.1 Response Spectrum

When the forcing function is a random process, even the bound of the force is known. In this event, it is difficult to determine the bound of the response. The bound, or the peak value of responses, is important. The response spectrum is a tool in which to statistically determine the peak value.

Consider a realization of ground excitation:

$$m\ddot{x}(t) + c\dot{x}(t) + kx(t) = -m\ddot{x}_g(t)$$

In terms of the natural period and damping ratio, the ground excitation can be rewritten as

$$\ddot{x}(t) + \frac{4\pi\zeta}{T_n}\dot{x}(t) + \frac{4\pi^2}{T_n^2}x(t) = -\ddot{x}_g(t) \quad (7.103)$$

Now, consider the j import to a system whose natural period is T_i :

$$\ddot{x}_{ij}(t) + \frac{4\pi\zeta}{T_i}\dot{x}_{ij}(t) + \frac{4\pi^2}{T_i^2}x_{ij}(t) = -\ddot{x}_{g_j}(t) \quad (7.104)$$

where $\ddot{x}_{g_j}(t)$ is the realization of the j th ground excitation. Thus, Equation 7.101 can be solved to find the response $x_{ij}(t)$. Here, the subscript i indicates a system with period T_i .

Denote

$$x_{ij} = \max(|x_{ij}(t)|) \quad (7.105)$$

Taking the mean value pulse of one standard deviation results in

$$x_i = \bar{x}_i + \sigma_i = \frac{1}{N} \sum_{j=1}^N x_{ij} + \frac{1}{N} \sum_{j=1}^N \left[x_{ij}^2 - \left(\frac{1}{N} \sum_{j=1}^N x_{ij} \right)^2 \right] \quad (7.106)$$

This is illustrated by the line characterized by the symbol “***” in Figure 7.7.

At each period T_i , the mean value $\bar{x}_i$ and the standard deviation σ_i of the peak responses are computed, in which the subscript i corresponds to the statistics taken in accordance with the i th period T_i . The sum, $\bar{x}_i + \sigma_i$, is taken as raw data for the statistical response spectral value x_i .

Referring to Equation 7.106, N signifies the number of records used. Using this method, the number of measurements x_i is not limited and x_i , which is a function of T_i , will obtain a reasonable resolution of T_i . In this instance, T_i is the i th natural frequency. Note that for convenience, the subscript n is omitted from the term T_{ni} in the following text.

7.3.2.2 Design Spectra

Because the combined quantities of x_i form a nonsmooth curve, this result is not convenient to manage. Therefore, further measures should be taken to smooth the curve. Additional measurements may be achieved from the envelope of all x_i or other

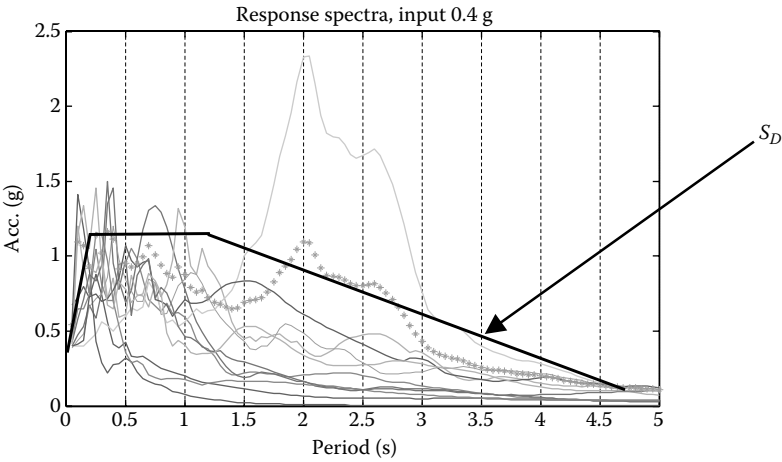


FIGURE 7.7 Earthquake response spectra, S_D , $\zeta = 5\%$.

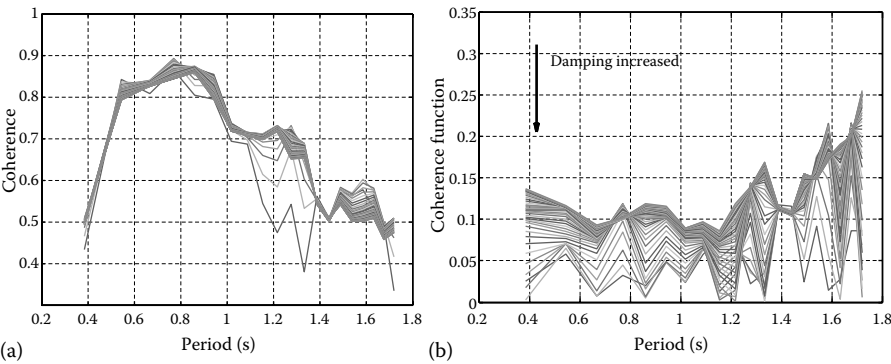


FIGURE 7.8 Coherence function (acceleration and displacement). (a) Coherence (linear system). (b) Correlation between acceleration and displacement.

measures, which are referred to as the *design spectra* (see Chopra [2003] and Liang et al. [2012], for instance). The design displacement response spectrum is denoted by S_D and is shown by the black line in Figure 7.8.

7.3.3 CRITERIA OF DESIGN VALUES

7.3.3.1 Pseudo Spectrum

To use the above description of response spectrum, it is seen that if the damping force is sufficiently small, then the initial force is approximately equal to the restoring force. That is, in Equation 6.123, let

$$c\dot{x}(t) = 0 \tag{7.107}$$

we can write

$$m\ddot{x}_A(t) = -kx_R(t) \quad (7.108)$$

where the subscript R is particularly used to emphasize the relative displacement or relative acceleration and

$$\ddot{x}_A(t) = \ddot{x}_R(t) + \ddot{x}_g(t) \quad (7.109)$$

and the subscript A stands for absolute accelerations.

Dividing m from both sides of Equation 7.108 yields

$$\ddot{x}_A(t) = -\frac{4\pi^2}{T_n^2} x_R(t) \quad (7.110)$$

In the literature, the spectrum of the peak absolute acceleration is generated through Equation 7.110 based on the response spectrum of relative displacement, which is referred to as the *pseudo spectrum*. From Equation 7.110, it seems that a designer can only consider either the absolute acceleration or the relative displacement because they are related by a proportional factor $\frac{4\pi^2}{T_n^2}$. In other words, in this circumstance, we have only a single parameter for design criterion. If the acceleration satisfies the design criterion, then the displacement can be checked accordingly, and no extra effort is needed.

However, the pseudo spectrum is used with the condition described in Equation 7.107. When damping becomes large, Equation 7.107 will no longer hold. In the following examples, we consider the correlation between the absolute acceleration and the relative displacement, when the system is subjected to random excitations.

7.3.3.2 Correlation of Acceleration and Displacement

It is understandable that, if Equation 7.110 holds, then the peak values of the absolute acceleration and the relative displacement satisfies the following:

$$|\ddot{x}_A(t)| \propto |x_R(t)| \quad (7.111)$$

In these cases, we can see that the correlation of $|\ddot{x}_A(t)|$ and $|x_R(t)|$ is unity.

However, due to possible larger damping, Equation 7.107 is no longer valid, the relationship between the peak values $|\ddot{x}_A(t)|$ and $|x_R(t)|$ is no longer unity. In Figure 7.8a and b, we plot the coherence functions versus the natural period T_n with different damping ratios. From Figure 7.8a, where the damping ratio is from 0.01 to 0.05, it can be seen that when the period is greater than 1.6 seconds, when damping ratio becomes larger and close to 5%, the coherence becomes smaller and smaller. In Figure 7.8b, the damping ratios are chosen from 0.1 to 0.5; when the damping ratio is larger than 10%, the coherence function is even smaller than 50%. In other words,

the peak values of the absolute acceleration and the relative displacement become uncorrelated.

This indicates that Equations 7.110 or 7.111 will indeed no longer be valid. In this circumstance, the acceleration and the displacement become independent; the design parameters become two, instead of the single parameter with small damping. To understand the physical significance, let us further discuss coherence function in Section 7.4.

7.4 COHERENCE ANALYSES

In vibration analysis and testing, the major error often comes from the transfer function measurement. In this section, a method to ensure the accuracy of transfer functions is discussed.

7.4.1 ESTIMATION OF TRANSFER FUNCTION

In any realistic test, measurement noise is unavoidable. Using noise-contaminated data to generate transfer functions and extract modal parameters may result in inaccurate modal estimation. In certain cases, a mode may not be properly measured. In other cases, we will have the so-called “noise mode.” That is, certain peak values look like transfer function magnifications at some frequencies and seem to be vibration modes. However, these peak values are caused by measurement noises.

To make sure a peak value measured is a true mode, we need an approach to evaluate the results of the transfer functions. In the previous chapter, we mentioned a method using orthogonality to check the reciprocal modal vector. The procedure to check or evaluate the modal estimation is often referred to as the procedure of modal confidence.

In the frequency domain, coherence function is a commonly used method, which can be more convenient and accurate. In Chapter 6, we mentioned that transfer function can be calculated as the Fourier transform ratios of the output and input, that is,

$$H(s) = \frac{X(s)}{F(s)} \quad (7.112)$$

In vibration testing, however, we can have two different methods to measure the transfer functions, estimated through $H_1(\omega)$ and $H_2(\omega)$ given by

$$H_1(\omega) = \frac{S_{fx}(\omega)}{S_f(\omega)} \quad (7.113)$$

and

$$H_2(\omega) = \frac{S_x(\omega)}{S_{yf}(\omega)} \quad (7.114)$$

Equations 7.113 and 7.114 are mathematically correct. In practical measurements, auto- and cross-spectral density functions are obtained through the fast Fourier transform signal picked up during the test, and a great number of their averages, that is, the cross-PSD functions of the i th measurement, can be calculated as

$$\hat{S}_{f_i x_i}(\omega) = X_i(\omega, T) F_i(\omega, T)^* \quad (7.115)$$

and

$$\hat{S}_{x_i f_i}(\omega) = F_i(\omega, T) X_i(\omega, T)^* \quad (7.116)$$

As mentioned previously, T is the sample length. The auto-PSD function of the i th output is given by

$$\hat{S}_{x_i x_i}(\omega) = X_i(\omega, T) X_i(\omega, T)^* \quad (7.117)$$

The auto-PSD function of the i th input is given by

$$\hat{S}_{f_i f_i}(\omega) = F_i(\omega, T) F_i(\omega, T)^* \quad (7.118)$$

Here, to emphasize the i th measurement and the complex conjugate, the subscripts are written as $x_i x_i$ and $f_i f_i$, instead of x_i and f_i only.

In the above equations, $X_i(\omega)$ and $F_i(\omega)$ are the Fourier transforms of response and the force obtained from the i th test.

Furthermore, suppose we have conducted a total of n tests, for which we can have the average of the cross-PSD function of the i th measurement as given by

$$\hat{S}_{fx} \approx \frac{1}{n} \sum_{i=1}^n \hat{S}_{f_i x_i} \quad (7.119)$$

or the unbiased average written as

$$\hat{S}_{fx} \approx \frac{1}{n-1} \sum_{i=1}^n \hat{S}_{f_i x_i} \quad (7.120)$$

We also have the average of the cross-PSD function of the i th test as

$$\hat{S}_{xf} \approx \frac{1}{n} \sum_{i=1}^n \hat{S}_{x_i f_i} \quad (7.121)$$

or the unbiased average written as

$$\hat{S}_{xf} \approx \frac{1}{n-1} \sum_{i=1}^n \hat{S}_{x_i f_i} \quad (7.122)$$

In addition, the average of the auto-PSD function of the i th output is given by

$$\hat{S}_x \approx \frac{1}{n} \sum_{i=1}^n \hat{S}_{x_i x_i} \quad (7.123)$$

or the unbiased average written as

$$\hat{S}_x \approx \frac{1}{n-1} \sum_{i=1}^n \hat{S}_{x_i x_i} \quad (7.124)$$

and the average of the auto-PSD function of the i th input is given by

$$\hat{S}_f \approx \frac{1}{n} \sum_{i=1}^n \hat{S}_{f_i f_i} \quad (7.125)$$

or the unbiased average written as

$$\hat{S}_f \approx \frac{1}{n-1} \sum_{i=1}^n \hat{S}_{f_i f_i} \quad (7.126)$$

In the sense of statistics, one should use the unbiased averages. However, in the following equation, we can realize that because the estimated transfer functions are the ratios of PSD functions, either n or $n - 1$ will be finally canceled.

Let us consider the input-system-output relationship shown in Figure 7.9. As mentioned in Chapter 4, in many cases, the noise $M(\omega)$ and $N(\omega)$ are not correlated with the forcing function, $F(\omega)$, nor with the response $X(\omega)$. Therefore, we can approximately have

$$S_{mf}(\omega) = S_{fm}(\omega) = S_{nx}(\omega) = S_{xn}(\omega) \approx 0 \quad (7.127)$$

In addition, the input and output noises are not correlated,

$$S_{nm}(\omega) = S_{mn}(\omega) \approx 0 \quad (7.128)$$

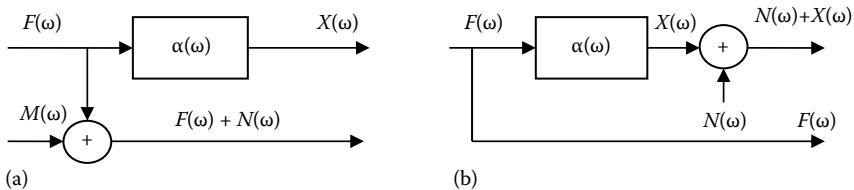


FIGURE 7.9 Noises and transfer function calculation. (a) Input and (b) output with significant noise.

That is, from Figure 7.9, we can see that

$$\begin{aligned}
 \hat{S}_f(\omega) &= [F(\omega) + M(\omega)][F(\omega) + M(\omega)]^* = F(\omega)F(\omega)^* \\
 &\quad + M(\omega)F(\omega)^* + F(\omega)M(\omega)^* + M(\omega)M(\omega)^* = S_f(\omega) + S_{fm}(\omega) \\
 &\quad + S_{mf}(\omega) + S_m(\omega) \approx S_f(\omega) \left(1 + \frac{S_m(\omega)}{S_f(\omega)} \right)
 \end{aligned} \quad (7.129)$$

and

$$\begin{aligned}
 \hat{S}_x &= [X(\omega) + N(\omega)][X(\omega) + N(\omega)]^* = X(\omega)X(\omega)^* \\
 &\quad + N(\omega)X(\omega)^* + X(\omega)N(\omega)^* + N(\omega)N(\omega)^* = S_x(\omega) + S_{xn}(\omega) \\
 &\quad + S_{nx}(\omega) + S_n(\omega) \approx S_x(\omega) \left(1 + \frac{S_n(\omega)}{S_x(\omega)} \right)
 \end{aligned} \quad (7.130)$$

Here, for convenience, we omit the sample length T . Furthermore,

$$\begin{aligned}
 \hat{S}_{xf}(\omega) &= [F(\omega) + M(\omega)][X(\omega) + N(\omega)]^* \\
 &= F(\omega)X(\omega)^* + M(\omega)X(\omega)^* + F(\omega)N(\omega)^* \\
 &\quad + M(\omega)N(\omega)^* = S_{xf}(\omega) + S_{xm}(\omega) + S_{nf}(\omega) \\
 &\quad + S_{nm}(\omega) \approx S_{xf}(\omega)
 \end{aligned} \quad (7.131)$$

$$\begin{aligned}
 \hat{S}_{fx}(\omega) &= [X(\omega) + N(\omega)][F(\omega) + M(\omega)]^* \\
 &= X(\omega)F(\omega)^* + N(\omega)F(\omega)^* + X(\omega)M(\omega)^* \\
 &\quad + N(\omega)M(\omega)^* = S_{fx}(\omega) + S_{fn}(\omega) + S_{mx}(\omega) \\
 &\quad + S_{mn}(\omega) \approx S_{fx}(\omega)
 \end{aligned} \quad (7.132)$$

With the help of the above approximations, the estimated transfer function can be calculated through the averaged results, that is,

$$\hat{H}_1(\omega) = \frac{\hat{S}_{fx}(\omega)}{\hat{S}_f(\omega)} = H(\omega) \left(1 + \frac{S_m(\omega)}{S_f(\omega)} \right)^{-1} \quad (7.133)$$

and

$$\hat{H}_2(\omega) = \frac{\hat{S}_x(\omega)}{\hat{S}_y(\omega)} = H(\omega) \left(1 + \frac{S_n(\omega)}{S_x(\omega)} \right) \quad (7.134)$$

where $S_m(\omega)$ and $S_n(\omega)$ are the auto-PSD functions of the input and output noises.

7.4.2 COHERENCE FUNCTION

Introduced in Chapter 4, the coherence function can be seen as a ratio of transfer functions H_2 and H_1 . To measure the coherence function practically, we consider the values of H_1 and H_2 .

At the resonance point,

$$\hat{H}_1(\omega) \Big|_{\text{resonance}} = \frac{\hat{S}_f(\omega)}{\hat{S}_f(\omega)} = H(\omega) \left(1 + \frac{S_m(\omega)}{S_f(\omega)} \right)^{-1} \quad (7.135)$$

It is seen that

$$\left| \hat{H}_1(\omega) \Big|_{\text{resonance}} \right| < |H(\omega)|$$

and

$$\hat{H}_2(\omega) \Big|_{\text{resonance}} = \frac{\hat{S}_x(\omega)}{\hat{S}_y(\omega)} = H(\omega) \left(1 + \frac{S_n(\omega)}{S_x(\omega)} \right)^{-1} \approx H(\omega) \quad (7.136)$$

At the antiresonance point,

$$\hat{H}_1(\omega) \Big|_{\text{antiresonance}} = \frac{\hat{S}_f(\omega)}{\hat{S}_f(\omega)} = H(\omega) \left(1 + \frac{S_m(\omega)}{S_f(\omega)} \right)^{-1} \approx H(\omega) \quad (7.137)$$

and

$$\hat{H}_2(\omega) \Big|_{\text{antiresonance}} = \frac{\hat{S}_x(\omega)}{\hat{S}_y(\omega)} = H(\omega) \left(1 + \frac{S_n(\omega)}{S_x(\omega)} \right) \quad (7.138)$$

It is seen that

$$\left| \hat{H}_2(\omega) \Big|_{\text{antiresonance}} \right| > |H(\omega)|$$

Now, with the measured data, let us define the *coherence function* $\gamma_{fx}^2(\omega)$ as follows:

$$\gamma_{fx}^2(\omega) = \frac{\hat{H}_1(\omega)}{\hat{H}_2(\omega)} \quad (7.139)$$

It is seen that

$$\gamma_{fx}^2(\omega) = \frac{|\hat{S}_{fx}(\omega)|^2}{\hat{S}_x(\omega)\hat{S}_f(\omega)} \quad (7.140)$$

Using the coherence function, we can evaluate the transfer functions by checking the level of $\gamma_{fx}^2(\omega)$. From the discussion of the values of α_1 and α_2 , it can be realized that

$$100\% \geq \gamma_{fx}^2(\omega) \geq 0\% \quad (7.141)$$

In addition, we can also see that the higher the value of the coherence function is, the better accuracy we can have.

Generally speaking, if

$$\gamma_{fx}^2(\omega) > 70\% \quad (7.142)$$

then the corresponding peak value can belong to a mode. However, in certain cases, a mode is recognized when the coherence function is greater than 50%.

In Figure 7.10a, a sample transfer function $\hat{H}_1(\omega)$ is plotted. From this plot, we can see that there exist three modes. However, we do not know how well the transfer

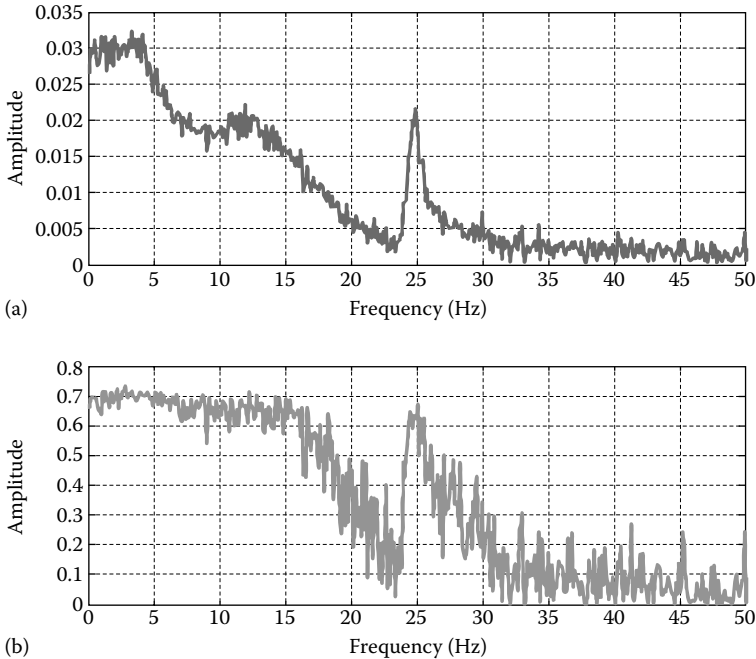


FIGURE 7.10 Measurement accuracy of (a) a transfer and (b) a coherence function.

function is measured. In Figure 7.10b, we also plot the corresponding coherence function $\gamma_{fx}^2(\omega)$. It is seen that in the neighborhood of 4, 12, and 25 Hz, which are the resonant regions of these three modes, the corresponding values of the coherence function are about 70%, 67%, and 65%, respectively. Therefore, at least the transfer function of the first mode can be seen as an accurate measurement.

7.4.3 IMPROVEMENT OF COHERENCE FUNCTIONS

From the previous discussion, it is seen that, to improve the measurement accuracy, we need higher values of coherence functions. Practically speaking, during a test, we need methods to improve the value of coherence functions. First of all, measurement location plays an important role. If the location is close to the nodal point of a certain mode, then the corresponding coherence will have a relatively lower value. In this case, we need to vary the location. Additionally, to judge if the measurement of that mode is improved, we can check the coherence.

Second, if the location cannot be changed, then we need to have a more defined frequency resolution. With limited measurement buffer size, the zoom Fourier transform method can be used. A detailed discussion of the zoom Fourier transform is beyond the scope of this manuscript. Interested readers may consult Clark (2005) for instance.

Generally speaking, the zoom Fourier transform or zoom fast Fourier transform (zoom FFT) is a signal processing technique used to analyze a portion of a spectrum at high resolution. The basic steps to apply the zoom FFT to this frequency region are as follows:

1. Frequency translate to shift the frequency range of interest down to near 0 Hz (DC).
2. Low-pass filter to prevent aliasing when subsequently sampled at a lower sample rate.
3. Resample at a lower rate.
4. FFT the resampled data. Multiple blocks of data are needed to have an FFT of the same length. The resulting spectrum will now have a much smaller resolution bandwidth, compared with an FFT of nontranslated data.

Third, more pieces of measurement can be used to increase the average. It may be found that, if 30 averages are not sufficient, then averages around 300 may be needed.

7.5 TIME SERIES ANALYSIS

The previous discussion described that when the input of a linear system is a random process, the output will also be a random process. When we attempt to measure such output, however, additional noise is often inevitable so the measure time history will be a sum of two pieces of random series. Our goal now is to reduce the effect on the measurement due to noise contaminations.

Time series analysis can be used to account for the measurement noise. It is an important method in signal processing. In this section, we briefly discuss the modeling of time series and the corresponding key characters.

7.5.1 TIME SERIES

The time series is essentially a random process, generated by the response of a system with discrete time due to excitation of white noise processes. There are many publications about time series analysis; among these, for example, readers may consult Box et al. (1994), Hamilton (1994), and Shumway and Stoffer (2011), especially, Ludeman (2003).

7.5.1.1 General Description

Due to noise contamination, the measured vibration signal, no matter if it is deterministically or randomly excited, will be considered a random process. Because of the data acquisition, which turns analogue signals into digital signals, the measured time history is in the format of time sequence; this is one of the major features of the time series.

More importantly, the time series is arranged in the order of occurrence in time. Namely, the earlier measured data is arranged in an earlier position and so on. Happening in order is the second feature of time series. This feature means that the process is not mutually independent because the independent sequence of the order is not an issue. Ordered sequences often possess strong correlations.

Randomness is the third, but most important feature. In the following, we will focus on randomness. The main purpose of time series analysis is to find the pattern and corresponding statistical parameters of the sequence. The basic method of the analysis is to establish a correct model, in most cases, one of the typical models of time series, and then analyze the model.

Three common applications of time series analysis include

1. *Description.* By using a specific model, the statistical properties of the measured random process can be described, such as covariance functions for unveiling the features of the correlation, the PSD functions for understanding the frequency components, the Green functions to measure the eigenparameters of a system, and others.
2. *Prediction.* Based on established models and data picked up in measurements, we may predict future values of the time series, and the tendency of the time-varying development.
3. *Control.* Based on modeling and prediction, we can further adjust the parameters of system and control its output with the proper amount of feedback.

7.5.1.2 Useful Models of Time Series

Suppose $w(n)$ is an input and $x(n)$ is the corresponding output of a discrete system and there exists the following function:

$$x(n) = -\sum_{k=1}^p a_k x(n-k) + \sum_{k=0}^q b_k w(n-k), \quad n \geq 0, p > q \quad (7.143)$$

where both $x(n)$ and $w(n)$ are generic terms, which can be time points of random process, or realizations of random process.

If $w(n)$ is a stationary white noise series, and to all j and k , we have

$$E[w(n)] = 0 \quad (7.144)$$

and

$$E[w(j)w(k)] = \sigma^2 \delta_{ij} \quad (7.145)$$

then the output is called the *autoregressive–moving-average* (ARMA) process with p autoregressive terms and q moving-average terms, denoted by ARMA(p, q). The general ARMA model was first described in the 1951 thesis of Whittle (also see Box and Jenkins [1971]). Here, δ_{ij} is a Kronecker delta function (Leopold Kronecker, 1823–1891), and σ^2 is the variance of $w(n)$.

If all $k = 1, 2, \dots, q, a_k = 0$, the output is called q th *moving-average* (MA) process, denoted by MA(q). In this case,

$$x(n) = \sum_{k=0}^q b_k w(n-k) \quad (7.146)$$

If all $k = 1, 2, \dots, q, b_k = 0$, the output is called p th *autoregressive* (AR) process, denoted by AR(p). In this case,

$$x(n) = -\sum_{k=1}^p a_k x(n-k) + b_0 w(n) \quad (7.147)$$

In the following, let us first discuss the characters of the ARMA process, such as mean, variance and correlation functions, and probability density function, which are decided by the statistical properties of input $w(n)$.

We note that if the time variation of $w(n)$ and $x(n)$ are defined in the entire time domain, the corresponding system will have a steady state output. If they are only defined when $n \geq 0$, we will have transient processes.

7.5.2 CHARACTERS OF ARMA MODELS

7.5.2.1 Moving-Average Process MA(q)

The moving-average process MA(q) is modeled by Equation 7.146, where $w(n)$ is a stationary white noise process. It is not necessarily Gaussian.

7.5.2.1.1 Mean

The mean of MA(q) can be calculated as

$$\begin{aligned} E[x(n)] &= E\left\{\sum_{k=0}^q b_k w(n-k)\right\} \quad n \geq 0 \\ &= \sum_{k=0}^q b_k E[w(n-k)] \end{aligned} \quad (7.148)$$

Because

$$E\{w(n-k)\} = 0, \quad k = 1, 2, \dots \quad (7.149)$$

we have

$$E[x(n)] = 0, \quad n \geq 0 \quad (7.150)$$

7.5.2.1.2 Variance

Now, consider the variance of MA(q). We can write

$$\begin{aligned} \sigma_x^2(n) &= E[x^2(n)] = E\left[\left\{\sum_{k=1}^q b_k w(n-k)\right\}^2\right] \\ &= E\left[\left\{\sum_{k=0}^q b_k w(n-k)\right\}\left\{\sum_{j=0}^q b_j w(n-j)\right\}\right] \end{aligned} \quad (7.151)$$

Consider $n \geq q$, we have

$$\sigma_x^2(n) = \sum_{k=0}^q \sum_{j=1}^q b_k b_j E[w(n-k)w(n-j)] = \sigma^2 \sum_{k=0}^q b_k^2, \quad n \geq q \quad (7.152)$$

This is because in $E[w(n-k)w(n-j)]$ the only nonzero term is obtained when $j = k$. Also note that when $n \geq q$ the variance of $w(n)$ is constant σ^2 . Now, in the case of $0 \leq n < q$, we need to reconsider the result so that we have only nonzero terms $w(n-k)$ and $w(n-j)$. In this case, the variance becomes a function of n , that is,

$$\sigma_x^2(n) = \sigma^2 \sum_{k=0}^n b_k^2, \quad 0 \leq n < q \quad (7.153)$$

Note that, in this case, we will witness a transient process.

7.5.2.1.3 Autocorrelation Function

When $k < 0$ and $j < 0$, we have zero autocorrelation functions, that is,

$$R_x(j, k) \equiv 0, \quad k, j < 0 \quad (7.154)$$

Generally speaking, the value of autocorrelation function depends on the time variable $k + m$ and k , where both k and m are integers.

That is,

$$R_X(k+m, k) = E[x(k+m)x(k)] \\ = E \left[\left\{ \sum_{i=0}^q b_k w(k+m-i) \right\} \left\{ \sum_{j=0}^q b_k w(k-j) \right\} \right], \quad k, m \geq 0 \quad (7.155)$$

In the case that $k > q$ and for such m , that $0 \leq m < q$, we can rewrite Equation 7.155 as

$$R_X(k+m, k) = E[\{b_0 w(k+m) + b_1 w(k+m-1) + \dots + b_q w(k+m-q)\} \{b_0 w(k) + b_1 w(k-1) + \dots + b_q w(k-q)\}] \quad (7.156)$$

Examine the resulting products, such as $\{b_0 w(k+m)\} \{b_0 w(k)\}$ as well as $E[\{b_g w(k+m-g)\} \{b_h w(k-h)\}]$. It is seen that if $m = 0$, then $k+m = k$ so that

$$E[\{b_0 w(k+m)\} \{b_0 w(k)\}] = \sigma^2 b_0^2$$

However, if $m \neq 0$, then

$$E[\{b_0 w(k+m)\} \{b_0 w(k)\}] = 0$$

Similarly, we see that if $k+m-g = k-h$, or $h = g-m$, then

$$E[\{b_g w(k+m-g)\} \{b_h w(k-h)\}] = E[\{b_g w(k+m-g)\} \{b_{g-m} w(k+m-g)\}] \\ = \sigma^2 b_g b_h = \sigma^2 b_g b_{g-m} = E[\{b_{h+m} w(k-h)\} \{b_h w(k-h)\}] = \sigma^2 b_{h+m} b_h \quad (7.157)$$

and if $k+m-g \neq k-h$, or $h+m \neq g$, then

$$E[\{b_g w(k+m-g)\} \{b_h w(k-h)\}] = 0 \quad (7.158)$$

Therefore, Equation 7.156 can be replaced by

$$R_X(k+m, k) = \sum_{h=0}^{q-m} b_h b_{h+m} E[w^2(k-h)] = \sigma^2 \sum_{h=0}^{q-m} b_h b_{h+m}, \quad 0 \leq m \leq q \quad (7.159)$$

Based on the same observation described in Equation 7.158, when $m > q$,

$$R_X(k+m, k) = 0 \quad (7.160)$$

It is seen that when $k > q$, then the autocorrelation function of random process $MA(q)$ is independent from that at time point k , and is only a function of time lag m . Therefore, after time point q , because the corresponding mean is zero and is therefore a constant, the process is weakly stationary.

7.5.2.2 Autoregressive Process AR(p)

When $n \geq 0$, the excitation of the p th autoregressive process AR(p) is a stationary white noise. Let us focus on this case, namely,

$$x(n) = -\sum_{k=1}^p a_k x(n-k) + b_0 w(n), \quad n \geq 0 \quad (7.161)$$

Note that for $w(n)$, Equation 7.145 must be satisfied. Now, let us consider the mean, variance, and autocorrelation functions of the autoregressive process AR(p) given by Equation 7.161.

7.5.2.2.1 Mean

When $n < 0$, $E[x(n)] = 0$ and $x(n) = 0$, which leads to $E[x(n-k)] = 0$. In addition, $E[w(n)] = 0$, therefore, the mean of this autoregressive process can be obtained as follows:

$$E[x(n)] = -\sum_{k=1}^p a_k E[x(n-k)] + b_0 E[w(n)] = 0, \quad n \geq 0 \quad (7.162)$$

7.5.2.2.2 Variance

Because the mean is zero, the variance can be calculated as

$$\sigma_x^2(n) = E[x^2(n)] = \left[\left\{ -\sum_{k=1}^p a_k x(n-k) + b_0 w(n) \right\} \left\{ -\sum_{k=1}^p a_k x(n-k) + b_0 w(n) \right\} \right] \quad (7.163)$$

Similar to the observation to examine the products discussed previously, we see that the term $x(n-k)$ occurs before $w(n)$ so that $x(n-k)$ is not a function of $w(n)$, thus $E[x(n-k)w(n)] = 0$

Therefore,

$$E \left[\left\{ -\sum_{k=1}^p a_k x(n-k) w(n) \right\} \right] = 0 \quad (7.164)$$

With the help of Equation 7.164, we further write

$$\sigma_x^2(n) = E \left[\left\{ -\sum_{k=1}^p a_k x(n-k) \right\} \left\{ -\sum_{j=1}^p a_j x(n-j) \right\} \right] + b_0^2 \sigma^2 \quad (7.165)$$

Furthermore, let us take the mathematical expectation and write

$$\sigma_x^2(n) = \sum_{k=1}^p \sum_{j=1}^p a_k a_j R_x(n-k, n-j) + b_0^2 \sigma^2 \quad (7.166)$$

7.5.2.2.3 Autocorrelation

Generally speaking, even if a signal has a zero initial condition, the autocorrelation still has a transient process. This point can be realized by observing the existence of the transient solution $x_p(t)$ in Chapter 6 (see Equation 6.100), although in this case $x(t)$ is deterministic. We now consider that the transient portion in the autocorrelation function becomes negligible, that is, when $n > p$. In this case, first consider $R_X(n, n-1)$, given by

$$\begin{aligned} R_X(n, n-1) &= E[x(n)x(n-1)] \\ &= E\left[\left\{-\sum_{k=1}^p a_k x(n-k) + b_0 w(n)\right\}x(n-1)\right] \quad n > p \end{aligned} \quad (7.167)$$

Because $x(n-1)$ is not a function of $w(n)$, and $w(n)$ is white noise, we have

$$E(w(n)x(n-1)) = 0 \quad (7.168)$$

Therefore, Equation 7.167 can be replaced by

$$R_X(n, n-1) = -\sum_{k=1}^p a_k R_X(n-1, n-k) \quad (7.169)$$

Now, with the same idea, we can multiply $x(n-j)$, $j = 2, 3, \dots, p$ on both sides of Equation 7.161 and take the corresponding mathematic expectation, that is, we will have

$$R_X(n, n-j) = -\sum_{k=1}^p a_k R_X(n-j, n-k), \quad j = 2, 3, \dots, p \quad (7.170)$$

We should also consider the case of $R_X(n, n)$, which is equal to $\sigma_X^2(n)$. From Equation 7.166, it is seen that

$$R_X(n, n) = -\sum_{j=1}^p a_j R_X(n, n-j) + b_0^2 \sigma^2 \quad (7.171)$$

Equations 7.169 through 7.171 provide the formulae to calculate the autocorrelation function for time lag from 0 to p . Note that Equation 7.170 will also be valid when $j > p$.

Now, when AR(p) has already reached steady state, then the corresponding autocorrelation $R_X(r, s)$ is only a function of the time lag

$$j = r - s \quad (7.172)$$

In this case, Equation 7.170 can be rewritten as

$$R_X(j) = -\sum_{k=1}^p a_k R_X(k-j), \quad j = 1, 2, 3, \dots, p \quad (7.173)$$

For autocorrelation functions, $R_X(-j) = R_X(j)$, we can rewrite Equation 7.173 in the format of a matrix equation called the *Yule-Walker equation* (George Udny Yule [1871–1951], Gilbert Thomas Walker [1868–1958]), that is,

$$\begin{bmatrix} R_X(0) & R_X(1) & \dots & R_X(p-1) \\ R_X(1) & R_X(0) & \dots & R_X(p-2) \\ & & \dots & \\ R_X(p-1) & R_X(p-2) & \dots & R_X(0) \end{bmatrix} \begin{Bmatrix} a_1 \\ a_2 \\ \dots \\ a_p \end{Bmatrix} = \begin{Bmatrix} R_X(1) \\ R_X(2) \\ \dots \\ R_X(p) \end{Bmatrix} \quad (7.174)$$

If these autocorrelation functions $R_X(0) \dots R_X(p)$ can be obtained, then we can calculate these coefficients $a_1 \dots a_p$ to determine the process AP(q).

On the other hand, if these coefficients $a_1 \dots a_p$ are known to obtain the autocorrelation functions, we can use the following matrix equation:

$$\begin{bmatrix} 1 & a_1 & a_2 & \dots & a_{p-2} & a_{p-1} & a_p \\ a_1 & 1+a_2 & a_3 & \dots & a_{p-1} & a_p & 0 \\ a_2 & a_1+a_3 & 1+a_4 & \dots & a_p & 0 & 0 \\ & & & \dots & & & \\ 0 & a_p & a_{p-1} & \dots & a_3 & 1+a_2 & a_1 \\ a_p & a_{p-1} & a_{p-2} & \dots & a_2 & a_1 & 1 \end{bmatrix} \begin{Bmatrix} R_X(0) \\ R_X(1) \\ R_X(2) \\ \dots \\ R_X(p-1) \\ R_X(p) \end{Bmatrix} = \begin{Bmatrix} -b_0\sigma^2 \\ 0 \\ 0 \\ \dots \\ 0 \\ 0 \end{Bmatrix} \quad (7.175)$$

7.5.2.3 ARMA(p, q)

Now consider the process of ARMA(p, q), which satisfies Equations 7.143 through 7.145. For convenience, Equation 7.145 is replaced by

$$E(w^2(n)) = \sigma^2 \quad (7.176)$$

Let us consider the mean, variance, and autocorrelation functions.

7.5.2.3.1 Mean

Taking the mathematic expectation of Equation 7.143, we have

$$E[x(n)] = -\sum_{k=1}^p a_k E[x(n-k)] + \sum_{k=0}^q b_k E[w(n-k)], \quad n \geq 0 \quad (7.177)$$

Because

$$E[w(n-k)] = 0$$

Equation 7.177 can be replaced by

$$E[x(n)] = - \sum_{k=1}^p a_k E[x(n-k)], \quad n \geq 0$$

Assume zero initial conditions, we further have

$$E[x(n)] = 0, \quad n \geq 0 \quad (7.178)$$

7.5.2.3.2 Variance

Because the mean is zero, for the variance, we can write

$$\begin{aligned} \sigma_x^2(n) &= E[x^2(n)] = R_x(n, n) \\ &= E \left[\left\{ - \sum_{k=1}^p a_k x(n-k) + b_0 \sum_{k=0}^q w(n-k) \right\} \{x(n)\} \right] \\ &= - \sum_{k=1}^p a_k R_x(n, n-k) + \sum_{k=0}^q b_k E[w(n-k)x(n)] \\ &= - \sum_{k=1}^p a_k R_x(n, n-k) + \sum_{k=0}^q b_k R_{xw}(n, n-k) \end{aligned} \quad (7.179)$$

7.5.2.3.3 Autocorrelation Function

Similar to the case of $AR(p)$, the autocorrelation will have a transient process. We can consider that the transient portion in the autocorrelation function becomes negligible, namely, $n > p$. In addition, we also consider $R_x(n, n-1)$, first, given by

$$\begin{aligned} R_x(n, n-1) &= E[x(n)x(n-1)] \\ &= E \left[\left\{ - \sum_{k=1}^p a_k x(n-k) + \sum_{k=0}^q b_k w(n-k) \right\} x(n-1) \right] \\ &= - \sum_{k=1}^p a_k R_x(n-1, n-k) + \sum_{k=0}^q b_k E[w(n-k)x(n-1)] \end{aligned} \quad (7.180)$$

Similar to the above treatment, considering $E[x(n)x(n-j)]$, $j = 2, 3, \dots, p$, we have

$$\begin{aligned} R_X(n, n-j) &= E[x(n)x(n-j)] \\ &= -\sum_{k=1}^p a_k R_X(n-j, n-k) + \sum_{k=0}^q b_k E[w(n-k)x(n-j)], \quad j = 2, 3, \dots, p \end{aligned} \quad (7.181)$$

Equations 7.180 and 7.181 provide formulae to calculate the autocorrelation function for time lag from 1 to p .

When ARMA(p, q) has already reached steady state, then the corresponding autocorrelation $R_X(s)$ is only a function of the time lag given by Equation 7.172. In this case, we replace Equations 7.180 and 7.181 by

$$R_X(j) = -\sum_{k=1}^p a_k R_X(k-j) + \Phi_j(\mathbf{a}, \mathbf{b}) \quad j = 1, 2, 3, \dots, p \quad (7.182)$$

where $\Phi_j(\mathbf{a}, \mathbf{b})$ is the term of the second summations in Equations 7.180 and 7.181, which are a nonlinear functions. Whereas,

$$\mathbf{a} = [a_1, a_2, \dots, a_p], \quad \mathbf{b} = [b_1, b_2, \dots, b_q] \quad (7.183)$$

By using the equation in matrix form, we have

$$\begin{bmatrix} R_X(0) & R_X(1) & \dots & R_X(p-1) \\ R_X(1) & R_X(0) & \dots & R_X(p-2) \\ & & \dots & \\ R_X(p-1) & R_X(p-2) & \dots & R_X(0) \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ \dots \\ a_p \end{bmatrix} + \begin{bmatrix} \Phi_1(\mathbf{a}, \mathbf{b}) \\ \Phi_2(\mathbf{a}, \mathbf{b}) \\ \dots \\ \Phi_p(\mathbf{a}, \mathbf{b}) \end{bmatrix} = \begin{bmatrix} R_X(1) \\ R_X(2) \\ \dots \\ R_X(p) \end{bmatrix} \quad (7.184)$$

Note that Equation 7.184 is nonlinear.

Example 7.2

Find the mean, variance, and autocorrelation of the following process ARMA(1, 1):

$$x(n) = -a_1 x(n-1) + b_0 w(n) + b_1 w(n-1), \quad n \geq 0$$

MEAN

We can see from Equation 7.178, $E[x(n)] = 0$.

AUTOCORRELATION FUNCTION

With zero mean, the variance at time n is equal to the value of autocorrelation $R_X(0, 0)$. That is,

$$\begin{aligned} \sigma^2(n) &= E[x^2(n)] = R_X(n, n) = E[x(n)\{-a_1 x(n-1) + b_0 w(n) + b_1 w(n-1)\}] \\ &= -a_1 R_X(n, n-1) + b_0 E[x(n)w(n)] + b_1 E[x(n)w(n-1)] \end{aligned}$$

The right-hand side of the above equation has two means that need to be evaluated. Consider the last one first. Substituting the expression of $x(n)$ into the above equation and rearranging the results, and further taking the mathematical expectation, we can write

$$E[x(n)w(n-1)] = E\{-a_1x(n-1) + b_0w(n) + b_1w(n-1)\}w(n-1) = -a_1E[x(n-1)w(n-1)] + b_1\sigma^2 = -a_1E[a_1x(n-2) + b_0w(n-1) + b_1w(n-2)]w(n-1) + b_1\sigma^2 = (-a_1b_0 + b_1)\sigma^2$$

The reason we can have the above result is because $x(n-1)$ is not a function of $w(n)$, and therefore $E[x(n-2)w(n-1)] = 0$. Because $w(n)$ is white noise, $E[w(n-2)w(n-1)] = 0$.

Another mean we need can be calculated as

$$E[(x(n)w(n))] = E[-a_1x(n-1) + b_0w(n) + b_1w(n-1)]w(n) = b_0\sigma^2$$

Therefore, the variance is

$$\sigma^2(n) = -a_1 R_x(n, n-1) + (b_0^2 - a_1b_0b_1 + b_1^2)\sigma^2$$

The autocorrelation function can be written as

$$R_x(n, n-1) = E[x(n)x(n-1)] = E[[-a_1x(n-1) + b_0w(n) + b_1w(n-1)]x(n-1)] = -a_1R_x(n-1, n-1) + b_0E[x(n-1)w(n)] + b_1E[x(n-1)w(n-1)]$$

Again, because $x(n-1)$ is not a function of $w(n)$, $E[x(n-2)w(n-1)] = 0$. Consider, $E[x(n-1)w(n-1)] = E[w(n-1)\{-a_1x(n-2) + b_0w(n-1) + b_1w(n-2)\}] = b_0\sigma^2$

We thus have

$$R_x(n, n-1) = -a_1R_x(n-1, n-1) + b_0b_1\sigma^2$$

To have the autocorrelation function for the steady state response, when the time lag $j = 0$ and $j = 1$, we have

$$R_x(0) = \sigma^2(0) = -a_1R_x(1) + (b_0^2 - a_1b_0b_1 + b_1^2)\sigma^2$$

and

$$R_x(1) = -a_1R_x(0) + b_0b_1\sigma^2$$

Solving the above two equations, we can further write

$$R_x(0) = \frac{(b_0^2 - 2b_0b_1 + b_1^2)\sigma^2}{1 - a_1^2}$$

and

$$R_x(1) = \frac{(a_1^2b_0b_1 - a_1b_0^2 - a_1b_1^2 + b_0b_1)\sigma^2}{1 - a_1^2}$$

In addition, let us consider when time lag $j = 2$.

Because $E[x(n-2)w(n)] = 0$ and $E[x(n-2)w(n-1)] = 0$, we can write

$$\begin{aligned} R_x(n, n-j) &= E[x(n)x(n-j)] = E[[-a_1x(n-1) + b_0w(n) + b_1w(n-1)]x(n-j)] \\ &= -a_1R_x(n-1, n-2) \end{aligned}$$

Similarly, when $j > q$, for any $j > 2$, we can write

$$\begin{aligned} R_x(n, n-2) &= E[x(n)x(n-2)] = E[[-a_1x(n-1) + b_0w(n) + b_1w(n-1)]x(n-2)] \\ &= -a_1R_x(n-1, n-j+1) \end{aligned}$$

Furthermore, for steady state process of ARMA(1, 1) with time lag > 1 , we can write

$$R_x(j) = (-a_1)^{j-1} R_x(1), \quad j \geq 2$$

VARIANCE

Because the mean of ARMA(1, 1) = 0, the variance can be obtained as

$$\sigma_x^2(n) = R_x(0) = \frac{(b_0^2 - 2b_0b_1 + b_1^2)\sigma^2}{1 - a_1^2}$$

It is noted that when the orders of p and q are greater than unity, it is difficult to write the autocorrelation function and variance in closed forms.

7.5.3 ANALYSES OF TIME SERIES IN THE FREQUENCY DOMAIN

7.5.3.1 Z-Transform

Fourier transform transfers a signal in the continuous time domain into the continuous frequency domain. To account for the signals in the discrete time domain, it is known that the discrete Fourier transform (DFT) can be used (see Chapter 4). However, DFT is not convenient for analytical analysis; rather it was developed for computational applications. It is needed to have an analytical transform that handles signals in the discrete time domain, such as the above-mentioned time series like ARMA models. Z transform transfers a signal from the discrete time domain into the z -domain. From a mathematical viewpoint, the Z-transform can be seen as a Laurent series in which the sequence of numbers under consideration are the Laurent expansion of an analytic function. The basic idea now known as the Z-transform was known to Laplace, and reintroduced in 1947 by W. Hurewicz as a tractable way to solve linear, constant-coefficient difference equations. It was later dubbed “the z -transform” by Ragazzini and Zadeh in a sampled data control group in 1952. In the following, we briefly introduce the z transform without a detailed discussion on its main properties and existing conditions.

7.5.3.2 Sampling of Signals

With the help of delta functions, a signal in the continuous time domain denoted by $x(t)$ can be sampled using the following treatment:

$$x_d(t) = \sum_{k=0}^{\infty} x(k) \delta(t - k\Delta t) \quad (7.185)$$

where Δt is the sampling time interval. The subscript d denotes $x_d(t)$ is in discrete form.

Note that although $x_d(t)$ can only have nonzero values at the moment of sampling, it is still in the continuous time domain.

Taking the Laplace transform of $x_d(t)$, we have

$$\begin{aligned} X_d(s) &= \mathcal{L}[x_d(t)] = \mathcal{L}\left[\sum_{k=0}^{\infty} x(k) \delta(t - k\Delta t)\right] \\ &= \sum_{k=0}^{\infty} x(k) \mathcal{L}[\delta(t - k\Delta t)] = \sum_{k=0}^{\infty} x(k) e^{-sk\Delta t} \end{aligned} \quad (7.186)$$

Now, let a variable z be

$$z = e^{s\Delta t} \quad (7.187)$$

and substitution of Equation 7.187 into 7.186, we have

$$X_d(s) \Big|_{z=e^{s\Delta t}} = X(z) = \sum_{k=0}^{\infty} x(k) z^{-k} \quad (7.188)$$

In Equation 7.188, the series $X(z)$ is the function of variable z . When $z = e^s$ (especially $z = e^{j\omega}$), $X(z)$ is referred to as the z transform and denoted by

$$X(z) = \mathcal{Z}[x(t)] \quad (7.189)$$

Here, we omit the subscript d for $X(z)$ is obviously a discrete series. The physical meaning of z will be discussed in Section 7.5.4.3.

The inverse z transform, denoted by $\mathcal{Z}^{-1}[X(z)]$ can be calculated by

$$x(t) = \mathcal{Z}^{-1}[X(z)] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega t} d\omega \quad (7.190)$$

7.5.3.3 Transfer Function of Discrete Time System

Now, we discuss the case in which an SDOF system is being excited by a time series of white noise.

First, let $f(n)$ and $x(n)$ be the input and output of a general system, with zero initial conditions. We can write

$$x(n) = -\sum_{k=1}^p a_k f(n-k) + \sum_{k=0}^q b_k x(n-k), \quad n \geq 0, p > q \quad (7.191)$$

Taking z transform on both sides on Equation 7.191, we have

$$X(z) = -\sum_{k=1}^p a_k z^{-k} F(z) + \sum_{k=0}^q b_k z^{-k} X(z), \quad n \geq 0, p > q \quad (7.192)$$

where

$$F(z) = \mathcal{Z}[f(t)] \quad (7.193)$$

is the z transform of the input forcing function $f(t)$; and $X(z)$ is the z transform of the output given by Equation 7.192. From Equation 7.192, we can write the transfer function in the z domain as

$$H(z) = \frac{X(z)}{F(z)} = \frac{\sum_{k=0}^q b_k z^{-k}}{1 + \sum_{k=1}^p a_k z^{-k}}, \quad n \geq 0, p > 2 \quad (7.194)$$

From Equation 7.194, it is seen that the transfer function $H(z)$ described in the z domain is a rational function of z^{-1} . If the excitation is white noise, then the response will be a random process ARMA(p, q). Taking the inverse z transform of the transform function, we can have the unit impulse response function, namely,

$$h(n) = \mathcal{Z}^{-1}[H(z)] \quad (7.195)$$

If the coefficient b_k is zero except b_0 , that is,

$$b_k = 0, k = 1, 2, \dots, q \quad (7.196)$$

then, the transfer function takes the following form,

$$H(z) = \frac{X(z)}{F(z)} = \frac{b_0}{1 + \sum_{k=1}^p a_k z^{-k}}, \quad n \geq 0, p > q \quad (7.197)$$

From Equation 7.197, we see that if the input to the system is a white noise, then the output is an autoregressive process $AR(p)$. Another interesting case is when

$$a_k = 0, \quad k = 1, 2, \dots, p \quad (7.198)$$

then, the transfer function takes the following form,

$$H(z) = \sum_{k=1}^q b_k z^{-k} \quad (7.199)$$

In this case, from Equation 7.199, if the input to the system is a white noise, then the output is a moving-average process $MA(q)$.

7.5.3.4 PSD Functions

7.5.3.4.1 PSD Function of $MA(q)$

Based on the transfer function given by Equation 7.199, we can calculate the auto-PSD function for the process of $MA(q)$. That is,

$$\begin{aligned} S_X(\omega) &= S_X(z) \Big|_{z=e^{j\omega}} = H(z) S_F(z) H(z^{-1}) \Big|_{z=e^{j\omega}} \\ &= \left[\sum_{k=0}^q b_k z^{-k} (\sigma^2) \sum_{k=0}^q b_k z^k \right] \Big|_{z=e^{j\omega}} = \left| \sum_{k=0}^q b_k e^{-jk\omega} \right|^2 \sigma^2 \end{aligned} \quad (7.200)$$

7.5.3.4.2 PSD Function of $AR(p)$

Here, suppose the process $AR(p)$ has input $f(n)$ with zero mean and the autocorrelation function can be written as

$$R_X(k+m, k) = \sigma^2 \delta(k) \quad (7.201)$$

Based on the transfer function given by Equation 7.197, we have

$$\begin{aligned} S_X(\omega) &= S_X(z) \Big|_{z=e^{j\omega}} = H(z) S_F(z) H(z^{-1}) \Big|_{z=e^{j\omega}} \\ &= \left[\frac{b_0}{1 + \sum_{k=1}^p a_k z^{-k}} (\sigma^2) \frac{b_0}{1 + \sum_{k=1}^p a_k z^k} \right] \Big|_{z=e^{j\omega}} \\ &= \frac{b_0^2 \sigma^2}{\left| 1 + \sum_{k=1}^p a_k e^{-jk\omega} \right|^2} \end{aligned} \quad (7.202)$$

7.5.3.4.3 PSD Function of ARMA(p, q)

Assume the input $f(n)$ is zero mean and the autocorrelation function is given as

$$R_X(k+m, k) = \sigma^2 \delta(k) \quad (7.203)$$

The auto-PSD function for ARMA(p, q) is (see Equation 7.194)

$$\begin{aligned} S_X(\omega) &= S_X(z) \Big|_{z=e^{j\omega}} = H(z) S_F(z) H(z^{-1}) \Big|_{z=e^{j\omega}} \\ &= \left[\frac{\sum_{k=0}^q b_k z^{-k}}{1 + \sum_{k=1}^p a_k z^{-k}} (\sigma^2) \frac{\sum_{k=0}^q b_k z^k}{1 + \sum_{k=1}^p a_k z^k} \right]_{z=e^{j\omega}} \\ &= \frac{\left| \sum_{k=0}^q b_k e^{-jk\omega} \right|^2}{\left| 1 + \sum_{k=1}^p a_k e^{-jk\omega} \right|^2} \sigma^2 \end{aligned} \quad (7.204)$$

7.5.4 TIME SERIES OF SDOF SYSTEMS

7.5.4.1 Difference Equations

It is known that we can use difference equations to approximate differential equations. With sufficiently small time intervals, Δt , the velocity may be written as

$$\dot{x}(n) = \frac{x(n+1) - x(n)}{\Delta t} \quad (7.205)$$

and the acceleration can be written as

$$\ddot{x}(n) = \frac{\dot{x}(n+1) - \dot{x}(n)}{\Delta t} = \frac{x(n+2) - 2x(n+1) + x(n)}{\Delta t^2} \quad (7.206)$$

Suppose the excitation of an SDOF system can be written as $f(n)$. Substitution of Equations 7.205 and 7.206 into Equation 7.1 yields

$$m \frac{x(n+2) - 2x(n+1) + x(n)}{\Delta t^2} + c \frac{x(n+1) - x(n)}{\Delta t} + kx(n) = f(n) \quad (7.207)$$

Dividing $m/\Delta t^2$ on both sides of Equation 7.207 and rearranging the resulting equation, we can write

$$x(n+2) = -\left(-2 + \frac{c\Delta t}{m}\right)x(n+1) - \left(1 - \frac{c\Delta t}{m} + \frac{k\Delta t^2}{m}\right)x(n) + \frac{\Delta t^2}{m}f(n) \quad (7.208)$$

Equation 7.207 is the governing equation of motion of an SDOF system. Similarly, for the case of ground excitation, we have

$$m\ddot{x} + c\dot{x} + kx = -m\ddot{x}_g$$

in which the variable x is a relative displacement, and x_g is a ground displacement.

Substitution of Equations 7.205 and 7.206 into the above equation yields

$$\begin{aligned} m \frac{x(n+2) - 2x(n+1) + x(n)}{\Delta t^2} + c \frac{x(n+1) - x(n)}{\Delta t} + kx(n) \\ = -m \frac{x_g(n+2) - 2x_g(n+1) + x_g(n)}{\Delta t^2} \end{aligned}$$

or

$$\begin{aligned} x(n+2) = -\left(-2 + \frac{c\Delta t}{m}\right)x(n+1) - \left(1 - \frac{c\Delta t}{m} + \frac{k\Delta t^2}{m}\right)x(n) \\ + x_g(n+2) - 2x_g(n+1) + x_g(n) \end{aligned} \quad (7.209)$$

Furthermore, for ground excitation, we also have

$$m\ddot{x} + c\dot{x} + kx = c\dot{x}_g + kx_g$$

in which the variable x is an absolute displacement, and x_g is also a ground displacement.

Substitution of Equations 7.205 and 7.206 into the above equation yields

$$\begin{aligned} m \frac{x(n+2) - 2x(n+1) + x(n)}{\Delta t^2} + c \frac{x(n+1) - x(n)}{\Delta t} + kx(n) \\ = c \frac{x_g(n+1) - x_g(n)}{\Delta t} + kx_g(n) \end{aligned}$$

or

$$\begin{aligned}
 x(n+2) = & -\left(-2 + \frac{c\Delta t}{m}\right)x(n+1) - \left(1 - \frac{c\Delta t}{m} + \frac{k\Delta t^2}{m}\right)x(n) \\
 & + \frac{c\Delta t}{m}x_g(n+1) + \left(-\frac{c\Delta t}{m} + \frac{k\Delta t^2}{m}\right)x_g(n)
 \end{aligned} \quad (7.210)$$

7.5.4.2 ARMA Models

The above difference equations can be written in the form of typical ARMA models. That is, consider the case of the SDOF system excited by forcing function $f(n)$, Equation 7.208 can be written as

$$x(n) = -a_1x(n-1) - a_2x(n-2) + b_2f(n-2) \quad (7.211)$$

with

$$a_1 = -2 + \frac{c\Delta t}{m} \quad (7.212a)$$

$$a_2 = 1 - \frac{c\Delta t}{m} + \frac{k\Delta t^2}{m} \quad (7.212b)$$

and

$$b_2 = \frac{\Delta t^2}{m} \quad (7.212c)$$

In Equation 7.210, without loss of generality, we can use n to replace $n+2$, which will also be used in the following examples.

Equation 7.211 established excitation due to ground acceleration can be written as

$$x(n) = -a_1x(n-1) - a_2x(n-2) + b_0x_g(n) + b_1x_g(n-1) + b_2x_g(n-2) \quad (7.213)$$

where a_1 and a_2 are given by Equations 7.212a and 7.212b. However,

$$b_0 = 1 \quad (7.214a)$$

$$b_1 = -2 \quad (7.214b)$$

and

$$b_2 = 1 \quad (7.214c)$$

Furthermore, Equation 7.210, describing the excitation based on ground damping and restoring force can be written as

$$x(n) = -a_1x(n-1) - a_2x(n-2) + b_1x_g(n-1) + b_2x_g(n-2) \quad (7.215)$$

whereas a_1 and a_2 are given by Equations 7.212a and 7.212b and

$$b_1 = \frac{c\Delta t}{m} \quad (7.216a)$$

and

$$b_2 = -\frac{c\Delta t}{m} + \frac{k\Delta t^2}{m} \quad (7.216b)$$

It is also seen that

$$\mathcal{Z}[x(n-k)] = X(z)z^{-k} \quad (7.217)$$

7.5.4.3 Transfer Functions

We now consider the transfer function based on the model for the excitation $f(n)$. Suppose the forcing function is white noise. Take z -transform of Equation 7.210,

$$X(z) = -a_1X(z)z^{-1} - a_2X(z)z^{-2} + b_2F(z)z^{-2} \quad (7.218)$$

The transfer function can then be written as

$$H(z) = \frac{X(z)}{F(z)} = \frac{b_2z^{-2}}{1 + a_1z^{-1} + a_2z^{-2}} = \frac{b_2}{z^2 + a_1z + a_2} \quad (7.219)$$

Substitution of Equations 7.212 into 7.219 yields

$$\begin{aligned} H(z) &= \frac{\frac{\Delta t^2}{m}}{z^2 + \left(-2 + \frac{c\Delta t}{m}\right)z + 1 - \frac{c\Delta t}{m} + \frac{k\Delta t^2}{m}} \\ &= \frac{\Delta t^2}{m} \frac{1}{z^2 + (-2 + 2\zeta\omega_n\Delta t)z + 1 - 2\zeta\omega_n\Delta t + \omega_n^2\Delta t^2} \end{aligned} \quad (7.220)$$

Consider the poles of $H(z)$, which are the zeroes of the denominator in Equation 7.220. That is, let

$$z^2 + (-2 + 2\zeta\omega_n\Delta t)z + 1 - 2\zeta\omega_n\Delta t + \omega_n^2\Delta t^2 = 0 \quad (7.221)$$

which has solutions given by

$$z = 1 - \zeta \omega_n \Delta t \pm j (1 - \zeta^2)^{1/2} \omega_n \Delta t \quad (7.222)$$

Note that Δt can be sufficiently small. Seen in Equation 7.187, $z = e^{s\Delta t}$, by letting

$$e^{s\Delta t} = 1 - \zeta \omega_n \Delta t \pm j (1 - \zeta^2)^{1/2} \omega_n \Delta t \quad (7.223)$$

for sufficiently small Δt , we have

$$s = -\zeta \omega_n \pm j (1 - \zeta^2)^{1/2} \omega_n \quad (7.224)$$

The above shows that the Laplace variable s , on the condition that the transfer function reaches its poles, is equivalent to the eigenvalues of the SDOF system. Furthermore, we can prove that the transfer function using the Laplace variable s , $H(s)$, and the transfer function using variable z , $H(z)$, has the same values, provided Δt is sufficiently small. That is

$$\lim_{\Delta t \rightarrow 0} H(z) = H(s) \quad (7.225)$$

For the cases expressed by Equations 7.73 and 7.75, we can have the same observations. Therefore, we use the time series of the ARMA model to describe an SDOF system. In Chapter 8, we will further discuss the utilization of different functions for MDOF systems.

Example 7.3

Given an SDOF system with $m = 1$, $c = 4$, and $k = 1000$, plot transfer functions of $H(s)$ as well as $H(z)$ based on Equation 7.220 with $\Delta t = 0.0005$, 0.0010 , and 0.0050 s.

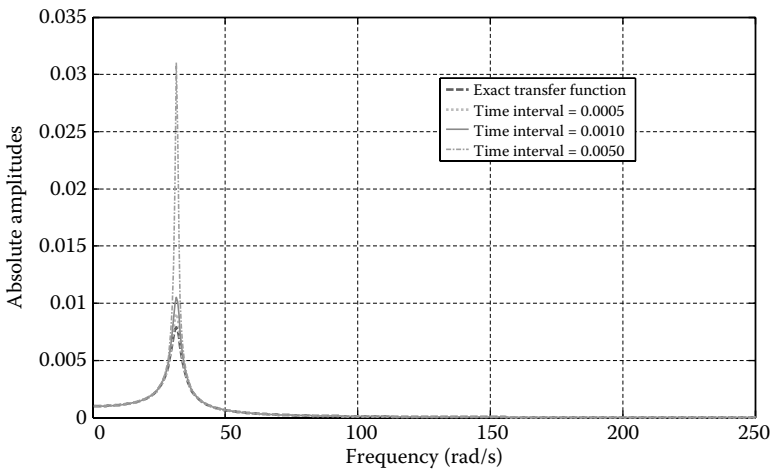


FIGURE 7.11 Exact and approximated transfer functions.

The results are plotted in Figure 7.11. From these curves, it is seen that when Δt is sufficiently small, $H(z)$ can be a good approximation of the exact calculation of $H(s)$. However, when $\Delta t = 0.005$ s, which is often used in practical measurement, we will have larger errors especially in the resonant region.

We note that in this example the damping ratio is 6.3%. When the damping ratio becomes large or different natural frequencies are used (or both), the situation will not be improved. Therefore, sufficiently small time intervals need to be carefully chosen when using time series to directly analyze an SDOF system.

7.5.4.4 Stability of Systems

7.5.4.4.1 General Description

In Chapter 6, we showed that for an SDOF system, we need $c \geq 0$ and $k > 0$ to achieve stable vibrations. The condition of $c \geq 0$ is equivalent to having a nonpositive portion of the system's eigenvalue, namely,

$$\text{Re}(\lambda) \leq 0 \quad (7.226)$$

Now, we examine the ARMA model to establish criterion for the system's stability. Recall the definitions of ARMA(p, q) and AR(p), respectively, described in Equations 7.143 and 7.161. It can be seen that both are difference equations with constant coefficients.

For convenience, let us define a lag operator (backshift operator) B such that

$$B[x(n)] = x(n-1) \quad (7.227)$$

It is seen that

$$B^k[X(n)] = x(n-k) \quad (7.228)$$

Rewrite an ARMA(p, q) model as

$$\sum_{k=0}^p a_k x(n-k) = \sum_{k=0}^q b_k w(n-k), \quad a_0 = 1, n \geq 0, p > q \quad (7.229)$$

Consider the corresponding homogeneous difference equation, we write

$$\sum_{k=0}^p a_k x(n-k) = 0, \quad a_0 = 1, n \geq 0 \quad (7.230)$$

It can be seen that the homogeneous difference equation for an AR(p) model can also be described by Equation 7.230. With the help of the lag operation, Equation 7.230 can be further written as

$$\sum_{k=0}^p a_k x(n-k) = \mathcal{A}_p(B)[x(n)] = 0, \quad a_0 = 1 \quad (7.231)$$

where $\mathcal{A}_p(B)$ is an operator polynomial, which will be discussed shortly after the introduction of characteristic equation. In addition, the corresponding characteristic equation of the ARMA(p, q) and AP(p) is written as (recall characteristic equation described in Equation 6.39)

$$\lambda^p + a_1\lambda^{p-1} + \dots + a_{p-1}\lambda + a_p = \sum_{k=0}^p a_k\lambda^{p-k} = 0, \quad a_0 = 1 \quad (7.232)$$

where the solutions λ of the characteristic equation are the eigenvalues of ARMA(p, q) or AR(p). By using these eigenvalues, we can write the operator polynomial as

$$\mathcal{A}_p(B) = (1 - \lambda_1 B)(1 - \lambda_2 B) \dots (1 - \lambda_p B) \quad (7.233)$$

Letting $\mathcal{A}_p(B) = 0$, we can find the relationship between coefficients a_k and eigenvalues λ_i . That is, these eigenvalues must satisfy the factorization of the operator polynomial equivalent to zero.

7.5.4.4.2 Stability of Process AR(2)

Now considering a special case of the homogeneous equation of AR(2), we have

$$x(n) + a_1x(n-1) + a_2x(n-2) = 0 \quad (7.234)$$

The factorization of the operator polynomial is

$$(1 + a_1B + a_2B^2) = (1 - \lambda_1B)(1 - \lambda_2B) = 0 \quad (7.235)$$

Therefore, we have

$$\lambda_1 + \lambda_2 = -a_1 \quad \text{and} \quad \lambda_1\lambda_2 = a_2 \quad (7.236)$$

The characteristic equation is

$$\lambda^2 + a_1\lambda + a_2 = 0 \quad (7.237)$$

Thus, we can solve Equation 7.237 to obtain the eigenvalues

$$\lambda_{1,2} = \frac{-a_1 \pm j\sqrt{a_1^2 - 4a_2}}{2} \quad (7.238)$$

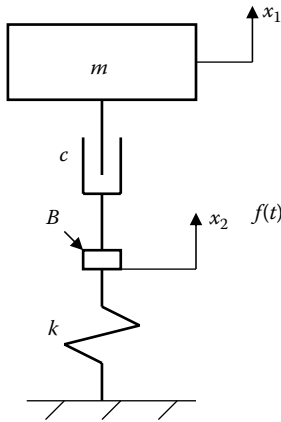
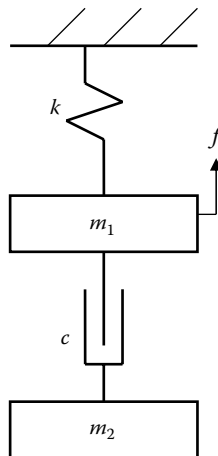
It can be proven if the following criterion is satisfied

$$|\lambda_i| < 1 \quad (7.239)$$

the system is stable.

PROBLEMS

1. A system is shown in Figure P7.1 with white noise excitation $f(t)$.
 - a. Find the equation of motion for this system.
 - b. What is the transfer function of this system?
 - c. Find the PSD matrix.
 - d. Find the RMS value of the response of x_1 . (Hint: block B is massless.)
2. A white noise force is applied on the first mass of the system given in Figure P7.2. Find its governing equation and transfer function. What are the transfer functions measured at the first and the second mass? Find the standard deviation of the response x_1 .

**FIGURE P7.1****FIGURE P7.2**

3. A system shown in Figure P7.2 is excited by ground white noise motion. Denote the stiffness connecting m_1 and the ground to be k_1 and there is an additional stiffness k_2 connecting m_1 and m_2 . Write the equation of motion. If mass m_1 is sufficiently small ($m_1 = 0$), then, what is the transfer function of ground displacement and the relative displacement? What are the power density functions of the relative displacement and the absolute acceleration of the responses measured at the second mass?
4. An SDOF system with mass m , stiffness k , and damping ratio ζ , vibrates due to a white noise excitation with level W_0 .
 - a. Compute the RMS absolute acceleration by applying a quasistatic dynamic load.
 - b. Find the relative displacement and calculate the RMS restoring force and with your calculations, compute the RMS absolute acceleration again. Then, explain any difference in the results.
5. A white noise force $F(t)$ with level $15 \text{ N}^2/\text{Hz}$ is applied to the system shown in Figure P7.3. The mass is 3.5 kg . Design a steel rod to find the minimum b so that the strength $R = 360 \text{ MPa}$ will be greater than three times the RMS stress. The damping ratio is 0.05 . The modulus of elasticity of steel is 207 GPa . Determine the resulting natural frequency as well.
6. An SDOF system is shown in Figure P7.4 with mass $= 0.259 \text{ lb sec}^2/\text{in.}$, stiffness $k = 63.0 \text{ lb/in.}$, and damping ratio $\zeta = 0.064$, excited by white noise acceleration $= 0.02 \text{ g}^2/\text{Hz}$ with occurrence $= 10 \text{ s period}$ ($g = 386 \text{ in/s}^2$).

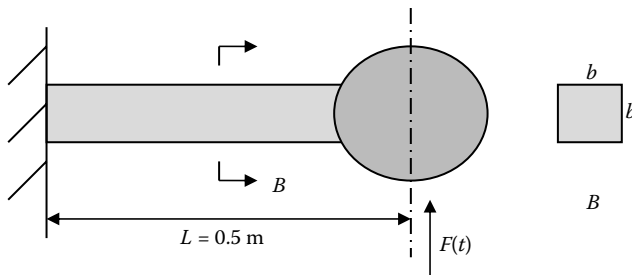


FIGURE P7.3

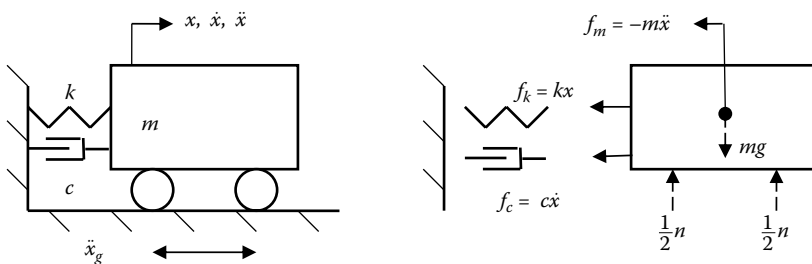


FIGURE P7.4

- a. Find the RMS of the absolute acceleration of the mass.
 - b. The response of mass m is a Gaussian narrow band. Assume all the peaks and valleys are independent. Find the median maximum acceleration of the mass.
 - c. Calculate the RMS of the relative displacement.
 - d. Compute the median maximum ground force.
7. A camera is mounted on a base with white noise motion, the base acceleration is $0.04 \text{ g}^2/\text{Hz}$. The camera has natural frequency = 27.4 Hz and damping ratio = 12% . In a sufficiently long period with more than 1000 peaks, the expected maximum peak will be approximately four times the RMS. Estimate the peaks of the absolute acceleration and relative displacement of the camera.
8. Find the mean, variance, and autocorrelation functions of stationary process MA (1), MA (2), and MA (3), which are all with white noise zero mean input $w(n)$. Suppose $k > q$.
9. Calculate the mean, variance, and autocorrelation functions of stationary process AR(1), given by

$$x(n) = -a_1x(n-1) + b_0w(n) \quad n \geq 0$$

10. Prove that for AR(2) described by

$$x(n) + a_1x(n-1) + a_2x(n-2) = 0$$

is stable when $|\lambda_i| < 1$, where

$$\lambda_{1,2} = \frac{-a_1 \pm j\sqrt{a_1^2 - 4a_2}}{2}$$

8 Random Vibration of MDOF Linear Systems

The random responses of multi-degree-of-freedom (MDOF) systems are discussed in this chapter. General references can be found in Clough and Penzien (1993), Wirsching et al. (2006), Cheng (2001), Cheng and Truman (2001), Chopra (2003), Inman (2008), and Liang et al. (2012).

8.1 MODELING

In real-world applications, modeling is often the logical starting point in gaining an understanding of a system. Therefore, in the study of MDOF systems, similar to the previous chapter about SDOF systems, a model is first discussed.

8.1.1 BACKGROUND

Many vibration systems are too complex to be modeled as SDOF systems. For instance, a moving car will encounter vertical bumps, as well as swaying in the horizontal direction. One cannot use the measure of vertical bumping to determine the degree of rotational rocking because they are responses to independent events. In this case, the vertical motion and horizontal rotation is described by unique degrees of freedom represented by two independent displacements of the front and rear wheels.

An MDOF system, with n independent displacements, can have n natural frequencies and n linearly independent vibration shape functions.

8.1.1.1 Basic Assumptions

We consider first the following assumptions:

1. Linear system with time-invariant physical parameters
Assume the MDOF system is linear. The previous discussed requirements of a linear system also apply to MDOF vibrations.
2. Forcing function
For purposes of simplification, the excitation forcing function may be assumed to be stationary and, in many cases, ergodic, as well as Gaussian.
 - a. Stationary
Occurs when the first and second moment are independent of time t .
 - b. Ergodic
The system responds impartially to the initial states due to the significant lapse in time from that initial state. In other words, the system “forgets” its initial conditions.

c. Gaussian

This results in linear combinations that yield normal distributions.

8.1.1.2 Fundamental Approaches

One of the following approaches may be used in dealing with an MDOF system:

1. Direct method direct integration
2. Modal analysis

An MDOF system can be decoupled into n second-order SDOF vibrations, or it can be decoupled into $2n$ first-order SDOF subsystems. The decoupled vibrator is often referred to as vibration modes. Each mode will have a single natural frequency and a single damping ratio, equivalent to an SDOF system. Conversely, in each mode, there exist n locations of displacement. The ratio of this displacement is fixed, which can be described by an $n \times 1$ vector, referred to as a mode shape.

Normal mode. If the mode shape vector can be written as real-valued, then it is a normal mode. If an MDOF system has real mode only, then the following conditions are mutually necessary and sufficient:

- The damping is proportional
- The N-DOF system decouples to n normal modes in model space
- Each mode is a second-order SDOF vibration system

Complex mode. If the mode shape vector cannot be written as real-valued, then it is a complex mode. In this case, the following conditions are mutually necessary and sufficient:

- The damping is nonproportional
- The N-DOF system decouples to $2n$ modes in state space
- At least two modes are complex valued and are of the first-order system

In each case, there is model superposition: the total solution is a linear combination of the modal solutions.

8.1.2 EQUATION OF MOTION

The modeling of MDOF a system is examined in the following.

8.1.2.1 Physical Model

Figure 8.1 shows a typical model of a 2-DOF system.

The equilibrium of force of the first mass is

$$\sum F_1 = m_1 \ddot{x}_1 + c_1 \dot{x}_1 + c_2 (\dot{x}_1 - \dot{x}_2) + k_1 x_1 + k_2 (x_1 - x_2) - f_1 = 0 \quad (8.1)$$

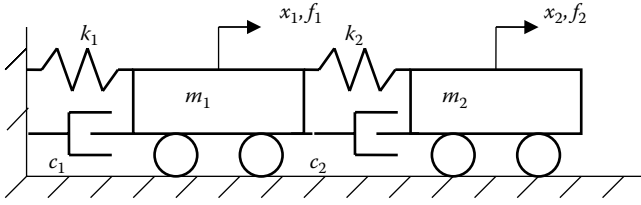


FIGURE 8.1 DOF system.

The balance of force of the second mass is

$$\sum F_2 = m_2 \ddot{x}_2 + c_2 (\dot{x}_2 - \dot{x}_1) + k_2 (x_2 - x_1) - f_2 = 0 \quad (8.2)$$

Rearranging both equations yields

$$m_1 \ddot{x}_1 + (c_1 + c_2) \dot{x}_1 + (-c_2) \dot{x}_2 + (k_1 + k_2) x_1 + (-k_2) x_2 = f_1 \quad (8.3)$$

$$m_2 \ddot{x}_2 + (-c_2) \dot{x}_1 + (c_2) \dot{x}_2 + (-k_2) x_1 + (k_2) x_2 = f_2 \quad (8.4)$$

In matrix form, this is generally written as

$$[m]\{\ddot{x}\} + [c]\{\dot{x}\} + [k]\{x\} = \{f\} \quad (8.5a)$$

or

$$\mathbf{M}\ddot{\mathbf{x}}(t) + \mathbf{C}\dot{\mathbf{x}}(t) + \mathbf{K}\mathbf{x}(t) = \mathbf{f}(t) \quad (8.5b)$$

Here, for the example shown in Figure 8.1, $\mathbf{M}$ is the mass matrix.

$$\mathbf{M} = \begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix} \quad (8.6a)$$

In general, $\mathbf{M}$ is defined as

$$\mathbf{M} = \begin{bmatrix} m_{11} & m_{12} & \dots & m_{1n} \\ m_{21} & m_{22} & \dots & m_{2n} \\ & & \dots & \\ m_{n1} & m_{n2} & \dots & m_{nn} \end{bmatrix} \quad (8.6b)$$

$\mathbf{C}$ is the damping matrix, for the example shown in Figure 8.1

$$\mathbf{C} = \begin{bmatrix} c_1 + c_2 & -c_2 \\ -c_2 & c_2 \end{bmatrix} \quad (8.7a)$$

Likewise, $\mathbf{C}$ is defined as

$$\mathbf{C} = \begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1n} \\ c_{21} & c_{22} & \cdots & c_{2n} \\ & & \cdots & \\ c_{n1} & c_{n2} & \cdots & c_{nn} \end{bmatrix} \quad (8.7b)$$

$\mathbf{K}$ is the stiffness matrix, for the example shown in Figure 8.1

$$\mathbf{K} = \begin{bmatrix} k_1 + k_2 & -k_2 \\ -k_2 & k_2 \end{bmatrix} \quad (8.8a)$$

In the same way, $\mathbf{K}$ is defined as

$$\mathbf{K} = \begin{bmatrix} k_{11} & k_{12} & \cdots & k_{1n} \\ k_{21} & k_{22} & \cdots & k_{2n} \\ & & \cdots & \\ k_{n1} & k_{n2} & \cdots & k_{nn} \end{bmatrix} \quad (8.8b)$$

As a general rule, $\mathbf{x}$ corresponds to the displacement vector.

$$\mathbf{x} = \begin{Bmatrix} x_1 \\ x_2 \\ \cdots \\ x_n \end{Bmatrix} \quad (8.9)$$

Lastly, $\mathbf{f}$ signifies the force vector.

$$\mathbf{f} = \begin{Bmatrix} f_1 \\ f_2 \\ \cdots \\ f_n \end{Bmatrix} \quad (8.10)$$

8.1.2.2 Stiffness Matrix

Beginning with the stiffness matrix, the physical parameters are considered.

8.1.2.2.1 Determination of $\mathbf{K}$ Matrix (Force Method, Displacement Method)

Consider static force, defined by

$$\mathbf{f} = \mathbf{K}\mathbf{x} \quad (8.11)$$

The j th row can be written as

$$f_j = k_{j1}x_1 + k_{j2}x_2 + \dots + k_{ji}x_i + \dots + k_{jn}x_n \quad (8.12)$$

From the above equation, this can be perceived as

$$k_{ji} = f_j \Big|_{x_i=1, x_p=0(p \neq i)} \quad (8.13)$$

Equation 8.13 can then be used to construct a stiffness matrix.

8.1.2.2.2 Property of Stiffness Matrix

The following are the major properties of a stiffness matrix.

8.1.2.2.2.1 Symmetry The stiffness matrix is symmetric

$$k_{ji} = k_{ij} \quad (8.14a)$$

$$\mathbf{K}^T = \mathbf{K} \quad (8.14b)$$

8.1.2.2.2.2 Full Rank For full rank to exist, the following conditions must hold true:

1. $\text{rank}(\mathbf{K}) = n$ (8.15)
and
2. $\mathbf{K}$ is nonsingular
also
3. $\mathbf{K}^{-1}$ exists

8.1.2.2.2.3 Positive Definite The stiffness matrix is positive definite, that is,

$$\mathbf{K} > 0 \quad (8.16)$$

in which the “>” symbol for a matrix is used to denote the matrix as being positive definite, meaning all eigenvalues are greater than zero. This is denoted by

$$\text{eig}(\mathbf{K}) > 0 \quad (8.17)$$

8.1.2.2.2.4 Flexibility Matrix For a matrix to be a flexibility matrix, all of the following conditions must hold true:

$$\mathbf{K}^{-1} = \mathbf{S} \quad (8.18)$$

The flexibility matrix is symmetric

$$\mathbf{S}^T = \mathbf{S} \quad (8.19)$$

The flexibility matrix is full rank

$$\text{rank}(\mathbf{S}) = n \quad (8.20)$$

8.1.2.3 Mass and Damping Matrices

8.1.2.3.1 Mass Matrix

$\mathbf{M}$ is the mass coefficient matrix with the following characteristics:

1. $\mathbf{M}$ is full rank, where

$$\text{rank}(\mathbf{M}) = n \quad (8.21)$$

2. $\mathbf{M}$ is positive definite, where

$$\mathbf{M} > 0 \quad (8.22)$$

3. $\mathbf{M}$ is symmetric, where

$$\mathbf{M}^T = \mathbf{M} \quad (8.23)$$

8.1.2.3.1.1 Lumped Mass

$$\mathbf{M} = \text{diag}(m_i) \quad (8.24)$$

8.1.2.3.1.2 Consistent Mass Exists when $\mathbf{M}$ and $\mathbf{K}$ share the same “shape function” $\mathbf{D}$, namely,

$$\mathbf{K} = \mathbf{D}\mathbf{K}_\Delta\mathbf{D}^T \quad (8.25)$$

and

$$\mathbf{M} = \mathbf{D}\mathbf{M}_\Delta\mathbf{D}^T \quad (8.26)$$

In Equations 8.25 and 8.26, $\mathbf{D}$ is an $n \times n$ square matrix describing the displacement “shape functions.” $\mathbf{K}_\Delta$ and $\mathbf{M}_\Delta$ are diagonal matrices.

8.1.2.3.2 Damping Matrix

$\mathbf{C}$ is the damping coefficient matrix, which exhibits the following characteristics:

1. $\mathbf{C}$ is not necessarily full rank

$$\text{rank}(\mathbf{C}) \leq n \quad (8.27)$$

2. $\mathbf{C}$ is positive semidefinite

$$\mathbf{C} \geq \mathbf{0} \quad (8.28)$$

Here, the symbol “ $\geq$ ” for a matrix is used to denote the matrix as being positive semidefinite, whose eigenvalues are all greater than or equal to zero. This is denoted by

$$\text{eig}(\mathbf{C}) \geq \mathbf{0} \quad (8.29)$$

where $\text{eig}(\cdot)$ stands for operation of calculating the eigenvalues of matrix $(\cdot)$, which will be discussed in Sections 8.4.1 and 8.4.5 for proportionally and nonproportionally damped systems, respectively (also see Wilkinson 1965). The algebraic eigenvalue problem.

$\mathbf{C}$ is symmetric.

$$\mathbf{C}^T = \mathbf{C} \quad (8.30)$$

8.1.2.3.2.1 Proportional Damping (Caughey Criterion) If the following matrices commute, then the system is proportionally damped. This is referred to as the Caughey criterion (Thomas K. Caughey, 1927–2004)

$$\mathbf{C}\mathbf{M}^{-1}\mathbf{K} = \mathbf{K}\mathbf{M}^{-1}\mathbf{C} \quad (8.31)$$

8.1.2.3.2.2 Rayleigh Damping The following proportional combination of $\mathbf{M}$ and $\mathbf{K}$ is one of the commonly used forms of proportional damping, where α and β are scalar.

$$\mathbf{C} = \alpha\mathbf{M} + \beta\mathbf{K} \quad (8.32)$$

8.1.2.3.2.3 Nonproportional Damping If the Caughey criterion does not hold, namely,

$$\mathbf{C}\mathbf{M}^{-1}\mathbf{K} \neq \mathbf{K}\mathbf{M}^{-1}\mathbf{C} \quad (8.33)$$

then the system is nonproportionally (nonclassically) damped.

Example 8.1

A system has mass, damping, and stiffness matrices given by

$$\mathbf{M} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}, \mathbf{C} = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}, \text{ and } \mathbf{K} = \begin{bmatrix} 30 & -10 \\ -10 & 50 \end{bmatrix}$$

Find if the system is proportionally damped.

$\mathbf{CM}^{-1}\mathbf{K} = \begin{bmatrix} 65 & -45 \\ -35 & 35 \end{bmatrix}$, $\mathbf{KM}^{-1}\mathbf{C} = \begin{bmatrix} 65 & -35 \\ -45 & 35 \end{bmatrix}$ so that $\mathbf{CM}^{-1}\mathbf{K} \neq \mathbf{KM}^{-1}\mathbf{C}$ the system is nonproportionally damped.

8.1.3 IMPULSE RESPONSE AND TRANSFER FUNCTIONS

Consider impulse forcing functions being applied at a certain location of an MDOF system.

8.1.3.1 Scalar Impulse Response Function and Transfer Function

Figure 8.2a shows an input at location j and the response at location i . Denote $h_{ij}(t)$ as a unit impulse response function for coordinate i due to a unit force at j . Furthermore, denote $H_{ij}(\omega)$ as a transfer function, with the ratio of output, $X_i(\omega)$, and input $F_j(\omega)$. This is illustrated in Figure 8.2b. From Figure 8.2b, the transfer function can be defined as

$$H_{ij}(\omega) = X_i(\omega)/F_j(\omega) \quad (8.34)$$

Note that $X_i(\omega)$ is caused by $F_j(\omega)$ only.

$H_{ij}(\omega)$ and $h_{ij}(t)$ are Fourier pair denoted by

$$H_{ij}(\omega) \Leftrightarrow h_{ij}(t) \quad (8.35)$$

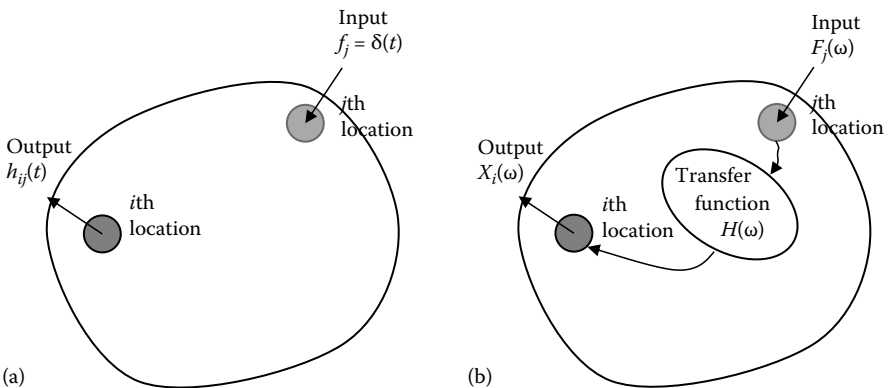


FIGURE 8.2 Relationship between input and output. (a) The time domain. (b) The frequency domain.

8.1.3.2 Impulse Response Matrix and Transfer Function Matrix

The input–output unit impulse responses can be collected and arranged in $\mathbf{h}$ matrix as

$$\mathbf{h}(t) = \begin{bmatrix} h_{11}(t) & h_{12}(t) & \dots & h_{1n}(t) \\ h_{21}(t) & h_{22}(t) & \dots & h_{2n}(t) \\ \dots & \dots & \dots & \dots \\ h_{n1}(t) & h_{n2}(t) & \dots & h_{nn}(t) \end{bmatrix} \quad (8.36)$$

It is seen that the $\mathbf{h}$ matrix is symmetric

$$\mathbf{h}(t) = \mathbf{h}(t)^T \quad (8.37)$$

The Fourier transform matrix is denoted by $\mathbf{H}$ as

$$\mathbf{H}(\omega) = \begin{bmatrix} H_{11}(\omega) & H_{12}(\omega) & \dots & H_{1n}(\omega) \\ H_{21}(\omega) & H_{22}(\omega) & \dots & H_{2n}(\omega) \\ \dots & \dots & \dots & \dots \\ H_{n1}(\omega) & H_{n2}(\omega) & \dots & H_{nn}(\omega) \end{bmatrix} \quad (8.38)$$

which is also symmetric.

$$\mathbf{H}(\omega) = \mathbf{H}(\omega)^T \quad (8.39)$$

8.1.3.3 Construction of Transfer Functions

Let the forcing function vector be of a harmonic excitation written as

$$\mathbf{f}(t) = \mathbf{f}_0 e^{j\omega t} \quad (8.40)$$

Furthermore, let us denote the response vector as

$$\mathbf{x}(t) = \mathbf{x}_0 e^{j\omega t} \quad (8.41)$$

Substitution of Equations 8.40 and 8.41 into Equation 8.5b results in

$$\mathbf{V}(\omega)\mathbf{x}_0 = \mathbf{f}_0 \quad (8.42)$$

where

$$\mathbf{V}(\omega) = -\omega^2 \mathbf{M} + j\omega \mathbf{C} + \mathbf{K} \quad (8.43)$$

$\mathbf{V}(\omega)$ is referred to as the *impedance matrix*. The impedance matrix is of full rank and symmetric. Its inverse matrix is denoted as

$$\mathbf{V}^{-1}(\omega) = \mathbf{H}(\omega) \quad (8.44)$$

Here, $\mathbf{H}(\omega)$ is the transfer function matrix (frequency response function matrix), where

$$\mathbf{x}_0 = \mathbf{H}(\omega) \mathbf{f}_0 \quad (8.45)$$

Example 8.2

A system has mass, damping, and stiffness matrices given by $\mathbf{M}$, $\mathbf{C}$, and

$$\mathbf{K} = \begin{bmatrix} 30 & -10 \\ -10 & 50 \end{bmatrix}$$

If one can measure the amplitude of the displacement as $\mathbf{x}_0 = [1 \ 0.5]^T$, find the vector of forcing function.

$$\mathbf{f}_0 = \mathbf{H}^{-1}(0) \mathbf{x}_0 = \mathbf{V}(0) \mathbf{x}_0 = \mathbf{K} \mathbf{x}_0 = [25 \ 15]^T$$

8.1.3.4 Principal Axes of Structures

In many cases, the responses of a structure can be decoupled into two perpendicular directions physically, say, the east–west and the south–north directions. In this case, the system is said to have principal axes, that is, the x -axis and the y -axis. The input along the x -axis will not have any response along the y -axis, and vice versa. The decoupling of structural responses into principal axes can reduce the computation burden so that less degree-of-freedom should be considered.

Note that when a structure is nonproportionally damped, it will not have principal axes. However, even if a structure is proportionally damped, it may also have no principal axes (Liang and Lee 2002, 2003). For structures without principal axes, one cannot decouple the responses in X or Y directions.

8.2 DIRECT MODEL FOR DETERMINING RESPONSES

The direct method is used when the number of degrees-of-freedom is not significantly large. In the following, we will find the statistical properties by ensemble averaging, in which the assumption of temporal averages can be used. That is, the input forcing functions are ergodic, which will be required in specific cases.

8.2.1 EXPRESSION OF RESPONSE

For simplicity, consider that the output is measured at a single location only. Denote the response at that location due to the i th input $F_i(t)$ as $X_i(t)$, as seen in Figure 8.3.

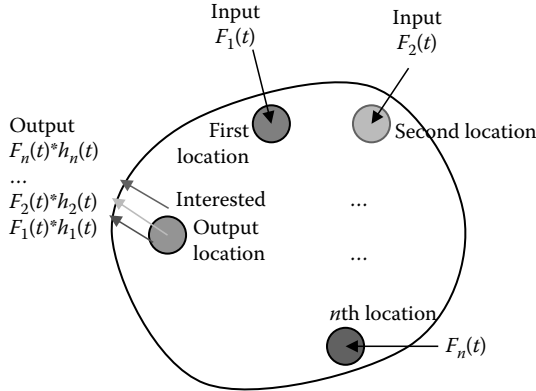


FIGURE 8.3 Multiple-input, single-output.

$$X_i(t) = F_i(t) * h_i(t) \quad (8.46)$$

In Equation 8.46, $h_i(t)$ is the unit impulse response at the location along the x -axis due to the specific force $F_i(t)$. Given that the system is linear, the total response is the total sum of $X_i(t)$. That is,

$$X(t) = \sum_{i=1}^n X_i(t) \quad (8.47)$$

Furthermore, substitution of Equation 8.46 into Equation 8.47 will yield

$$X(t) = \sum_{i=1}^n F_i(t) * h_i(t) = \sum_{i=1}^n \left[\int_0^t F_i(t-\tau) h_i(\tau) d\tau \right] = \sum_{i=1}^n \left[\int_{-\infty}^{\infty} F_i(t-\tau) h_i(\tau) d\tau \right] \quad (8.48)$$

In Figure 8.3, the total solution is the sum of all n terms of $\int_{-\infty}^{\infty} F_i(t-\tau) h_i(\tau) d\tau$. However, these terms are not calculated individually. In the following, how to compute the corresponding numerical characteristics will be described.

8.2.2 MEAN VALUES

The mean value of the responses is first considered for multiple input and single output, as shown in Figure 8.3. The case of multiple input–multiple output will be further described later.

8.2.2.1 Single Coordinate

If only a single response at a certain location is considered and the corresponding integration can be carried out, its mean can be calculated as

$$\mu_X = E[X(t)] = E\left\{\sum_{i=1}^n \left[\int_{-\infty}^{\infty} F_i(t-\tau) h_i(\tau) d\tau \right]\right\} \quad (8.49)$$

Here, $X(t)$ is the response and $F_i(t)$ is the stationary excitation at the i th location, with a mean value of

$$E[F_i(t)] = \mu_{Fi} \quad (8.50)$$

Thus,

$$\mu_X = \sum_{i=1}^n \left\{ \int_{-\infty}^{\infty} E[F_i(t-\tau)] h_i(\tau) d\tau \right\} = \sum_{i=1}^n \left\{ \mu_{Fi} \int_{-\infty}^{\infty} h_i(\tau) d\tau \right\} \quad (8.51)$$

Finally,

$$\mu_X = \sum_{i=1}^n \{\mu_{Fi} H_i(0)\} \quad (8.52)$$

8.2.2.2 Multiple Coordinates

Now, the response of all n coordinates is considered. In this case, we have multiple inputs and multiple outputs.

8.2.2.2.1 Vector of Mean

In the case of multiple coordinates, the mean values are collected into a vector. The mean of the first location will be the first element of the vector, determined by

$$\mu_{X1} = \sum_{i=1}^n \{\mu_{Fi} H_{1i}(0)\} \quad (8.53)$$

The second location is determined by

$$\mu_{X2} = \sum_{i=1}^n \{\mu_{Fi} H_{2i}(0)\} \quad (8.54)$$

Similarly, this is repeated to the n th location, which can be calculated by

$$\mu_{Xn} = \sum_{i=1}^n \{\mu_{Fi} H_{ni}(0)\} \quad (8.55)$$

In Equations 8.53 through 8.55, the term $H_{ji}(0)$ is the transfer function of i th input and j th output when $\omega = 0$;

The mean values, written in matrix form, are represented by

$$\boldsymbol{\mu}_X = \mathbf{H}(0)\boldsymbol{\mu}_F \quad (8.56)$$

where matrix $\mathbf{H}(0)$ is defined in Equation 8.38 when $\omega = 0$

$$\boldsymbol{\mu}_X = \begin{Bmatrix} \mu_{X1} \\ \mu_{X2} \\ \dots \\ \mu_{Xn} \end{Bmatrix} \quad (8.57)$$

and

$$\boldsymbol{\mu}_F = \begin{Bmatrix} \mu_{F1} \\ \mu_{F2} \\ \dots \\ \mu_{Fn} \end{Bmatrix} \quad (8.58)$$

Example 8.3

A system has mass, damping, and stiffness matrices given by $\mathbf{M}$, $\mathbf{C}$, and

$$\mathbf{K} = \begin{bmatrix} 30 & -10 \\ -10 & 50 \end{bmatrix}$$

The mean of input is $\boldsymbol{\mu}_F = [0 \quad -5]^T$, find the vector of the output mean.

$$\boldsymbol{\mu}_X = \mathbf{H}(0)\boldsymbol{\mu}_F = \mathbf{K}^{-1}\boldsymbol{\mu}_F = [-0.036 \quad -0.107]^T$$

It is seen that, although $F_1(t)$ is zero mean, the mean of output does not contain zeros.

8.2.2.2.2 Zero Mean

For a zero mean process of forcing function

$$\boldsymbol{\mu}_F = \mathbf{0} \quad (8.59)$$

where $\mathbf{0}$ is a $n \times 1$ null vector. Correspondingly, it is seen that

$$\boldsymbol{\mu}_X = \mathbf{0} \quad (8.60)$$

In general, a zero mean response can always be achieved. That is, if the forcing functions are not equivalent to zero mean, then a zero mean response can be achieved using

$$\mathbf{f}(t) = \mathbf{f}_{\text{non}}(t) - \boldsymbol{\mu}_F \quad (8.61)$$

In the above equation, $\mathbf{f}(t)$ and $\mathbf{f}_{\text{non}}(t)$ are, respectively, zero mean and nonzero mean random process vectors of forcing functions.

The response of this is as follows:

$$\mathbf{x}(t) = \mathbf{x}_{\text{non}}(t) - \boldsymbol{\mu}_x \quad (8.62)$$

In this instance, $\mathbf{x}(t)$ and $\mathbf{x}_{\text{non}}(t)$ are, respectively, zero mean and nonzero mean random process vectors of responses. Note that, the corresponding forcing function vector is

$$\mathbf{f}(t) = \begin{Bmatrix} F_1(t) \\ F_2(t) \\ \dots \\ F_n(t) \end{Bmatrix} \quad (8.63)$$

and the response vector is

$$\mathbf{x}(t) = \begin{Bmatrix} X_1(t) \\ X_2(t) \\ \dots \\ X_n(t) \end{Bmatrix} \quad (8.64)$$

Both vectors of Equations 8.63 and 8.64 are random processes. Namely, at least one of the elements in $F_i(t)$ or $X_j(t)$ are considered to be random; whereas both $\mathbf{f}(t)$ and $\mathbf{x}(t)$ are random.

8.2.3 CORRELATION FUNCTIONS

Next, the correlation functions are considered. The autocorrelation function of the response measured at a single location due to multiple inputs is written as

$$\begin{aligned} R_X(\tau) &= E[X(t)X(t+\tau)] = E \left[\sum_{i=1}^m X_i(t) \sum_{j=1}^m X_j(t+\tau) \right] \\ &= E \left[\sum_{i=1}^m \sum_{j=1}^m \int_{-\infty}^{\infty} h_i(\xi) F_i(t-\xi) d\xi \int_{-\infty}^{\infty} h_j(\eta) F_j(t+\tau-\eta) d\eta \right] \quad (8.65) \\ &= \sum_{i=1}^m \sum_{j=1}^m \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h_i(\xi) h_j(\eta) E[F_i(t-\xi) F_j(t+\tau-\eta)] d\xi d\eta \end{aligned}$$

Here, $m \leq n$, indicating the forces applied at m locations, and n may be less than m .

Given that the forcing functions are stationary, the cross-correlation function of $F_i(t)$ and $F_j(t)$ can be denoted as

$$R_{F_i F_j}(\tau) = E[F_i(t)F_j(t + \tau)] \quad (8.66)$$

Substitution of Equation 8.66 into Equation 8.20 results in

$$R_X(\tau) = \sum_{i=1}^m \sum_{j=1}^m \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h_i(\xi) h_j(\eta) R_{F_i F_j}(\xi - \eta + \tau) d\xi d\eta \quad (8.67)$$

8.2.4 SPECTRAL DENSITY FUNCTION OF RESPONSE

8.2.4.1 Fourier Transforms of $\mathbf{f}(t)$ and $\mathbf{x}(t)$

Suppose the Fourier transforms of $\mathbf{f}(t)$ and $\mathbf{x}(t)$ exist. One may consider how and under what conditions these will occur.

Suppose,

$$\mathbf{F}(\omega) = \begin{Bmatrix} F_1(\omega) \\ F_2(\omega) \\ \dots \\ F_n(\omega) \end{Bmatrix} \quad (8.68)$$

with a response vector of

$$\mathbf{X}(\omega) = \begin{Bmatrix} X_1(\omega) \\ X_2(\omega) \\ \dots \\ X_n(\omega) \end{Bmatrix} \quad (8.69)$$

8.2.4.2 Power Spectral Density Function

The cross-power spectral density (PSD) function matrix of input can be written as

$$\begin{aligned} \mathbf{S}_F(\omega) &= \begin{bmatrix} S_{F_1}(\omega) & S_{F_1 F_2}(\omega) & \dots & S_{F_1 F_n}(\omega) \\ S_{F_2 F_1}(\omega) & S_{F_2}(\omega) & & S_{F_2 F_n}(\omega) \\ & & \dots & \\ S_{F_n F_1}(\omega) & S_{F_n F_2}(\omega) & & S_{F_n}(\omega) \end{bmatrix} \\ &= \mathcal{F}^{-1} \begin{bmatrix} R_{F_1}(\tau) & R_{F_1 F_2}(\tau) & \dots & R_{F_1 F_n}(\tau) \\ R_{F_2 F_1}(\tau) & R_{F_2}(\tau) & & R_{F_2 F_n}(\tau) \\ & & \dots & \\ R_{F_n F_1}(\tau) & R_{F_n F_2}(\tau) & & R_{F_n}(\tau) \end{bmatrix} \end{aligned} \quad (8.70)$$

where $S_{F_j F_k}(\omega)$ and $R_{F_j F_k}(\tau)$ are, respectively, the cross-PSD and correlation function of input forcing functions F_j and F_k . If $j = k$, we obtain auto-PSD and autocorrelation functions.

Unlike SDOF systems, expect off-the-diagonal entries in Equation 8.70; the off-diagonal entries contain cross-power PSD among the input locations. In this case,

$$S_{F_i F_j}(\omega) = \lim_{T \rightarrow \infty} \frac{1}{2\pi T} \sum_p [F_{ip}(\omega, T) * F_{jp}(\omega, T)] \quad (8.71)$$

where $F_{ip}(\omega, T)$ is the Fourier transform of the p th measurement of a forcing function applied at location i , with a measurement length of T .

$$S_{F_j F_k}(\omega) = cE[F_j(\omega) * F_k(\omega)] \quad (8.72)$$

Now, consider the cross-PSD function matrix of output, the j th entry can be written as

$$S_{X_j X_k}(\omega) = cE[X_j(\omega) * X_k(\omega)] \quad (8.73)$$

In Equations 8.72 and 8.73, c is a constant.

Consider the equation,

$$\mathbf{X}(\omega) = \mathbf{H}(\omega)\mathbf{F}(\omega) \quad (8.74)$$

Postmultiplying the Hermitian transposes on both sides of Equation 8.74 results in

$$\mathbf{X}(\omega)\mathbf{X}(\omega)^H = \mathbf{H}(\omega)\mathbf{F}(\omega)[\mathbf{H}(\omega)\mathbf{F}(\omega)]^H \quad (8.75)$$

Here, the Hermitian transposes of a complex valued matrix $\mathbf{A}$ is given by

$$\mathbf{A}^H = (\mathbf{A}^*)^T = (\mathbf{A}^T)^* \quad (8.76)$$

Next, take the expected value of both sides and multiply by a constant, represented by c .

$$cE[\mathbf{X}(\omega)\mathbf{X}(\omega)^H] = cE\{\mathbf{H}(\omega)\mathbf{F}(\omega)[\mathbf{H}(\omega)^*\mathbf{F}(\omega)^*]^T\} = \mathbf{H}(\omega)cE\{\mathbf{F}(\omega)\mathbf{F}(\omega)^*\}[\mathbf{H}(\omega)^*]^T \quad (8.77)$$

With the input of Equations 8.72 and 8.73, the result becomes

$$\mathbf{S}_X(\omega) = \mathbf{H}(\omega)\mathbf{S}_F(\omega)\mathbf{H}(\omega)^H \quad (8.78)$$

Here, $\mathbf{S}_X(\omega)$ is the cross-PSD matrix given by

$$\begin{aligned} \mathbf{S}_X(\omega) &= \begin{bmatrix} S_{X_1}(\omega) & S_{X_1X_2}(\omega) & \dots & S_{X_1X_n}(\omega) \\ S_{X_2X_1}(\omega) & S_{X_2}(\omega) & & S_{X_2X_n}(\omega) \\ & & \dots & \\ S_{X_nX_1}(\omega) & S_{X_nX_2}(\omega) & & S_{X_n}(\omega) \end{bmatrix} \\ &= \mathcal{F}^{-1} \begin{bmatrix} R_{X_1}(\tau) & R_{X_1X_2}(\tau) & \dots & R_{X_1X_n}(\tau) \\ R_{X_2X_1}(\tau) & R_{X_2}(\tau) & & R_{X_2X_n}(\tau) \\ & & \dots & \\ R_{X_nX_1}(\tau) & R_{X_nX_2}(\tau) & & R_{X_n}(\tau) \end{bmatrix} \end{aligned} \quad (8.79)$$

Both cross-PSD matrices $\mathbf{S}_F(\omega)$ and $\mathbf{S}_X(\omega)$ given by Equations 8.70 and 8.79 are useful. Because we will study not only the auto-PSD of S_{F_i} and S_{X_i} but also the relationships of singles between location j and k .

8.2.4.3 Mean Square Response

The mean square response of the i th location is

$$E[x_i^2(t)] = X_{ii}^2 = \int_{-\infty}^{\infty} S_{X_i}(\omega) d\omega \quad (8.80)$$

where X_{ii}^2 is constant.

8.2.4.4 Variance

When $x_i(t)$ is of zero mean, then

$$\sigma_{X_i}^2 = X_{ii}^2 \quad (8.81)$$

8.2.4.5 Covariance

$$\sigma_{X_iX_j}(0) = \int_{-\infty}^{\infty} S_{X_iX_j}(\omega) d\omega \quad (8.82)$$

8.2.5 SINGLE RESPONSE VARIABLE: SPECTRAL CASES

8.2.5.1 Single Input

As an example, if a single input of $f_k(t)$ exists, then

$$S_{F_iF_j}(\omega) = \begin{cases} S_{F_k}(\omega), & i = j = k \\ 0, & \text{elsewhere} \end{cases} \quad (8.83)$$

$$S_X(\omega) = H_k^*(\omega)H_k(\omega)S_{F_k}(\omega) \quad (8.84)$$

or

$$S_X(\omega) = |H_k(\omega)|^2 S_{F_k}(\omega) \quad (8.85)$$

8.2.5.2 Uncorrelated Input

If all inputs, $f_i(t)$, are uncorrelated, then

$$S_{F_i F_j}(\omega) = \begin{cases} S_{F_k}(\omega), & i = j = k = 1, 2, \dots, n \\ 0, & \text{elsewhere} \end{cases} \quad (8.86)$$

and

$$S_X(\omega) = \sum_{k=1}^n \left[|H_k(\omega)|^2 S_{F_k}(\omega) \right] \quad (8.87)$$

$$\sigma_X^2 = \sum_{k=1}^n \sigma_i^2 \quad (8.88)$$

In the above instance, σ_i^2 is the variance of $x(t)$ attributed to $f_i(t)$ solely.

8.3 NORMAL MODE METHOD

When the degrees-of-freedom for an MDOF system are considerably large, the above computation can be extensive. In this case, modal analysis can greatly reduce the computation burden while maintaining a reasonable accuracy. First, we consider the proportionally damped system and normal mode. Although in the real world, such systems rarely exist. When the damping ratio is comparatively small, in the region of less than 5%, the normal mode method can provide reliable approximations (see Cheng [2001] and Chopra [2003]).

8.3.1 PROPORTIONAL DAMPING

As mentioned previously, for the Caughey criterion to be satisfied, a system must be proportionally damped. In this section, the mathematical and physical meaning of the Caughey criterion will be discussed first.

8.3.1.1 Essence of Caughey Criterion

Recall the Caughey criterion:

$$\mathbf{C}\mathbf{M}^{-1}\mathbf{K} = \mathbf{K}\mathbf{M}^{-1}\mathbf{C} \quad (8.89)$$

Multiply by $\mathbf{M}^{-1}$ on both sides of the equation.

$$[\mathbf{M}^{-1}\mathbf{C}][\mathbf{M}^{-1}\mathbf{K}] = [\mathbf{M}^{-1}\mathbf{K}][\mathbf{M}^{-1}\mathbf{C}] \quad (8.90)$$

Equation 8.90 implies that the two matrices $[\mathbf{M}^{-1}\mathbf{C}]$ and $[\mathbf{M}^{-1}\mathbf{K}]$ commute, if and only if the matrices share the identical eigenvector matrix Φ . Matrices $[\mathbf{M}^{-1}\mathbf{C}]$ and $[\mathbf{M}^{-1}\mathbf{K}]$ are, respectively, referred to as generalized damping and stiffness matrices. The eigenvector matrix Φ is the mode shape of the $\mathbf{M}$ - $\mathbf{C}$ - $\mathbf{K}$ system, as discussed further in Section 8.3.5.4.

Physically, if the distribution of the individual dampers and springs are identical and the amount of individual damping and stiffness are proportional, then both the generalized damping and stiffness matrices share identical eigenvectors. In qualitative terms, this means that both damping and stiffness are “regularly” distributed.

8.3.1.2 Monic System

To have generalized damping and stiffness matrices, the monic system must be generated first.

8.3.1.2.1 Concept of Monic Mass

By multiplying $\mathbf{M}^{-1}$ on both sides of Equation 8.5b, the homogeneous form can be considered.

$$\mathbf{I}\ddot{\mathbf{x}}(t) + \mathbf{M}^{-1}\mathbf{C}\dot{\mathbf{x}}(t) + \mathbf{M}^{-1}\mathbf{K}\mathbf{x}(t) = \mathbf{0} \quad (8.91)$$

In which, the mass matrix becomes the identity matrix. It is referred to as a monic system.

Note that, in Equation 8.91, the monic MDOF vibration system has newly formed generalized damping matrix, $\mathbf{M}^{-1}\mathbf{C}$, and stiffness matrix, $\mathbf{M}^{-1}\mathbf{K}$.

8.3.1.2.2 Solution of Monic Systems

Similar to previous examples, the semidefinite method is used.

Assume that

$$\mathbf{x}(t) = \Phi e^{\lambda t} \quad (8.92)$$

The characteristic equation is written as

$$\lambda^2\Phi + \lambda\mathbf{M}^{-1}\mathbf{C}\Phi + \mathbf{M}^{-1}\mathbf{K}\Phi = \mathbf{0} \quad (8.93)$$

or

$$[\mathbf{I}\lambda^2 + \lambda\mathbf{M}^{-1}\mathbf{C} + \mathbf{M}^{-1}\mathbf{K}]\Phi = \mathbf{0} \quad (8.94)$$

Because $\Phi \neq \mathbf{0}$, otherwise the solution is trivial, then

$$\det[\mathbf{I}\lambda^2 + \lambda\mathbf{M}^{-1}\mathbf{C} + \mathbf{M}^{-1}\mathbf{K}] = 0 \quad (8.95)$$

Matrix $[\mathbf{I} \lambda^2 + \lambda \mathbf{M}^{-1}\mathbf{C} + \mathbf{M}^{-1}\mathbf{K}]$ is referred to as a λ matrix, whose determinant is a $2n$ polynomial, with solutions of the $2n$ complex conjugate pair λ_i and λ_i^* . In general, the $2n$ corresponding vectors, $\boldsymbol{\Phi}_i$ and $\boldsymbol{\Phi}_i^*$, will also exist. However, if the Caughey criterion is satisfied, then the vector $\boldsymbol{\Phi}_i$ will be real valued. In this case, it is referred to as normal mode shape.

Consider the SDOF system:

$$\lambda = -\zeta\omega_n + j\sqrt{1-\zeta^2}\omega_n \quad (8.96)$$

Here, we further have

$$\lambda_i = -\zeta_i\omega_{ni} + j\sqrt{1-\zeta_i^2}\omega_{ni} \quad i = 1, \dots, n \quad (8.97a)$$

and

$$\lambda_i^* = -\zeta_i\omega_{ni} - j\sqrt{1-\zeta_i^2}\omega_{ni} \quad i = 1, \dots, n \quad (8.97b)$$

The variables, λ_i , ζ_i , and ω_{ni} are, respectively, referred to as eigenvalue, damping ratio, and natural frequency of the i th normal mode. The triple $\langle \omega_{ni}, \zeta_i, \boldsymbol{\Phi}_i \rangle$ is called the i th normal modal parameter. The phrase “normal” means that the eigenvalues are calculated from the proportionally damped system.

8.3.2 EIGEN-PROBLEMS

In Equations 8.97a and 8.97b, the damping ratios and the natural frequencies are parts of eigenvalues. It is of importance that these eigen-problems be further explored.

8.3.2.1 Undamped System

First, consider an undamped system, where

$$\mathbf{C} = \mathbf{0} \quad (8.98)$$

and

$$\zeta_i = 0 \quad (8.99)$$

Thus,

$$\lambda_i = j\omega_{ni} \quad (8.100)$$

Furthermore,

$$-\omega_{ni}^2 \boldsymbol{\Phi}_i + \mathbf{M}^{-1}\mathbf{K}\boldsymbol{\Phi}_i = \mathbf{0}$$

or

$$\omega_{ni}^2 \boldsymbol{\phi}_i = \mathbf{M}^{-1} \mathbf{K} \boldsymbol{\phi}_i \quad (8.101)$$

Equation 8.101 is referred to as the eigen-problem of matrix $\mathbf{M}^{-1} \mathbf{K}$ with scalar eigenvalue ω_{ni}^2 and eigenvector $\boldsymbol{\phi}_i$. A square matrix multiplied by its eigenvector is equal to a scalar, the corresponding eigenvalue multiplied by the same eigenvector.

8.3.2.2 Underdamped Systems

Similarly, $\mathbf{M}^{-1} \mathbf{C}$ is also a square matrix and will contain eigenvectors and eigenvalues. Because $\mathbf{M}^{-1} \mathbf{C}$ and $\mathbf{M}^{-1} \mathbf{K}$ share the same eigenvector, the corresponding eigenvalue can be denoted as $2\zeta_i \omega_{ni}$, with

$$2\zeta_i \omega_{ni} \boldsymbol{\phi}_i = \mathbf{M}^{-1} \mathbf{C} \boldsymbol{\phi}_i \quad i = 1, \dots, n \quad (8.102)$$

Example 8.4

A system has mass, damping, and stiffness matrices given by

$$\mathbf{M} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}, \mathbf{C} = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}, \text{ and } \mathbf{K} = \begin{bmatrix} 30 & -10 \\ -10 & 30 \end{bmatrix}$$

Check whether this system is proportionally damped and find the corresponding eigenvalues and eigenvectors.

It is seen that $\mathbf{C} \mathbf{M}^{-1} \mathbf{K} = \mathbf{K} \mathbf{M}^{-1} \mathbf{C}$ so that the system is proportionally damped. From Equation 8.101, it is seen that, $\omega_{n1} = 3.4917$ and $\omega_{n2} = 5.7278$. The corresponding eigenvectors are $\boldsymbol{\phi}_1 = [0.4896 \ 0.8719]^T$ and $\boldsymbol{\phi}_2 = [0.9628 \ -0.2703]^T$.

From Equation 8.102, we see that $\mathbf{M}^{-1} \mathbf{C} \boldsymbol{\phi}_1 = [0.1073 \ 0.1911]^T$. Dividing the first element by 0.4896, namely, the first element in $\boldsymbol{\phi}_1$ results in 0.2192 (the same result can be found by dividing the first element by 0.8719). Furthermore, damping ratio $\zeta_1 = 0.2192/(2\omega_{n1}) = 0.0314$. Similarly, the damping ratio $\zeta_2 = 0.1991$.

Therefore, the eigenvalues are

$$\lambda_1 = -\zeta_1 \omega_{n1} \pm j \sqrt{1 - \zeta_1^2} \omega_{n1} = -0.1096 \pm 3.4900j$$

and

$$\lambda_2 = -1.1404 \pm 5.6131j$$

8.3.3 ORTHOGONAL CONDITIONS

The eigenvector can be further used to decouple the MDOF system. Before completing this calculation, first, consider why this is possible. The answer to this is based on the orthogonal conditions.

8.3.3.1 Weighted Orthogonality

Equation 8.101 can be rewritten as

$$\omega_{ni}^2 \mathbf{M} \boldsymbol{\phi}_i = \mathbf{K} \boldsymbol{\phi}_i \quad (8.103)$$

which is referred to as the i th generalized eigen-equation.

Multiplying $\boldsymbol{\phi}_j^T$ on both sides of Equation 8.103 results in

$$\omega_{ni}^2 \boldsymbol{\phi}_j^T \mathbf{M} \boldsymbol{\phi}_i = \boldsymbol{\phi}_j^T \mathbf{K} \boldsymbol{\phi}_i \quad (8.104)$$

Further multiplying $\boldsymbol{\phi}_i^T$ on both sides of the j th generalized eigen-equation will yield

$$\omega_{nj}^2 \boldsymbol{\phi}_i^T \mathbf{M} \boldsymbol{\phi}_j = \boldsymbol{\phi}_i^T \mathbf{K} \boldsymbol{\phi}_j \quad (8.105)$$

Because $\mathbf{M}$ and $\mathbf{K}$ are symmetric, then take transpose on both sides of Equation 8.104,

$$\omega_{ni}^2 (\boldsymbol{\phi}_j^T \mathbf{M} \boldsymbol{\phi}_i)^T = (\boldsymbol{\phi}_j^T \mathbf{K} \boldsymbol{\phi}_i)^T \quad (8.106)$$

Here note that

$$(\boldsymbol{\phi}_j^T \mathbf{M} \boldsymbol{\phi}_i)^T = \boldsymbol{\phi}_i^T \mathbf{M} \boldsymbol{\phi}_j, \quad (\boldsymbol{\phi}_j^T \mathbf{K} \boldsymbol{\phi}_i)^T = \boldsymbol{\phi}_i^T \mathbf{K} \boldsymbol{\phi}_j \quad (8.107)$$

Substitution of Equations 8.107 into Equation 8.106 and subtracting the subsequent result from Equation 8.105, results in

$$(\omega_{nj}^2 - \omega_{ni}^2) \boldsymbol{\phi}_i^T \mathbf{M} \boldsymbol{\phi}_j = 0 \quad (8.108)$$

Because in general,

$$\omega_{ni}^2 \neq \omega_{nj}^2 \quad (8.109)$$

Thus,

$$\boldsymbol{\phi}_i^T \mathbf{M} \boldsymbol{\phi}_j = 0, \quad \text{when } i \neq j \quad (8.110)$$

and

$$\boldsymbol{\phi}_i^T \mathbf{M} \boldsymbol{\phi}_i = m_i, \quad \text{when } i = j \quad (8.111)$$

where m_i is called the i th *modal mass*.

Combining Equations 8.110 and 8.111, results in the orthogonal condition:

$$\boldsymbol{\phi}_i^T \mathbf{M} \boldsymbol{\phi}_j = \begin{cases} m_i, & i = j \\ 0, & i \neq j \end{cases} \quad (8.112)$$

Similarly, the following can also be proved

$$\boldsymbol{\phi}_i^T \mathbf{K} \boldsymbol{\phi}_j = \begin{cases} k_i, & i = j \\ 0, & i \neq j \end{cases} \quad (8.113)$$

and likewise

$$\boldsymbol{\phi}_i^T \mathbf{C} \boldsymbol{\phi}_j = \begin{cases} c_i, & i = j \\ 0, & i \neq j \end{cases} \quad (8.114)$$

Here, k_i and c_i are called the *ith modal stiffness* and *model damping* coefficient, respectively; similar to modal mass, we use italic letters to denote these modal parameters. Equations 8.112, 8.113, and 8.114 are referred to as weighted orthogonal conditions.

8.3.3.2 Modal Analysis

8.3.3.2.1 Characteristic Equation

Using the orthogonal conditions, the eigenvector or mode shape can be used to obtain the SDOF vibration systems mode by mode. In doing so, first consider the homogeneous equation:

$$\mathbf{M}\ddot{\mathbf{x}}(t) + \mathbf{C}\dot{\mathbf{x}}(t) + \mathbf{K}\mathbf{x}(t) = \mathbf{0} \quad (8.115)$$

Next, assume the following:

$$\mathbf{x}(t) = \boldsymbol{\phi}_i e^{\lambda_i t} \quad (8.116)$$

Substituting Equation 8.116 into Equation 8.115 and premultiplying $\boldsymbol{\phi}_i^T$ on both sides yields

$$\lambda_i^2 \boldsymbol{\phi}_i^T \mathbf{M} \boldsymbol{\phi}_i e^{\lambda_i t} + \lambda_i \boldsymbol{\phi}_i^T \mathbf{C} \boldsymbol{\phi}_i e^{\lambda_i t} + \boldsymbol{\phi}_i^T \mathbf{K} \boldsymbol{\phi}_i e^{\lambda_i t} = 0, \quad i = 1, \dots, n \quad (8.117)$$

The characteristic equation for n SDOF systems has now been obtained.

$$m_i \lambda_i^2 + c_i \lambda_i + k_i = 0, \quad i = 1, \dots, n \quad (8.118)$$

In comparing the characteristic equation for n SDOF systems to that of an SDOF system, it is determined that

$$\omega_{ni} = \sqrt{\frac{k_i}{m_i}}, \quad \text{for } i = 1, \dots, n \quad (8.119)$$

and

$$\zeta_i = \frac{c_i}{2\sqrt{m_i k_i}}, \quad \text{for } i = 1, \dots, n \quad (8.120)$$

Similar to an SDOF system, when

$$\zeta_i < 1 \quad (8.121)$$

the i th mode is underdamped.

When

$$\zeta_i = 1 \quad (8.122)$$

the system is critically damped. Lastly, when

$$\zeta_i > 1 \quad (8.123)$$

the system is overdamped. In the case of critically damped and overdamped systems, the i th mode reduces to two real valued subsystems. Thus, the system will no longer contain vibration modes. Note that again, for a stable system, we need all damping ratios to be nonnegative, which can be guaranteed by $\mathbf{M} > 0$, $\mathbf{C} \geq 0$ and $\mathbf{K} > 0$. This will also be true for nonproportionally damped systems.

8.3.3.2.2 Vibration Modes

8.3.3.2.2.1 The Essence of Equation 8.116, Separation of Variables In Equation 8.116 the assumption $\mathbf{x}(t) = \boldsymbol{\Phi}_i$ multiplied by $e^{\lambda_i t}$ implies the separation of variables. In comparing Equation 8.116 to Equation 6.36, the term $e^{\lambda_i t}$ can be seen as a free decay response of an SDOF vibration system with an amplitude of unity. Thus, $e^{\lambda_i t}$ is a scalar temporal function, which describes the i th modal vibration. The unity vibration response can then be denoted as follows:

$$q_i(t) = e^{\lambda_i t} \quad (8.124)$$

Here, $q_i(t)$ is called the i th modal response of the free decay vibration. Furthermore, in looking at Φ_i , it is seen that Φ_i contains spatial variable only, written as

$$\Phi_i = \begin{Bmatrix} \phi_{1i} \\ \phi_{2i} \\ \dots \\ \phi_{ni} \end{Bmatrix} \quad (8.125)$$

Here, Φ_i distributes the modal response $q_i(t)$ to different mass from 1 through n .

Equation 8.116 can be rewritten as

$$\mathbf{x}_i(t) = \Phi_i q_i(t) \quad (8.126)$$

where the subscript i in the physical domain $\mathbf{x}_i$ stands for the response due to the i th mode only. Substituting Equation 8.126 into Equation 8.115 and premultiplying Φ_i^T on both sides of the result will yield

$$\Phi_i^T \mathbf{M} \Phi_i \ddot{q}_i(t) + \Phi_i^T \mathbf{C} \Phi_i \dot{q}_i(t) + \Phi_i^T \mathbf{K} \Phi_i q_i(t) = 0, \quad i = 1, \dots, n \quad (8.127a)$$

or

$$m_i \ddot{q}_i(t) + c_i \dot{q}_i(t) + k_i q_i(t) = 0, \quad i = 1, \dots, n \quad (8.127b)$$

Suppose the system is excited by initial conditions

$$\mathbf{v}_0 = \begin{Bmatrix} v_{01} \\ v_{02} \\ \dots \\ v_{0n} \end{Bmatrix} \quad \text{and} \quad \mathbf{x}_0 = \begin{Bmatrix} x_{01} \\ x_{02} \\ \dots \\ x_{0n} \end{Bmatrix}$$

The modal initial conditions can be found as

$$\dot{\mathbf{q}}(0) = [\dot{q}_1(0), \dot{q}_2(0), \dots, \dot{q}_n(0)]^T = [\Phi_1 \ \Phi_2, \dots, \Phi_n]^{-1} \mathbf{v}_0 \quad (8.128a)$$

$$\mathbf{q}(0) = [q_1(0), q_2(0), \dots, q_n(0)]^T = [\Phi_1 \ \Phi_2, \dots, \Phi_n]^{-1} \mathbf{x}_0 \quad (8.128b)$$

As a result, an n SDOF vibration system has been attained. This procedure is referred to as *modal decoupling* or *modal analysis*.

Example 8.5

In the previous example in Section 8.3.2, the eigenvectors were calculated to be $\phi_1 = [0.4896 \ 0.8719]^T$ and $\phi_2 = [0.9628 \ -0.2703]^T$. Find the model response functions of the system and calculate the free decay vibration of the modal response due to initial velocity $\mathbf{v}_0 = [1, 2]^T$ and initial displacement $\mathbf{x}_0 = [-2, 2]^T$.

The modal initial conditions are

$$\dot{\mathbf{q}}(0) = \begin{bmatrix} 0.4896 & 0.9628 \\ 0.8719 & -0.2703 \end{bmatrix}^{-1} \begin{Bmatrix} 1 \\ 2 \end{Bmatrix} = \begin{Bmatrix} 2.2595 \\ -0.1105 \end{Bmatrix}$$

$$\mathbf{q}(0) = \begin{bmatrix} 0.4896 & 0.9628 \\ 0.8719 & -0.2703 \end{bmatrix}^{-1} \begin{Bmatrix} 1 \\ 2 \end{Bmatrix} = \begin{Bmatrix} 1.4250 \\ -2.8021 \end{Bmatrix}$$

The first modal mass, damping, and stiffness are, respectively,

$$\phi_1^T \mathbf{M} \phi_1 = 1.7603, \phi_1^T \mathbf{C} \phi_1 = 0.3859 \text{ and } \phi_1^T \mathbf{K} \phi_1 = 21.4615$$

Thus, the first modal equation is

$$1.7603 \ddot{q}_1(t) + 0.3859 \dot{q}_1(t) + 21.4615 q_1(t) = 0$$

with an initial modal velocity of 2.2595 and a modal displacement of 1.4250. Similarly, we have the second modal equation written as

$$1.0731 \ddot{q}_2(t) + 2.4474 \dot{q}_2(t) + 35.2052 q_2(t) = 0$$

with an initial modal velocity of -0.1105 and a modal displacement of -2.8021 .

8.3.4 MODAL SUPERPOSITION

Because the system is linear, once the modal responses are obtained, we can summarize them to construct the response in the physical domain, letting $\mathbf{x}_i(t) = \phi_i q_i(t)$ we have

$$\mathbf{x}(t) = \mathbf{x}_1(t) + \mathbf{x}_2(t) + \dots + \mathbf{x}_n(t) = \phi_1 q_1(t) + \phi_2 q_2(t) + \dots + \phi_n q_n(t) = [\phi_1 \ \phi_2 \ \dots \ \phi_n] \mathbf{q}(t) \quad (8.129)$$

Accordingly, the response denoted by $\mathbf{x}(t)$ is called the physical response, compared with the modal responses denoted by $q_i(t)$. Note that, at a given time t , $\mathbf{x}(t)$ is a vector, the j th element is the displacement measured at the j th location, whereas $q_i(t)$ is a scalar, which is the response of the i th mode.

Example 8.6

In the example from Section 8.3.3, we calculated the modal response $q_1(t)$ and $q_2(t)$. Find the response in the physical domain.

$$\mathbf{x}(t) = [\boldsymbol{\phi}_1 \ \boldsymbol{\phi}_2 \ \dots \ \boldsymbol{\phi}_n] \mathbf{q}(t)$$

The results are plotted in Figure 8.4b, as a comparison, the modal response calculated in the previous example are plotted in Figure 8.4a. Additionally, because Equation 8.126 only contains the i th mode, it can be rewritten as follows:

$$\mathbf{x}_i(t) = \boldsymbol{\phi}_i q_i(t) \quad (8.130)$$

Here, the italic symbol $\boldsymbol{\phi}_i$ is used to denote the normalized mode shape. Note that

$$\boldsymbol{\phi}_i = \frac{\boldsymbol{\phi}_i}{\sqrt{m_i}} \quad (8.131)$$

In other words, the mode shape $\boldsymbol{\phi}_i$ can be normalized so that the following product is unity:

$$\boldsymbol{\phi}_i^T \mathbf{M} \boldsymbol{\phi}_i = 1$$

Given that the system is linear, the linear combination can be obtained as

$$\mathbf{x}(t) = \sum_{i=1}^n a_i \mathbf{x}_i(t) = \sum_{i=1}^n a_i \boldsymbol{\phi}_i q_i(t) \quad (8.132)$$

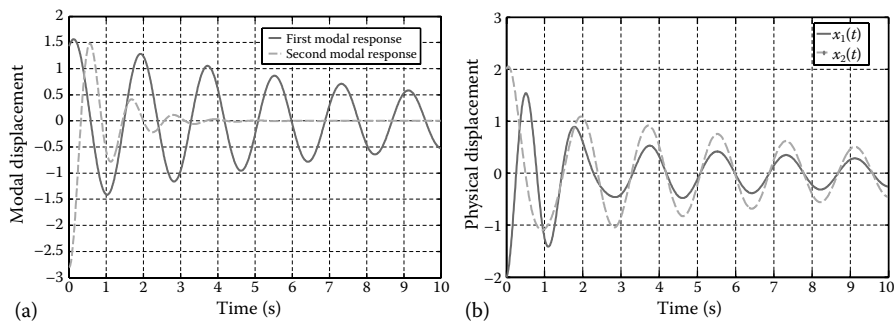


FIGURE 8.4 Modal and physical responses.

Equation 8.132 is also referred to as *modal superposition* with the normalized mode shape of ϕ_i . The scalar a_i is called the *modal participating factor* for the free decay vibrate mode. And

$$a_i = \frac{1}{\sqrt{m_i}} \quad (8.133)$$

It is noted that there can be several different types of normalization for the mode shape ϕ_i . Equation 8.131 is only one of the normalizations.

8.3.5 FORCED RESPONSE AND MODAL TRUNCATION

8.3.5.1 Forced Response

The concept of modal superposition can also be used in representing the solutions of forced responses.

Again, it is assumed that

$$\mathbf{x}_i(t) = q_i(t) \phi_i$$

where $q_i(t)$ is the i th modal response. In the case of forced vibration, it is no longer equal to $e^{\lambda_i t}$ as described in Equation 8.124. Rather, it becomes the forced modal response. The modal response $q_i(t)$ can be solved as follows.

In solving for $q_i(t)$, first substitute Equation 8.133 into Equation 8.5b, the equation of forced vibration for an **M-C-K** system. In the same way, premultiplying ϕ_i^T on both sides of the resulting equation will yield

$$\phi_i^T \mathbf{M} \phi_i \ddot{q}_i(t) + \phi_i^T \mathbf{C} \phi_i \dot{q}_i(t) + \phi_i^T \mathbf{K} \phi_i q_i(t) = \phi_i^T \mathbf{f}(t) \quad (8.134)$$

The scalar forcing function is denoted as

$$\phi_i^T \mathbf{f}(t) = g_i(t) \quad (8.135)$$

This will result in a typical equation of motion for an SDOF vibration system:

$$m_i \ddot{q}_i(t) + c_i \dot{q}_i(t) + k_i q_i(t) = g_i(t) \quad (8.136)$$

where $g_i(t)$ is the i th modal forcing function.

Based on our knowledge of SDOR systems, $q_i(t)$ is now solvable from Equation 8.136.

8.3.5.2 Rayleigh Quotient

Dividing $\phi_i^T \mathbf{M} \phi_i$ from both sides of Equation 8.134 results in the monic modal equation:

$$\ddot{q}_i(t) + \frac{\phi_i^T \mathbf{C} \phi_i}{\phi_i^T \mathbf{M} \phi_i} \dot{q}_i(t) + \frac{\phi_i^T \mathbf{K} \phi_i}{\phi_i^T \mathbf{M} \phi_i} q_i(t) = \frac{\phi_i^T \mathbf{f}(t)}{\phi_i^T \mathbf{M} \phi_i} \quad (8.137)$$

It is seen that

$$\frac{\phi_i^T \mathbf{C} \phi_i}{\phi_i^T \mathbf{M} \phi_i} = \frac{\phi_i^T \mathbf{C} \phi_i}{\phi_i^T \phi_i} = 2\zeta_i \omega_{ni} \quad (8.138)$$

and

$$\frac{\phi_i^T \mathbf{K} \phi_i}{\phi_i^T \mathbf{M} \phi_i} = \frac{\phi_i^T \mathbf{K} \phi_i}{\phi_i^T \phi_i} = \omega_{ni}^2 \quad (8.139)$$

Now, consider a generic notation denoted by

$$R = \frac{\phi^T \mathbf{A} \phi}{\phi^T \phi} \quad (8.140)$$

In this instance, $\mathbf{A}$ is an $n \times n$ square positive or a positive semidefinite matrix and ϕ is an $n \times 1$ vector.

The ratio described in Equation 8.140 is referred to as the *Rayleigh quotient*, denoted by R . When the vector ϕ varies, the Rayleigh quotient R will vary accordingly. It can be proven that, only if ϕ becomes the eigenvector of $\mathbf{A}$ will R reach a standing point. The value of R at a standing point will be the corresponding eigenvalue.

In Equations 8.138 and 8.139, the terms of $\frac{\phi_i^T \mathbf{C} \phi_i}{\phi_i^T \phi_i}$ and $\frac{\phi_i^T \mathbf{K} \phi_i}{\phi_i^T \phi_i}$ are all Rayleigh quotients. Because ϕ_i is the i th eigenvector, the terms $2\zeta_i \omega_i$ and ω_i^2 are, respectively, the eigenvalues of the damping and stiffness matrices of the monic system $\mathbf{I}-\mathbf{C}-\mathbf{K}$.

Furthermore, the terms $\frac{\phi_i^T \mathbf{C} \phi_i}{\phi_i^T \mathbf{M} \phi_i}$ and $\frac{\phi_i^T \mathbf{K} \phi_i}{\phi_i^T \mathbf{M} \phi_i}$ are generalized Rayleigh quotients. When ϕ reaches the i th mode shape ϕ_i , the corresponding generalized Rayleigh quotients $2\zeta_i \omega_i$ and ω_i^2 reach, respectively, the eigenvalues of the damping matrix $\mathbf{M}^{-1}\mathbf{C}$ and stiffness matrices $\mathbf{M}^{-1}\mathbf{K}$ of system $\mathbf{M}-\mathbf{C}-\mathbf{K}$.

Also note that, ϕ will reach the i th eigenvector of $\mathbf{M}^{-1}\mathbf{C}$ and $\mathbf{M}^{-1}\mathbf{K}$ simultaneously. Therefore, the system $\mathbf{M}-\mathbf{C}-\mathbf{K}$ can be decoupled mode by mode. In other words, the Rayleigh quotient is the base of normal modal analysis.

8.3.5.3 Ground Excitation and Modal Participation Factor

For the case of ground excitation,

$$\mathbf{f}(t) = -\mathbf{M}\mathbf{J}\ddot{x}_g(t) \quad (8.141)$$

and $\mathbf{x}(t)$ becomes the relative displacement vector (see Chapter 6, Base excitation and Equation 6.127). Here, in Equation 8.141

$$\mathbf{J} = \begin{Bmatrix} 1 \\ 1 \\ \dots \\ 1 \end{Bmatrix} \quad (8.142)$$

Thus, the generic modal force in Equation 8.137 can be replaced by

$$g_i(t) = \frac{\boldsymbol{\phi}_i^T \mathbf{f}(t)}{\boldsymbol{\phi}_i^T \mathbf{M} \boldsymbol{\phi}_i} = - \frac{\boldsymbol{\phi}_i^T \mathbf{M} \mathbf{J} \ddot{x}_g(t)}{\boldsymbol{\phi}_i^T \mathbf{M} \boldsymbol{\phi}_i} \quad (8.143)$$

The scalar Γ_i can be denoted as

$$\Gamma_i = \frac{\boldsymbol{\phi}_i^T \mathbf{M} \mathbf{J}}{\boldsymbol{\phi}_i^T \mathbf{M} \boldsymbol{\phi}_i} \quad (8.144)$$

In Equation 8.143, the term $\Gamma_i \ddot{x}_g(t)$ is defined as the modal participation factor for the i th mode, whereas Γ_i is the unit acceleration load for the i th mode. In the following, for convenience, Γ_i is also referred to as the *modal participation factor*. It will be shown that the value of Γ_i will depend on the normalization of $\boldsymbol{\phi}_i$.

8.3.5.4 Modal Superposition, Forced Vibration

8.3.5.4.1 Eigenvalue and Eigenvector Matrices

Combine the eigenvalues and eigenvector in matrix form. This is denoted as

$$\boldsymbol{\Phi} = [\boldsymbol{\phi}_1 \ \boldsymbol{\phi}_2 \ \dots \ \boldsymbol{\phi}_n] \quad (8.145)$$

It can be shown that

$$\boldsymbol{\Phi}^T \mathbf{M} \boldsymbol{\Phi} = \text{diag}(m_i), \quad \text{for } i = 1, \dots, n \quad (8.146)$$

$$\boldsymbol{\Phi}^T \mathbf{C} \boldsymbol{\Phi} = \text{diag}(c_i), \quad \text{for } i = 1, \dots, n \quad (8.147)$$

and

$$\boldsymbol{\Phi}^T \mathbf{K} \boldsymbol{\Phi} = \text{diag}(k_i), \quad \text{for } i = 1, \dots, n \quad (8.148)$$

Furthermore,

$$\mathbf{\Phi}^{-1}[\mathbf{M}^{-1}\mathbf{C}]\mathbf{\Phi} = \text{diag}(2\zeta_i\omega_{ni}), \quad \text{for } i = 1, \dots, n \quad (8.149)$$

and

$$\mathbf{\Phi}^{-1}[\mathbf{M}^{-1}\mathbf{K}]\mathbf{\Phi} = \text{diag}(\omega_{ni}^2), \quad \text{for } i = 1, \dots, n \quad (8.150)$$

Here, the $\text{diag}(2\zeta_i\omega_{ni})$ and the $\text{diag}(\omega_{ni}^2)$ are eigenvalue matrices of the matrices $\mathbf{M}^{-1}\mathbf{C}$ and $\mathbf{M}^{-1}\mathbf{K}$, respectively. Additionally, $\mathbf{\Phi}$ is the eigenvector matrix.

In Equations 8.146 through 8.150, the eigenvector matrix can be normalized in many respects. When calculating the solution of a system specifically, once $\mathbf{\Phi}$ is chosen to have fixed modal mass, damping, and stiffness as defined in Equations 8.146 through 8.148, the value of any ϕ_i should not be changed. Thus, the value of the modal participation factor Γ_i is fixed, as well as the modal response $q_i(t)$. The solution in the physical domain can then be determined.

8.3.5.4.2 Solution of Forced $\mathbf{M-C-K}$ System

Using the matrix form, with a fixed value of $\mathbf{\Phi}$, the solution of the forced MDOF system in the physical domain can be written as

$$\mathbf{x}(t) = \mathbf{\Phi}\mathbf{q}(t) \quad (8.151)$$

Equation 8.151 can be seen as a linear transform or mapping. In this instance, the modal response $\mathbf{q}(t)$ is transferred by the mode shape matrix $\mathbf{\Phi}$ to a physical response. Generally, $\mathbf{q}(t)$ is said to be in the *modal domain* or *modal space*, whereas $\mathbf{x}(t)$ is in the *physical domain* or *physical space*.

If the degree-of-freedom has an order of n , then it is an n -dimensional modal domain.

In Equation 8.151,

$$\mathbf{q}(t) = \begin{Bmatrix} q_1(t) \\ q_2(t) \\ \dots \\ q_n(t) \end{Bmatrix} \quad (8.152)$$

Therefore, Equation 8.151 can be rewritten as

$$\mathbf{x}(t) = \begin{Bmatrix} x_1(t) \\ x_2(t) \\ \dots \\ x_n(t) \end{Bmatrix} = \begin{bmatrix} \phi_{11} & \phi_{12} & \dots & \phi_{1n} \\ \phi_{21} & \phi_{22} & & \phi_{2n} \\ & \dots & & \\ \phi_{n1} & \phi_{n2} & \dots & \phi_{nn} \end{bmatrix} \begin{Bmatrix} q_1(t) \\ q_2(t) \\ \dots \\ q_n(t) \end{Bmatrix} \quad (8.153)$$

and

$$x_j(t) = \sum_{i=1}^n \phi_{ji} q_i(t) \quad (8.154)$$

8.3.5.5 Modal Truncation

Higher modes will contain much less energy. Thus, it is practical to use the first S modes in an approximation of the solution, written as

$$x_j(t) = \sum_{i=1}^S \phi_{ji} q_i(t) \quad (8.155)$$

Typically, the number of modes, S , will be considerably smaller than the number of total modes, n , that is,

$$S \ll n \quad (8.156)$$

Specifically, this is expressed as

$$\mathbf{x}(t)_{n \times 1} \approx \begin{bmatrix} \boldsymbol{\phi}_1 & \boldsymbol{\phi}_2 & \dots & \boldsymbol{\phi}_S \end{bmatrix}_{n \times S} \begin{Bmatrix} q_1(t) \\ q_2(t) \\ \dots \\ q_S(t) \end{Bmatrix}_{S \times 1} \quad \text{for } S < n \quad (8.157)$$

In matrix form,

$$\mathbf{x}(t) \approx \boldsymbol{\Phi}_C \mathbf{q}_C(t) \quad (8.158)$$

where $\boldsymbol{\Phi}_C = [\boldsymbol{\phi}_1 \ \boldsymbol{\phi}_2 \ \dots \ \boldsymbol{\phi}_S]_{n \times S}$ is the truncated mode shape matrix and $\mathbf{r}_C$ is the truncated modal response.

Additionally,

$$\mathbf{q}_C(t) = \begin{Bmatrix} q_1(t) \\ q_2(t) \\ \dots \\ q_S(t) \end{Bmatrix}_{S \times 1}$$

In many cases, only the first modal response will be used. This is called the fundamental modal response, which is used to represent the displacement. This is written as

$$\mathbf{x}(t) \approx \boldsymbol{\phi}_1 q_1(t) \quad (8.159)$$

8.3.6 RESPONSE TO RANDOM EXCITATIONS

Next, we consider random responses. Generally speaking, it is rare to have a rigorously defined stationary response for two reasons even though our vibration system is stable. First, when the forcing function is not stationary. Second, even the input is stationary, for a limited operating time, we will have transient response, which is not stationary. Therefore, in practical applications, a stationary process of vibration responses should not be assumed until we can prove it. In addition, many random processes are not zero mean either. In such circumstances, we consider covariance function of (t_1, t_2) , instead of using the correlation function of (τ) , as a general approach. Furthermore, for averaging, we should consider ensemble average instead of temporal average in many practical applications. On the other hand, however, many engineering random processes are mean square integrable. In the following, we assume all the signals are mean square integrable throughout the chapter, and we use lowercase letters f , g , q , and x , etc., for random processes for simplicity.

8.3.6.1 Modal and Physical Response

We now consider the mean and covariance of the response of proportionally damped systems through normal mode decoupling. Suppose that in Equation 8.5b, the forcing function is a random process. After decoupling, we can have (see Equation 8.136)

$$m_i \ddot{q}_i(t) + c_i \dot{q}_i(t) + k_i q_i(t) = g_i(t) = \boldsymbol{\phi}_i^T \mathbf{f}(t) = \boldsymbol{\phi}_i^T \begin{Bmatrix} f_1(t) \\ f_2(t) \\ \dots \\ f_n(t) \end{Bmatrix} \quad (8.160)$$

with modal initial velocity $\dot{q}_i(0)$ and modal initial displacement $q_i(0)$ (see Equation 8.128a,b).

Here $f_j(t)$ is the physical forcing at the j th location, whereas $g_i(t)$ is the i th modal force. If the forcing function $\mathbf{f}(t)$ is Gaussian, it is easy to see that the modal forcing functions should also be Gaussian. For stable MDOF systems, the i th modal response is also Gaussian. Furthermore, both the j th responses given by the complete or truncated modal superposition (see Equations 8.154 and 8.155) should also be Gaussian. Thus, the output responses will be completely characterized by the means and covariance functions. In the following examples, let us use the complete response for convenience.

Therefore, we first consider the modal response $q_i(t)$, which can be written as

$$q_i(t) = q_i(0) e^{-\zeta_i \omega_{ni} t} \left(\cos \omega_{di} t + \frac{\zeta_i \omega_{ni}}{\omega_{di}} \sin \omega_{di} t \right) + \dot{q}_i(0) h_i(t) + \int_0^T h_i(t-\tau) g_i(\tau) d\tau \quad (8.161)$$

Here, $h_i(t)$ is the i th unit impulse response function with damping ratio ζ_i and natural frequency ω_{ni} . In addition,

$$\omega_{di} = \sqrt{1 - \zeta_i^2} \omega_{ni} \quad (8.162)$$

is the i th damped natural frequency. With the help of modal superposition, we further have

$$\mathbf{x}(t) = \Phi \mathbf{q}(t) \quad (8.163)$$

8.3.6.2 Mean

The mean of $\mathbf{x}(t)$ is given by

$$\begin{aligned} \boldsymbol{\mu}_x(t) = \Phi \operatorname{diag} \left[q_i(0) e^{-\zeta_i \omega_{ni} t} \left(\cos \omega_{di} t + \frac{\zeta_i \omega_{ni}}{\omega_{di}} \sin \omega_{di} t \right) + \dot{q}_i(0) h_i(t) \right] \\ + \int_0^T \Phi \operatorname{diag} [h_i(t-\tau) \mu_{g_i}(\tau)] d\tau \end{aligned} \quad (8.164)$$

where $\operatorname{diag}[(\cdot)]_i$ is a diagonal matrix with its i th entry equivalent to $(\cdot)_i$. Furthermore

$$\mu_{g_i}(t) = E[g_i(t)] \quad (8.165a)$$

It should be note that the mean vector of the force in the physical domain is given by

$$\boldsymbol{\mu}_f(t) = E[\mathbf{f}(t)] \quad (8.165b)$$

8.3.6.3 Covariance

The covariance matrix of the random response $\mathbf{x}(t)$ is given by

$$\boldsymbol{\sigma}_{xx}(t_1, t_2) = E[\{\mathbf{x}(t_1) - \boldsymbol{\mu}_x(t_1)\} \{\mathbf{x}(t_2) - \boldsymbol{\mu}_x(t_2)\}^T]$$

Substituting from Equation 8.163 yields

$$\sigma_{\mathbf{xx}}(t_1, t_2) = \int_0^{t_1} d\tau_1 \int_0^{t_2} d\tau_2 \mathbf{\Phi} \mathbf{H}(t_1 - \tau_1) \mathbf{\Phi}^T E \left(\text{diag} \left[\left\{ \mathbf{g}(\tau_1) - \boldsymbol{\mu}_{g_i}(\tau_1) \right\} \left\{ \mathbf{g}(\tau_2) - \boldsymbol{\mu}_{g_i}(\tau_2) \right\}^T \right] \right) \mathbf{\Phi} \mathbf{H}(t_2 - \tau_2) \mathbf{\Phi}^T \quad (8.166)$$

Here

$$\mathbf{g}(\tau_1) - \boldsymbol{\mu}_g(\tau_1) = \begin{Bmatrix} g_1(\tau_1) - \mu_{g_1}(\tau_1) \\ g_2(\tau_1) - \mu_{g_2}(\tau_1) \\ \dots \\ g_n(\tau_1) - \mu_{g_n}(\tau_1) \end{Bmatrix} \quad (8.167a)$$

and

$$\begin{aligned} & \text{diag} \{ [\mathbf{g}(\tau_1) - \boldsymbol{\mu}_g(\tau_1)] [\mathbf{g}(\tau_2) - \boldsymbol{\mu}_g(\tau_2)]^T \} \\ &= \text{diag} \{ [g_i(\tau_1) - \mu_{g_i}(\tau_1)] [g_i(\tau_2) - \mu_{g_i}(\tau_2)] \} \end{aligned} \quad (8.167b)$$

In the examples above, we denote the covariance of the modal forcing process to be

$$\sigma_{\mathbf{ff}}(t_1, t_2) = E[\{\mathbf{f}(\tau_1) - \boldsymbol{\mu}_f(\tau_1)\} \{\mathbf{f}(\tau_2) - \boldsymbol{\mu}_f(\tau_2)\}^T] \quad (8.168)$$

Substitution of Equation 8.167b into Equation 8.166 results in

$$\sigma_{\mathbf{xx}}(t_1, t_2) = \int_0^{t_1} d\tau_1 \int_0^{t_2} d\tau_2 \mathbf{\Phi} \mathbf{H}(t_1 - \tau_1) \mathbf{\Phi}^T \sigma_{\mathbf{ff}}(t_1, t_2) \mathbf{\Phi} \mathbf{H}(t_2 - \tau_2) \mathbf{\Phi}^T \quad (8.169)$$

For convenience, in Equations 8.166 and 8.169, a special diagonal H matrix is given by

$$H[(.)] = \text{diag}[h_i(.)] \quad (8.170)$$

8.3.6.4 Probability Density Function for $x_i(t)$

If the forcing function is Gaussian, the response is also Gaussian, then the PDF of $x_i(t)$ is given by

$$f_{X_i}(x_i) = \frac{1}{\sqrt{2\pi}\sigma_{X_i}} e^{-\frac{(x_i - \mu_{X_i})^2}{2\sigma_{X_i}^2}} \quad (8.171)$$

where the variance $\sigma_{X_i}^2$ is the i th entry of the covariance matrix $\sigma_{XX}(t, t)$.

If the $f_j(t)$, the forcing function applied at j th location, etc., are jointly normally distributed, then $x_j(t)$ are also jointly normally distributed. The PDF can be given by

$$f_X(x_1, x_2, \dots, x_n) = \frac{1}{\sqrt[2]{2\pi} \sqrt{\det[\sigma_{FF}(t, t)]}} e^{-\frac{1}{2}(\mathbf{x} - \mu_X)^T \sigma_{FF}(t, t) (\mathbf{x} - \mu_X)} \quad (8.172)$$

Example 8.7

Suppose an automobile suspension system can be modeled as shown in Figure 8.5. For the ground excitation system, we have the equation of motion written as

$$\mathbf{M}\ddot{\mathbf{x}} + \mathbf{C}\dot{\mathbf{x}} + \mathbf{K}\mathbf{x} = -\mathbf{M}\mathbf{J} \ddot{x}_g$$

where $\mathbf{x} = [x_1 \ x_2]^T$ is the vector of relative displacement. Find the mean and covariance of the displacement by using the normal mode method.

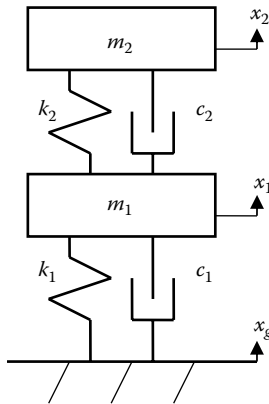


FIGURE 8.5 Model of automobile suspension system.

Suppose this system is proportionally damped. The equation of motion can be decoupled as

$$\ddot{q}_1(t) + 2\zeta_1\omega_{n1}\dot{q}_1(t) + \omega_{n1}^2 q_1(t) = g_1(t)$$

$$\ddot{q}_2(t) + 2\zeta_2\omega_{n2}\dot{q}_2(t) + \omega_{n2}^2 q_2(t) = g_2(t)$$

If the ground acceleration $\ddot{x}_g(t)$ is a stationary white noise with auto-PSD S_0 , then both the modal force $g_1(t)$ and $g_2(t)$ will be proportional to $\ddot{x}_g$ and

$$\sigma_{FF}(t_1, t_2) = \mathbf{D}S_0\delta(\tau)$$

where $\mathbf{D}$ is a 2×2 diagonal matrix with corresponding proportional factors. In this case, we can have

$$\mathbf{D} = \begin{bmatrix} \frac{m_1}{\Phi_1^T \mathbf{M} \Phi_1} & \\ & \frac{m_2}{\Phi_2^T \mathbf{M} \Phi_2} \end{bmatrix}^2$$

and S_0 is the auto-PSD of the ground acceleration

$$\sigma_{xx}(t_1, t_2) = S_0 \int_0^{t_1} d\tau_1 \int_0^{t_2} d\tau_2 \Phi H(t_1 - \tau_1) \Phi^T \mathbf{D} \Phi H(t_2 - \tau_2) \Phi^T$$

Suppose we have $m_1 = 1400$ (kg); $m_2 = 120$ (kg); $c_1 = 12.571$ (kN/m-s); $c_2 = 1.429$ (kN/m-s); $k_1 = 2200$ (kN/m); $k_2 = 250$ (kN/m). We thus can calculate the natural frequencies, damping ratios, and damped natural frequencies as $\omega_{n1} = 35.838$ (rad/s), $\omega_{n2} = 50.487$ (rad/s); $\zeta_1 = 0.1024$, $\zeta_2 = 0.1442$; $\omega_{d1} = 35.650$ (rad/s), $\omega_{d2} = 49.960$ (rad/s); the modal mass of the first and second mode are, respectively, $m_1 = 284.1181$ and $m_2 = 180.8987$.

The diagonal H matrix is

$$H = 1000 \text{ diag}([e^{-3.6696t} \sin(35.6497t)/10.129 \quad e^{-7.2828t} \sin(49.9593t)/9.038])$$

Suppose $S_0 = 25.26$, the matrix $\mathbf{D} S_0$ is

$$\mathbf{D} = \text{diag}([D_1 \quad D_2]) = \text{diag}([613.3, \quad 11.11])$$

The mode shape matrix is

$$\Phi = \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix} = \begin{bmatrix} 0.3581 & 0.2181 \\ 0.9337 & -0.9759 \end{bmatrix}$$

Based on the above computation, the variance can be calculated (see Equation 8.169). For example, the first entry of $\sigma_{xx}(t, t)$ is

$$\begin{aligned}
 & \frac{\phi_{11}^4 D_1 + \phi_{11}^2 \phi_{21}^2 D_2}{\omega_{d1}^2 m_1^2} \int_0^\infty e^{-2\zeta_1 \omega_{n1}(t-\tau)} \sin^2 \omega_{d1}(t-\tau) d\tau \\
 & + \frac{2\phi_{11}^2 \phi_{21}^2 D_1 + 2\phi_{11} \phi_{12} \phi_{21} \phi_{22} D_2}{\omega_{d1} \omega_{d2} m_1 m_2} \int_0^\infty e^{-(\zeta_1 \omega_{n1} + \zeta_2 \omega_{n2})(t-\tau)} \sin \omega_{d1}(t-\tau) \sin \omega_{d2}(t-\tau) d\tau \\
 & + \frac{\phi_{12}^4 D_1 + \phi_{12}^2 \phi_{22}^2 D_2}{\omega_{d2}^2 m_2^2} \int_0^\infty e^{-2\zeta_2 \omega_{n2}(t-\tau)} \sin^2 \omega_{d2}(t-\tau) d\tau \\
 & = \left[11.03 \int_0^\infty e^{-7.339(t-\tau)} \sin^2 35.65(t-\tau) d\tau \right. \\
 & \quad - 148.05 \int_0^\infty e^{-10.952(t-\tau)} \sin 35.65(t-\tau) \sin 49.96(t-\tau) d\tau \\
 & \quad \left. + 2.32 \int_0^\infty e^{-14.566(t-\tau)} \sin^2 49.96(t-\tau) d\tau \right] \times 10^{-8}
 \end{aligned}$$

8.4 NONPROPORTIONALLY DAMPED SYSTEMS, COMPLEX MODES

If the Caughey criterion cannot be satisfied, then the system is *nonproportionally damped*, or *generally damped*. In this case, the mode shape function can no longer be used to decouple the system. However, modal analysis can still be carried out in a $2n$ space. Generally, this will result in the mode shape being complex in value. The corresponding decoupling is referred to as the *complex mode* method (Liang and Inman 1990; Liang and Lee 1991b).

8.4.1 NONPROPORTIONAL DAMPING

Given that the complex mode is the result of damping, the damping will be considered first.

8.4.1.1 Mathematical Background

The following are both mutually sufficient and necessary:

1. The Caughey criterion is not satisfied (Caughey and O'Kelly 1965; Ventura 1985)

$$\mathbf{C}\mathbf{M}^{-1}\mathbf{K} \neq \mathbf{K}\mathbf{M}^{-1}\mathbf{C} \quad (8.173)$$

2. The Rayleigh quotients of $\mathbf{M}^{-1}\mathbf{C}$ and $\mathbf{M}^{-1}\mathbf{K}$ do not reach the standing point simultaneously for, at minimum, one mode
3. $\mathbf{M}^{-1}\mathbf{C}$ and $\mathbf{M}^{-1}\mathbf{K}$ do not share the same eigenvector matrix
4. The mode shape is not the eigenvector of $\mathbf{M}^{-1}\mathbf{K}$ or the eigenvector of $\mathbf{M}^{-1}\mathbf{C}$

5. The **M-C-K** system cannot be decoupled in n -dimensional modal space
6. At least two modal energy transfer ratios are nonzero

In the event that all of the above is true, *nonproportional damping* exists.

8.4.1.2 The Reality of Engineering

1. It is very rare to have proportional damping (Liang and Lee 1991a).
2. If the damping force is small, then portional damping can be used as a good approximation.
3. If the damping force is sufficiently large, then using proportional damping can introduce large error.

8.4.2 STATE VARIABLE AND STATE EQUATION

Remembering that nonproportionally damped systems require the use of modal analysis, a $2n$ space must be generated (Warburton and Soni 1977; Villaverde 1988).

Rewriting Equation 8.5b:

$$\mathbf{M}\ddot{\mathbf{x}}(t) + \mathbf{C}\dot{\mathbf{x}}(t) + \mathbf{K}\mathbf{x}(t) = \mathbf{f}(t) \quad (8.174)$$

Equation 8.174 is modified into a matrix equation, referred to as the *state equation*

$$\begin{Bmatrix} \ddot{\mathbf{x}} \\ \dot{\mathbf{x}} \end{Bmatrix} = \begin{bmatrix} -\mathbf{M}^{-1}\mathbf{C} & -\mathbf{M}^{-1}\mathbf{K} \\ \mathbf{I} & 0 \end{bmatrix} \begin{Bmatrix} \dot{\mathbf{x}} \\ \mathbf{x} \end{Bmatrix} + \begin{Bmatrix} \mathbf{M}^{-1}\mathbf{f} \\ 0 \end{Bmatrix} \quad (8.175)$$

Furthermore, with the help of the state and the input matrices $\mathcal{A}$ and $\mathcal{B}$, Equation 8.175 can be expressed as

$$\dot{\mathbf{Y}}(t) = \mathcal{A}\mathbf{Y}(t) + \mathcal{B}\mathbf{f}(t) \quad (8.176)$$

Here, the dimension of the vector $\mathbf{Y}$ is $2n \times 1$, referred to the *state vector*, specifically

$$\mathbf{Y}(t) = \begin{Bmatrix} \dot{\mathbf{x}}(t) \\ \mathbf{x}(t) \end{Bmatrix}_{2n \times 1} \quad (8.177)$$

Remembering

$$\mathbf{x}(t) = \begin{Bmatrix} x_1(t) \\ x_2(t) \\ \dots \\ x_n(t) \end{Bmatrix}_{n \times 1} \quad (8.178)$$

the *state matrix* in time can be written as

$$\mathcal{A} = \begin{bmatrix} [-\mathbf{M}^{-1}\mathbf{C}]_{n \times n} & [-\mathbf{M}^{-1}\mathbf{K}]_{n \times n} \\ \mathbf{I}_{n \times n} & \mathbf{0}_{n \times n} \end{bmatrix}_{2n \times 2n} \quad (8.179)$$

Here, $\mathbf{I}$ and $\mathbf{0}$ are the identity and null matrices, respectively, with the dimension $n \times n$. Note that the state matrix is not necessarily expressed in Equation 8.179, another form can be seen in the example in Section 8.4.4. Finally, the *input matrix* $\mathcal{B}$ is

$$\mathcal{B} = \begin{bmatrix} \mathbf{M}^{-1} \\ \mathbf{0} \end{bmatrix} \quad (8.180)$$

In Equation 8.180, $\mathbf{0}$ is also the $n \times n$ null matrix.

8.4.3 EIGEN-PROBLEM OF NONPROPORTIONALLY DAMPED SYSTEM

8.4.3.1 State Matrix and Eigen-Decomposition

The homogeneous form of Equation 8.176 is (Tong et al. 1994)

$$\dot{\mathbf{Y}}(t) = \mathcal{A}\mathbf{Y}(t) \quad (8.181)$$

To evaluate the eigen-properties of these systems, it must first be assumed that

$$\mathbf{Y} = \mathbf{P}_{2n \times 1} e^{\lambda t} \quad (8.182)$$

where, $\mathbf{Y}$ is the solution of the homogeneous Equation 8.181. Specifically expressed, this may result in one of the following being true:

$$\lambda \mathbf{P}_{2n \times 1} e^{\lambda t} = \mathcal{A} \mathbf{P}_{2n \times 1} e^{\lambda t} \quad (8.183)$$

or

$$\lambda \mathbf{P}_{2n \times 1} = \mathcal{A} \mathbf{P}_{2n \times 1} \quad (8.184a)$$

where λ is a scalar and $\mathbf{P}_{2n \times 1}$ is a $2n \times 1$ vector.

If $\mathbf{Y} = \mathbf{P}_{2n \times 1} e^{\lambda t}$ is a solution of Equation 8.181, then Equation 8.184a should hold. Additionally, if Equation 8.184a holds, then $\mathbf{Y} = \mathbf{P}_{2n \times 1} e^{\lambda t}$ will also be a solution of Equation 8.181. Thus, from the theory of linear algebra and the theory of vibration systems, it can be proved that these necessary and sufficient conditions are maintained.

Further assume the system to be underdamped. Because $\mathcal{A}$ is an asymmetric matrix, this will conventionally result in λ and $\mathbf{P}_{2n \times 1}$ being complex valued. Taking the complex conjugate of Equation 8.184a yields

$$\lambda^* \mathbf{P}_{2n \times 1}^* = \mathcal{A} \mathbf{P}_{2n \times 1}^* \quad (8.184b)$$

Equations 8.184a and 8.184b form the typical eigen-problem. To be exact, if both equations result in $\mathbf{Y} = \mathbf{P}_{2n \times 1} e^{\lambda t}$ as a solution of Equation 8.181, then Equations 8.184 implies that λ is one of the eigenvalues of the matrix $\mathcal{A}$ with $\mathbf{P}$ as the corresponding eigenvector.

Suppose a system that has n DOFs, thus yielding n pairs of eigenvalues and eigenvectors in the complex conjugates. Accordingly, Equations 8.184a and 8.184b can be further expanded as

$$\lambda_i \mathbf{P}_i = \mathcal{A} \mathbf{P}_i \quad i = 1, \dots, n \quad (8.185a)$$

Taking the complex conjugate of both sides of the above equation will yield

$$\lambda_i^* \mathbf{P}_i^* = \mathcal{A} \mathbf{P}_i^* \quad i = 1, \dots, n \quad (8.185b)$$

It is known that the eigen-problem as described in Equations 8.185a and 8.185b implies that all the eigenvectors, $\mathbf{P}_i$'s and $\mathbf{P}_i^*$'s are linearly independent. Additionally, each eigenvector is individually associated with a unique eigenvalue λ_i (or λ_i^*). This is expressed as

$$\lambda_i = -\zeta_i \omega_i + j\sqrt{1 - \zeta_i^2} \omega_i, \quad i = 1 \dots n \quad (8.186a)$$

and

$$\lambda_i^* = -\zeta_i \omega_i - j\sqrt{1 - \zeta_i^2} \omega_i, \quad i = 1 \dots n \quad (8.186b)$$

In the case of a nonproportionally damped system, the natural frequency ω_i and the damping ratio ζ_i are derived from the above equation. Thus, for a nonproportionally damped system:

$$\omega_i = |\lambda_i|, \quad i = 1, \dots, n \quad (8.187)$$

and

$$\zeta_i = -\text{Re}(\lambda_i)/\omega_i \quad (8.188)$$

Up to now, the natural frequency (or angular natural frequency) was all obtained through the square root of the stiffness k over m or the square root of k_i over m_i . In general, this method of calculation cannot be used to obtain the natural frequency for damped systems. The natural frequency must instead be calculated through Equations 8.186a,b. To distinguish the natural frequency calculated from Equations 8.186a,b from the previously defined quantities, the italic symbol, ω_i is used. In addition, the normal symbol ω_{ni} stands for the i th natural frequency of the corresponding undamped **M-O-K** system.

8.4.3.2 Eigenvectors and Mode Shapes

The eigenvalue λ_i (or λ_i^*) can have an infinite number of eigenvectors. That is, suppose $\mathbf{P}_i$ is one of the corresponding eigenvectors proportional to vector $\mathbf{R}_i$, then for vector $\mathbf{R}_i$,

$$\mathbf{R}_i = \alpha \mathbf{P}_i \quad (8.189)$$

will also be the eigenvector associated with that λ_i , where α is an arbitrary nonzero scalar. The eigenvector, $\mathbf{P}_i$ is also associated with the eigenvalue λ_i . Because $\mathbf{P}_i$ is a $2n \times 1$ vector, it is seen that through the assumption described in Equation 8.182, $\mathbf{P}_i$ can be written as

$$\mathbf{P}_i = \begin{Bmatrix} \lambda_i \mathbf{p}_i \\ \mathbf{p}_i \end{Bmatrix} \quad (8.190)$$

where $\mathbf{p}_i$ is an $n \times 1$ vector. Given that the system is linear, the solution can have all the linear combinations of $\mathbf{p}_i$'s and $\mathbf{p}_i^*$'s as follows:

$$\begin{aligned} \mathbf{x}(t) &= \mathbf{p}_1 e^{\lambda_1 t} + \mathbf{p}_2 e^{\lambda_2 t} + \dots + \mathbf{p}_n e^{\lambda_n t} + \mathbf{p}_1^* e^{\lambda_1^* t} + \mathbf{p}_2^* e^{\lambda_2^* t} + \dots + \mathbf{p}_n^* e^{\lambda_n^* t} \\ &= [\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_n] \begin{Bmatrix} e^{\lambda_1 t} \\ e^{\lambda_2 t} \\ \dots \\ e^{\lambda_n t} \end{Bmatrix} + [\mathbf{p}_1^*, \mathbf{p}_2^*, \dots, \mathbf{p}_n^*] \begin{Bmatrix} e^{\lambda_1^* t} \\ e^{\lambda_2^* t} \\ \dots \\ e^{\lambda_n^* t} \end{Bmatrix} \end{aligned} \quad (8.191)$$

Denote

$$\mathbf{P} = [\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_n] \quad (8.192)$$

and

$$\mathbf{E}(t) = \begin{Bmatrix} e^{\lambda_1 t} \\ e^{\lambda_2 t} \\ \dots \\ e^{\lambda_n t} \end{Bmatrix} \quad (8.193)$$

From this, the following can be obtained:

$$\mathbf{x}(t) = \mathbf{P} E(t) + \mathbf{P}^* E^*(t) \quad (8.194)$$

$$\dot{\mathbf{x}}(t) = \mathbf{P} \Delta E(t) + \mathbf{P}^* \Delta^* E^*(t) \quad (8.195)$$

and

$$\ddot{\mathbf{x}}(t) = \mathbf{P} \Delta^2 E(t) + \mathbf{P}^* \Delta^{*2} E^*(t) \quad (8.196)$$

or

$$\begin{Bmatrix} \dot{\mathbf{x}}(t) \\ \mathbf{x}(t) \end{Bmatrix} = \begin{bmatrix} \mathbf{P} \Delta & \mathbf{P}^* \Delta^* \\ \mathbf{P} & \mathbf{P}^* \end{bmatrix} \begin{Bmatrix} E(t) \\ E^*(t) \end{Bmatrix} \quad (8.197)$$

and

$$\begin{Bmatrix} \ddot{\mathbf{x}}(t) \\ \dot{\mathbf{x}}(t) \end{Bmatrix} = \begin{bmatrix} \mathbf{P} \Delta & \mathbf{P}^* \Delta^* \\ \mathbf{P} & \mathbf{P}^* \end{bmatrix} \begin{bmatrix} \Delta & \\ & \Delta^* \end{bmatrix} \begin{Bmatrix} E(t) \\ E^*(t) \end{Bmatrix} \quad (8.198)$$

In this instance, Δ is defined as the diagonal $n \times n$ matrix, which contains all the n -sets of eigenvalues. In addition, Δ can be written as follows:

$$\Delta = \text{diag}(\lambda_i) = \text{diag} \left(-\zeta_i \omega_i + j \sqrt{1 - \zeta_i^2} \omega_i \right) \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \dots & \\ & & & \lambda_n \end{bmatrix}_{n \times n} \quad (8.199)$$

Substitution of Equation 8.198 into Equation 8.180 with the aid of Equation 8.177 results in

$$\begin{bmatrix} \mathbf{P} \Delta & \mathbf{P}^* \Delta^* \\ \mathbf{P} & \mathbf{P}^* \end{bmatrix} \begin{bmatrix} \Delta & \\ & \Delta^* \end{bmatrix} \begin{Bmatrix} E(t) \\ E^*(t) \end{Bmatrix} = \mathcal{A} \begin{bmatrix} \mathbf{P} \Delta & \mathbf{P}^* \Delta^* \\ \mathbf{P} & \mathbf{P}^* \end{bmatrix} \begin{Bmatrix} E(t) \\ E^*(t) \end{Bmatrix} \quad (8.200)$$

The $2n \times u$ matrix $\mathbf{E}$ can further be defined as

$$\mathbf{E} = \left[\begin{Bmatrix} E(t) \\ E^*(t) \end{Bmatrix}, \begin{Bmatrix} E(t + \Delta t) \\ E^*(t + \Delta t) \end{Bmatrix}, \dots, \begin{Bmatrix} E[t + (u-1)\Delta t] \\ E^*[t + (u-1)\Delta t] \end{Bmatrix} \right], \quad u \geq 2n \quad (8.201)$$

which can be shown to have full rank $2n$. Furthermore, this can be written as

$$\begin{bmatrix} P\Delta & P^* \Delta^* \\ P & P^* \end{bmatrix} \begin{bmatrix} \Delta & \\ & \Delta^* \end{bmatrix} \mathbf{E} = \mathcal{A} \begin{bmatrix} P\Delta & P^* \Delta^* \\ P & P^* \end{bmatrix} \mathbf{E} \quad (8.202)$$

Given that

$$\mathbf{E}\mathbf{E}^+ = \mathbf{I}_{2n \times 2n} \quad (8.203)$$

both sides of Equation 8.202 can be postmultiplied by $\mathbf{E}^+$, where the superscript $+$ stands for the pseudo inverse.

$$\begin{bmatrix} P\Delta & P^* \Delta^* \\ P & P^* \end{bmatrix} \begin{bmatrix} \Delta & \\ & \Delta^* \end{bmatrix} = \mathcal{A} \begin{bmatrix} P\Delta & P^* \Delta^* \\ P & P^* \end{bmatrix} \quad (8.204)$$

Equation 8.204 indicates that the state matrix $\mathcal{A}$ can be decomposed by the eigenvalue matrix

$$\mathbf{\Lambda} = \begin{bmatrix} \Delta & \\ & \Delta^* \end{bmatrix} \quad (8.205)$$

and the eigenvector matrix $\mathcal{P}$

$$\mathcal{P} = \begin{bmatrix} P\Delta & P^* \Delta^* \\ P & P^* \end{bmatrix} = [\mathbf{P}_1, \mathbf{P}_2, \dots, \mathbf{P}_{2n}] \quad (8.206)$$

Note that the eigenvector matrix is now arranged to have the form of a complex conjugate pair

$$\begin{bmatrix} P\Delta \\ P \end{bmatrix} \text{ and } \begin{bmatrix} P\Delta \\ P \end{bmatrix}^* \quad (8.207)$$

Accordingly,

$$\mathcal{P} = [\mathbf{P}_1, \mathbf{P}_2, \dots, \mathbf{P}_n], [\mathbf{P}_1, \mathbf{P}_2, \dots, \mathbf{P}_n]^* \quad (8.208)$$

and,

$$\begin{bmatrix} P\Delta \\ P \end{bmatrix}^* = \begin{bmatrix} P^* \Delta^* \\ P^* \end{bmatrix} \quad (8.209)$$

That is,

$$\mathcal{A} = \mathcal{P} \Lambda \mathcal{P}^{-1} \quad (8.210)$$

or

$$\Lambda = \mathcal{P}^{-1} \mathcal{A} \mathcal{P} \quad (8.211)$$

The matrix Λ in Equation 8.211 will maintain the same eigenvalue format as a proportionally damped system. Conversely, the eigenvector $\mathcal{P}$ can obtain a different form from the proportionally damped case due to the nonuniqueness of the eigenvector $\mathbf{P}_i$ as previously discussed.

In general, the submatrix $\mathbf{P}$ in the eigenvector matrix $\mathcal{P}$ is complex valued. Equations 8.210 and 8.211 can therefore be used to define modal analysis in the $2n$ complex modal domain. In this case, $\mathbf{P}$ is called the *mode shape matrix*. Note that, $\mathbf{P}$ contains n vectors as expressed in Equation 8.192.

In this instance, there will be n set of triples $\langle \mathbf{p}_i, \zeta_i, \omega_i \rangle$, along with n set of its complex conjugates. It is apparent that the damping ratio ζ_i and the natural frequency ω_i can be obtained through Equation 8.199. In this situation, the triple $\langle \mathbf{p}_i, \zeta_i, \omega_i \rangle$ and its complex conjugate define the i th complex mode.

8.4.3.3 Modal Energy Transfer Ratio

In a generally damped MDOF system, if by design and letting $\mathbf{C} = 0$, then the corresponding natural frequency ω_{ni} can be calculated through the eigen-equation as mentioned in Chapter 7, which is repeated as follows:

$$\omega_{ni} \boldsymbol{\Phi}_i = \mathbf{M}^{-1} \mathbf{K} \boldsymbol{\Phi}_i \quad (8.212)$$

The modal energy transfer ratio (ETR) ξ_i can be approximated as (Liang et al. 1992; Liang et al. 2012)

$$\xi_i = \ln \left(\frac{\omega_i}{\omega_{ni}} \right) \quad (8.213)$$

From the discussion of SDOF systems, we have established that the natural frequency denotes the corresponding modal energy. Assume that an MDOF system is originally undamped and then a particular type of damping is gradually added to the system. If the damping remains proportional, then the natural frequency ω_{ni} will remain unchanged.

Otherwise, it will be changed to ω_i . In the event the natural frequency changes, one of two events will occur. Either a certain amount of energy will be transferred into the mode, when

$$\xi_i = \frac{E_{T_i}}{4\pi E_K} > 0 \quad (8.214)$$

or, a certain amount of energy will be transferred out of the mode, when

$$\xi_i = \frac{E_{T_i}}{4\pi E_K} < 0 \quad (8.215)$$

In Equations 8.214 and 8.215, E_{T_i} is the i th modal energy transferred during a cycle and E_K is the maximum conservative energy. Comparing this to the modal damping ratio which relates to energy dissipation E_{D_i} yields

$$\zeta_i = \frac{E_{D_i}}{4\pi E_{K_i}} \geq 0 \quad (8.216)$$

The modal energy transfer ratio can be used in identifying whether a specific mode is complex. Namely, a nonproportionally damped system may encompass both complex and normal modes. This scenario cannot be distinguished through the Caughey criterion because the Caughey criterion can only provide global judgment.

In a nonproportionally damped system, the natural frequency of the first complex mode will always be greater than that of the undamped one, that is,

$$\omega_1 > \omega_{n1} \quad (8.217)$$

For the i th mode with $\xi_i \neq 0$, the mode shape $\mathbf{p}_i$ will be complex valued, that is,

$$\mathbf{p}_i = \begin{Bmatrix} p_{1i} \\ p_{2i} \\ \cdots \\ p_{ni} \end{Bmatrix} = \begin{Bmatrix} |p_{1i}| e^{j\theta_{1i}} \\ |p_{2i}| e^{j\theta_{2i}} \\ \cdots \\ |p_{ni}| e^{j\theta_{ni}} \end{Bmatrix} \quad (8.218)$$

Here θ_{ji} is the corresponding phase angle.

Example 8.8

A 2-DOF system is given by

$$\mathbf{M} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} (\text{kg}), \mathbf{C} = \begin{bmatrix} 6 & -3 \\ -3 & 3 \end{bmatrix} (\text{N/m-s}), \text{ and } \mathbf{K} = \begin{bmatrix} 50 & -10 \\ -10 & 30 \end{bmatrix} (\text{N/m})$$

Find the natural frequencies, modal energy transfer ratios, damping ratios, and mode shapes.

Decoupled from the corresponding state matrix, we have eigenvalues $-0.4158 \pm 3.7208j$ and $-3.3342 + 6.2306j$. The natural frequency of the first mode is $\omega_1 =$

$[(-0.4158)^2 + (3.7208)^2]^{1/2} = 3.7440$ (rad/s), and that of the second mode is $\omega_2 = 7.0666$ (rad/s).

Note that the natural frequencies of the undamped **M-O-K** system is $\omega_{n1} = 3.6913$ and $\omega_{n2} = 7.1676$. Thus, the corresponding modal energy transfer ratios are

$\xi_i = \ln\left(\frac{\omega_1}{\omega_{n1}}\right) = 0.0142$ and $\xi_2 = -0.0142$. It is seen that, due to the nonproportional damping, the first natural frequency becomes larger by receiving energy so that the ETR is positive, whereas for the second mode, the natural frequency becomes smaller by giving up energy and the ETR is thus negative. In addition, for energy conservation, we see that $\xi_1 + \xi_2 = 0$.

The damping ratio of the first mode is $\zeta_1 = 0.4158/3.7440 = 0.1111$, and that of the second mode is $\zeta_2 = 0.4119$ (rad/s).

From the state matrix, we also have the eigenvectors

$$\mathcal{P} = \begin{bmatrix} -0.9596 & -0.9596 & 0.3061+0.1255i & 0.3061-0.1255j \\ 0.2138+0.1176j & 0.2138-0.1176j & 0.9077 & 0.9077 \\ 0.0641+0.1197j & 0.0641-0.1197j & 0.0242-0.0850j & 0.0242+0.0850j \\ 0.0004-0.0345j & 0.0004+0.0345j & -0.0269-0.2409j & -0.0269+0.2409j \end{bmatrix}$$

Note that the first and the second columns are for the second mode, and the third and the fourth columns are for the first mode. So that the first mode shapes are given by

$$[0.0242 \mp 0.0850j, \quad 0.0269 \mp 0.2409j]^T$$

In addition, the second mode shape is $[0.0641 \pm 0.00197j \quad 0.0004 \mp 0.00345j]^T$.

8.4.4 RESPONSE TO RANDOM EXCITATIONS

Similar to a proportionally damped system, we now consider the response to random excitations for nonproportionally damped systems. Here, for convenience, we rewrite the equation of motion of a generally damped MDOF system as

$$\begin{Bmatrix} \dot{\mathbf{x}}(t) \\ \ddot{\mathbf{x}}(t) \end{Bmatrix} = \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{M}^{-1}\mathbf{K} & -\mathbf{M}^{-1}\mathbf{C} \end{bmatrix} \begin{Bmatrix} \mathbf{x}(t) \\ \dot{\mathbf{x}}(t) \end{Bmatrix} + \begin{Bmatrix} \mathbf{0} \\ \mathbf{M}^{-1} \end{Bmatrix} \mathbf{f}(t) \quad (8.219)$$

Or

$$\dot{\mathbf{Y}}(t) = \mathcal{A} \mathbf{Y}(t) + \mathcal{B} \mathbf{f}(t) \quad (8.220)$$

where

$$\mathbf{Y}(t) = \begin{Bmatrix} \mathbf{x}(t) \\ \dot{\mathbf{x}}(t) \end{Bmatrix}, \quad \mathcal{A} = \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{M}^{-1}\mathbf{K} & -\mathbf{M}^{-1}\mathbf{C} \end{bmatrix} \text{ and } \mathcal{B} = \begin{Bmatrix} \mathbf{0} \\ \mathbf{M}^{-1} \end{Bmatrix} \quad (8.221)$$

where **I** and **0** are, respectively, identity and null submatrices with proper dimensions.

In this case, the state matrix can also be decoupled as indicated in Equation 8.210 so that the state equation can also be decoupled as mentioned previously.

8.4.4.1 Modal and Physical Response

If the input is Gaussian, the forcing function $\mathcal{F}(t) = \mathcal{B}f(t)$ of the equation of motion described in the state equation (Equation 8.220), can have a mean and $\mu_{\mathcal{F}}(t)$ and covariance function given by

$$\sigma_{\mathcal{F}\mathcal{F}}(t_1, t_2) = E \left[\{ \mathcal{F}(t_1) - \mu_{\mathcal{F}}(t_1) \} \{ \mathcal{F}(t_2) - \mu_{\mathcal{F}}(t_2) \}^T \right] \quad (8.222)$$

Furthermore, we know that the state equation can be decoupled by premultiplying $\mathcal{P}^{-1}$ on both sides of Equation 8.220, that is, by denoting

$$\mathbf{U}(t) = \mathcal{P}^{-1} \mathbf{Y}(t) \quad (8.223)$$

We can have

$$\begin{cases} \dot{\mathbf{U}}(t) = \Lambda \mathbf{U}(t) + \mathcal{P}^{-1} \mathcal{F}(t) \\ \mathbf{U}(0) = \mathcal{P}^{-1} \begin{Bmatrix} \mathbf{x}(0) \\ \dot{\mathbf{x}}(0) \end{Bmatrix} \end{cases} \quad (8.224)$$

The solution of Equation 8.223 can be written as

$$\mathbf{U}(t) = \mathbf{e}^{\Lambda t} \mathbf{U}(0) + \int_0^t \mathbf{e}^{\Lambda(t-\tau)} \mathcal{P}^{-1} \mathcal{F}(\tau) d\tau \quad (8.225)$$

In Equation 8.225, $\mathbf{e}^{\Lambda t}$ is a $2n \times 2n$ diagonal matrix, such that

$$\mathbf{e}^{\Lambda t} = \text{diag}[\exp(\lambda_i t)] \quad (8.226)$$

From Equation 8.223, we may write the response in the physical domain as

$$\mathbf{Y}(t) = \mathcal{P} \mathbf{U}(t) = \mathcal{P} \mathbf{e}^{\Lambda t} \mathcal{P}^{-1} \mathbf{Y}(0) + \int_0^t \mathcal{P} \mathbf{e}^{\Lambda(t-\tau)} \mathcal{P}^{-1} \mathcal{F}(\tau) d\tau \quad (8.227)$$

It can be proven that

$$\mathcal{P} \mathbf{e}^{\Lambda t} \mathcal{P}^{-1} = e^{\mathcal{A}t} \quad (8.228)$$

Therefore, the response $\mathbf{Y}(t)$ can be further written as

$$\mathbf{Y}(t) = \mathbf{e}^{\mathcal{A}t} \mathbf{Y}(0) + \int_0^t \mathbf{e}^{\mathcal{A}(t-\tau)} \mathcal{F}(\tau) d\tau \quad (8.229)$$

In Equations 8.228 and 8.229, the term $e^{\mathcal{A}t}$ is a $2n \times 2n$ state transition matrix.

To see the relationship between the state transition matrix and the characteristic matrix of a generally damped MDOF system, defined by $s\mathbf{I} - \mathcal{A}$, taking the inverse Laplace transform of the inverse of characteristic matrix yields

$$\begin{aligned} \mathcal{L}^{-1}[(s\mathbf{I} - \mathcal{A})^{-1}] &= \mathcal{L}^{-1}(s^{-1}\mathbf{I} + s^{-2}\mathcal{A} + s^{-3}\mathcal{A}^2 + \dots) \\ &= \mathbf{I} + \mathcal{A}t + \mathcal{A}t^2/2! + \mathcal{A}t^3/3! + \dots = e^{\mathcal{A}t} \end{aligned} \quad (8.230)$$

Note that

$$(s\mathbf{I} - \mathcal{A})^{-1} = \frac{\text{adj}(s\mathbf{I} - \mathcal{A})}{\det[(s\mathbf{I} - \mathcal{A})]} = \frac{\text{adj}(s\mathbf{I} - \mathcal{A})}{\prod_{i=1}^{2n} (s - \lambda_i)} \quad (8.231)$$

The jk th entry of the inverse matrix $(s\mathbf{I} - \mathcal{A})^{-1}$ can be written as $\frac{\sum_{i=1}^n {}_i a_{jk}}{s - \lambda_i} + \frac{\sum_{i=1}^n {}_i a_{jk}^*}{s - \lambda_i^*}$.

Therefore, the jk th entry of $\mathcal{L}^{-1}[(s\mathbf{I} - \mathcal{A})^{-1}]$ is $\sum_{i=1}^n {}_i a_{jk} e^{\lambda_i t} + \sum_{i=1}^n {}_i a_{jk}^* e^{\lambda_i^* t}$.

For the steady state response, the free decay solution dissipated to zero. We thus have

$$\mathbf{Y}(t) = \int_0^t e^{\mathcal{A}(t-\tau)} \mathcal{F}(\tau) d\tau = \int_{-\infty}^t e^{\mathcal{A}(t-\tau)} \mathcal{F}(\tau) d\tau = \int_0^\infty e^{\mathcal{A}\tau} \mathcal{F}(t-\tau) d\tau \quad (8.232)$$

8.4.4.2 Mean

First consider the total response. Similar to the proportionally damped systems, the vector of mean $\boldsymbol{\mu}_Y(t)$ is given by

$$\boldsymbol{\mu}_Y(t) = \mathcal{P} e^{\Lambda t} \mathcal{P}^{-1} \mathbf{Y}(0) + \int_0^t \mathcal{P} e^{\Lambda(t-\tau)} \mathcal{P}^{-1} \boldsymbol{\mu}_{\mathcal{F}}(t-\tau) d\tau \quad (8.233)$$

Or using the state transition matrix, we have

$$\boldsymbol{\mu}_Y(t) = \mathbf{e}^{\mathcal{A}t} \mathbf{Y}(0) + \int_0^t \mathbf{e}^{\mathcal{A}\tau} \boldsymbol{\mu}_{\mathcal{F}}(t-\tau) d\tau \quad (8.234)$$

8.4.4.3 Covariance

8.4.4.3.1 General Covariance

The covariance of the nonproportionally damped system is given by

$$\boldsymbol{\sigma}_{\mathbf{Y}\mathbf{Y}}(t_1, t_2) = E[\{\mathbf{Y}(t_1) - \boldsymbol{\mu}_{\mathbf{Y}}(t_1)\}\{\mathbf{Y}(t_2) - \boldsymbol{\mu}_{\mathbf{Y}}(t_2)\}^T] \quad (8.235)$$

Substituting from Equation 8.233 yields

$$\begin{aligned} \boldsymbol{\sigma}_{\mathbf{Y}\mathbf{Y}}(t_1, t_2) = & \int_0^{t_1} d\tau_1 \int_0^{t_2} d\tau_2 E \left[\left\{ \mathcal{P} \mathbf{e}^{\Lambda \tau_1} \mathcal{P}^{-1} [\mathcal{F}(t_1 - \tau_1) - \boldsymbol{\mu}_{\mathcal{F}}(t_1 - \tau_1)] \right\} \left\{ \mathcal{P} \mathbf{e}^{\Lambda \tau_2} \mathcal{P}^{-1} [\mathcal{F}(t_2 - \tau_2) - \boldsymbol{\mu}_{\mathcal{F}}(t_2 - \tau_2)] \right\}^T \right] \\ & (8.236) \end{aligned}$$

with the help of Equation 8.219, the above equation can be written as

$$\boldsymbol{\sigma}_{\mathbf{Y}\mathbf{Y}}(t_1, t_2) = \int_0^{t_1} d\tau_1 \int_0^{t_2} d\tau_2 \{ \mathcal{P} \mathbf{e}^{\Lambda \tau_1} \mathcal{P}^{-1} \boldsymbol{\sigma}_{\mathcal{F}\mathcal{F}}(t_1 - \tau_1, t_2 - \tau_2) \mathcal{P}^{-T} \mathbf{e}^{\Lambda \tau_2} \mathcal{P}^T \} \quad (8.237)$$

Otherwise, using the state transition matrix, we have

$$\boldsymbol{\sigma}_{\mathbf{Y}\mathbf{Y}}(t_1, t_2) = \int_0^{t_1} d\tau_1 \int_0^{t_2} d\tau_2 \{ \mathbf{e}^{\mathcal{A} \tau_1} \boldsymbol{\sigma}_{\mathcal{F}\mathcal{F}}(t_1 - \tau_1, t_2 - \tau_2) \mathbf{e}^{\mathcal{A} \tau_2} \} \quad (8.238)$$

To evaluate Equation 8.238, we need to calculate the covariance function of $\boldsymbol{\sigma}_{\mathcal{F}\mathcal{F}}(t_1, t_2)$, which might not always be easy. In the following examples, let us consider a special case when the excitation is stationary and the responses reach steady states.

8.4.4.3.2 Steady State Covariance under White Noise Excitation

Based on Equation 8.232, for the steady state response with zero mean, the covariance can be written as

$$\begin{aligned} \boldsymbol{\sigma}_{\mathbf{Y}\mathbf{Y}}(t_1, t_2) &= E[\mathbf{Y}(t_1) \mathbf{Y}(t_2)^T] \\ &= E \left[\left\{ \int_0^\infty \mathbf{e}^{\mathcal{A} \tau_1} \mathcal{F}(t_1 - \tau_1) d\tau_1 \right\} \left\{ \int_0^\infty \mathbf{e}^{\mathcal{A} \tau_2} \mathcal{F}(t_2 - \tau_2) d\tau_2 \right\}^T \right] \\ &= E \left[\int_0^\infty \int_0^\infty \mathbf{e}^{\mathcal{A} \tau_1} \mathcal{F}(t_1 - \tau_1) \{ \mathcal{F}(t_2 - \tau_2) \}^T \mathbf{e}^{\mathcal{A}^T \tau_2} d\tau_1 d\tau_2 \right] \\ &= \int_0^\infty \int_0^\infty \mathbf{e}^{\mathcal{A} \tau_1} E[\{ \mathcal{F}(t_1 - \tau_1) \} \{ \mathcal{F}(t_2 - \tau_2) \}^T] \mathbf{e}^{\mathcal{A}^T \tau_2} d\tau_1 d\tau_2 \end{aligned} \quad (8.239)$$

Thus far, Equations 8.239 and 8.238 are essentially the same formulae for the integration limits in Equation 8.238 and can be expanded to infinity. Now, if the force $\mathbf{F}(t)$ in $\mathcal{F}(t) = \mathcal{B}\mathbf{f}(t)$ is n -dimensional independent Gaussian, then

$$E[\{\mathcal{F}(t_1 - \tau_1)\}\{\mathcal{F}(t_2 - \tau_2)\}^T] = \mathcal{B}\mathbf{D}\delta(\tau)\mathcal{B}^T \quad (8.240)$$

where

$$\mathbf{D}\delta(\tau) = \text{diag}(d_i)_{n \times n} \delta(\tau) \quad (8.241)$$

is the covariance matrix of the forcing function.

Example 8.9

Suppose a 3-DOF system is excited by a forcing function $\mathbf{f}(t) = w(t) [g_1 \ g_2 \ g_3]^T$, where g_i is the amplitude of force $f_i(t)$ applied on the i th mass, and $w(t)$ is a white noise process with PSD equal to S_0 .

The matrix $\mathbf{D}\delta(\tau)$ can be written as $\text{diag}(d_i)_{n \times n} \delta(\tau)$ and $d_i = 2\pi S_0 g_i^2$.

When $t_1 = t_2$, substitution of Equation 8.240 into Equation 8.239 yields

$$\begin{aligned} \sigma_{\mathbf{y}}(t_1, t_1) &= \int_0^\infty \int_0^\infty \mathbf{e}^{\mathcal{A}\tau_1} \mathcal{B} \mathbf{D} \delta(\tau_2 - \tau_1) \mathcal{B}^T \mathbf{e}^{\mathcal{A}^T \tau_2} d\tau_1 d\tau_2 \\ &= \int_0^\infty \mathbf{e}^{\mathcal{A}\tau_1} \mathcal{B} \mathbf{D} \int_0^\infty \delta(\tau_2 - \tau_1) \mathcal{B}^T \mathbf{e}^{\mathcal{A}^T \tau_2} d\tau_2 d\tau_1 \\ &= \int_0^\infty \mathbf{e}^{\mathcal{A}\tau} \mathcal{B} \mathbf{D} \mathcal{B}^T \mathbf{e}^{\mathcal{A}^T \tau} d\tau = \boldsymbol{\sigma}_{\mathbf{y}}(\tau) \end{aligned} \quad (8.242)$$

For convenience, denote

$$\mathbf{e}^{\mathcal{A}\tau} \mathcal{B} \mathbf{D} \mathcal{B}^T \mathbf{e}^{\mathcal{A}^T \tau} = \mathbf{G}(\tau) \quad (8.243)$$

Take the derivative of $\mathbf{G}(\tau)$ with respect to τ ,

$$\begin{aligned} \frac{d\mathbf{G}(\tau)}{d\tau} &= \mathcal{A} \mathbf{e}^{\mathcal{A}\tau} \mathcal{B} \mathbf{D} \mathcal{B}^T \mathbf{e}^{\mathcal{A}^T \tau} + \mathbf{e}^{\mathcal{A}\tau} \mathcal{B} \mathbf{D} \mathcal{B}^T \mathbf{e}^{\mathcal{A}^T \tau} \mathcal{A}^T \\ &= \mathcal{A} \mathbf{G}(\tau) + \mathbf{G}(\tau) \mathcal{A}^T \end{aligned} \quad (8.244)$$

The integral of Equation 8.244, with the limit from 0 to ∞ , can be written as

$$\int_0^\infty \frac{d\mathbf{G}(\tau)}{d\tau} d\tau = \mathbf{G}(\infty) - \mathbf{G}(0) = \mathcal{A} \int_0^\infty \mathbf{G}(\tau) d\tau + \int_0^\infty \mathbf{G}(\tau) d\tau \mathcal{A}^T \quad (8.245)$$

It is seen that

$$\mathbf{G}(\infty) = 0 \quad (8.246)$$

and

$$\mathbf{G}(0) = \mathcal{B} \mathbf{D} \mathcal{B}^T \quad (8.247)$$

Therefore, we have

$$\mathcal{A} \boldsymbol{\sigma}_{\mathbf{Y}\mathbf{Y}}(0) + \boldsymbol{\sigma}_{\mathbf{Y}\mathbf{Y}}(0) \mathcal{A}^T = -\mathcal{B} \mathbf{D} \mathcal{B}^T \quad (8.248)$$

Equation 8.248 can be seen as an algebraic equation with unknown of the covariance matrix $\boldsymbol{\sigma}_{\mathbf{Y}\mathbf{Y}}(0)$. Because the input and output are stationary, we have $\boldsymbol{\sigma}_{\mathbf{Y}\mathbf{Y}}(0) = \boldsymbol{\sigma}_{\mathbf{Y}\mathbf{Y}}(t, t)$.

8.4.4.3.3 Computations of Covariance

To solve $\boldsymbol{\sigma}_{\mathbf{Y}\mathbf{Y}}(0)$ through Equation 8.248, consider the definition of the covariance, namely,

$$\boldsymbol{\sigma}_{\mathbf{Y}\mathbf{Y}}(0) = E[\mathbf{Y}(t) \mathbf{Y}^T(t)] = E \begin{bmatrix} \mathbf{x}\mathbf{x}^T & \mathbf{x}\dot{\mathbf{x}}^T \\ \dot{\mathbf{x}}\mathbf{x}^T & \dot{\mathbf{x}}\dot{\mathbf{x}}^T \end{bmatrix} = \begin{bmatrix} \boldsymbol{\sigma}_{\mathbf{x}\mathbf{x}} & \boldsymbol{\sigma}_{\mathbf{x}\dot{\mathbf{x}}} \\ \boldsymbol{\sigma}_{\dot{\mathbf{x}}\mathbf{x}} & \boldsymbol{\sigma}_{\dot{\mathbf{x}}\dot{\mathbf{x}}} \end{bmatrix} \quad (8.249)$$

Note that

$$\boldsymbol{\sigma}_{\mathbf{x}\mathbf{x}} = \boldsymbol{\sigma}_{\mathbf{x}\mathbf{x}}^T \quad (8.250)$$

$$\boldsymbol{\sigma}_{\mathbf{x}\dot{\mathbf{x}}} = \boldsymbol{\sigma}_{\dot{\mathbf{x}}\mathbf{x}}^T \quad (8.251)$$

and

$$\boldsymbol{\sigma}_{\dot{\mathbf{x}}\dot{\mathbf{x}}} = \boldsymbol{\sigma}_{\dot{\mathbf{x}}\dot{\mathbf{x}}}^T \quad (8.252)$$

With the help of Equations 8.250 through 8.251, and substituting Equation 8.249 into Equation 8.248, these four partition submatrices may be written as

$$\boldsymbol{\sigma}_{\mathbf{x}\dot{\mathbf{x}}} + \boldsymbol{\sigma}_{\dot{\mathbf{x}}\mathbf{x}} = 0 \quad (8.253)$$

From Equations 8.251 and 8.253, we can see that the diagonal entries of matrices $\boldsymbol{\sigma}_{\mathbf{x}\dot{\mathbf{x}}}$ and $\boldsymbol{\sigma}_{\dot{\mathbf{x}}\mathbf{x}}$ are zeros.

$$\boldsymbol{\sigma}_{\dot{\mathbf{x}}\dot{\mathbf{x}}} - \boldsymbol{\sigma}_{\mathbf{x}\mathbf{x}} \mathbf{K}^T \mathbf{M}^{-T} - \boldsymbol{\sigma}_{\mathbf{x}\mathbf{x}} \mathbf{C}^T \mathbf{M}^{-T} = 0 \quad (8.254)$$

Taking the transpose of Equation 8.254 results in

$$\mathbf{M}^{-1}\mathbf{K}\boldsymbol{\sigma}_{\dot{\mathbf{x}}\mathbf{x}} - \mathbf{M}^{-1}\mathbf{C}\boldsymbol{\sigma}_{\dot{\mathbf{x}}\mathbf{x}} - \boldsymbol{\sigma}_{\dot{\mathbf{x}}\mathbf{x}} = 0 \quad (8.255)$$

In addition, we can further obtain

$$\mathbf{M}^{-1}\mathbf{K}\boldsymbol{\sigma}_{\dot{\mathbf{x}}\mathbf{x}} + \mathbf{M}^{-1}\mathbf{C}\boldsymbol{\sigma}_{\dot{\mathbf{x}}\mathbf{x}} + \boldsymbol{\sigma}_{\dot{\mathbf{x}}\mathbf{x}}\mathbf{K}\mathbf{M}^{-1} + \boldsymbol{\sigma}_{\dot{\mathbf{x}}\mathbf{x}}\mathbf{C}\mathbf{M}^{-1} = \mathbf{M}^{-1}\mathbf{D}\mathbf{M}^{-T} \quad (8.256)$$

From Equations 8.250 and 8.253, we have

$$\boldsymbol{\sigma}_{\dot{\mathbf{x}}\mathbf{x}} = -\boldsymbol{\sigma}_{\mathbf{x}\dot{\mathbf{x}}} \quad (8.257)$$

in which the $\boldsymbol{\sigma}_{\mathbf{x}\dot{\mathbf{x}}}$ is an antisymmetric matrix. Furthermore, from Equations 8.254 or 8.255 and Equation 8.256, we can write

$$\boldsymbol{\sigma}_{\mathbf{x}} = \boldsymbol{\sigma}_{\dot{\mathbf{x}}}\mathbf{M}^T\mathbf{K}^{-T} - \boldsymbol{\sigma}_{\dot{\mathbf{x}}\mathbf{x}}\mathbf{C}^T\mathbf{K}^{-T} \quad (8.258)$$

or

$$\boldsymbol{\sigma}_{\mathbf{x}\mathbf{x}} = \mathbf{K}^{-1}\mathbf{M}\boldsymbol{\sigma}_{\dot{\mathbf{x}}\mathbf{x}} - \mathbf{K}^{-1}\mathbf{C}\boldsymbol{\sigma}_{\dot{\mathbf{x}}\mathbf{x}} \quad (8.259)$$

Therefore, $\boldsymbol{\sigma}_{\mathbf{x}\mathbf{x}}$ can be obtained through $\boldsymbol{\sigma}_{\dot{\mathbf{x}}\mathbf{x}}$, $\boldsymbol{\sigma}_{\mathbf{x}\dot{\mathbf{x}}}$ and the corresponding matrix productions.

It is noted that the mass, damping, and stiffness matrices are symmetric. With the help of Equations 8.258 and 8.259, we can write

$$\mathbf{C}\boldsymbol{\sigma}_{\mathbf{x}\mathbf{x}}\mathbf{K} + \mathbf{K}\boldsymbol{\sigma}_{\mathbf{x}\mathbf{x}}\mathbf{C} + \mathbf{M}\boldsymbol{\sigma}_{\dot{\mathbf{x}}\mathbf{x}}\mathbf{K} - \mathbf{K}\boldsymbol{\sigma}_{\dot{\mathbf{x}}\mathbf{x}}\mathbf{M} = 0 \quad (8.260)$$

and premultiply and postmultiply $\mathbf{M}$ on both sides of Equation 8.256 yielding

$$\mathbf{K}\boldsymbol{\sigma}_{\dot{\mathbf{x}}\mathbf{x}}\mathbf{M} - \mathbf{M}\boldsymbol{\sigma}_{\dot{\mathbf{x}}\mathbf{x}}\mathbf{K} + \mathbf{M}\boldsymbol{\sigma}_{\dot{\mathbf{x}}\mathbf{x}}\mathbf{C} + \mathbf{C}\boldsymbol{\sigma}_{\dot{\mathbf{x}}\mathbf{x}}\mathbf{M} = \mathbf{D} \quad (8.261)$$

Equations 8.260 and 8.261 have a total of $(3n^2 + n)/2$ independent unknown variables and they provide $(3n^2 + n)/2$ totally independent equations. Therefore, in the $2n \times 2n$ matrix $\boldsymbol{\sigma}_{\mathbf{Y}\mathbf{Y}}(\tau)$ is solvable.

Example 8.10

Consider a 2-DOF system shown in Figure 8.1, where $f_2(t) = 0$.

In this case, $\mathbf{f}(t) = [f_1(t) \quad 0]^T$ and

$$\mathbf{D} = \begin{bmatrix} S_0 & 0 \\ 0 & 0 \end{bmatrix}$$

Denote

$$\begin{aligned}\boldsymbol{\sigma}_{xx} &= \begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{12} & \sigma_{22} \end{bmatrix} \boldsymbol{\sigma}_{\dot{x}\dot{x}} = \begin{bmatrix} 0 & -\sigma_{23} \\ \sigma_{23} & 0 \end{bmatrix} \\ \boldsymbol{\sigma}_{\dot{x}\dot{x}} &= \begin{bmatrix} 0 & \sigma_{23} \\ -\sigma_{23} & 0 \end{bmatrix} \boldsymbol{\sigma}_{xx} = \begin{bmatrix} \sigma_{33} & \sigma_{34} \\ \sigma_{34} & \sigma_{44} \end{bmatrix}\end{aligned}$$

Substitution of the above equations into Equations 8.260 and 8.261 results in the following four equations

$$(k_1 c_2 + k_2 c_1) \sigma_{23} + m_1 c_2 \sigma_{33} + [m_1(k_1 + k_2) - k_2 m_1] \sigma_{34} - m_2 k_2 \sigma_{44} = 0$$

$$-m_1 k_2 \sigma_{23} + 2m_1(c_1 + c_2) \sigma_{33} - 2m_1 c_2 \sigma_{34} = S_0$$

$$[m_1 k_2 - m_2(k_1 + k_2)] \sigma_{23} - m_1 c_2 \sigma_{33} + [m_1 c_2 + m_2(c_1 + c_2)] \sigma_{34} - m_2 c_2 \sigma_{44} = 0$$

as well as

$$2m_2 k_2 \sigma_{23} - 2m_2 c_2 \sigma_{34} + 2m_1 c_2 \sigma_{44} = 0$$

Specifically, denote

$$w_1 = \sqrt{\frac{k_1}{m_1}}, w_2 = \sqrt{\frac{k_2}{m_2}}, \zeta_1 = \frac{c_1}{2\sqrt{k_1 m_1}}, \zeta_2 = \frac{c_2}{2\sqrt{k_2 m_2}}, \mu = \frac{m_2}{m_1} \text{ and } r = \frac{w_2}{w_1}$$

in which w_i and ζ_i are used for mathematical convenience: they are not exactly the natural frequency and damping ratio. The above equations can be replaced by

$$2rw_1(r\zeta_1 + \zeta_2)\sigma_{23} + r^2\sigma_{33} + (\mu - 1)r^2\sigma_{34} - \mu r^2\sigma_{44} = 0$$

$$-w_1\mu r^2\sigma_{23} + 2(\zeta_1 + \zeta_2 r\mu)\sigma_{33} - 2\mu r\zeta_2\sigma_{34} = S_0/2m_2 w_2$$

$$-w_1[1 + (\mu - 1)r^2]\sigma_{23} - 2\zeta_2 r\sigma_{33} + 2[\zeta_1 + (1 + \mu)r\zeta_2]\sigma_{34} - 2\mu r\zeta_2\sigma_{44} = 0$$

$$w_1 r\sigma_{23} - 2\zeta_2\sigma_{34} + 2\zeta_2\sigma_{44} = 0$$

We therefore have

$$\sigma_{23} = -A[2rw_1\zeta_2(r\zeta_1 + \zeta_2)]$$

$$\sigma_{33} = -Aw_1^2 \left\{ \mu r^3 \zeta_1 + \zeta_2 [1 - 2r^2 + (1 + \mu)r^4] + 4r\zeta_1 \zeta_2^2 (1 + r^2) + 4r^2 \zeta_1^2 \zeta_2 + 4r^2 \zeta_2^3 \right\}$$

$$\sigma_{34} = -Ar^2w_1^2 \left\{ \zeta_2 \left[(1+\mu)r^2 - 1 \right] + 4r\zeta_1\zeta_2^2(1+r^2) + 4\zeta_2^3 \right\}$$

and

$$\sigma_{44} = -Ar^2w_1^2 \left[r\zeta_1 + \zeta_2(1+\mu)r^2 + 4r\zeta_1\zeta_2^2 + 4\zeta_2^3 \right]$$

where

$$A = \frac{S_0/m_1}{4w_1^3B}$$

and

$$B = \mu r^3\zeta_1^2 + \mu r\zeta_2^2 + \left[1 - 2r^2 + (1+\mu)^2r^2 \right] \zeta_1\zeta_2 + 4r^2\zeta_1\zeta_2 \left[\zeta_1^2 + (1+\mu)^2\zeta_2^2 \right] + 4r\zeta_1^2\zeta_2^2 \left[1 + (1+\mu)r^2 \right]$$

Substituting the above solutions into Equation 8.256, we can finally solve σ_{11} , σ_{12} , and σ_{13} as given below:

$$\sigma_{11} = (\sigma_{11} + \mu\sigma_{11})/w_1^2$$

$$\sigma_{12} = (\sigma_{34} + \mu\sigma_{44})/w_1^2 - 2\zeta_1\sigma_{23}/w_1$$

$$\sigma_{13} = [\sigma_{34} + (1+\mu r^2)/r^4\sigma_{44}]/w_1^2 + 2(\zeta_2 - \mu\zeta_2)\sigma_{23}/(w_1r)$$

8.4.4.3.4 Steady State Covariance under Color Noise Excitation

Next, consider the excitation to be stationary color noise, which can be seen as a solution of Equation 7.100 in Chapter 7. In this case, for multiple input, the scalar equation is extended to a vector equation given by

$$\frac{d}{dt}\mathbf{f}(t) = -\boldsymbol{\eta}\mathbf{f}(t) + \boldsymbol{\Gamma}w(t) \quad (8.262)$$

where

$$\boldsymbol{\eta} = \text{diag}(\eta_i) \text{ and } \boldsymbol{\Gamma} = \text{diag} [(2\eta_i)^{1/2}] \quad (8.263)$$

Equations 8.262 together with the state equation of the motion given by Equation 8.219 can be combined as

$$\begin{Bmatrix} \dot{\mathbf{x}}(t) \\ \dot{\mathbf{x}}(t) \\ \dot{\mathbf{f}}(t) \end{Bmatrix} = \begin{bmatrix} \mathbf{0} & \mathbf{I} & \mathbf{0} \\ -\mathbf{M}^{-1}\mathbf{K} & -\mathbf{M}^{-1}\mathbf{C} & \mathbf{M}^{-1} \\ \mathbf{0} & \mathbf{0} & -\boldsymbol{\eta} \end{bmatrix} \begin{Bmatrix} \mathbf{x}(t) \\ \dot{\mathbf{x}}(t) \\ \mathbf{f}(t) \end{Bmatrix} + \begin{Bmatrix} \mathbf{0} \\ \mathbf{0} \\ \boldsymbol{\Gamma} \end{Bmatrix} w(t) \quad (8.264)$$

where $\mathbf{I}$ and $\mathbf{0}$ are, respectively, identity and null submatrices with proper dimensions.

Denoting

$$\mathbf{Z}(t) = \begin{Bmatrix} \mathbf{x}(t) \\ \dot{\mathbf{x}}(t) \\ \mathbf{f}(t) \end{Bmatrix}, \hat{\mathcal{A}} = \begin{bmatrix} \mathbf{0} & \mathbf{I} & \mathbf{0} \\ -\mathbf{M}^{-1}\mathbf{K} & -\mathbf{M}^{-1}\mathbf{C} & \mathbf{M}^{-1} \\ \mathbf{0} & \mathbf{0} & -\boldsymbol{\eta} \end{bmatrix} \text{ and } \hat{\mathcal{B}} = \begin{Bmatrix} \mathbf{0} \\ \mathbf{0} \\ \boldsymbol{\Gamma} \end{Bmatrix} \quad (8.265)$$

we have the state equation written as

$$\dot{\mathbf{Z}}(t) = \hat{\mathcal{A}}\mathbf{Z}(t) + \hat{\mathcal{B}}w(t) \quad (8.266)$$

The covariance matrix in this circumstance is given by

$$\boldsymbol{\sigma}_{\mathbf{ZZ}} = E[\mathbf{Z}(t)\mathbf{Z}^T(t)] = E \begin{bmatrix} \mathbf{xx}^T & \mathbf{x}\dot{\mathbf{x}}^T & \mathbf{x}\mathbf{f}^T \\ \dot{\mathbf{x}}\mathbf{x}^T & \dot{\mathbf{x}}\dot{\mathbf{x}}^T & \dot{\mathbf{x}}\mathbf{f}^T \\ \mathbf{f}\mathbf{x}^T & \mathbf{f}\dot{\mathbf{x}}^T & \mathbf{f}\mathbf{f}^T \end{bmatrix} \quad (8.267)$$

Because $w(t)$ is white noise, we can directly use the result obtained in the above subsection, that is,

$$\hat{\mathcal{A}}\boldsymbol{\sigma}_{\mathbf{ZZ}} + \boldsymbol{\sigma}_{\mathbf{YY}}\hat{\mathcal{A}}^T = -\hat{\mathcal{B}}\hat{\mathbf{D}}\hat{\mathcal{B}}^T \quad (8.268)$$

where $\hat{\mathbf{D}} = \text{diag}(S_{0i})$, and S_{0i} is the corresponding magnitude of PSD of white noise applied at the i th location.

8.4.4.4 Brief Summary

Nonproportional damping is caused by the different distributions of damping and stiffness, which is common in practical engineering applications. In this circumstance, certain or total modal shapes will become complex-valued. In addition, we will witness nonzero modal energy transfer ratios.

Nonproportionally damped structures will not have principal axes, which may alter the peak responses of a structure under seismic ground excitations and, in many cases, the peak responses will be enlarged (Liang and Lee 1998). In this subsection, how to account for nonproportionally damped systems is discussed. The basic idea is to use the state space and the state matrix (Gupta and Jaw 1986; Song et al. 2007a,b).

8.5 MODAL COMBINATION

8.5.1 REAL VALUED MODE SHAPE

8.5.1.1 Approximation of Real Valued Mode Shape

The following computations of variance and RMS are based on modal combinations. The modes can be either normal or complex. For normal modes, ϕ_{ji} represents the j th element of the i th mode shape.

For complex modes,

$$\phi_{ji} = (-1)_{ji}^{\delta} |p_{ji}| \quad (8.269)$$

where

$$\delta_{ji} = \begin{cases} 0, & -\frac{\pi}{2} < \theta_{ji} \leq \frac{\pi}{2} \\ 1, & \frac{\pi}{2} < \theta_{ji} \leq \frac{3\pi}{2} \end{cases} \quad (8.270)$$

As a result, the newly simplified i th mode shape vector can be written as

$$\phi_i = \begin{Bmatrix} \phi_{1i} \\ \phi_{2i} \\ \dots \\ \phi_{ni} \end{Bmatrix} \quad (8.271)$$

The newly simplified mode shape matrix can be written as

$$\Phi = [\phi_1 \quad \phi_2 \quad \dots \quad \phi_n] \quad (8.272)$$

8.5.1.2 Linear Dependency and Representation

8.5.1.2.1 Linear Independent

Using the above approximation will result in n mode shape functions. If the MDOF system is proportionally damped, then the corresponding mode shape matrix is of full rank. This is represented as

$$\text{rank}(\Phi) = n \quad (8.273)$$

It is noted that, when using the approximation as described in Equation 8.269, the full rank condition represented in Equation 8.273 is not necessarily satisfied. This is practically true when a complex mode shape is simplified into a real valued approximation. In this situation, the full rank condition should be checked to see if Equation 8.273 holds.

8.5.1.2.2 Full Rank Representation

Suppose Equation 8.273 is satisfied, then all the column vector ϕ_i will be linear independent and the inverse of matrix Φ will exists. Specifically, there will always be a matrix Ψ such that

$$\Psi = \Phi^{-1} \quad (8.274)$$

In this case, any $n \times 1$ vector can be represented by matrix Φ . Now suppose there is a vector $\mathbf{y}$, such that

$$\mathbf{y} = \begin{Bmatrix} y_1 \\ y_2 \\ \dots \\ y_n \end{Bmatrix} \quad (8.275)$$

Here, $\mathbf{y}$ can be written as

$$\mathbf{y} = a_1 \phi_1 + a_2 \phi_2 \dots + a_n \phi_n \quad (8.276)$$

In the above equation, instances of a_i are scalars to be determined. Thus, to use this mode shape vector to represent Y , all values of scalar a_i must be determined. Namely,

$$\begin{Bmatrix} a_1 \\ a_2 \\ \dots \\ a_n \end{Bmatrix} = \Psi \mathbf{y} \quad (8.277)$$

8.5.1.2.3 Truncated Representation

In special cases, we may not have n mode shapes. In that event, only m modes exist and $m < n$, then Equation 8.276 will become (Liang et al. 2012)

$$\mathbf{y} \approx a_1 \phi_1 + a_2 \phi_2 \dots + a_m \phi_m \quad (8.278)$$

or

$$\mathbf{y} \approx \Phi_C \mathbf{a}_C \quad (8.279)$$

where the truncated mode shape matrix is

$$\Phi_C = [\phi_1, \phi_2 \dots \phi_m] \quad (8.280)$$

and

$$\mathbf{a}_C = \begin{Bmatrix} a_1 \\ a_2 \\ \dots \\ a_m \end{Bmatrix} \quad (8.281)$$

In the above case, all the parameters a_i can be determined through a least square approach. Explicitly, denote

$$\mathbf{e} = \mathbf{y} - a_1\boldsymbol{\Phi}_1 + a_2\boldsymbol{\Phi}_2 \dots + a_m\boldsymbol{\Phi}_m \quad (8.282)$$

and let

$$\mathbf{e}^T \mathbf{e} = \min \quad (8.283)$$

To achieve Equation 8.280, consider the equation

$$\frac{\partial \mathbf{e}^T \mathbf{e}}{\partial a_i} = 0, \quad i = 1, \dots, m \quad (8.284)$$

Equation 8.281 will result in

$$\mathbf{a}_C = (\boldsymbol{\Phi}_C^T \boldsymbol{\Phi}_C)^{-1} \boldsymbol{\Phi}_C^T \mathbf{y} \quad (8.285)$$

8.5.2 NUMERICAL CHARACTERISTICS

In the following examples, general references can be found in the book of Wirsching et al. (1995).

8.5.2.1 Variance

Through modal analysis, the variance of randomly forced response can be estimated by

$$\sigma_{x_j}^2 = D(x_j) = D \left(\sum_{i=1}^S \phi_{ji} q_i \right) \quad (8.286)$$

In this instance, S is the number of truncated modes.

8.5.2.2 Root Mean Square

In the following, specific formulae will be listed to estimate the root mean square value of the response taken from a signal location j , without detailed derivations.

8.5.2.2.1 Square Root of the Sum of Square

Suppose all modal response are widely separated. That is, they do not interfere with each other, then (Singh 1980)

$$\sigma_{x_j} = \sqrt{\sum_{i=1}^S (\phi_{ji} q_i)^2} \quad (8.287)$$

or

$$\sigma_{x_j} = \sqrt{\sum_{i=1}^S (\phi_{ji})^2 \sigma_i^2} \quad (8.288)$$

In this instance, σ_i^2 is the mean square value of the i th modal response $q_i(t)$.

8.5.2.2.2 Absolute Method

Let us assume that all modal responses have a correlation coefficient represented by

$$\rho = \pm 1 \quad (8.289)$$

Equation 8.279 is used to imply the case of linear dependency. Here, $q_i(t_1)$ and $q_i(t_2)$ are correlated, meaning that they vary with the same pattern.

$$\sigma_{x_j} = \sum_{i=1}^n |\phi_{ji} \sigma_i| \quad (8.290)$$

Equation 8.290 is the equation of standard deviation, where σ_i is the standard deviation of the i th modal response.

8.5.2.2.3 Naval Research Laboratory Method

Assuming the first modal response can be correlated with the square root of the sum of square (SRSS) of the rest of the modes, then

$$\sigma_{x_j} = |\phi_{j1} \sigma_1| + \sqrt{\sum_{i=2}^S (\phi_{ji} \sigma_i)^2} \quad (8.291)$$

8.5.2.2.4 Closed-Space Modes

In certain systems, some modes are in a relatively closed space. These are referred to as closed-space modes. The difference between closed-space modes and modes that are separated is conceptually shown in Figure 8.6. Consider the case where there is one set of closed-space modes. This case can be represented in the equation:

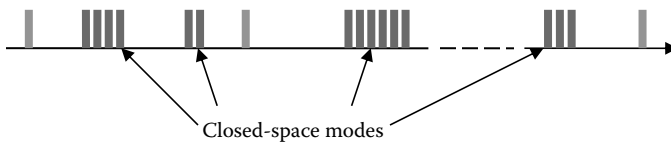


FIGURE 8.6 Closed space modes.

$$\sigma_{X_j} = \sum_{j=1}^Z |\phi_{ji} \sigma_i| + \sqrt{\sum_{j=Z+1}^{n-Z} (\phi_{ji} \sigma_i)^2} \quad (8.292)$$

Here, Z is the total number of modes deemed to be close.

8.5.2.2.5 Modified Square Root of the Sum of Square Method

If more than one group of closed-space modes exists, then the following equation is used:

$$\sigma_{X_j} = \sqrt{\sum_{p=1}^H \left[\sum_{i=1}^p |\phi_{ji} \sigma_i| \right]^2} + \sum_{i=S+1}^{n-S} (\phi_{ji} \sigma_i)^2 \quad (8.293)$$

In this instance, H is the number of sets of equal or close eigenvalues and p is the number of close modes in a given set (Richard et al. 1988).

8.5.3 COMBINED QUADRATIC COMBINATION

If the cross-effect among modes needs to be taken into account, then the method of combined quadratic combination (CQC) should be used (Der Kiureghian 1980, 1981; Sinha and Igusa 1995):

$$\sigma_{X_j} = \sqrt{\sum_{i=1}^n \sum_{k=1}^n (\sigma_{ji} \rho_{ik} \sigma_{kj})} \quad (8.294)$$

For Equation 8.294,

$$\sigma_{ji} = \phi_{ji} \sigma_i \quad (8.295)$$

and

$$\rho_{ik} = \frac{8\sqrt{\zeta_i \zeta_k} (\zeta_i + r \zeta_k) r^{3/2}}{(1-r^2)^2 + 4\zeta_i \zeta_k r(1+r^2) + 4(\zeta_i^2 + \zeta_k^2) r^2} \quad (8.296)$$

In the event that the i th and k th modes are normal, then

$$r = \frac{\omega_{nk}}{\omega_{ni}}, \quad k > i \quad (8.297)$$

For the case of complex modes,

$$r = \frac{\omega_k}{\omega_i}, \quad k > i \quad (8.298)$$

PROBLEMS

1. A dynamic absorber can be used to reduce the vibrations of an SDOF system subjected to sinusoidal excitation $f(t) = F_0 \cos(\omega t)$. Shown in Figure P8.1, the blue $m-k$ system is the primary SDOF system and the red m_a and k_a are additional mass and stiffness. Therefore, with the additional mass, the system becomes 2-DOF.

Denote $\omega_p = (k/m)^{1/2}$, $\omega_a = (k_a/m_a)^{1/2}$, and $\mu = m_a/m$.

- a. Show that the dynamic magnification factor β_{dyn} for the primary displacement can be written as

$$\beta_{\text{dyn}} = \frac{Xk}{F_0} = \frac{1 - \frac{\omega^2}{\omega_a^2}}{\left[1 - \mu \left(\frac{\omega_a^2}{\omega_p^2} \right) - \frac{\omega^2}{\omega_p^2} \right] \left[1 - \frac{\omega^2}{\omega_p^2} \right] - \mu \left(\frac{\omega_a^2}{\omega_p^2} \right)}$$

- b. Suppose $m = 6000$ kg and the resonant driving frequency = 60 Hz. The mass of the absorber is chosen to be 1200 kg. Determine the range of frequencies within which the displacement $x(t)$ is less with the additional mass than without the additional mass.

2. Given

$$\mathbf{M} = \begin{bmatrix} 4 & & & \\ & 4.6 & & \\ & & 3 & \\ & & & 5 \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} 3 & -2 & 0 & 0 \\ -2 & 4 & -2 & \\ & -2 & 5 & -3 \\ & & -3 & 5 \end{bmatrix}, \text{ and}$$

$$\mathbf{K} = \begin{bmatrix} 500 & -200 & & \\ -200 & 600 & -400 & \\ & -400 & 550 & -150 \\ & & -150 & 500 \end{bmatrix}$$

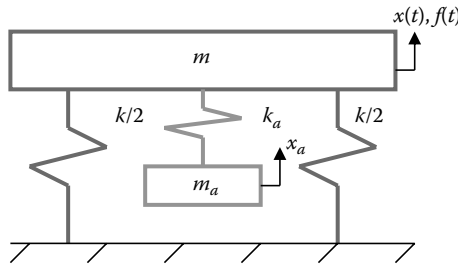


FIGURE P8.1

- Check if the system is proportionally damped and calculate the damping ratios and natural frequencies
- Generate a proportionally damped system with damping matrix $\mathbf{C}_{\text{new}}$ such that it can have the identical natural frequencies and damping ratio as the system by using the eigenvector matrix of $\mathbf{K}$ matrix
- Suppose this system is excited by ground acceleration $10 \sin(\omega_1 t) + 2 \sin(\omega_2 t + 0.5)$ and have the displacement resonant at the first and the acceleration second natural frequencies, what will these driving frequencies be?
- To reduce the response of displacement of x_1 of 30%, how should you increase the damping matrix by adding $\Delta \mathbf{C}$, namely, find c in

$$\mathbf{C} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & c \end{bmatrix}$$

- With the mass, damping, and stiffness matrices given by Problem 2, using MATLAB to generate 30 random ground motions: $t = (0:1:1,999) * 0.01$; $xga = \text{randn}(2000,30)$ with zero initial condition. If, on average, the displacement of x_4 is needed to reduce 30%, how can you choose additional stiffness $\Delta \mathbf{K}$, namely, find k in

$$\Delta \mathbf{K} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & k \end{bmatrix}$$

- A 2DOF system is shown in Figure P8.2 with a force being white noise process applied on mass 1. Knowing $m_1 = m_2 = 1$, $k_1 = k_2 = 100$, $c_1 = c_2 = 2$, find the equation of motion, the transfer function; using the normal mode method to find the mean and covariance of the displacement.
- The system shown in Figure P8.2 is excited by ground white noise motion, where $m_1 = m_2 = 1$, $k_1 = k_2 = 100$, $c_1 = 18$ and $c_2 = 0$. Find the transfer

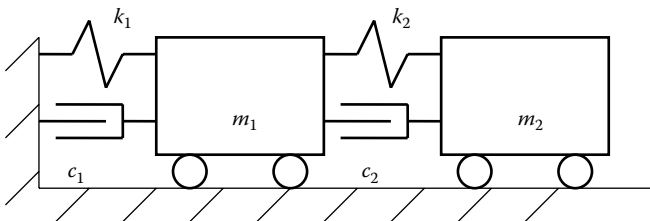


FIGURE P8.2

function. Calculate the mean and covariance by using the complex mode method.

6. Derive a general formula of mean value for a nonproportional system under excitation of random initial conditions only.
7. In the system given by Figure P8.3; c_1 and c_2 are, respectively, zero k_1 and k_2 are, respectively, 500 and 200; $m_1 = 1500$. $m_2 = 50 + \Delta m$ where Δm is a random variable with the following distribution:

Δm	0	30	60	200
p	1/2	2/6	1/6	1/6

Suppose the ground excitation is white noise with PSD = 1, find the distribution of the RMS absolute acceleration of m_2 .

8. Prove that for proportionally damped systems with white noise excitations, the first entry of $\sigma_{xx}(t, t)$ can be calculated as

$$\begin{aligned} & \frac{\phi_{11}^4 D_1 + \phi_{11}^2 \phi_{21}^2 D_2}{\omega_{d1}^2 m_1^2} \int_0^\infty e^{-2\zeta_1 \omega_{n1}(t-\tau)} \sin^2 \omega_{d1}(t-\tau) d\tau \\ & + \frac{2\phi_{11}^2 \phi_{21}^2 D_1 + 2\phi_{11} \phi_{12} \phi_{21} \phi_{22} D_2}{\omega_{d1} \omega_{d2} m_1 m_2} \int_0^\infty e^{-(\zeta_1 \omega_{n1} + \zeta_2 \omega_{n2})(t-\tau)} \sin \omega_{d1}(t-\tau) \sin \omega_{d2}(t-\tau) d\tau \\ & + \frac{\phi_{12}^4 D_1 + \phi_{12}^2 \phi_{22}^2 D_2}{\omega_{d2}^2 m_2^2} \int_0^\infty e^{-2\zeta_2 \omega_{n2}(t-\tau)} \sin^2 \omega_{d2}(t-\tau) d\tau \end{aligned}$$

9. For a system given by Figure P8.3 with
 - a. Using the method of SRSS to calculate the variance of x_1
 - b. Using the method of CQC to calculate the variance of x_1
 - c. Using the method of closed-space modes to calculate the variance of x_1
 - d. Compare and explain your results

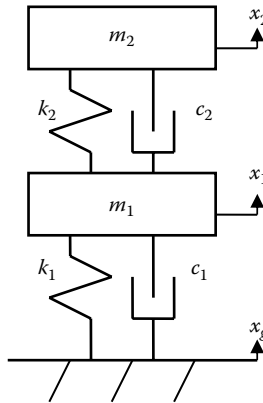


FIGURE P8.3

10. An MDOF system with $\mathbf{M} = \text{diag} ([1 \quad 1.5 \quad 2])$

$$\mathbf{C} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 3 & -2 \\ 0 & -2 & 2 \end{bmatrix} \text{ and } \mathbf{K} = \begin{bmatrix} 150 & -50 & 0 \\ -50 & 100 & -50 \\ 0 & -50 & 70 \end{bmatrix}$$

is excited by forcing function

$$\mathbf{f}(t) = \begin{Bmatrix} 0 \\ 2 \\ 1.5 \end{Bmatrix} w(t)$$

where $w(t)$ is a white noise with PSD = 10. Calculate the covariance matrix $\boldsymbol{\sigma}_{YY}(0)$.

Section IV

*Applications and
Further Discussions*

9 Inverse Problems

In this chapter and Chapters 10 and 11, we present several topics by applying the knowledge gained previously. These topics do not cover the total applications of random process and random vibrations. They may be considered as “practical” applications and utilizations of the concept of the random process. We will also discuss methods to handle engineering problems that are difficult to be treated as closed-form mathematical models and/or difficult to be approximated as stationary processes.

Inverse problems are a relatively broad topic. The field of inverse problems was first discovered and introduced by Viktor Ambartsumian (1908–1996). The inverse problem is a general framework used to convert measured data into information about a physical object or system, which has broad applications. One of the difficulties in solution of inverse problems is due to the existence of measurement noises. In other words, when working with inverse problems, both random variables as well as random processes must be considered. In this chapter, inverse problems related to vibration systems will be briefly outlined. Additionally, key issues in system identifications as well as vibration testing will be discussed. Special emphasis will be given to measurement uncertainties.

For more detailed description of inverse problems, readers may consult the works of Chadan and Sabatier (1977), Press et al. (2007), and Aster et al. (2012).

9.1 INTRODUCTION TO INVERSE PROBLEMS

9.1.1 CONCEPT OF INVERSE ENGINEERING

In Chapter 6, inverse problems as related to vibration systems were introduced. In engineering applications, inverse problems can be rather complex. It is necessary to identify projects with inverse problems in order to avoid treating forward problems as inverse problems. However, in certain applications, it may be necessary to solve inverse problems.

9.1.1.1 Key Issues

The key issues involved in inverse problems are listed as follows.

9.1.1.1.1 Modeling Modeling as a fundamental approach is often a starting point. Generally, the following should be considered:

1. The type of models—Is it static or dynamic? Is it linear or nonlinear?
2. The number of degrees of freedom (DOFs) or order of models.
3. The type of damping—Is it proportionally or nonproportionally damped?

9.1.1.1.2 *Boundary Conditions*

Boundary conditions are key issues, yet are difficult to be accurately defined. Identical models with different boundary conditions can have very different modal parameters and physical responses. The types of boundary conditions and the time varying and nonlinearity of the conditions need to be identified.

9.1.1.1.3 *Testing*

To solve an inverse problem, conducting vibration testing is often needed. In general, vibration testing includes (1) actuation and measurement, (2) S/N ratios, (3) data management and analysis, and (4) test repeatability and reliability.

9.1.1.1.4 *Regression and Realization*

Quite often, parameters are obtained through regression procedures for analytical and empirical models. When using regression procedures, the following must be considered: (1) the criteria of regression or realizations, (2) computational algorithms, (3) the stability and robustness of the identified model, and (4) error and sensitivity analysis involved in the regression.

9.1.1.2 **Error**

In solving inverse problems, a range of errors are inevitable. It is essential that these errors be reduced. To rigorously define errors is difficult. Therefore, errors can approximately be classified as follows:

1. Bias or systematic errors
 - Repeated errors regardless of the test conditions
 - Errors due to incorrect calibrations
 - Errors due to inaccurate modeling
 - Errors due to loading nonlinearity
 - Errors due to insufficient measurement resolution
 - Errors due to limited measurement dynamic range
 - Errors due to insufficient testing duration or insufficient frequency response
2. Precision or random errors
 - Errors due to random ambient conditions
 - Human errors
 - Errors due to power leakage
 - Errors due to insufficient analog-to-digital convention (A/D)
 - Errors due to environmental noises
3. Accident or illegitimate errors
 - Incorrect modeling
 - Mistakes
 - Unexpected accident
 - Chaotic measurement conditions
 - Incorrect numerical simulations

The action needed to improve the procedure is dependent on the type of error that occurred. In the following, the nature of errors and the corresponding improvement will be discussed as it mainly relates to random problems.

9.1.1.3 Applications

Inverse problems have multiple applications such as

1. System identification
2. Trouble shooting
3. Design modification
4. Model confirmation

9.1.2 ISSUES OF INVERSE PROBLEMS

We now briefly consider important issues of inverse problems.

9.1.2.1 Modeling

Quite often, for one natural phenomenon, there may be more than one model. For example, suppose that 50 random data values y are measured and indexed from 1 to 50 by the variable x . It can be assumed that the relationship between y and x is linear or of first order, i.e., $y = ax + b$. Through the first-order regression, the parameters a and b can be determined. The quadratic form or second order, i.e., $y = ax^2 + bx + c$, can also be used, finding the parameters a , b , and c . This can be repeated for the third order, fourth order, and so on.

In Figure 9.1a, plots of the original model and several regressed models are shown, including the first-order (linear) regression, the second-order (quadratic) regression, and the third-order regression. While the models are dissimilar, all of them regressed using the same original data. In Figure 9.1a, it is seen that the second- and third-order regressions are rather close. This, however, does not necessarily mean that when the regression order is chosen to be sufficiently high, the models will converge.

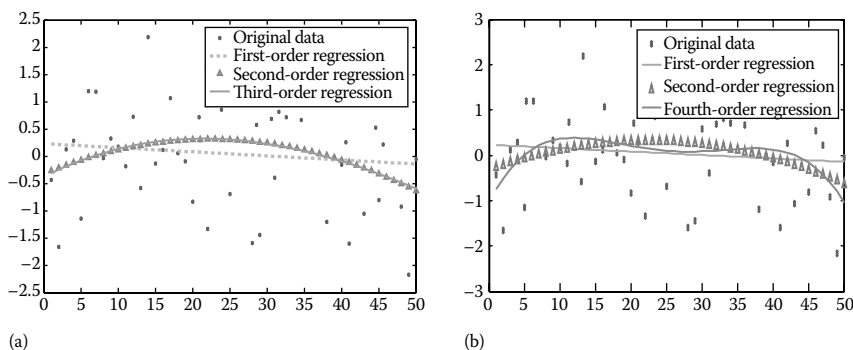


FIGURE 9.1 Directly measured data and regressed models. (a) Regression including third-order approach. (b) Regression including fourth-order approach.

In Figure 9.1b, the same original data are shown, with the first- and second-order regressions. However, instead of the third-order regressed model, the fourth-order model is plotted. Instead of the fourth-order regressed model being rather close to the second-order regression model, it is shown to be significantly different.

The above example indicates that one should be very careful to use an a priori model.

9.1.2.1.1 *Physical Model*

There are several kinds of models. The most commonly used model is the physical model, which is established based on mechanical analysis.

A static model, which can be either linear or nonlinear, is often described by algebraic equations, which is time invariable.

A dynamic model, which can also be either linear or nonlinear, is commonly described by differential equations. For example, the M-C-K model, with either lumped or finite-element parameters, is a second-order ordinary differential equation in matrix form.

The first inverse problem is completely solved after the physical model is confirmed.

9.1.2.1.2 *Response Model*

The second most commonly used model is the response model. The direct response, by definition, consists of the response time histories. As described in previous chapters (started from Section 4.2.2.1.2), a response is resulted from a convolution of an impulse response function and a forcing function. Therefore, the response time history will contain information of both the dynamic behavior of a given vibration system and information of excitations. However, because the amount of data of response time histories is often overwhelmingly large, this model is easier to measure but difficult to execute.

When excitation is a random process, the response model is also a random process. When the excitation is deterministic, mathematically, the response is often modeled as a deterministic process. Furthermore, due to various levels of noise contaminations, the time history will likely be a random process as well.

9.1.2.1.3 *Modal Model*

Chapter 8 has shown the transfer of the M-C-K model into the modal domain using eigenvector matrices, and the use of the set of natural frequencies, damping ratios, and mode shapes to represent a multidegree-of-freedom (MDOF) system (the modal model). There are multiple advantages in using the modal model. The amount of information in the modal model, compared to the physical model, particularly the response model, is greatly reduced. Additionally, in computing the possible response of the MDOF system, the modal model typically provides a more accurate result. Also, in many cases, for an existing MDOF vibration system, an exact physical model will not exist. Consequently, through modal testing, a significantly accurate modal model can be achieved.

The modal model can be extracted from a physical model; however, a physical model often cannot be obtained from the modal model. In other words, in comparison to the physical and response models, the modal model is a dimension-reducing model.

Generally speaking, the modal model is only practical for a linear system, which can be either in normal mode or complex mode, dependent upon the natural of damping.

9.1.2.1.4 *Input–Output (Force–Response) Model*

In addition to the above-mentioned three basic models, there also exists the input–output model. From the viewpoint of system theory, a response model is an output model. Practically, due to inevitable noises, solely using the response to precisely determine the modal or physical parameters is a challenging task. By also using the input, the transfer function can be measured, which, in turn, provides a greater understanding of the dynamic behaviors of a system. Lastly, it is noted that the input–output model can be either static or dynamic.

9.1.2.1.5 *Geometric Model*

The geometric model used to describe the topology configuration of an engineering object mainly consists of dimensions and drawings, i.e., an AutoCAD drawing. In some instances, the motion mechanisms of the geometric model may need to be specified.

9.1.2.1.6 *Statistical Models*

A statistical model is established through statistical survey. A commonly used model consists of simple statistical parameters, such as numerical characteristics (mean values, variance, etc.) and distribution functions. Neural networks and fuzzy logic are other examples of the statistical model.

9.1.2.1.7 *Other Models*

In addition to the above models, other types of models exist, such as the pseudo-dynamic model and the real-time hybrid model, among others.

9.1.2.2 **Identification, Linear System**

The first inverse problem is to identify the best-fit model. For linear MDOF vibration systems, system identification can be either in the physical or the modal model. The targets include (1) deterministic parameters, such as M , C , and K , modal parameters, transfer functions, and coherence functions; (2) degree of linearity of the system; (3) order of the system; and (4) stability of the system, such as time invariant or variant.

9.1.2.2.1 *Input Identifications*

The second inverse problem is to identify the best-fit input, such as forcing functions, ground excitations in addition to possible noises.

9.1.2.2.2 *System Monitoring*

Three major applications, (1) system health monitoring, (2) nondestructive evaluation, and (3) damage assessment of system monitoring, have received more attention in the past decades.

9.1.2.3 **Identification, General System**

A system can often be nonlinear and/or time variant. Identification of such systems may consist of (1) linearity range, (2) similitude analysis such as dimensionless analysis and scaling effect, and (3) nonlinear models such as nonlinear stiffness, nonlinear damping, and geometric nonlinearity.

9.1.2.3.1 *Nonlinear Dynamics*

Nonlinear dynamics, specifically random dynamics, is one of the most difficult fields to handle. Nevertheless, it is often inevitable. Linearization is commonly used in nonlinear dynamics analysis. Separation of the total duration of a nonstationary process into limited pieces of “relative” stationary process is another commonly used measure. This separation is not limited to the time domain. A nonlinear deformation can also be divided into piecewise linear ranges.

9.1.2.3.2 *Material Identification*

One more inverse problem is that of material identification. For this case, the strength of materials is measured and identified to determine elastic or inelastic stress–strain relations, fatigue and aging problems, the surface and contact mechanics, as well as other properties. For example, properties such as chemical components, chemical functions, and conditions of rust and corrosion can be determined.

9.1.2.4 **Simulations**

Simulation technologies are also based on inverse problems, although a simulation is generally considered a forward problem.

Numerical simulations and physical simulations are two basic approaches. In general, the success of a simulation will not depend solely upon the stability and effectiveness of computational algorithms and the precision of test apparatus; it will also depend on the accuracy of the models.

9.1.2.5 **Practical Considerations**

In engineering applications, the following issues are important.

9.1.2.5.1 *Accuracy of Models*

Accuracy of models is perhaps the most important issue in modeling. It is not only the starting point but also the critical task through the entire problem-solving project. Very often, an iterative approach, known as a trial-and-error approach, is needed. The criterion in judging that iteration converges as well as verges to the right point is often needed. This should be independent to establishing the model.

9.1.2.5.2 *Measurement S/N Ratio*

In vibration testing, having a sufficiently high S/N ratio is often an unavoidable step in solving inverse problems. Correct modeling, effective testing and precise measurement, sophisticated signal processing, and accurate parameter extraction all contribute to an increased S/N ratio.

9.1.2.5.3 *Randomness and Uncertainty*

Measurements must be taken in order to manage the randomness of data and the uncertainty of parameter identification. This is another key issue in solving inverse problems. Adequate sampling sizes and correct statistical surveys are necessary to eliminate this uncertainty. Methods to judge the randomness and uncertainty are often needed.

In the following, the above issues will be considered in more detail.

9.1.3 THE FIRST INVERSE PROBLEM OF DYNAMIC SYSTEMS

The measurement of transfer functions, the key issue of identification for dynamic systems, is examined first.

9.1.3.1 General Description

The first inverse problem is often solved through a two-step approach. The first step is to obtain the transfer function. The second step is to extract the modal or physical model through the transfer function.

Figure 9.2 repeats the expression of the relationship among the input–system–output, the fundamental approach of the inverse problem of systems.

The formula can be repeated to calculate the transfer function, which can be obtained by one of two ways. The transfer function can be obtained through the ratio of Fourier transforms of output and input:

$$H(\omega) = \frac{X(\omega)}{F(\omega)} = \frac{\text{Fourier transform of output}}{\text{Fourier transform of input}} \quad (9.1)$$

As noted earlier, the transfer functions can also be obtained through the power spectral density functions:

$$H(\omega) = \frac{X(\omega)X(\omega)^*}{F(\omega)X(\omega)^*} = \frac{S_X(\omega)}{S_{XF}(\omega)} = \frac{\text{auto PSD of output}}{\text{cross PSD of input–output}} \quad (9.2)$$

and

$$H(\omega) = \frac{X(\omega)F(\omega)^*}{F(\omega)F(\omega)^*} = \frac{S_{FX}(\omega)}{S_F(\omega)} = \frac{\text{cross PSD of output–input}}{\text{auto PSD of input}} \quad (9.3)$$

Here, the uppercase letters represent the Fourier transforms. The temporal pairs of the Fourier transforms are not necessarily random.

Equation 9.1, the definition of transfer function, is seldom used practically, especially in the case of random excitations. The more practical choice is the method through power spectral density functions, such as H_1 and H_2 , since it can provide more accurate and stable estimations.

Extraction of modal and/or physical parameters requires certain in-depth knowledge, which is beyond the scope of random process and vibration. Interested readers may

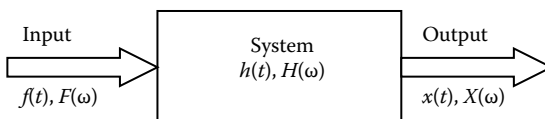


FIGURE 9.2 System and input–output.

consult the work of Ewins (2000) or He and Fu (2004) for more detailed descriptions. In this chapter, a list of fundamental formulas in the frequency and time domains will be provided. The most commonly used method for estimations of transfer functions is first summarized.

9.1.3.2 Impulse Response

Consider the impulse response for the signal degree-of-freedom (SDOF) system:

$$H(\omega) = \mathcal{F}[h(t)] \quad (9.4)$$

Now consider the MDOF system with an input at the j th location that is measured at the i th location:

$$H_{ij}(\omega) = \mathcal{F}[h_{ij}(t)] \quad (9.5)$$

Note that the impulse response function is a normalized response $x(t)$ with respect to the amplitude of the impact force given by

$$h(t) = \frac{x(t)}{f_{\max}} \quad (9.6)$$

9.1.3.3 Sinusoidal Response

For sinusoidal excitation, $f(\omega, t) = f_0 \sin(\omega t)$ with sweeping frequency ω , the transfer function for the SDOF system is given by

$$H(\omega) = \frac{x(\omega, t)}{f(\omega, t)} \quad (9.7)$$

For the MDOF system, the transfer function is denoted by

$$H_{ij}(\omega) = \frac{x_i(\omega, t)}{f_j(\omega, t)} \quad (9.8)$$

9.1.3.4 Random Response

Again, considering random excitations, repeat the process discussed previously as follows (see Equation 4.35) in order to practically measure the transfer functions.

9.1.3.4.1 Fourier Transform of Output and Input

$$X(\omega, T) = \mathcal{F}[x_T(t)] \quad (9.9)$$

and

$$F(\omega, T) = \mathcal{F}[F_T(t)] \quad (9.10)$$

In this instance, the lowercase letters represent the temporal functions or measured values in the physical domain. Note that these temporal functions are taken from random sets; once measured, they become deterministic “realizations.”

9.1.3.4.2 Power Spectral Density Functions

The auto- and cross-power spectral density functions are repeated as follows:

$$S_{FX}(\omega) = \Sigma X_k(\omega, T) F_k(\omega, T)^* \quad (9.11)$$

and

$$S_{XF}(\omega) = \Sigma F_k(\omega, T) X_k(\omega, T)^* \quad (9.12)$$

Also

$$S_X(\omega) = \Sigma |X_k(\omega, T)|^2 \quad (9.13)$$

and furthermore

$$S_F(\omega) = \Sigma |F_k(\omega, T)|^2 \quad (9.14)$$

9.1.3.4.3 Transfer Functions

The transfer functions are estimated as

$$H_1(\omega, T) = \frac{\sum_{k=1}^n X_k(\omega, T) F_k(\omega, T)^*}{\sum_{k=1}^n |F_k(\omega, T)|^2} \quad (9.15)$$

when the output contains a high level of noise and as

$$H_2(\omega, T) = \frac{\sum_{k=1}^n |X_k(\omega, T)|^2}{\sum_{k=1}^n F_k(\omega, T) X_k(\omega, T)^*} \quad (9.16)$$

when the input contains a high level of noise.

In both cases, n should be a fairly large number to effectively reduce the noise contaminations. For most cases,

$$n > 30 \quad (9.17)$$

suffices.

9.1.3.4.4 Coherence Function

The coherence function is used to reject the “noise” mode, with present criteria mentioned previously.

$$\gamma_{FX}^2(\omega) = \frac{H_1(\omega, T)}{H_2(\omega, T)} \quad (9.18)$$

In the following, for the sake of simplicity, we will omit the notation T . However, it is noted that, practically, we will use Equation 9.18 for transfer function measurement.

9.1.3.5 Modal Model

Modal analysis is referred to as the extraction of the modal parameter from the transfer function once measured.

9.1.3.5.1 Frequency Domain Method

In the frequency domain, the transfer function $H_{ij}(\omega)$ is directly used to provide information about natural frequencies and damping ratios through the relationships of the following curves: amplitude/phase vs. frequencies, real/imaginary vs. frequencies, and real vs. imaginary. This is possible because the transfer function $H_{ij}(\omega)$ can be written as a function of the corresponding natural frequencies and damping ratios, namely,

$$H_{ij}(\omega) = f(\omega_i, \zeta_i) \quad (9.19)$$

Here, $f(\cdot)$ stands for a function of $(\cdot)$, ω_i and ζ_i are respectively the i th natural frequency and damping ratio of the system. The i th mode shape Φ_i can be determined from the complex-valued amplitude of $\mathbf{H}_i(\omega)$, where

$$\mathbf{H}_i(\omega_i) = \begin{Bmatrix} H_{1i}(\omega_i) \\ H_{2i}(\omega_i) \\ \dots \\ H_{ni}(\omega_i) \end{Bmatrix} \quad (9.20)$$

Furthermore,

$$\Phi_i = a_i \mathbf{H}_i(\omega_i) \quad (9.21)$$

where a_i is a proportional factor.

9.1.3.5.2 Time Domain Method

In an ideal case, if the measurements of the acceleration, velocity, and displacement vectors ($\ddot{\mathbf{x}}$, $\dot{\mathbf{x}}$, and $\mathbf{x}$) could be determined, then the state matrix would exist:

$$\mathcal{A} = \dot{\mathbf{Y}}\mathbf{Y}^{-1} \quad (9.22)$$

In the above equation,

$$\dot{\mathbf{Y}} = \left[\begin{array}{c} \left\{ \begin{array}{c} \ddot{\mathbf{x}}(t_1) \\ \dot{\mathbf{x}}(t_1) \end{array} \right\}, \left\{ \begin{array}{c} \ddot{\mathbf{x}}(t_2) \\ \dot{\mathbf{x}}(t_2) \end{array} \right\}, \dots, \left\{ \begin{array}{c} \ddot{\mathbf{x}}(t_{2n}) \\ \dot{\mathbf{x}}(t_{2n}) \end{array} \right\} \end{array} \right] \quad (9.23)$$

and

$$\mathbf{Y} = \left[\begin{array}{c} \left\{ \begin{array}{c} \dot{\mathbf{x}}(t_1) \\ \mathbf{x}(t_1) \end{array} \right\} \left\{ \begin{array}{c} \dot{\mathbf{x}}(t_2) \\ \mathbf{x}(t_2) \end{array} \right\} \dots \left\{ \begin{array}{c} \dot{\mathbf{x}}(t_{2n}) \\ \mathbf{x}(t_{2n}) \end{array} \right\} \end{array} \right] \quad (9.24)$$

where $\mathbf{x}(t_i)$ is the displacement measured at the first time point, etc.

For normal modes, the generalized damping and stiffness matrices $\mathbf{M}^{-1}\mathbf{C}$ and $\mathbf{M}^{-1}\mathbf{K}$ can be found from the state matrix. Furthermore, the natural frequencies, damping ratios, and mode shapes can also be obtained.

Note that the mass matrix $\mathbf{M}$ remains unknown. Having the measured general damping and stiffness matrices does not mean that the physical parameters can be obtained.

For complex modes, the eigendecomposition can be directly carried out from the state matrix and the modal parameters can be obtained. Recall the eigenproblem:

$$\lambda_i \mathbf{P}_i = \mathcal{A} \mathbf{P}_i \quad (9.25)$$

From the eigenvalue λ_i , the natural frequency is

$$\omega_i = |\lambda_i|, \quad i = 1, \dots, n \quad (9.26)$$

and the damping ratio is

$$\zeta_i = -\text{Re}(\lambda_i)/\omega_i \quad (9.27)$$

From the eigenvector $\mathbf{P}_i$, the mode shape $\mathbf{p}_i$ can be calculated from:

$$\mathbf{P}_i = \begin{Bmatrix} \lambda_i \mathbf{p}_i \\ \mathbf{p}_i \end{Bmatrix} \quad (9.28)$$

Practically, it is not always possible to simultaneously measure the acceleration, velocity, and displacement. Unless the signals from all n -locations can be measured concurrently, Equation 9.22 exists only mathematically.

The signal can be approximately measured in a limited number of locations, represented by

$$\mathbf{z}(t_1) = \begin{Bmatrix} z_1(t_1) \\ z_2(t_1) \\ \dots \\ z_m(t_1) \end{Bmatrix}_{m \times 1} \quad (9.29)$$

where m is the total number of measurement locations, and $z_i(t)$ is a generic term of displacement, velocity, or acceleration. Note that, we use lowercase letters to denote the measured values, including that measured from random signals; and we continue to use uppercase letters to denote generic random set.

Construct two Hanckel matrices:

$$\dot{\mathbf{Y}} = \begin{bmatrix} \begin{Bmatrix} \mathbf{z}(t_2) \\ \mathbf{z}(t_3) \\ \dots \\ \mathbf{z}(t_{p+1}) \end{Bmatrix} \begin{Bmatrix} \mathbf{z}(t_3) \\ \mathbf{z}(t_4) \\ \dots \\ \mathbf{z}(t_{p+2}) \end{Bmatrix} \dots \begin{Bmatrix} \mathbf{z}(t_{q+1}) \\ \mathbf{z}(t_{q+2}) \\ \dots \\ \mathbf{z}(t_{p+q}) \end{Bmatrix} \end{bmatrix}_{(mp) \times q} \quad (9.30)$$

and

$$\mathbf{Y} = \begin{bmatrix} \begin{Bmatrix} \mathbf{z}(t_1) \\ \mathbf{z}(t_2) \\ \dots \\ \mathbf{z}(t_p) \end{Bmatrix} \begin{Bmatrix} \mathbf{z}(t_2) \\ \mathbf{z}(t_3) \\ \dots \\ \mathbf{z}(t_{p+1}) \end{Bmatrix} \dots \begin{Bmatrix} \mathbf{z}(t_q) \\ \mathbf{z}(t_{q+1}) \\ \dots \\ \mathbf{z}(t_{p+q-1}) \end{Bmatrix} \end{bmatrix}_{(mp) \times q} \quad (9.31)$$

Here,

$$t_{i+1} - t_i = \Delta t, \quad i = 1, \dots, p + q \quad (9.32)$$

where Δt is the sampling time interval. Integers p and q are such that

$$mp \geq 2n \quad (9.33)$$

and

$$q \geq mp \quad (9.34)$$

It can be proven that

$$\dot{\mathbf{Y}}\mathbf{Y}^+ = \exp(\mathcal{A}\Delta t) \quad (9.35)$$

and

$$\text{eig}(\exp(\mathcal{A}\Delta t)) = \text{diag}(e^{\lambda_i \Delta t}) \quad (9.36)$$

Based on Equation 9.36, the natural frequencies and damping ratios can be calculated. However, the eigenvectors of matrix $\exp(\mathcal{A}\Delta t)$ does not contain the full information of mode shapes, unless

$$m \geq n \quad (9.37)$$

9.1.4 THE SECOND INVERSE PROBLEM OF DYNAMIC SYSTEMS

In specific cases, the second inverse problem of dynamic systems is used to identify an unknown input.

9.1.4.1 General Background

From Equation 9.1, the following can be written:

$$F(\omega) = H(\omega)^{-1}X(\omega) \quad (9.38)$$

The autopower spectral density function of the input is calculated as

$$S_F(\omega) = |H(\omega)|^{-2}S_X(\omega) \quad (9.39)$$

Since

$$\sigma_F(\omega) = \int_{-\infty}^{\infty} S_F(\omega) d\omega \quad (9.40)$$

this will yield

$$\sigma_F(\omega) = \int_{-\infty}^{\infty} |H(\omega)|^{-2} S_X(\omega) d\omega \quad (9.41)$$

In most cases, the above integral of Equation 9.41 does not exist. Therefore, Equation 9.41 cannot be directly used. However, in selected cases, the autopower spectral density functions do exist.

9.1.4.2 White Noise

Consider the case of white noise by recalling Equation 7.50. The white noise input can be written as

$$S_F = \frac{kc\sigma_X^2}{\pi} \quad (9.42)$$

9.1.4.3 Practical Issues

9.1.4.3.1 Sampling Frequency and Cutoff Band

To satisfy the Nyquist sampling theorem, the signal must be low-pass filtered and the cutoff frequency ω_c or f_c is given by

$$\omega_c = 1/2\omega_s \quad (9.43)$$

and

$$f_c = 1/2f_s \quad (9.44)$$

Here, ω_s or f_s is the sampling frequency.

In addition, the total length of sampling is given by T , yielding

$$x(t) = x(\omega_c, T) = \mathcal{F}^{-1}[X(\omega_c, T)] \quad (9.45)$$

9.1.4.3.2 Estimation of RMS Value

$$\sigma_F(f) = \int_0^{f_c} |H(f)|^2 W_X(f) df \quad (9.46)$$

For SDOF systems,

$$|H(\omega)|_{\omega_c}^2 = (k - m\omega^2)^2 - (c\omega)^2 \Big|_{\omega_c} = (k - 4\pi^2 m f^2)^2 - 4\pi^2 (cf)^2 \Big|_{f_c} \quad (9.47)$$

9.2 SYSTEM PARAMETER IDENTIFICATION

System identification is an important subfield of inverse problems, which establishes mathematical models of dynamical systems from measured data. System identification also includes the optimal design of experiments for efficiently generating informative data for fitting such models as well as model reduction. System parameter identification is the identification of required properties, parameters, and models through the measurement of random input and output of the system. Due to the randomness and uncertainties of “measured” data, the carrying out of system identification is to use statistical analysis (see Goodwin and Payne 1977; Walter and Pronzato 1997).

9.2.1 PARAMETER ESTIMATION, RANDOM SET

First, the estimate of numerical characteristics of a random set will be considered.

9.2.1.1 Maximum Likelihood

In estimating statistical parameters, there are certain criteria that are used to deal with random variables and processes based on the condition that all possible variables in a given set will not be exhausted. It must be decided under what condition the estimation will be satisfied. Maximum likelihood estimation (MLE) provides a commonly used criterion. It had been used earlier by Gauss and Laplace, and was popularized by Fisher between 1912 and 1922. Reviews of the development of maximum likelihood have been provided by a number of authors (for instance, see LeCam 1990). This method determines the parameters that maximize the probability or likelihood of the sample data. MLE is considered to be more robust, with minimal exceptions, and yields estimators with more accurate statistical properties. They are versatile and can be applied to most models and types of data. They are also efficient for quantifying uncertainty using confidence bounds. In this section, the focus will be on MLE, and as a comparison, the method of moments will be briefly discussed, which is among the estimation methods other than MLE.

9.2.1.1.1 Mean and Averaging

Consider the mean value given by

$$\bar{x} = \frac{1}{n} \sum_{j=1}^n x_j \quad (9.48)$$

In Equation 9.48, $\bar{x}$ is the first moment about the origin of all the samples $[x_j]$. Generally, the parameter needed to be estimated is simply a certain moment, such as the mean or the RMS values. A simple method to estimate the unknown parameter is referred to as the *moment estimation*.

Example 9.1

Suppose $X \sim N(\mu, \sigma)$, where the mean and the standard deviation μ and σ , respectively, are unknown and need to be estimated.

Consider the first and second moments about the origin:

$$\hat{\mu} = E[X] = \bar{x}$$

and

$$\hat{\mu}^2 + \hat{\sigma}^2 = E[X^2] = \overline{x^2}$$

In the above, and hereafter, the overhead symbol “ $\hat{\cdot}$ ” represents the estimated value.

Therefore, Equation 9.48 can be used to estimate the mean $\hat{\mu}$ and further calculate

$$\hat{\sigma}^2 = \overline{x^2} - \hat{\mu}^2 = \frac{1}{n} \sum_{j=1}^n x_j^2 - \bar{x}^2 = S^2$$

and

$$\sigma = \sqrt{S^2} = S$$

Here, S^2 and S stand for respectively the sample variance and standard deviation. Equation 9.48 provides the basic approach of averaging. It is noted that $\bar{x}$ is the sample mean of all measured x_j , which are the samples taken from the random set X . Typically, n samples will be taken and the total number of variables of X will be much larger than n . Therefore, a reasonable question would be, can $\bar{x}$ be used to estimate the mean value of all variables in X ? Besides the mean value, the variance and the moments will also be considered. An additional reasonable question would be, is there any bias in these parameter estimations based on the average described in Equation 9.48 or, more generally, is there any bias in the moment vestimation?

These questions are analyzed in the following.

9.2.1.1.2 Probability Density Function

Suppose that n independent observations $x_1, x_2, \dots, x_n$ are measured, whose probability density (or probability mass) is unknown. Both the observed variables x_i and the parameters $\mathbf{p}_i$ can be vectors. The probability density function (PDF) $f_x(\cdot)$ is considered to belong to a certain family of distributions $\{f_{X_j}(x_j, \mathbf{p}), \mathbf{p} \in P\}$, which is called the parametric model. Thus, f_x corresponds to $\mathbf{p} = \mathbf{p}_1$, which is called the *true value* of the parameter. It is desirable to find the value of $\hat{\mathbf{p}}$, referred to as the *estimator*, since it would be as close to the true value of $\mathbf{p}_1$ as possible.

The parameter vector of the first PDF is explicitly denoted by $\mathbf{p}_1$, while the parameter vector of the second PDF is explicitly denoted by $\mathbf{p}_n$. Since x_j is independent, the joint PDF can be written as

$$f_{X_1 \dots X_n}(x_1, \dots, x_n, \mathbf{p}_n) = \prod_{j=1}^n f_{X_j}(x_j, \mathbf{p}_1) = \prod_{j=1}^n f_X(x_j, \mathbf{p}_1) \quad (9.49)$$

In Equation 9.49, $\mathbf{p}_1$ is the parameter to be calculated. Under the condition that the unknown parameter is indeed $\mathbf{p}_1$, and $x_1, x_2, \dots, x_n$ are independent, Equation 9.49 is in the form of the total production of $f_{X_j}(x_j, \mathbf{p}_1)$.

9.2.1.1.3 Likelihood Function

Example 9.2

A machine is used to manufacture certain specimens. Of the specimens produced, some are usable, while others are not. The random variable X is used to denote these two cases. When useable $X = 0$; otherwise $X = 1$. Therefore, X is 0–1 distributed.

The probability of $X = 1$ is p : $P(X = 1) = p$; thus, $P(X = 0) = 1 - p$. This can be written by the uniform formula (see the Bernoulli distribution described in Equation 1.70):

$$P(X = k) = p^k(1 - p)^k, \quad k = 0, 1$$

In checking the effectiveness of this machine, five specimens are chosen. It is found that the first two are unusable, while the remaining three are usable. The probability p is estimated as follows.

It is known that $0 < p < 1$. Thus, p can be any value between 0 and 1. The following values of p will be calculated: 0.2, 0.4, 0.6, and 0.8. The chance of $P(x_1 = 1, x_2 = 1, x_3 = 0, x_4 = 0, \text{ and } x_5 = 0) = p^2(1 - p)^3$ will also be calculated. The results are listed in Table 9.1.

From this table, it is seen that $p = 0.4$ is the largest calculated chance of making unusable specimens. This implies that, among 0.2, 0.4, 0.6, and 0.8, $p = 0.4$ is most likely the targeted probability. This is an example of the maximum likelihood method.

A joint PDF can be studied by viewing the observed values $x_1, x_2, \dots, x_n$ to be fixed “parameters,” whereas the value of $\mathbf{p}_n$ is allowed to vary freely. From this point of view, this PDF is called the *likelihood* function. In using this method, the focus is on $\mathbf{p}_n$. In other words, the value of $\mathbf{p}_n$ is found such that the PDF is likely to be the “best” estimation.

The joint PDF is thus called the *likelihood function*, denoted as

$$\mathcal{L}(\mathbf{p}_1) = \prod_{j=1}^n f_X(x_j, \mathbf{p}_1) \tag{9.50}$$

The next step is to maximize the likelihood.

TABLE 9.1
Calculated Chance of Making Unusable Specimens

p	$P(x_1 = 1, x_2 = 1, x_3 = 0, x_4 = 0, \text{ and } x_5 = 0) = p^2(1 - p)^3$
0.2	$0.2^2(0.8)^3 = 0.02048$
0.4	$0.4^2(0.6)^3 = 0.03456$
0.6	$0.6^2(0.4)^3 = 0.02304$
0.8	$0.8^2(0.2)^3 = 0.00512$

9.2.1.1.4 Log-Likelihood Function

For mathematical convenience, the log-likelihood function is used.

$$\mathcal{L}(\mathbf{p}_1) = \ln[L(\mathbf{p}_1)] \quad (9.51)$$

9.2.1.1.5 Maximum Likelihood Method

In this method, the derivative is taken with respect to the unknown variable $\mathbf{p}_1$ and the result is set equal to zero. By solving this equation, the proper value of $\mathbf{p}_1$ is found. That is,

$$\frac{d \ln(L)}{d\mathbf{p}_1} = 0 \quad (9.52)$$

Note that from Equation 9.52, the vector $\mathbf{p}_1$ may contain n elements, explicitly,

$$\mathbf{p}_1 = \begin{Bmatrix} p_1 \\ p_2 \\ \dots \\ p_n \end{Bmatrix} \quad (9.53)$$

In this case, Equation 9.52 is realized by

$$\begin{aligned} \frac{\partial \ln(L)}{\partial p_1} &= 0 \\ \frac{\partial \ln(L)}{\partial p_2} &= 0 \\ &\dots \\ \frac{\partial \ln(L)}{\partial p_n} &= 0 \end{aligned} \quad (9.54)$$

Example 9.3

Consider the above-mentioned case of the random variable X with a 0–1 distribution, expressed as

$$P(X = k) = p^k(1 - p)^k, \quad k = 0, 1$$

In this instance, p is the unknown parameter to be estimated through the maximum likelihood method. The samples of X are $(x_1, x_2, \dots, x_n)$.

First, the likelihood function is given by

$$\mathcal{L} = \prod_{j=1}^n P(X = x_j) = \prod_{j=1}^n p^{x_j} (1-p)^{1-x_j} = p^{\sum_{i=1}^n x_i} (1-p)^{n - \sum_{i=1}^n x_i}$$

Second, take the logarithm on both sides of the above equation:

$$\ln(\mathcal{L}) = \sum_{i=1}^n x_i \ln(p) + \left(n - \sum_{i=1}^n x_i \right) \ln(1-p)$$

Third, take the derivative with respect to the unknown parameter p and let the result be equal to zero.

In this case, " $\mathbf{p}_1$ " contains only one parameter p ; therefore,

$$\frac{d\ln(\mathcal{L})}{dp} = \frac{1}{p} \sum_{i=1}^n x_i - \frac{1}{1-p} \left(n - \sum_{i=1}^n x_i \right) = 0$$

It is found that, when

$$p = \frac{1}{n} \sum_{i=1}^n x_i$$

$\ln(\mathcal{L})$ reaches the maximum value. Therefore, the estimation of p based on the maximum likelihood method is, when comparing with Equation 9.48,

$$\hat{p} = \frac{1}{n} \sum_{i=1}^n x_i = \bar{X}$$

Example 9.4

Suppose that a random variable X is evenly distributed in between 0 and a with the PDF

$$f_X(X) = \begin{cases} \frac{1}{a}, & 0 \leq x \leq a \\ 0, & \text{otherwise} \end{cases}$$

Estimate the unknown parameter a .

First, the likelihood function is given by

$$\mathbf{L} = \prod_{i=1}^n f_X(x_i) = \begin{cases} \prod_{i=1}^n \frac{1}{a}, & 0 \leq x_i \leq a \\ 0, & \text{otherwise} \end{cases} = \begin{cases} a^{-n}, & 0 \leq \min_i x_i \leq \max_i x_i \leq a \\ 0, & \text{otherwise} \end{cases}$$

Second, consider the case $\mathbf{L} \neq 0$ and take the logarithm:

$$\ln(\mathbf{L}) = -n \ln(a)$$

Third, take the derivative with respect to a , yielding

$$\frac{d \ln(\mathbf{L})}{da} = -\frac{n}{a} = 0$$

The above equation has no meaningful solutions. This implies that, when $\mathbf{L} \neq 0$, the derivative is equal to

$$\frac{d \ln(\mathbf{L})}{da} \neq 0$$

However, the above inequality does not necessarily mean that the likelihood function $\mathbf{L}$ has no maximum value in between 0 and a . In fact, a and $\mathbf{L}$ have an inverse relationship. The smaller the value of a is, the larger the value of $\mathbf{L}$ will be. However, a cannot be smaller than $(\max_i x_i)$. This is written as

$$\hat{a} = \max_i x_i$$

Example 9.5

Consider the sample set $(x_1, x_2, x_3, x_4, x_5, x_6) = (1, 2, 3, 5, 4, 9)$. Given the above condition, $\hat{a} = 9$.

Now, compare the estimation through the maximum likelihood method and through the moment method.

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx = \int_0^a x \frac{1}{a} dx = \frac{a}{2}$$

However, it is also true that

$$\frac{\hat{a}}{2} = E[X]$$

For this example,

$$\hat{a} = 2E[X] = 2\bar{x} = 2 \times 4 = 8 \neq 9$$

This result implies that the estimation through moment about the origin has a bias.

9.2.1.2 Bias and Consistency

Statistical estimation is the estimation of certain parameters and/or distribution functions of random sets through samples, whose size can be considerably smaller than the entire space of the random sets. As a result, the estimation can be biased and/or inconsistent.

To judge if an estimator is unbiased, check whether the expected value of the estimation is equal to the “true” value. This maximum likelihood estimator should be unbiased. Additionally, it should also be consistent. That is, with a sufficiently large number of observations n , it should be possible to find the value of $\mathbf{p}$ with arbitrary precision. This means that as n approaches to infinity, the estimator $\mathbf{p}_i$ converges in probability to its true value.

The following is true for the maximum likelihood estimator (MLE):

1. MLE is consistent: estimation converges.
2. MLE is asymptotically unbiased: it converges to a “correct” value.
3. MLE is efficient: the correct value yields the minimum variance among all unbiased estimates.
4. MLE is sufficient: it uses all measured data.
5. MLE is invariant.
6. MLE is asymptotically normally distributed.

9.2.1.2.1 Mean Estimator

Consider the formula used to calculate the mean value of random set X , here X_j etc. is a conceptual realization taken from generic random set,

$$\bar{X} = \frac{1}{n} \sum_{j=1}^n X_j \quad (9.55)$$

To check if it is unbiased, take the mathematical expectation of Equation 9.55 as follows:

$$\begin{aligned} E[\bar{X}] &= E\left[\frac{1}{n} \sum_{j=1}^n X_j\right] = \frac{1}{n} E\left[\sum_{j=1}^n X_j\right] = \frac{1}{n} \sum_{j=1}^n E[X_j] \\ &= \frac{1}{n} \sum_{j=1}^n \mu_X = \mu_X \end{aligned} \quad (9.56)$$

Equation 9.56 implies that the estimation of Equation 9.55 is indeed *unbiased*. To see if the estimation is consistent, consider

$$\begin{aligned} D[\bar{X}] &= D\left[\frac{1}{n} \sum_{j=1}^n X_j\right] = \frac{1}{n^2} D\left[\sum_{j=1}^n X_j\right] = \frac{1}{n^2} \left(\sum_{j=1}^n D[X_j] + \sum_{j \neq k} \sum \text{cov}[X_j, X_k] \right) \\ &= \frac{1}{n^2} \sum_{j=1}^n \sigma_X^2 = \frac{\sigma_X^2}{n} \end{aligned} \quad (9.57)$$

Equation 9.57 implies that, when n is sufficiently large, the variance of $\bar{X}$ tends to zero. This implies that the estimation is *consistent*.

9.2.1.2.2 Variance Estimator

We now consider the bias of the variance estimator.

$$\hat{\Sigma}_X^2 = \frac{1}{n} \sum_{j=1}^n (X_j - \mu_X)^2 \quad (9.58)$$

Here, $\hat{\Sigma}_X^2$ is the random variable from which the variance is estimated. It can be proven that the mean of $\hat{\Sigma}_X^2$ is $\hat{\sigma}_X^2$. This implies that when the mean μ_X is known, Equation 9.58 provides unbiased estimation of the variance.

If the mean μ_X is unknown, then first analyze the following formula, which is used to estimate the variance:

$$S_X^2 = \frac{1}{n-1} \sum_{j=1}^n (X_j - \bar{X})^2 \quad (9.59)$$

Consider the corresponding mean value:

$$\begin{aligned} E[S_X^2] &= E\left[\frac{1}{n-1} \sum_{j=1}^n (X_j - \bar{X})^2\right] = E\left[\frac{1}{n-1} \sum_{j=1}^n [(X_j - \mu_X) - (\bar{X} - \mu_X)]^2\right] \\ &= \frac{1}{n-1} \sum_{j=1}^n E[(X_j - \mu_X)^2] - 2E[(X_j - \mu_X)(\bar{X} - \mu_X)] + E[(\bar{X} - \mu_X)^2] \quad (9.60) \\ &= \frac{1}{n-1} \sum_{j=1}^n \left(\sigma_X^2 - \frac{2}{n} \sigma_X^2 + \frac{1}{n} \sigma_X^2 \right) = \sigma_X^2 \end{aligned}$$

Consequently, Equation 9.59, with the factor $(n - 1)$ instead of n , provides an unbiased estimation.

$$S_X^2 = \frac{1}{n-1} \sum_{j=1}^n (x_j - \bar{x})^2 \quad (9.61)$$

9.2.2 CONFIDENCE INTERVALS

9.2.2.1 Estimation and Sampling Distributions

Because estimation is based on samples, not entire variable sets, therefore, the estimator itself is a random variable, which also has distributions.

9.2.2.1.1 Probability of Correct Estimation

9.2.2.1.1.1 Mean The mean must fall within all possible values. Because it is impossible to obtain all the random variables, there exists a chance that the “true” mean is outside the sampling space. We now consider the probability of correct estimation.

Assume that the distribution of the mean estimator $\bar{X}$ is normal, and the variance of $\bar{X}$ is known; in the form of standard normal distribution, the following is true:

$$P\left(-\infty < \frac{\bar{X} - \mu_X}{\sigma_X / \sqrt{n}} \leq z_{1-\alpha}\right) = \Phi(z_{1-\alpha}) = 1 - \alpha \quad (9.62)$$

9.2.2.1.1.2 Variance We next consider the variance estimator by writing

$$(n-1) \frac{S_X^2}{\sigma_X^2} = \frac{1}{\sigma_X^2} \sum_{j=1}^n (X_j - \bar{X})^2 = \sum_{j=1}^n \left(\frac{X_j - \mu_X}{\sigma_X} \right)^2 - \left(\frac{\bar{X} - \mu_X}{\sigma_X / \sqrt{n}} \right)^2 \quad (9.63)$$

From Equation 9.63, the first term on the right-hand side is chi-square with n DOF, while the second term is chi-square with one DOF. Due to the regenerative character of chi-square random variables, the left-hand side is seen to be chi-square with $(n - 1)$ DOF. Consequently, this can be rewritten as

$$P\left[(n-1) \frac{S_X^2}{\sigma_X^2} > x_{1-\alpha}\right] = 1 - F_{\chi_{n-1}^2}(x_{1-\alpha}) = 1 - \alpha \quad (9.64)$$

9.2.2.1.2 Confidence Intervals

9.2.2.1.2.1 Confidence Interval of Mean The $(1 - \alpha)$ percentage, two-sided confidence interval for mean can be obtained by solving the double inequalities in the

argument on the left-hand side of Equation 9.62. Using the establishments of probabilities of correct estimation yields

$$\left(\bar{x} - \frac{z_{1-\alpha/2} \sigma_X}{\sqrt{n}}, \bar{x} + \frac{z_{1-\alpha/2} \sigma_X}{\sqrt{n}} \right) \quad (9.65)$$

9.2.2.1.2.2 Confidence Interval of Variance Similar to Equation 9.64, the $(1 - \alpha)$ percentage confidence interval for the variance can be written as

$$\left(0, \frac{(n-1)S_X^2}{x_{1-\alpha}} \right) \quad (9.66)$$

9.2.2.1.2.3 Mean with Unknown Variance The variance in Equation 9.66 is assumed to be known. However, it is in fact unknown. In this specific instance, the variance estimator S_X^2 is used to approximate σ_X^2 . For this reason, there is a need to study the distribution of

$$t_{n-1} = \frac{\bar{X} - \mu_X}{S_X / \sqrt{n}} \quad (9.67)$$

Note that Equation 9.67 indicates a t -distribution, also known as a Student distribution. This was previously discussed in Chapter 2. The two-sided confidence interval for the mean is given as

$$P \left[-b_{1-\alpha/2} < \frac{\bar{X} - \mu_X}{S_X / \sqrt{n}} \leq b_{1-\alpha/2} \right] = F_{t_{n-1}}(b_{1-\alpha/2}) - F_{t_{n-1}}(-b_{1-\alpha/2}) = 1 - \alpha \quad (9.68)$$

By solving the double inequalities in the argument on the left-hand side for the mean μ_X , the confidence interval is determined to be

$$\left(\bar{x} - \frac{b_{1-\alpha/2} S_X}{\sqrt{n}}, \bar{x} + \frac{b_{1-\alpha/2} S_X}{\sqrt{n}} \right) \quad (9.69)$$

9.2.3 PARAMETER ESTIMATION, RANDOM PROCESS

Based on the discussion of the parameter of a random process, the random process will now be further described in the following (see Stigler 1986, 1999).

9.2.3.1 General Estimation

An unknown random process should not be assumed stationary before analyzing its statistical characteristics. Unless the process is known to be stationary, ensemble average must be used.

9.2.3.1.1 Mean and Variance

The following formulae are used to estimate the mean and the variance of a random process.

9.2.3.1.1.1 *Mean* Suppose that M samples are taken; then

$$\bar{x}_j = \frac{1}{M} \sum_{m=1}^M x_{mj} \quad (9.70)$$

At first glance of Equation 9.70, the mean of a random process appears to be identical to that of a random variable. Conversely, the mean $\bar{x}_j$ has a subscript j , indicating the j th realization of the random process $X(t)$. Namely, the average described in Equation 9.70 is an ensemble average.

9.2.3.1.1.2 *Variance* Similarly, the unbiased estimation of variance is given by

$$S_X^2(t_j) = \frac{1}{M-1} \sum_{m=1}^M (x_{mj} - \bar{x}_j)^2, \quad j = 0, 1, \dots, n-1 \quad (9.71)$$

9.2.3.1.1.3 *Standard Deviation* From Equation 9.71, the standard deviation is determined to be

$$S_X(t_j) = \sqrt{S_X^2(t_j)}, \quad j = 0, 1, \dots, n-1 \quad (9.72)$$

9.2.3.1.2 Correlation

Another fundamental difference between a random variable and a random process is in the analysis of correlations. We now consider the correlation of processes X and Y .

9.2.3.1.2.1 *Joint PDF* First, consider the joint distribution of X and Y by denoting the cross-correlation function R as

$$R = E[XY] \quad (9.73)$$

The joint PDF can then be written as

$$f_{XY}(x, y) = \frac{1}{2\pi\sqrt{\sigma_X^2\sigma_Y^2 - R^2}} \exp\left[-\frac{1}{2(\sigma_X^2\sigma_Y^2 - R^2)}(\sigma_Y^2x^2 - 2Rxy + \sigma_X^2y^2)\right]$$

where

$$-\infty < x, y < \infty \quad (9.74)$$

9.2.3.1.2.2 *Likelihood Function* Next, consider the likelihood function of n pairs of realizations of X and Y , where the likelihood function is

$$L(\sigma_X^2, \sigma_Y^2, R) = \frac{1}{(2\pi)^n \sqrt[n]{\sigma_X^2 \sigma_Y^2 - R^2}} \times \exp \left[\frac{1}{-2(\sigma_X^2 \sigma_Y^2 - R^2)} \sum_{j=1}^n (\sigma_Y^2 x_j^2 - 2R x_j y_j + \sigma_X^2 y_j^2) \right] \quad (9.75)$$

9.2.3.1.2.3 *Log-Likelihood Function* Furthermore, the log-likelihood function is

$$\begin{aligned} \mathcal{L}(\sigma_X^2, \sigma_Y^2, R) = & -n \ln(2\pi) - \frac{n}{2} \ln \left(\sqrt[n]{\sigma_X^2 \sigma_Y^2 - R^2} \right) \\ & - \frac{1}{2(\sigma_X^2 \sigma_Y^2 - R^2)} \sum_{j=1}^n (\sigma_Y^2 x_j^2 - 2R x_j y_j + \sigma_X^2 y_j^2) \end{aligned} \quad (9.76)$$

9.2.3.1.2.4 *Maximum Log-Likelihood Function* The maximum log-likelihood function can be used to estimate the variances and joint PDF. First, maximize the log-likelihood function by letting

$$\frac{\partial}{\partial \sigma_X^2} [\mathcal{L}(\sigma_X^2, \sigma_Y^2, R)] = 0 \quad (9.77)$$

thus yielding the following equation:

$$n + \frac{1}{\hat{\sigma}_Y^2} \sum_{j=1}^n y_j^2 = \frac{1}{\hat{\sigma}_X^2 \hat{\sigma}_Y^2 - \hat{r}^2} \sum_{j=1}^n (\hat{\sigma}_Y^2 x_j^2 - 2\hat{r} x_j y_j + \hat{\sigma}_X^2 y_j^2) \quad (9.78)$$

Next, let

$$\frac{\partial}{\partial \sigma_Y^2} [\mathcal{L}(\sigma_X^2, \sigma_Y^2, R)] = 0 \quad (9.79)$$

The following second equation results:

$$n + \frac{1}{\hat{\sigma}_X^2} \sum_{j=1}^n x_j^2 = \frac{1}{\hat{\sigma}_X^2 \hat{\sigma}_Y^2 - \hat{r}^2} \sum_{j=1}^n (\hat{\sigma}_Y^2 x_j^2 - 2\hat{r} x_j y_j + \hat{\sigma}_X^2 y_j^2) \quad (9.80)$$

Additionally, let

$$\frac{\partial}{\partial R}[\mathcal{L}(\sigma_X^2, \sigma_Y^2, R)] = 0 \quad (9.81)$$

We can obtain a third equation as given below:

$$n\hat{r} + \sum_{j=1}^n x_j y_j = \frac{\hat{r}}{\hat{\sigma}_X^2 \hat{\sigma}_Y^2 - \hat{r}^2} \sum_{j=1}^n (\hat{\sigma}_Y^2 x_j^2 - 2\hat{r} x_j y_j + \hat{\sigma}_X^2 y_j^2) \quad (9.82)$$

Adding Equations 9.78 and 9.80 and dividing the result by 2 will further yield in

$$n + \frac{1}{2} \left[\frac{1}{\hat{\sigma}_Y^2} \sum_{j=1}^n y_j^2 + \frac{1}{\hat{\sigma}_X^2} \sum_{j=1}^n x_j^2 \right] = \frac{\hat{r}}{\hat{\sigma}_X^2 \hat{\sigma}_Y^2 - \hat{r}^2} \sum_{j=1}^n (\hat{\sigma}_Y^2 x_j^2 - 2\hat{r} x_j y_j + \hat{\sigma}_X^2 y_j^2) \quad (9.83)$$

Therefore, we can write

$$\sum_{j=1}^n x_j y_j = \frac{\hat{r}}{2} \left(\frac{1}{\hat{\sigma}_X^2} \sum_{j=1}^n x_j^2 + \frac{1}{\hat{\sigma}_Y^2} \sum_{j=1}^n y_j^2 \right) \quad (9.84)$$

To use Equation 9.84, we need to have both the variances of processes X and Y , which can be respectively estimated as

$$\hat{\sigma}_X^2 = \frac{1}{n} \sum_{j=1}^n x_j^2 \quad (9.85)$$

and

$$\hat{\sigma}_Y^2 = \frac{1}{n} \sum_{j=1}^n y_j^2 \quad (9.86)$$

Thus, the correlation function is

$$\hat{r} = \frac{1}{n} \sum_{j=1}^n x_j y_j \quad (9.87)$$

which is generally written as

$$\hat{R} = \frac{1}{n} \sum_{j=1}^n X_j Y_j \quad (9.88)$$

The following equation can be used to check if the estimator for the correlation R is biased:

$$E[\hat{R}] = E\left[\frac{1}{n} \sum_{j=1}^n X_j Y_j\right] = \frac{1}{n} \sum_{j=1}^n E[X_j Y_j] = R \quad (9.89)$$

Equation 9.89 indicates that the value of this estimation is unbiased.

Furthermore, to see if the estimator is consistent, we consider the following:

$$D[\hat{R}] = D\left[\frac{1}{n} \sum_{j=1}^n X_j Y_j\right] = \frac{1}{n^2} \sum_{j=1}^n D[X_j Y_j] = \frac{1}{n} (R^2 + \sigma_X^2 \sigma_Y^2) \quad (9.90)$$

For this example, the estimator of the correlation is consistent.

9.2.3.1.2.5 Autocorrelation Function The estimator of the autocorrelation function is

$$\hat{r}_X(t_i, t_j) = \frac{1}{M} \sum_{m=1}^M x_{mi} x_{mj}, \quad t_i, t_j \in T \quad (9.91)$$

It can be proven that

$$E[\hat{r}_X(t_i, t_j)] = R_X(t_i, t_j) \quad (9.92)$$

Namely, the estimator described in Equation 9.92 is unbiased.

In addition,

$$D[\hat{r}_X(t_i, t_j)] = \frac{R_X^2(t_i, t_j) - \sigma_X^2(t_j) \sigma_X^2(t_i)}{n} \quad (9.93)$$

which can be used to check the consistency.

9.2.3.1.2.6 *Cross-Correlation Function* The estimator of a cross-correlation function is

$$\hat{r}_{XY}(t_i, t_j) = \frac{1}{M} \sum_{m=1}^M x_{mi} y_{mj}, \quad t_i, t_j \in T \quad (9.94)$$

It can be derived from Equation 9.94 that

$$E[\hat{r}_{XY}(t_i, t_j)] = R_{XY}(t_i, t_j) \quad (9.95)$$

and

$$D[\hat{r}_{XY}(t_i, t_j)] = \frac{R_X^2(t_i, t_j) - \sigma_X^2(t_j)\sigma_Y^2(t_j)}{n} \quad (9.96)$$

Thus, Equation 9.94 has an unbiased and consistent estimator.

9.2.3.1.2.7 *Covariance* The cross-covariance function is estimated based on the maximum likelihood method as

$$\hat{C}_{XY}(t_i, t_j) = \frac{1}{M} \sum_{m=1}^M (x_{mi} - \bar{x}_i)(y_{mj} - \bar{y}_j), \quad t_i, t_j \in T \quad (9.97)$$

Note that, for this case,

$$E[\hat{C}_{XY}(t_i, t_j)] \neq C_{XY}(t_i, t_j) \quad (9.98)$$

or explicitly speaking, Equation 9.97 is a biased estimator.

9.2.3.1.2.8 *Correlation Coefficient Function* The estimator of the cross-correlation coefficient can be written as

$$\hat{\rho}_{XY}(t_i, t_j) = \frac{\hat{C}_{XY}(t_i, t_j)}{s_X(t_i)s_Y(t_j)} \quad (9.99)$$

Due to the inequality in Equation 9.98, this estimator is also biased.

9.2.3.2 Stationary and Ergodic Process

The above estimators were for general process and based on the MLE and ensemble average. In the following, stationary processes and ergodic processes are examined.

9.2.3.2.1 Mean

9.2.3.2.1.1 *Non-Ergodic* For a non-ergodic process, the mean can be written as

$$\bar{x} = \frac{1}{n} \sum_{j=0}^{n-1} \bar{x}_j = \frac{1}{n} \sum_{j=0}^{n-1} \left(\frac{1}{M} \sum_{m=1}^M x_{mj} \right) = \frac{1}{Mn} \sum_{j=0}^{n-1} \sum_{m=1}^M x_{mj} \quad (9.100)$$

9.2.3.2.1.2 *Ergodic* For an ergodic process, the mean is represented in a much simpler formula:

$$\bar{x} = \frac{1}{n} \sum_{j=0}^{n-1} x_j \quad (9.101)$$

9.2.3.2.2 Variance

9.2.3.2.2.1 *Non-Ergodic* Similar to the mean value, if the process is non-ergodic, the variance can be written as

$$S_X^2 = \frac{1}{n} \sum_{j=0}^{n-1} S_X^2(t_j) = \frac{1}{n(M-1)} \sum_{j=0}^{n-1} \sum_{m=1}^M (x_{mj} - \bar{x}_j)^2 \quad (9.102)$$

9.2.3.2.2.2 *Ergodic* If the process is ergodic, the variance is

$$S_X^2 = \frac{1}{n} \sum_{j=0}^{n-1} (x_j - \bar{x})^2 \quad (9.103)$$

9.2.3.2.3 Standard Deviation

Once the variance is estimated, the standard deviation can be calculated based on the following relationship:

$$S_X = \sqrt{S_X^2} \quad (9.104)$$

9.2.3.2.4 Autocorrelation

Next, the autocorrelation function of a stationary process will be described.

9.2.3.2.4.1 *Non-Ergodic* For a non-ergodic process,

$$\hat{r}_X(\tau_j) = \frac{1}{n-j} \sum_{i=0}^{n-1-j} \hat{r}_X(t_i, t_i + \tau_j), \quad 0 \leq t_i, \tau_j \leq T \quad (9.105)$$

For the discussion on variance, the expression of $\hat{r}_X(t_i, t_i + \tau_j)$ was used. One can proceed from this point to obtain the expressions.

9.2.3.2.4.2 *Ergodic* For an ergodic process,

$$\hat{r}_X(\tau_j) = \frac{1}{n-j} \sum_{i=0}^{n-1-j} x_i x_{i+j}, \quad 0 \leq \tau_j \leq T \quad (9.106)$$

9.2.3.2.5 *Cross-Correlation, Ergodic*

We now consider two ergodic processes X and Y ; the cross-correlation function is estimated as

$$\hat{r}_{XY}(\tau_j) = \frac{1}{n-j} \sum_{i=0}^{n-1-j} x_i y_{i+j}, \quad 0 \leq \tau_j \leq T \quad (9.107)$$

9.2.3.2.6 *Covariance, Ergodic*

For two ergodic processes X and Y , the covariance function is estimated as

$$\hat{C}_{XY}(\tau_j) = \frac{1}{n-j} \sum_{i=0}^{n-1-j} (x_i - \bar{x})(y_{i+j} - \bar{y}), \quad 0 \leq \tau_j \leq T \quad (9.108)$$

9.2.3.2.7 *Cross-Correlation Coefficient, Ergodic*

For two ergodic processes X and Y , the cross-correlation coefficient is estimated as

$$\hat{\rho}_{XY}(\tau_j) = \frac{\hat{C}_{XY}(\tau_j)}{s_X s_Y}, \quad 0 \leq \tau_j \leq T \quad (9.109)$$

9.2.3.3 Nonstationary Process

In real-world applications, processes are often nonstationary. Nonstationary processes will be discussed next.

9.2.3.3.1 *Direct Analysis*

The first method for working with nonstationary processes is the direct method, in which a nonstationary process is made “stationary.”

9.2.3.3.1.1 *Product of Deterministic and Random Processes* The product of deterministic and random processes is achieved by allowing

$$X(t) = a(t)U(t) \quad -\infty < t < \infty \quad (9.110)$$

Here, $a(t)$ is deterministic and $U(t)$ is approximately stationary, but potentially nonstationary, random process with zero mean and unity variance.

In many cases, the process $a(t)$ is also a function of driving frequency, which can be seen as a systematic error in measurement.

9.2.3.3.1.2 *Autocorrelation* In Equation 9.110, if $a(t)$ and $U(t)$ can be successfully separated from a nonstationary process $X(t)$, then the autocorrelation function can be written as

$$R_X(t,s) = a(t)a(s)R_U(t,s) \quad \text{for } -\infty < t,s < \infty \quad (9.111)$$

The autocorrelation function is

$$R_X(t,s) = a(t)a(s)R_U(s-t) \quad \text{for } -\infty < t,s < \infty \quad (9.112)$$

The mean square value for $t = s$ is

$$\sigma_X^2(t) = a^2(t) \quad (9.113)$$

9.2.3.3.1.3 *Mean Square Value* The estimator of the mean square value for a nonstationary process can be written as

$$\hat{\sigma}_X^2(t_j) = \sum_{k=-N}^N w_k x_{j+k}^2, \quad j = N, \dots, n-1-N \quad (9.114)$$

where w_k is an even weight function and, n and N are respectively the numbers of temporal points and the numbers of pieces of measurements, and we have

$$\sum_{k=-N}^N w_k = 1 \quad (9.115)$$

The weight function is used to emphasize certain values. In most cases, a denser distribution will exist near the central point. In this specific case, the denser distribution is near $k = 0$. This allows the bias to be minimized. Thus, the expected value of the estimator of variance may be written as

$$E[\hat{\sigma}_X^2(t_j)] = \sum_{k=-N}^N w_k \sigma^2(t_{j+k}), \quad j = N, \dots, n-1-N \quad (9.116)$$

From Equation 9.116, it is seen that when $\sigma^2(t_{j+k})$ varies evenly (symmetrically) for k in between the values of $-N$ and N and the variation is linear, the expected value is close to zero. As a result, the estimator is unbiased. Otherwise, it will be biased. Furthermore, an increase in the value of the weight function w_k near zero will lower the possible bias.

9.2.3.3.1.4 *Variance of Mean-Square Estimator* The corresponding variance of the mean-square estimator is

$$D[\hat{\sigma}_X^2(t_j)] = 2 \sum_{k=-N}^N \sum_{m=-N}^N w_k w_m R_X^2(j+k, j+m), \quad j = N, \dots, n-1-N \quad (9.117)$$

where $R_X(j, k)$ is the autocorrelation function of the process $X(t)$ at $t = j\Delta t$ and $s = k\Delta t$.

When $t = s$, $R_X(t, s)$ will be at its maximum. Thus, to minimize the variance of the estimator $\hat{\sigma}_X^2(t_j)$, the value of weight function w_k near zero must be reduced, and certainly not larger. This is contrary to the case of the mean estimator. Thus, a different set of weight functions should be used to deal with the estimation of the mean and mean-square values.

9.2.3.3.2 *Indirect Analysis of Shock Response Spectrum*

Assessing the severity and accounting the effects of nonstationary random processes can be achieved through the shock response spectrum. This is contradictory with the power spectral density function. The response spectrum is given in Section 7.3.2. The shock response spectrum is discussed in the following.

Recall the base excitation:

$$\ddot{x} + 2\zeta\omega\dot{x} + \omega^2x = 2\zeta\omega\dot{z} + \omega^2z \quad (9.118)$$

where $z(t)$ is the ground displacement and $\ddot{x}(t)$ is the absolute acceleration. If $\ddot{z}(t)$ is a shock with a given amplitude, the peak response of $\ddot{x}$ will depend upon the natural frequency ω or $f = \omega/2\pi$, with a given value of damping ratio ζ . Thus, the shock spectrum can be written as

$$B(f) = \max_t |\ddot{x}(t)| \quad (9.119)$$

Denote a series of shocks measured from the ensemble of a single nonstationary random process as $\ddot{z}_{mj}$, $m = 1, \dots, M$, $j = 0, \dots, n-1$, where m indexes the number of the measurement from the ensemble, while j represents time. From Equation 9.119, a collection of shock spectra denoted by $B_m(f)$, $m = 1, \dots, M$, is used to obtain a representative shock response spectrum $B_c(f)$. This representative shock response spectrum is employed to indicate a possible spectrum with a controlled level of conservatism. Recall that in the discussion given in Chapter 7, the earthquake response spectrum is a sum of the mean value pulse with one standard deviation.

The mean of the shock spectrum can be written as

$$\bar{B}(f) = \frac{1}{M} \sum_{m=1}^M B_m(f) \quad (9.120)$$

and the standard deviation can be written as

$$S_B(f) = \left\{ \frac{1}{M-1} \sum_{m=1}^M [B_m(f) - \bar{B}(f)]^2 \right\}^{1/2} \quad (9.121)$$

The representative shock response spectrum is now given by

$$B_c(f) = \bar{B}(f) + K S_B(f) \quad (9.122)$$

where

$$K > 0 \quad (9.123)$$

to stay on the conservative side.

9.2.4 LEAST SQUARES APPROXIMATION AND CURVE FITTING

While the MLE is most commonly used, the least squares approximation is also a universal approach. It is often used as criteria for data regression or curve fitting.

This method was the culmination of several advances in the eighteenth century. The combination of different observations is the best estimate of the true value. The errors can be decreased with aggregation. This method was first expressed by Cotes in 1722. A clearer and concise exposition was published by Legendre in 1805, which described the method as an algebraic procedure for fitting linear equations to data, and Legendre demonstrates this method by analyzing the same data as Laplace for the shape of the earth (see Charnes et al. 1976).

9.2.4.1 Concept of Least Squares

First, let us consider the concept of least squares.

9.2.4.1.1 Sum of Squares

Suppose that there exist n data samples denoted as r_i , for $i = 1, \dots, n$. The sum S of squares is given by

$$S = \sum_{i=1}^n r_i^2 \quad (9.124)$$

9.2.4.1.2 Residue

The residue r_i is the difference between the measured value y_i and the function f . The function f is generated by all the y_i values.

$$r_i = y_i - f(x_i, \mathbf{p}) \quad (9.125)$$

In Equation 9.125, f is a function of the independent variable x_i and the parameter $\mathbf{p}$.

9.2.4.1.3 Linear Least Squares

Consider f as a linear function of x_i ; then

$$\mathbf{p} = \{\alpha_0, \alpha_1\} \quad (9.126)$$

and

$$f(x, \mathbf{p}) = \alpha_0 + \alpha_1 x \quad (9.127)$$

9.2.4.1.4 Nonlinear Least Squares

A polynomial may be used to describe the function f as

$$f(x, \mathbf{p}) = \alpha_0 + \alpha_1 x + \alpha_2 x^2 + \dots, \quad (9.128)$$

when

$$\mathbf{p} = \{\alpha_0, \alpha_1, \alpha_2, \dots\} \quad (9.129)$$

The residue can be minimized as

$$\frac{\partial S}{\partial \alpha_j} = 2 \sum_{i=1}^n r_i \frac{\partial r_i}{\partial \alpha_j} = 0, \quad j = 1, \dots, m \quad (9.130)$$

or

$$\sum_{i=1}^n r_i \frac{\partial f(x_i, \mathbf{p})}{\partial \alpha_j} = 0 \quad (9.131)$$

By solving Equation 9.131, the parameter $\mathbf{p}$, which enables the residue minimum, can be determined.

9.2.4.2 Curve Fitting

Curve fitting is often expressed by a mathematical function. The aim is to best fit the measured data points, which are possibly subject to constraints. Interpolation technology will allow for an exact fit to the data when it is required. Additionally, the fitted function may be smoothed to result in a “better looking” curve.

Regression is often used for curve fitting through measured data pairs, which are believed to be independent variables and their corresponding functions. Statistical

inference is often used to deal with any uncertainties and randomness. Extrapolation can be used to predict results beyond the range of the observed data, although this implies a greater degree of uncertainty.

We now consider the following function of x and y :

$$y = f(x) \quad (9.132)$$

which will be curve-fitted through least squares regression.

9.2.4.3 Realization of Least Squares Method

9.2.4.3.1 Linear Function

In the situation when the function is linear,

$$y = f(x) = a_1 x + a_0 \quad (9.133)$$

9.2.4.3.2 Nonlinear Function

In the situation when the function is nonlinear,

$$y = f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots a_0 \quad (9.134)$$

In general, y_i can be written as

$$y_i = a_n x_i^n + a_{n-1} x_i^{n-1} + \dots a_0 x_i, \quad i = 1, \dots, p, p > n \quad (9.135)$$

In matrix form, this can be written as

$$\begin{bmatrix} x_1^n & x_1^{n-1} & \dots & x_1 \\ x_2^n & x_2^{n-1} & \dots & x_2 \\ & & \dots & \\ x_p^n & x_p^{n-1} & \dots & x_p \end{bmatrix} \begin{Bmatrix} a_n \\ a_{n-1} \\ \dots \\ a_0 \end{Bmatrix} = \begin{Bmatrix} y_1 \\ y_2 \\ \dots \\ y_p \end{Bmatrix} \quad (9.136)$$

where the parameter $\mathbf{p}$ is equal to

$$\mathbf{p} = \begin{Bmatrix} a_n \\ a_{n-1} \\ \dots \\ a_0 \end{Bmatrix} \quad (9.137)$$

From Equation 9.136, the nonlinear function can be determined as given below:

$$\begin{Bmatrix} a_n \\ a_{n-1} \\ \dots \\ a_0 \end{Bmatrix} = \begin{bmatrix} x_1^n & x_1^{n-1} & \dots & x_1 \\ x_2^n & x_2^{n-1} & \dots & x_2 \\ & & \dots & \\ x_p^n & x_p^{n-1} & \dots & x_p \end{bmatrix}^+ \begin{Bmatrix} y_1 \\ y_2 \\ \dots \\ y_p \end{Bmatrix} \quad (9.138)$$

In Equation 9.138, the superscript + stands for the pseudo inverse of a matrix, sat matrix $\mathbf{A}$, written as

$$\mathbf{A}^+ = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \quad (9.139)$$

9.3 VIBRATION TESTING

Vibration testing is an important measure in inverse dynamic problems. The focus of this section is on random vibration-related issues, as opposed to general vibration testing. Generally in vibration testing, the amount of randomness is fairly small in comparison to the desired signals and the measurement noises. For this reason, randomness is typically ignored. However, in some instances, the randomness decreases the signal-to-noise ratio significantly. In this section, randomness and uncertainty will be discussed only in qualitative terms, rather than quantitative details.

Strictly speaking, randomness and uncertainty have separate concepts. For random variables or processes, even though individual events cannot be predicted, moments and distributions can be estimated from the corresponding pattern. In contrast, uncertain events are unable to be measured. However, for application to engineering problems, randomness and uncertainty will not differentiate in most situations.

For a more systematic approach, readers may consult McConnell (1995) for instance.

9.3.1 TEST SETUP

Test setup is the beginning step of vibration testing. To physically install a structure that will simulate the system being tested, the system must first be correctly modeled.

9.3.1.1 Mathematical Model

Mathematical models are rather fundamental and are often referred to as an analytic formulation or a closed-form solution. For example, the SDOF vibration system is represented by the mathematical model

$$m\ddot{x}(t) + c\dot{x}(t) + kx(t) = f(t) \quad (9.140)$$

with a solution of

$$x(t) = h(t) * f(t) \quad (9.141)$$

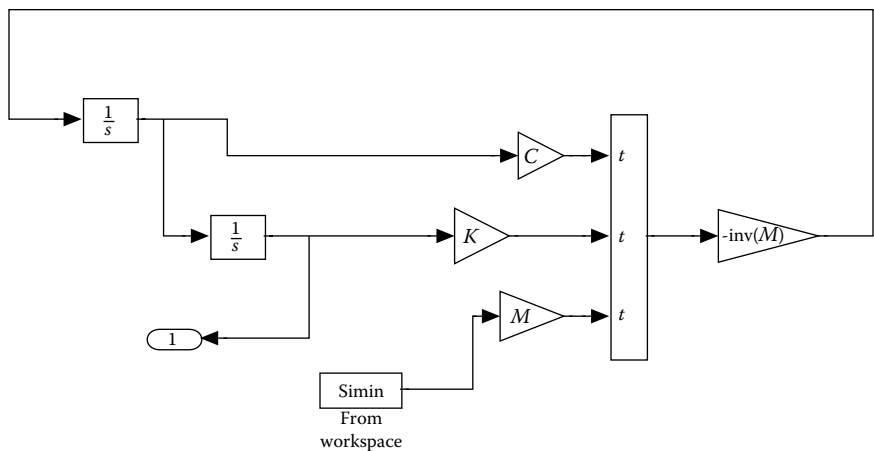


FIGURE 9.3 Simulink model of an SDOF system.

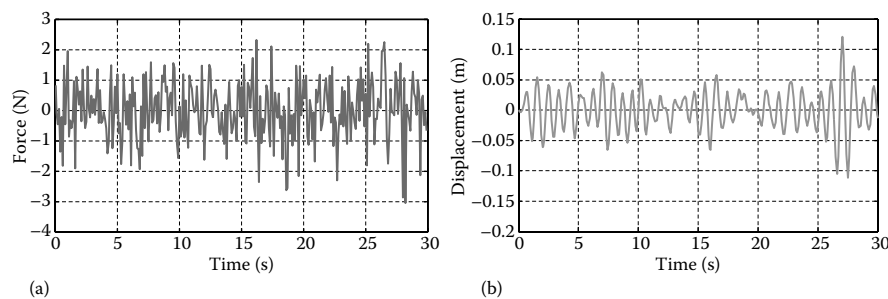


FIGURE 9.4 (a) Input plots for a “random” excitation; (b) output plots for a “random” excitation.

9.3.1.2 Numerical Model

The use of a computer is often necessary to establish a numerical or computational model. For example, consider the SDOF vibration model from Simulink as shown in Figure 9.3.

Figure 9.4 shows the input and output of the numerical model.

9.3.1.3 Experimental Model

If the models described in Section 9.3.1 or Figure 9.3 have no or acceptable error, then it is not necessary to conduct a vibration test. However, these two models can be exceedingly “deterministic.” Namely, there are many uncertain aspects in the real world that cannot be precisely represented by mathematical or numerical models. Individual examples include boundary conditions, specific properties of elements or members in vibration systems, the “correct” number of degrees of freedom, and the distributions of mass, damping, and stiffness, among others.



FIGURE 9.5 Photo of bridge test.

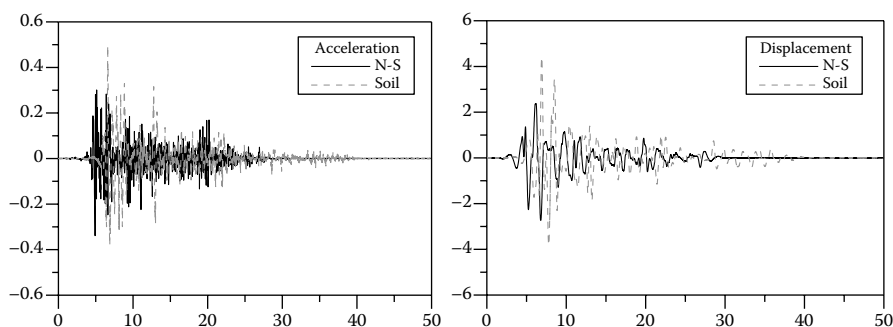


FIGURE 9.6 Recorded time histories.

In this situation, the setup of a physical test model to deal with the uncertainties becomes necessary. During the test setup, it becomes imperative to isolate the test targets in order to best minimize the test uncertainty.

Figures 9.5 and 9.6 show an example of experimental testing on a highway bridge and recorded time histories on a model bridge, respectively.

9.3.2 EQUIPMENT OF ACTUATION AND MEASUREMENT

To use vibration testing as a measure of system identification, the input and output are needed. The input is often a force, or a displacement, which is referred to as an actuation. Both the input and output responses of the test model must be measured. In using this method, it becomes possible to reduce the degree of uncertainties.

9.3.2.1 Actuation

Actuation is completed by using actuators, force hammers, or other measures. Figure 9.7 shows a typical hydraulic actuator, while Figure 9.9 shows an electromagnetic actuator and Figure 9.10 shows an impact hammer.

One of the advantages of using actuators is that we can directly apply “random” excitations for testing, such as white and/or color noises. These artificial “random”



FIGURE 9.7 Photos of actuators and shaking tables.

processes, often called pseudo-random, can simulate true random environments. Examples are experimental studies by using earthquake shaking tables to provide simulated earthquake ground motions.

It is noted that impact hammers cannot provide random process excitations. The response of impulse excitation is a free-decay vibration.

However, no matter using actuators or hammers, noise contaminations are unavoidable, which are random processes in nature, and they should be accounted for vibration testing.

9.3.2.1.1 Actuators

Due to uneven frequency responses of actuators, the input demand and the output force/displacement will not be 100% proportional. Furthermore, due to possible control loops that can be nonlinear as well as time varying, this phenomenon can be magnified.

9.3.2.1.1.1 Transfer Function of an Actuator Therefore, in experimental studies using an actuator, its transfer functions must first be considered. Practically speaking, there are two kinds of functions: the first is the *transfer function without load*, which is expressed in Equation 9.142 and shown in Figure 9.8. The second is the *transfer function with load*, which is described in Equation 9.143.

$$H_o(\omega) = \frac{F(\omega)}{S(\omega)} \quad (9.142)$$

$$H_L(\omega) = \frac{F(\omega)}{S(\omega)} \quad (9.143)$$

In initial comparisons of Equations 9.142 and 9.143, the transfer functions appear identical. However, the curves of magnitude of force (in kips) vs. frequency are significantly different between the two transfer functions. Even in the case of zero load

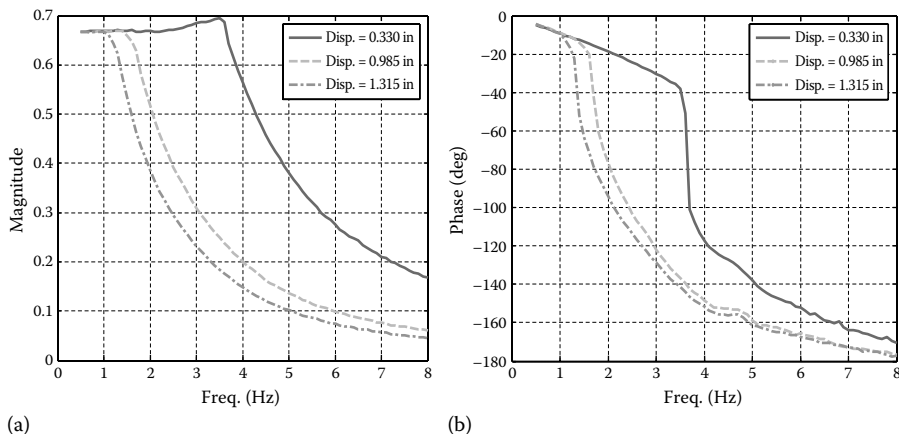


FIGURE 9.8 Measured transfer functions: (a) magnitude vs. frequency; (b) phase vs. frequency.

and different displacement will the curves be dissimilar. This is shown in Figure 9.8a. In this case, the main reason is the limitation of the maximum velocity for the given actuators and secondarily nonlinear control.

For given demands of force and/or displacement, deterministic or random, the actual force/displacement will be understandably different. Note that the transfer function shown in Figure 9.8 is measured through the sine sweep test. When random signals are used, the transfer function should be measured accordingly.

9.3.2.1.1.2 Harmonic Distortion Most actuators (see Figure 9.9a and b) will have different degrees of harmonic distortion. The ideal forcing function can be written as

$$f(t) = f_0 \sin(\omega_0 t) \quad (9.144)$$

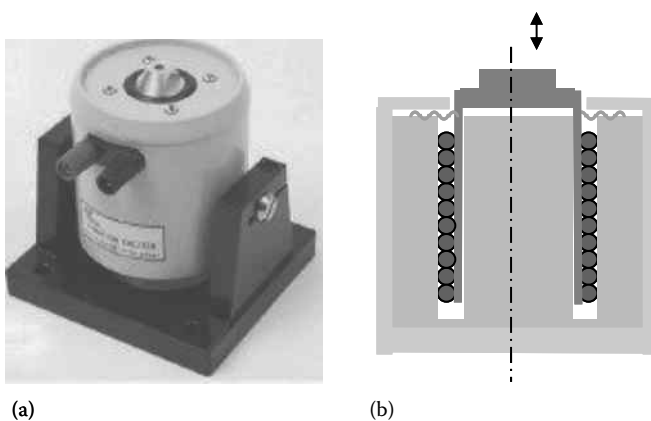


FIGURE 9.9 Electromagnetic actuator. (a) Photo of an actuator and (b) conceptual drawing.

This can be written in a more practical form:

$$f(t) = f_0 \sin(\omega_0 t) + f_1 \sin(2\omega_0 t) + f_2 \sin(3\omega_0 t) + \dots n(\Omega t + \Theta) \quad (9.145)$$

In Equation 9.145, $n(\Omega t + \Theta)$ denotes uncertainty and unwanted noises.

In comparing Equation 9.145 with Equation 9.144, it is observed that given a pure sinusoidal signal for the forcing function to an actuator, the real-world signal is distorted to have other frequency components, such as $f_1 \sin(2\omega_0 t)$, $f_2 \sin(3\omega_0 t)$, etc. This signifies that the actuator is behaving nonlinearly.

9.3.2.1.1.3 Tracking Filter The nonlinearity can be minimized by using tracking filters, which have the following transfer function:

$$H_T(\omega) = \begin{cases} 1, & \omega = \omega_0 \\ 0, & \text{elsewhere} \end{cases} \quad (9.146)$$

In the ideal case, the output of the actuator $F_{\text{practice}}(\omega)$ in the frequency domain will be

$$F_{\text{practice}}(\omega) = F(\omega)H_T(\omega) \quad (9.147)$$

9.3.2.1.2 Impact Hammers

Impulse-like forces are introduced by an impact hammer. An example of this is shown in Figure 9.10.

9.3.2.1.2.1 Forcing Function An example of the forcing function of an impact hammer for an ideal case is

$$f(t) = f_0 \delta(t) \quad (9.148)$$



FIGURE 9.10 Impact hammer. (Photo courtesy of PCB Piezotronics.)

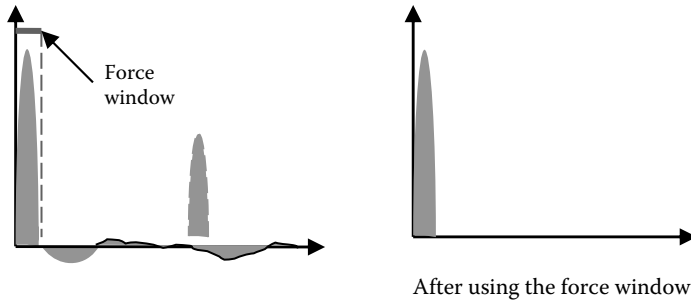


FIGURE 9.11 Recorded forces.

In practical applications, the half-sine-like time history of the impact force and the impact duration, denoted by frequency ω , can be varied by the softness of the hammer head as well as the surface of the test object. This is represented by the following equation:

$$f(t) = f_0 \sin(\omega t) + n(\Omega t + \Theta), \quad 0 < \omega t < \pi/2 \quad (9.149)$$

In Equation 9.149, $n(\Omega t + \Theta)$ denotes uncertainty and unwanted noises.

9.3.2.1.2.2 Force Window The force window can be used to minimize unwanted noise. The idealized function of the force window in the time domain is

$$w(t) = \begin{cases} 1, & 0 < t < \frac{\pi}{2\omega} \\ 0, & \text{elsewhere} \end{cases} \quad (9.150)$$

For an ideal case in which the force is generated by the hammer $f_{\text{practice}}(t)$ in the time domain is given by

$$f_{\text{practice}}(t) = f(t)w(t) \quad (9.151)$$

This is generally shown in Figure 9.11.

9.3.2.2 Measurement

Both the input forces and the output responses during a vibration test need to be measured. The necessary instrumentation for these measurements typically contains a sensory system, a data acquisition system, and signal processing, which may introduce some degree of randomness and uncertainty. The basic method to address randomness is averaging. For most vibration testing, randomness is ignored and averaging is not carried out. However, to acquire more precise measurements, it will be necessary (see Bendat and Piersol 2011).

9.3.2.2.1 Sensors

We now consider the basic concept of sensors and analyze the possible randomness.

9.3.2.2.1.1 Sensitivity The *sensitivity* of a sensor is defined as the output of the unit input of the sensor. Typically, the signals are measured at the lowest frequency point.

$$S_{(\cdot)} = \frac{\text{output}}{\text{input}} = \text{output} \big|_{\text{unit input}} \quad (9.152)$$

Once a sensor is manufactured, its sensitivity should be calibrated and likely recorded and included in its packaging. However, the sensitivity will likely experience drifts and/or fluctuations over time due to many factors, such as temperature, pressure of the atmosphere, mounting, cabling, and grounding, among others.

For different driving frequencies, the ratio of output and input will vary. This will be further explained in Section 9.3.2.2.1.5.

9.3.2.2.1.2 Resolution, q Resolution is the smallest possible measurement that determines the ability to observe fine details in the measurement.

$$q = S_{v-q} e_{\min} \quad (9.153)$$

S_{v-q} refers to the sensitivity that transfers the electronic signal to the required measurement quantity. Furthermore, $e_{\min}$ is the minimum electric signal that a measuring device can put out without being contaminated by noises.

9.3.2.2.1.3 Dynamic Range The dynamic range describes the minimum and the maximum measurable ranges in decibels, as denoted in Equation 9.154.

$$D_D = 20 \log(e_{\max}/e_{\min}) \text{ (dB)} \quad (9.154)$$

Here, $e_{\max}$ is the maximum electric signal that a measuring device can output.

9.3.2.2.1.4 Time Constant A time constant is described as

$$\frac{e(t) - e(\infty)}{e(0) - e(\infty)} = e^{-\frac{t}{T}} \quad (9.155)$$

where $e(0)$ is the initial signal and $e(\infty)$ is the signal measured after a sufficiently long period. Also, $e(t)$ is the signal picked up at time t . In Equation 9.155, the parameter T specifically denotes the *time constant* given by

$$T = RC \quad (9.156)$$

where R in ohms and C in farads are respectively the resistance and capacitance of the measurement circuit and T is in seconds.

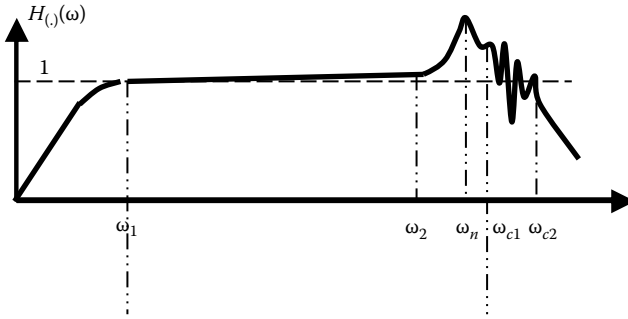


FIGURE 9.12 Frequency response function.

9.3.2.2.1.5 Frequency Response It is advantageous that Equation 9.158 is able to maintain the measuring frequency range. This is shown in Figure 9.12.

$$H_{(j)}(\omega) = \text{const} \quad (9.157)$$

In the above, the corresponding frequency band is referred to as the working band. Characteristically for the working band, the phase ϕ vs. the frequency ω plot of the frequency response function is a straight line. This is represented in the following:

$$\phi = a\omega \quad (9.158)$$

In this instance, a is the slope of the line.

However, as seen in Figure 9.12, the dynamic magnification curves will not be an exact horizontal line. Denoting the normalized frequency response function as $H_{(j)}(\omega)$, then

$$H_{(j)}(\omega) = \frac{H_{(j)}(\omega)}{S_{(j)}} \neq 1 \quad (9.159)$$

9.3.2.2.1.6 Linearity The measurement linearity can be defined by Figure 9.13.

Here,

$$L_s = \frac{e_U - e_m}{e_m} \quad (9.160a)$$

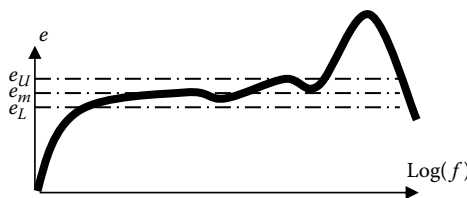


FIGURE 9.13 Linearity.

and/or

$$L_s = \frac{e_m - e_L}{e_m} \quad (9.160b)$$

where e_U and e_L are the upper and lower limits of the measured signal, respectively, and e_m is the mean value, with

$$L_s \leq 5\% \quad (9.161)$$

9.3.2.2.1.7 *Cross-Sensitivity* The cross-sensitivity is described by (see Figure 9.14)

$$s_c = \frac{S_\theta}{S_z} = \tan(\phi) \quad (9.162)$$

This is shown in Figure 9.14.

Characteristically, the cross-sensitivity is required to be less than or equal to 5%, that is,

$$s_c \leq 5\% \quad (9.163)$$

An ideal sensor should have the linear capability to pick up physical signals and output electric signals proportionally. Unfortunately, due to the limitation of measurement ranges, natures of nonlinearity in both the frequency and time domains, the output signal will not be purely proportional to the physical signal. Additionally, unwanted noise can contaminate the sensory output, including nonzero transfer signals.

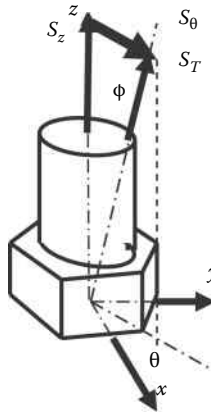


FIGURE 9.14 Cross-sensitivity.

As a result, the actual sensitivity of a sensor ($\cdot$) can be written as

$$S_{(\cdot)}(\omega, t) = S_{(\cdot)} H_{(\cdot)}(\omega) N_{(\cdot)}(\omega, t) \quad (9.164)$$

Here, $H_{(\cdot)}(\omega)$ is the normalized frequency response function of the sensor, which describes the frequency range and the dynamic magnification. The function, $N_{(\cdot)}(\omega, t)$, covers all the randomness due to sensitivity drift and noises at the moment of signal pickup, among others.

9.3.3 SIGNAL AND SIGNAL PROCESSING

9.3.3.1 Data-Acquisition System

Figure 9.15 shows a typical data-acquisition system, which consists of several portions of instrumentation.

As shown in Figure 9.15, there are many links in the signal chain. In each link, there is a probability that the signals will be distorted by the above-mentioned non-linear response or contaminated by external noises.

The gain of each link can be denoted by $S_i(\omega, t)$. An example of such a gain is the case of amplifiers. Similar to the case of the above-mentioned sensor, the corresponding gain is also a function of the driving frequency and furthermore a random process.

$$S_i(\omega, t) = S_i H_i(\omega) N_i(\omega, t), \quad i = 1, \dots \quad (9.165)$$

The signal measured and stored in data memories denoted by $Y(\omega, t)$ will be different from the signal to be measured, denoted by $x(t)$. This is due to the sensitivity

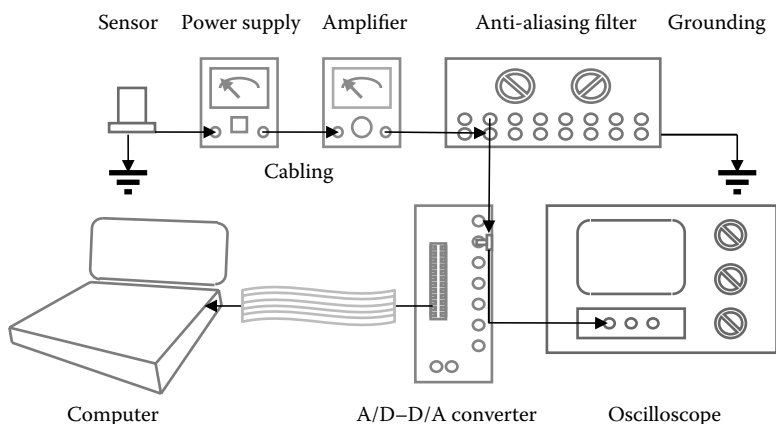


FIGURE 9.15 Example of data acquisition systems.

of the sensor as described in Equation 9.164 and the gain as described in Equation 9.165. The expression of randomness is simplified and is given by

$$Y(\omega, t) = N_p(\omega, t) \prod_i S_i \int_0^t x(\tau) h_s(t - \tau) d\tau + N_a(t) \quad (9.166)$$

In Equation 9.166, $h_s(t)$ is the impulse response function of the total measurement system, which can be regarded as an inverse Fourier transform of the normalized transfer function of the total measurement system. That is,

$$h_s(t) \Leftrightarrow H_s(\omega) = \prod_i H_i(\omega) \quad (9.167)$$

where $H_1(\omega)$ is the transfer function of the first link of the measurement system, namely, the sensor's.

Using the knowledge of transfer functions, each $H_i(\omega)$ can be measured instrument by instrument, resulting in $H_s(\omega)$. Nevertheless, a more accurate and effective method is to measure $H_s(\omega)$ of the total system as a whole. The objective is to determine the linear range and allowed error range (refer to Figure 9.13).

$$H_s(\omega) = \prod_i S_i H_i(\omega) = S_1 \prod_i H_i(\omega) \quad (9.168)$$

In an ideal case,

$$\prod_i H_i(\omega) = 1 \quad (9.169)$$

and

$$H_s(\omega) = S_1 = \text{const} \quad (9.170)$$

Equation 9.170 is often given in many test menus. However, Equation 9.170 can only be used in the event when the measurement of randomness proves to be negligible.

$N_p(\omega, t)$ is a random process that modifies the output convolution caused by the *noise contamination of production*. Furthermore, $N_a(t)$ is an additional random process due to the *noise contamination of addition*. A simplified expression of $N_p(\omega, t)$ only can be observed due to gain drift, which is often modeled by normal distributions. This is written as

$$N_p(\omega, t) \sim N(\mu_N(t), \sigma_N(t)) \quad (9.171)$$

where $\mu_N(t)$ and $\sigma_N(t)$ are the corresponding mean and standard deviation, respectively. Equation 9.70 can be used to estimate the sample average $\bar{x}_j$ to approximate $\mu(t)$, while Equation 9.71 can be used to estimate the sample variance $S_X^2(t_j)$ to approximate $\sigma^2(t)$.

According to Equation 9.71, by increasing the number of tests n ,

$$S_X^2(t_j) \rightarrow 0 \quad (9.172)$$

so that

$$\sigma(t) \rightarrow 0 \quad (9.173)$$

In so doing, the influence of $N_p(t)$ is significantly reduced. Nevertheless, *systematic error* will certainly remain given that it cannot be removed through averages (see Equation 9.110).

A simplified expression of $N_a(t)$ is the result of zero-drift measurement, among many sources of noises. Averaging can reduce the influence of $N_a(t)$. For the first and second types of transfer functions, $H_1(\omega)$ and $H_2(\omega)$ are the specific available methods to treat the additive noises.

Figure 9.16a shows a measured gain of a data acquisition system whose minimal value would be 5 if the system is not contaminated by both noise $N_p(\omega, t)$ and $N_a(t)$.

Through averaging, the noise $N_a(t)$ can be reduced as shown in Figure 9.16b. On the other hand, $N_p(\omega, t)$ is a systematic error, which cannot be removed through averaging. However, if the fluctuation is deterministic, then the previously mentioned curve fit method can be used to obtain the approximated function.

9.3.3.2 Single Processing and Window Functions

In the situation when a measurement error due to $N_p(t)$ exists, further steps will be necessary.

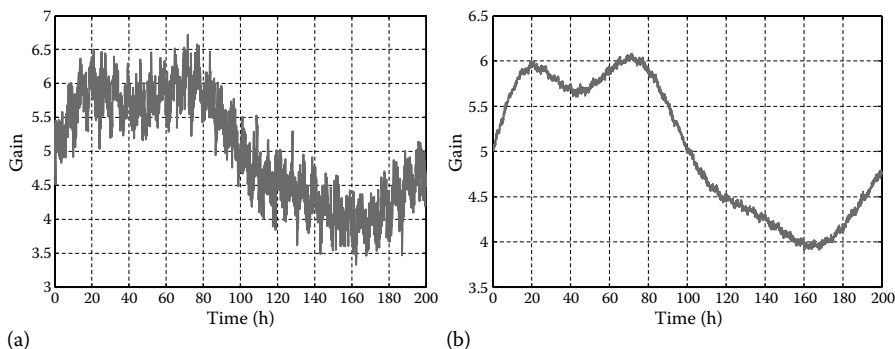


FIGURE 9.16 (a) Raw and (b) averaged time histories.

9.3.3.2.1 Power Leakage and Frequency Windows

Consider a system excited by the driving frequency ω_0 :

$$\omega_0 = 2\pi f_0 = \frac{2\pi}{T_0} \quad (9.174)$$

Often, the signal is sampled by an integer not a multiple of ω_0 . When the signal is not sampled with period T_0 , then there will be discontinuities in the magnitude and slope, which can be seen in Figure 9.17.

Since the total power of the signal should not be changed, the smaller spectra lines, illustrated in Figure 9.18b, will shear the power with the central line. As a result, the length of the line is reduced, which describes the concept of power leakage.

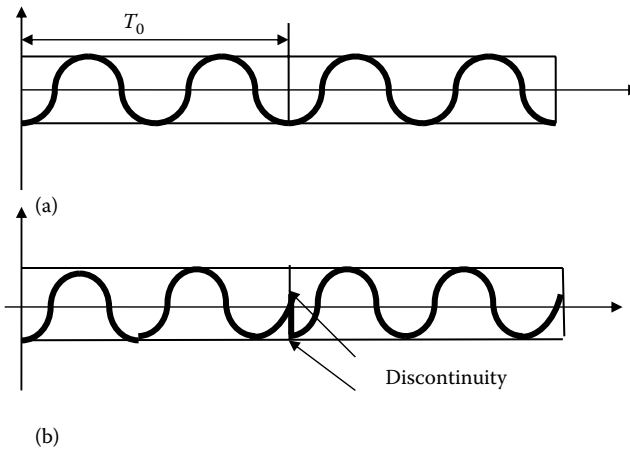


FIGURE 9.17 Total sampling period. (a) Signal with a frequency multiple of the sampling period. (b) Signal with a frequency not a multiple of the sampling period.

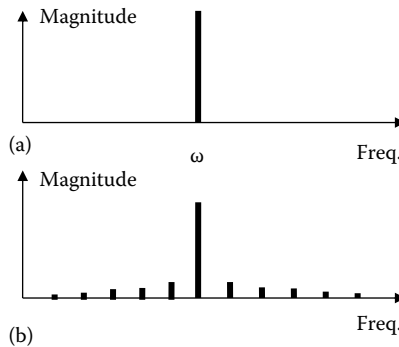


FIGURE 9.18 (a) Original spectrum. (b) Power leakage.

Because of the particular buffering size and sampling rate, the power leakage is a random event. Such random error cannot be minimized by the averaging of repeated tests. Such error is referred to as systematic error.

9.3.3.2.2 Window Functions, Minimizing Power Leakage

In order to minimize the effect of power leakage, window functions can be used. Window functions are also referred to as apodization functions or tapering functions and are functions zero valued outside of some chosen time interval.

A function that is constant inside the interval and zero elsewhere is referred to as a *rectangular* window, also known as a *boxcar* window or Dirichlet window, because the shape of its graphical representation is rectangular. In fact, when samples from the signals within a certain measurement period are taken, the signal is shown to be multiplied by the rectangular window function. In other words, the product is also zero valued outside the time interval. All that means is the “view” through the window.

To minimize the effectiveness of power leakage, window functions should be nonnegative smooth “bell-shaped” curves, which greatly reduces the discontinuity.

Unlike the low-pass filter used for anti-aliasing, which is used before sampling (A/D conversion), window function, also known as windowing, is typically used after sampling. Denoting the total number of samples as N , the commonly used window functions in modal testing can be listed as follows:

1. Rectangular window

$$w(n) = \begin{cases} 1, & 0 < n < N \\ 0, & \text{elsewhere} \end{cases} \quad (9.175)$$

2. Hamming window (Richard W. Hamming, 1915–1998)

$$w(n) = \begin{cases} 0.53836 - 0.46164 \cos\left(\frac{2n\pi}{N-1}\right), & 0 < n < N \\ 0, & \text{elsewhere} \end{cases} \quad (9.176)$$

3. Hanning window (Julius F. von Hann, 1839–1921)

$$w(n) = \begin{cases} 0.5 \left[1 - \cos\left(\frac{2n\pi}{N-1}\right) \right] & 0 < n < N \\ 0, & \text{elsewhere} \end{cases} \quad (9.177)$$

The Hanning and Hamming windows are both known as “raised cosine” windows.

4. Cosine window (sine window)

$$w(n) = \begin{cases} \sin\left(\frac{n\pi}{N-1}\right) & 0 < n < N \\ 0, & \text{elsewhere} \end{cases} \quad (9.178)$$

5. Gauss window ($\sigma \leq 0.5$)

$$w(n) = \begin{cases} e^{-\frac{1}{2} \left[\frac{n-(N-1)/2}{\sigma(N-1)/2} \right]^2} & 0 < n < N \\ 0, & \text{elsewhere} \end{cases} \quad (9.179)$$

6. Bartlett–Hann window

$$w(n) = \begin{cases} 0.62 - 0.48 \left| \frac{n}{N-1} - 0.5 \right| - 0.38 \cos\left(\frac{2n\pi}{N-1}\right), & 0 < n < N \\ 0, & \text{elsewhere} \end{cases} \quad (9.180)$$

7. Blackman window ($\alpha = 0.16$) (Blackman and Tukey, 1959)

$$w(n) = \begin{cases} \frac{1-a}{2} - 0.5 \cos\left(\frac{2n\pi}{N-1}\right) + \frac{a}{2} \cos\left(\frac{4n\pi}{N-1}\right), & 0 < n < N \\ 0, & \text{elsewhere} \end{cases} \quad (9.181)$$

8. Kaiser–Bessel window (Kaiser and Schafer 1980; Friedrich W. Bessel, 1784–1846)

$$w(n) = \begin{cases} 0.3929 - 0.51 \cos\left(\frac{2n\pi}{N-1}\right) + 0.0959 \cos\left(\frac{4n\pi}{N-1}\right) \\ -0.0012 \cos\left(\frac{6n\pi}{N-1}\right), & 0 < n < N \\ 0, & \text{elsewhere} \end{cases} \quad (9.182)$$

In using the above windows, the original solitary $y(t)$ is managed by multiplying it by a window function, resulting in the function $y_w(t)$ denoted by

$$y_w(t) = y(t)w(t) \quad (9.183)$$

In the discrete time domain, this becomes

$$y_w(n) = y(n)w(n) \quad (9.184)$$

9.3.4 NYQUIST CIRCLE (HARRY NYQUIST, 1889–1976)

In Section 9.3.3, we discussed the transfer function by using the absolute value vs. frequency plot, as well as the angle vs. frequency plot.

If there exists no noise, theoretically, we can use the above-mentioned modal testing to measure the natural frequency and the damping ratio. However, due to noises, such curves will be altered so that to find more accurate data, we need the curve fit of the Nyquist plot for two reasons. First, the Nyquist plot is very close to a pure circle, whose equation is simple and straightforward. Second, because the resonant region of the Nyquist plot occupies the entire half circle, the corresponding curve fit can provide more accurate results.

9.3.4.1 Circle and Nyquist Plot

The transfer function consists of both real and imaginary parts; both are functions of driving frequencies. The Nyquist plot is a special form of the transfer function, where the imaginary vs. real part is plotted. Figure 9.19 shows a typical Nyquist plot.

From Figure 9.19, it is seen that the Nyquist plot of an SDOF system is quite close to a circle, which is referred to as the Nyquist circle, so that we can use the function of a circle to fit the Nyquist plot.

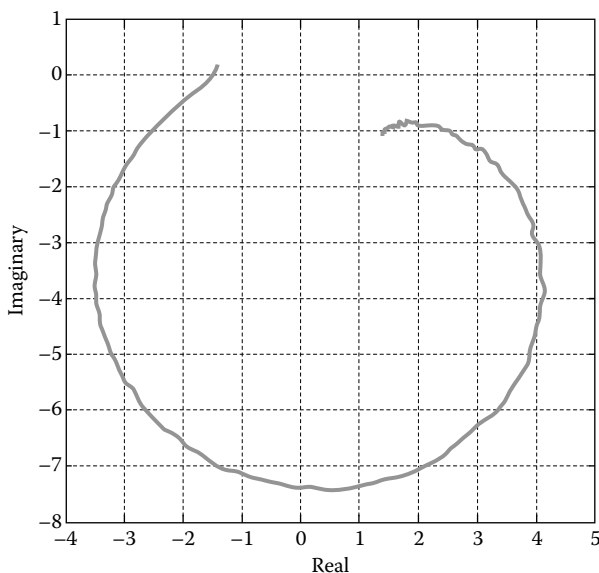


FIGURE 9.19 Nyquist plot.

Denote R as

$$R = \frac{1}{m\omega_d} \quad (9.185)$$

We have

$$H(\omega) = \frac{R}{2j[\zeta\omega_n + j(\omega - \omega_d)]} - \frac{R}{2j[\zeta\omega_n + j(\omega + \omega_d)]} \quad (9.186)$$

Note that, in the neighborhood of the natural frequency ω_d , the first term on the right-hand side of Equation 9.186 will be significantly larger than the second term. Thus, we can rewrite Equation 9.186 as

$$H(\omega) \approx \frac{R}{2j[\zeta\omega_n + j(\omega - \omega_d)]} \quad (9.187)$$

That is, we can just use the first term to carry out the curve fit.

Rewrite Equation 9.187 as

$$H(\omega) \approx \text{Re}[H(\omega)] + j \text{Im}[H(\omega)] = \frac{R}{2} \frac{\omega_d - \omega}{[(\omega_d - \omega)^2 + \zeta^2\omega_n^2]} + j \frac{R}{2} \frac{-\zeta\omega_n}{[(\omega_d - \omega)^2 + \zeta^2\omega_n^2]} \quad (9.188)$$

It is seen that the following equation

$$\{\text{Re}[H(\omega)]\}^2 + \left\{ \text{Im}[H(\omega)] + \frac{R}{4\zeta\omega_n} \right\}^2 = \left\{ \frac{R}{4\zeta\omega_n} \right\}^2 \quad (9.189)$$

is an equation of a circle. The center is at $(0, \frac{R}{4\zeta\omega_n})$, and the diameter is $\frac{R}{2\zeta\omega_n}$.

Using this circle to fit the Nyquist plot, the natural frequency is the cross-point of the circle and the imaginary axis (see Figure 9.20).

The Nyquist plot as shown in Figure 9.20, is not exactly a circle. However, in the resonant region, the Nyquist plot is very close to a circle; we thus can use the circle fit to identify the corresponding modal parameters.

When accelerance frequency response function (FRF) is used, it can be proven that

$$A(\omega) \approx \text{Re}[A(\omega)] + j \text{Im}[A(\omega)] = \frac{R}{2} \frac{-(\omega_d - \omega)\omega^2}{[(\omega_d - \omega)^2 + \zeta^2\omega_n^2]} + j \frac{R}{2} \frac{\zeta\omega_n\omega^2}{[(\omega_d - \omega)^2 + \zeta^2\omega_n^2]} \quad (9.190)$$

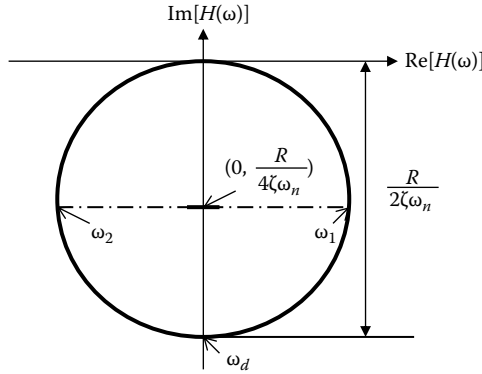


FIGURE 9.20 Using a circle to approximate receptance Nyquist plot.

In this case, we have the circle equation written as

$$\{\text{Re}[A(\omega)]\}^2 + \left\{ \text{Im}[A(\omega)] - \frac{R\omega^2}{4\zeta\omega_n} \right\}^2 = \left\{ \frac{R\omega^2}{4\zeta\omega_n} \right\}^2 \quad (9.191)$$

This circle can be conceptually shown in Figure 9.20.

9.3.4.2 Circle Fit

To carry out the circle fit for the receptance Nyquist plot, assume that the center is located at point $(0, r_{jk})$, and the circle starts from the origin $(0,0)$. The starting point of the Nyquist plot is not at the origin $(0,0)$. However, for convenience, we name the starting point of the Nyquist circle at exactly the origin for the normal mode of a mode, which is not affected by any other modes of the system. Now, rewrite Equation 9.191 as

$$x^2 + (y + r_{jk})^2 = r_{jk}^2 \quad (9.192)$$

and furthermore

$$x^2 + y^2 + 2r_{jk}y = 0 \quad (9.193)$$

Denote

$$x_i = \text{Re}[_p\alpha_{jk}(\omega_i)] \quad (9.194)$$

and

$$y_i = \text{Im}[_p\alpha_{jk}(\omega_i)] \quad (9.195)$$

with m measurement points of the FRF; we can have

$$2 \begin{Bmatrix} y_1 \\ y_2 \\ \dots \\ y_m \end{Bmatrix} r_{jk} = - \begin{Bmatrix} x_1^2 + y_1^2 \\ x_2^2 + y_2^2 \\ \dots \\ x_m^2 + y_m^2 \end{Bmatrix} \quad (9.196)$$

Therefore, the parameter r can be calculated as

$$r_{jk} = -\frac{1}{2} \begin{Bmatrix} y_1 \\ y_2 \\ \dots \\ y_m \end{Bmatrix}^+ \begin{Bmatrix} x_1^2 + y_1^2 \\ x_2^2 + y_2^2 \\ \dots \\ x_m^2 + y_m^2 \end{Bmatrix} = -\frac{\begin{Bmatrix} y_1 \\ y_2 \\ \dots \\ y_m \end{Bmatrix}^T \begin{Bmatrix} x_1^2 + y_1^2 \\ x_2^2 + y_2^2 \\ \dots \\ x_m^2 + y_m^2 \end{Bmatrix}}{2 \begin{Bmatrix} y_1 \\ y_2 \\ \dots \\ y_m \end{Bmatrix} \begin{Bmatrix} y_1 \\ y_2 \\ \dots \\ y_m \end{Bmatrix}} \quad (9.197)$$

From the Nyquist plot, if we can further measure the half power frequencies ω_1 and ω_2 , and furthermore the damped natural frequency ω_{dp} , for the p th mode, then the natural frequency ω_p and the damping ratio ζ_p can be obtained. And, with the help of the parameter r_{jk} , the mode shape can also be determined (Figure 9.21).

Theoretically speaking, the center of the Nyquist circle is exactly located at the imaginary axis. Once the resonant point ω_{dp} is known, which must also be located at the imaginary axis, the center can be found at the halfway point from the origin to the resonant point. However, in Section 9.3.4.3, we will show that, due to the influences of

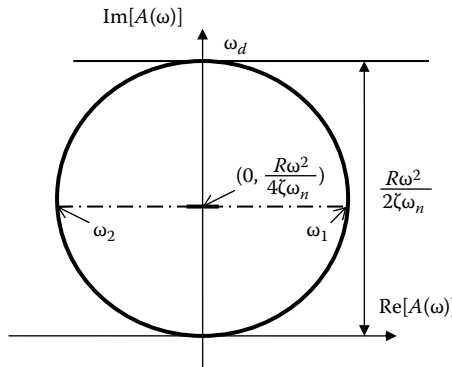


FIGURE 9.21 Using a circle to approximate acceleration Nyquist plot.

other modes, the starting point of the Nyquist circle will move away from the origin. Therefore, it is better to use the above method to locate the center and the origin.

9.3.4.3 Natural Frequency and Damping Ratio

As shown in the Nyquist plot, if we can measure the half-power frequencies ω_1 and ω_2 , and furthermore the damped natural frequency ω_d , then the natural frequency ω_n and the damping ratio ζ can be obtained.

It is known that

$$\zeta = \frac{\omega_2 - \omega_1}{2\omega_n} = \frac{f_2 - f_1}{2f_n} \quad (9.198)$$

and

$$\omega_n = \frac{\omega_d}{\sqrt{1 - \zeta^2}} \quad (9.199)$$

Therefore,

$$(1 - \zeta^2)\omega_n^2 = \omega_d^2 \quad (9.200)$$

Substituting Equation 9.198 into Equation 9.199, we have

$$\omega_p = \left[\omega_{dp}^2 - \frac{(\omega_2 - \omega_1)^2}{4} \right]^{1/2} \quad (9.201)$$

Using Equations 9.198 and 9.201, the natural frequency and the damping ratio can be calculated.

PROBLEMS

1. Given an SDOF system with $m = 10$ kg, $k = 100$ N/m, and its damping ratio is a random variable with a uniform distribution between 0.01 and 0.1. Suppose that this system is excited by random initial velocity $v_0 \sim N(1, 0.2)$ m/s.
 - a. Calculate the free decay responses and identify the corresponding damping coefficient.
 - b. Use the least squares curve fit to find the relationship of the identified damping coefficient and generated damping ratio. Explain your results.
2. With the given six earthquake records (El Centro, Kobe, Northridge, in both S-N and E-W directions):
 - a. Calculate the means, RMS values, and standard deviations of each record.

- b. Generate the time histories of ground displacement through the ground accelerations. At $t = 0.5, 1.0,$ and 10.0 s, find the distributions of the instantaneous ground accelerations. Suppose that the earthquake ground displacements are random processes. Check your results at the three instances. If at a given place, say, an earthquake shaking table, these time histories can be seen as realizations of a random process, is such a process stationary and/or ergodic? Explain your judgment.
3. Use the given six earthquake records (El Centro, Kobe, Northridge, in both S-N and E-W directions).
 - a. To generate your own pseudo response spectra of acceleration, with $\text{PGA} = 0.4$ g, damping ratio = 0.05 and 0.20.
 - b. To generate your own response spectra of absolute acceleration, with $\text{PGA} = 0.4$ g, damping ratio = 0.05 and 0.20. Compare your results with the pseudo spectra.
 - c. Plot the standard deviations at each period vs. the period for the pseudo spectra.
4. Use the maximum likelihood method to write formulae for estimation of
 - a. An exponentially distributed variable with a PDF given by

$$f_X(x) = \lambda e^{-\lambda x} \quad \lambda > 0$$

- b. A normally distributed random variable with a PDF given by

$$f_N(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]$$

5. A 3-DOF structure has mass, stiffness and damping matrices given by $\mathbf{M} = \text{diag}([500, 500, 2000]) \cdot 10^3$ kg

$$\mathbf{C} = \begin{bmatrix} 1050 & -1000 & 0 \\ -1000 & 2050 & -1000 \\ 0 & -1000 & 3200 \end{bmatrix} \text{ kN/m-s}$$

$$\text{and } \mathbf{K} = \begin{bmatrix} 1000 & -1000 & 0 \\ -1000 & 2000 & -1000 \\ 0 & -1000 & 3200 \end{bmatrix} \text{ MN/m}$$

- a. Calculate the natural frequencies and damping ratios.
- b. Use MATLAB® ">>t=(0:1:1999)*0.02; Ag = 4 *randn(2000,30);" to generate ground excitation and compute the absolute acceleration $\mathbf{x}_A = [\mathbf{x}_{A1}; \mathbf{x}_{A2}; \mathbf{x}_{A3}]_{2000 \times 30}^T$.
- c. Calculate the corresponding transfer function $H_1, H_2,$ and H_3 through 30 averages.

6. Use the Nyquist circle fit to find the corresponding natural frequencies and damping ratios. Compare your results with that obtained in Problem 5a.
7. a. Estimate the mean and the standard deviation of the following data:

$X = \begin{bmatrix} -0.1011 & 0.1918 & 0.3092 & 0.5513 & 0.5584 \\ 0.5436 & 0.7443 & 0.6915 & 0.8632 & 0.9183 \\ 0.9654 & 0.9594 & 1.1095 & 0.8044 & 1.0095 \\ 1.0146 & 0.9408 & 0.8533 & 0.7138 & 0.6796 \\ 0.5605 & 0.3612 & 0.2255 & 0.1057 & -0.1425 \end{bmatrix}$

$Y = \begin{bmatrix} -0.1011 & 0.0614 & 0.0508 & 0.1692 & 0.0591 \\ -0.0644 & 0.0380 & -0.1009 & -0.0020 & -0.0048 \\ 0.0000 & -0.0318 & 0.1095 & -0.1874 & 0.0428 \\ 0.0896 & 0.0731 & 0.0578 & 0.0040 & 0.0677 \\ 0.0569 & -0.0256 & -0.0377 & -0.0296 & -0.1475 \end{bmatrix}$

- b. Use MATLAB to generate $t = (0:1:24) * 0.1307$. Use MATLAB to plot $(t, x, t, x-y, t, y)$. Explain your results.
8. Use the given time history of ground excitation to study the peak and RMS values of the vibration of an automobile; its basic weight is 15.0 kN, with additional weight W_p to be a uniformly distributed random variable from 0.5 up to 5.0 kN. Assume that the stiffness of the suspension system is 2.0 MN/m. Assume that the damping coefficient is 12.5 kN/m-s. (Hint: the automobile is modeled as an SDOF system, $g = 10 \text{ m/s}^2$.)
9. Design an experimental test to investigate the relationship of the normal pressure vs. friction coefficient under a sinusoidal vibration environment of pairs of specimens with certain materials. The outputs of this study are
 - a. The curve fitted PDF of the friction coefficient under a given amplitude of vibration with fixed normal pressure
 - b. The curve fitted PDF of the friction coefficient under a given amplitude of normal pressure with a fixed level of vibration
 - c. The curve fitted joint PDF of the friction coefficient under given amplitudes of normal pressure and vibration
10. Design an experimental test to investigate the relationship of the normal pressure vs. friction coefficient under a white noise random vibration environment of pairs of specimens with certain materials. The outputs of this study are
 - a. The curve fitted PDF of the friction coefficient under a given bound of vibration with fixed normal pressure
 - b. The curve fitted joint PDF of the friction coefficient under given amplitudes of normal pressure and vibration
 - c. The curve fitted PDF of the friction coefficient that is greater than a given level μ_m , under a given bound of vibration with fixed normal pressure.

10 Failures of Systems

In most cases, a system, such as a structure, a machine, a vehicle, etc., is designed to be reliable. That is, the structure and/or components will not fail under given conditions. The conditions include given levels of loads, deformations, environmental conditions, etc., and the service periods. However, systems do fail and the failures are likely to be random processes. In addition, these random processes are likely to be nonstationary.

The fundamental approach of reliability-based design for structures is to determine the correct size of an element given its material and shape so that the failure probability is at an acceptable level.

There are six different failure modes of structural components:

1. Yielding
2. Excessive deformation
3. Brittle fracture
4. Ductile fracture
5. Buckling
6. Fatigue

Failure modes 1–5 involve “level crossing” (level exceeding), while failure mode 6 is primarily for high-cycle fatigue.

10.1 3 σ CRITERION

When a quantitative description of either crossing levels or fatigue cycles is addressed, probability is involved. For most cases, 0% failure is not achievable within reasonable cost. For this reason, a small percentage of failure probability can be allowed. This does, however, result in two issues. The first is how to establish the allowed level of failure probability, and the second is how to calculate the failure probability.

10.1.1 BASIC DESIGN CRITERIA

A commonly used criterion is to allow the design strength R be greater than a specified level, for example,

$$R \geq 3\sigma_s \quad (10.1)$$

In this instance, σ_s is the standard deviation (RMS value) of a stationary Gaussian stress process $S(t)$. Note that the strength of a material is its ability to withstand an

applied stress without failure, which is essentially a random variable, and it is treated as deterministic.

When the mean value of the stress is nonzero, namely, μ_S , then

$$R \geq \mu_S + 3\sigma_S \quad (10.2)$$

10.1.2 3σ CRITERION

Given that $S(t)$ is Gaussian and has zero mean, then

$$P[|S(t)| > 3\sigma_S] = 0.0026 \quad (10.3)$$

or for nonzero mean stress,

$$P[S(t) > \mu_S + 3\sigma_S] = 0.0026 \quad (10.4)$$

Both Equations 10.3 and 10.4 indicate that the failure probability of the stress level being greater than $3\sigma_S$ is less than 0.3%.

10.1.3 GENERAL STATEMENT

10.1.3.1 General Relationship between S and R

A more general case is when both the stress S and the strength R are random. The mean of R can then be expressed as

$$\mu_R \geq \xi \sigma_S \quad (10.5)$$

where the parameter ξ is a function of Q and C_R . This is illustrated in Figure 10.1.

Here, Q is the ratio of μ_S and σ_S (which are the mean and standard deviation of the stress process $S(t)$, respectively), which is the inverse of the coefficient of variation of $S(t)$.

$$Q = \frac{\mu_S}{\sigma_S} \quad (10.6)$$

In addition, C_R is the coefficient of variation of R , the strength:

$$C_R = \frac{\sigma_R}{\mu_R} \quad (10.7)$$

Note that R is a random variable with a mean of μ_R and a standard deviation of σ_R . The parameter ξ can be determined as

$$\xi = 3\eta \quad (10.8)$$

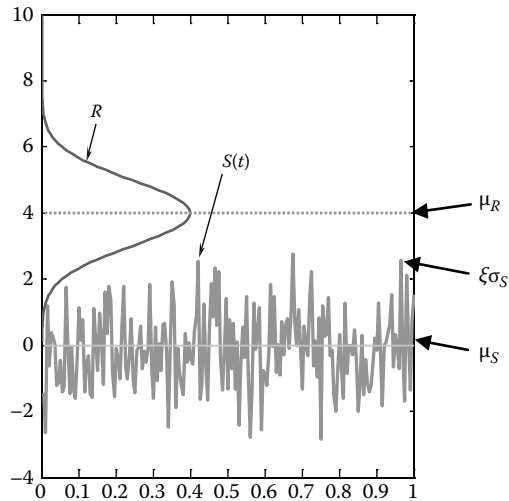


FIGURE 10.1 Variable stress S and strength R .

where η is a safety factor and usually

$$\eta = 1.0 - 2.0$$

(10.9)

A larger value of the safety factor is taken when larger values of Q (larger than 0.5) and C_R (larger than 0.1) are used. Table 10.1 lists the values of η .

Example 10.1

A stress $S(t)$ is Gaussian with $\sigma_S = 70$ MPa and $\mu_S = 35$ MPa. Design the mean strength μ_R of a rod that satisfies the generalized 3σ rule, assuming that $C_R = 0.1$.

$$Q = \frac{\mu_S}{\sigma_S} = 35/70 = 0.5$$

TABLE 10.1
Safety Factor η

C_R	Q									
	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	
0.00	1.0000	1.0033	1.0167	1.0333	1.0500	1.0833	1.1333	1.1600	1.2000	
0.07	1.0267	1.0367	1.0667	1.0833	1.1000	1.1167	1.1433	1.1700	1.2100	
0.10	1.0433	1.0667	1.0833	1.1000	1.1167	1.1500	1.1600	1.1733	1.2200	
0.15	1.1167	1.1267	1.1400	1.1667	1.2000	1.2167	1.2500	1.3000	1.4733	
0.20	1.2233	1.2500	1.2667	1.2833	1.3600	1.3750	1.4167	1.4700	1.5333	

From Table 10.1, when $C_R = 0.1$, the value of η is found to be 1.15 and ξ is calculated to be 3.45.

The corresponding strength is

$$\mu_R = \xi \sigma_S = 3.45 \times 70 = 241.5 \text{ MPa}$$

For a comparison, consider R to be deterministic; then

$$R \geq \mu_S + 3\sigma_S = 35 + 3 \times 70 = 245 \text{ MPa}$$

Now, suppose that the strength of the rods to be designed is random for aspects such as size, yielding stress, etc. To further reduce the chance of failure of the rods, let

$$\mu_R - 1.0\sigma_R \geq \mu_S + 3\sigma_S$$

This results in

$$\mu_R = 245/0.9 = 272 \text{ MPa}$$

10.1.3.2 System Failure, Further Discussion

In Chapter 1, when the concept of failure probability, p_f , is first introduced, it is shown that the value of p_f depends upon two probability density function (PDF) curves: the PDF of load and the PDF of resistance. In Chapter 2, the concept of p_f was further extended to the examination of another set of two PDFs: the PDFs of realistic and expected maximum frictions. The probabilities described in Equations 10.3 and 10.4 are actually failure probabilities. Although in Equations 10.3 and 10.4, only the value of $S(t)$ varies, but the values of 3σ and/or $\mu_S + 3\sigma$ are deterministic. Thus, both stress S and strength R can be random variables or even random processes.

Generally speaking, the failure probability is the collection of all the chances that the real demands, Q_D in Equation 1.130, S in Equations 10.3 and 10.4, μ_{Q_i} in Equation 1.153, etc., exceed the expected resistance, R_D in Equation 1.130, R in Equations 10.1 and 10.2, μ_R in Equation 1.153, etc. Since both the demands and the resistance can be random processes, the basic equation of the double integration, see Equation 1.139a for example, is the key model to compute the failure probability. However, the double integration such as that given in Equation 1.139a can only handle random variables and not a random process. In Sections 10.2 and 10.4, we will further discuss how to estimate the failure probability when at least one of the demands or resistances is time variable, namely, a random process.

10.2 FIRST PASSAGE FAILURE

As mentioned previously, there are six types of failures in general. Most of them (except failure mode 4) can be modeled as a certain breaking of the limit state when the demand S reaches the resistance R , namely, $R = S$. The first passage failure means that the failure occurs the first time that S exceeds R . In the literature, Crandall et al. (1966), Lin (1970), Vanmarcke (1975, 1984), and Marley (1991) provide excellent

contributions to study the first passage failure. In this section, the probability of the first passage failure will be further considered.

10.2.1 INTRODUCTION

In general, once the stress $S(t)$ is greater than the strength R , failure will occur. That is,

$$S(t) > R \quad (10.10)$$

For example, the first passage failure can be seen in the brittle failure mode.

Now, let T and T_s be the time to failure and the duration of service life, respectively. The failure probability, p_f , of the first passage failure can then be expressed by

$$p_f = P(T < T_s) \quad (10.11)$$

Using Y to denote the maximum value of the peaks of the stress process, it can be seen that the event $Y > R$ implies a failure mode. Thus, the failure probability can be written as

$$p_f = P(Y > R) = 1 - F_Y(R) \quad (10.12)$$

where $F_Y(R)$ is the cumulative distribution function (CDF) of R .

10.2.2 BASIC FORMULATION

10.2.2.1 General Formulation

Assume that the strength $R(t)$ is characteristically large in comparison to the stress $S(t)$, so that the probability of $S(t) > R(t)$ is rare, under the assumption that up-crossing with a rate of ν_{R^+} is a Poisson process.

10.2.2.1.1 Failure Probability

Recall from Chapter 5 that

$$P(\text{no up-crossing in } \Delta t) = \exp[-\nu_{R^+}(t)\Delta t] \quad (10.13)$$

For service life T_s , this becomes

$$P(\text{no up-crossing in } T_s) = \prod_{i=1}^k \exp[-\nu_{R^+}(t_i)\Delta t] = \exp\left[-\sum_{i=1}^k \nu_{R^+}(t_i)\Delta t\right] \quad (10.14)$$

From Equation 10.14, when

$$\Delta t \rightarrow 0 \quad (10.15)$$

Equation 10.14 can be reduced to

$$P(\text{no up-crossing in } T_s) = \exp \left[- \int_0^{T_s} v_{R^+}(t) dt \right] \quad (10.16)$$

Thus, the failure probability of the level R being crossed is

$$p_f = 1 - \exp \left[- \int_0^{T_s} v_{R^+}(t) dt \right] \quad (10.17)$$

10.2.2.1.2 Rate of Crossing

To take advantage of Equation 10.17, the formula of v_{R^+} must first be obtained. This, in general, is not easy. However, in the case in which the process $S(t)$ is Gaussian, the formula of v_{R^+} can be achieved by recalling Equations 5.24 and 5.26.

$$v_{R^+}(t) = v_{0^+} \exp \left[- \frac{1}{2} \left(\frac{R(t) - \mu_s}{\sigma_s} \right)^2 \right] \quad (10.18)$$

Here, $v_{0^+}(t)$ is the rate of zero up-crossing, and μ_s and σ_s are the mean and the standard deviation of the stress, respectively.

10.2.2.2 Special Cases

Now consider several special cases of failure probability.

10.2.2.2.1 v_{R^+} Being Constant

When R is constant, then from Equation 10.18, v_{R^+} will also be constant and

$$p_f = v_{R^+} T_s \quad (10.19)$$

10.2.2.2.2 Gaussian and Narrowband

If $S(t)$ is Gaussian and a narrowband process, then

$$v_{0^+} = f_0 \quad (10.20)$$

In this case, f_0 is the center frequency of the narrowband process, and the failure probability is

$$p_f = (f_0 T_s) \exp \left[- \frac{1}{2} \left(\frac{R(t) - \mu_s}{\sigma_s} \right)^2 \right] \quad (10.21)$$

10.2.2.2.3 Constant Symmetric Bounded with Zero Mean

If the process has constant boundary and it is symmetric, namely, has zero mean, then the following are required.

$$|S(t)| = R \quad (10.22)$$

$$S(t)|_{\max} = R \quad (10.23)$$

and

$$S(t)|_{\min} = -R \quad (10.24)$$

Since there is zero mean

$$\mu_s = 0 \quad (10.25)$$

and $S(t)$ is a narrowband, consequently,

$$\nu_{0^+} = 2f_0 \quad (10.26)$$

In this particular case, the failure probability is

$$p_f = (2f_0T_s) \exp \left[-\frac{1}{2} \left(\frac{R}{\sigma_s} \right)^2 \right] \quad (10.27)$$

10.2.3 LARGEST AMONG INDEPENDENT PEAKS

To consider the first passage failure, an alternative approach is to examine the PDF of the peaks of the stress process Z_i by assuming the strength R to be constant.

10.2.3.1 Exact Distribution

The exact distribution is an additional approach for when R is not a function of time. The peak Z_i , however, forms a random process.

Next, consider the case of a distribution $F_Z(z)$.

Suppose that Z_i are mutually independent and B_i denotes the i th event for $Z_i < R$; then

$$B_i = Z_i < R \quad (10.28)$$

B is the event for which Y , the largest peak in a sample of size n , is less than R . This is denoted by

$$B = Y < R \quad (10.29)$$

Furthermore, B is found to be the intersection of all values of B_i 's, denoted by

$$B = B_1 \cap B_2 \cap \cdots \cap B_n = \bigcap_{i=1}^n B_i \quad (10.30)$$

All B_i 's are independent in general. Thus, $P(B)$ can be written as

$$P(B) = \prod_{i=1}^n P(B_i) = [P(B_i)]^n \quad (10.31)$$

Conversely, $P(B)$, when it is a CDF, can be written as

$$P(B) = P(Y < R) = F_Y(R) \quad (10.32)$$

Additionally, $P(B_i)$ as a CDF is

$$P(B_i) = P(Z \leq R) = F_Z(R) \quad (10.33)$$

Note that

$$F_Y(R) = [F_Z(R)]^n \quad (10.34)$$

Therefore,

$$P(\text{no peak exceeds } R) = P(B) = [F_Z(R)]^n \quad (10.35)$$

or, alternatively, Equation 10.35 can be written as

$$p_f = 1 - [F_Z(R)]^n = 1 - F_Y(R) \quad (10.36)$$

If $S(t)$ is narrowbanded, then peak Z will have a Rayleigh distribution such that

$$F_Z(z) = 1 - \exp \left[-\frac{1}{2} \left(\frac{z}{\sigma_s} \right)^2 \right] \quad (10.37)$$

Similar to the process of obtaining a CDF described in Equation 10.35, it is determined that

$$F_Y(R) = \left\{ 1 - \exp \left[-\frac{1}{2} \left(\frac{R}{\sigma_s} \right)^2 \right] \right\}^n \quad (10.38)$$

Furthermore, since the failure probability is presumed small, the exponential term in Equation 10.38 must also be small. Through the binomial series expansion and by allowing

$$n = 2f_0T_s \quad (10.39)$$

p_f can be approximated as

$$p_f \approx (2f_0T_s) \exp \left[-\frac{1}{2} \left(\frac{R}{\sigma_s} \right)^2 \right] \quad (10.40)$$

Comparing Equation 10.40 with Equation 10.30, we see that the failure probability can be obtained through alternative approaches. Apparently, larger resistance strength R will result in smaller failure probability.

10.2.3.2 Extreme Value Distribution

10.2.3.2.1 CDF

A stationary peak process has a distribution of peak heights $F_Z(z)$. As $n \rightarrow \infty$, $F_Y(y)$ will approach extreme-value distributions. For most commonly used distributions, $F_Y(y)$ approaches to the extreme value type I distribution (EVI); for further reference, refer to Equation 2.198, which is repeated as

$$F_Y(y) = P(Y \leq y) = e^{-e^{-\alpha(y-\beta)}} \quad (10.41)$$

10.2.3.2.2 Mean and Standard Deviation

The mean and RMS of Y can be written as

$$\mu_Y = \beta + \frac{0.577}{\alpha} \quad (10.42)$$

$$\sigma_Y = \frac{1.283}{\alpha} \quad (10.43)$$

10.2.3.2.3 Parameters of EVI

To take advantage of Equations 10.42 and 10.43, the parameters α and β must first be calculated. These parameters are determined through the distribution of the individual peak Z from the following equations:

$$\alpha = nf_Z(\beta) \quad (10.44)$$

$$\beta = F_Z^{-1} \left(1 - \frac{1}{n} \right) \quad (10.45)$$

10.2.3.2.4 Special Case of Narrowband Gaussian

In the unusual case, the stress process $S(t)$ is a narrowband Gaussian; it follows that the mean is given by

$$\mu_Y = \left(\sqrt{2 \ln n} + \frac{0.577}{\sqrt{2 \ln n}} \right) \sigma_S \quad (10.46)$$

the standard deviation is given by

$$\sigma_Y = \frac{0.577 \sigma_S}{\sqrt{2 \ln n}} \quad (10.47)$$

and the coefficient of variation (COV) C_Y is

$$C_Y = \frac{\sigma_Y}{\mu_Y} = \frac{1}{1.5588 \ln n + 0.4497} \quad (10.48)$$

Furthermore, the parameters α and β are calculated by

$$\alpha = \frac{\sqrt{2 \ln n}}{\sigma_S} \quad (10.49)$$

and

$$\beta = \sqrt{2 \ln n} \sigma_S \quad (10.50)$$

Example 10.2

Consider the case with the EVI where $n = 1000$ and $\sigma_s = 50$. Calculate the mean and the standard deviation and plot the corresponding PDF of the EVI.

The PDF of the EVI can be obtained by the derivative of $F_Y(y)$ (see Equation 10.41):

$$f_Y(y) = \frac{d[F_Y(y)]}{dy} = \alpha e^{-\alpha(y-\beta)} e^{-\alpha(y-\beta)}$$

With $n = 1000$ and $\sigma_s = 50$, we have $\alpha = 0.074$ and $\beta = 185.85$. The mean is 193.61 and the standard deviation is 48.40. The PDF is plotted in Figure 10.2.

From Figure 10.2, the plot of the PDF is asymmetric and has larger right tails, which has been discussed in Chapter 2 (recall Figure 2.9). Since the EVI has larger left tails, the chance to have a larger value of y , the extreme value, is comparatively greater.

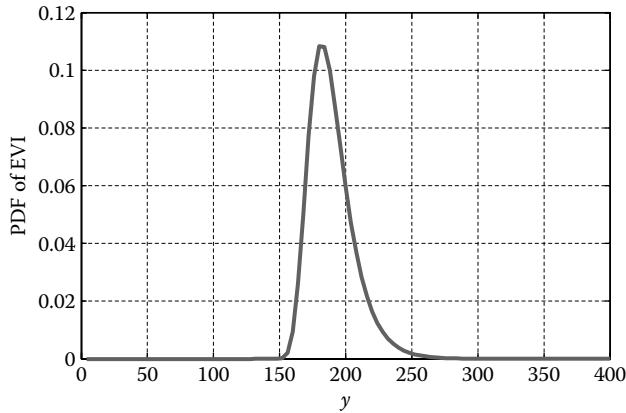


FIGURE 10.2 PDF of EVI.

10.2.3.3 Design Value Based on Return Period

For the design value S_0 , $F_Z(S_0)$ is the CDF for a stationary peak process, where

$$\text{design life} \geq \text{service life} \quad (10.51)$$

The total services period of S_0 is T_n such that

$$T_n = n \quad (10.52)$$

When evenly distributed, the peak having a probability of expience equal to $1/n$ is

$$P(Z > S_0) = 1 - F_Z(S_0) = \frac{1}{n} \quad (10.53)$$

10.3 FATIGUE

In Chapter 5, we described the issue of fatigue by classifying it into two categories: high cycle and low-cycle fatigue. The low-cycle test is further classified as type C and type D low-cycle tests. However, the focus of those studies is on the time-varying development, the model of random processes, and the corresponding statistical parameters. In this section, we will further discuss the issue of fatigue, with the focus on fatigue failures.

10.3.1 PHYSICAL PROCESS OF FATIGUE

In the following, we emphasize high-cycle fatigue, first introduced in Chapter 5. The physical process of fatigue occurs, under a high cycle of oscillatory tensile stress with a large amplitude, when a small crack initiates at a certain location with stress concentration. In the event that the total cycles make the material fatigue greater than 10^3 , generally 10^5 , then it is said to be high-cycle fatigue (see Figure 10.3 and also Figure 5.14).

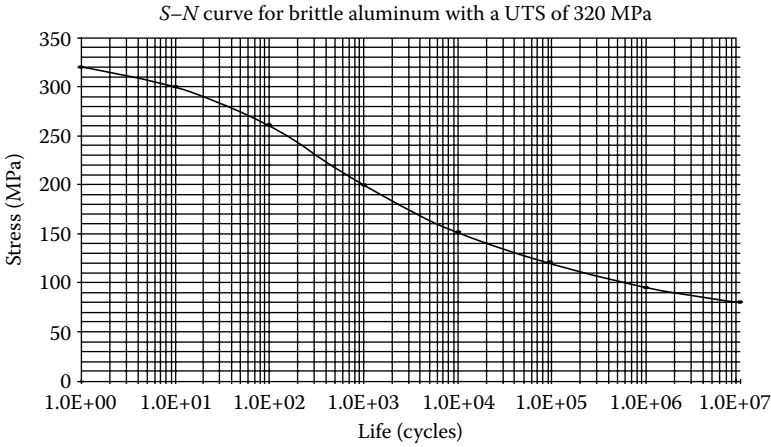


FIGURE 10.3 *S-N* curve.

Fatigue is a special failure mode, perhaps the most important failure mode in mechanical design characterized by the following:

1. Random and inherent unpredictability
2. Difficulty in using laboratory test data to determine practical material behavior
3. Challenging to establish an accurate model of mechanical environments
4. Environmental effects producing fatigue-sensitive stress distributions

Let N denote the cycles of fatigue and S the amplitude of stress. The *S-N* curve can be used to describe the phenomenon of fatigue through

$$N = N(S; \mathbf{A}) \quad (10.54)$$

Here, $\mathbf{A}$ is a vector of parameters related to the *S-N* curve.

10.3.2 STRENGTH MODELS

In the literature, there are several fatigue models available. Fundamentally, the stress-based approach, the strain-based approach, and the fracture mechanics approach are the basic considerations.

10.3.2.1 High-Cycle Fatigue

The key parameter obtained through fracture mechanics is the stress intensity factor range given by

$$\Delta K = Y(a) S \sqrt{\pi a} \quad (10.55)$$

Here, ΔK is the stress intensity factor range; S is the applied stress range; a is the crack depth for a surface flaw or half-width for a penetration flaw; and $Y(a)$ is the geometry correction factor.

The crack growth rate da/dn can be represented as

$$\frac{da}{dn} = C(\Delta K)^m \quad (10.56)$$

Equation 10.56 is referred to as the Paris law (Paris 1964). In this instance, C and m are empirical constants. Therefore, integrating Equation 10.56 will yield

$$N = \frac{1}{S^m C \pi^{m/2}} \int_{a_0}^{a_f} \frac{da}{Y^m(a) a^{m/2}} \quad (10.57)$$

where a_0 is an initial crack length and a_f is the failure crack length.

Note that when the level of crack is less than a certain threshold, ΔK_{th} , the crack will not propagate. Wirsching and Chen (1988) developed a model to estimate the random fatigue with $\Delta K_{th} > 0$.

10.3.2.2 Miner's Rule, More Detailed Discussion

10.3.2.2.1 Constant-Amplitude Tests vs. Random Amplitude

In Chapter 5, we briefly mentioned Miner's rule. Now, let us further study it by specifying the stress of a test specimen. Recall the S - N curve shown in Figure 5.14; for practicality, the stress S is assumed to be constant, as in fatigue tests. In reality, S varies as a random process. Thus, there is a need to simplify the description of S in order to obtain a workable formula. In Figure 10.4, we show the Miner's rule (Figure 10.4a) with conceptual plots describing variations of stress S_i (Figure 10.4b).

10.3.2.2.2 Assumption

A stress process can be described by discrete events, such as number cycles. The spectrum of amplitude for stress cycles can be defined as

$$D = \sum_{i=1}^k \frac{n_i}{N_i} \quad (10.58)$$

Equation 10.58 is a more general description than Equation 5.145. Again, D signifies the damage index, where if

$$D \geq 1.0 \quad (10.59)$$

then the specimen is considered to be damaged or be in a failure mode.

In reality, the event that fatigue failure occurs is random. That is, we may see a failure even if $D < 1$, or when $D > 1$, no failure occurs. In Chapter 5, we used a

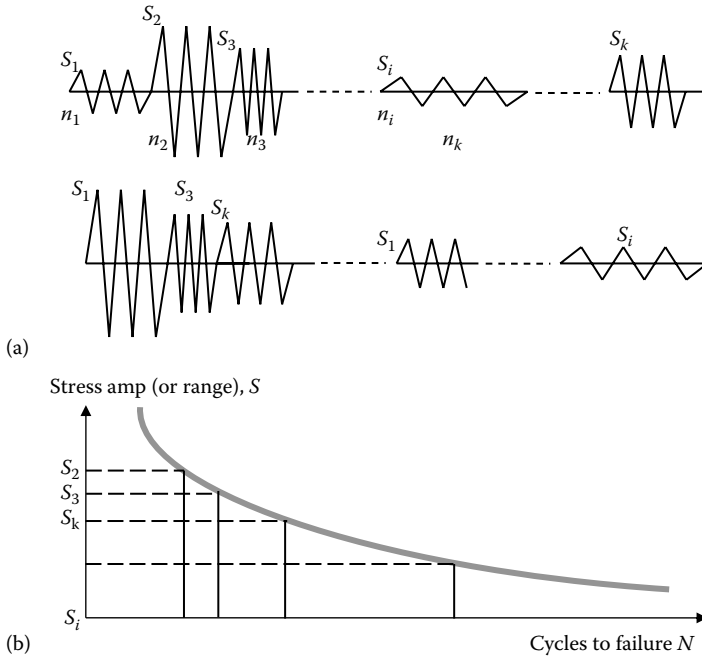


FIGURE 10.4 Miner's rule. (a) Variation of stress. (b) Stress level vs. failure cycles.

Markov chain to study the chance of failure at a different level (in fact, a different level of D). In the following, we examine the fatigue failure from another angle: since whenever Equation 10.59 is reached, the failure deterministically occurs. We now consider the chance to reach $D \geq 1.0$.

10.3.3 FATIGUE DAMAGES

The concept of the damage model is important because of its difficulties in modeling structural damages. One specific reason is the fact that excitation is often a random process. Here, some specific models will be described for better understanding model random damages.

10.3.3.1 Narrowband Random Stress

In the following discussion, we assume that the stress $S(t)$ is a narrowband random process.

10.3.3.1.1 Discrete Stress Spectrum, Fundamental Case

The following method reduces the random process to a discrete random variable. Generally speaking, the amplitude of $S(t)$ is a continuous number. By artificially drawing lines as shown in Figure 10.5a, the stress within the two lines is denoted as S_i and the corresponding window as ΔS_i . In the range ΔS_i , the number of peaks is counted as n_i (see Figure 10.5b).

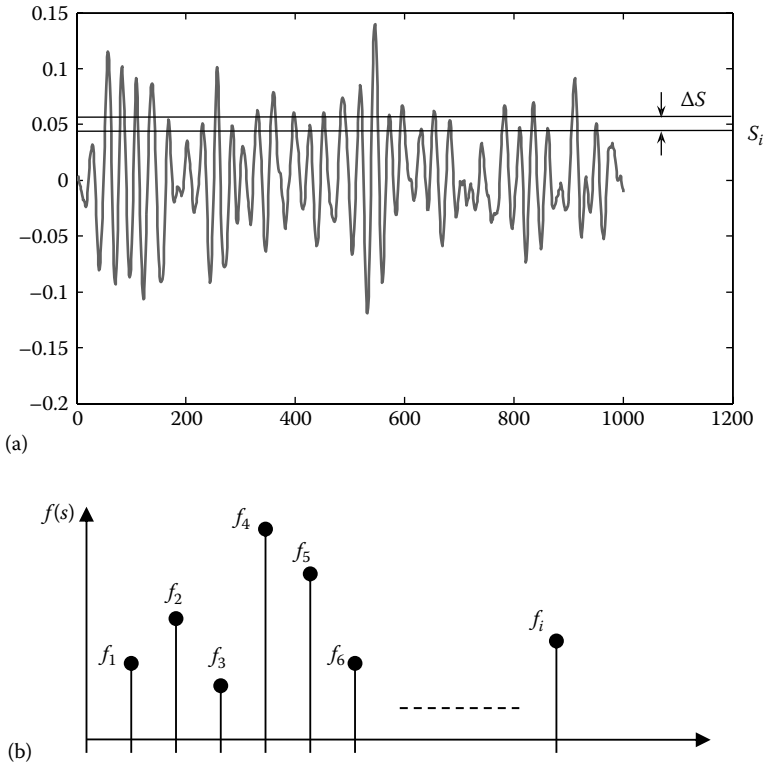


FIGURE 10.5 Probability mass function. (a) Range of ΔS_i . (b) PMF vs. ΔS_i .

The fraction of stress at level S_i is written as

$$f_i = \frac{n_i}{n} \quad (10.60)$$

where

$$n = \sum_{i=1}^k n_i \quad (10.61)$$

As a result, f_i is shown as the probability mass function of the random variable S_i . The total fatigue damage can now be written as

$$D = \sum_{i=1}^k \frac{n_i}{N_i} = n \sum_{i=1}^k \frac{f_i}{N_i} \quad (10.62)$$

10.3.3.1.2 Discrete Stress Spectrum, Linear Model for S–N Curve

By assuming the fatigue strength to be a linear function in the log–log plot, the fatigue analysis can be significantly simplified. In so doing, the fatigue strength A is equated to

$$NS^m = A \quad (10.63)$$

Comparing Equations 10.57 and 10.63 will yield

$$A = \frac{1}{C\pi^{m/2}} \int_{a_0}^{a_f} \frac{da}{Y^m(a)a^{m/2}} \quad (10.64)$$

With the linear relationship of Equation 10.63, damage D can be written as

$$D = \frac{n}{A} \sum_{i=1}^k f_i S_i^m \quad (10.65)$$

In Equation 10.65, S_i indicates the i th amplitude or range. As a result, S is a discrete random variable. Thus, consider that its expected value may be written as

$$E(S^m) = \sum_{i=1}^k f_i S_i^m \quad (10.66)$$

The substitution of Equation 10.66 into Equation 10.65 will result in

$$D = \frac{n}{A} E(S^m) \quad (10.67)$$

In the case that S is constant, Equation 10.65 will further be reduced to

$$D = \frac{n}{A} S^m \quad (10.68)$$

An equivalent constant-amplitude stress is obtained by comparing Equations 10.67 and 10.68 yielding

$$S_e = [E(S^m)]^{1/m} \quad (10.69)$$

Here, S_e is referred to as *Miner's stress* or *equivalent stress*.

10.3.3.1.3 Continuous Stress Spectra

Previously, S was taken to be discrete. However, for a continuous model, the probability mass function will become a PDF, such as a Rayleigh distribution. For example, the probability mass function in the range $(s, s + \Delta S)$ is

$$f_i \approx f_s(s)\Delta s \quad (10.70)$$

In this example, the total fatigue damage is the sum of all incremental damages in each window ΔS , since

$$D \approx n \sum_{i=1}^k \frac{f_s(s)\Delta s}{N(s)} \quad (10.71)$$

Now, consider a continuous stress as

$$\Delta s \rightarrow 0 \quad (10.72)$$

The damage will be equated to

$$D = n \int_0^{\infty} \frac{f_s(s)ds}{N(s)} \quad (10.73)$$

For a linear S – N curve, Equation 10.73 becomes

$$D = \frac{n}{A} \int_0^{\infty} s^m f_s(s) ds \quad (10.74)$$

The solution of Equation 10.74 is given by

$$D = \frac{n}{A} E(S^m) \quad (10.75)$$

Comparing Equations 10.67 and 10.75, the expected value is calculated by using the integral described in Equation 10.74.

10.3.3.1.4 Continuous Spectra, Special Cases

Now, suppose that $S(t)$ is stationary and a narrowband Gaussian.

10.3.3.1.4.1 Rayleigh Distribution of $S(t)$ When the stress amplitude or range is of Rayleigh distribution, then based on the amplitude of the stress S and the strength A , we have

$$E(S^m) = S_e^m = (\sqrt{2}\sigma)^m \Gamma\left(\frac{1}{2}m + 1\right) \quad (10.76)$$

and based on range, we can write

$$E(S^m) = S_e^m = (2\sqrt{2}\sigma)^m \Gamma\left(\frac{1}{2}m + 1\right) \quad (10.77)$$

where $\Gamma(\cdot)$ is the gamma function.

Fatigue damage at time t is obtained by combining Equations 10.76 and 10.77 with Equation 10.75 yielding the following, when S and A are based on amplitude,

$$D = \frac{\nu_{0^+}\tau}{A} (\sqrt{2}\sigma)^m \Gamma\left(\frac{1}{2}m + 1\right) \quad (10.78)$$

and S and A are based on range,

$$D = \frac{\nu_{0^+}\tau}{A} (2\sqrt{2}\sigma)^m \Gamma\left(\frac{1}{2}m + 1\right) \quad (10.79)$$

Here,

$$n = \nu_{0^+}\tau \quad (10.80)$$

where $\nu_{0^+}\tau$ is the zero up-crossing rate of the positive slope.

10.3.3.1.4.2 Weibull Distribution of $S(t)$ (Waloddi Weibull, 1887–1979) When stress amplitude or range is *Weibull*, the CDF of S is written as

$$F_S(s) = 1 - e^{-\left(\frac{s}{\delta}\right)^\xi} \quad (10.81)$$

Here ξ is dimensionless and is referred to as the Weibull modulus or shape parameters, often used to describe variability in measured material strength of brittle materials. When measurements show high variation, ξ will be small, which indicates that flaws are clustered inconsistently and the measured strength is generally weak and variable. Products made from components of low Weibull modulus will exhibit low reliability and their strengths will be broadly distributed.

The parameter δ is the scale parameter of the distribution and to be later eliminated. In this case, the expected value of S^m is

$$E(S^m) = \delta^m \Gamma\left(\frac{m}{\xi} + 1\right) \quad (10.82)$$

Let the design stress for static failure modes be S_0 . The probability of a “once-in-a-lifetime” failure is represented by

$$P(S > S_0) = \frac{1}{N_s} \quad (10.83)$$

Here, N_s is the total number of stress applications in the service life. Substitution of Equation 10.81 into Equation 10.83 will yield

$$S_0 = [\ln(N_s)]^{1/\xi} \delta \quad (10.84)$$

The elimination of variable δ is achieved by substituting Equation 10.84 into Equation 10.82, producing

$$D = \frac{N_s}{A} S_0^m [\ln(N_s)]^{-m/\xi} \Gamma\left(\frac{m}{\xi} + 1\right) \quad (10.85)$$

10.3.3.1.5 Blocks of Continuous Spectra

In the previous example, the Weibull distribution is used to model long-term fatigue stress. An alternative approach is to “create” a nonstationary process stationary by separating the total process into k blocks. For each block, the duration is to become comparatively short. Thus, the variation can be sufficiently small so that the stationary process can be approximately used.

In each block, the Rayleigh peak distributions can be assumed. The RMS of each block is σ_i , the rate of zero up-crossing is v_{i^+} , and the time of application is τ_i . The total damage can then be written in the following in which A is the given amplitude:

$$D = \sum_{i=1}^k \frac{v_{i^+} \tau_i}{A} [\sqrt{2}\sigma]^m \Gamma\left(\frac{m}{2} + 1\right) \quad (10.86)$$

When A is the range (peak-to-valley), the damage becomes

$$D = \sum_{i=1}^k \frac{v_{i^+} \tau_i}{A} [2\sqrt{2}\sigma]^m \Gamma\left(\frac{m}{2} + 1\right) \quad (10.87)$$

10.3.3.1.6 Mean Stress

In the previous example, the stress process was assumed to be zero mean. Nevertheless, cases with a nonzero mean of μ_s do occur. In this instance, the coefficient of fatigue strength A is adjusted by

$$A = A_0 \left(1 - \frac{\mu_s}{S_u}\right)^m, \quad \mu_s \geq 0 \quad (10.88)$$

Here, A_0 is the coefficient of fatigue strength based on zero-mean tests, while S_u is the ultimate strength of the material.

10.3.3.2 Wideband Random Stress

10.3.3.2.1 Equivalent Narrowband Process

In real occurrences, processes are frequently non-narrowbanded, although it could be treated as narrowband. This is seen in Figure 10.6a and b.

In engineering practice, a judgment described by the following equation can be made on the amount of jittering:

$$W(f_0) > 20 \sum W(f_h) \quad (10.89)$$

In the above, $W(f_0)$ is the power special density function of the fundamental frequency f_0 , and $\sum W(f_h)$ is the sum of the power spectral densities (PSDs) of the rest of the frequency components.

Equation 10.89 can be used as a criterion for using the following linear S – N model:

$$NS^m = A \quad (10.90)$$

To treat the process as an equivalent narrowband, let

$$D = \frac{v_{0+} \tau}{A} (\sqrt{2} \sigma)^m \Gamma\left(\frac{1}{2}m + 1\right), \quad S \text{ and } A \text{ based on amplitude} \quad (10.91)$$

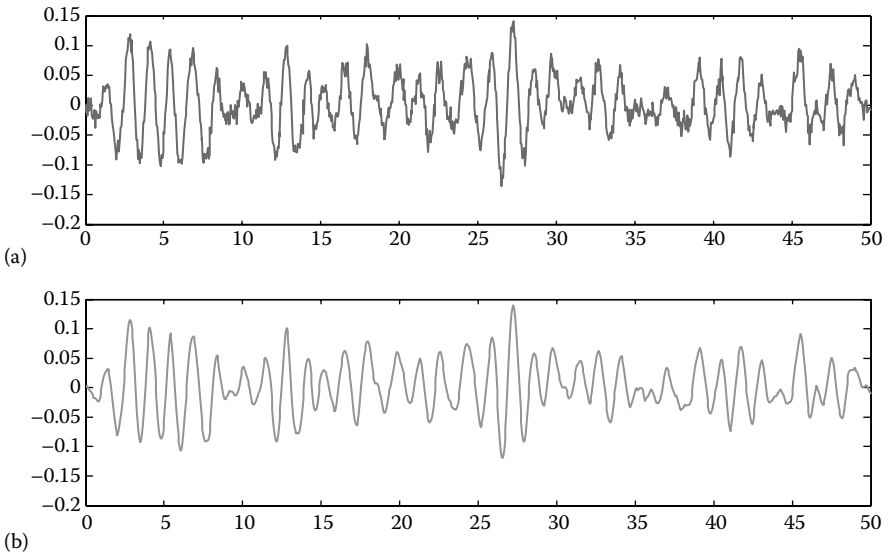


FIGURE 10.6 (a) Non-narrowband process that can be treated as narrowband. (b) Narrowband process.

$$D = \frac{v_0^+ \tau}{A} (2\sqrt{2}\sigma)^m \Gamma\left(\frac{1}{2}m + 1\right) \quad S \text{ and } A \text{ based on range} \quad (10.92)$$

10.3.3.2.2 Rainflow Algorithm

In actual processes, there exist many non-narrowbanded processes that cannot be treated as narrowband processes. When the Miner's rule is used, we see that the key issue is to determine the amplitude of stress S_i and to count the number of cycles n_i associated with S_i . For narrowband responses, there is only one “up and down” in one cycle, so that the amplitude between the up and the down is obvious and easy to count. For a broadband system, the amplitude is more difficult to realize. We thus need certain methods or algorithms to further explore these S_i and n_i . Among the algorithms for identifying stress cycles in a given wideband record, the rainflow method is widely used to estimate the stress range and mean values (see Dowling 1972).

Figure 10.7a and b conceptually plots the narrowband and broadband systems, respectively. These responses are calculated by assuming $m = 10$ and $k = 400$. When the damping ratio is small, say, $\zeta = 0.04$, the response is narrowband. When the damping ratio is large, say, $\zeta = 0.8$, the response becomes broadband.

The rainflow algorithm, also referred to as the “rainflow counting method,” is used in the analysis of fatigue data in order to reduce a spectrum of varying stress into a set of simple stress reversals. It allows the application of Miner's rule to consequently assess the fatigue life of a structure subject to complex loading.

This algorithm is applied by first transforming the stress $S(t)$ to a process illustrated with dotted lines as shown in Figure 10.7b and the solid line in Figure 10.7a with peaks and troughs. In Figures 10.7b and 10.8a, at $t = 0$, the direction of the process trajectory is shown by the thick dotted arrow. Then, the graph is rotated 90° as illustrated in Figure 10.8b. Note that the original graph is shown in Figure 10.8a. In Figure 10.8 at $t = 0$, the direction of the process trajectory is also shown by the thick dotted arrow. The tensile stresses are expressed as the water sources at both peaks. The compressive stresses are expressed as the troughs. The downward water flows are considered according to the following rules (note that in order to

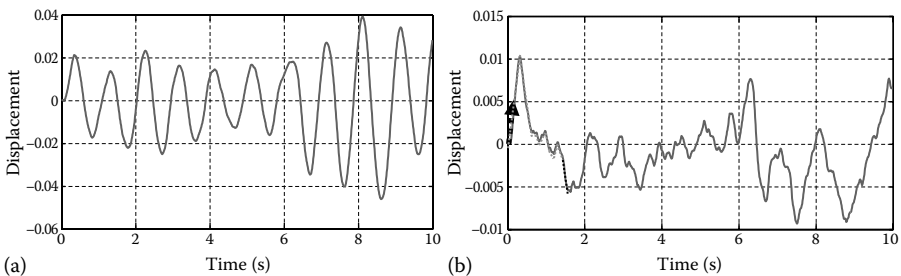


FIGURE 10.7 (a) Narrowband and (b) broadband systems.

show the concept of the rainflow method, the responses in Figures 10.7 and 10.8 are different):

1. A rainflow path starts at a trough, continuing down the roof until it encounters a trough that is more negative than the origin. (For example, the path starts at 1 and ends at 5.)
2. A rainflow path is terminated when it encounters a flow from a previous path. (For example, the path begun at 3 and was terminated as shown in Figure 10.8.)
3. A new path is not started until the path under consideration is stopped.
4. Trough-generated half-cycles are defined for the entire record. In each cycle, the stress range S_i is the vertical excursion of a path. The mean μ_{S_i} is the midpoint. (For example, see S_1 and S_2 in Figure 10.8b.)
5. The process is repeated in reverse with peak-generated rainflow paths. For sufficiently long records, each trough-generated half-cycle is matched to the peak-generated half-cycle to form a whole cycle. One may choose only to analyze a record for peak (through) generated half-cycles, thus assuming that each cycle is a full cycle.

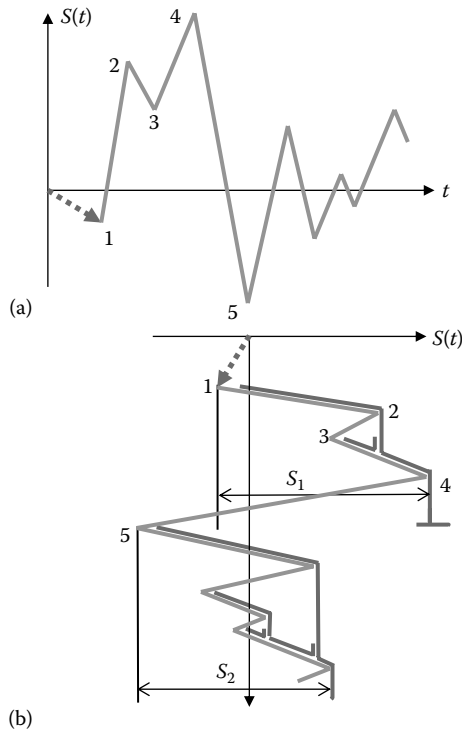


FIGURE 10.8 (a) Process of stress and (b) rainflow.

10.3.3.2.3 Closed-Form Expression for Rainflow Damages

The closed-form formula for fatigue damage under a wideband stress process can be developed through the rainflow algorithm.

10.3.3.2.3.1 Empirical Model Wirsching and Light (1980) developed an empirical model of a general expression for fatigue damage, which can be written as

$$D = \lambda(\epsilon, m) D_{\text{NB}} \quad (10.93)$$

Here, $\lambda(\epsilon, m)$ is a rainflow correct factor, and ϵ is the spectral width parameter. D_{NB} is the damage estimated in Equations 10.94 and 10.95 using a narrowband process.

Based on the amplitude, the coefficient of fatigue strength A is

$$D_{\text{NB}} = \frac{\nu_{0+} \tau}{A} \left[\sqrt{2} \sigma_s \right]^m \Gamma \left(\frac{m}{2} + 1 \right) \quad (10.94)$$

where ν_{0+} is the equivalent frequency, the rate of zero up-crossing.

Based on the range, the coefficient of fatigue strength A is

$$D_{\text{NB}} = \frac{\nu_{0+} \tau}{A} \left[2\sqrt{2} \sigma_s \right]^m \Gamma \left(\frac{m}{2} + 1 \right) \quad (10.95)$$

The rainflow correction factor can be obtained by simulating processes containing a range of spectral shapes such as

$$\lambda(\epsilon, m) = a(m) + [1 - a(m)](1 - \epsilon)^{b(m)} \quad (10.96)$$

where empirically

$$a(m) = 0.926 - 0.033m \quad (10.97)$$

and

$$b(m) = 1.587m - 2.323 \quad (10.98)$$

10.3.3.2.3.2 Refined Model of Effective Stress Range H A more refined model is achieved by counting the effective stress range H , considered as the Rayleigh distribution, (see Ortiz and Chen 1987). The CDF of H is then written as

$$F_H(h) = 1 - \exp \left[-\frac{1}{2} \left(\frac{h}{2\beta_k \sigma_s} \right)^2 \right] \quad (10.99)$$

Here, β_k is the generalized spectral bandwidth given by

$$\beta_k = \sqrt{\frac{M_2 M_k}{M_0 M_{k+2}}} \quad (10.100)$$

and the parameter k is

$$k = \frac{2.0}{m} \quad (10.101)$$

In Equation 10.100, M_j is the j th moment of the one-sided spectral density function:

$$M_j = \int_0^\infty f^j W_s(f) df \quad (10.102)$$

In this approach, the general expression for the wideband damage is

$$D = \lambda_k D_{NB} \quad (10.103)$$

where

$$\lambda_k = \frac{\beta_k^m}{\alpha} \quad (10.104)$$

Here, α is the irregular factor, as defined in Equation 5.40. It is repeated as follows:

$$\alpha = \frac{v_{0+}}{v_p} = \frac{\omega_{0+}}{\omega_p} \quad (10.105)$$

10.3.3.2.3.3 Alternative Approach An alternative approach (Lutes and Larson 1990) can be written as

$$D = \lambda_L D_{NB} \quad (10.106)$$

where

$$\lambda_L = \frac{M_{2/m}^{m/2}}{v_{0+}} \quad (10.107)$$

A typical value for the fracture mechanics model or for welded joints is $m = 3$.

10.3.4 DAMAGES DUE TO TYPE D LOW CYCLE

In the above discussion, two types of failure are considered: the first passage and the high-cycle fatigue. The generality of these two cases is that once the critical point

is reached, the failures occur. For instance, either the stress $S(t)$ exceeds the allowed strength R or the damage index D reaches 1.0. Explicitly, before these critical points, no artificially defined damage occurred.

From the viewpoint of a random process, the process is “memoryless.” The process is often used to describe the stress time history but not the history of damage growth.

In real-world applications, damages that experience growth over time must be dealt with. Examples of such damages include crack propagations, incidents of aging and corruptions, gradual loss of pre-stress in reinforced concrete elements, etc. In these situations, the earlier damage will not be self-cured and will affect the remaining damage process. Thus, it will no longer be “memoryless.”

Very often, such a damage process is random and is caused by the nonlinear stiffness of systems. While in Chapter 5, we discussed the low-cycle development by using a Markov model on the type D fatigue, in the following, we will further consider the corresponding failure mode and its related parameters.

10.3.4.1 Fatigue Ductility Coefficient

As mentioned in Chapter 5, when the stress becomes sufficiently high such that the material yields, the deformation becomes plastic. In this case, low-cycle fatigue may occur and the failure mode is defined differently. In general, the cycles to fatigue are less, or considerably less, than 1000. It can be a loss of the entire function of systems, the collapse of the structure, or the introduction of severe structural damage. When low-cycle fatigue occurs, the account in terms of stress is less revealing and the strain in the material will offer more enhanced information. In Chapter 11, this point of concept will be discussed in a more detailed fashion. Low-cycle fatigue is regularly characterized by the aforementioned Coffin–Manson relation (see Sornette et al. 1992) defined by

$$\frac{\Delta \epsilon_p}{2} = \epsilon'_f (2N)^c \quad (10.108)$$

In this instance, $\Delta \epsilon_p/2$ refers to the amplitude of the plastic strain and ϵ'_f denotes an empirical constant called the *fatigue ductility coefficient* or the failure strain for a single reversal. Furthermore, $2N$ is the number of reversals to failure of N cycles, while c is an empirical constant named the *fatigue ductility exponent*, commonly ranging from -0.5 to -0.7 for metals in time-independent fatigue.

10.3.4.2 Variation of Stiffness

In the course of type D low-cycle fatigue, a structure will have decreased stiffness, which can be seen when the inelastic range is reached and then repeated. Rzhnevsky and Lee (1998) studied the decreasing stiffness and found that the stiffness decrease can be seen as a function of the accumulation of inelastic deformation. This type of accumulative damage is more complex than pure low-cycle fatigue. This is because during certain cycle strokes, the deformation will be sufficiently too large to yield the structure, whereas in other cycles, the deformations can respectively be small. Empirically, the stiffness and the accumulation can be expressed as

$$k_n = k_0 e^{-a_0 sh^{-1}(\gamma_n)} \quad (10.109)$$

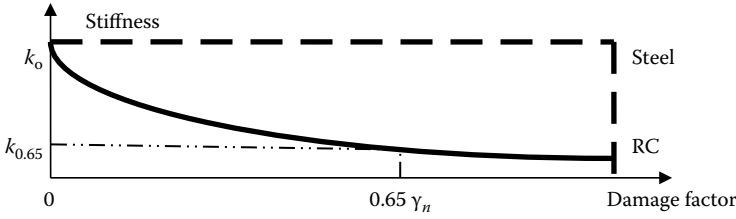


FIGURE 10.9 Stiffness versus accumulation of inelastic deformation.

Here, k_0 and k_n are, respectively, the original stiffness and the stiffness after n semicycle inelastic deformations; a_0 is the peak value of the first inelastic deformation. The subscripts 0, 1, and n denote the cycles without inelastic, the first, and the n th cycle of inelastic deformations, respectively. Additionally, the term γ_n is defined as

$$\gamma_n = \left| \sum_{i=1}^n a_i \right| \quad (10.110)$$

The term γ_n is called the *damage factor*, which is the summation of the absolute value of all the peak values of the inelastic deformations.

For an initial stiffness of k_0 , the values of the inelastic deformation a_i can be specified in an experimental study of certain types of reinforced concrete (RC) components and structures, comparing with steel, whose stiffness is virtually constant. Figure 10.9 conceptually shows the relation between the decreased stiffness and the damage factor.

For a comparison, both the constant and the decrease stiffness are plotted in Figure 10.9. Conceptually, both are allowed to have the same value initially. It is then seen that as the inelastic formations accumulate, the constant stiffness will maintain its value until its final failure stage. However, the decreased stiffness will begin to diminish from the first semicycle, although the decreased stiffness will eventually have a total failure. It is also conceptually plotted at the point when the constant stiffness fails.

Example 10.3

At 65% of the accumulation of the total inelastic deformation, an individual stiffness, denoted by $k_{0.65}$, is considered to be the minimum allowed value. This is written as

$$k_{0.65} = k_0 e^{-a_0 s h^{-1}(\gamma_{0.65})} \quad (10.111)$$

where the term $\gamma_{0.65}$ is defined as

$$\gamma_{0.65} = \left| \sum_{i=1}^{0.65n} a_i \right| \quad (10.112)$$

10.4 CONSIDERATIONS ON RELIABILITY DESIGN

In Section 10.2, the random relationship between the stress process $S(t)$ and the random strength R was studied. It was indicated that the probability of $S > R$ is seldom zero. Chapter 2 exemplified the fundamental approach of the probability- or reliability-based design to ensure that the probability of $S > R$ is smaller than the allowed level. Practically, the stress S can be the result of more than one load, all of which can be random in nature. In this section, the formula of calculating the failure probability and the surface of load combinations will be further examined.

In the following, we use a case study to explain the necessary procedure of reliability design on systems whose loads are combinations of random processes. The case study is a reliability design of highway bridges, which are subjected to multi-hazards (MHs). The basic idea is to use the method of variable separation to approximate random processes by random variables.

10.4.1 FURTHER DISCUSSION OF PROBABILITY-BASED DESIGN

10.4.1.1 Random Variable vs. Random Process

Recalling the concept of the limit state introduced in Chapter 1 (see Equation 1.152), we can write (also see Ellingwood et al. 1980, Nowak 1999)

$$F = R - L \quad (10.113)$$

where we use L to denote the realistic load. That is, when $F = 0$ is reached, failure occurs. Similar to Chapter 1, Equation 1.154, the failure probability is given as

$$p_f = P(F \leq 0) = P(R - L \leq 0), \text{ for most cases.} \quad (10.114)$$

Here, instead of demand Q , symbol L is used to denote multiple loads.

In Equation 10.114, both R and L are random variables, which are essentially time invariant. However, consider the term L in reality, which can be MH loads, and these loads are likely time varying in nature. That is, we often have

$$L = L_1 + L_2 + L_3 + \dots \quad (10.115)$$

And each hazard load L_i is likely the time variable, that is,

$$L_i = Q_i(t) \quad (10.116)$$

Therefore, it can be very difficult to have the PDF of the combined load because they are actually a random process. In most cases, the loads are not addible.

In developing probability-based designs of bridges subjected to MH extreme loads, researchers have pursued the following. Wen et al. (1994) first provided a comprehensive view of multiple loads and divided the total failure probability into several partial terms. Although there is no detailed model of how to formularize these partial failure probabilities, Wen et al.'s work pointed to a direction to establish closed-form analytical results for limit state equations, instead of using empirical or semi-empirical

approaches. Nowak and Collins (2000) discussed alternative ways for several detailed treatments in load combinations, instead of partial failure probabilities. Ghosn et al. (2003) provided the first systematic approach on MH loads for establishing the design limit state equations. They first considered three basic approaches: (1) Turkstra's rule, (2) the Ferry–Borges model, and (3) Wen's method. In addition, they also used Monte Carlo simulations (which will be discussed in Chapter 11). Among these approaches, Ghosn et al. focused more on the Ferry–Borges method, a simplified model for MH load combinations. Hida (2007) believed that Ghosn's suggestion could be too conservative and discussed several engineering examples in less common load combination.

In the above-mentioned major studies, the loads including common live (truck) load and extreme loads are assumed independent. Moreover, no matter they occur with or without combinations, the distributions of their intensity are assumed to remain unchanged. These methods unveiled a key characteristic of MH loads—the challenge of establishing multihazard load and resistance factor design (MH-LRFD) is the time-variable load combination. With simplified models, these approaches provided possible procedures to calculate the required load and resistance factors for design purposes.

Because of the lack of sufficient statistical data, simplifying assumptions may either underestimate or overestimate certain factors. Cases in which the assumptions are accurate can be shown, while in other cases, the distributions may vary and the independency may not exist. In this study, we have pursued both theoretical and simplified approaches so that results may be compared and evaluated.

10.4.1.2 Necessity of Distinguishing Time-Invariant and Time-Variable Loads

To illustrate that there is no problem to combine one time-invariant with time-independent loads but that it is difficult to consider more than one time-variable loads, Figure 10.10 plots the conceptual PDFs of time-invariant and time-variable loads and their combinations. Figure 10.10a shows the combination of L_1 , a time variable load,

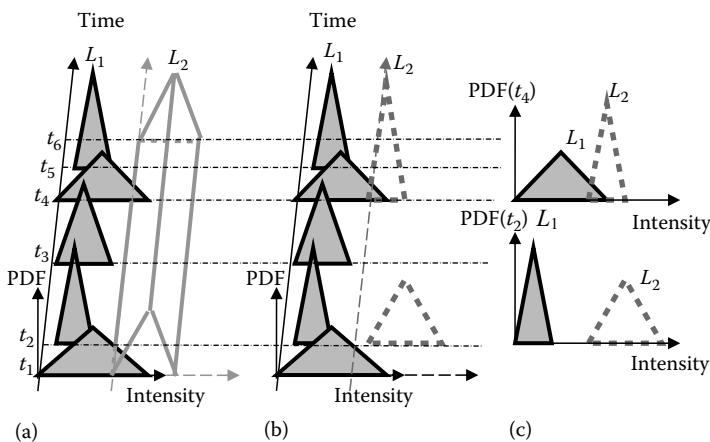


FIGURE 10.10 PDF of time-invariant and time-variable loads and their combinations. (a) L_2 : constant load; (b) L_2 : time varying load; (c) time varying load combination.

and L_2 , a time-invariant load. Whenever L_1 occurs, L_2 is there “waiting” for the combination. For example, if we have truck and dead loads only, then we can treat the combined load as a regular random variable without considering the time variations.

In Figure 10.10b, when combining two (or more) time variables, we may have the chance L_1 occurs without L_2 . There is also the chance when larger-valued L_1 occurs with the occurrence of smaller-valued L_2 , as illustrated in Figure 10.10c. We may also have the chance when smaller-valued L_1 occurs with the occurrence of larger-valued L_2 , for example, at t_4 , which is specifically shown in Figure 10.10c. In these cases, we may have different sample spaces in calculating the corresponding distributions, as well as L_1 and L_2 distributions.

10.4.1.3 Time-Variable Load at a Given Time Spot

Suppose a bridge is subjected to two time-variable loads L_1 and L_2 only. If a larger-valued load, L_2 , occurs before the smaller load, L_1 , then in order to count the maximum load effect, L_1 must be rejected. In other words, to count L_1 , we need to consider a precondition: in terms of time sequence, at the amplitude of L_1 , the load effect due to L_2 as well as that due to the combination L_1 and L_2 cannot occur before L_1 . These situations are discussed in more detail in the following.

Truck, vessel and vehicle collisions, earthquakes, as well as wind gusts are time variables (random process). In many cases, these loads cannot be directly combined; when one load, L_1 , occurs, others may not occur simultaneously. In general, we have

$$L_1 + L_2 + L_3 = L_1 \quad (10.117)$$

If, at a given moment, all these loads occur simultaneously, then

$$L_1 + L_2 + L_3 \neq L_1 \quad (10.118)$$

The distributions of the combined load $L_1 + L_2 + L_3$ described in Equations 10.117 and 10.118 can be different. To obtain the correct corresponding random distributions, both single load and combined load distributions, several methods are available. We will use a method of event separation. That is, the failure probability is separated under the condition of a certain “pure” load or “pure” load combination only.

In what follows, the combination of a single time-variable load and the time-invariable dead load is taken as a *single time-variable load*. When L is only a single time-variable load and it is given a special value x , then there is no concern of another load. However, under MH loads, in Equation 10.114, the event $(L \geq R)$ must also be analyzed by examining the load L .

10.4.1.4 Combination of Time-Variable Loads in a Given Time Period

Consider now at a given time duration in which there exists two different kinds of time-variable loads simultaneously,

$$\ell(t) = \ell_1(t) + \ell_2(t) \quad (10.119)$$

where $\ell(t)$, $\ell_1(t)$, and $\ell_2(t)$ are the sum and the special time varying loads, respectively. With a given time t , they are deterministic.

In most cases, the maximum value of the combined load $\ell(t)$ does not equal to the sums of the maximum or amplitude values of $\ell_1(t)$ and $\ell_2(t)$. That is,

$$\|\ell(t)\| \neq \|\ell_1(t)\| + \|\ell_2(t)\| \quad (10.120)$$

In some previous studies, the amplitudes of $\ell_1(t)$ and $\ell_2(t)$ are treated as constant when the time duration Δt is taken to be sufficiently small. However, when Δt is indeed sufficiently small for one load at a certain moment, it may not be sufficiently small for the second load. Thus, such a treatment, such as the Ferry–Borges model, may yield inaccurate results.

10.4.1.5 Additivity of Distribution Functions

Consider again the failure probability of $P(L \geq R)$. Similar to Equation 1.139b, if both L and R are normally distributed, then the distribution of $L-R$ can be calculated by considering the PDF of $R-L$, denoted by $f_{R-L}(x)$. Namely, all the cases of $R-L \leq 0$ mean that there is a chance of failure. That is,

$$p_f = \int_{-\infty}^0 f_{R-L}(x) dx \quad (10.121)$$

For a standardized variable, Equation 10.121 can be rewritten as

$$p_f = \Phi(Z|_{Z=-\beta}) = \Phi(-\beta) \quad (10.122)$$

where $\Phi(-\beta)$ is the CDF of the normally distributed standardized variable defined in Chapter 1 (Nowak and Collins 2000).

Turkstra and Madsen (1980) suggested a simplified method for load combinations. In many cases, this model is an oversimplified treatment on random process because it does not handle the load combination in the random process level at all. Instead, it directly assumes that whenever two loads are combined, the “30% rule” can always be used. On the other hand, this simplifying assumption can sometimes be rather conservative.

Another method is the Ferry Borges–Castanheta model. Detailed description of the Ferry–Borges’s model can be found in Ghosn (2003). In order to handle time-variable loads, the Ferry–Borges’s model breaks down the entire bridge life into significantly short time periods, in which these loads can be assumed constant so that the variables can be added. Based on the discussion of extreme distribution in Chapter 2, the cumulative probability function, $F_{X_{1,2,\max},T}$, of the maximum value of load combinations, $X_{1,2,\max,T}$, in time T is obtained by

$$F_{X_{1,2,\max},T}(x) = [F_{X_{1,2}}(x)]^{T/t} \quad (10.123)$$

Here, $F_{X_{1,2}}(x)$ is the CDF in the short period. In Equation 10.122, the value t is the average duration of a single event, say, several seconds; therefore, the integer ratio T/t can be a very large value. If a certain error exists, generated by using Equation 10.123,

which is unavoidable in many cases, then the resulting CDF $F_{X_{1,2}, \max.T}$ can contain unacceptable errors.

To use the Ferry–Borges's model, one needs to make several simplifying assumptions, which introduce errors. The largest error, generally speaking, comes from Equation 10.123, which is based on the assumption that the distributions in each individual period t are identical. In addition, the load combinations in a different period t must be independent; that is, the random process of the load combination must be an independent stationary process, which is not true. Because the ratio T/t can be very large, large errors may result.

10.4.2 FAILURE PROBABILITY UNDER MH LOAD

10.4.2.1 Failure Probability Computation

Below, two distinct approaches to establish the design limit state equations are described.

The first approach is as follows: (1) formulation of closed-form partial failure probabilities consisting of two portions of a term, namely, the conditional probabilities of failure and the probability of corresponding conditions; (2) formulation of the probability of conditions; (3) computation of the conditional probabilities; (4) establishment of a criterion to reject unnecessary loads; (5) normalization of load effects; and (6) determination of limit state equations. These five tasks are logically related in sequence.

The second approach is to compute directly the partial failure probabilities. Since each direct computation will involve single and load combinations, simplifications, such as Turkstra's and/or Ferry–Borges's models, will be necessary. With design examples, the limit state equations will be established by the least squares method. The first approach is described in the following.

10.4.2.2 Time-Invariant and -Variable Loads

Failure probabilities of a structure are the sums of the chances whenever load effects exceed resistance. Under MH loads, the total failure probability of a bridge consists of several partial failure probabilities due to dependent load effects and different PDFs. To evaluate the partial failure probability of combined loads exceeding bridge resistance, one needs to calculate the effects of load combinations. Besides dead load, most bridge loads are time variables so that the probabilities for these loads with specific magnitudes occurring simultaneously are likely not 100% but will depend on the nature of these loads, which is a major condition of load occurrences. Furthermore, given that the amplitude of a load effect exceeds the resistance should have an additional restricting condition, which requires that before such a load occurs, no effects caused by other hazards reach the same level. This method is defined as the separation of partial bridge failure probabilities.

10.4.2.3 Principles of Determining Load and Load Combination

As mentioned above, the dead load L_D can be reasonably assumed time invariant. When a single time-varying load occurs, it can be directly added to the dead load. The method to deal with such load combination has been well established. In the following, we will consider the load combination for time-varying loads only.

10.4.2.3.1 Maximum Possible Loads

When considering loads, no matter how frequent or rare they occur, we only consider their maximum possible values, which are random variables and roughly have bell-type distributions. In a Cartesian coordinate system, the x -axis of the random distribution curve is the intensity of the maximum possible load effects, instead of individual load effects.

To extract maximum valued loads from a load pool, a method or a criterion of picking up the loads is needed. In general, the maximum value of a load is picked up from the pool of total possible loads occurring in the bridge lifespan. For certain loads with the same type that can occur simultaneously, an additional consideration is the maximum possible combination of the simultaneous load, for example, two or more trucks crossing a given bridge at the same time. These trucks must load the bridge at different locations of the bridge, i.e., the combined load effect, which is often different from the direct sum of the load effects due to the peak value of these truck loads.

Consider the event that the combined load effect is greater than resistance:

$$L_1 + L_2 + L_3 + \dots \geq R \quad (10.124)$$

There are three cases for which Equation 10.124 holds. The first is that individual load effects are greater than or equal to R . In this case, when this single load occurs, we need to consider only the peak level of this load. The second is that none of these individual load effects are greater than or equal to R , but their combined effect is. In this case, only the peak level of a load may not be sufficient. The third case is the combination of the first and second cases, which will be considered in Section 10.4.2.5.

10.4.2.4 Total and Partial Failure Probabilities

We now discuss the method to divide the total failure probability into several terms. For each term, we will have a comparatively simpler relationship between a single type of load (or load combination) and resistance. Therefore, it is easier to not underestimate or overestimate the corresponding chance of bridge reliability. Details of how to reassemble these partial terms together to formulate the entire limit state equation for all possible loads will not be discussed here.

10.4.2.5 Independent Events

Let us continue the discussion on Equation 10.124. Besides the two cases mentioned in Section 10.4.2.3.1, the third possibility is the contribution of both of these cases. Suppose that we now have three different kinds of loads, L_1 , L_2 , and L_3 . From Equation 10.124, we can write

$$P(L_1 + L_2 + L_3 \geq R) = p_f \quad (10.125)$$

To add up the load effects of Equation 10.125 without miscounting, Wen et al. (1994) suggested the use of total probability. That is, the entire case is dissected into several mutually exclusive subcases. Each subcase only contains a single kind of load

(or load combinations). This way, we can deal with the time-variable loads and load combinations more systematically and reduce the chance of miscounting.

Thus, Equation 10.124 can be further rewritten as follows:

$$\begin{aligned}
 &P(L_1 \geq R|_{L_1 \text{ only}})P(L_1 \text{ only}) + \\
 &P(L_2 \geq R|_{L_2 \text{ only}})P(L_2 \text{ only}) + \\
 &P(L_3 \geq R|_{L_3 \text{ only}})P(L_3 \text{ only}) + \\
 &P(L_1 + L_2 \geq R|_{L_1 \cap L_2 \text{ only}})P(L_1 \cap L_2 \text{ only}) + \\
 &P(L_1 + L_3 \geq R|_{L_1 \cap L_3 \text{ only}})P(L_1 \cap L_3 \text{ only}) + \\
 &P(L_2 + L_3 \geq R|_{L_2 \cap L_3 \text{ only}})P(L_2 \cap L_3 \text{ only}) + \\
 &P(L_1 + L_2 + L_3 \geq R|_{L_1 \cap L_2 \cap L_3 \text{ only}})P(L_1 \cap L_2 \cap L_3 \text{ only}) = p_f
 \end{aligned} \tag{10.126}$$

Here, p_f is the failure probability; $P(\cdot)$ means the probability of event $(\cdot)$ happening; the normal letter of $L_{(\cdot)}$ means the load effect due to load $L_{(\cdot)}$. The symbol “ $|_{(\cdot)}$ ” stands for condition $(\cdot)$. The symbol “ $\cap$ ” stands for occurring simultaneously. $L_1 \cap L_2$ stands for the condition that loads L_1 and L_2 occur simultaneously, but that there are only these two loads; in other words, no other loads (except the dead load L_D) show up during this time interval.

This formula of total probability is theoretically correct but is difficult to realize practically. The main reason is that, again, these loads L_1 , L_2 , and L_3 are time variables and that at a different level of load effect, $P(L_i \text{ only})$ will be different. In the following, we introduce the concept of partial failure probability to deal with these difficulties.

For the sake of simplicity, first consider two loads only. Denote

$$P(L_1 \geq R|_{L_1 \text{ only}}) \equiv P(L_1) \tag{10.127}$$

The probability of only having load L_1 is given as

$$P(L_1 \text{ only}) \equiv p_{L_1} \tag{10.128}$$

Therefore, the failure probability due to load L_1 only can be written as

$$p_f^{L_1} = P(L_1 \geq R|_{L_1 \text{ only}})P(L_1 \text{ only}) = P(L_1)p_{L_1} \tag{10.129}$$

Similarly, we can rewrite $p_f^{L_1 L_2}$ and $p_f^{L_2}$ in the same format as described in Equation 10.129. Here, $p_f^{L_1}$, $p_f^{L_1 L_2}$, and $p_f^{L_2}$ are used to respectively denote the failure probabilities caused by L_1 only, L_1 and L_2 simultaneously, and L_2 only.

Thus, we can write

$$p_f^{L_1} + p_f^{L_1 L_2} + p_f^{L_2} =$$

$$P(L_1)p_{L_1} + P(L_1 L_2)p_{L_1 L_2} + P(L_2)p_{L_2} = p_f \quad (10.130)$$

Equation 10.130 is defined as *the separation of partial bridge failure probabilities*. The intensity distributions of a load can be different in different regions. If the occurrence of certain loads, say, truck load L_1 , is very frequent, then we may assume that the distribution of L_1 remains the same, no matter L_1 occurs with or without L_2 . If the occurrence of other loads, say, earthquake load L_2 , is very rare, then we may assume that the distribution of L_2 when it is together with L_1 will be different from the distribution of L_2 when it occurs solely without L_1 .

The essence of Equation 10.130 is that the failure probability p_f can be written as a sum of several *partial failure probabilities*, $p_f^{L_1}$, $p_f^{L_1 L_2}$, and $p_f^{L_2}$. It will be further seen that, to realize Equation 10.130, we will have rather complicated situations, because the PDFs of the load effects in events L_1 , $L_1 \cap L_2$, and L_2 are not identical. In different events, the random distribution functions can be rather different. In addition, although the time variable loads L_1 and L_2 are independent, the corresponding load effects on a given bridge will no longer be independent. This is the major difference between the proposed method and that described in Equation 10.126. That is, the terms $P(L_1)$ and p_{L_1} will vary due to different levels.

Furthermore, in Equation 10.126, each partial failure probability is a product of two terms. For example, $p_f^{L_1} = P(L_1)p_{L_1}$. Here, the first term, $P(L_1)$, is referred to as the *conditional probability*, and the second term, p_{L_1} , is referred to as the *probability of condition*.

In the following, the general principle of formulation on these partial probabilities will be discussed. The focus is on the conditional probability. Detailed computation of the second term will be discussed in Section 10.4.4 that follows the description of conditional probability.

10.4.2.6 Mutually Exclusive Failures, the Uniqueness Probabilities

For the sake of simplicity, in the following discussion of load combination, we purposely ignore the dead load because it is time invariant.

In Figure 10.11, we use a Venn diagram to show time-variable load combinations and use earthquake and truck load effect as an example to describe the concept. In the figure, p_f and $p_{\bar{f}}$ are the probability of failure and nonfailure of a bridge ($p_f + p_{\bar{f}} = 1$), respectively, subjected to earthquake and truck loads. In Equation 10.130, let p_f^E , p_f^{ET} , and p_f^T denote the failure probabilities due to the earthquake effect only, the combined effect of both earthquake and truck, and the truck effect only, respectively. The event of the earthquake effect, the combined effect, and the truck effect greater than or equal to the resistance are mutually exclusive events.

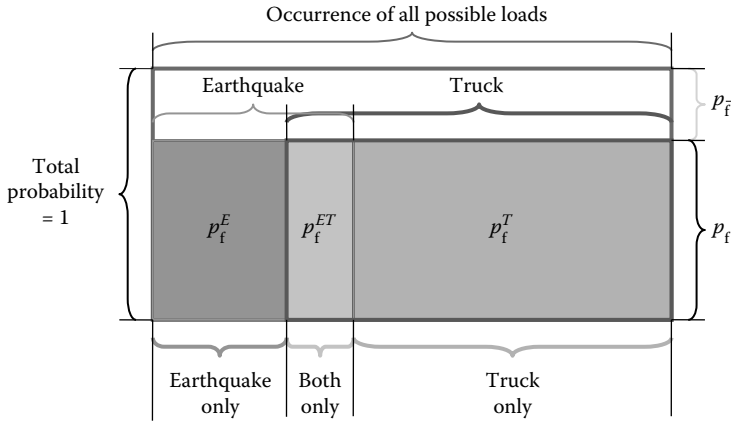


FIGURE 10.11 Venn diagram of failure probabilities.

That is,

$$p_f^E + p_f^{ET} + p_f^T = p_f \quad (10.131)$$

In addition, when these individual failure probabilities $p_f^{(.)}$ are calculated for each case of the maximum load effect of (.) exceeding the resistance, we need to also consider a restricting condition: no other effect reaches the same level. The detail of this will be discussed in the following.

10.4.2.6.1 Condition of Maximum Load Effect

In Equation 10.130, those first probabilities, for example, $P(L_i)$, exist only under rigorous conditions. To count the maximum load effect, we need to evaluate the condition that guarantees that what is counted is a true maximum effect. As seen in Figure 10.12, the x -axis of the probability density curve of the load effect is the maximum value of the load effect. Consider the case that a bridge is subjected to the time-invariant dead load and a single time-variable load only. When the sum of the time-variable load and the dead load reaches a certain maximum level, say, x , there is no chance for other loads to reach this specific level.

However, if other loads exist when the first time-variable load reaches a certain level, then there can be a chance that the second time-variable load had already reached and/or exceeded the same level. In this case, the effect due to the first load cannot be recognized as the maximum value. In other words, for a time-variable load effect to be recognized as the maximum value at intensity x , we need a restricting condition that no other loads can reach the same level.

Figure 10.12 conceptually shows this condition. In the figure, the solid and dotted lines are the PDFs of the first load effect and the resistance, respectively. The dash-dotted line represents the second time-variable load effect, for convenience, which is placed below the first load effect curve in the figure. The shadowed area represents

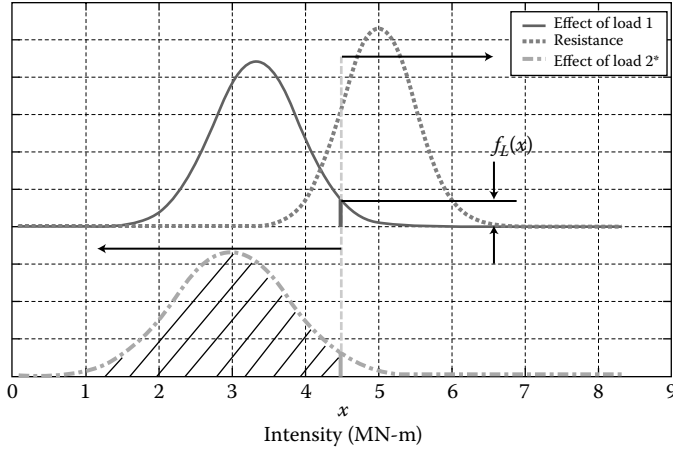


FIGURE 10.12 Condition of load 1 reaches a certain maximum level x .

all the chances that the second load is smaller than this given level x . In the figure, for the sake of simplicity, we assume that loads 1 and 2 have no combinations.

In general, the sum of all these chances can be expressed as the integral of the PDF of the second load effect, $f_s(z, x)$. Here, the subscript “S” denotes the second load, and in the following, we use the font of “Ruling Script LT Std” to denote the conditional PDF, whose condition is the main PDF taking the value of x . That is,

$$P_s(x) = \int_{-\infty}^x f_s(z, x) dz \leq 1 \quad (10.132)$$

Based on the above, when calculating the failure probability, we cannot use the PDF of the first load only. In other words, the PDF of the first load, $f_L(x)$, shown in Figure 10.12, must be modified as $f_L(x)P_s(x)$. In general, $P_s(x) < 1$, so that the resulted failure probability due to the first load only should be smaller than the case with the first load only.

The additional possibility that the value of combined loads 1 and 2 must also be smaller than level x is given by

$$P_c(x) = \int_{-\infty}^x f_c(w, x) dw \leq 1 \quad (10.133)$$

Here, the subscript C denotes the combined load effect of both the first and second loads, and the PDF of the combined load effect is denoted by f_c . In this circumstance, the PDF of the first load must be further modified as $f_L(x)P_s(x)P_c(x)$.

Therefore, the conditional failure probability due to load 1 should be written as

$$\begin{aligned}
 P(L_1) &= \left[\int_{-\infty}^{\infty} f_R(x) \int_x^{\infty} f_{L_1}(y) P_{L_2}(x) P_{C_2}(x) dy dx \right] \\
 &= \int_{-\infty}^{\infty} f_R(x) \int_x^{\infty} f_{L_1}(y) \int_{-\infty}^y f_{L_2}(z) \int_{-\infty}^y f_C(w) dw dz dy dx
 \end{aligned} \tag{10.134}$$

where $f_{L_1}(y)$, $f_{L_2}(z, x)$, and $f_C(w, x)$ are the PDFs of effect due to load 1, due to load 2, and due to the combination of loads 1 and 2, respectively; $P_{L_2}(x)$ and $P_{C_2}(x)$ are the condition probability $P_s(x)$ due to load 2 and the condition probability $P_c(x)$ due to the combination of load 1 and load 2, respectively. In the circumstance of MHs, a specific load, say, load 1, is divided into two portions. The first portion is based on the case of load 1 only, and the second case is that both loads occur simultaneously. We use $f_{L_1}(y)$ and $f_{L_2}(z, x)$ to respectively denote load 1 in the first case and load 2 in the second case.

Similarly, the failure probability due to combined load 1 and load 2 should be written as

$$\begin{aligned}
 p(L_1 L_2) &= \left[\int_{-\infty}^{\infty} f_R(x) \int_x^{\infty} f_c(y) P_{L_1}(x) P_{L_2}(x) dy dx \right] \\
 &= \int_{-\infty}^{\infty} f_R(x) \int_x^{\infty} f_c(y) \int_{-\infty}^y f_{L_1}(z, x) \int_{-\infty}^y f_{L_2}(w, x) dw dz dy dx
 \end{aligned} \tag{10.135}$$

where f_c is the PDF of the combined load (L_1 and L_2 occur simultaneously); f_{L_1} and f_{L_2} are the PDFs of the effects due to loads 1 and 2 in the first case, respectively; $P_{L_1}(x)$ and $P_{L_2}(x)$ are the condition probability $P_s(x)$ due to loads 1 and 2, respectively. Note that, generally speaking, $f_{L_1} \neq f_{L_1}$ and $f_{L_2} \neq f_{L_2}$.

Similarly, the failure probability due to load 2 should be written as

$$\begin{aligned}
 p(L_2) &= \left[\int_{-\infty}^{\infty} f_R(x) \int_x^{\infty} f_{L_2}(y) P_{L_1}(x) P_{C_1}(x) dy dx \right] \\
 &= \int_{-\infty}^{\infty} f_R(x) \int_x^{\infty} f_{L_2}(y) \int_{-\infty}^y f_{L_1}(z, x) \int_{-\infty}^y f_C(w, x) dw dz dy dx
 \end{aligned} \tag{10.136}$$

10.4.3 GENERAL FORMULATIONS

10.4.3.1 Total Failure Probability

With the help of Equations 10.134–136, the total failure probability can then be written as

$$\begin{aligned}
 p_f &= p_f^{L_1} + p_f^{L_1 L_2} + p_f^{L_2} \\
 &= \left[\int_{-\infty}^{\infty} f_R(x) \int_x^{\infty} f_{L_1}(y) \int_{-\infty}^y f_{L_2}(z, x) \int_{-\infty}^y f_G(w, x) p_{L_1} dw dz dy dx \right] \\
 &\quad + \left[\int_{-\infty}^{\infty} f_R(x) \int_x^{\infty} f_G(y) \int_{-\infty}^y f_{L_1}(z, x) \int_{-\infty}^y f_{L_2}(w, x) p_{L_1 L_2} dw dz dy dx \right] \\
 &\quad + \left[\int_{-\infty}^{\infty} f_R(x) \int_x^{\infty} f_{L_2}(y) \int_{-\infty}^y f_{L_1}(z, x) \int_{-\infty}^y f_G(w, x) p_{L_2} dw dz dy dx \right]
 \end{aligned} \tag{10.137}$$

In Equation 10.137, the total failure probability is all-inclusive if only two loads are present, which is referred to as the comprehensive failure probability.

10.4.3.2 Occurrence of Loads in a Given Time Duration

In Equation 10.137, we purposely write the terms p_{L_1} , $p_{L_1 L_2}$, and p_{L_2} at the end of each of the partial failure probabilities $p_f^{L_1}$, $p_f^{L_1 L_2}$, and $p_f^{L_2}$. These terms are occurrence probabilities of events L_1 , $L_1 \cap L_2$, and L_2 being greater than or equal to the value x , respectively, which denote the exclusive nature of occurrence of L_1 only, $L_1 \cap L_2$, and L_2 only when these load effects are taken at the value x . For convenience, these occurrences of loads are referred to as the probability of condition, as mentioned previously. These probabilities are functions of the value x . The detailed discussions of these occurrence probabilities will be presented in Section 10.4.4.1.

10.4.3.3 Brief Summary

In this subsection, we presented a proposed methodology on comprehensive bridge reliability based on the formulation of partial failure probability, in order to determine the design limit state equations under time-varying and infrequent loads. Specifically, if the designed value of failure probability is given, then each partial failure probability is uniquely determined. Then, we can calculate the partial reliability index one by one, according to the classification of the effect of either single load or combined loads.

For dead and live (truck) loads only, the relationships of the load and the resistance effects are treated as time-invariant random variables with normal distributions. The corresponding limit state is simply described by the reliability index, from which the load and resistance factors can be uniquely determined.

Multihazard loads are time variables. The limit state of loads not exceeding the resistance exists, but the total failure probability is far more complex to calculate.

In order to establish the American Association of State Highway and Transportation Officials (AASHTO) LRFD compatible MH-LRFD, the concept of comprehensive bridge reliability is introduced. To achieve the objective, one of the feasible treatments is to use the sum of several partial failure probabilities to represent the total failure probability by separation of mutually exclusive loading events. The partial failure probability results in a partial limit state used to form the total limit state. In this situation, each partial limit state defines the exceedance to the resistance due to a special kind of load, including two or more load simultaneous combinations.

This process requires that the partial limit states be mutually exclusive events. When dealing with one partial failure probability, the interference of other loads/load combinations need not be concerned. Thus, the consideration of the failure probability is simplified so that the evaluation of the total failure probability can be carried out with sufficient accuracy. However, the procedure of determination of the load and resistant factors remains a more complex issue. In order to determine the partial reliability index, the relationship between load and resistance must follow a normal distribution. The authors have shown quantitatively that such “normalization” is doable with acceptable accuracy for engineering practice. The details are not elaborated herein.

Once the distributions are normalized, with the help of partial reliability indices, the ratios of mean values among the loads/load combinations and resistance can be fully determined, uniquely and accurately. By means of using commonly used values of load and resistance biases, the load and resistant factors of the uniformed design limit state for highway bridges can be finally calculated. In order to be fully compatible with currently used AASHTO LRFD (2004) and with the experiences of bridge engineers, further calibrations may be needed to ensure the accuracy of these factors.

In this procedure, each partial failure probability is further formulated as a product of a conditional probability and a probability of condition. So far, the main focus is given to the conditional probability. The probability of condition will be discussed in Section 10.4.4.

10.4.4 PROBABILITY OF CONDITIONS

In the above, we have introduced the concept of partial failure probability to replace the random process with random variables.

To numerically calculate partial and total failure probabilities, it is necessary to carry out the aforementioned integrations by considering individual load value x . Thus, we can further write

$$P_f(x) = P_f^T(x) + P_f^E(x) + P_f^{ET}(x) = P(T \geq R|_{R=x, \text{ only } T \geq x})P(\text{only } T \geq x) \\ + P(E \geq R|_{E=x, \text{ only } E \geq x})P(\text{only } E \geq x) + P(T + E \geq R|_{T+R=x, T+E \geq x})P(T + E \geq x) \quad (10.138)$$

In the above, we use uppercase $P_f(x)$, etc., to denote the probability of event (x) , whereas we use the lower case p_f , etc., to denote the exact value of failure probabilities.

10.4.4.1 Condition for Occurrence of Partial Failure Probabilities

In Equation 10.138, there are three partial failure probabilities. Each term contains a condition, described as the probability of event (only the specific load $\geq x$), such as $P(\text{only } T \geq x)$, $P(\text{only } E \geq x)$, and $P(T + E \geq x)$. In Section 10.4.3, they were referred to as the probability of condition. For example, the probability of condition for a generic load L_i is denoted as p_{L_i} .

To further realize the general condition of load separation, we first consider the first special case of having a truck load only and the truck load effect T is greater than or equal to R , which is the partial failure probability of p_i^T .

In this circumstance, the condition of maximum truck load only, which exceeds the level x , consists of five independent events:

- a. There must be a load (excluding the case of no load).
- b. The load is truck load only, denoted by T only (no other load occurring simultaneously).
- c. The load is greater than or equal to x , denoted by $T \geq x$ (not considering the case of $T < x$).
- d. The load is the maximum load denoted by $T = \max$ (not considering other intensity).
- e. The total types of loads are truck and earthquake only (no other time variable loads occurring).

The above five conditions are mutually independent. Each condition can be seen as a probability, which can be denoted by

- a. $P(\text{there is a load})$
- b. $P(T \text{ only})$
- c. $P(T > x)$
- d. $P(T = \max)$, with
- e. $P(\text{only } T \geq x) + P(\text{only } E \geq x) + P(T + E \geq x) = 1$

In general, the condition of having a single load effect L_i (again, the combined loads are treated as special kinds of single loads) can be written as a product of $P(L_i \text{ only})P(L_i > x)P(L_i = \max)$ (see conditions b, c, and d). These three probabilities are relatively more complex, whereas the probabilities of conditions a and e are comparatively simpler. In the following, we will mainly focus on issues b, c, and d, with issues a and e only briefly described in a simplified manner.

10.4.4.2 Event of Single Type of Loads

10.4.4.2.1 Nature of Occurrences of Load T and E

In condition b, T and E are the maximum truck and earthquake loads, respectively. Therefore, the event (only $T \geq x$) should be the existence of truck loads only, which is greater than or equal to a certain value x . The value of truck load effect in event (only $T \geq x$) cannot be arbitrary. That is, among all possible truck load effects, we need to consider the value being greater than or equal to x .

Therefore, the above-mentioned events are not equal to the existence of *all* possible truck loads only (same for earthquake load, etc.). The event of existence of occurrence of such truck loads can be further divided as the intersection of event “the existence of all truck loads” and the event “those loads must be greater than or equal to value x .” That is,

$$(only\ T \geq x) = [(there\ exist\ only\ truck\ loads) \cap (truck\ load\ effect \geq x) \cap (truck\ load\ effects = the\ maximum\ value)] \mid [(there\ must\ be\ a\ load) \cap (there\ are\ only\ truck\ and\ earthquake\ loads)]$$

The events shown on the right-hand side of the above equation are independent. Therefore, we can have

$$\begin{aligned} P(only\ T \geq x) &= [P(there\ exist\ only\ truck\ loads)P(truck\ load\ effect \geq x)P(truck\ load\ effect = max)] \mid [P(there\ must\ be\ a\ load)P(there\ are\ only\ truck\ and\ earthquake\ loads)] \\ &= \{[P(there\ exist\ only\ truck\ loads) \mid P(there\ must\ be\ a\ load)][P(truck\ load\ effect \geq x) \mid P(truck\ load\ effect = max)] \mid P(there\ are\ only\ truck\ and\ earthquake\ loads)]\} \end{aligned}$$

In the following, we first discuss the probability of the event of only a single existing type of load, such as $P(there\ exist\ only\ truck\ loads)$.

10.4.4.2.2 Existence of All Possible Loads

We now discuss the uniqueness probability starting with single load and then continuing to load combinations. There can be several approaches. We chose only the Poisson distribution model here.

10.4.4.2.3 Modeling of the Probability of Condition

10.4.4.2.3.1 Occurrence of Loads in Given Duration, General Model If there is no given load, say L_1 , in period t , then the corresponding probability is $1 - e^{-\lambda t}$, which is equivalent to the sum of probabilities of the case $k = 1, 2, \dots, \infty$. That is,

$$p(L_1\ exists) = 1 - e^{-\lambda t} = e^{-\lambda t} \sum_{k=1}^{\infty} \frac{(\lambda t)^k}{k!} \quad (10.139)$$

If in a time span, there cannot be an infinite number of certain loads, say L_1 , using $1 - e^{-\lambda t}$ to calculate the probability of seeing that load L_1 can result in an overestimation.

10.4.4.2.3.2 Occurrence of Combined Time Variable Load Effect With two loads, L_1 and L_2 only, the corresponding hazards are likely to have different loading duration, denoted by t_{0i} . To simplify the computation, we assume that during the life span of a bridge, all the loading duration of L_{0i} can be represented by its average as a fixed value, t_{0i} . In general, the average loading duration due to different loads will be different. Without loss of generality, suppose L_2 has longer duration, denoted by t_{L_2} , and that L_1 is the load with the shorter duration. The basic idea on the chance of load

combination is as follows: First, assume that in the life span of a bridge there are a total of up to n occurrences of L_2 ; each has the same duration t_{L_2} . We thus have the possible cases of L_2 to be one occurrence (duration is $1 \times t_{L_2}$), two occurrences (duration is $2 \times t_{L_2}$), and so on. Secondly, in the duration t_{L_2} , we can calculate the probability of having up to m load effect L_1 , which is the sum of one L_1 , two L_1 , ... up to mL_1 . In the duration $2t_{L_2}$, we can calculate the probability of having up to $2 \times m$ load L_1 , etc.

Denote the quantity m to be the upper limit of the maximum possible number of loads L_1 . Since the physical length of a given bridge is fixed, there must be a limited number of vehicles "occurring" on the bridge. The probability that load L_1 agrees in t_{L_2} is denoted by ${}_1p_{s_{L_1}}$, where the subscript 1 in front of symbol $p_{s_{L_1}}$ stands for load L_1 occurring in one time interval t_{L_2} . The corresponding computation is given by

$${}_1p_{s_{L_1}} = e^{-\lambda_{L_1}t_{L_2}} \sum_{k=1}^m \frac{(\lambda_{L_1}t_{L_2})^k}{k!} \quad (10.140)$$

where λ_{L_1} is the average loading rate of L_1 .

If r time intervals exist, the total duration is rt_{L_2} . The probability of load L_1 showing up in rt_{L_2} is denoted by ${}_rp_{s_{L_1}}$, where the subscript r in front of symbol $p_{s_{L_1}}$ stands for load L_1 occurring in duration rt_{L_2} . We have

$${}_rp_{s_{L_1}} = e^{-\lambda_{L_1}(rt_{L_2})} \sum_{k=1}^{mr} \frac{[\lambda_{L_1}(rt_{L_2})]^k}{k!} \quad (10.141)$$

The quantity ${}_rp_{s_{L_1}}$ is, in fact, a conditional probability, which is the probability of the occurrence of L_1 under the condition of having L_2 . It is assumed that L_1 and L_2 are independent. Therefore, the unconditional occurrence probability of L_1 can be written as ${}_rp_{s_{L_1}}p_{s_{L_2i}}$, where $p_{s_{L_2i}}$ is the probability of occurrence of load L_{2i} , which will be discussed in the following.

In the service life span of a bridge, T_H , there may be up to n loads L_2 , with each having its own occurrence probability. More specifically, let the letter i denote the said level of load effect L_2 . We can use the symbol n_i to denote the maximum possible number of such load effects.

More detailed computation should distinguish the simultaneous occurrence of both loads L_1 and L_2 during different periods. For example, suppose an earthquake lasts 60 s. If this 60-s earthquake happens in a rush hour, then we may see more trucks. Another detailed consideration is to relate the duration t_{L_2} to the level of load L_2 . For example, a larger level of earthquake may have a longer duration, and so on. Here, for the sake of simplicity, we assume that the number m and the length t_{L_2} are constant. In this case, the total probability of occurrence of load L_{2i} , denoted by $p_{s_{L_{2i}}}$, is given by

$$p_{s_{L_{2i}}} = e^{-\lambda_{L_{2i}}T_H} \sum_{r=1}^{n_i} \frac{(\lambda_{L_{2i}}T_H)^r}{r!} \quad (10.142)$$

where the first subscript s of $p_{s_{L_{2i}}}$ stands for the segment of the uniqueness probability; n_i is the total number of load L_{2i} occurring in T_H (75 years); and $\lambda_{L_{2i}}$ is the rate of occurrence or the reciprocal of the return period of the specific load L_{2i} .

A more simplified estimation of the occurrence of load L_2 is to take its average value without distinguishing the detailed levels. In this case, we use n to denote the average number of occurrences of load L_2 . The life span of the bridge is denoted by T_H . The probability of occurrence of such load effect L_2 in the bridge life span is

$$p_{s_{L_2}} = e^{-\lambda_{L_2} T_H} \sum_{r=1}^n \frac{(\lambda_{L_2} T_H)^r}{r!} \quad (10.143)$$

where n is the total number of loads L_2 occurring in T_H (75 years). Now, the probability of simultaneous occurrence of both loads L_1 and L_2 with level i is the product denoted by $p_{L_1 L_{2i}}$ given by

$$p_{L_1 L_{2i}} = e^{-\lambda_{L_{2i}} T_H} \sum_{r=1}^{n_i} p_{s_{L_1}} \frac{(\lambda_{L_{2i}} T_H)^r}{r!} = e^{-\lambda_{L_{2i}} T_H} \left\{ \sum_{r=1}^{n_i} \left[e^{-\lambda_{L_1} (r T_{L_2})} \sum_{k=1}^{mr} \frac{(\lambda_{L_1} r T_{L_2})^k}{k!} \right] \frac{(\lambda_{L_{2i}} T_H)^r}{r!} \right\} \quad (10.144)$$

where n_i is the number of occurrences of load effect L_2 with level i in the duration of T_H .

With the simplified treatment described in Equation 10.144, we have simultaneous occurrence of both loads L_1 and L_2 , denoted by $p_{L_1 L_2}$ given by

$$p_{L_1 L_2} = e^{-\lambda_{L_2} T_H} \left\{ \sum_{r=1}^n \left[e^{-\lambda_{L_1} (r T_{L_2})} \sum_{k=1}^{mr} \frac{(\lambda_{L_1} r T_{L_2})^k}{k!} \right] \frac{(\lambda_{L_2} T_H)^r}{r!} \right\} \quad (10.145)$$

10.4.4.2.3.3 Uniqueness Probability of Truck and Earthquake Loads We now consider first the existence of all possible loads of the same kind, for example, all truck loads only, all earthquake loads only, and all the load combinations only.

Based on Equations 10.144 and/or 10.145, we use truck load T and earthquake load E as examples to replace loads L_1 and L_2 . Figure 10.13 shows the Venn diagram of situation 1: no trucks, no earthquakes; situation 2: occurrence of trucks; and situation 3: occurrence of earthquakes. We are interested in three mutually exclusive cases, namely, trucks only (no earthquake), denoted by $T \cap \bar{E}$; trucks and earthquake simultaneously, denoted by $T \cap E$; and earthquake only (no trucks), denoted by $\bar{T} \cap E$.

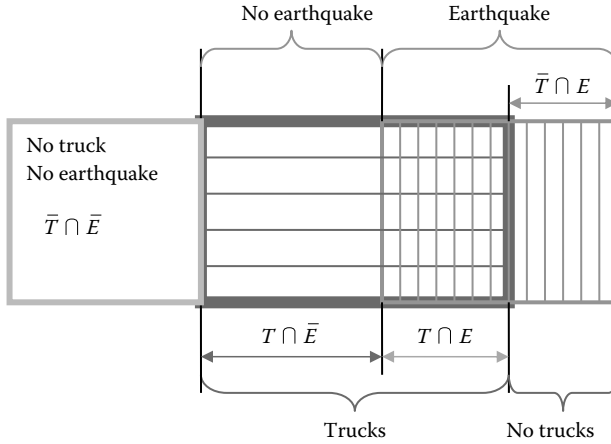


FIGURE 10.13 Venn diagram of occurrence events of trucks and earthquakes.

In Figure 10.13 and in the following discussion, the overhead bar stands for “no existence.” Including situation 1, denoted by $\bar{T} \cap \bar{E}$, the total probability is unity. Therefore, we can write

$$p(T \cap \bar{E})/[1 - p(\bar{T} \cap \bar{E})] + p(T \cap E)/[1 - p(\bar{T} \cap \bar{E})] + p(\bar{T} \cap E)/[1 - p(\bar{T} \cap \bar{E})] = 1 \quad (10.146)$$

Equation 10.146 implies that once the probabilities $p(T \cap \bar{E})$, $p(T \cap E)$, and $p(\bar{T} \cap E)$ are calculated, they can be normalized under the condition of having a load, denoted by $[1 - p(\bar{T} \cap \bar{E})]$. We now consider these probabilities. In order to simplify the notations, the necessary nomenclatures are listed as follows:

Total time, T_H : The period of 75 years measured by seconds; $T_H = 75 \times 365.25 \times 24 \times 3600$

Truck crossing time, t_T : average duration of a truck crossing a bridge

Average duration of earthquakes, t_E

Number of trucks in T_H , n_T

Number of earthquakes in T_H , n_E

Total time of having trucks, T_T

Total time of not having trucks, $T_{\bar{T}}$

Total time of having quakes, T_E

Total time of not having quakes, $T_{\bar{E}}$

Occurrence probability of trucks in T_H , p_t

Occurrence probability of earthquakes in T_H , p_e

Occurrence probability of both truck and earthquakes in T_H , p_{te}

Probability of having both trucks and quakes in T_H , p_{te}

Probability of having trucks in T_H without earthquakes, $p_{t\bar{e}}$

Probability of having quakes in T_H without trucks, $p_{\bar{t}e}$

Probability of having neither trucks nor quakes in T_H , $p_{\bar{t}\bar{e}}$

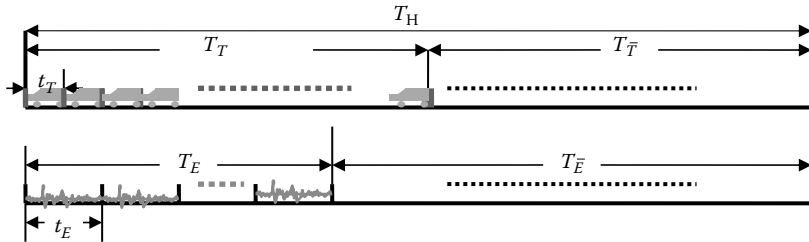


FIGURE 10.14 Load durations.

Figure 10.14 shows a special arrangement of the duration T_T , T_E vs. T_H , which is the base of the following analysis.

10.4.4.2.3.4 Uniqueness Probability of Simultaneous Truck and Earthquake Loads Consider the simultaneous loads of both trucks and earthquakes.

Note that we may consider different levels of earthquakes instead of the overall occurrence of earthquakes. Suppose that earthquakes with the same i th effect, denoted by E_i , have the average duration t_{Ei} . First, to simplify the analysis, we use the average duration of all t_{Ei} , denoted by t_E , in order to study the probability of trucks showing up in t_E , denoted by ${}_1P_{s_T}$. Suppose that in the duration t_E , there are up to m trucks crossing the bridge,

$${}_1P_{s_T} = e^{-\lambda_T t_E} \sum_{k=1}^m \frac{(\lambda_T t_E)^k}{k!} \quad (10.147)$$

where λ_T is the average truck rate.

The conditional probability of trucks showing up in r time intervals, rt_E , denoted by ${}_rP_{s_T}$ with mr trucks crossing the bridge is given by

$${}_rP_{s_T} = e^{-\lambda_T (rt_E)} \sum_{k=1}^{mr} \frac{[\lambda_T (rt_E)]^k}{k!} \quad (10.148)$$

Next, we consider the above-mentioned particular time interval. The probability of occurrence of the non-exceedance earthquake E_i in T_H is

$$p_{s_{Ei}} = e^{-\lambda_{Ei} T_H} \sum_{k=1}^{n_{Ei}} \frac{(\lambda_{Ei} T_H)^k}{k!} \quad (10.149)$$

where n_{Ei} is the total number of earthquakes E_i occurring in 75 years; λ_{Ei} is the rate of earthquakes with an effect level i in T_H .

In this case, the simultaneous occurrence probability of both earthquakes with the effect amplitude E_i and truck loads can be written as

$$p_{TE_i} = e^{-\lambda_{E_i} T_H} \left\{ \sum_{r=1}^{n_i} \left[e^{-\lambda_T (rt_E)} \sum_{k=1}^{mr} \frac{(\lambda_T rt_E)^k}{k!} \right] \frac{(\lambda_{E_i} T_H)^r}{r!} \right\} \quad (10.150)$$

The term p_{TE_i} is a uniqueness probability denoting the unique chance of the simultaneous occurrence of a truck load and an earthquake load with level E_i . At the specific moment, only these two loads are acting on the bridge.

Now, if we use the simplified approach without considering the level of earthquake effects, we have the probability of occurrence of the non-exceedance earthquake E in T_H , denoted by p_{s_E} and with n_E earthquakes. Then

$$p_{s_E} = e^{-\lambda_E T_H} \sum_{r=1}^{n_E} \frac{(\lambda_E T_H)^r}{r!} \quad (10.151)$$

where λ_E is the average rate or the reciprocal of the average earthquake return period. In this case, we have the average occurrence probability given as

$$p_{te} = e^{-\lambda_E T_H} \left\{ \sum_{r=1}^{n_E} \left[e^{-\lambda_T (rt_E)} \sum_{k=1}^{mr} \frac{(\lambda_T rt_E)^k}{k!} \right] \frac{(\lambda_E T_H)^r}{r!} \right\} \quad (10.152)$$

Suppose that we have r earthquakes in 75 years. The probability of having no trucks in the duration rt_E is

$$p_{\bar{T}} = e^{-\lambda_T (rt_E)} \quad (10.153)$$

The probability of having an earthquake only (without trucks) can be written as

$$p_{\bar{te}} = e^{-\lambda_E T_H} \left\{ \sum_{r=1}^{n_E} \left[e^{-\lambda_T (rt_E)} \right] \frac{(\lambda_E T_H)^r}{r!} \right\} \quad (10.154)$$

Note that one of the disadvantages of using the above equations is the difficulty of computing the factorial. If the number of k is large, say, more than 150 for most personal computers, then the term $k!$ cannot be calculated. Therefore, let us consider an alternative approach.

In one duration t_E , suppose we have the following event: an earthquake is occurring while simultaneously up to m trucks are crossing the bridge. The probability of that event is

$${}_1 p_{te} = \left[e^{-\lambda_T t_E} \sum_{r=1}^m \frac{(\lambda_T t_E)^r}{r!} \right] \left[e^{-\lambda_E T_H} (\lambda_E T_H)^r \right] \quad (10.155)$$

In that duration, the probability of no such event occurring is

$$1 - {}_1p_{te} = 1 - e^{-\lambda_T t_E - \lambda_E T_H} (\lambda_E T_H) \sum_{r=1}^m \frac{(\lambda_T t_E)^r}{r!} \quad (10.156)$$

Suppose that in the life span, a total of n_E earthquakes occurred. Therefore, at least the event that an earthquake occurs simultaneously with up to m trucks can have the uniqueness occurrence probability written as

$$p_{te} = 1 - (1 - {}_1p_{te})^{n_E} = 1 - \left[1 - e^{-\lambda_T t_E - \lambda_E T_H} (\lambda_E T_H) \sum_{r=1}^m \frac{(\lambda_T t_E)^r}{r!} \right]^{n_E} \quad (10.157)$$

and the probability when earthquakes occur with no trucks crossing the bridge is given as

$$p_{\bar{t}e} = \left[1 - e^{-\lambda_T t_E - \lambda_E T_H} (\lambda_E T_H) \right]^{n_E} \quad (10.158)$$

Furthermore, instead of directly calculating the occurrence probabilities of $p_{i\bar{e}}$, p_{te} , and $p_{e\bar{t}}$ using the above equations, a method based on Figure 10.15 may be used. The figure gives the three probabilities, of which two of them are determined through the above approach.

When the number n or n_i in the Poisson distribution is too large, the corresponding occurrence probabilities cannot be easily computed. We thus offer the following simplification.

In general, the number of occurrences of earthquakes is considerably smaller than that of trucks. Thus, assume that the probability of having both trucks and earthquakes and the probability of having earthquakes only can be calculated for the

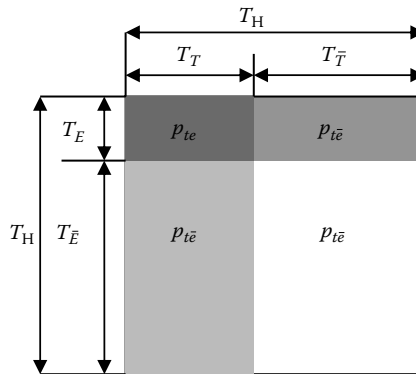


FIGURE 10.15 Probabilities of occurrences.

number n_E is sufficiently small. Based on the concepts described in Figures 10.13 and 10.15, the probability of having an earthquake can be calculated as

$$p_e = p_{te} + p_{e\bar{t}} \quad (10.159)$$

Therefore,

$$\begin{aligned} p_e &= e^{-\lambda_E T_H} \left\{ \sum_{r=1}^{n_E} \left[e^{-\lambda_T (rt_E)} \sum_{k=1}^{mr} \frac{(\lambda_T rt_E)^k}{k!} \right] \frac{(\lambda_E T_H)^r}{r!} \right\} + e^{-\lambda_E T_H} \left\{ \sum_{r=1}^{n_E} [e^{-\lambda_T (rt_E)}] \frac{(\lambda_E T_H)^r}{r!} \right\} \\ &= e^{-\lambda_E T_H} \left\{ \sum_{r=1}^{n_E} e^{-\lambda_T (rt_E)} \left[1 + \sum_{k=1}^{mr} \frac{(\lambda_T rt_E)^k}{k!} \right] \frac{(\lambda_E T_H)^r}{r!} \right\} \end{aligned} \quad (10.160)$$

Thus, the probability of having trucks is

$$p_t = \frac{p_{te}}{p_e} \quad (10.161)$$

Therefore,

$$p_t = \frac{p_{te}}{p_{te} + p_{e\bar{t}}} \quad (10.162)$$

Furthermore, the probability of having trucks only can be written as

$$p_{t\bar{e}} = p_t - p_{te} \quad (10.163)$$

To calculate the occurrence probabilities further under the condition of having a load only, the relationship among these conditional occurrence probabilities can be described by Figure 10.12. That is, by only considering the dark “areas” in the Venn diagram shown in Figure 10.15, we can let the sum of all these three conditional probabilities equal to unity. This Poisson model can be shown by using the following numerical example.

Example 10.4

Based on the Poisson model, the probabilities of truck load only, earthquake load only, and simultaneously having both loads in the period of t_E seconds can be obtained. Suppose that the daily average truck rate is 1000.

In a given duration, $\lambda_T = 1000/(24 \times 3600) = 0.0116$. In the duration of an earthquake, $t_E = 80$ s. The probability of having up to 100 trucks is

$$e^{-\lambda_T t_E} \sum_{k=1}^{100} \frac{(\lambda_T t_E)^k}{k!} = 0.6038$$

Similarly, during an earthquake, the chance of only two trucks crossing a bridge is 0.5366. It is seen that, comparing 100 trucks with the probability = 0.6038, the difference is not very large.

Note that T_H is 2.3668×10^9 (s). Suppose that in 75 years, there are a total of 15 earthquakes ($n_E = 15$) so that $\lambda_E = 15/T_H = 6.3376 \times 10^{-9}$. Suppose that the average number of trucks crossing a bridge during an earthquake is $m = 2$. The chance of simultaneously having a truck and an earthquake is

$$p_{te} = e^{-\lambda_E T_H} \left\{ \sum_{r=1}^{n_E} \left[e^{-\lambda_T (rt_E)} \sum_{k=1}^{mr} \frac{(\lambda_T rt_E)^k}{k!} \right] \frac{(\lambda_E T_H)^r}{r!} \right\} = 0.5677$$

The probability of the occurrence of an earthquake without a truck is

$$p_{\bar{te}} = e^{-\lambda_E T_H} \left\{ \sum_{r=1}^{n_E} \left[e^{-\lambda_T (rt_E)} \right] \frac{(\lambda_E T_H)^r}{r!} \right\} = 1.1615 \times 10^{-4}$$

and

$$p_e = p_{\bar{te}} + p_{te} = 0.5678$$

Therefore,

$$p_{\bar{te}} = p_{te} \frac{1 - p_e}{p_e} = 0.4321$$

The normalized probabilities of having a truck only, having both a truck and an earthquake, and having an earthquake only are denoted as $p_{\bar{te}}$, p_{te} , and $p_{\bar{te}}$, which can be further defined and calculated according to the next subsections.

Note that, besides the above approach, by using a mixed Poisson distribution, we could also describe a slightly different formulation of the probability of conditions.

10.4.4.2.4 Existence of All Possible Loads under the Conditions with Loads Only

We are now interested in the situation of having loads. Thus,

$$p_{te} + p_{\bar{te}} + p_{\bar{te}} = 1 - p_{\bar{te}} \quad (10.164)$$

Here, the term $(1 - p_{\bar{t}e})$ is a probability of the event (there must be loads), that is,

$$P(\text{there must be loads}) = 1 - p_{\bar{t}e} \quad (10.165)$$

Denoting the probability of having both trucks and earthquakes in T_H under the condition of having a load only as p_{TE} , we have

$$p_{TE} = \frac{p_{te}}{1 - p_{\bar{t}e}} \quad (10.166)$$

Moreover, denoting the probability of having trucks in T_H without earthquakes, under the condition of with load only, as $p_{T\bar{E}}$, we also have

$$p_{T\bar{E}} = \frac{p_{t\bar{e}}}{1 - p_{\bar{t}e}} \quad (10.167)$$

In addition, denote the probability of having quakes in T_H without trucks, under the condition having a load only, as $p_{E\bar{T}}$. We have

$$p_{E\bar{T}} = \frac{p_{\bar{t}e}}{1 - p_{\bar{t}e}} \quad (10.168)$$

Thus,

$$p_{TE} + p_{T\bar{E}} + p_{E\bar{T}} = 1 \quad (10.169)$$

That is, consider the probability of the event (truck load only), under the condition of “there must be load(s),” we have

$$P(\text{truck load only}) = P(\text{truck load only} \mid \text{there must be loads}) / P(\text{there must be loads}) \quad (10.170)$$

10.4.4.2.5 Maximum Loads

Now consider the probability of a specific load being the maximum load. This requires the load effect L to be the maximum value. It is a value extracted from all the possible loads applied to the entire spectrum of possible bridges to be designed under this load. For example, let L be the truck load T only. T must contain two conditions: First, among all events of a truck crossing, T must be equal to or exceed a certain level x , so that we have the term $P(\text{truck load effect} \geq x)$. Second, among all the peak values, T must reach the maximum value, as described by the term $P(\text{truck load effect} = \max)$.

10.4.4.2.5.1 Exceeding Values The probability $P(\text{truck load effect} \geq x)$ implies the chance that the load effect exceeds the level of the chosen resistance effect, that is,

$$P(\text{truck load effect} \geq x) = \int_x^{\infty} f_t(u) du \quad (10.171)$$

where f_t is the PDF of the truck load (see Figure 10.16). It shall be noted that this particular PDF is different from the term $\int_x^{\infty} f_{L_1}(y) dy$ in Equation 10.137, where

$$f_{L_1} \equiv f_T \quad (10.172)$$

which is the PDF of the maximum value of the truck load effect.

10.4.4.2.5.2 Maximum Values Consider now the probability of the event when a specific load is a possible maximum load, which has the probability $P(\text{load effect} = \text{max})$.

Using the truck load effect as an example, the conceptual plots of the PDF of the distributions of the truck loads, denoted by f_T , is shown in Figure 10.17, where the coordinate is the value of the PDF and the abscissa is the intensity of the conceptual load effect.

From Figure 10.17, the total probability from zero to a given maximum load level with intensity up to x (such as the one shown in Figure 10.16 with a moment of 25,000 kps ft.) can be written as

$$P(\text{truck load} = \text{max}) = \int_0^x f_T(v) dv \quad (10.173)$$

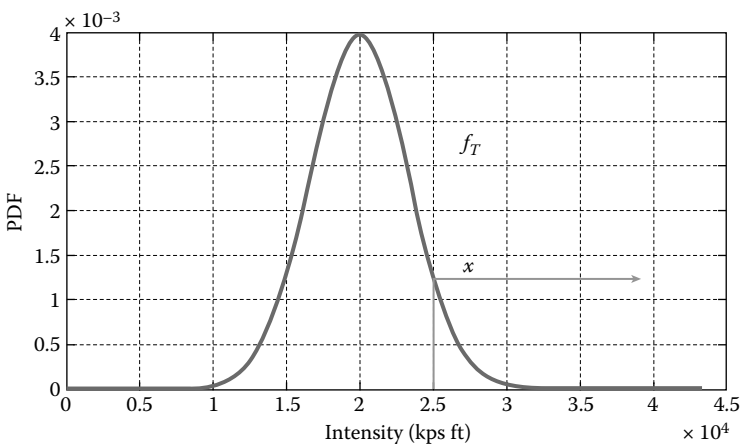


FIGURE 10.16 Loads $\geq$ level x .

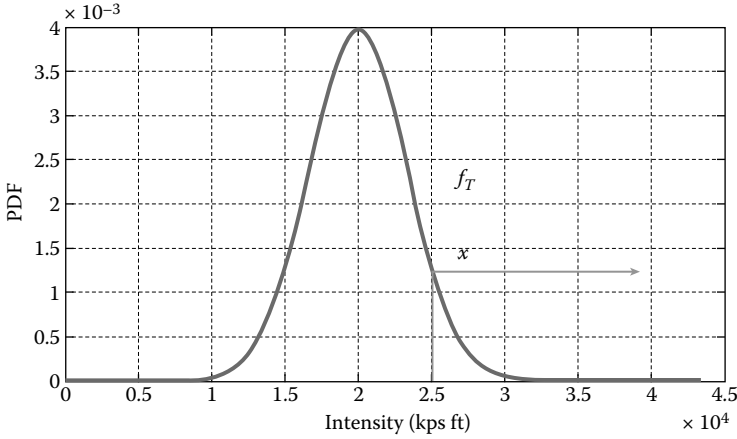


FIGURE 10.17 PDF of maximum load.

From Figure 10.17, it is seen that this probability is the CDF of the distribution of the maximum load up to level x .

10.4.4.2.5.3 Probability of Total Conditions, Individual Cases By using the example of truck loads, and considering $P(T$ only) denoted by p_t , the above discussion on the probability of conditions is summarized as follows:

$$\begin{aligned} p_t &= P(\text{truck load only})P(\text{truck load} \geq x)P(\text{truck load} = \max) \\ &= p_{TE} \int_x^\infty f_i(u) du \int_0^x f_T(v) dv \end{aligned} \quad (10.174)$$

Similarly, consider $P(E \geq x$ only), denoted by p_e :

$$\begin{aligned} p_e &= P(\text{earthquake load only})P(\text{earthquake load effect} \geq x)P(\text{earthquake load} = \max) \\ &= p_{TE} \int_x^\infty f_e(u) du \int_0^x f_E(v) dv \end{aligned} \quad (10.175)$$

In the above, f_e and f_E are the PDF of the regular and the maximum earthquake load effects, respectively.

Furthermore, the probability $P(T \cap E \geq x)$, denoted by $p_{t \cap e}$, can be written as

$$\begin{aligned} p_{t \cap e} &= P(\text{combined truck and quake load})P(\text{combined load effect} \geq x) \\ P(\text{combined load} = \max) &= p_{TE} \int_x^\infty f_c(u) du \int_0^x f_C(v) dv \end{aligned} \quad (10.176)$$

where f_c and f_e are the PDF of the regular and maximum combined truck and earthquake load effect, respectively.

The sum of p_t , p_e , and $p_{t \cap e}$ is not necessarily equal to unity. This fact is rather inconvenient for calculating the total probability. Thus, the terms p_t , p_e , and $p_{t \cap e}$ as probability of conditions with individual effect values are considered in the following.

10.4.4.2.5.4 Probability of Conditions, Unity Space As a total probability, consider all the terms of the partial failure probabilities p_t^T , p_t^E , and p_t^{TE} . All the loads, T , E , and/or $T + E$ are compared with a unified value, the resistant value R . That is, in Equation 10.1, when R is given, the values of T , E , and $T + E$ cannot be chosen individually. Accordingly, in these conditional probabilities, the amplitudes of T , E , and $T + E$ are actually of the same value, respectively. In other words, the sum of all three conditional probabilities must be unity. That is,

$$P(\text{only } T \geq x) + P(\text{only } E \geq x) + P(T \cap E \geq x) = 1 \quad (10.177)$$

Equation 10.177 means that the terms p_t , p_e , and $p_{t \cap e}$ should be normalized according to unity. This can be done by

$$p_t/(p_t + p_e + p_{t \cap e}) + p_e/(p_t + p_e + p_{t \cap e}) + p_{t \cap e}/(p_t + p_e + p_{t \cap e}) = 1 \quad (10.178)$$

We thus have the normalized or uniformed conditional probabilities as follows:

$$P(\text{only } T \geq x) = p_t/(p_t + p_e + p_{t \cap e}) \quad (10.179)$$

which is the normalized conditional probability of the maximum truck load only,

$$P(\text{only } E \geq x) = p_e/(p_t + p_e + p_{t \cap e}) \quad (10.180)$$

which is the normalized conditional probability of the maximum earthquake load only, and

$$P(T \cap E \geq x) = p_{t \cap e}/(p_t + p_e + p_{t \cap e}) \quad (10.181)$$

which is the uniformed conditional probability of the maximum combined load only.

10.4.5 BRIEF SUMMARY

In the above, the probability of condition for having a specific load only by using truck and earthquake loads as examples is formulated. (The generic formulations for other loads are identical.) Since the dead load is time invariant, whenever a time-variable load occurs, the dead load is “waiting” there for the load combination. Therefore, the dead load is omitted in this discussion.

Suppose that we have only two load L_1 and L_2 . There exist three distinct cases: Case 1 is only having load effect L_1 ; case 2 is only having load effect L_2 ; and case 3 is only having the combined load effect $L_1 + L_2$, which is denoted as L_3 ($L_3 = L_1 + L_2$). Each case is treated as a single kind of load effect.

The probability of condition for having the case ($L_i \geq$ certain level x only, $i = 1, 2, 3$) consists of the following events:

- a. There must be a load (the corresponding probability is denoted as p_A).
- b. The load is L_i only (the corresponding probability is denoted as p_{Bi}).
- c. The load effect L_i is greater than or equal to x (the corresponding probability is denoted as p_{Ci}).
- d. The load L_i is of the maximum value (the corresponding probability is denoted as p_{Di}).
- e. The total types of loads only include L_1, L_2 , and L_3 (the corresponding probability is denoted as p_E).

The resulting condition probability for load L_i can then be written as

$$P(L_i \geq x) = [(p_{Bi}/p_A)(p_{Ci}p_{Di})]/p_E \quad (10.182)$$

To calculate the occurrence of single and/or simultaneous loads, we can use either Poisson or mixed distributions.

Generally speaking, we introduced a methodology in this section to develop reliability-based bridge design under the conditions of MH loads. The first step is to formulate the total bridge failure probability using the approach of partial failure probabilities. In Section 10.4.2, we introduced the concept of comprehensive bridge reliability, the essence of which is an all-exclusive approach to address all necessary loads, time-invariant and time-variable, regular and extreme, on the same platform. In so doing, all loads, as long as they contribute accountably to bridge failures, are included. The basic approach to realize the comprehensive reliability is to break down these loads into separate cases, referred to as partial failure probability.

Each partial failure probability contains only a “pure” load or a “pure” load combination. Technically speaking, these pure loads can be treated as time-invariant random variables, although most loadings are time-varying random processes. The key for such separation of variables is to find the condition when the load occurs on its own, which is referred to as the probability of condition, and this is addressed in Section 10.4.4. It is seen that, to realize this term, we need to further break down the probability into five independent subconditions.

Once we calculate all the partial failure probabilities by having the partial conditional probabilities and the probabilities of conditions, the summation will give us the total bridge failure probability.

To form the design limit equations and to extract the load and resistant factors for practical bridge design, however, the formulation of the total probability is only the first step. To obtain the load and resistance factors, additional efforts are necessary, which are outside the scope of this manuscript.

PROBLEMS

1. The bearing of a type of simply supported bridges is subjected to dead load (DL) and live load (LL); both can be modeled as normally distributed random variables. Suppose that $DL \sim N(150 \text{ kps}, 45 \text{ kps})$ and $LL \sim N(235 \text{ kps},$

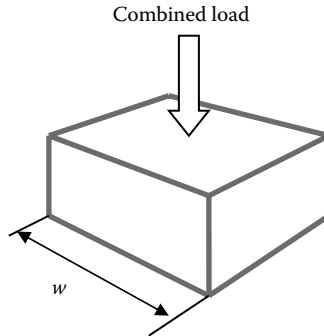


FIGURE P10.1

- 480 kps). If the strength of the bearing is considered as deterministic, use the 3σ criterion to calculate the design load acting on the bearing.
2. A rectangular-shaped bridge bearing is subjected to the loads given in Problem 1. The size of w is 5.5 in. The resistance stress R is considered to be normally distributed random variables with $\text{COV} = 0.15$. Use Equation 10.5 and Table 10.1 to determine the required strength (see Figure P10.1).
 3. A rectangular-shaped bridge bearing is subjected to loads so that the demanding load is $L \sim N(1000 \text{ kpi}, 300 \text{ kps})$. The resistance stress S_R is considered to be a normally distributed random variable also, and $S_R \sim N(40 \text{ ksi}, 3 \text{ ksi})$. Suppose that we need to let the failure probability be less than 0.0002. Calculate the size w of the bearing (see Figure P10.1).
 4. A shaft is subjected to a cyclic load up to 10^6 cycles. The corresponding standard deviation of the stress of the shaft is 3.8 ksi. Suppose that the mean value of the design strength of the shaft is 12.5 ksi. Use the 3σ criterion to design the shaft and calculate the failure probability by using the model of EVI. (Hint: The load is a narrowband process.)
 5. A cantilever beam subjected to stationary Gaussian narrowband excitation $f(t)$ is shown in Figure P10.2. The auto-PSD of $f(t)$ is also given in Figure P10.2. The central frequency of the excitation is 5 Hz. The loading duration is 20 min. It is required to design the minimum dimension b . The material is 6061-T6 aluminum with a mean yield strength of 40 ksi and a mean

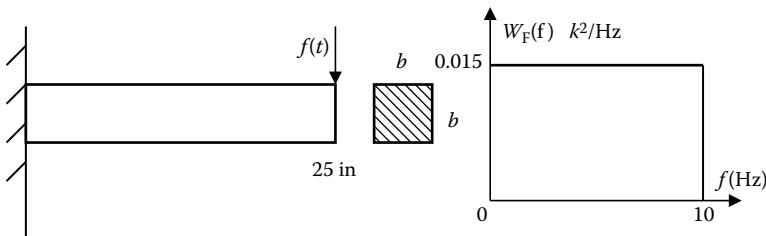


FIGURE P10.2

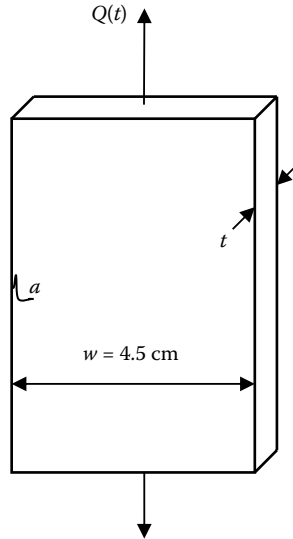


FIGURE P10.3

- ultimate strength of 45 ksi, both with a COV of 0.1. The natural frequency is sufficiently high so that there is no dynamic magnification.
- Design this beam with the 3σ criterion.
 - Design this beam for a first passage failure, with respect to ultimate. The reliability goal is 98% for the service life.
- A stationary force $Q(t)$ is applied to the plate shown in Figure P10.3. The mean and the standard deviation of $Q(t)$ are 15 and 20 kN, respectively. The failure mode is brittle fracture. Fracture toughness is given as $K_C = 26 \text{ MPa}\sqrt{\text{m}}$. The crack size is $a = 1.3 \text{ mm}$. No subcritical crack propagation (fatigue) is assumed. The geometry factor is $Y = 1.25$. Determine the minimum value required for t using the 3σ criterion. Failure occurs when the stress intensity factor $K = YS(\pi a)^{1/2} > K_C$ with stress S .
 - Reconsider the plate shown in Figure P10.3. Assume that $Q(t)$ is narrow-band with a central frequency of 1.5 Hz. The applying duration is 120 s. Using the first passage criterion, design the plate so that the probability of failure is less than 0.002.
 - Suppose that a type of bridge will be constructed in a river where scour hazards may occur with an average duration of 2 days. In a year, on average, vessel collision on the bridge may occur three times. Calculate the probability of two vessel collisions when a bridge scour occurs.
 - Suppose that in 75 years, there are a total of 100 bridge scours. Calculate the chance of simultaneously having a scour and a vessel collision under the condition given in Problem 8. Calculate the probability of up to three vessel collisions.

11 Nonlinear Vibrations and Statistical Linearization

In previous chapters, we often assumed that a system under consideration is linear. However, practically speaking, there are many nonlinear systems that are subjected to random and time-varying loads and deformations. In such a situation, it can be difficult to treat the response of a nonlinear system as a stationary process. In fact, analyzing such a random nonlinear system can be rather complex.

In this chapter, the basics of nonlinear dynamic systems are introduced. Monte Carlo simulation is used as a tool to address the complexity of such problems.

11.1 NONLINEAR SYSTEMS

In this chapter, it is assumed that all random processes are integrable. This assumption is applicable for most engineering systems.

Generally speaking, we have the following reasons for a dynamic system to be nonlinear:

1. Nonlinear damping, that is, the damping force, is not proportional to velocity
2. Nonlinear spring, the restoring force, is not proportional to displacement
3. Nonlinear boundary conditions
4. Nonlinear feedback control
5. Nonlinear complex systems, such as fluid–structure interaction

Within the above scope of nonlinear vibration, we mainly focus on the first two cases.

11.1.1 EXAMPLES OF NONLINEAR SYSTEMS

If the following equation is satisfied:

$$g[\alpha X(t) + \beta Y(t)] = \alpha g[X(t)] + \beta g[Y(t)] \quad (11.1)$$

then the system is linear. Otherwise, the system is nonlinear. Here, $g(\cdot)$ denotes a function of variable $(\cdot)$ and α and β are scalars.

The essence of Equation 11.1 is twofold. The first is *additivity*, that is,

$$g(x + y) = g(x) + g(y)$$

The second is homogeneity, that is,

$$g(\alpha x) = \alpha g(x) \text{ for all } \alpha$$

11.1.1.1 Nonlinear System

If Equation 11.1 does not hold, we will then have a nonlinear system. In the following, we first theoretically consider several typical nonlinear systems. Furthermore, only certain examples are discussed. These models will be referenced in order to model a bilinear system.

11.1.1.1.1 Bilinear Model

First consider a general bilinear relationship between $Y(t)$ and $X(t)$ given as

$$Y(t) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h_2(\tau_1, \tau_2) X(t - \tau_1) X(t - \tau_2) d\tau_1 d\tau_2 \quad (11.2)$$

Here $h_2(\tau_1, \tau_2)$ is referred to as the *temporal kernel function*.

If

$$\tau_1, \tau_2 < 0 \quad (11.3)$$

then

$$h_2(\tau_1, \tau_2) = 0 \quad (11.4)$$

When the input is bounded, the output will also be bounded.

$$B = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |h_2(\tau_1, \tau_2)| d\tau_1 d\tau_2 \quad (11.5)$$

That is, a scalar $B < \infty$ exists.

The double Fourier transform of $h_2(\tau_1, \tau_2)$ is referred to as the frequency kernel

$$H_2(\omega_1, \omega_2) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h_2(\tau_1, \tau_2) e^{-j(\omega_1 \tau_1 + \omega_2 \tau_2)} d\tau_1 d\tau_2 \quad (11.6)$$

Thus, in order to describe a bilinear system, either $h_2(\tau_1, \tau_2)$ or $H_2(\omega_1, \omega_2)$ must be known. In the following, several examples of bilinear systems as well as the corresponding kernel functions will be considered.

11.1.1.1.2 Quadratic System

The quadratic system (see Figure 11.1) is defined as

$$Y(t) = X^2(t) \quad (11.7)$$

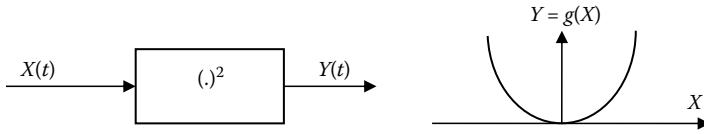


FIGURE 11.1 Quadratic system.

To see why a quadratic system is bilinear, denote

$$X(t) = \int_{-\infty}^{\infty} X(t - \tau_1) \delta(\tau_1) d\tau_1 \quad (11.8)$$

Substitution of Equation 11.8 into Equation 11.2 yields

$$\begin{aligned} Y(t) = X^2(t) &= \left(\int_{-\infty}^{\infty} X(t - \tau_1) \delta(\tau_1) d\tau_1 \right) \left(\int_{-\infty}^{\infty} X(t - \tau_2) \delta(\tau_2) d\tau_2 \right) \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \delta(\tau_1) \delta(\tau_2) X(t - \tau_1) X(t - \tau_2) d\tau_1 d\tau_2 \end{aligned} \quad (11.9)$$

Consequently, a quadratic system is bilinear and the kernel function is

$$h_2(\tau_1, \tau_2) = \delta(\tau_1) \delta(\tau_2) \quad (11.10)$$

In this case, it is seen that

$$H_2(\omega_1, \omega_2) = 1 \quad (11.11)$$

Thus,

$$\mathcal{F}[Y(t)] = \mathcal{F}[X^2(t)] = X(\omega) * X(\omega) \quad (11.12)$$

From Equation 11.12, it is shown that a process $X(t)$ through a quadratic system varies its frequency component.

11.1.1.1.3 Linear and Quadratic Systems in Series

When a linear and a quadratic system are in series, another bilinear system may exist (see Figure 11.2).

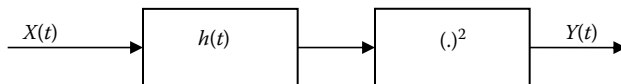


FIGURE 11.2 Linear and quadratic systems in series.

Denote

$$\begin{aligned}
 Y(t) &= \left[\int_{-\infty}^{\infty} h(\tau) X(t - \tau) d\tau \right]^2 \\
 &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h_1(\tau_1) h_2(\tau_2) X(t - \tau_1) X(t - \tau_2) d\tau_1 d\tau_2
 \end{aligned}
 \tag{11.13}$$

From Equation 11.13, it can be determined that

$$h_2(\tau_1, \tau_2) = h_1(\tau_1) h_2(\tau_2) \tag{11.14}$$

Additionally, it can be proven that

$$H_2(\omega_1, \omega_2) = H(\omega_1)H(\omega_2)$$

11.1.1.2 Memoryless Nonlinear System

In the following, examples of memoryless systems are used to show the probability density function (PDF) and correlation functions.

11.1.1.2.1 Definition

Denote

$$Y(t) = g(X(t)) \tag{11.15}$$

In Equation 11.15, g is a real function with a single variable only. That is, for a given moment t , $Y(t)$ is defined by $g(X(t))$ only. In other words, the system is memoryless.

11.1.1.2.2 PDF of Quadratic System

The aforementioned quadratic system is in fact memoryless. Consider the PDF of

$$Y(t) = X^2(t)$$

Suppose that $X(t)$ is Gaussian; then

$$f_X(x_t) = \frac{1}{\sqrt{2\pi}\sigma_t} e^{-\frac{x_t^2}{2\sigma_t^2}} \tag{11.16}$$

Respectively denote x_t and y_t as the random variables of input and output process at time t . It can be proven that

$$f_Y(y_t) = \frac{1}{\sqrt{2\pi y_t} \sigma_t} e^{-\frac{y_t}{2\sigma_t^2}} u(y_t) \tag{11.17}$$

where $u(\cdot)$ is a step function.

11.1.1.2.3 Correlation Function

$$\begin{aligned}
 R_Y(t_1, t_2) &= E[Y(t_1)Y(t_2)] = E[g(X(t_1))g(X(t_2))] \\
 &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(X(t_1))g(X(t_2))f(x_1, x_2, t_1, t_2)dx_1 dx_2
 \end{aligned}
 \quad (11.18)$$

11.1.2 GENERAL NONLINEAR SYSTEM, VOLTERRA MODEL (VITO VOLTERRA, 1860–1940)

In Section 11.1.1, we have introduced several nonlinear systems. To have a general description of system nonlinearity, let us consider the Volterra series, which provides a general model for nonlinear analytical systems that are both time-invariable and with limit memory (Volterra 1959). The Volterra series for system analysis was originated by Wiener (1958). The series was used for an approximate analysis on radar noise in a nonlinear receiver circuit.

$$\begin{aligned}
 Y(t) &= k_0 + \int_0^{\infty} k_1(\tau_1)X(t-\tau_1)d\tau_1 + \int_0^{\infty} \int_0^{\infty} k_2(\tau_1, \tau_2)X(t-\tau_1)X(t-\tau_2)d\tau_1 d\tau_2 \\
 &+ \dots + \int_0^{\infty} \int_0^{\infty} \dots \int_0^{\infty} k_n(\tau_1, \tau_2, \dots, \tau_n)X(t-\tau_1)X(t-\tau_2) \dots X(t-\tau_n)d\tau_1 d\tau_2 \dots d\tau_n
 \end{aligned}
 \quad (11.19)$$

The output is a sum of the constant k_0 and those integrals taken from one dimension to n dimensions.

If only the first term, the constant, and second term, the convolution, exist, the system is linear with h_1 being the impulse–response function; otherwise, it is nonlinear with multiple convolutions. For example, with h_2 in Equation 11.2 is the second order convolution.

11.1.3 STRUCTURE NONLINEARITY

11.1.3.1 Deterministic Nonlinearity

11.1.3.1.1 Nonlinear Spring

11.1.3.1.1.1 Softening Spring Two types of commonly used nonlinear springs in engineering systems modeling are softening and hardening springs. Figure 11.3 illustrates a softening spring. In Chapter 10, Equation 10.108 defined one kind of spring softening mechanism using the failure strain ε_f' to denote the yielding point (d_y, f_y) . Overloading the spring can result in softening the spring's stiffness. In this instance, the stress passes the yielding point, which is commonly seen in structural damage. In Figure 11.3b, the spring is shown to be below the yielding point (d_y, f_y) .

In Figure 11.3a, f_m and x_0 are the maximum force and deformation, respectively. When the load f increases, the corresponding stiffness will decrease continuously. As a result, a certain point, for example, $0.6f_m$ and $d_{0.6}$, may be set, with an unloading stiffness k_u and an effective stiffness k_{eff} at the maximum deformation. Another

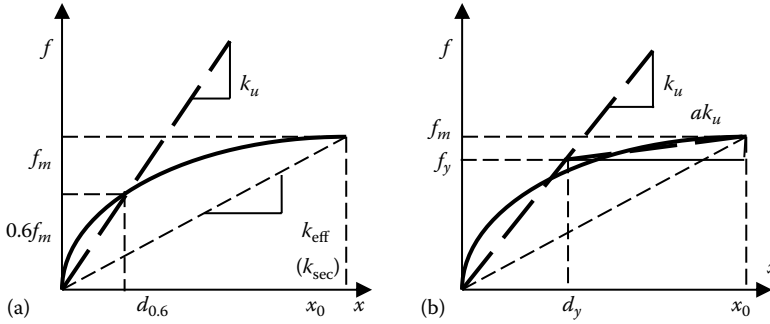


FIGURE 11.3 Softening stiffness. (a) Yielding point: $0.6 f_m$. (b) Otherwise defined yielding point.

commonly used action is shown in Figure 11.3b. At the yielding point f_y and d_y , the unloading stiffness is defined by k_u , and beyond that point, the loading stiffness is defined as $k_l = ak_u$.

Generally speaking, when the loading force is sufficiently small, the stiffness is close to linear and the force and deformation are close to proportional. (Recall Hooke's law.) In this case, the use of either stress or strain to denote the deformation is fairly equivalent, and both stress and force are commonly used. However, beyond the yielding point, a rather large deformation occurs even under a rather small force. Consequently, using strain or deformation becomes more convenient (see, for instance, FEMA 2009, Figure c12.1-1).

11.1.3.1.1.2 Estimation of Effective Stiffness In the literature and as accepted by many building codes, the effective stiffness is calculated by using the secant line shown in Figure 11.3, namely,

$$k_{\text{eff}} = k_{\text{sec}} = f_m/x_0 \quad (11.20)$$

However, it must be noted that when an effective linear system is used to represent a nonlinear vibration system, the effective stiffness should satisfy the following:

$$k_{\text{eff}} = 2E_p/x_0^2 \quad (11.21a)$$

and

$$k_{\text{eff}} = f_c/x_0 \quad (11.21b)$$

Here, E_p is the potential energy restored by the system with displacement x_0 ; f_c is the conservative force, and when the system yields and reaches a displacement of x_0 , the maximum force f_m will contain both the conservative force and the dissipative force f_d . Specifically, this is written as

$$f_m = f_c + f_d \quad (11.22)$$

As a result, the effective stiffness k_{eff} will be smaller than the secant stiffness k_{sec} .

In Chapter 6, it was shown that a vibration is typically caused by the energy exchange between potential and kinetic energies. The natural frequency of a linear system can then be obtained by letting the maximum potential energy to be equal to the maximum kinetic energy, that is,

$$\frac{kx_0^2}{2} = \frac{mv_0^2}{2} = \frac{m\omega_n^2 x_0^2}{2} \quad (11.23)$$

For a nonlinear system, the above equation should be modified as

$$\frac{k_{\text{eff}}x_0^2}{2} = \frac{m\omega_n^2 x_0^2}{2} \quad (11.24)$$

Here, k_{eff} is defined in Equations 11.21. The effective frequency ω_n can be calculated by

$$\omega_n = \sqrt{\frac{k_{\text{eff}}}{m}} \quad (11.25)$$

However, it is seen that (Liang et al. 2012)

$$\sqrt{\frac{k_{\text{eff}}}{m}} = \sqrt{\frac{f_c/x_0}{m}} < \sqrt{\frac{f_m/x_0}{m}} \quad (11.26)$$

On the right-hand side of Equation 11.26, the term f_m/x_0 is often used to estimate the effective stiffness, referred to as the *secant stiffness*, specifically $k_{\text{sec}} = f_m/x_0$. Considering the dynamic property of a nonlinear system, the secant stiffness should not be used as the effective stiffness. Following this logic, the effective stiffness should therefore be defined differently.

In the case structurally bilinear, denoted by the shadowed regions in Figure 11.4, when the system moves from 0 to x_0 , the potential energy is given by

$$E_p = 1/2 \left[k_u d_y^2 + k_d (x_0 - d_y)^2 \right] \quad (11.27)$$

Defining

$$k_{\text{eff}} = 2E_p/x_0^2 = \left[k_u d_y^2 + k_d (x_0 - d_y)^2 \right] / x_0^2 \quad (11.28)$$

and by using the displacement ductility, Equation 11.28 can be rewritten as

$$k_{\text{eff}} = \mu^{-2} [k_u + k_d(\mu - 1)^2] \quad (11.29)$$

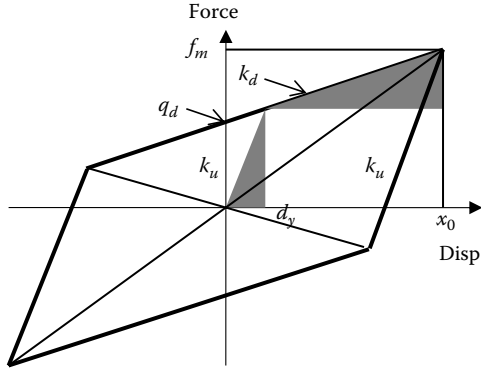


FIGURE 11.4 Maximum potential energy of a bilinear system.

where the ductility is

$$\mu = x_0/d_y \quad (11.30)$$

Further calculations yield an effective stiffness of

$$k_{\text{eff}} = \frac{1 + a(\mu - 1)^2}{\mu^2} k_u \quad (11.31)$$

In Equation 11.31, a is the ratio of the yield stiffness k_d and the unload stiffness k_u . Accordingly, the corresponding effective period is

$$T_{\text{eff}} = \sqrt{\frac{\mu^2}{1 + a(\mu - 1)^2}} T_1 = 2\pi\mu \sqrt{\frac{m}{[1 + a(\mu - 1)^2]k_u}} \quad (11.32)$$

Comparing the above effective period from Equation 11.32 to the period obtained through secant stiffness k_{sec} :

$$T'_{\text{eff}} = 2\pi \sqrt{\frac{f_m m}{x_0}} \quad (11.33)$$

By comparison, it is observed that T'_{eff} is a more concise estimate than T_{eff} .

11.1.3.1.2 Nonlinear Damping

11.1.3.1.2.1 Timoshenko Damping (Stepan P. Timoshenko, 1878–1972) By using the approach Timoshenko damping (see Liang et al. 2012), it is possible to derive the effective damping ratio of the entire bilinear system by using several different approximations of the effective stiffness. It can be shown that the steady-state

response of a linear system under sinusoidal excitation, the damping ratio, can be calculated through the following equation:

$$\zeta = \frac{E_d}{4\pi E_K} \quad (11.34)$$

where E_d and E_K are, respectively, the energy dissipated during a cycle and the maximum potential (kinetic) energy. For nonlinear damping, the damping ratio will be denoted by ζ_{eff} in the subsequent sections.

11.1.3.1.2.2 Bilinear Damping For the Timoshenko damping, initially the approach based on the lines k_u and ak_u can be used to obtain

$$\zeta_{\text{eff}} = \frac{E_d}{4\pi E_k} = \frac{2(\mu - 1)(1 - a)}{\pi\mu(1 + a\mu - a)} \quad (11.35)$$

Using Equation 11.27 of maximum potential energy, the damping ratio is written as

$$\zeta_{\text{eff}} = \frac{2q_d(x_0 - d_y)}{\pi[k_u x_0^2 + k_d(x_0 - d_y)^2 + kx_0^2]} \quad (11.36)$$

In the viewpoint of the entire system, this can be rewritten as

$$\zeta_{\text{eff}} = \frac{2q_d(x_0 - d_y)}{\pi[k_u x_0^2 + k_d(x_0 - d_y)^2]} \quad (11.37)$$

Given that the characteristic strength q_d for the bilinear system can be written as

$$q_d = (k_u - k_d)d_y \quad (11.38)$$

the damping ratio can be denoted as

$$\zeta_{\text{eff}} = \frac{E_d}{4\pi E_p} = \frac{2(\mu - 1)(1 - a)}{\pi[\mu^2 + a(\mu - 1)^2]} \quad (11.39)$$

11.1.3.1.2.3 Sublinear Damping When damping is viscous, the damping force will generally be expressed as

$$f_d(t) = c|\dot{x}|^\beta \text{syn}(\dot{x}) \quad (11.40)$$

where β is the damping exponent.

The energy dissipation by the damping force during a cycle is

$$E_d = c\omega_f^\beta x_0^{\beta+1} \int_0^{\frac{\pi}{2}} \cos^{\beta+1}(\omega_f t) d\omega_f t = c\omega_f^\beta x_0^{\beta+1} A_\beta \quad (11.41)$$

For Equation 11.41, A_β denotes

$$A_\beta = \int_0^{\frac{\pi}{2}} \cos^{\beta+1}(\omega_f t) d\omega_f t = \frac{2\sqrt{\pi}\Gamma\left(\frac{\beta+2}{2}\right)}{\Gamma\left(\frac{\beta+3}{2}\right)} \quad (11.42)$$

Through the use of Equation 11.33, the damping ratio can become

$$\zeta_{\text{eff}} = \frac{cx_o^{\beta-1}\omega_{\text{eff}}^\alpha A_\beta}{2\pi m\omega_{\text{eff}}^2} = \frac{cx_o^{\beta-1}\omega_{\text{eff}}^\alpha A_\beta}{2\pi k_{\text{eff}}} \quad (11.43)$$

When $\beta = 0$, the nonlinear viscous damping will be reduced to that of dry-friction damping. For this instance,

$$f_d(t) = c \sin(\dot{x}) \quad (11.44)$$

and

$$c = \mu N \quad (11.45)$$

As mentioned above, friction damping can be modeled as a special case of bilinear damping.

In the event $\beta = 1$, the viscous damping will reduce to that of linear damping.

11.1.3.1.2.4 Alternative Damping An additional way to calculate the effective damping ratio is through the force-based or alternative damping. Namely, this is achieved through the equation

$$\zeta_{\text{eff}} = \frac{f_d}{2f_m} \quad (11.46)$$

where f_d and f_m are the amplitude of the damping and maximum forces as previously defined.

In the following example, we consider the case of bilinear damping by using the alternative approach. To clearly define the dissipative and restoring forces in a bilinear system can be complex. A bilinear system may possibly have one of the

subsequent cases—case 1: friction dampers installed in a linear structure; case 2: bilinear dampers installed in a linear structure; case 3: friction dampers installed in a bilinear structure; and case 4: bilinear dampers installed in a bilinear structure. For each case, the dissipative force f_d may be different due to the nonlinearity of the total system.

In order to simplify the study, assume that the damping force is $f_d = q_d = f_y - a(k_u d_y)$, with an equivalent restoring force of

$$f_m = (k_u - k_d)d_y + k_d x_0 \quad (11.47)$$

and thus yielding a damping ratio of

$$\zeta_{\text{eff}} = \frac{f_d}{2f_m} = \frac{(1-a)}{2[1+a(\mu-1)]} \quad (11.48)$$

It is noted that, with different formulas for the dissipative force, the calculated damping ratio will be slightly varied.

11.1.3.2 Random Nonlinearity

11.1.3.2.1 Random Force and Displacement

In the above discussion of nonlinear stiffness and damping, two assumptions existed, the first one being that the maximum force, f_m , and displacement, x_0 , are fixed values, although in many cases, the maximum force and displacement will be a random process.

We now consider the probability of the maximum deformation, and note that the probability of maximum force will have similar results. (Referring back to Section 5.2.) This is understood as a problem of the distributions of extrema.

11.1.3.2.1.1 Joint PDF of Displacement and Velocity Consider the correlation between the displacement X and the velocity $\dot{X}$. Both are assumed to be stationary and narrow-banded with a zero mean. Among these three assumptions, the case of zero mean is the most realistic, while the case of narrow-banded is a sensible approximation. This will be further discussed in Section 11.1.3.2.1.7. The assumption of stationary is a rather rough estimation. However, a nonstationary process can be modified through separation of $a(t)$ and $U(t)$ (see Equation 9.108).

For a joint PDF,

$$E[X(t)\dot{X}(t)] = \left. \frac{dR_X(\tau)}{d\tau} \right|_{\tau=0} \quad (11.49)$$

Substituting

$$R_X(\tau) = \int_{-\infty}^{\infty} S_X(\omega) e^{i\omega\tau} d\omega \quad (11.50)$$

will yield

$$E[X(t)\dot{X}(t)] = \frac{d}{d\tau} \left[\int_{-\infty}^{\infty} S_X(\omega) e^{j\omega\tau} d\omega \right] \Big|_{\tau=0} = j\omega \int_{-\infty}^{\infty} S_X(\omega) d\omega = 0 \quad (11.51)$$

The integral for S_{XX} is zero given that S_{XX} is an even function. By further assuming that the displacement X and the velocity $\dot{X}$ be Gaussian, a joint PDF given by

$$f_{X\dot{X}}(x, \dot{x}) = \frac{1}{2\pi\sigma_X\sigma_{\dot{X}}} \exp \left[-\frac{1}{2} \left(\frac{x^2}{\sigma_X^2} + \frac{\dot{x}^2}{\sigma_{\dot{X}}^2} \right) \right] \quad (11.52)$$

is resulted. Equation 11.52 can be rewritten as

$$f_{X\dot{X}}(x, \dot{x}) = \frac{1}{\sqrt{2\pi}\sigma_X} \exp \left(-\frac{1}{2} \frac{x^2}{\sigma_X^2} \right) \frac{1}{\sqrt{2\pi}\sigma_{\dot{X}}} \exp \left(-\frac{1}{2} \frac{\dot{x}^2}{\sigma_{\dot{X}}^2} \right) = f_X(x) f_{\dot{X}}(\dot{x}) \quad (11.53)$$

11.1.3.2.1.2 Level and Zero Up Crossing Now, we consider the case where the rate of the displacement is larger than level a . From Equation 5.24, we can obtain

$$v_{a^+} = \frac{1}{2\pi} \frac{\sigma_{\dot{X}}}{\sigma_X} e^{-\frac{1}{2} \frac{a^2}{\sigma_X^2}} \quad (11.54)$$

When $a = 0$, this will result in a zero up-crossing rate. Reference Equation 5.26 for any additional explanations.

$$v_{0^+} = \frac{1}{2\pi} \frac{\sigma_{\dot{X}}}{\sigma_X} \quad (11.55)$$

11.1.3.2.1.3 Distribution of Peak Value Furthermore, again referring back to Chapter 5, the PDF of the peak value a of displacements is given by

$$f_A(a) = \frac{a}{\sigma_X^2} \exp \left[-\frac{1}{2} \left(\frac{a^2}{\sigma_X^2} \right) \right], \quad a > 0 \quad (11.56)$$

For additional explanation, see Equation 5.100. Figure 11.6 plots an example of the PDF.

11.1.3.2.1.4 Proportional Constant In order to calculate the above-mentioned effective stiffness and damping, the maximum displacement and therefore maximum force must be known. This is true by using either the energy method or the secant

stiffness. However, for random processes, the maximum displacement will only be reached a few times, and in most cases, the magnitude of displacements will be smaller than the maximum value. To estimate the effective stiffness and damping more realistically, a specific displacement d_p , which is smaller than the maximum value, must be found. Using a proportional coefficient p_c , the displacement d_p can be written as

$$d_p = p_c x_0 \quad (11.57)$$

To simplify engineering applications, a fixed constant p_c is more appropriate. Next, a proper value of this constant will be considered.

Using a simple bilinear model of nonlinear stiffness the elastic perfect plastic model, we can find the point d_p for this particular model. The nonlinear force–displacement relationship is shown in Figure 11.5, where f_m is the maximum force and d_y is the yielding deformation.

The above has shown that the nonlinear stiffness, denoted by k_n , is a function of the peak displacement a .

When the peak displacement a is less than d_y , the system remains linear and the stiffness k_n is equal to k_u . As the peak displacement reaches d_y , the total conservative energy becomes

$$E_p = 1/2 k_u d_y^2 = 1/2 f_m d_y \quad (11.58)$$

When the peak displacement is larger than d_y , then referencing Equation 11.28, the stiffness k_n is

$$k_n = 2E_p/a^2 = f_m d_y/a^2 \quad (11.59)$$

In between zero and the maximum peak displacements x_0 , the average stiffness k_{ave} is given by

$$k_{ave} = \int_0^{x_0} k_n f_A(a) da = \int_0^{d_y} k_u \frac{a}{\sigma_X^2} e^{-\frac{1}{2} \frac{a^2}{\sigma_X^2}} da + \int_{d_y}^{x_0} \frac{f_m d_y}{a^2} \frac{a}{\sigma_X^2} e^{-\frac{1}{2} \frac{a^2}{\sigma_X^2}} da \quad (11.60)$$

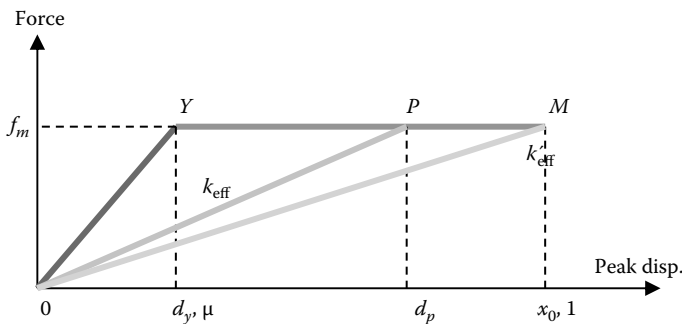


FIGURE 11.5 Elastic perfect plastic deformation.

Use the average stiffness to represent the effective stiffness:

$$k_{\text{eff}} = k_{\text{ave}} = \int_0^{d_y} \frac{f_m}{d_y} \frac{a}{\sigma_X^2} e^{-\frac{1}{2} \frac{a^2}{\sigma_X^2}} da + \int_{d_y}^{x_0} \frac{f_m d_y}{a \sigma_X^2} e^{-\frac{1}{2} \frac{a^2}{\sigma_X^2}} da \quad (11.61)$$

Figure 11.5 illustrates the relationship

$$k_u d_y = k_{\text{eff}} d_p \quad (11.62)$$

In this instance, the displacement d_p can be written as

$$d_p = k_u d_y / k_{\text{eff}} = \frac{f_m}{d_y} d_y / k_{\text{ave}} = \frac{f_m}{k_{\text{ave}}} \quad (11.63)$$

Furthermore, the proportional constant p_c can be determined by rearranging Equation 11.57 to

$$p_c = d_p / x_0 \quad (11.64)$$

Without loss of generality, the maximum possible value of the peak displacement can be normalized to unity so that

$$p_c = d_p \quad (11.65)$$

It is noted that the distribution of the peak displacement a is from zero to infinity. Allowing

$$x_0 = 1.0 \quad (11.66)$$

will result in errors ε in computing the probability of a being beyond 1 (the normalized displacement), denoted by

$$\varepsilon = \int_1^{\infty} \frac{a}{\sigma_X^2} e^{-\frac{1}{2} \frac{a^2}{\sigma_X^2}} da \quad (11.67)$$

Seemingly, the error is a function of the variance σ_X^2 of the process $X(t)$.

11.1.3.2.1.5 Allowed Uncertainty In using Equation 11.64, Equation 11.61 must first be evaluated, which contains an unknown variable d_y . It is found that the yielding deformation d_y is the property of the structure, which is virtually independent of the displacement process $X(t)$. Given that Equation 11.61 is a generic formula, the value of parameter d_y is *uncertain*. Equation 11.61 can be evaluated through statistical study of the random process $X(t)$ only.

In engineering applications, it is uncommon to thoroughly investigate the possible distribution of certain unknown variables, such as yielding deformations. Rather,

such unknown variables are treated as uncertain parameters. To determine how much error is present, the maximum possible value of the uncertain variables must be determined. In the case of unknown yielding deformations, when the value of d_y is small, less error will exist. Conversely, when the value of d_y is large, more error will exist.

We now consider the maximum allowable d_y when Equation 11.60 is used.

Refer again to Figure 11.5. Before the yielding point, the force is proportional to the displacement. Consequently, the distribution density function of the force vs. the displacement, denoted by $f_F(a)$, is exactly equal to the PDF of the displacement. Explicitly, this can be written as

$$f_F(a) = \frac{a}{\sigma_X^2} \exp \left[-\frac{1}{2} \left(\frac{a^2}{\sigma_X^2} \right) \right], \quad 0 \leq a < d_y \quad (11.68)$$

After yielding, the force f_m will remain constant, while the displacement will vary from d_y to 1. This is denoted by

$$f_F(a) = \frac{d_y}{\sigma_X^2} \exp \left[-\frac{1}{2} \left(\frac{d_y^2}{\sigma_X^2} \right) \right] \left(\frac{a}{1-d_y} \right), \quad d_y \leq a < 1 \quad (11.69)$$

As the peak displacement a varies from 0 to 1, given that x_0 is normalized to be 1, the cumulative distribution function (CDF) of the force will become unity, which is explicitly written as

$$\int_0^{d_y} \frac{a}{\sigma_X^2} e^{-\frac{1}{2} \frac{a^2}{\sigma_X^2}} da + \int_{d_y}^1 \frac{d_y}{\sigma_X^2} e^{-\frac{1}{2} \frac{d_y^2}{\sigma_X^2}} \left(\frac{a}{1-d_y} \right) da = 1 \quad (11.70)$$

Evaluating Equation 11.70 will result in

$$-e^{-\frac{1}{2} \frac{d_y^2}{\sigma_X^2}} + 1 + \frac{d_y}{\sigma_X^2} e^{-\frac{1}{2} \frac{d_y^2}{\sigma_X^2}} (1-d_y) = 1 \quad (11.71)$$

or

$$\sigma_X^2 = d_y(1-d_y) \quad (11.72)$$

Suppose a displacement of $X(t)$ is given and the variance σ_X^2 is fixed. Then, the “allowed value” of the yielding displacement d_y is given by

$$d_y = \frac{1 \pm \sqrt{1-4\sigma_X^2}}{2} \quad (11.73)$$

Note that, with x_0 normalized to be unity, the displacement ductility μ is

$$\mu = 1/d_y \quad (11.74)$$

At this instance, consider the error of letting $x_0 = 1$.

For example, suppose that

$$\sigma_x = 1/3 \quad (11.75)$$

Then

$$\int_0^1 f_A(a) da = \int_0^1 \frac{a}{\left(\frac{1}{3}\right)^2} \exp \left\{ -\frac{1}{2} \left(\frac{a^2}{\left(\frac{1}{3}\right)^2} \right) \right\} da \approx 99\%$$

or

$$\varepsilon = 1 - 99\% = 1\%$$

When considering the negative square root, Table 11.1 can be used to show the corresponding errors.

From Table 11.1, it is established that, when the standard deviation $\sigma_x = 0.3$, calculated through Equation 11.67, the error $\varepsilon = 0.4\%$, which is sufficiently small. However, based on Equation 11.72, the “allowed yielding displacement” is also small at a value of 0.1, with a ductility of 10. It is seen that when the allowed yielding displacement becomes increasingly larger, the error ε will also be larger. As an example, when $d_y = 0.2$, the error will become 4.4%.

Next we calculate the averaged effective stiffness. For example, when $\sigma_x = 0.4$ and $d_y = 0.2$, the first term of Equation 11.61 yields

$$\frac{f_m}{d_y} \int_0^{d_y} \frac{a}{\sigma_x^2} e^{-\frac{1}{2} \frac{a^2}{\sigma_x^2}} da = 0.59 f_m$$

TABLE 11.1
Errors versus Yielding Displacements

d_y	0.100	0.109	0.118	0.127	0.138	0.149	0.160	0.173	0.186	0.2000
μ	10.0	9.21	8.50	7.85	7.27	6.74	6.25	5.80	5.38	5
σ_x	0.30	0.31	0.32	0.33	0.34	0.36	0.37	0.38	0.39	0.4
ε	0.004	0.006	0.008	0.011	0.015	0.019	0.024	0.030	0.034	0.044

The second term will yield

$$\int_{d_y}^1 \frac{f_m}{a^2} \frac{a}{\sigma_X^2} e^{-\frac{1}{2} \frac{a^2}{\sigma_X^2}} da = 0.95 f_m$$

and the average or effective stiffness is given by

$$k_{\text{eff}} = k_{\text{ave}} = 0.59 f_m + 0.95 f_m = 1.54 f_m$$

As a result, the displacement is

$$d_p = \frac{f_m}{k_{\text{ave}}} = 0.65$$

Moreover, given that $x_0 = 1$,

$$p_c = 0.65 \quad (11.76)$$

Similarly, other values of d_y and σ_X can be calculated. Table 11.2 lists a number of computed results. From the table, as d_y varies from 0.13 to 0.20, p_c will vary from 0.6 to 0.65.

11.1.3.2.1.6 Simplified Approach Additionally from Tables 11.1 and 11.2, it is seen that for the simplified bilinear model, when the value of d_y is small, very large errors do not occur. Thus, a more simplified approach can be used by assuming that the equivalent displacement of d_p can be computed at

$$d_p = \mu_A + d_y \quad (11.77)$$

An additional approximation is done by using the sum of the mean plus 70% standard deviation represented by

$$d_p = \mu_A + 0.7\sigma_A \quad (11.78)$$

Here, the mean μ_A and the standard deviation σ_A will be determined as follows. Table 11.3 gives a comparison of the calculated results.

TABLE 11.2

Yielding Displacements d_y versus Proportional Constant p_c

d_y	0.13	0.14	0.15	0.16	0.17	0.18	0.19	0.20
σ_X	0.34	0.35	0.36	0.37	0.38	0.38	0.39	0.4
p_c	0.60	0.61	0.61	0.62	0.63	0.64	0.64	0.65

TABLE 11.3
Equivalent d_p

d_y	0.13	0.14	0.15	0.16	0.17	0.18	0.19	0.20
μ_A	0.42	0.43	0.45	0.46	0.37	0.48	0.49	0.50
σ_A	0.22	0.23	0.23	0.24	0.25	0.25	0.26	0.26
p_c	0.60	0.61	0.61	0.62	0.63	0.64	0.64	0.65
$\mu_A + d_y$	0.55	0.57	0.60	0.62	0.64	0.66	0.68	0.70
$\mu_A + 0.7\sigma_A$	0.58	0.59	0.61	0.63	0.64	0.66	0.67	0.68

Observe from Figure 11.6 that the peak of the PDF (often referred to as the distribution *mode*) of the Rayleigh distribution is at

$$a = \sigma_X \tag{11.79}$$

while the mean value is at

$$\mu_A = \sqrt{\frac{\pi}{2}} \sigma_X \tag{11.80}$$

Furthermore, the standard deviation is

$$\sigma_A = \sqrt{\frac{4 - \pi}{2}} \sigma_X \tag{11.81}$$

Since $a = 1$ denotes a normalized maximum peak, Equation 11.79 implies that the yielding displacement occurs when the corresponding PDF is at its maximum value.

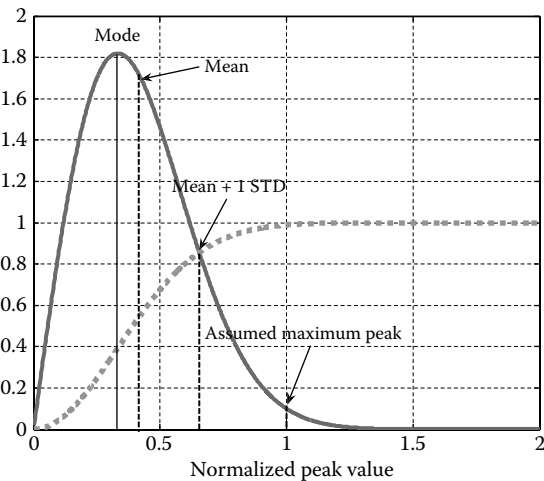


FIGURE 11.6 Rayleigh distribution.

11.1.3.2.1.7 Penzien Constant (Joseph Penzien, 1924–2011) When low cycle fatigue occurs, the unloading stiffness k_u may be reduced. When this occurs, the displacement at k_{eff} will be significant, resulting in a larger proportional constant.

Furthermore, for a displacement process that is not a narrow-band Gaussian, Equation 11.56 may not be sufficient to accurately describe the distribution of the peak value. For different models, different accuracy assignments and values of yielding points will result in varying values of the proportional parameter p_c . The proportional parameter p_c will also vary due to the reduction of the unloading stiffness. For simplification of engineering applications, however, Equation 11.76 is often used, namely,

$$p_c = 0.65$$

Professor J. Penzien (1927–2011) suggested that, during a random process, the chance of reaching x_0 is minimal. To provide a more realistic estimation of the effective stiffness and damping based on observation, 0.65 times x_0 should be used rather than 0.65 alone. The proportional constant 0.65 is accordingly referred to as the *Penzien constant*.

11.1.3.2.1.8 Effective Stiffness and Damping Estimation under Random Process In using the Penzien constant, the values x_0 are replaced by $(p_c x_0)$ or $(0.65x_0)$ in the aforementioned formulas to calculate the effective stiffness and damping, as well as the effective period, among others.

For example, refer to Figure 11.5 for the values of k_{eff} and k'_{eff} . To better estimate the effective stiffness substitute

$$k_{\text{eff}} = f_m / 0.65x_0 \quad (11.82)$$

in the equation

$$k'_{\text{eff}} = f_m / x_0 \quad (11.83)$$

Additionally, the equation

$$\zeta_{\text{eff}} = \frac{c(p_c x_0)^{\beta-1} \omega_{\text{eff}}^\alpha A_\beta}{2\pi m \omega_{\text{eff}}^2} = \frac{c(0.65x_0)^{\beta-1} \omega_{\text{eff}}^\alpha A_\beta}{2\pi A_{\text{eff}}} \quad (11.84)$$

is used to replace Equation 11.43 for a better estimation of effective damping.

11.1.3.2.2 Variable Path of Loading/Unloading

The above descriptions are for the cases when the loading and unloading paths are deterministic. In actuality, due to the stiffness deteriorations, as described by Equation 10.108, resulted by low cycle fatigue, these paths may be altered. In this case, if the loading process is random, then a memoryless random process will not exist. This is to say that previous stiffness damages in a system will affect the subsequent force–deformation relationship.

For Figure 11.7, suppose that (f_1, d_1) is the first yielding point and the corresponding effective stiffness $k_{\text{eff}1}$ is calculated based on the initial unloading stiffness k_{u1} .

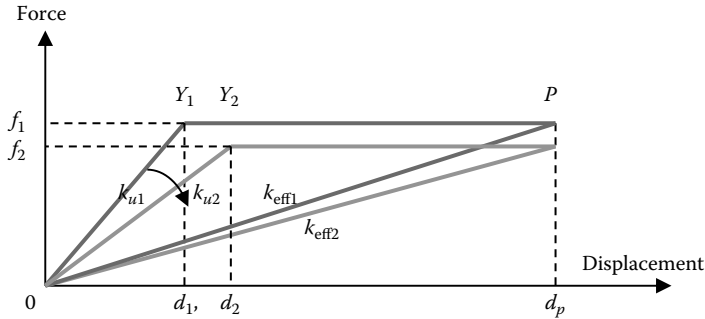


FIGURE 11.7 Reduction of effective stiffness.

The second yielding point is then denoted by (f_2, d_2) . If the stiffness is reduced (see Equation 10.108 for further reference), the corresponding effective stiffness will be denoted by k_{eff2} such that

$$k_{eff2} < k_{eff1} \quad (11.85)$$

Seemingly, the value of k_{eff2} will depend on not only the value of d_p but also by the degree of which the unloading stiffness k_{u2} is reduced. In general, the computations of k_{eff2} , as well as similar computations, are rather complex. The Monte Carlo simulation provides a practical method for carrying out this estimation. This will be briefly discussed in Section 11.3.

11.2 NONLINEAR RANDOM VIBRATIONS

In Chapters 7 and 8, we discussed the vibration problems of linear systems. Generally speaking, for time-varying excitations and responses, *stochastic differential equation* (SDE) will be used. The SDE is a differential equation in which one or more of the terms are a random process, so that their solution is also a random process. SDEs are used to model diverse phenomena, but not limited to typical vibration problems. In 1905, Einstein explanted that the Brownian motion is due to statistical fluctuations, which leads to a SDE and the derivative of the Wiener process (Bichteler 1998). Besides white noise excitations, we can have many other types of random processes as excitations. For more detailed information, readers may consult Monohar (1995), for instance.

11.2.1 GENERAL CONCEPT OF NONLINEAR VIBRATION

Relationships between two sets of variables, particularly between the input and the output of random processes, can be linear. Otherwise, the system is said to be nonlinear.

For linear systems, the transfer function has been the key link to deal with a linear input–output process. However, for a general nonlinear system, a transfer function does not exist. Therefore, an alternative approach to describe the relationship is needed.

While a linear system can be described through the classical method, nonlinear systems can have many different characteristics, most of which do not have

closed-form solutions. In the following, general approaches to study nonlinear random vibrations will be discussed briefly.

11.2.1.1 The Phase Plane

The phase plane plots velocity vs. displacement, which is a common tool to provide a general approach to study the nature of specific nonlinear vibrations, including system stability, bound of peak values, and others.

11.2.1.1.1 Energy Dissipation

Figure 11.8 shows the plot of viscous damping force vs. a deterministic displacement, where the smallest loop is obtained when $\beta = 1$ and the smaller the value of β is, the larger energy dissipation loop will be resulted. It is known that this form of plot directly indicates the energy dissipation of a system with steady-state responses 0.1 m. Because viscous damping forces relate directly to velocities, Figure 11.8 can be seen as several generalized phase plans, among which only the linear velocity vs. displacement has an exact ellipse curve.

As shown in Figure 11.9, given the same displacement for general viscous damping, the smaller the damping exponent β is, the larger the amount of energy that can be dissipated. However, this does not necessarily mean that a system with low β will dissipate more energy. As an example, consider a system with a mass of $m = 1$, a damping coefficient of $c = 1$, and a stiffness of $k = 50$, in which the system is excited by a deterministic sinusoidal force with a driving frequency of $\omega = 3$. Figure 11.10 shows the phase plane of each case when $\beta = 0.1$ and $\beta = 1.0$. This explains that the area enclosed in the displacement–velocity curve with $\beta = 1.0$ is larger than that in the case with $\beta = 0.1$.

11.2.1.1.2 Bounds

Figure 11.10 shows the phase plane for the two systems in Figures 11.8 and 11.9. Clearly, the curves will provide information for the bounds of the velocities and displacements.

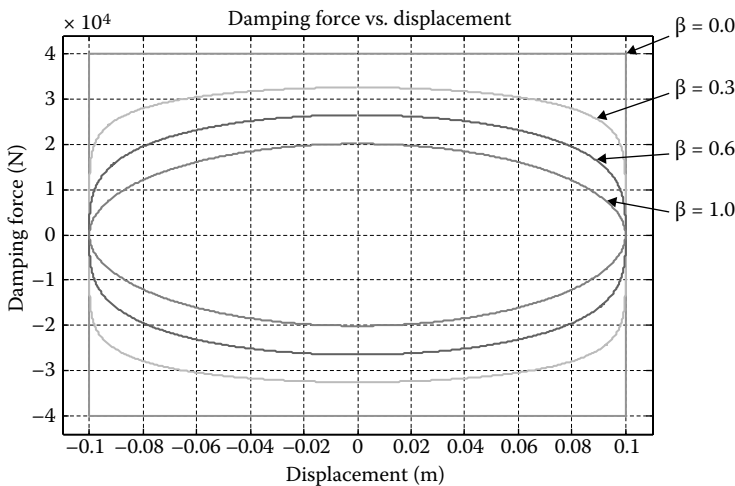


FIGURE 11.8 Viscous damping forces and generalized phase plans.

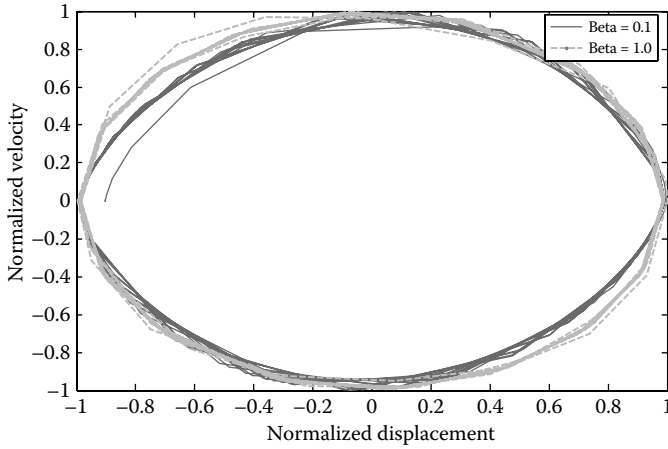


FIGURE 11.9 Phase plane.

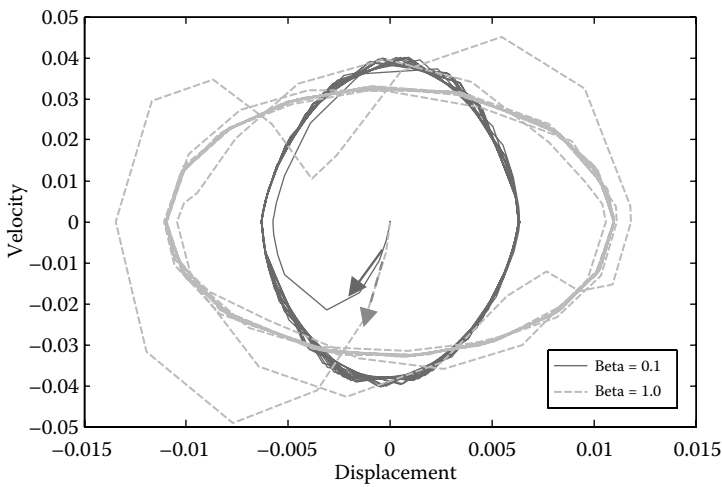


FIGURE 11.10 Simulated phase plane.

11.2.1.2 Example of Nonlinear Vibration with Closed-Form Solution

In order to see the fundamental difference between linear and nonlinear vibration, consider an example of the Duffing equation as described in the following.

11.2.1.2.1 Duffing Equation (Georg Duffing, 1861–1944)

The Duffing equation is used to describe a typical nonlinear vibration system such that (see Lin 1967)

$$m\ddot{x} + c\dot{x} + kx \pm \mu x^3 = f(t) \quad (11.86)$$

Here m , c , k , and μ are all constant. Let us first consider the undamped case with sinusoidal excitation given by

$$\ddot{x} + \omega_n^2 x \pm \alpha x^3 = p \cos(\omega t) \quad (11.87)$$

where $\omega_n = (k/m)^{1/2}$, $\alpha = \mu/m$ and p is a constant.

Equation 11.87 can be solved by the method of iteration, and the first solution is assumed to be

$$x^{(0)} = A \cos(\omega t) \quad (11.88)$$

Substitution of Equation 11.88 into Equation 11.87 and the integration of the second derivative expression twice yields the next iterative approximation of

$$x^{(1)} = \frac{1}{\omega^2} (\omega_n^2 A \pm 0.75 \alpha A^3 - p) \cos(\omega t) \pm \dots \quad (11.89)$$

Since the amplitude A in Equation 11.89 must be equal to that of Equation 11.88, ignoring the higher order terms will yield

$$\frac{0.75 \alpha A^3}{\omega_n^2} = \left(1 - \frac{\omega^2}{\omega_n^2} \right) A - \frac{p}{\omega_n^2} \quad (11.90)$$

11.2.1.2.2 Numerical Simulations

When the input is random, numerical simulations are often needed. Figure 11.11 illustrates a Simulink diagram for the Duffing equation.

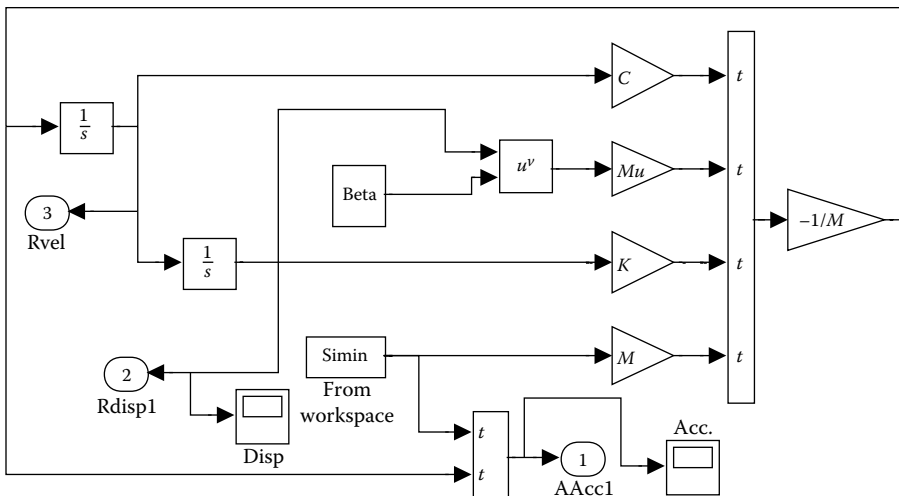


FIGURE 11.11 Simulink model (Beta = 3).

Figure 11.12 shows the phase plane of the above-mentioned Duffing equation, where $m = 1$, $k = 50$, $c = 0$, and $\mu = 2$. Figure 11.13 illustrates the same phase plane but for the unstable case when $c = -0.5$. Both phase planes are under sinusoidal excitation with values of $p = 1$ and $\omega = 3$.

Figure 11.14 demonstrates the phase plane of the above-mentioned Duffing equation under a random excitation, with Figure 11.15 showing the time history of the displacement. From Figure 11.15, the vibration is clearly seen as nonlinear, for the input has ω only and is deterministic but the response contains more frequency components. The time history of the response is comparable to that of a narrow-band process.

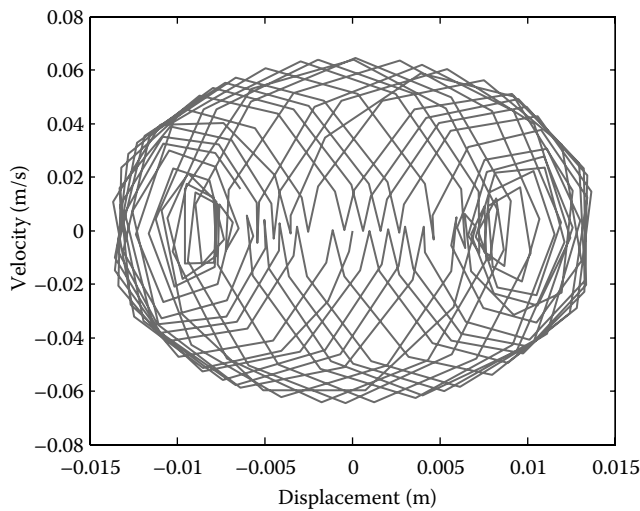


FIGURE 11.12 Duffing responses.

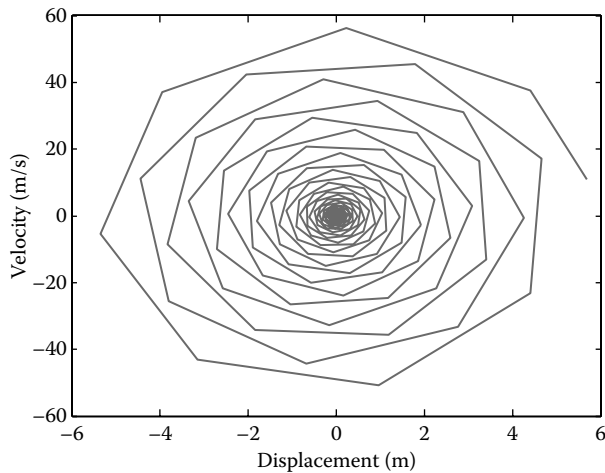


FIGURE 11.13 Unstable system.

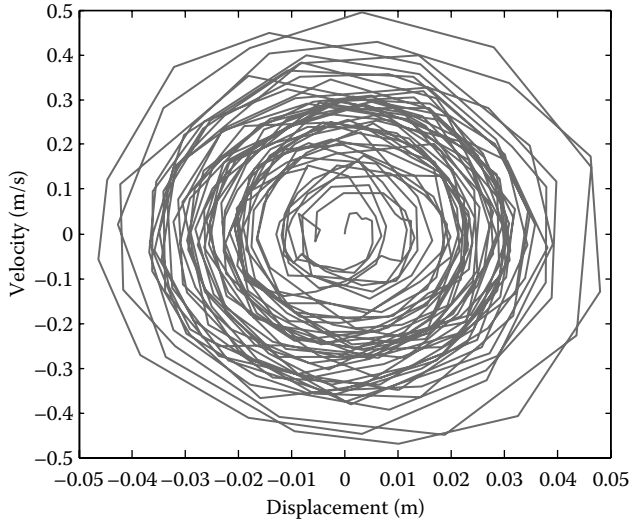


FIGURE 11.14 Random Duffing response.

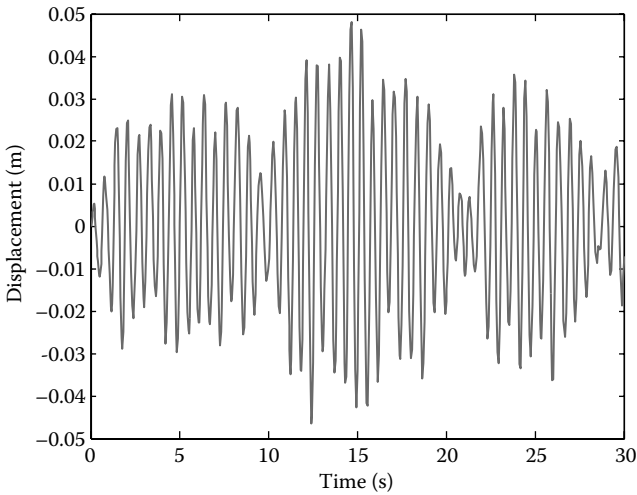


FIGURE 11.15 Duffing displacement.

Although the Duffing equation only depicts one kind of nonlinear vibration, from this example, the following can be approximately stated:

1. Nonlinear vibration is also a repetitive motion. However, while linear vibration will have a single equilibrium point, nonlinear vibration may have multiple points.
2. Similarly, nonlinear vibration has its own motion frequencies. However, the frequency will vary due to several factors, for instance, the input levels.

3. Nonlinear vibration also has the capability to dissipate energy. When the damping effective is sufficiently small, for example, $\zeta_{\text{eff}} < 0.3$, the response time history is similar to that of a narrow-band process. Nevertheless, the damping effect will not be fixed. In certain cases, nonlinear vibration has the potential to become unstable or chaotic.

11.2.1.2.3 Other Examples

Beside the Duffing equation, we provide several other nonlinear engineering vibration problems without detailed discussion.

11.2.1.2.3.1 Coulomb Damping (Charles-Augustin de Coulomb, 1736–1806)

Seen in Equation 11.40 when the damping exponent $\beta = 0$, we have *dry-friction damping*, which is also called *Coulomb damping*. A single degree-of-freedom (SDOF) system with dry-friction damping can be modeled as (see Ahmadi and Su 1987)

$$m\ddot{x} + \mu mg \operatorname{syn}(\dot{x}) + kx = f(t) \quad (11.91)$$

where μ is the friction coefficient.

11.2.1.2.3.2 Nonlinear Viscous and Sublinear Damping

In Equation 11.40, when damping exponent $\beta \neq 1$, we have nonlinear viscous damping, and especially, when $\beta < 1$, we have the sublinear damping. An SDOF system with nonlinear viscous damping can be modeled as

$$m\ddot{x} + c|\dot{x}|^\beta \operatorname{syn}(\dot{x}) + kx = f(t) \quad (11.92)$$

11.2.1.2.3.3 Bouc Oscillator

The bilinear stiffness shown in Figure 11.4 can be further generalized as a Bouc-Wen model (Wen 1989); an SDOF system with such nonlinear stiffness can be modeled as

$$\begin{cases} m\ddot{x} + c\dot{x} + \alpha kx + (1 - \beta)kz = f(t) \\ \dot{z} = \frac{1}{\eta} \left\{ A\dot{x} - \nu \left[\beta |\dot{x}| |z|^{n-1} z - \gamma \dot{x} |z|^n \right] \right\} \end{cases} \quad (11.93)$$

where α , η , ν , β , γ are parameters that describe the shape of the nonlinear stiffness; A is the amplitude and n is the number of cycles.

11.2.1.2.3.4 Rectangular Rocking

A rectangular rocking block is a vibration-impact system, which can be modeled as

$$\begin{cases} I\ddot{\theta} + WR \sin[\alpha \operatorname{syn}(\theta) - \theta][1 + f(t)] + WR \cos[\alpha \operatorname{syn}(\theta) - \theta]g(t) = 0 \\ \dot{\theta}(t_+^*) = c\dot{\theta}(t_-^*), \quad t^* : \dot{\theta}(t^*) = 0 \end{cases} \quad (11.94)$$

where θ is the rocking angle, W is the weight of the block, R and α are respectively the distance and angle of the center of gravity to a corner of the block, $f(t)$ and $g(t)$ are respectively excitations relating the horizontal and vertical accelerations (see Iyenger and Dash 1978).

11.2.1.2.3.5 Van der Pol Oscillator (Balthasar van der Pol, 1889–1959) The Van der Pol oscillator can be used to model chimney vibration due to a cross-wind (see Vickery and Basu 1983). That is,

$$m\ddot{x} + c\dot{x} + kx + \alpha\dot{x} + \beta\dot{x}^3 = f(t) \quad (11.95)$$

where m , c , k , α , and β are scalars.

11.2.1.2.3.6 Modified Morison Problem The modified Morison model can be used to describe the vibration of an offshore structure due to dynamic sea wave motions (see Taylor and Rajagopalan 1983). That is,

$$\begin{cases} \mathbf{M}\ddot{\mathbf{x}} + \mathbf{C}\dot{\mathbf{x}} + \mathbf{k}\mathbf{x} = \mathbf{f}(t) \\ \mathbf{f}_i(t) = \frac{1}{2}C_D\rho A_i[u_i(t) - \dot{x}_i(t)]|u_i(t) - \dot{x}_i(t)|^2 + C_M\rho V_i\dot{u}_i(t) - C_M\rho V_i\ddot{x}_i(t) \end{cases} \quad (11.96)$$

Here, C_D and C_M are the Morison drag and inertia coefficients; ρ is the fluid density; A_i and V_i are the projected area and volume associated with the i th node; $\dot{x}_i$, $\ddot{x}_i$, u_i , and $\dot{u}_i$, are respectively the structural and wave velocity and acceleration at node i .

11.2.1.3 System with Nonlinear Damping Only

When a dynamic system has linear stiffness with nonlinear damping, it will result in nonlinear vibration. However, in this case, we may make the following approximation:

$$\omega_i \approx \omega_{ni}, \quad i = 1, \dots, S \quad (11.97)$$

and

$$\zeta_i = \frac{E_{d_i}}{4\pi E_{k_i}} \quad i = 1, \dots, S \quad (11.98)$$

11.2.1.4 System with Nonlinear Spring

When a dynamic system has nonlinear stiffness, with or without nonlinear damping, it will result in nonlinear vibration. The effective natural frequency will vary such that

$$\omega_i \neq \omega_{ni} \quad (11.99)$$

11.2.2 MARKOV VECTOR

In the above section, we introduced nonlinear vibration with deterministic input. Now let us consider excitations as random processes. Specifically, a system due to a special input of Gaussian white noise process will have the response of a *diffusion process*. A diffusion process is the solution to an SDE, which is the net movement of a substance from a high concentration region to a region of low concentration. The aforementioned Brownian motion is only a good example of a diffusion process. In general, a diffusion process is a Markov process with continuous sample paths. (For more details, the Fokker–Planck–Kolmogorov [FPK] equation or Kolmogorov forward equation [Equation 11.105] may be considered.) Horsthemke and Lefever (1984) and Rishen (1989) describe these equations in detailed fashion.

11.2.2.1 Itô Diffusion and Kolmogorov Equations

An *Itô diffusion* (Kiyoshi Itô, 1915–2008) is a solution to a specific type of SDE. The associated transitional PDF will satisfy *Kolmogorov equations*. The *Itô type* equation of motion can be given as follows:

$$\begin{cases} d\mathbf{x}(t) = f\{\mathbf{x}(t), t\}dt + g\{\mathbf{x}(t), t\}d\mathbf{B}(t) \\ \mathbf{x}(t_0) = \mathbf{x}_0 \end{cases} \quad (11.100)$$

where $\mathbf{x}(t)$ is an $n \times 1$ response vector; $\mathbf{x}_0$ is an initial condition vector, independent of $\mathbf{B}(t)$; $f\{\mathbf{x}(t), t\} = [f_{ij}]$ and $g\{\mathbf{x}(t), t\} = [g_{ij}]$ are respectively $n \times n$ and $n \times m$ matrices. $\mathbf{B}(t)$ is an $m \times 1$ vector of the Brownian motion process, with the following properties:

$$E[B_j(t + \Delta t) - B_j(t)] = 0 \quad (11.101)$$

$$E[\Delta_i(t)\Delta_j(t)] = \delta_{ij}\Delta t \quad (11.102)$$

where δ_{ij} is a Kroneker delta function and $\Delta_j(t) = B_j(t + \Delta t) - B_j(t)$.

Equation 11.100 can be used for both multi-degree of freedom (MDOF) linear and nonlinear discrete systems. The excitations can be both stationary and nonstationary, both white noise and nonwhite noise, and both external and parameter excitations. The initial conditions can be random as well.

The transitional PDFs that satisfy Kolmogorov equations, denoted by $p(\mathbf{x}, t | \mathbf{x}_0, t_0)$, includes the following cases, as long as the response process is Markovian.

11.2.2.1.1 Chapman–Kolmogorov–Smoluckowski Integral (CKS) Equations

(Sydney Chapman, 1888–1970; Marian Smoluchowski, 1872–1917)

The CKS equation is an identity relating the joint probability distributions of different sets of coordinates on a stochastic process, which is given by

$$p(\mathbf{x}, t | \mathbf{x}_0, t_0) = \int_{-\infty}^{\infty} p(\mathbf{x}, t | \mathbf{z}, \tau) p(\mathbf{z}, \tau | \mathbf{x}_0, t_0) d\mathbf{z} \quad (11.103)$$

11.2.2.1.2 Fokker–Planck–Kolmogorov (FPK) Equation (Kolmogorov Forward Equation)

(Adriaan D. Fokker, 1887–1972; Max K.E.L. Planck, 1858–1947)

Suppose the state $\mathbf{x}$ of a system at time t . Namely a probability distribution $p(\mathbf{x}, t)$ is known. We want to know further the probability distribution of the state at a later time. The phrase “forward” means for that $p(\mathbf{x}, t)$ serves as the initial condition and the FPK equation is integrated forward in time. This equation is given by

$$\frac{\partial p(\mathbf{x}, t | \mathbf{x}_0, t_0)}{\partial t} = - \sum_{j=1}^n \frac{\partial [f_j(\mathbf{x}, t) p(\mathbf{x}, t | \mathbf{x}_0, t_0)]}{\partial x_j} + \sum_{i,j=1}^n \frac{\partial^2 [a_{ij} p(\mathbf{x}, t | \mathbf{x}_0, t_0)]}{\partial x_i \partial x_j} \quad (11.104)$$

where

$$a_{ij} = \sum_{k=1}^n g_{ik} g_{jk} \quad (11.105)$$

and the term a_{ij} will also be used in Equations 11.106, 11.112, and 11.117.

11.2.2.1.3 Kolmogorov Backward Equation

Assume that a system state $\mathbf{x}(t)$ evolves according to the SDE given by Equation 11.100, this equation is given by

$$\frac{\partial p(\mathbf{x}, t | \mathbf{x}_0, t_0)}{\partial t_0} = - \sum_{j=1}^n f_j(\mathbf{x}_0, t) \frac{\partial [p(\mathbf{x}, t | \mathbf{x}_0, t_0)]}{\partial x_{0j}} + \sum_{i,j=1}^n a_{ij} \frac{\partial^2 [p(\mathbf{x}, t | \mathbf{x}_0, t_0)]}{\partial x_{0i} \partial x_{0j}} \quad (11.106)$$

11.2.2.2 Solutions of FPK Equation

It is noted that the FPK equation discussed to date is the only approach to have exact solutions of nonlinear random vibrations. Besides the exact solutions, the FPK equation can also be used to carry out approximations. The drawback of this method is that the responses must be Markovian.

11.2.2.2.1 Exact Solutions

In the case of existence of stationary solutions of FPK equations, we can find the solutions work for all first-order systems and a limited set of higher order systems. Specifically, an SDOF and/or MDOF vibration system with the aforementioned nonlinear damping and nonlinear stiffness excited by white noise process can have exact solutions.

The basic idea is, in a time reversal between t and $-t$, the response vector can be classified as composed of either even or odd functions; the even components will not change their sign, but the odd variables will have the sign changed. Denote the i th response as

$$\tilde{x}_i = \delta_i x_i \quad (11.107)$$

where $\delta_i = 1$ and -1 ; we can have even and odd variables, respectively.

Furthermore, we define the state of detailed balance as

$$p(\mathbf{x}, t | \mathbf{x}_0, t_0) = p(\tilde{\mathbf{x}}_0, t | \tilde{\mathbf{x}}, t_0) \quad t > t_0 \quad (11.108)$$

For the steady-state responses, in terms of the drift coefficient A_i and diffusion coefficient B_{ij} , this condition can be written as

$$A_i(\mathbf{x})p(\mathbf{x}) + \delta_i A_i(\tilde{\mathbf{x}})p(\mathbf{x}) - \frac{\partial [B_{ij}(\mathbf{x})p(\mathbf{x})]}{\partial x_j} = 0 \quad (11.109)$$

and

$$B_{ij}(\mathbf{x}) - \delta_i \delta_j B_{ij}(\tilde{\mathbf{x}}) = 0 \quad (11.110)$$

In Equations 11.109 and 11.110, summation on repeated indices is necessary.

When the above conditions are satisfied, the stationary solution can be expressed as

$$p(\mathbf{x}) = Ce^{-U(\mathbf{x})} \quad (11.111)$$

where C is a constant, $U(\mathbf{x})$ is the generalized potential; by solving $U(\mathbf{x})$, we can have the solution $p(\mathbf{x})$ described in Equation 11.111.

11.2.2.2.2 Moment Solutions

When the FPK equation has input to be white noise arising out of a non-Gaussian process, the solution becomes the nondiffusive Markov process. The solution will still have Markovian property, but the motion equation of the transitional PDF will have an infinite number of terms.

Soong (1973) derived the governing equations for the moments based on the FPK equation. The moments of the function $h[\mathbf{x}(t), t]$ of the response solution $\mathbf{x}(t)$ of Equation 11.100 can be given by

$$\frac{dE[h(\mathbf{x}, t)]}{dt} = \sum_{j=1}^n E \left(f_j \frac{\partial h}{\partial x_j} \right) + \sum_{i,j=1}^n E \left[a_{ij} \frac{\partial^2 h}{\partial x_i \partial x_j} \right] + E \left(\frac{\partial h}{\partial t} \right) \quad (11.112)$$

Upon setting

$$h[\mathbf{x}(t), t] = \prod_{i=1}^n x_i^{k_i} \quad (11.113)$$

and choosing different values for k_i , we can further derive equations for most commonly seen moments.

When the Markov property of response is used to study the first passage properties, either the Kolmogorov forward or backward equations can be solved in conjunction with appropriate boundary conditions imposed along the critical barriers.

Another approximate solution is to start with the Kolmogorov backward equation, based on which we can derive equations for moments of the first passage time, recursively.

Denote the time required by the response trajectory of Equation 11.100 at the point

$$\mathbf{x} = \mathbf{x}_0 \quad (11.114)$$

in the phase space at time t_0 to cross a special safe domain for the first time as $\tilde{T}(\mathbf{x}_0)$.

The moments

$$M_0 = 1 \quad (11.115)$$

$$M_k = E[\tilde{T}^k], \quad k = 1, 2, \dots, n \quad (11.116)$$

can be shown to have the governing equation given by

$$-\sum_{j=1}^n f_j(\mathbf{x}_0, t) \frac{\partial M_k}{\partial x_{0j}} - \sum_{i,j=1}^n a_{ij} \frac{\partial^2 M_k}{\partial x_i \partial x_j} + k M_{k-1} = 0, \quad k = 0, 1, 2, \dots \quad (11.117)$$

These equations are referred to as the *generalized Pontriagin–Vitt (GPV) equations* (Lev Semenovich Pontriagin 1908–1988).

11.2.2.2.3 Approximate Solutions

For nonlinear random vibration problems, in general, it is difficult to obtain exact solutions. We often use the iterative approach based on the parametrix method to obtain the existence and uniqueness of the corresponding partial differential equations.

Numerical solutions through a computational approach can be a second approach to obtain approximate solutions. In so doing, we often can find certain results. However, one of the important tasks of using the numerical solutions is to ensure the uniqueness of the solutions.

11.2.3 ALTERNATIVE APPROACHES

Besides the above-mentioned methods, there are several other alternative approaches to solve the problems of nonlinear random vibrations in the literature. In the following, we only briefly discuss these basic ideas for the nondiffusion Markov process.

11.2.3.1 Linearization

In Section 11.1, several nonlinear damping and stiffness were discussed. It is shown that one of the approaches is to linearize the nonlinear coefficients. That is, by using

certain equivalent linear parameters, the systems become linear. In other words, this approach is to find a linear system to approximate the nonlinear responses. The criteria of these approaches include equal force, equal displacement, equal energy, and their weighted combinations.

The linearization method is the most popular approach to deal with nonlinear systems. It is especially applicable for nonlinear systems under random excitation. Many observations have shown that the random excitations themselves linearize the responses, particularly in terms of aforementioned homogeneity. That is, the bound of the random responses tends to be more proportional to the amplitudes of the input bound than that of the deterministic excitations.

11.2.3.2 Perturbation

When the equations of motion possess nonlinear coefficients, slightly apart from linear parameters, the solution can be expanded in a power series by small parameters. This will lead to a set of linear equations, which is suitable for handling polynomial nonlinearities.

The above-discussed small parameter describes the difference between nonlinear and linear systems. In most cases, this method requires that the random forcing function should be additive and/or multiplicative.

When the above-mentioned parameters are sufficiently small, however, large errors can be expected.

11.2.3.3 Special Nonlinearization

Compared with the method of linearization, this approach uses certain equivalent nonlinear parameters, instead of using linear parameters. However, the equivalent nonlinear system can have closed-form solutions, or it can be easier to solve. The criteria of this approach are similar to those of linearization—equal force, equal displacement, equal energy, and their weighted combinations.

11.2.3.4 Statistical Averaging

If the nonlinear system only contains low capacity of energy dissipation, statistical averages may be used to generate diffusion Markov responses. That is, by using averages, the equivalent FPK equation approximates the nondiffusion cases.

The average can be the amplitudes and phases, which are typically performed in the frequency domain. It can also be energy envelopes as well as the combinations of both amplitudes/phases and energy envelopes.

This method often requires broadband excitations.

11.2.3.5 Numerical Simulation

For nonlinear random vibrations, numerical simulations are always powerful tools. Many above-mentioned approaches also require numerical simulations. The computational vibration solvers as well as related developments on numerical simulations have been well established, and they will be continuously developed and/or improved. It is worth mentioning that computational tools, such as MATLAB® and Simulink, can be a good platform to carry out the numerical simulations. In Section 11.3, we will discuss a special numerical simulation.

11.3 MONTE CARLO SIMULATIONS

11.3.1 BASICS OF MONTE CARLO METHOD

Monte Carlo simulations, also known as *Monte Carlo methods*, or *Monte Carlo experiments*, use repeated random samples to obtain numerical results to simulate engineering problems as well as problems in many other fields, such as physical and mathematical problems. These simulations are carried out multiple times over so that the distribution of targeted probabilistic entity can be statistically determined.

In the 1930s, Fermi first experimented with the Monte Carlo method while studying neutron diffusion. Since it was being kept as a secret, the similar work of von Neumann and Ulam in the Los Alamos Scientific Laboratory required a code name so that von Neumann chose the name *Monte Carlo*. In many cases, if a dynamic system involves random variables or random process, due to the uncertainty, the closed-form expression can be difficult to obtain. Monte Carlo simulations become useful tools to gain numerically statistical results. In other cases, although objects are deterministic, it can be difficult and/or costly to carry out the corresponding experiments or computations; Monte Carlo simulations again may be used to provide good approximations.

Monte Carlo methods are mainly used in four distinct problem classes: pattern recognition, optimization, numerical integration, and generation of PDFs.

In engineering applications involving random variables and processes, determining the desired values can be exceedingly difficult. The above-mentioned estimation of effective stiffness due to random low cycle fatigue is an example. In this case, the value of the effective stiffness is based on the maximum displacement and unloading stiffness. The newly resulted unloading stiffness is a function of the yielding displacement, in which the displacement is also a function of the stiffness. Additionally, both the loading and the displacement are random. Thus, finding a closed-form formula to calculate both the stiffness and the displacement within the current methods studied is not a feasible approach. For such a complicated circumstance, either experimental study or numerical simulation is needed to determine the effective stiffness.

The Monte Carlo method is one of the most practical computational simulations based on repeated random samplings. This method can simulate physical as well as mathematical systems, such as the previously mentioned effective stiffness for nonlinear vibrations. Monte Carlo simulation methods are particularly useful in dealing with a large number of coupled degrees and nonlinear responses, in addition to random environments. These methods are also widely used in mathematics—classically used for the evaluation of definite integrals, particularly multidimensional integrals with complex boundary conditions. When compared with alternative methods or human intuition, it is shown to be a widely successful method in risk analysis and reliability designs.

We now use an example to explain how a Monte Carlo simulation works. It is the computation of the value π by using the Monte Carlo method.

1. Draw a square on the ground with the length of its side to be 2, and then inscribe a circle within it (see Figure 11.16). We know that the area of the square is $2 \times 2 = 4$, and the area of the circle is $(2/2)^2\pi = \pi$.
2. Uniformly scatter various objects of uniform size throughout the square, for example, grains of sand.

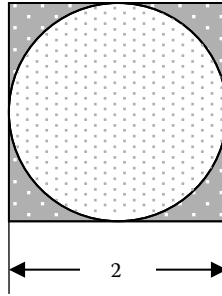


FIGURE 11.16 Square and circle.

3. Given that the two areas exhibit a ratio of $\pi/4$, the objects, when randomly scattered, should fall within the areas by approximately the same ratio. Thus, counting the number of objects in the circle and dividing by the total number of objects within the square should yield an approximated ratio of $\pi/4$.
4. Lastly, multiplying the result by 4 will then yield an approximation for π .

This method of π approximation describes the general pattern of Monte Carlo simulations. First, the domain of inputs is defined in the same way as the square that circumscribes the circle. Second, the inputs are generated randomly with specified distributions such as scatter individual grains within the square. In the above example, the distribution was uniform. Third, each input is counted as falling either within or outside of the circle. Finally, the results of each test are surveyed to determine the desired result, which is the approximation of π .

From this example, we can also realize three basic requirements. First, we need a comparatively large number of experiments: in the above example, inputting the grains. Second, these experiments should be uniformly carried out. That is, in the above example, the area of the grains should be the same, and the grains must be uniformly distributed. Third, the experiments must cover the entire area. In the above example, the area of the square must be fully covered.

Practically, the first two requirements are easier to realize. The third requirement needs a pre-knowledge of the targeted domain, which can be difficult to realize. In the above example, we knew the domain exactly, which is the square. However, in engineering applications, uncertainty on what is exactly the domain may exist.

Generally speaking, there is not a distinct Monte Carlo method but a large and widely used class of approaches. These common approaches for applying the Monte Carlo method consist of the following steps:

1. Define a domain of possible inputs or excitations.
2. Generate inputs randomly from the domain based on a specified probability distribution.
3. Perform a deterministic computation using the inputs.
4. Statistically aggregate the individual computations into the final result.

Although difficult, it is necessary to ensure that the large number of tests converge to the correct results.

11.3.1.1 Applications

Monte Carlo simulations have many applications, specifically in modeling with a significant number of uncertainties in inputs and system nonlinearities. In the following, some examples of applicable fields are briefly given.

11.3.1.1.1 Mathematics

In general, Monte Carlo simulations are used in mathematics in order to solve various problems by generating suitable random numbers and observing the fraction of the numbers that follow a specified property or properties. This method is useful for obtaining numerical solutions to problems that are too complicated to solve analytically. One of the most common applications of Monte Carlo simulations is the Monte Carlo integration.

11.3.1.1.1.1 Integration Let us first consider an integral expressed in Chapter 10—the total failure probability of a bridge subjected to multiple hazard loads. By just considering two loads, L_1 and L_2 , the formula is repeated as follows:

$$\begin{aligned}
 p_f &= p_f^{L_1} + p_f^{L_1 L_2} + p_f^{L_2} \\
 &= \left[\int_{-\infty}^{\infty} f_R(x) \int_x^{\infty} f_{L_1}(y) \int_{-\infty}^y f_{L_2}(z, x) \int_{-\infty}^y f_c(w, x) p_{L_1} dw dz dy dx \right] \\
 &\quad + \left[\int_{-\infty}^{\infty} f_R(x) \int_x^{\infty} f_c(y) \int_{-\infty}^y f_{L_1}(z, x) \int_{-\infty}^y f_{L_2}(w, x) p_{L_1 L_2} dw dz dy dx \right] \\
 &\quad + \left[\int_{-\infty}^{\infty} f_R(x) \int_x^{\infty} f_{L_2}(y) \int_{-\infty}^y f_{L_1}(z, x) \int_{-\infty}^y f_c(w, x) p_{L_2} dw dz dy dx \right] \quad (11.118)
 \end{aligned}$$

Equation 11.118 gives the total failure probability p_f when the calculation is deterministic, based on rigorously defining those PDFs and conditional probabilities, as well as establishing exact integral limits. However, both the analysis and the computation can be rather complex, even though numerical integration can be carried out.

Deterministic numerical integration is usually operated by taking a number of evenly spaced samples from a function. In general, such integration works well for functions of one variable. For multiple variables, there will be vector functions, for which deterministic quadrature methods may be very inefficient. For example, to numerically integrate a function of a two-dimensional vector, equally spaced grid points over a two-dimensional surface are required. In this case, a 100×100 grid requires 10,000 points. If the vector has 100 dimensions, the same spacing on the grid would require 100^{100} points. Note that, in many engineering problems, one dimension is actually a degree of freedom; an MDOF system with 100 degrees is considered a small-sized problem. A finite-element model can easily contain thousands of DOFs. Therefore, the corresponding computational burden is huge and impractical.

Monte Carlo simulations can sharply reduce the procedure of mathematical derivation on multiple integrals and the demands of integration over multiple dimensions. In many cases, the integral can be approximated by randomly selecting points within this 100-dimensional space and statistically taking the average of the function values at these selected points. It is well known that by the law of large numbers, Monte Carlo simulation will display N^{-2} convergence, which implies that regardless of the number of dimensions, quadrupling the number of sampled points will halve the error.

11.3.1.1.1.2 Improved Integration In many cases, the accuracy of numerical integration can be improved. Consider a Rayleigh distribution for example, where at point C, the CDF of the Rayleigh distribution reaches the peak value of 0.3033. The coordinates of A, B, D, and E are, respectively, $(A_x, A_y) = (0.25, 0.062)$, $(B_x, B_y) = (1.75, 0.2983)$, $(C_x, C_y) = (6.75, 0.053)$, and $(D_x, D_y) = (7.0, 0.038)$. Suppose that at each point, along the x -axis, we have an equal increment of 0.25. At point A, the change of the CDF is 20%. From point B to C, the change is 1.6%, and from point D to E, the change is 0.6%. This phenomenon is seen in all nonuniformly distributed probability curves. The integration accuracy may be improved by using unevenly spaced samples from the distribution function. In the neighborhood of point A, we may choose a smaller increase in the x -axis, and in the neighborhood of point D, we may choose a larger increase in the x -axis.

On the other hand, it is also seen that, at points A and/or D and E, the value of the PDF is comparatively much smaller than the region near points B and C. That is, the weighting factors of the integrand function will be different at different regions.

When performing Monte Carlo simulation, we can make the samples more likely to come from regions of high contribution to the integral, such as in the neighborhood of point B, as well as A, and use less samples from regions of low contribution, such as the neighborhood of point D. Namely, the points should be drawn from a distribution similar in form to the integrand. Note that the samples are still taken randomly.

However, accomplishing this improvement can be difficult for the PDF, or related functions are often unknown or uncertain. For this reason, approximate methods are used.

11.3.1.1.2 Physical Sciences

For computational physics, physical chemistry, and related applied fields, Monte Carlo simulations play an important part and have diverse applications, from complicated quantum chromodynamics calculations to designing heat shields and aerodynamic forms. Monte Carlo simulations are very useful in statistical physics; for example, Monte Carlo molecular modeling works as an alternative for computational molecular dynamics as well as for computation of statistical field theories of simple particle and polymer models (Baeurle 2009).

In experimental particle physics, these methods are used for designing detectors, understanding behavior, and comparing experimental data to theory and on a vastly large scale of galaxy modeling (MacGillivray and Dodd 1982). Monte Carlo methods are also used in ensemble models that form the basis of modern weather forecasting operations.

11.3.1.1.3 Monte Carlo Simulation versus “What If” Scenarios

From Figure 11.17, we can realize that the occurrences of samples are not equally weighted if the PDF of an occurrence is not uniformly distributed.

Consider deterministic modeling using single-point estimates, which can be seen at least philosophically as the opposite of the Monte Carlo simulation. In this deterministic modeling method, each uncertain variable within a model is assigned a “best estimation.” Various combinations of each input variable are manually chosen. For example, for each value, we can select the best, the worst, and the most likely cases. The results are then calculated for each so-called “what if” scenario (see Vose 2008).

On the other hand, Monte Carlo simulation utilizes random samples with PDFs as model inputs, which produce hundreds or even thousands of possible outcomes, comparing a few discrete “what if” scenarios. The results through Monte Carlo simulation therefore provide the possibilities of different outcomes occurring (also see Vose 2008).

11.3.1.1.4 Pattern Recognition

In many cases of a random process, it is necessary to identify certain patterns. In stricter terms, pattern recognition belongs to the category of system identification so that it is an inverse problem. While we will discuss the general inverse problem separately, specifically considered certain engineering patterns will be discussed.

In Chapter 7, Figure 7.7 shows an earthquake response spectrum and a design spectrum S_D . These spectra are actually generated by using the Monte Carlo pattern recognition. The domain of the input is all possible earthquake ground motions in an interested seismic zone with normalized peak amplitude, say, PGA of $0.4g$. This is the “definition of a domain of possible inputs or excitations.”

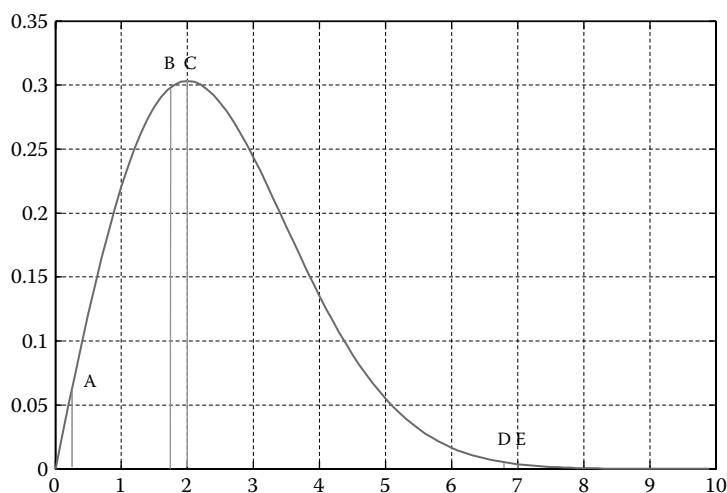


FIGURE 11.17 Rayleigh distribution.

If each ground motion time history is treated as a realization of a random process, then at a given natural period T_i , the output is the peak response, which can be seen as a sample through the “given” random excitation. Note that these time histories can be generated through numerical simulations; they can also be “randomly” picked up from the database of seismic records. From the mathematical point of view, both methods yield the same meaning, namely, “generation of inputs randomly from the domain based on a specified probability distribution.”

Then, each piece of ground excitation will be used to calculate the response of an SDOF system with period T_i , which is “to perform a deterministic computation using the input.”

To do the above computations, a large number of ground time histories will be used. Collection of all possible peak values under the given period T_i and “statistically aggregating the individual computations into the final result” will provide us the mean as well as the standard deviation of these responses. In addition, when the value of period T_i is varied in a proper region, we finally obtain a specific pattern, the response spectrum. In addition, based on the response spectrum, the design spectrum can also be generated.

Through this example, the procedure of pattern recognition can be explained. Practically speaking, we may have different kinds of patterns other than response spectra. However, the basic four-step procedure of Monte Carlo simulation essentially remains the same.

11.3.1.1.5 Optimization

Optimization can be computationally expensive. Therefore, Monte Carlo simulations are found to be very helpful for optimizations, especially for multiple dimensional optimizations. In most cases, Monte Carlo optimizations are based on random walks, which were briefly discussed in Chapter 5 (the Markov chains). Also in most cases, the optimization program will move around a marker in multidimensional space, seeking to move in directions that lead to a lower cost function of optimization. Although sometimes, it will move against the gradient, the average tendency is to gradually find the lowest value.

Compared to the “systematic” approach of optimization, the random walks may reduce the computational burden significantly.

11.3.1.1.6 Inverse Problems

In Chapter 9, we discussed the basic concepts and general procedure of engineering inverse problems and the need to consider randomness and uncertainty. This fact makes inverse problems more difficult to handle. One of the effective approaches is to use Monte Carlo simulations.

To solve inverse problems, Monte Carlo simulations are used to probabilistically formulate inverse problems. To do so, we need to define a probability distribution in the model space. This probability distribution combines a priori information with new information obtained by measuring and/or simulating selected observable parameters.

Generally speaking, the function linking data with model parameters can be nonlinear. Therefore, a posteriori probability in the model space may be complex to describe. In addition, the relationship can be multimodal, and some moments may not be well defined and/or have no closed-form explanations.

It is noted that to analyze an inverse problem, obtaining a maximum likelihood model is often insufficient. Typically, information on the resolution power of the data will also need to be known. In the general case, there may be a large number of model parameters. Furthermore, an inspection of the marginal probability densities of interest may be impractical or even useless. However, it is possible to pseudo-randomly generate a large collection of models according to the posterior probability distribution followed by analyzing the models in such a way that information on the relative likelihood of model properties is conveyed to the spectator.

This task can be accomplished by means of Monte Carlo simulation, especially in the case where no explicit formula for the a priori distribution is available.

In the literature, the best-known sampling method of significance is the Metropolis algorithm, which can be generalized, allowing the analysis of possible highly non-linear inverse problems with complex a priori information and data with an arbitrary noise distribution. For more detailed information, the work of Mosegaard and Tarantola (1995) or Tarantola (2005) may be consulted.

11.3.2 MONTE CARLO AND RANDOM NUMBERS

Intriguingly, Monte Carlo simulation methods do not always require true random numbers to be useful. However, for some applications, such as primality testing, unpredictability is vital (see Davenport 1992). Many of the most useful techniques use deterministic, pseudo-random sequences, making testing and re-running of simulations unproblematic. The only quality typically necessary to make good simulations is for the pseudo-random sequence to appear “random enough” in a certain sense.

Understanding this concept will depend on the application. In most cases, the pseudo-random sequence must pass a series of statistical tests. The most common requirement for statistical tests is to verify when a sufficient number of elements are present and that the numbers are uniformly distributed or follow an alternate desired distribution.

11.3.2.1 Generation of Random Numbers

As mentioned previously, Monte Carlo simulation needs to generate random samples. One of the key issues is to ensure that the occurrences of these samples are sufficiently “random.” To do so, we need to use artificially generated random numbers.

11.3.2.1.1 Linear Congruential Generator

The following formula can be used to generate uniformly distributed random variables between 0 and 1:

$$R_i = (aR_{i-1} + b) \text{ modulo } (c) \quad (11.119)$$

Here, a , b , and c are specific integers. The operation module indicates that (1) the variable $(aR_{i-1} + b)$ is divided by c , (2) the remainder is assigned to the value aR_i , and (3) the desired uniformly distributed random variable r_i is obtained as

$$r_i = R_i/d \quad (11.120)$$

The initial value R_0 is referred to as the seed.

For example, first let $a = 7$, $b = 8$, and $c = 9$; $R_0 = 1$.

$(7 \times 1 + 8) \text{ modulo } (9) \rightarrow 16/9 \rightarrow R_1 = 16 - 9 = 7$, $r_1 = 7/9 = 0.7778$; secondly, $(7 \times 7 + 8) \text{ modulo } (9) \rightarrow 57/9 \rightarrow R_2 = 57 - 9 \times 6 = 3$, $r_2 = 3/9 = 0.3333$; thirdly, $(7 \times 3 + 8) \text{ modulo } (9) \rightarrow 29/9 \rightarrow R_3 = 29 - 9 \times 3 = 2$, $r_3 = 2/9 = 0.2222$; fourthly, $(7 \times 2 + 8) \text{ modulo } (9) = 22/9 \rightarrow R_4 = 22 - 9 \times 2 = 4$, $r_4 = 4/9 = 0.4444$; etc.

To generate a random number between 0 and 1, use the MATLAB Code:

$$r = \text{rand}(n, 1) \quad (11.121)$$

11.3.2.2 Transformation of Random Numbers

To conduct a Monte Carlo simulation for a specific process, the statistical properties generated by the computers should be identical to this particular process. However, the above-mentioned random numbers are uniformly distributed. Therefore, they must be converted to the desired probability distribution.

11.3.2.2.1 Continuous Random Variables

11.3.2.2.1.1 Inverse Transformation Method If the inverse form of the desired CDF can be expressed analytically, then the inverse transformation method can be used.

First, a random set of uniformly distributed variables r are generated as mentioned above. Next, suppose that $F_Q(q)$ is the desired CDF, and its inverse F_Q^{-1} is known.

The variable Q can then be generated as

$$Q = F_Q^{-1}(r) \quad (11.122)$$

Example 11.1

Consider the CDF of the exponential distribution:

$$F_X(x) = 1 - e^{-\lambda x} \quad (11.123)$$

The inverse function is

$$x = F_X(r)^{-1} = -\ln(1 - r)/\lambda \quad (11.124)$$

For $r = \{0.7778, 0.3333, 0.2222, 0.4444, \dots\}$ and $\lambda = 2$, this will yield

$$Q = \{0.7521, 0.2027, 0.1256, 0.2939, \dots\}$$

11.3.2.2.1.2 Composition Method Certain random variables have composite probability distributions such as

$$F_X(x) = \sum_{i=1}^n w_i F_{X_i}(x) \quad (11.125)$$

where w_i are weighting functions, $F_{X_i}(x)$ are CDFs, and

$$\sum_{i=1}^n w_i = 1 \quad (11.126)$$

To obtain the corresponding distribution, two sets of uniformly distributed random numbers r_1 and r_2 must first be generated. The random numbers r_1 are then used to select a desired CDF of $F_{X_i}(x)$ for the generation of random numbers. To determine the variable set based on the selected CDF, the random variable r_2 is used.

Example 11.2

Generate a random variable set that satisfies the following PDF:

$$f_X(x) = 3/5 + x^3, \quad 0 \leq x \leq 1$$

The PDF can be decomposed as

$$f_X(x) = 3/5 f_1(x) + (1 - 3/5)f_2(x)$$

It is seen that

$$f_1(x) = 1 \text{ and } F_1(x) = x = r_2, \quad \text{for } 0 \leq x \leq 1$$

The variate is given by

$$x = F_1^{-1}(r_2) = r_2$$

for $f_2(x) = 5/2 x^3$, yielding $F_2(x) = 5/8 x^4 = r_2$, for $0 \leq x \leq 1$.
Thus, the resulting variate is

$$x = F_2^{-1}(r_2) = (8/5 r_2)^{1/4}$$

Subsequently, two uniformly distributed numbers are generated. Suppose that the numbers are randomly chosen as 0.530 and 0.181.

Given that $r_1 = 0.535 < 3/5 = 0.8$. By following $F_1(x)$, r_2 is used to generate the variate by

$$x_1 = F_1^{-1}(r_2) = 0.181$$

Next, let $r_1 = 0.722$ and $r_2 = 0.361$. Since $r_1 = 0.722 > 3/5$, r_2 is used to generate the variate by following $F_2(x)$ or explicitly

$$x_2 = F_2^{-1}(r_2) = (8 \times 0.361/5)^{1/2} = 0.870$$

This is similarly repeated.

11.3.2.2.1.3 Function-Based Method In specified distributions, random variables are expressed as functions of other random variables, which can be easier to generate. For example, the Gamma distribution is the sum of exponential distributions. If Y is a Gamma distribution with the parameter (k, λ) , then

$$Y_i = X_{1i} + X_{2i} + \dots X_{ki} \quad (11.127)$$

Here, X_{1i} , X_{2i} , ..., X_{ki} are k independent and exponentially distributed random variables with parameter λ and zero mean.

$$f_{X_j}(x) = e^{-\lambda x}/\lambda \quad (11.128)$$

Example 11.3

Generate a Gamma-distributed random variate with the parameters $k = 4$ and $\lambda = 2$.

First, generate uniformly distributed random numbers r within the range $[0, 1]$. Next, let

$$x = -\ln(r)/\lambda \quad (11.129)$$

As an example, the first four uniformly distributed random numbers are $r_1 = 0.7778$, $r_2 = 0.3333$, $r_3 = 0.2222$, and $r_4 = 0.4444$. The corresponding exponentially distributed random variates for $\lambda = 2$ will be as follows: 0.7521, 0.2027, 0.1256, and 0.2939.

Thus, the Gamma distributed variate is

$$g = \frac{1}{\lambda} \sum_{i=1}^4 \ln(r_i) = (0.7521 + 0.2027 + 0.1256 + 0.2939) = 1.3743$$

11.3.2.2.2 Discrete Random Variables

To generate discrete random variables is to transform a uniformly distributed random number to the desired probability mass function. One way to accomplish this is to handle the CDF as follows:

$$F_X(x_{j-1}) < r \leq F_X(x_j) \quad (11.130)$$

11.3.2.3 Random Process

11.3.2.3.1 Stationary Process

Now we use the aforementioned methods to generate a random process, which is weakly stationary.

First, generate a random variate with the desired distribution. Then, index the variate with the temporal parameter t . For example,

$$F(t) = A \cos(\Omega t + \Theta) \quad (11.131)$$

Here, A , Ω , and Θ are random variables, and $F(t)$ is a random process.

11.3.2.3.2 Nonstationary Process

Change the stationary process with certain desired modifications. For example,

$$G(t) = B(t)F(t) + \mu_Y(t) \quad (11.132)$$

11.3.3 NUMERICAL SIMULATIONS

Monte Carlo simulations generate samples from which the PDFs can be statistically determined. These samples are often obtained through numerical simulations.

Although the totality of numerical simulations is beyond the scope of this textbook, in the following, we briefly discuss several issues related to random vibrations and Monte Carlo simulations.

11.3.3.1 Basic Issues

In order to successfully render a Monte Carlo simulation, several basic issues include the establishment of proper models, the mathematical tools for modeling, the criteria used to judge if the simulation is acceptable, and the possible error or error bound between exact resulted and Monte Carlo simulations.

11.3.3.1.1 Models

In many engineering projects, proper modeling is a starting point, and it is an important issue, specifically for the case that relates to engineering vibration and the need to construct proper models. In what follows, we will use vibration systems to realize the issue of modeling.

For a vibration system, the complete model is the physical model, that is, use the M-C-K vibration model to directly treat the Monte Carlo simulation as a forward problem. For such a model, we need to determine the order of the system, the mass, the damping and stiffness coefficient matrices, as well as the forcing functions. We also need to decide whether they are linear or nonlinear models. Furthermore, we also need to consider certain randomness and uncertainty of models. Both the response model and the modal model are incomplete models. That is, to establish a proper model, the physical model, if possible, should be the first choice.

In many cases, the physical model is difficult to obtain. Practically, response models can be directly measured. Under certain signal processing, we can further obtain

frequency spectra, as well as other parameters such as damping ratios, through analyses in both time and frequency domains, which were mentioned in Chapter 9, in Section 9.3.1.3 of vibration testing. Based on these measured parameters, we can further simulate the response models. The reason that the response model is incomplete is that only from this model the physical parameters cannot be determined, for example, the corresponding mass.

A modal model can also be used. Since each mode is actually an SDOF system, with modal mass, damping, and stiffness, the discussion on physical models also applies. However, if the modal model is used, the system is assumed to be linear.

11.3.3.1.2 Criteria

In most cases, the reason for using Monte Carlo simulation is that it is difficult to directly compute the results, such as computation of integrations. In this case, how to judge whether the simulated result is valid is an important issue.

11.3.3.1.2.1 Consistency Consistency means that the results obtained through different groups of Monte Carlo simulations agree with each other with acceptable difference.

The consistency criterion is comparatively easier to realize, and in many cases, it is necessary to check if different groups of Monte Carlo simulations agree with each other.

For convenience, denote a specific parameter of a system obtained through the i th Monte Carlo simulation to be $\hat{\pi}_i$. Then the criterion of consistency of parameter $\hat{\pi}_i$ in total m simulations can be written as

$$\frac{\left| \hat{\pi}_i - \sum_{i=1}^m \hat{\pi}_i \right|}{\sum_{i=1}^m \hat{\pi}_i} \leq \epsilon_{\hat{\pi}} \quad (11.133)$$

where $\epsilon_{\hat{\pi}}$ is a preset small number specifically for parameter $\hat{\pi}_i$.

The choice of a different group can simply be a different run of Monte Carlo simulations with the same samples. It can also be the Monte Carlo simulation with a different number of generated samples, etc.

11.3.3.1.2.2 Unbiased Results Unbiased results mean that the statistical parameter $\hat{\pi}_i$ must be as close as the corresponding exact parameter π , that is,

$$\frac{\left| \pi - \sum_{i=1}^m \hat{\pi}_i \right|}{\pi} \leq \epsilon_{\pi} \quad (11.134)$$

where ϵ_{π} is a preset small number specifically for parameter $\hat{\pi}_i$.

Generally speaking, we often do not know the exact value of π , so that Equation 11.134 often cannot be realized.

11.3.3.2 Deterministic Systems with Random Inputs

11.3.3.2.1 Response Computation

Suppose that the mass, damping, and stiffness matrices of a system are given. With the generated forcing process $F_i(t)$,

$$\mathbf{M}\ddot{\mathbf{X}}_i + \mathbf{C}\dot{\mathbf{X}}_i + \mathbf{K}\mathbf{X}_i = \mathbf{F}_i(t) \quad (11.135)$$

The response can be calculated. Note that, since the generated forcing function is in fact known, Equation 11.135 can be rewritten as

$$\mathbf{M}\ddot{\mathbf{x}}_i + \mathbf{C}\dot{\mathbf{x}}_i + \mathbf{K}\mathbf{x}_i = \mathbf{f}_i(t) \quad (11.136)$$

Equation 11.136 can be solved as discussed in Chapter 8.

When the system is linear, the statistical properties of the responses can have a closed-form expression as mentioned in Chapters 7 and 8. However, in many cases, $\mathbf{C}$ and $\mathbf{K}$ can be nonlinear as previously mentioned. Moments and additional statistical studies may be needed to complete estimations on the PDF.

11.3.3.2.2 Statistical Estimation of Random Parameters

11.3.3.2.2.1 Simulated Random Process Once the responses $\mathbf{x}_i(t)$ are calculated through the generated $\mathbf{f}_i(t)$, the responses can be obtained as a collection of random processes, that is,

$$\mathbf{X}(t) = \{\mathbf{x}_i(t)\} \quad (11.137)$$

11.3.3.2.2.2 PDF By means of $\mathbf{X}(t)$, the distribution function can be studied. As previously discussed, a proper model of the PDF or the CDF must first be specified. Then, through the maximum likelihood estimation of parameters and curve fitting, the exact function can be determined.

11.3.3.2.2.3 Crossing Rate, Extrema Distribution The study of the rates of level up-crossing, zero up-crossing, the distribution of extrema, and the possible bounds of responses, for example, is possible through $\mathbf{X}(t)$.

11.3.3.2.2.4 Mean, Variance, and Correlations The first and second moment estimations can be calculated as previously discussed.

11.3.3.3 Random Systems

11.3.3.3.1 Random Distribution of Modal Parameters

Consider a dynamic system with a set of parameters denoted by π_i , $i = 1, 2, \dots$. For certain cases, the i th parameter of π_i for the system may vary within a relatively small range, approximately $\epsilon_{\pi_i}\%$. Note that a small variation of parameter π_i does not

necessarily indicate a small variation for the resulted response. In fact, the response variation depends upon the stability of the system. When the variation of π_i is random, the random variable Π_i can be written as

$$\Pi_i = \pi_i (1 + \varepsilon_\pi \%) \quad (11.138)$$

Here, the resulted variation in π_i forms a set of random variables Π_i .

Assume that the system being analyzed remains linear; for unspecified reasons, its modal parameters may vary. For the corresponding natural frequency and damping ratio, Monte Carlo simulations can be used as follows:

$$\Omega_i = \omega_i (1 + \varepsilon_\omega \%) \quad (11.139)$$

Here, Ω_i is the i th natural frequency and is considered to be random, whereas ω_i is the i th designed natural frequency and is considered to have a desired value. Additionally, ε_ω is a random variable with a proper distribution. For example,

$$\varepsilon_\omega \sim N(\mu, \sigma) \quad (11.140)$$

Given that the damping ratios are also assigned as random variables, then Z_i can be written as

$$Z_i = \zeta_i (1 + \varepsilon_\zeta \%) \quad (11.141)$$

In this instance, Z_i is the i th damping ratio and is considered to be random. Similarly, ζ_i is the i th designed damping ratio and is considered to have a desired value, while ε_ζ is a random variable with proper distributions.

Given a system with artificially assigned variable modal parameters, the responses $\mathbf{X}(t)$ can be calculated, and further statistical study can be completed as mentioned above.

11.3.3.3.2 Random Distribution of Physical Parameters

For random physical parameters, the following can be written:

$$K_{ij} = k_{ij} (1 + \varepsilon_k \%) \quad (11.142)$$

and

$$C_{ij} = c_{ij} (1 + \varepsilon_c \%) \quad (11.143)$$

Here, K_{ij} and C_{ij} are the ij th entries of the stiffness and damping matrices, respectively, which are considered to be random. Likewise, k_{ij} and c_{ij} are the ij th designed stiffness and damping, respectively. They are considered to have desired values. Lastly, ε_c are random variables with proper distributions.

The variation of k_{ij} and c_{ij} can be relatively large. Mathematically speaking, the range of random variables K_{ij} and C_{ij} can be $-\infty$ to ∞ . In this case, a normal distribution to model the corresponding variation distributions may be used. In engineering

applications, variables K_{ij} and C_{ij} cannot be smaller than zero. However, in this case, the use of a lognormal distribution to model the corresponding variation distributions is acceptable.

Example 11.4

Consider the stiffness of a steel shaft with length ℓ and a cross section A . Given deterministic design, this example is described by

$$k = EA/\ell \quad (11.144)$$

Suppose that the Young's modulus E , area A , and length ℓ are all random variables with a normal distribution as follows:

$$E \sim N(E_0, 0.2E_0) \quad (11.145)$$

$$A \sim N(A_0, 0.05A_0) \quad (11.146)$$

$$\ell \sim N(\ell_0, 0.08\ell_0) \quad (11.147)$$

Note for this instance that $(\cdot)_0$ is the mean value of $(\cdot)$. Then, the resulted stiffness is also a random variable, thus indicating that it is not normal any more.

PROBLEMS

1. A nonlinear vibration SDOF system is shown in Figure P11.1, with m , c , and k and the dry friction coefficient μ . Suppose that it is excited by a force $f(t) = \sin(\omega t)$.
 - a. Find the governing equation for the system.
 - b. Calculate the energy loss and determine the amplitude and phase of the forced response of the equivalent linear viscous system.
2. An SDOF system is given by

$$10\ddot{x} + \mu |\dot{x}|^\beta \operatorname{syn}(\dot{x}) + 40x = -10\ddot{x}_g(t)$$

where the mass is 10 kg; $k = 40$ N/m, $\ddot{x}_g(t) = -A \sin(2.1t)$ m/s², $\beta = 0.3$, and $\mu = 15$ N (s/m) ^{β} .

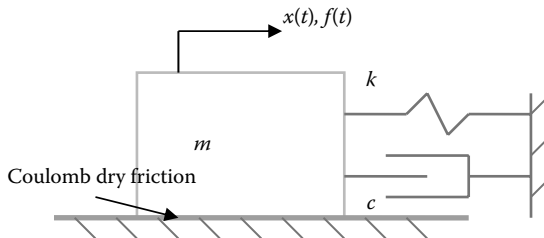


FIGURE P11.1

- a. Calculate the responses by using Simulink; plot the absolute accelerations, relative displacements, and the energy dissipation loops when $A = 0.2, 0.15, 0.1$, and 0.05 m/s^2 . Plot the steady-state displacement vs. the bound of ground acceleration.
- b. Calculate the effective damping ζ_{eff} through Timoshenko damping (Equation 11.34) when $A = 0.2, 0.15, 0.1$, and 0.05 m/s^2 . Plot the damping ratio vs. ground acceleration.
- c. Use the effective damping ratios obtained on (b) to linearize the system and calculate the responses, comparing the results with (a) when $A = 0.2, 0.15, 0.1$, and 0.05 m/s^2 .
3. Use $x = \text{randn}(\text{SampNum}, 1) * \text{Amp}$ and let $\text{Amp} = 10, 20, 30$, and 40 m/s^2 to generate 300 pieces of excitations acting on the system given in 11.2.
 - a. Calculate the relative displacements accordingly. Find the mean and standard deviation of these displacements.
 - b. Use the above-mentioned random input to study the relationship between the input amplitude and the mean peak responses.
4.
 - a. Calculate the linearized FRF by using $H(\omega) = \sum X_i(\omega)/F_i(\omega)$. The input is $\text{randn}(\text{SampNum}, 1) * \text{Amp}$ and let $\text{Amp} = 5, 10, 20, 30$, and 40 m/s^2 to generate 300 pieces of excitations acting on the system given in 11.2.
 - b. Using the half-power point method to calculate the linearized damping ratio, compare the results with those obtained in 11.2.
 - c. Examine the linearity.
5.
 - a. Find the effective stiffness k_{eff} by using the approach of secant stiffness of the linearized model for the system given in Problem 2 with linearized damping ratio $\zeta_{\text{eff}} = 0.29$.
 - b. Discuss the method of using secant stiffness to approximate the effective stiffness of a sublinear system, such as the one given in Problem 2 with $50 \text{ randn}(\text{SampNum}, 1)$.
6. Use the MATLAB code of “lsim” to compute the absolute acceleration and relative displacement through this linearized model based on parameters k_{eff} and ζ_{eff} , with the El Centro ground acceleration (E-W) as the excitation. The unit of the mass, damping, and stiffness is kg, N/m s, and N/m, respectively. The peak ground acceleration (PGA) of the El Centro earthquake is normalized to 4 m/s^2 .
7. An SDOF system is given by

$$10\ddot{x}(t) + c\dot{x}(t) + f_R(t) = 10\ddot{x}_g(t)$$

where $f_R(t)$ is the nonlinear restoring force. If $\ddot{x}_g(t) = 10 \sin(6t)$, then the restoring force can be plotted in Figure P11.2, with $c = 0$, $d_y = 0.02$, $q = 7$, $f_m = 10$, and $x_0 = 0.1$.

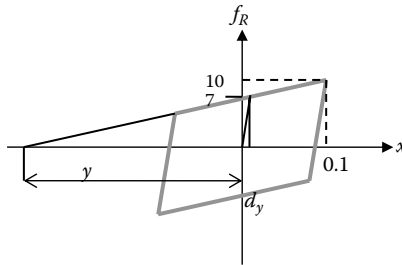


FIGURE P11.2

- a. Calculate the effective stiffness by using secant stiffness and the energy approach given by Equation 11.28.
- b. Calculate the effective damping ζ_{eff} through Timoshenko damping (Equation 11.34) and alternative damping (Equation 11.46).
- c. Use the MATLAB code of “lsim” to compute the relative displacement through this linearized model with k_{eff} and ζ_{eff} , with the El Centro ground acceleration (E-W) as the excitation. The unit of the mass, damping, and stiffness is kg, N/m s, and N/m, respectively. The PGA of the El Centro earthquake is normalized to 10 m/s². The linearized models include the following:

Case 1: Let the effective stiffness and the damping ratio calculated through the approach of using the secant stiffness be $k_{\text{eff}} = 100$ and $\zeta_{\text{eff}} = 0.14$, respectively.

Case 2: Let the effective stiffness and the damping ratio calculated through the energy approach be $k_{\text{eff}} = 34.4$ and $\zeta_{\text{eff}} = 0.35$, respectively.
8.
 - a. Create a Simulink model for the system shown in Problem 7.
 - b. Use the El Centro ground acceleration again to compute the absolute acceleration and relative displacement based on your Simulink model with PGA = 10 m/s² for the following cases:

Case 1: Let the effective stiffness and the damping ratio calculated through the approach of using the secant stiffness be $k_{\text{eff}} = 100$ and $\zeta_{\text{eff}} = 0.14$.

Case 2: Let the effective stiffness and the damping ratio calculated through the energy approach be $k_{\text{eff}} = 34.4$ and $\zeta_{\text{eff}} = 0.35$.

Case 3: nonlinear response by Simulink.
 - c. Use the Simulink model and let the PGA be 2.5, 5, 10, 15, 20, 25, and 30 m/s² to study the linearity of the peak responses of both the accelerations and the displacements.
9. Write a MATLAB code to carry out a Monte Carlo simulation for determining the upper-left area inside a square with each side = 6 cm, shown in Figure P11.3, where the red curve has the function $y = 0.24x^2$.

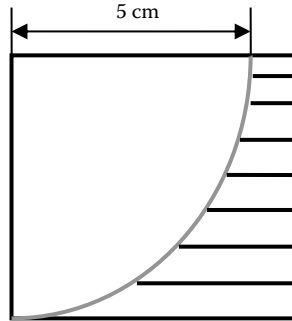


FIGURE P11.3

10.

- a. Write a MATLAB program for a random number generator based on the following Rayleigh distribution:

$$f_H(y) = \frac{y}{\sigma^2} e^{-\frac{1}{2}\left(\frac{y}{\sigma}\right)^2}, \quad 5 > y > 0, \quad \sigma = 1.5 \text{ m/s}^2$$

- b. Use Monte Carlo simulation to generate 500 pieces of random ground accelerations

$$xga(t) = y \text{ xgn}(t)$$

where y is the amplitude and xgn is the normalized earthquake records. The normalized time histories are chosen to be El Centro.

- c. With the generated ground recalculation to calculate the peak responses (both absolute acceleration and relative displacement) of the system shown in Problem 7, with the linearized model (k_{eff} , z_{eff}) of Case 2 obtained in Problem 7.
- d. With the generated ground recalculation to calculate the peak responses (both the absolute acceleration and the relative displacement) of the system shown in Problem 7, with the Simulink model obtained in Problem 7.

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RANDOM VIBRATION

Mechanical, Structural, and Earthquake Engineering Applications

Focuses on the Basic Methodologies Needed to Handle Random Processes

After determining that most textbooks on random vibrations are mathematically intensive and often too difficult for students to fully digest in a single course, the authors of ***Random Vibration: Mechanical, Structural, and Earthquake Engineering Applications*** decided to revise the current standard. This text incorporates more than 20 years of research on formulating bridge design limit states. Utilizing the authors' experience in formulating real-world failure probability-based engineering design criteria and their discovery of relevant examples using the basic ideas and principles of random processes, the text effectively helps students readily grasp the essential concepts. It eliminates the rigorous math-intensive logic training applied in the past, greatly reduces the random process aspect, and works to change a knowledge-based course approach into a methodology-based course approach. This approach underlies the book throughout, and students are taught the fundamental methodologies of accounting for random data and random processes as well as how to apply them in engineering practice.

Gain a Deeper Understanding of the Randomness in Sequences

Presented in four sections, the material discusses the scope of random processes, provides an overview of random processes, highlights random vibrations, and details the application of the methodology. Relevant engineering examples, included throughout the text, equip readers with the ability to make measurements and observations, understand basic steps, validate the accuracy of dynamic analyses, and master and apply newly developed knowledge in random vibrations and corresponding system reliabilities.

Random Vibration: Mechanical, Structural, and Earthquake Engineering Applications effectively integrates the basic ideas, concepts, principles, and theories of random processes. This enables students to understand the basic methodology and establish their own logic to systematically handle the issues facing the theory and application of random vibrations.

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